

# Intelligent Security API

## Developer Guide

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# 1 Reading Guide

| Chapter | Description |
| --- | --- |
| Overview | Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history. |
| ISAPI Framework | Read the chapter to take a quick look at the ISAPI framework and basic functions. |
| Quick Start Guide | Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading. |
| API Reference | Start programming according to API definitions. |
| How-To Video Guidance | How-to videos demonstrate detailed steps of different integration tasks. |

# 2 Overview

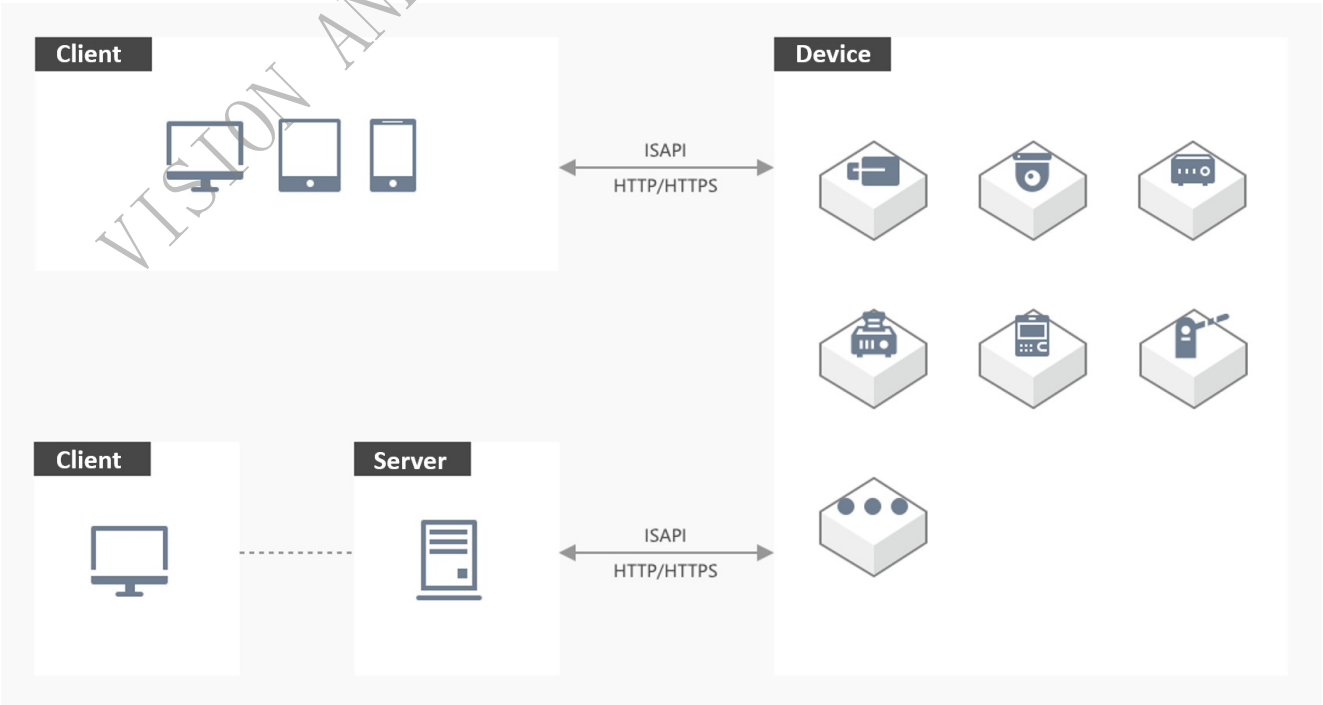
## 2.1 Introduction

Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

### 2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.




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### 2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

| | | | |
|---|---|---|---|
| Application Layer | Device Discovery<br>SADP | Signaling Interaction<br>ISAPI (HTTP) | Media Stream<br>RTSP/RTP |
| Transport Layer | MCAST | TCP | TCP/UDP |
| Network Layer | IP | | |
| Network Interface | Hardware Drive Interface | | |

## 2.2 Product Scope

- Traffic Cameras
  - Capture Cameras

    iDS-2CD9036-ESU, iDS-2CD9036-GSU, iDS-2CD9056-BIS, iDS-2CD9056-GIS, iDS-2CD9096-BIS, iDS-2CD9096-GIS, iDS-2CD9126-ASU, iDS-2CD9136-AES, iDS-2CD9136-AIS, iDS-2CD9136-AS, iDS-2CD9136-BIS, iDS-2CD9396-AES, iDS-2CD9396-AIS, iDS-2CD9396-AS, iDS-2CD9396-BES, iDS-2CD9396-BIS, iDS-2CD9396-BS, iDS-2CD9396-FS, iDS-2CD9396-HES, iDS-2CD9396-HIPS, iDS-2CD9396-HIS, iDS-2CD9525-ASD, iDS-2CD9545-BRS, iDS-2CD9545-BSZ, iDS-2CD9796-FEMFS, iDS-2CD9796-FEMS, iDS-2CD9796-HEMFS, iDS-2CD9796-HEMS, iDS-2CD9796-HMS, iDS-2CD97C6-HEMFS, iDS-2CD9856-BES, iDS-2CD9856-BIS, iDS-2CD9856-BS, iDS-2CD9856-HES, iDS-2CD9856-HIPS, iDS-2CD9856-HIS, iDS-2CD9885-CSS, iDS-2CD9896-CESS, iDS-2CD9896-CISS, iDS-2CD9896-FESS, iDS-2CD9896-FISS, iDS-2CD9896-HESS, iDS-2CD9896-HISS, iDS-2CD98C6-CILS, iDS-2CD98C6-HES, iDS-2CD98C6-HIS, iDS-2CD98G6-FES, iDS-2CD98G6-FIS, iDS-2CD98G6-HES, iDS-2CD98G6-HIS, iDS-2CD98K6-FES, iDS-2CD98K6-FIS, iDS-2CD98K6-FS, iDS-2CD98K6-HES, iDS-2CD98K6-HIS, iDS-TCH402-B

## 2.3 Terms And Definitions

### 2.3.1 Full Power Consumption Mode

Device operates in full consumption mode.

### 2.3.2 Low Power Consumption Mode

For multi-chip devices, only the main chip is powered on; for single-ship devices, the video image quality, frame rate, bit rate, etc. is not affected; 4G/5G chips work normally.

### 2.3.3 Real-Time Low Power Consumption Mode

For multi-chip devices, only the main chip is powered on; for single-ship devices, the video image quality, frame rate, bit rate, etc. might be affected; 4G/5G chips work in low power consumption keepalive mode (the device sends a heartbeat packet every 5 minutes).

### 2.3.4 Standby Mode

If no operation is performed on the device within 5 minutes or the device battery is lower than 20%, the device will go to sleep.

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### 2.3.5 AOV (Always-On Video) Mode

If there is no device operation for 5 minutes or the battery is below 20%, the device will enter sleep mode. In the sleep mode, if recording is triggered, the device will save the videos by extracting live view frames.

### 2.3.6 Event

Event refers to the information uploaded by the device actively. Event needs to be uploaded by the device in real time for the immediate response from the platform. If the device is offline, the event can be stored in the cache first and then be uploaded again when the connection is restored.

### 2.3.7 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

### 2.3.8 Traffic Enforcement Event Detection

The traffic enforcement event detection supports the function of violation enforcement, which can automatically detect and capture the evidence of events, including illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, and motor vehicle on non-motor vehicle lane, on the roads with no more than four lanes. The enforcement results will be uploaded as alarms in real time. The difference between traffic incident detection and traffic enforcement event detection is that the former only detects events which are not always illegal (e.g., congestion, roadblock, etc.) and the latter detects events of violating the traffic rules. For the traffic enforcement event detection (e.g., illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, etc.), the detailed violation information needs to be recorded, including the background picture, close-up picture, vehicle picture, and license plate number.

### 2.3.9 Traffic Incident Detection

AID (Automatic (Traffic) Incident Detection) is an intelligent traffic function for detecting traffic incidents that occurred on the roads with no more than four lanes and triggering alarms in real time. Supported incidents include thrown object detection, pedestrian detection, congestion detection, roadblock detection, traffic accident detection, fog detection, etc. AID is also known as traffic incidents of vehicles. The difference between traffic incident detection and traffic enforcement event detection is that the former only detects events which are not always illegal (e.g., congestion, roadblock, etc.) and the latter detects events of traffic rule violation. For the traffic enforcement event detection (e.g., illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, etc.), the detailed violation information needs to be recorded, including the background picture, close-up picture, vehicle picture, and license plate number.

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## 2.4 Symbols And Acronyms

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

HTTP: Hypertext Transfer Protocol.

OSD: On-screen display

PTZ: Pan-Tilt-Zoom

ANPR: Automatic Number Plate Recognition

TFS: Traffic Enforcement Event Detection (TFS is the abbreviation for Traffic Forensics System)

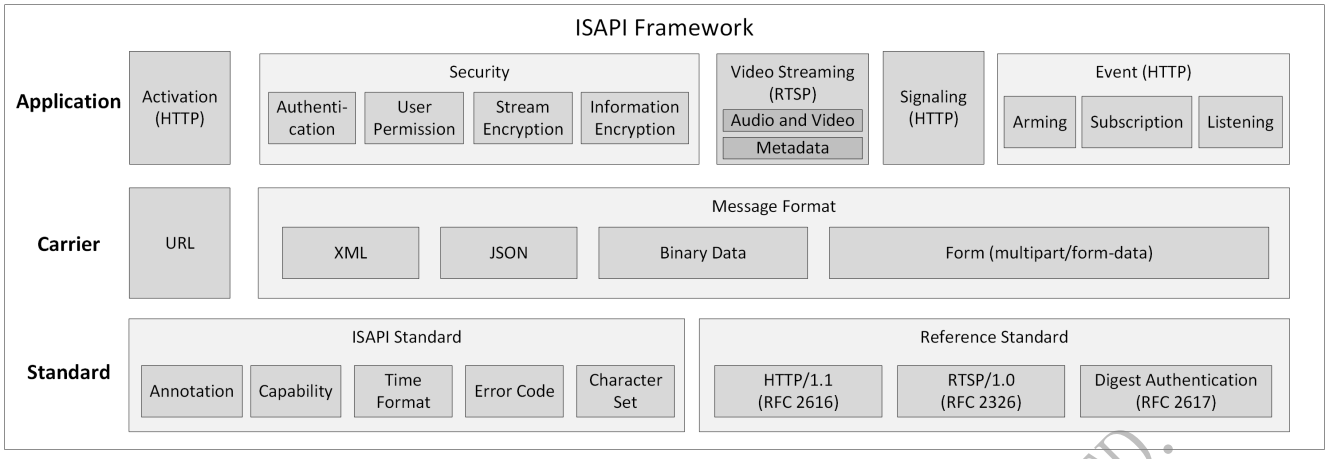
AID: Automatic (Traffic) Incident Detection

## 2.5 Update History

No update record

# 3 ISAPI Framework

## 3.1 Overview




**Notes**:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

## 3.2 Activation

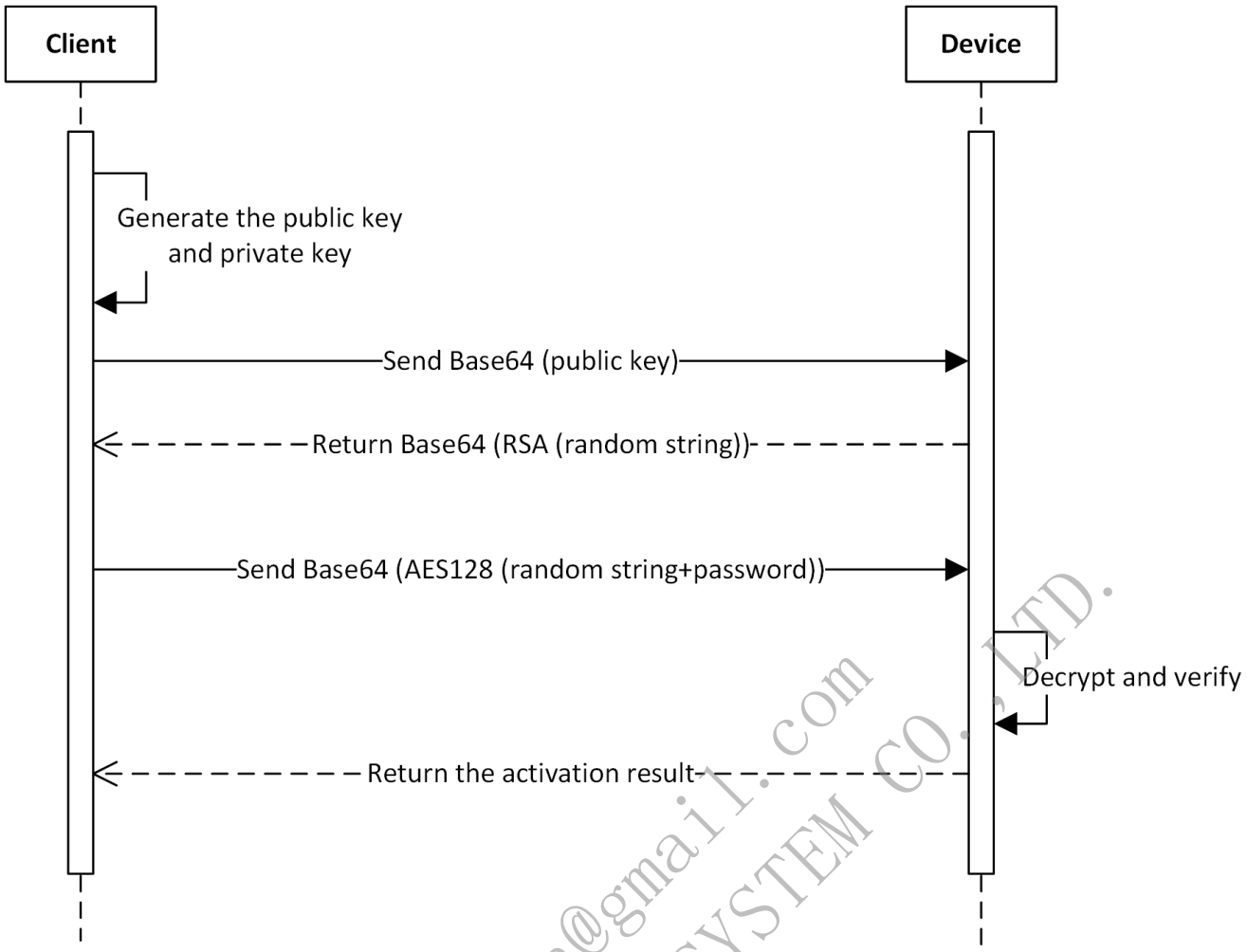
The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.

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Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.

7. The device verifies the password and returns the activation result.

**Notes**:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

# 3.3 Security Mechanism

## 3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

## 3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

**Administrator**: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

**Operator**: Has the permission to access general resources and a part of advanced resources.

**Normal User**: Only has the permission to access general resources.

## 3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

# 3.4 Video Streaming

## 3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

## 3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

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# 4 Quick Start Guide

## 4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

### 4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application
reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

### 4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

### 4.1.3 Java (HttpClient)

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```java
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

### 4.1.4 Python (requests)

```python
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

## 4.2 Message Parsing

### 4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

#### 4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

#### 4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

#### 4.2.1.3 Binary Data

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For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

#### 4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

**Notes**

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

| Node Name | Form Field | Description |
|---|---|---|
| pid | name | pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers. |
| contentid | Content-ID | contentid in XML/JSON messages corresponds to Content-ID in form headers. |
| filename | filename | filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers. |

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```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```

### 4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#Level 1,level2#Level 2,level3#Level 3]-->level1
    </level>
  </Node>
</NodeList>
```

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```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

| Annotation | Description | Remark |
| --- | --- | --- |
| ro | Attribute: Read-Only | This field can only be obtained and cannot be edited. |
| wo | Attribute: Write-Only | This field can only be edited and cannot be obtained. |
| req | Attribute: Required | This field is required for request messages sent to the device and response messages returned from the device. |
| opt | Attribute: Optional | This field is optional for request messages sent to the device and response messages returned from the device. |
| dep | Attribute: Dependent | This field is valid and required when specific conditions are satisfied. |
| object | Field Type: Object | The field of type object contains multiple sub-fields. |
| list | Field Type: List | The subType following it refers to the data type of sub-items in the list. |
| subType | Field Type: String | The range following it refers to the maximum and the minimum string size of the field. |
| int | Field Type: Int | The range following it refers to the maximum and the minimum value of the field. |
| float | Field Type: Float | The range following it refers to the maximum and the minimum value of the field. |
| bool | Field Type: Boolean | The value can be true or false. |
| enum | Field Type: Enumeration | The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators. |
| subType | Sub-Type of Field | When the type of field is list or enum, the value of subType is the data type of each sub-object. |
| desc | Field Description | The detailed description of the field. |

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### 4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

**Whether the device supports a function**: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

**The value range of the field**: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

**Note**: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

### 4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

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Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

**Example**: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

**Example**: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

**Notes**:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

### 4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

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| No. | Name | Special Character | No. | Name | Special Character |
|---|---|---|---|---|---|
| 1 | Open Parenthesis | ( | 18 | Dollar Sign | $ |
| 2 | Close Parenthesis | ) | 19 | Percent Sign | % |
| 3 | Plus Sign | + | 20 | Ampersand | & |
| 4 | Comma | , | 21 | Close Single Quotation Mark | ' |
| 5 | Minus Sign | - | 22 | Asterisk | * |
| 6 | Period | . | 23 | Slash | / |
| 7 | Semicolon | ; | 24 | Smaller Than | < |
| 8 | Equal Sign | = | 25 | Greater Than | > |
| 9 | At Sign | @ | 26 | Question Mark | ? |
| 10 | Open Square Bracket | [ | 27 | Caret | ^ |
| 11 | Close Square Bracket | ] | 28 | Open Single Quotation Mark | ' |
| 12 | Underscore | _ | 29 | Vertical Bar | \| |
| 13 | Open Brace | { | 30 | Tilde | ~ |
| 14 | Close Brace | } | 31 | Double Quotation Marks | " |
| 15 | Space | | 32 | Colon | : |
| 16 | Exclamation Mark | ! | 33 | Backslash | \| |
| 17 | Octothorpe | # | | | |

The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

### 4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

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```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

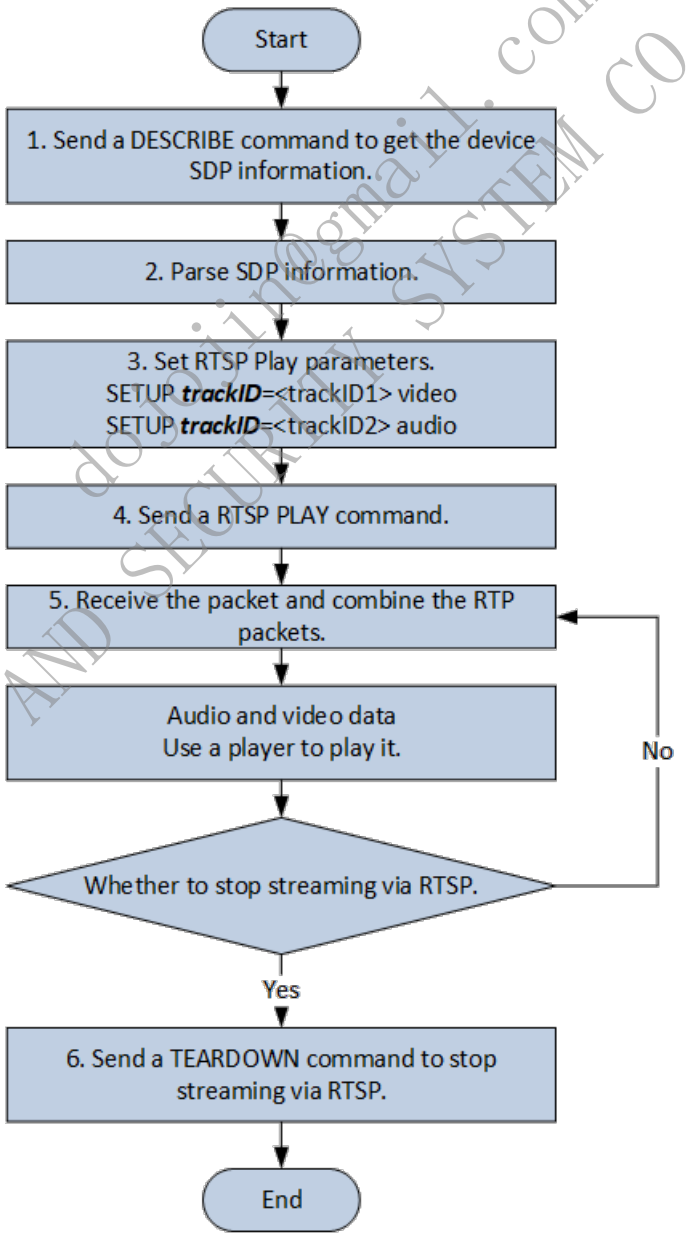
# 4.3 Real-Time Live View

## 4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

## 4.3.2 API Calling Flow




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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step. 2. The client parses the media SDP information returned by the device. 3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For

example, trackID=1 indicates videos while trackID=2 indicates audios. 4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`. 5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed. 6. The client sends the command RTSP TEARDOWN to stop streaming.

**Notes**:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

### 4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.22.126
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040lF000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

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```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

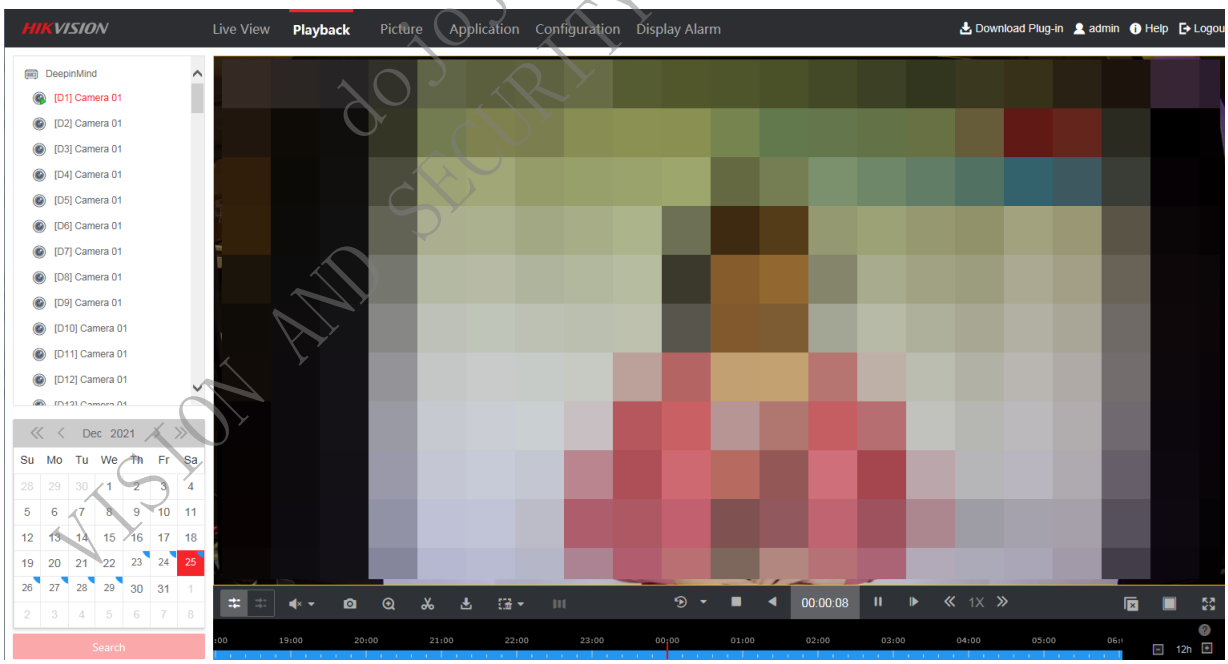
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

# 4.4 Playback

## 4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.




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## 4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in RFC 7826 (Real Time Streaming Protocol).

**Notes**:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

## 4.4.3 Example

### 4.4.3.1 Search for Videos According to Calendar

**Request from Client**

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

**Response from Device**

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```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

### 4.4.3.2 Search for Videos

### Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

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### Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

### 4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

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```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

## 4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

### 4.5.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

**Notes**

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

#### 4.5.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

**Event Message Parsing:**

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```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

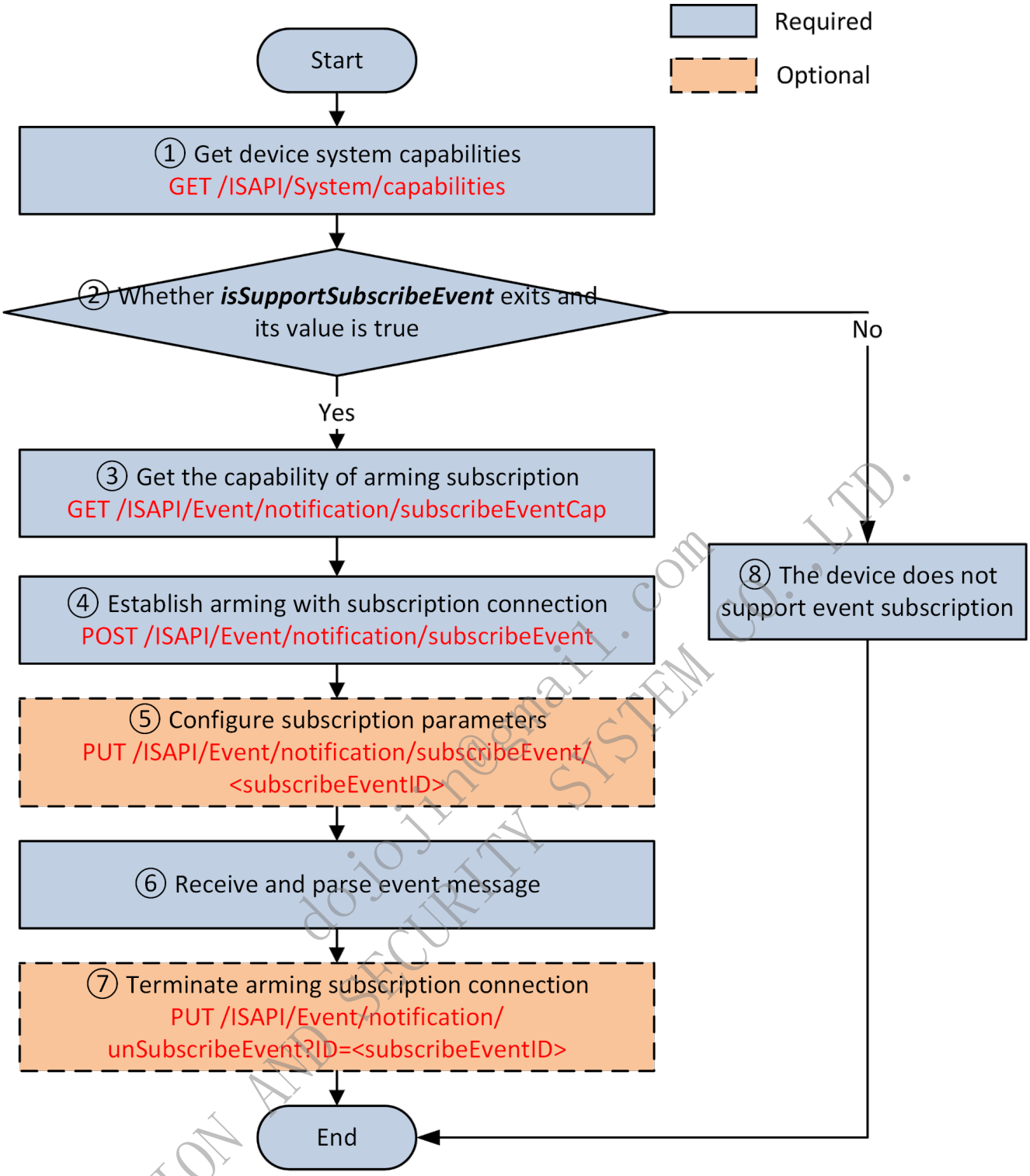
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

**Note:** `<data_gateway_ip>` and `<frontier>` are variables, `[Picture Data]` indicates the raw data of a picture.

#### 4.5.1.2 Arming with Subscription

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

**Note:**

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

**Event Messages Parsing:**

**Client Creates Arming with Subscription**

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

**Server Responds to Request and Push Event Message**

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the `Content-Type` is `multipart/form-data, boundary=AaB03x`, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark `--AaB03x--` will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Picture Data
--AaB03x--
```

The description of some keywords are as follows:

| Keyword | Example | Description |
|---|---|---|
| Content-Type | multipart/form-data; boundary=AaB03x | Content type. multipart/form-data means the message is in form format. |
| boundary | AaB03x | Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`. |
| Content-Disposition | form-data; name="ANPR.xml"; filename="ANPR.xml"; | Content description. `form-data` is a piece of form data. |
| name | "ANPR.xml" | Form name. |
| filename | "ANPR.xml" | File name of the form. |
| Content-Length | 9 | Content length, starting from the next `\r\n` to the next `--boundary`. |

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# 5 Device (General)

## 5.1 Arming and Subscription

### 5.1.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

### 5.1.2 API Calling Flow

#### 5.1.2.1 Without Subscription

1. Establish a connection for arming: `GET /ISAPI/Event/notification/alertStream`. You need to set `Connection: keep-alive` in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

##### 5.1.2.1.2 Syntax

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

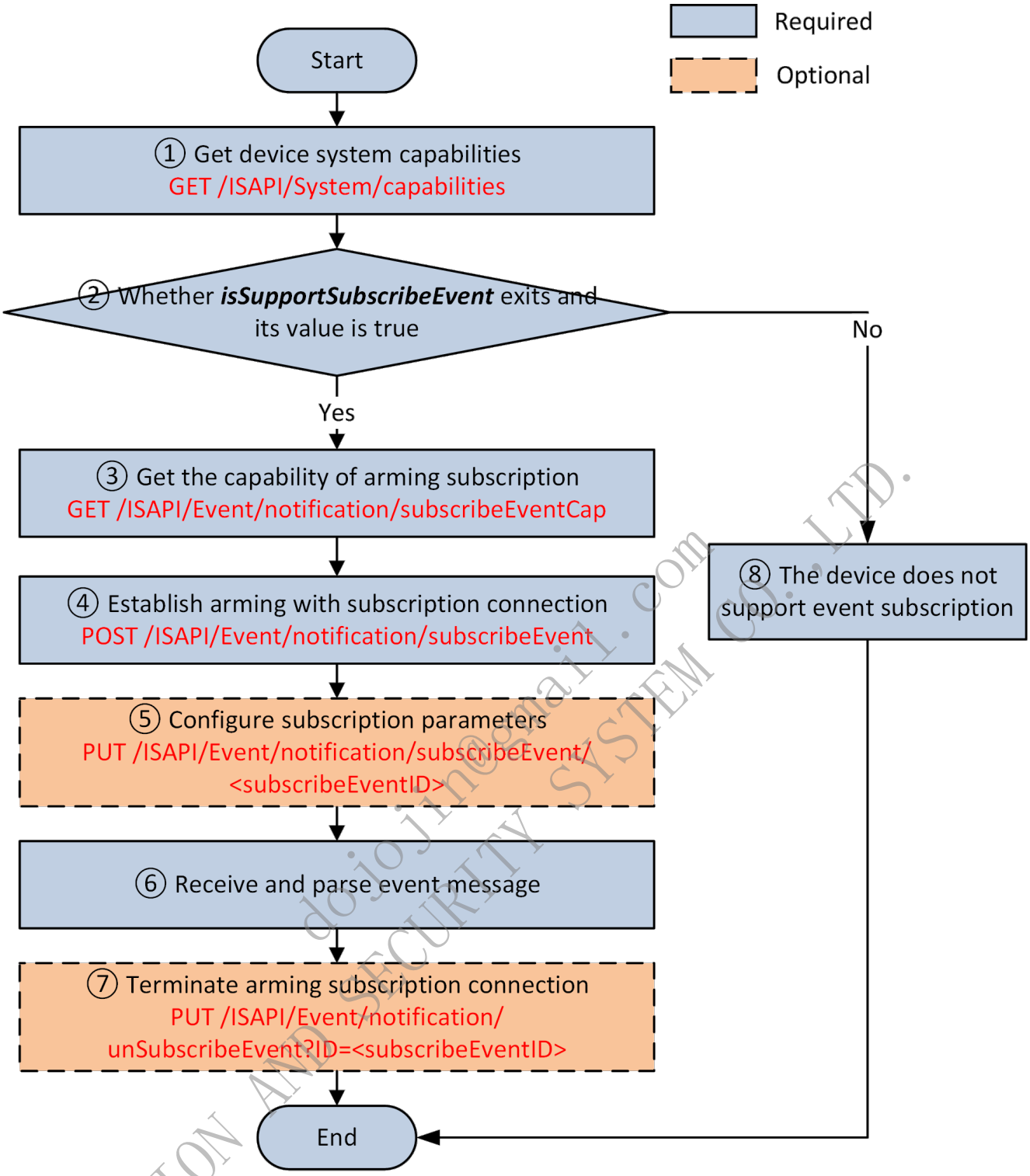
[Image Data]
--<frontier>
```

**Note:** `<data_gateway_ip>` and `<frontier>` are variables, and `[Image Data]` is an abbreviated representation indicating the raw data of an image at this location.

#### 5.1.2.2 Subscription

##### 5.1.2.2.1 API Calling Flow

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

**Note：**

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

#### 5.1.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

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### 5.1.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, `Content-Type` in Headers is usually `multipart/form-data, boundary=AaB03x`, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Note the dash `--` before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol `--AaB03x--` on the arming connection.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--
```

The description of some keywords are as follows:

| Keyword | Example | Description |
| --- | --- | --- |
| Content-Type | multipart/form-data; boundary=AaB03x | Content type. multipart/form-data means the message is in form format. |
| boundary | AaB03x | Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message. |
| Content-Disposition | form-data; name="ANPR.xml"; filename="ANPR.xml"; | Content description. |
| name | "ANPR.xml" | Form name. |
| filename | "ANPR.xml" | File name of the form. |
| Content-Length | 9 | Content length, starting from the next \r\n to the next --boundary. |

### 5.1.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

### 5.1.4 Sample Messages

#### 5.1.4.1 Establish Arming Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

#### 5.1.4.2 The Device Responses and Uploads an Event Message

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```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <ipAddress>10.17.133.46</ipAddress>
    <portNo>80</portNo>
    <protocol>HTTP</protocol>
    <macAddress>44:19:b6:6d:24:85</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-05-04T11:20:02+08:00</dateTime>
    <activePostCount>0</activePostCount>
    <eventType>heartBeat</eventType>
    <eventState>active</eventState>
    <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>
```

# 5.2 Calling Flow of Device Packet Capture

## 5.2.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

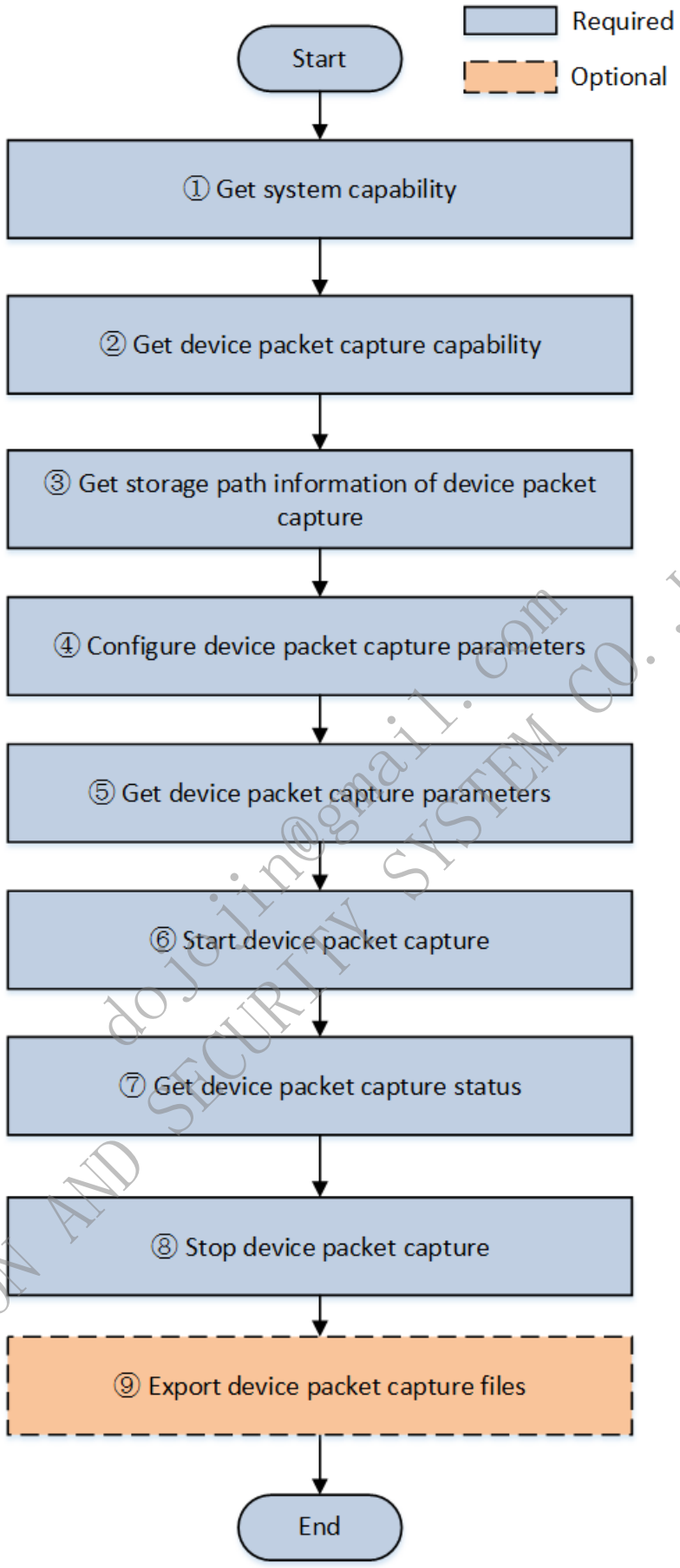
1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

## 5.2.2 API Calling Flow

### 5.2.2.1 Device Packet Capture

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

**Note**:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

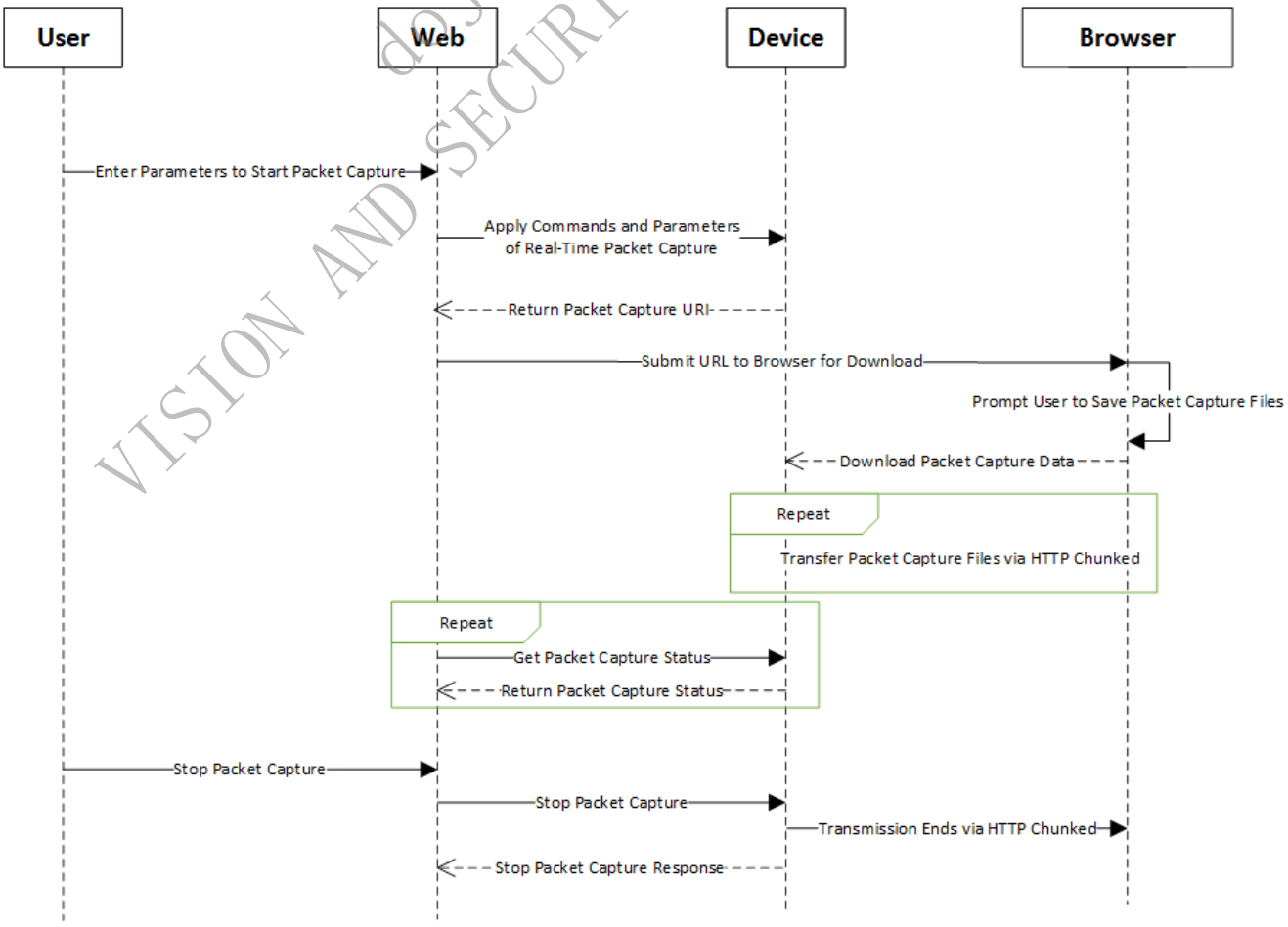
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

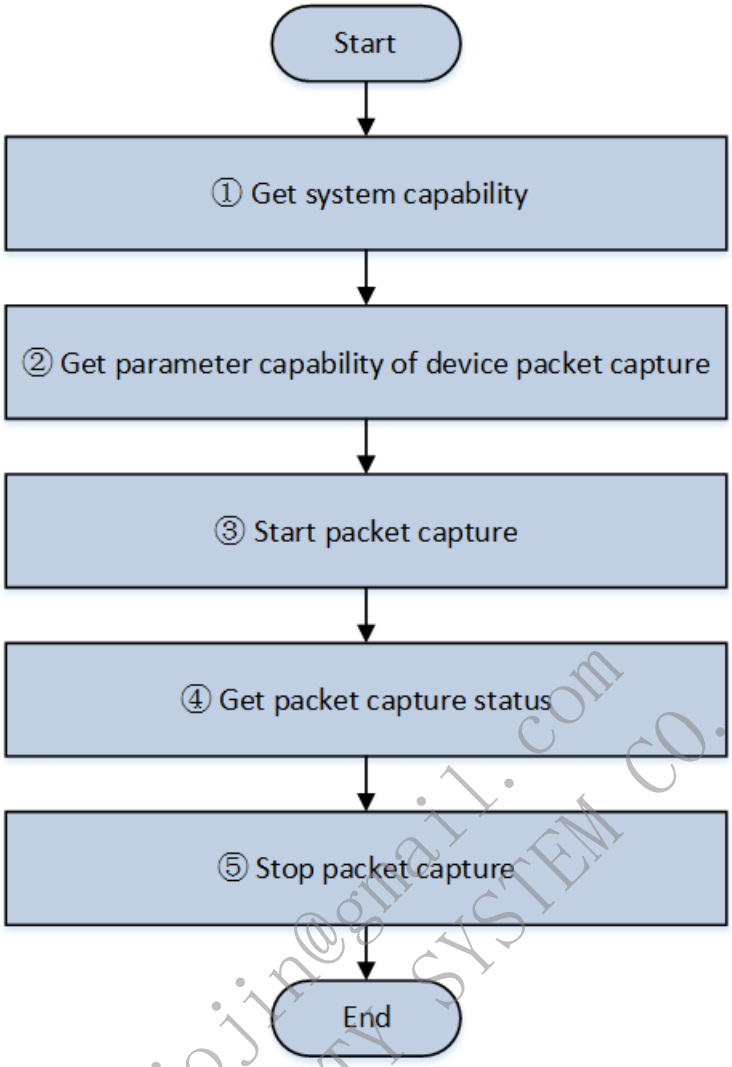
#### 5.2.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:




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See the following figure for the calling flow:




1. Get device system capabilities: `GET /ISAPI/System/capabilities`. `<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

**Note**:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

## 5.3 Device Peripherals Upgrade

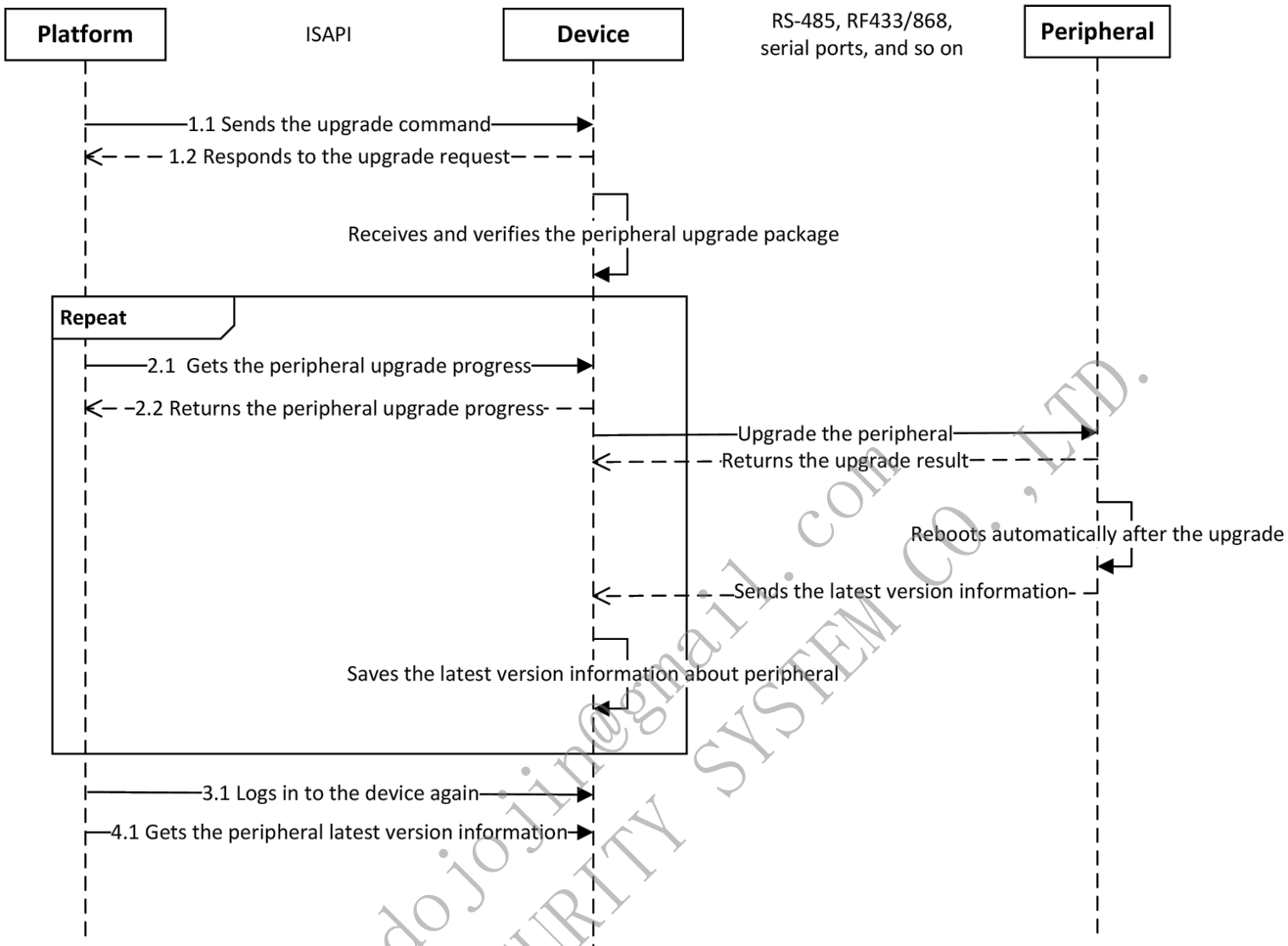
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### 5.3.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

### 5.3.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.




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1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.
2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

## 5.4 Device Time Sync

### 5.4.1 Introduction to the Function

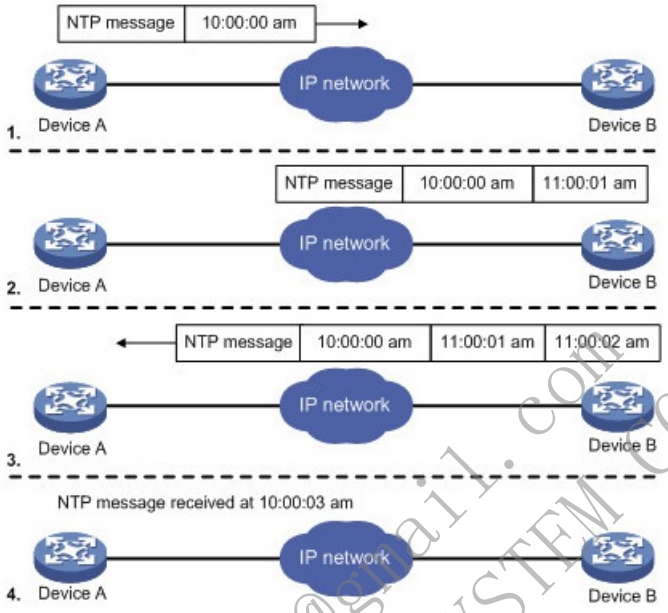
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

#### 5.4.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.




The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

## 5.4.2 API Calling Flow

### 5.4.2.1 Time Sync Configuration

**1. Get the Capability of Device Time synchronization Management**

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

**2. Set device time synchronization management parameters**

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`、`timeZone`.

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Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

**Note：** For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

#### 5.4.2.2 Time Zone Configuration

**1. Get device time zone configuration capability**

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

**2. Configure time zone parameters**

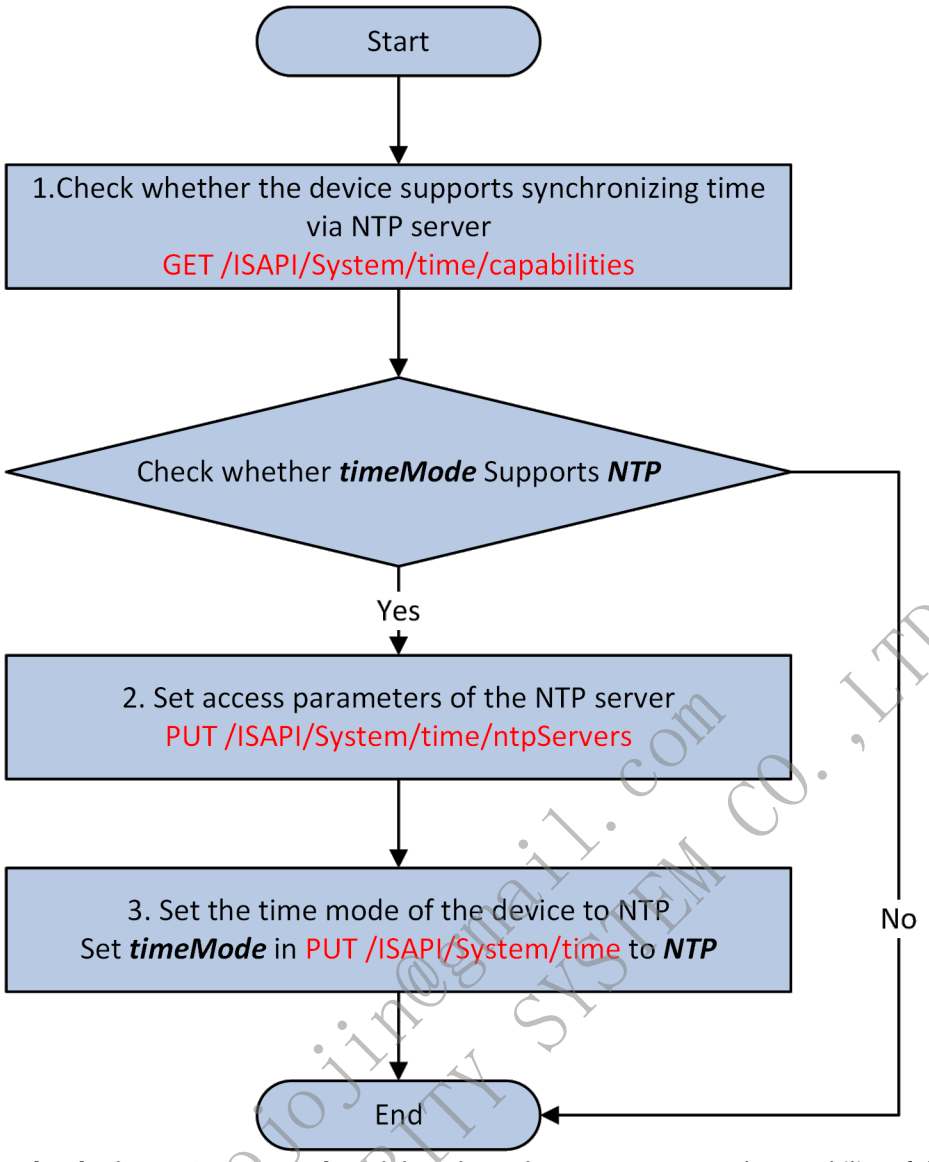
Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

#### 5.4.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):

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**1. Check whether the device supports synchronizing time via NTP server** Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

**2. Set access parameters of the NTP server**

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

**3. Set the time mode of the device to NTP**

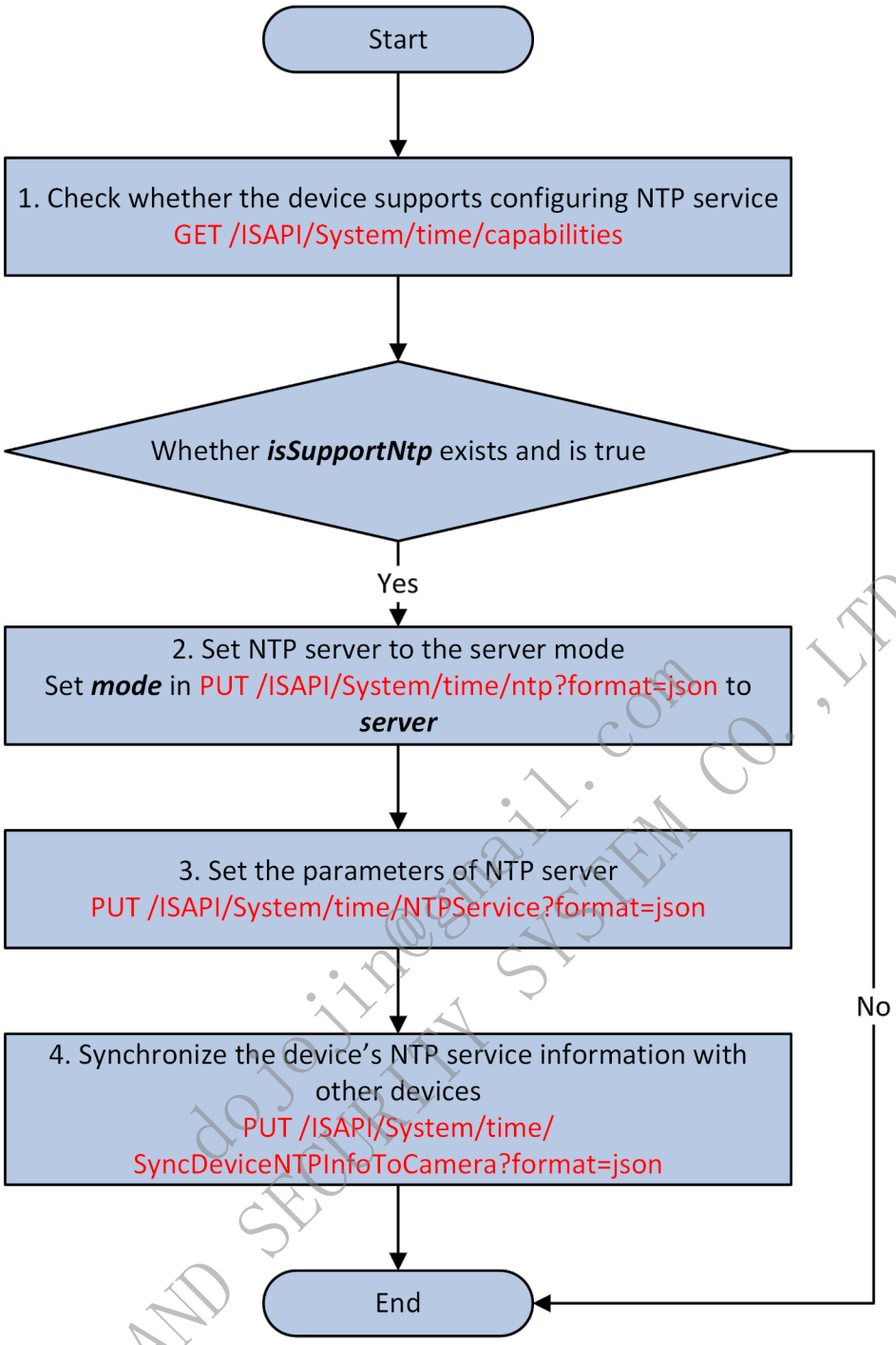
Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

#### 5.4.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):

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**1. Check whether the device supports configuring NTP service** Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

**2. Set NTP server to the server mode**

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

**3. Set the parameters of NTP server**

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

**4. Synchronize the device's NTP service information with other devices**

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

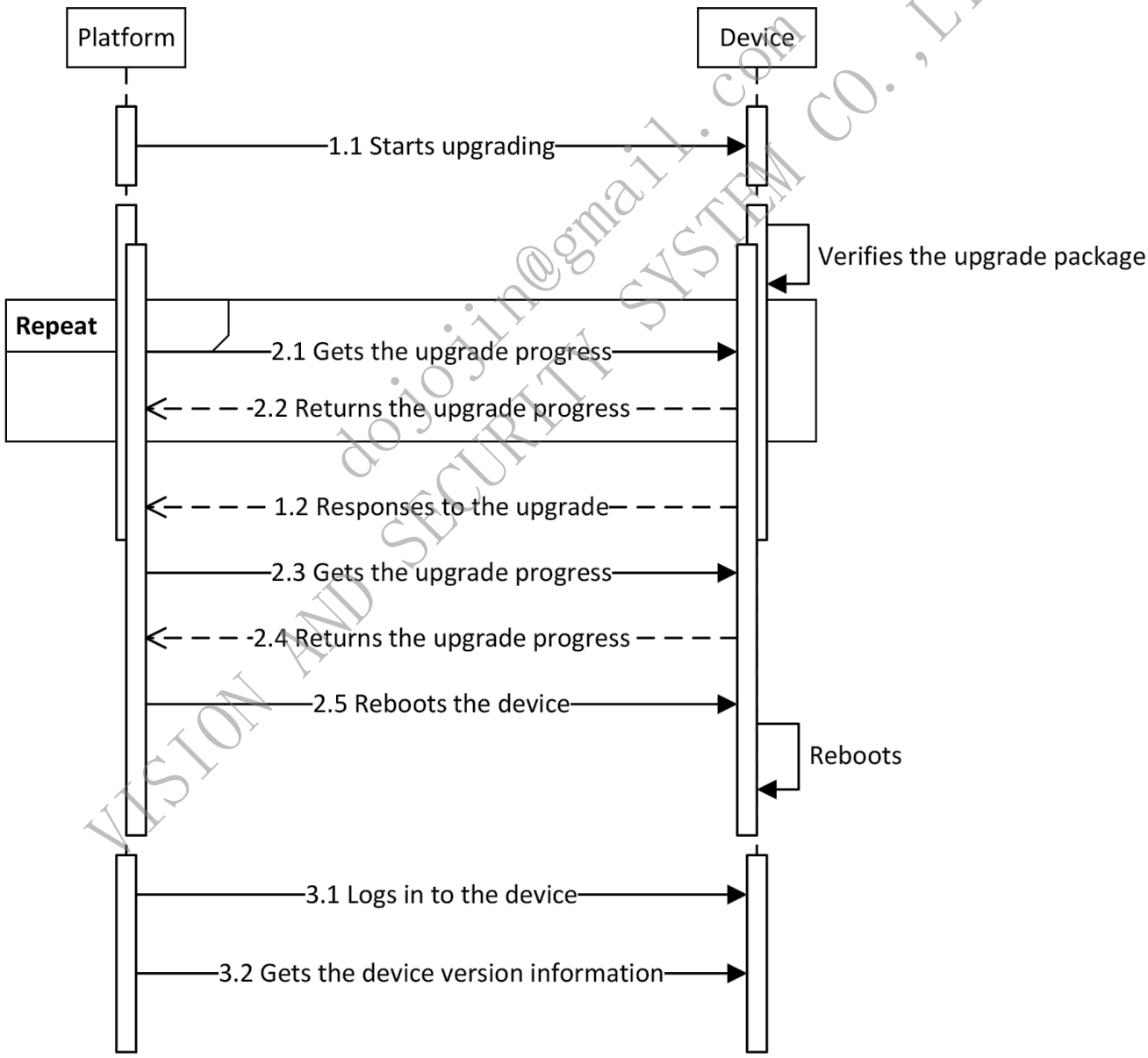
## 5.5 Device Upgrade

### 5.5.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

### 5.5.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.




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1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

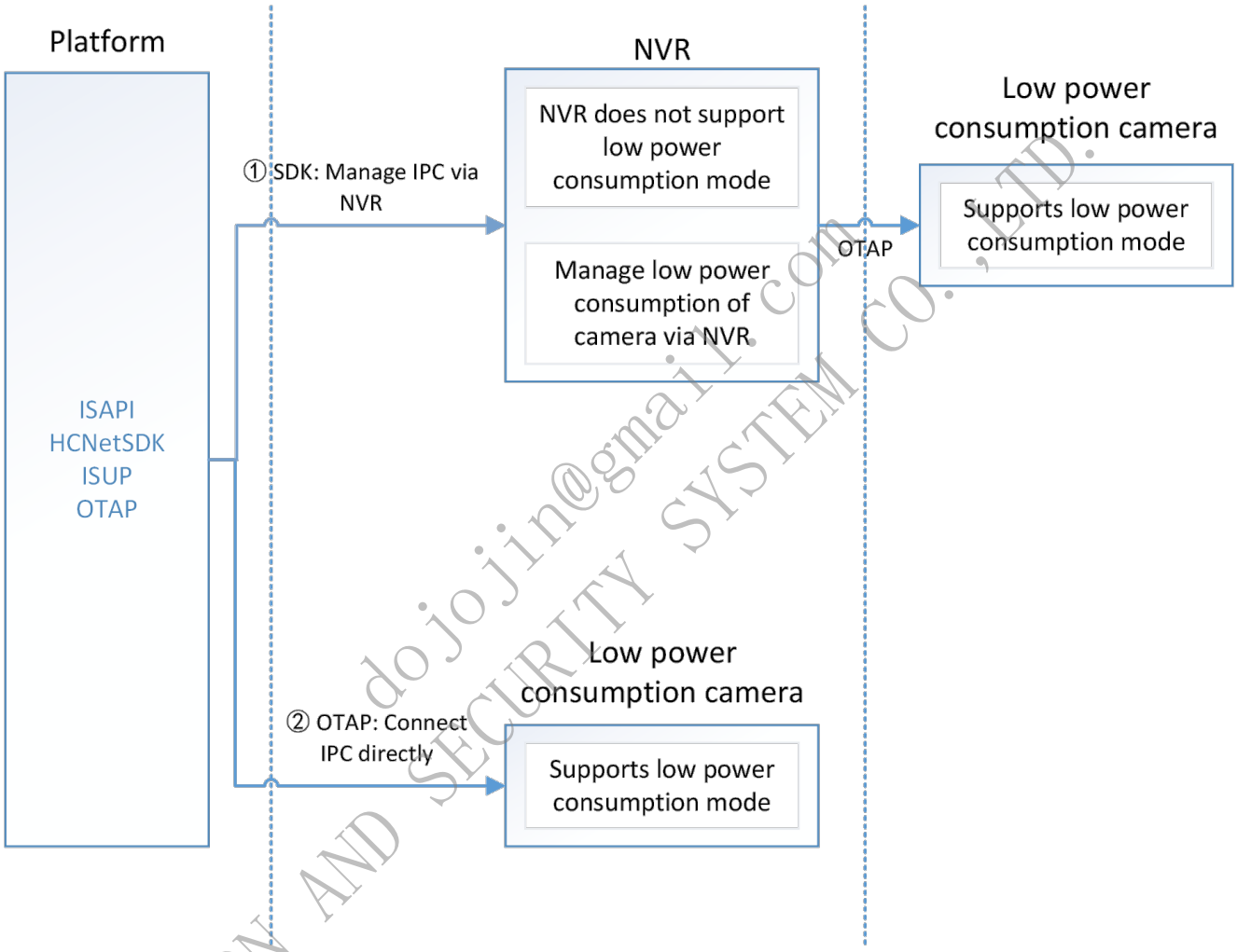
3. **Reboot devices.**

   Reboot devices: `PUT /ISAPI/System/reboot`.

## 5.6 Low Power Consumption Management of Network Camera via NVR

### 5.6.1 Introduction to the Function

For NVRs without low power consumption functions, they can be accessed by low power consumption cameras and manage the low power consumption functions including low power consumption management configuration, low power consumption mode, sleep/wakeup operation, etc. via the gateway sub-device solution.




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### 5.6.2 API Calling Flow

Sub-device access management: See details of sub-device transparent transmission in Sub-device Transparent Transmission.

1. Check whether the NVR device supports sub-device: `GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities`. If the `isSupportDevIndex` is returned and its value is true, it indicates that the NVR supports the transparent transmission function of sub-device.

2. Get the sub-device Information accessed by NVR: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. Each digital channel includes the parameter devIndex and digital channels from the same physical device have the same `devIndex`.

3. Get the low power consumption function supported by network camera: `GET /ISAPI/System/consumptionMode/capabilities?format=json&devIndex=<devIndex>`. Check whether the functions are supported according to the returned nodes in the response message.

**Note:**

- For operations on sub-device via NVR by the platform, the transparent transmission command carries the sub-device information `devIndex` in the request information, i.e., the URL parameter devIndex is suffixed to all ISAPI URLs.
- During ISAPI packaging, the URL parameter `devIndex=<devIndex>` suffixed to the URL represents a direct command to the sub-device; without `devIndex=<devIndex>`, it is a command to the NVR.

## 5.7 Low Power Consumption Mode Management

### 5.7.1 Introduction to the Function

In low power consumption mode, the device power consumption is lowered to improve the chip reliability and extend the battery life. For devices deployed in the wild or overhead that cannot be maintained frequently, the low power consumption management helps to extend the device service life and reduce the maintenance cost.

As the low power consumption devices are usually deployed in the wide without stable or available wired network but mostly cellular data without fixed IP address, the ISUP service should be enabled on deployment for device monitoring and management. Besides, when the device is sleeping and manual wakeup is required, it can only be woken up via the ISUP protocol. Therefore, it is recommended to integrate the ISUP protocol first for integration of low power consumption management function. So far the low power consumption management includes multiple management modes and can be selected according to the specific device and scenario. See details of consumption modes in Terms and Definitions.

**Note:**

- In sleeping mode, the device only maintain the heartbeat keepalive via the ISUP protocol (V5.0). Live view, playback, two-way audio, configuration, and other operations are not supported.
- The device will remain asleep for at most 5 minutes.
- If there are online users accessing functions like live view and playback, the device will not go to sleep.

### 5.7.2 API Calling Flow

1. Power consumption management: configure power consumption mode.
2. Sleep management: configure low power consumption sleep mode; configure sleeping schedule; configure auto-sleep countdown.
3. When the device is sleeping, operations such as live view and playback will fail.
4. Wakeup management: send a wakeup command to device via the ISUP protocol (V5.0); configure the wakeup schedule.

## 5.8 Module Sleep/Wakeup

### 5.8.1 Introduction to the Function

Besides the system-level chip, devices also consist of other modules (e.g., fall detection radar and vital sign detection radar consist of the system-level chip and radar module). When the integrator sends a sleep command to device, the device puts the module to sleep but the system-level chip is still operating normally. The integrator can wake up the module via the ISAPI protocol.

### 5.8.2 API Calling Flow

1. (Client) Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportConsumptionMode` is returned and its value is `true`, it indicates that the device supports switching power consumption modes.
2. (Client) Get the power consumption mode capability: `GET /ISAPI/System/consumptionMode/capabilities?format=json`. If `isSupportSleep` is returned and its value is `true`, it indicates that the device supports sleep; if `isSupportWakeUp` is returned and its value is `true`, it indicates that the device supports wakeup; if `isSupportWakeUp` is returned and its value is `true`, it indicates that the device supports sleeping mode.
3. Put the module to sleep immediately: `PUT /ISAPI/System/consumptionMode/Sleep?format=json`.
4. Wake up the module immediately: `PUT /ISAPI/System/consumptionMode/WakeUp?format=json`.
5. Get the current sleep status: `GET /ISAPI/System/consumptionMode/SleepStatus?format=json`.

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## 5.9 Mutually Exclusive Functions

### 5.9.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

### 5.9.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).

2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.

3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

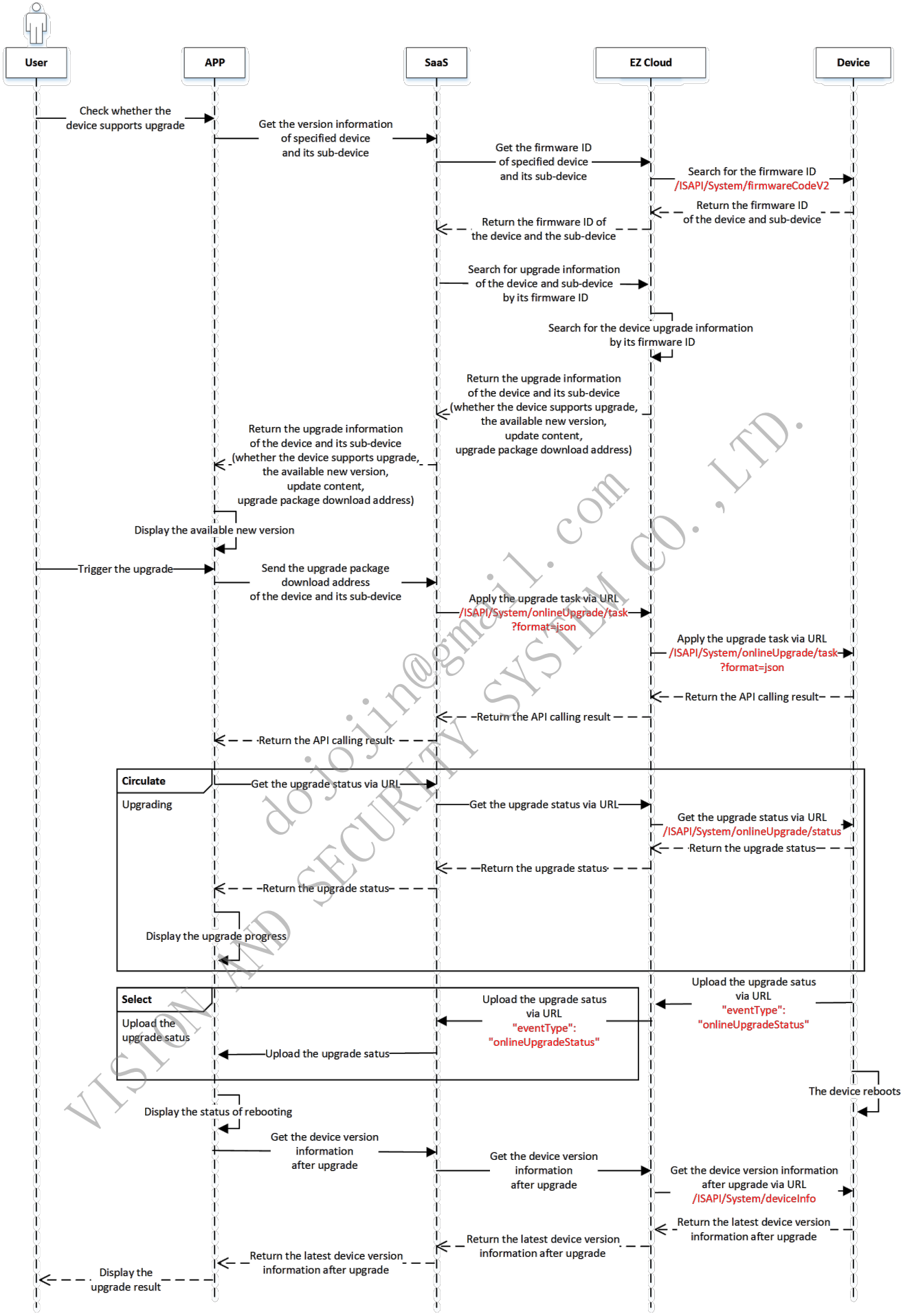
## 5.10 Remote Online Upgrade Based on The EZ2.0

### 5.10.1 Introduction to the Function

Considering that the device is connected to EZ via EZ 2.0 Protocol, EZ OTA upgrade solution is not supported. This solution applies to specified version upgrade.

### 5.10.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by its firmware ID. If there is an available new version, apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade devices (e.g., NVR) or sub-devices (e.g., IPC) via the node

device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.

3. During the device upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` and reboot automatically.
5. After both upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

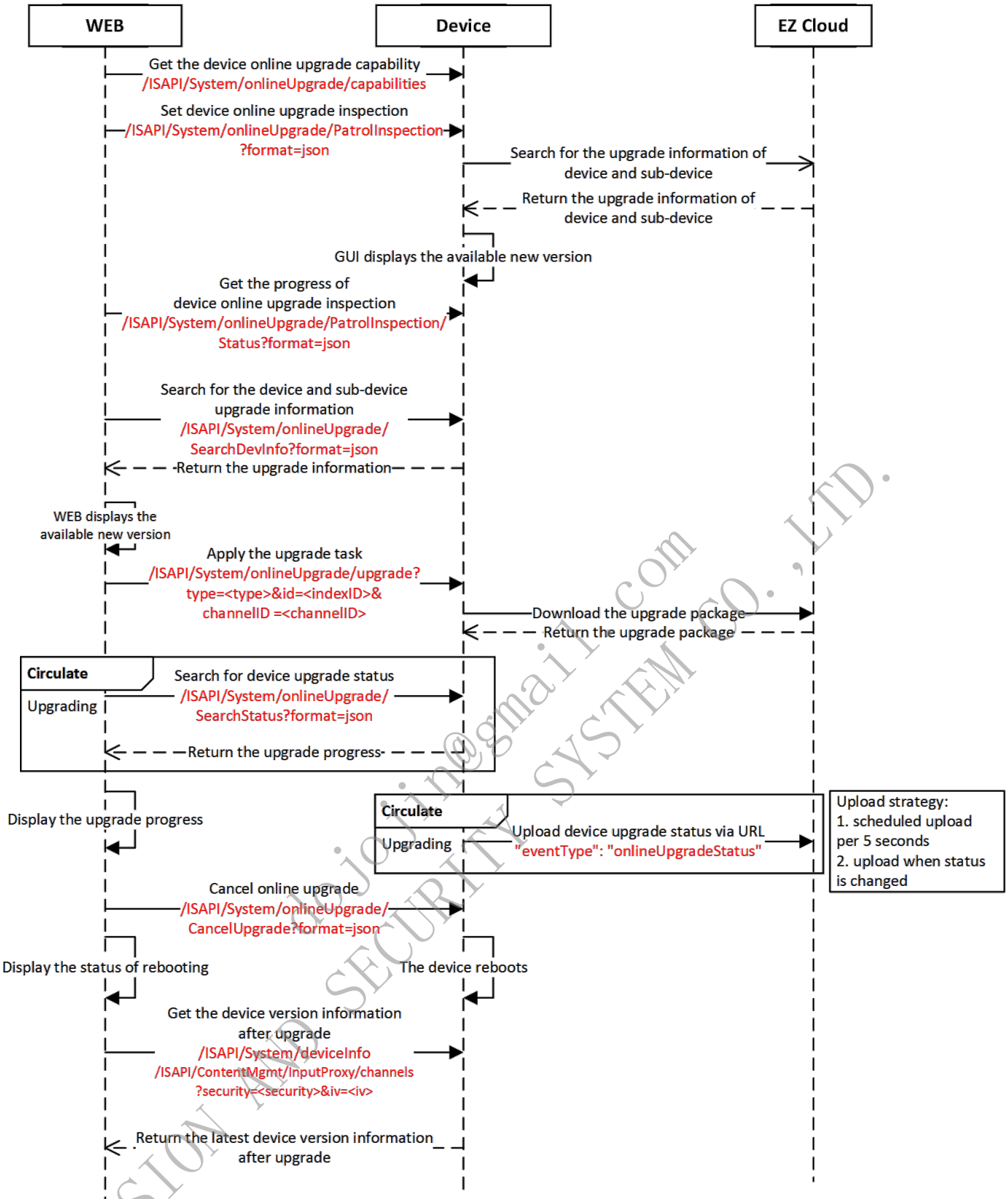
## 5.11 Remote Online Upgrade Based on the WEB Client

### 5.11.1 Introduction to the Function

This solution applies to remote online upgrade of the devices and sub-devices connected to WEB client.

### 5.11.2 API Calling Flow

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1. Get device online upgrade capability: `GET /ISAPI/System/onlineUpgrade/capabilities`.
2. Manually trigger the device online upgrade inspection: `POST /ISAPI/System/onlineUpgrade/PatrolInspection?format=json`. After it is enabled, the device will search for the available new version.
3. During the inspection, the WEB client can get the real-time inspection progress: `GET /ISAPI/System/onlineUpgrade/PatrolInspection/Status?format=json`.
4. After the device inspection is finished, search for the device and sub-device upgrade information: `GET /ISAPI/System/onlineUpgrade/SearchDevInfo?format=json`.
5. Apply the upgrade task: `PUT /ISAPI/System/onlineUpgrade/upgrade?type=<type>&id=<indexID>&channelID =<channelID>`. You can upgrade a specified device via the node `<channelID>` in the URL. Specify sub-device upgrade by configuring `<type>`.

6. You can search for online upgrade status during the upgrade: `GET /ISAPI/System/onlineUpgrade/SearchStatus?format=json`.
7. You can cancel the upgrade during the upgrade: `PUT /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json`.
8. After the upgrade and reboot are finished, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

## 5.12 Sensor-Triggered Wakeup

### 5.12.1 Introduction to the Function

Connect sensors to the low power consumption device via RS-485 interface. When the device is sleeping, if data collected by the sensors exceeds the configured threshold condition, the device will wake up in full power consumption mode and upload alarm information and relevant pictures/videos according to the triggering rules. This function is commonly applied to the wild environment.

### 5.12.2 API Calling Flow

1. (Client) Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportConsumptionMode` is returned, it indicates that the device supports switching power consumption modes.
2. (Client) Get the power consumption mode capability: `GET /ISAPI/System/consumptionMode/capabilities?format=json`. If `isSupportSensorTriggeringWakeParams` is returned, it indicates that the device supports configuring sensor-triggered wakeup rules; if `isSupportSensorTriggeringWakeRange` is returned, it indicates that the device supports getting the sensor-triggered wakeup range.
3. (Client) Get the sensor-triggered wakeup range: `GET /ISAPI/System/consumptionMode/SensorTriggeringWakeRange?format=json`; `sensorDataType` is the sensor-triggered wakeup sensor data supported by device. See details of sensor data in the dictionary file obtained from `GET /ISAPI/IoTGateway/SensorDataFlowMgr/SearchSensorDictFile?format=json`.
4. (Client) Configure sensor-triggered wakeup rules for device:

   Get capability: `GET /ISAPI/System/consumptionMode/SensorTriggeringWakeParams/capabilities?format=json`. Pay attention to the range of `@size` and `ruleID` in the response message; other nodes represents the variable types commonly used for sensor-triggered wakeup rules. See details of sensor data in the dictionary file obtained from Step 3.

   Get sensor-triggered wakeup rule parameters: `GET /ISAPI/System/consumptionMode/SensorTriggeringWakeParams?format=json`.

   Configure sensor-triggered wakeup rules: `PUT /ISAPI/System/consumptionMode/SensorTriggeringWakeParams?format=json`.

**Note:** If sensors with data exception are not disconnected, the device will wake up frequently. Therefore, after 3 times of rules being triggered for the same sensor within a day, the device will not wake up anymore.

## 5.13 Serial Port Accessed External Device Management

### 5.13.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

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### 5.13.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information od devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port:`POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

## 5.14 Serial Port Data Transparent Transmission

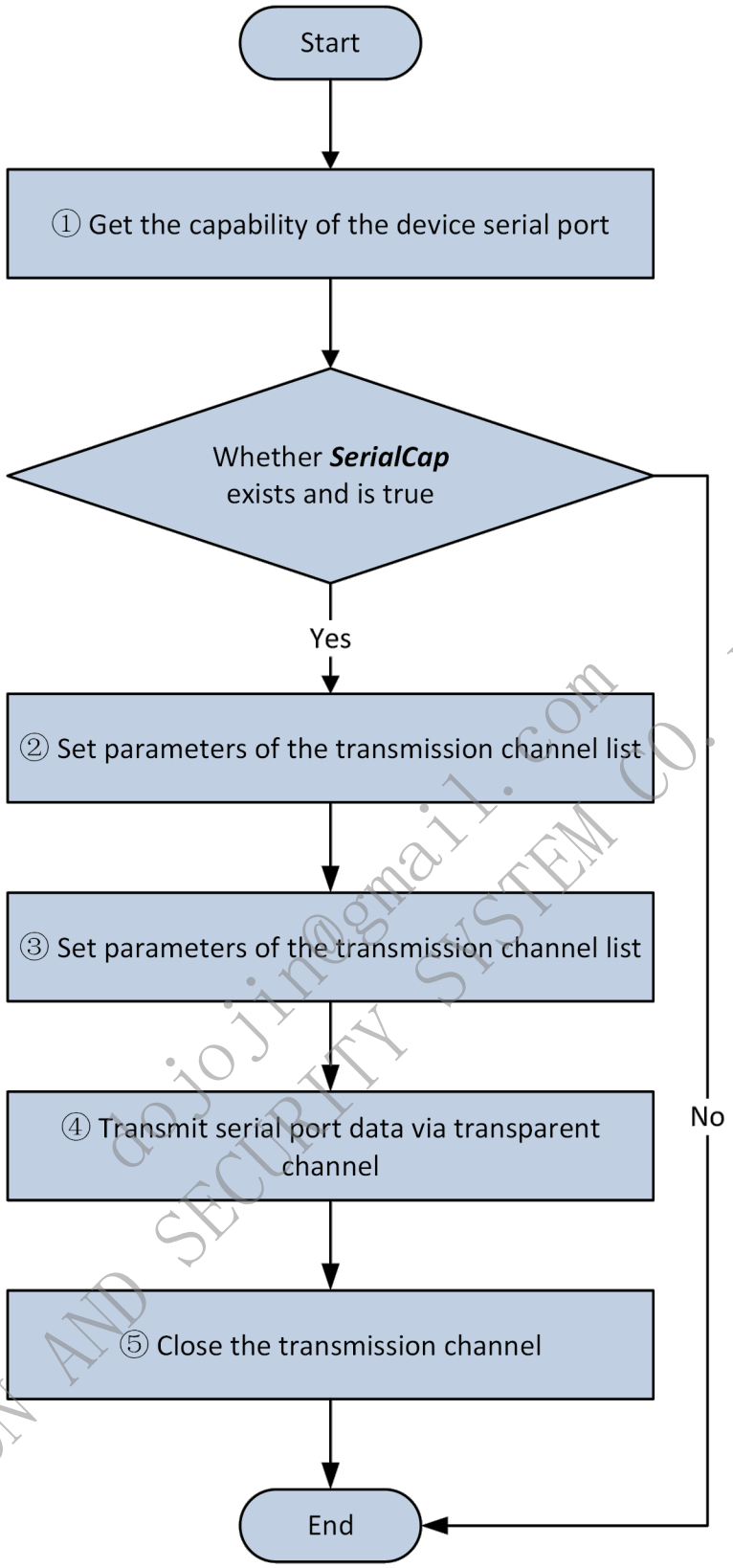
### 5.14.1 Introduction to the Function

RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of transparently transmitted serial data.

### 5.14.2 API Calling Flow

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1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open`;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.
- Send data to device serial port through transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.

5. Close the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close`.

# 5.15 Serial Port Parameter Configuration

## 5.15.1 Introduction to the Function

Serial port parameter configuration.

## 5.15.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: `GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: `GET /ISAPI/System/Serial/capabilities`.
- Get control parameters of all serial ports: `GET /ISAPI/System/Serial/ports?permissionController=<indexID>`.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: `GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.
- Configure control parameters of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.

5. Get the status of single serial port:`GET /ISAPI/System/Serial/ports/<portID>/status`.

# 5.16 Sleep

In full power consumption mode or real-time low power consumption mode, you can configure the sleep conditions. Once the conditions are met, the device will go to sleep.

## 5.16.1 Low Battery Sleep

### 5.16.1.1 Introduction to the Function

In full power consumption mode or real-time low power consumption mode, the device goes to sleep when the battery is lower than the sleep threshold.

### 5.16.1.2 API Calling Flow

1. (Client) Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportConsumptionMode` is returned, it indicates that the device supports switching power consumption modes.
2. (Client) Get the capability of switching power consumption modes: `GET /ISAPI/System/consumptionMode/capabilities?format=json`. If `Sleep` is returned, it indicates that the device supports low battery sleep.
3. (Client) Configure the low battery sleep threshold `powerThreshold`: `PUT /ISAPI/System/consumptionMode?`

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`format=json`.

#### 5.16.1.3 Exception Handling

Error Code:

| statusCode | statusString | subStatusCode | errorCode | errorMsg | Description | Remarks |
|---|---|---|---|---|---|---|
| 4 | Invalid Operation | deviceSleep | 0x40002097 | Please wake up device. | The device is sleeping. | |

### 5.16.2 Sleep Schedule

#### 5.16.2.1 Introduction to the Function

In full power consumption mode or real-time low power consumption mode, the device goes to sleep according to the configured sleeping time periods, wakes up and captures pictures at intervals, and goes back to sleep afterwards.

#### 5.16.2.2 API Calling Flow

1. (Client) Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportConsumptionMode` is returned, it indicates that the device supports switching power consumption modes.
2. (Client) Get the capability of switching power consumption modes: `GET /ISAPI/System/consumptionMode/capabilities?format=json`. If `TimingSleep` is returned, it indicates that the device supports sleep schedule,.
3. (Client) Configure the sleep schedule `TimingSleep` according to supported response parameters: `PUT /ISAPI/System/consumptionMode?format=json`.

**Note:**

- Besides the general device sleep, a certain module in device can be specified to sleep, e.g., the thermal imaging camera module (`thermalMovement`) and visible light camera module (`visibleMovement`) of dual-channel thermal imaging devices. Wake up the thermal imaging camera module manually: `PUT /ISAPI/System/consumptionMode/manualArouseThermalMovement?format=json`.

#### 5.16.2.3 Exception Handling

Error Code:

| statusCode | statusString | subStatusCode | errorCode | errorMsg | Description | Remarks |
|---|---|---|---|---|---|---|
| 4 | Invalid Operation | deviceSleep | 0x40002097 | Please wake up device. | The device is sleeping. | |
| 4 | Invalid Operation | restoreThermalimagingVideoFailed | 0x40002150 | The device is under over-temperature protection and cannot restore the thermal imaging video. | The device is under over-temperature protection and cannot restore the thermal imaging video. | |

### 5.16.3 Auto Sleep Countdown

#### 5.16.3.1 Introduction to the Function

After the set time, the device will go to sleep automatically.

#### 5.16.3.2 API Calling Flow

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In full power consumption mode, the device can perform auto sleep via the following 2 methods. Integrate the function according to the capability returned by device.

Method 1:

1. (Client) Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportConsumptionMode` is returned, it indicates that the device supports switching power consumption modes.
2. (Client) Get the capability of power consumption modes: `GET /ISAPI/System/consumptionMode/capabilities?format=json`. If `AutoSleep` is returned, it indicates that the device supports auto sleep.
3. (Client) Configure auto sleep `AutoSleep` according to supported response parameters: `PUT /ISAPI/System/consumptionMode?format=json`.

Method 2:

1. (Client) Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportSleep` is returned, it indicates that the device supports sleep.
2. (Client) Get the capability of sleep: `GET /ISAPI/System/sleep/capabilities?format=json`.
3. (Client) Configure sleep according to supported response parameters: `PUT /ISAPI/System/sleep?format=json`.

#### 5.16.3.3 Exception Handling

Error Code:

| statusCode | statusString | subStatusCode | errorCode | errorMsg | Description | Remarks |
|---|---|---|---|---|---|---|
| 4 | Invalid Operation | deviceSleep | 0x40002097 | Please wake up device. | The device is sleeping. | |

### 5.16.4 AOV Mode

#### 5.16.4.1 Function Introduction

On the basis of the standby mode, the AOV mode further reduces power consumption by recording at a fixed interval when recoring is triggered.

#### 5.16.4.2 API Calling Flow

1. Get the device capability. URL: `GET /ISAPI/System/capabilities`. If `isSupportConsumptionMode` is returned, the device supports switching power consumption mode.
2. Get the capability of device power consumption.
   URL: `GET /ISAPI/System/consumptionMode/capabilities?format=json`. If the returned `devWorkMode` contains `AOV`, the AOV mode is supported.
3. Configure AOV mode parameters.
   URL: `PUT /ISAPI/System/consumptionMode?format=json`. Set the recording frame-extracting rate (`AOVRecordFrameRate`) under AOV mode.

### 5.16.5 Exception Handling

Error Code

| statusCode | statusString | subStatusCode | errorCode | errorMsg |
|---|---|---|---|---|
| 4 | Invalid Operation | deviceSleep | 0x40002097 | Please wake up device |

## 5.17 Sub-device Batch Upgrade

### 5.17.1 Introduction to the Function

**Sub-device Batch Upgrade (Single Task):** applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

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**Management of Sub-device Batch Upgrade Tasks:** There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

### 5.17.2 API Calling Flow

**Sub-device Batch Upgrade (Single Task)**

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

**Sub-device Batch Upgrade Task Management**

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

**Note:** 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

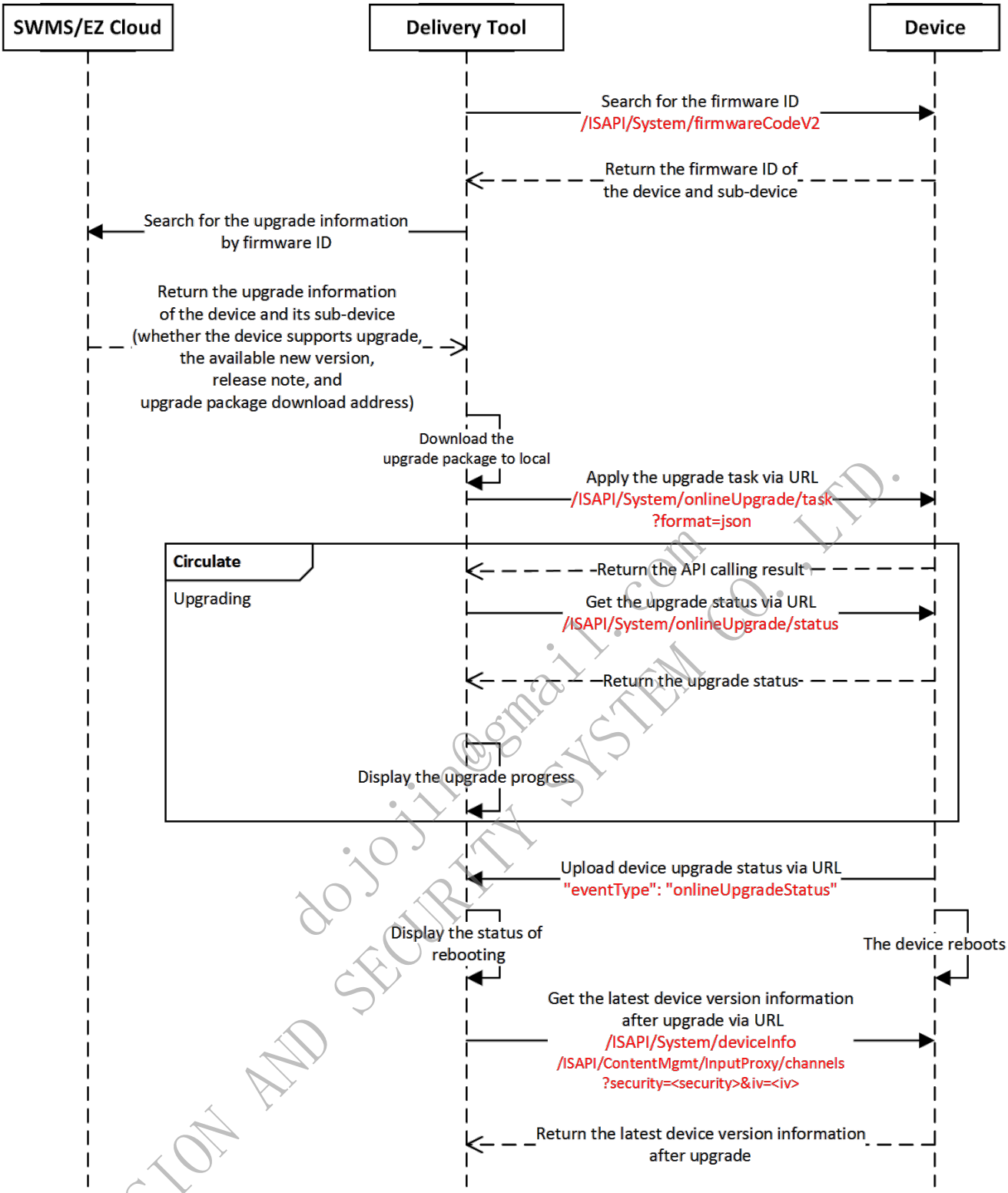
## 5.18 Upgrade Based on Interactive Tools Local Deployment

### 5.18.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI. If the device is connected via ISAPI protocols, it is required to set up HTTP(s) service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK, you can implement the device upgrade via transmitting ISAPI.

### 5.18.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by firmware ID on the SWMS/EZ Cloud, download upgrade package to local storage, and apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., camera) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress: `GET /ISAPI/System/onlineUpgrade/status`.
4. After upgrade is completed, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and reboot automatically.
5. After the device rebooting, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=`

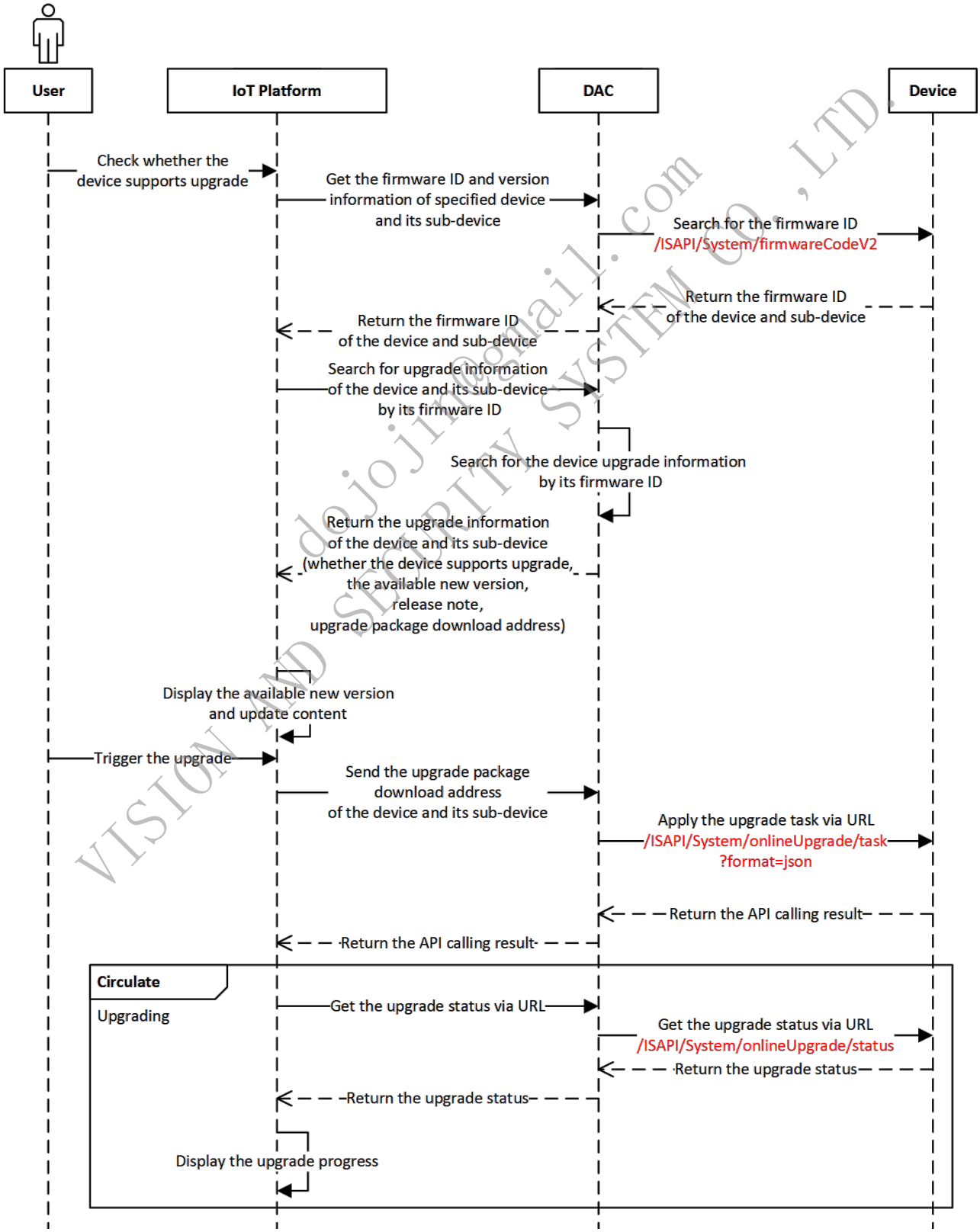
`<security>&iv=<iv>`.

## 5.19 Upgrade Based on Platform Local Deployment

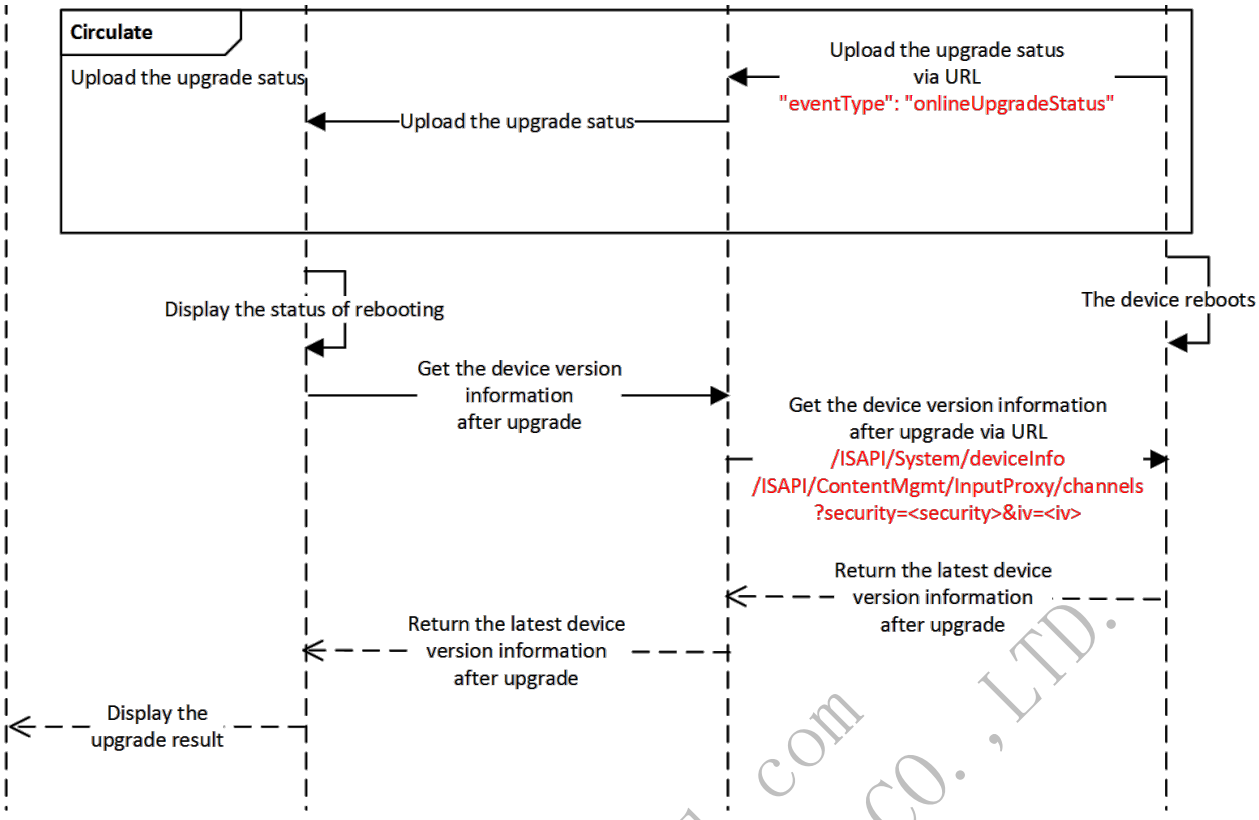
### 5.19.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI and ISUP. If via ISAPI, it is required to set up HTTP(s) service or transfer binary upgrade package files so as to store device upgrade package and generate URL for downloading upgrade package. If via ISUP, it is required to set up FTP service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK protocols, you can realize device upgrade via transmitting ISAPI.

### 5.19.2 API Calling Flow




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1. Search for device firmware ID on the platform: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information in DAC by device firmware ID on the platform. If there is an available new version, DAC will trigger an upgrade task and apply it to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and will reboot automatically.
5. After the upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

## 5.20 Wakeup Schedule

### 5.20.1 Introduction to the Function

So far there are 4 methods to wake up devices in sleeping mode:

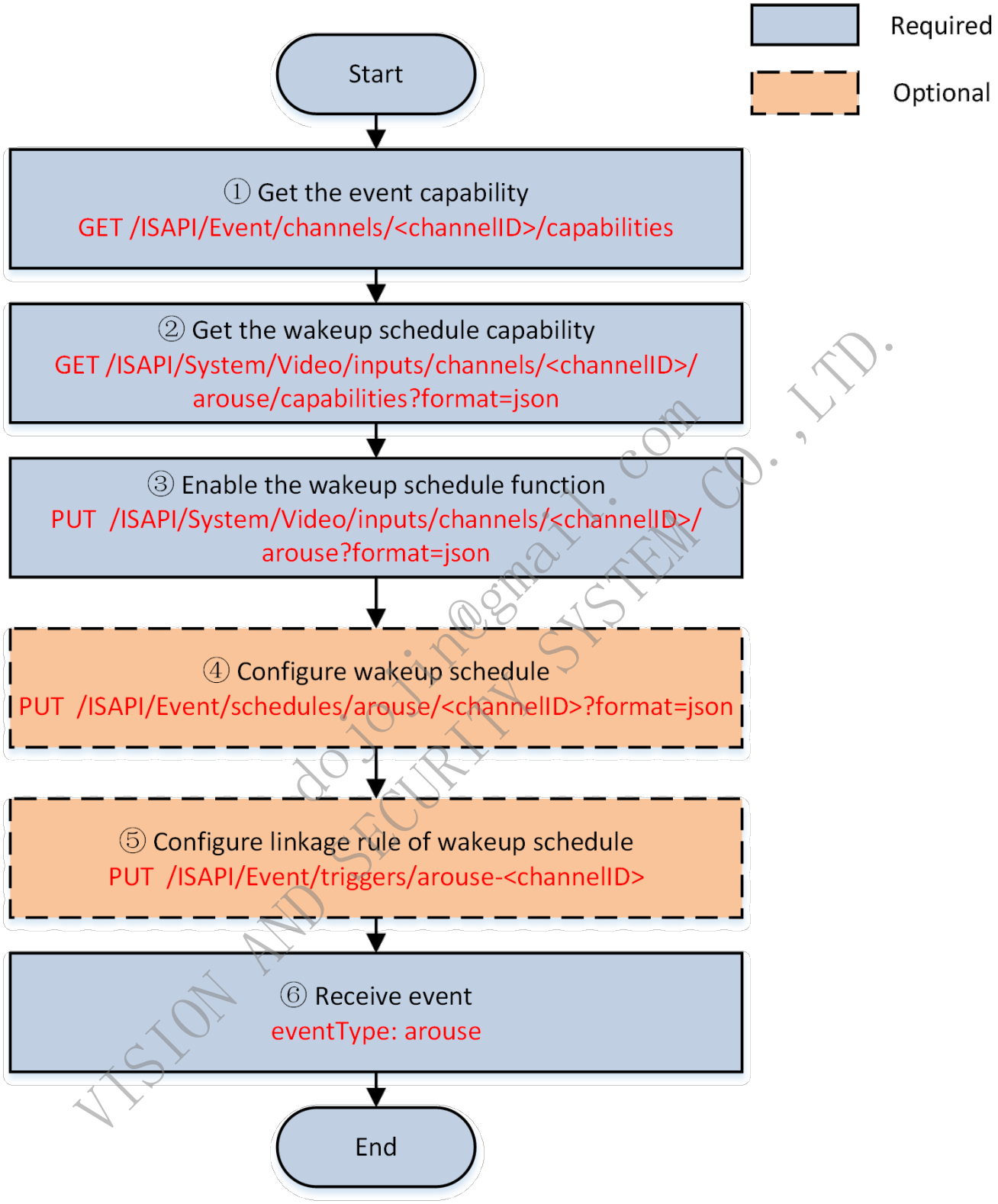
1. After the sleep schedule is configured, the device auto wakes up in wakeup mode, or auto wakes up according to the configured capture interval in sleeping mode.
2. For devices registered with ISUP service (V5.0), they wake up by applying ISUP command to themselves in sleeping mode.
3. (Client) Configure the device wakeup schedule and the device will wake up at set time.
4. Connect the low power consumption device with sensors. (Client) Configures the threshold condition for sensor-triggered device wakeup. Once the data collected by sensor exceeds the threshold, the device will wake up.

If the integration scenario requires configuring regular sleep schedule and capturing pictures as scheduled, Method 1 is recommended and see details in the API calling flow of sleep schedule; for client manually waking up the device, Method 2 is recommended and see details in the user manual of HCISUPSDK; if the integration scenario requires

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waking up device regularly to capture pictures when it is consistently sleeping, Method 3 is recommended and shown below.

### 5.20.2 API Calling Flow




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1. (Client) Get the event capability: `GET /ISAPI/Event/channels/<channelID>/capabilities`. If eventType is returned and its value includes "arouse", it indicates the channel supports wakeup schedule.
2. (Client) Get the wakeup schedule capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/arouse/capabilities?format=json`.
3. (Client) Enable the wakeup schedule function: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/arouse?format=json`.
4. (Optional) (Client) Configure wakeup schedule: `PUT /ISAPI/Event/schedules/arouse/<channelID>?format=json`.
5. (Optional) (Client) Configure linkage rule of wakeup schedule: `PUT /ISAPI/Event/triggers/arouse-<channelID>`.

6. The device wakes up as scheduled and uploads the event `eventType：arouse`. If the capture function is configured in Step 3, the uploaded event will include pictures captured when the device is awake.

**Note:**

- Check if single-channel devices support the wakeup schedule function: If Step 1 and 2 are both not supported, the function is not supported.
- When the wakeup schedule is enabled on the client and the sleep schedule is configured (scheduled capture enabled), the scheduled wakeup event will be triggered on scheduled capture.
- Cancel the device low battery sleep via Connect: `PUT /ISAPI/System/consumptionMode/cancelSleep?format=json`.

# 6 Video (General)

## 6.1 . OSD

### 6.1.1 Introduction to the Function

On-screen display is applied in the display interface of cameras, and is used to display some characters or graphics on the video to provide some information to the audience. After OSD configurations on the client software, the configured information text will be overlaid on the video stream of cameras and displayed on the screen.

The typical application scenarios of dynamic OSD: elevator floor number overlay for elevator monitoring, characters overlay for toll stations on highway/road, notification information overlay during live video programs. The typical application scenarios of static OSD: fixed information (such as camera position, date, time) overlay on the video signal.

### 6.1.2 API Calling Flow

#### 6.1.2.1 OSD

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`.

2. The client software configures OSD parameters, including whether to enable OSD, custom text overlay, time overlay, and overlay format, for a specified channel.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Set: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   **Remarks:**

   - The configuration URL supports various parameters, including custom text overlay, channel name overlay, time overlay, overlay format, and so on. Some of these parameters can also be configured by other URLs, as shown below: `TextOverlayList` can also be configured via URL of custom text overlay.

     `channelNameOverlay` can also be configured via URL of channel name overlay.

     `DateTimeOverlay` can also be configured via URL of channel time overlay.

   - The overlaid information alignment mode (defined by `alignment`) refers to the alignment mode of custom text:

     `customize`: custom mode. The client software sends the overlay position coordinates of custom text to the device, and the device will overlay the characters on the stream for display according to the position coordinates.

     `alignRight`: right align. The device will display custom text on the right boundary of screen.

     `alignLeft`: left align. The device will display custom text on the left boundary of screen.

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`allRight`: all right align. The device will display all overlay contents, including custom text, the channel name, the time on the right boundary of screen.

`allLeft`: all left align. The device will display all overlay contents, including custom text, the channel name, the time on the left boundary of screen.

#### 6.1.2.2 Custom Text Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `TextOverlayList` is returned, it indicates that custom text overlay is supported and `size` defines the maximum number of custom texts.
2. The client software configures custom text OSD. `textID` refers to the custom text No., and the value range is: [1, `size`]:

   Create a custom text for a specified channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Get overlay parameters of a custom text for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Set overlay parameters of a custom text for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Delete a custom text: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Get overlay parameters of all custom texts for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Set overlay parameters of all custom texts for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Delete all custom texts for a specified channel: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

#### 6.1.2.3 Channel Name Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `channelNameOverlay` is returned, it indicates that channel name overlay is supported.
2. The client software configures the channel name overlay, including whether to enable channel name overlay, overlay position, and so on.

   Get channel name overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

   Set channel name overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

#### 6.1.2.4 Channel Time Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `DateTimeOverlay` is returned, it indicates that channel time overlay is supported.
2. The client software configures the channel time overlay, including whether to enable channel time overlay, overlay position, date format, and so on.

   Get channel time overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

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Set channel time overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

## 6.2 Async Capturing Manually

### 6.2.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately responds by returning the URL of the storage server where the captured picture is stored. The client can request image data from the storage server based on the received URL. It is suitable for the limited network bandwidth between the client and the device or the client centrally manages a large number of devices. The client can choose to use the image URL address carried in the alarm content to request pictures from the storage server when the network conditions improve or there is no concurrency of a large number of requests, so as to reduce the probability of image download failure and the risk of network congestion.

### 6.2.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportSnapshotAsync` is returned and its value is `true`, async manual capturing shall be supported.
2. (Optional) The client gets the async manual capturing capability of a specified channel: `GET /ISAPI/Streaming/channels/<channelID>/picture/async/capabilities?format=json`.
3. The client controls the specified channel to manually and asynchronously capture pictures: `GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>`, and the device returns the URL of the storage server for capturing pictures.

**Remarks:**

- If the step 2 returns a failure during the integration，it is recommended that you check whether the channel supports async manual capturing according to steps 1 and 3.

## 6.3 Live View

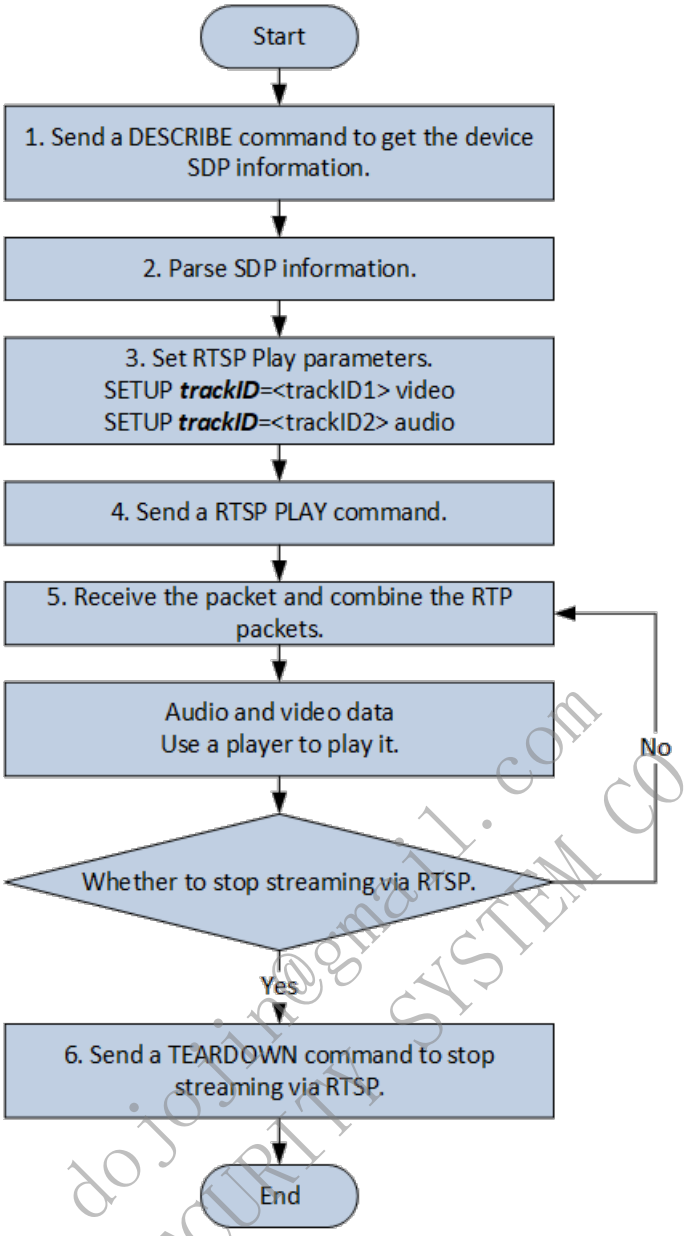
### 6.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

### 6.3.2 API Calling Flow

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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`; Digest authentication with devices is required.
2. The client parses the media SDP information returned by the device.
3. Set RTSP play parameters, set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP package sent by device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address, `[:port]` is optional, and 554 by default, `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream); for example, the IP address of the target device is `172.7.203.11`, and if the channel number is 172, the streaming address will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

### 6.3.3. Example

1. A client sends an RTSP DESCRIBE request.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s=Media Server V4.22.126
i=Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

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```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J.............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

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# 6.4 Manual Capturing Events

## 6.4.1 Introduction to the Function

According to the preset information and capture interval configured by the client, the device manually captures images at the specified preset and reports the image information in the form of events. It is used for the client to identify emergency situations and perform an emergency patrol on some presets for analysis. It has a higher priority than the daily scheduled patrol capture. Common usage scenarios: according to the scheduled capturing event (eventType: imageCapture) rules, the store camera regularly audits in a number of key areas. If in the event of an emergency, and a worker wants to immediately re-inspect each area, you can set a manual capture rule to complete an independent audit. After the audit is completed, the camera will continue to audit according to the scheduled capturing rules.

## 6.4.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualImageCapture` returns `true`, manual capturing shall be supported.
2. The client gets manual capturing capability for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture/capabilities?format=json`.
3. The client sets the parameters of manual capture events for a specified channel: Get parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`. Set parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`.
4. Receive the event uploaded by the device in arming or listening mode: `eventType:manualImageCapture`.

**Remarks:**

- Each time the client sets manual capture event parameters, the channel immediately takes a patrol capture of the relevant presets. When the last preset capture is finished, the manual capture event ends.

## 6.5 Preset Dedicated Focusing

### 6.5.1 Function Introduction

The preset dedicated focusing algorithm is designed for specific scenarios (such as instrument and gauge images) where the image is not clear at a certain position, to achieve clearer and faster focusing. The dedicated focusing algorithm includes foreground focusing, background focusing, and area focusing.

### 6.5.2 API Calling Flow

1. The client gets the device's PTZ capabilities via URL `GET /ISAPI/PTZCtrl/capabilities`. If `isSupportPresetsProFocus` is returned with `true` value, the device supports preset dedicated focusing.
2. The client gets the capability of preset dedicated focusing for the specified channel via URL `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg/capabilities?format=json`.
3. The client batch configures presets' dedicated focusing parameters for the specified channel.
   - Get parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
4. The client configures the dedicated focusing parameters of a single preset for the specified channel. If a preset is not clear under the batch configuration in step 3, it can be configured separately.
   - Add dedicated focusing parameters of a preset for a specified channel: `POST /ISAPI/PTZCtrl/channels/<channelID>/AddPresetsProFocusCfg?format=json`
   - Get parameters: `POST /ISAPI/PTZCtrl/channels/<channelID>/GetPresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/ModifyPresetsProFocusCfg?format=json`
   - Delete: `POST /ISAPI/PTZCtrl/channels/<channelID>/DeletePresetsProFocusCfg?format=json`
   - Get parameters of multiple presets: `GET /ISAPI/PTZCtrl/channels/<channelID>/GetMultiPresetsProFocusCfg?format=json`

### 6.5.3 Exception Handling

Error Codes

| statusCode | statusString | subStatusCode | errorCode | errorMsg | Chinese Description |
|---|---|---|---|---|---|
| 4 | Invalid Operation | numberReachedLlimit | 0x40001119 | The number reached limit. | The configured items have reached the limit and cannot be added further. Please delete unnecessary ones and try again. |

## 6.6 Sync Capturing Manually

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### 6.6.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately returns the captured picture data in respond. It is often used in scenarios with high network bandwidth between clients and devices to facilitate the rapid transmission of image data, such as a LAN built on a company campus.

### 6.6.2 API Calling Flow

1. (Optional) The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualSnapPicture` returns `true`, sync manual capturing shall be supported.
2. (Optional) The client gets sync manual capturing capability for a specified channel: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json`.
3. The client controls the specified channel to perform sync manual capturing: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>`, and the device returns the captured binary picture data.

**Remarks:**

- If you only need to capture a JEPG picture, you can jump to step 3.
- If you need to capture pictures other than JEPG or capture in the designated area of the video, you need to perform steps 1 and 2 to determine whether the device supports it. If it does, go to step 3.
- In step 3, the value ranges of `videoResolutionWidth` and `videoResolutionHeight` are the same as `<videoResolutionWidth> and <videoResolutionHeight>` returned by `GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities`.

## 6.7 Video Privacy Mask

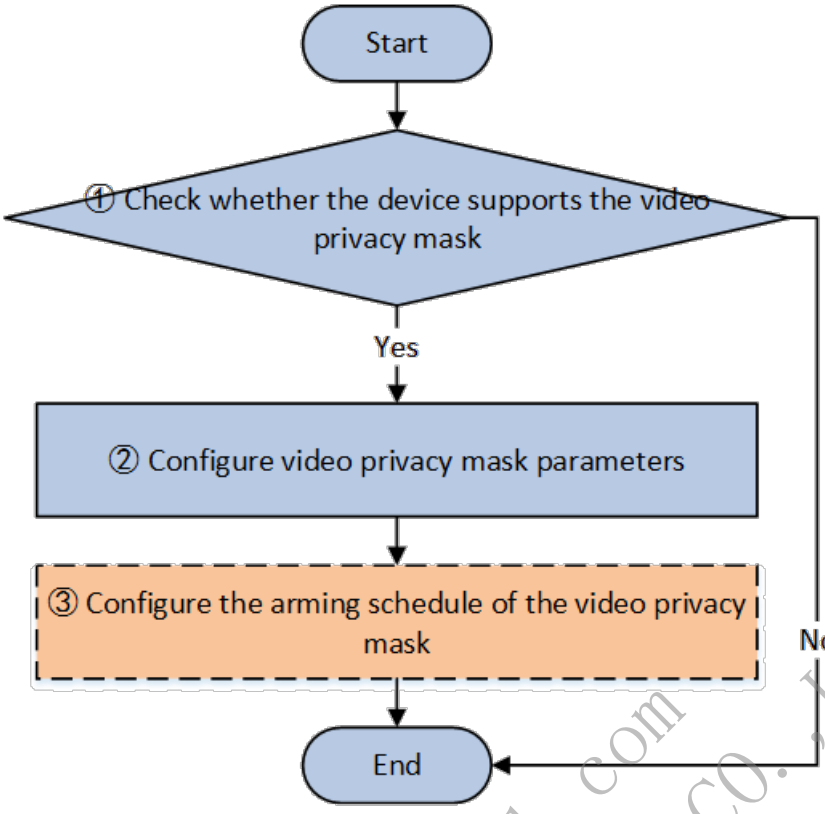
### 6.7.1 Introduction to the Function

Video privacy mask is to cover certain areas on the live image to protect personal privacy from being live viewed and recorded. You can use the masking method such as a mosaic or black frame as needed to make the area invisible. The device supporting privacy mask can specify the area(s) to be masked permanently on the live image, as a result of which the live view and recorded video during the live view will be influenced and the image cannot be restored. The following image shows the effect of masking an area with the mosaic.

### 6.7.2 API Calling Flow

#### 6.7.2.1 Video Privacy Mask

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**1. Check whether the device supports the video privacy mask.** Get the device's system capability: `GET /ISAPI/System/capabilities`. If `isSupportPrivacyMask` is returned and its value is true, it indicates that the video privacy mask is supported by the device.

**2. Configure the video privacy mask parameters.** You can specify sensitive area(s) to be masked on the live image (related field: `PrivacyMaskRegionList`), and up to 8 areas are supported. You can also configure the masking type, such as mosaic, black frame, and red frame (related field: `maskType`). Get the video privacy mask capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/privacyMaskCap`. Configure the video privacy mask parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`. Get the video privacy mask parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`.

**3. (Optional) Configure the arming schedule of the video privacy mask.** You can configure the privacy mask schedule as needed. If not configured, the privacy mask is performed 24 hours a day by default. Get the video privacy mask schedule: `GET /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`. Set the video privacy mask schedule:`PUT /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`.

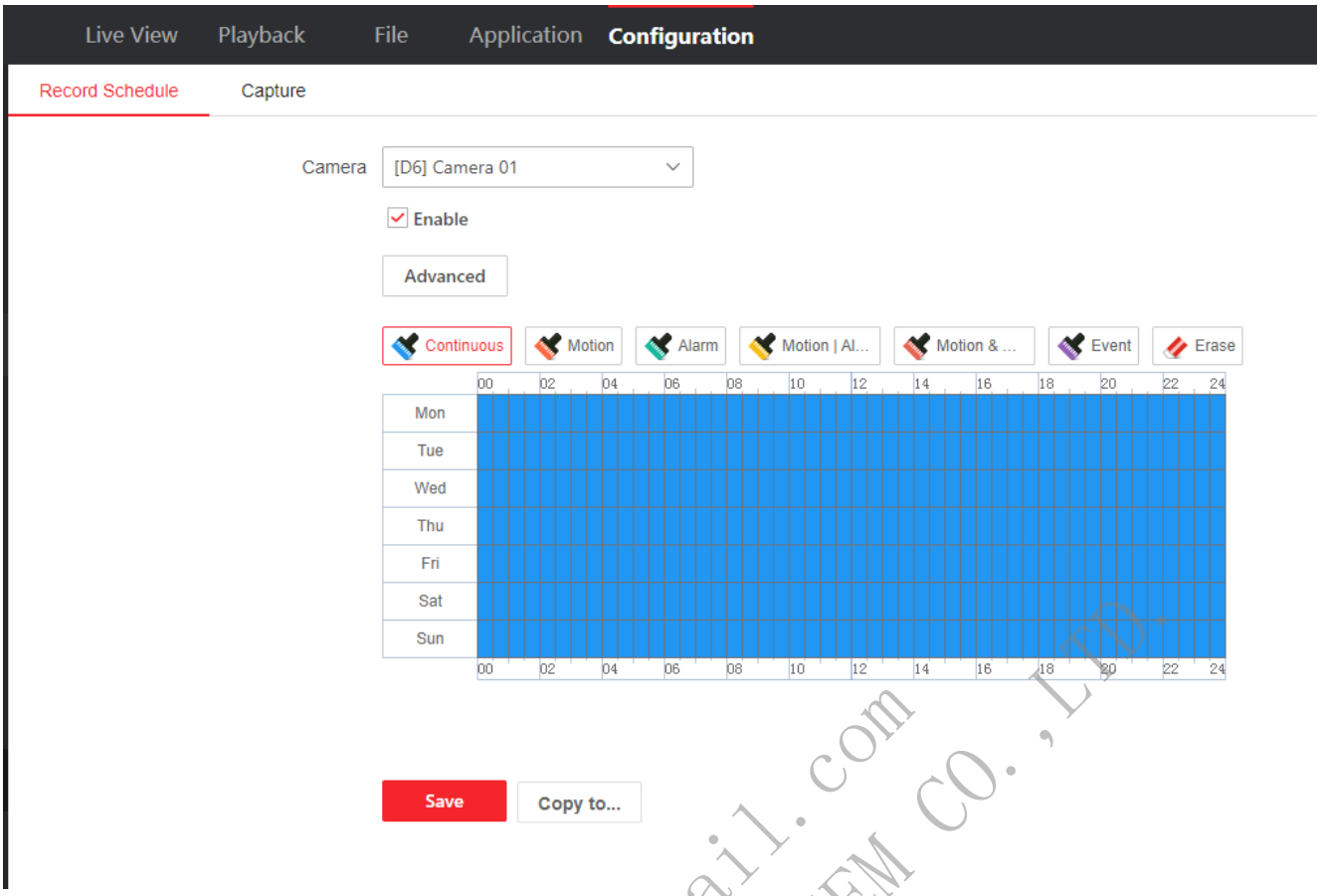
# 7 Video Management

## 7.1 . Scheduled Recording

### 7.1.1 Introduction to the Function

The web interface for reference is shown below.

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### 7.1.2 API Calling Flow

1. The client software gets the device system capability: `GET /ISAPI/System/capabilities`. When `isSupportScheduledRecordUpload` is returned and its value is `true`, it indicates that scheduled recording is supported.

2. The client software configure schedule parameters of scheduled recording. You can configure recording schedules for all channels or configure the recording schedule for a specified channel.

- Recording schedule management for all channels: Add a recording schedule: `POST /ISAPI/ContentMgmt/record/tracks`. Get all recording schedules parameters: `GET /ISAPI/ContentMgmt/record/tracks`. Set all recording schedules: `PUT /ISAPI/ContentMgmt/record/tracks`.

- Recording schedule management for a specified channel: Get the capability of configuring the recording schedule for a specified channel: `GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/capabilities?security=<security>&iv=<iv>`. Get the recording schedule parameters for a specified channel: `GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`. Set the recording schedule for a specified channel: `PUT /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`. Delete the recording schedule for a specified channel: `DELETE /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`.

## 7.2 Search for Event Video

### 7.2.1 Introduction to the Function

This function supports searching for the videos of linked cameras when an event occurs.

### 7.2.2 API Calling Flow

1. Check whether the device supports searching for event videos.

Get the device sysgem capability: `GET /ISAPI/System/capabilities`. If the node `<SerialCap>` is returned and its value is true, the device supports searching for event videos.

2. Search for event videos: `POST /ISAPI/ContentMgmt/eventRecordSearch?format=json`.

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# 7.3 Search for Videos and Pictures

## 7.3.1 Introduction to the Function

This function supports searching for the video/picture data stored in the device. Video data source includes scheduled recording, manual recording and event recording, and picture data source includes scheduled capture and event linkage capture.

## 7.3.2 API Calling Flow

1. Check the device's supported time format. Get device's system capabilities: `GET /ISAPI/System/capabilities`.
   - If node `isSupportTimeTypeSTD` is returned with value `true`, it indicates that the device supports the ISO 8601 standard time, and URLs of functions related to ISO 8601 time parameters support additional query parameter `timeType=STD`.
   - If node `isSupportTimeTypeSTD` is not returned. Call `GET /ISAPI/System/time/timeType?format=json` to get the time type. If the `type` field returns `local`, it indicates that the device video search and playback use local time. If the `type` field returns `UTC`, it indicates that the device video search and playback use ISO 8601 time format.
   - If the device does not support API `GET /ISAPI/System/time/timeType?format=json`. Call `GET /ISAPI/System/deviceInfo` and check the time type of device via field `deviceType`. If `deviceType` (device type) is CVR, NVR, DVR, or ITCCAM, the device video search and playback use local time. Otherwise, follow the protocol specifications and use the TZ format, for example, `2024-05-13T16:00:00Z`.

2. Check whether the device supports video/picture search. Get the device's video search capability via `GET /ISAPI/ContentMgmt/capabilities`. If `<isSupportRecordSearch>` is returned, it indicates that the device supports video/picture search.

3. Search for videos and pictures:
   i. Get the capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/capabilities`. Get the picture search types that the device supports searching via the node `<pictureSearchType>` returned in the message, including pictures of event capture, scheduled capture and manual capture. Get the video types that the device supports searching for via the node `<recordSearchType>` returned in the message.
   ii. Get the parameters capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/search/capabilities`. Get the periods when the device supports searching for videos and pictures via the node `<timeSpanList>` returned in the message. Get the file tags that the device supports searching via the node `<labelSearch>` returned in the message.
   iii. Search for videos and pictures: `POST /ISAPI/ContentMgmt/search?security=<security>&iv=<iv>`.
   - The URL containing `timeType=STD` indicates that the time in the input and output messages are according to the ISO 8601 time format. The URL without `timeType=STD` indicates to use device's original time.
   - Check whether it is picture or video to be searched via the node `<trackID>` in the request message `<CMSearchDescription>`. "101" (search for the main stream videos of channel 1), "103" (search for the pictures of channel 1), and it is channel 1 by default .

   **Note**:

   - The value of `CMSearchDescription.trackIDList.trackID` is obtained from the `id` value returned in the response message of `GET /ISAPI/ContentMgmt/record/tracks`. The meaning of `id`: `101` is the main stream of channel 1, `102` is the sub-stream of channel 1, `103` is the capture schedule of channel 1, `201` is the main stream of channel 2, and so on. When the device saves only one copy of channel videos, it does not distinguish between the main and sub stream, and only supports setting `trackID` to `xx1`.
   - If `<trackIDList>` is returned in response message of `GET /ISAPI/ContentMgmt/search/capabilities`, it indicates that the device supports accurate search, and the returned value in `<trackIDList>` indicates the channels that support independent video search.

# 7.4 Search for Videos by Target Type

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### 7.4.1 Introduction to the Function

This function realizes persons, vehicles or mixed targets distinguishing when searching for motion detection event videos. Searching for videos by target type is newly added. The search conditions include target type information, and the returned result of the search contains the start and end time as well as the file name of the video.

### 7.4.2 API Calling Flow

1. Check whether the device supports searching by target type:

Get the device capability: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports searching by target type. Currently, the device supports searching by person and by vehicle.

2. Get the capability of searching by target type: `GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json`.

3. Search for videos by target type: `POST /ISAPI/ContentMgmt/SearchByTargetType?format=json`.

## 7.5 Video Download

### 7.5.1 Introduction to the Function

After completing searching for videos (refer to the calling flow of video search), users can download the searched videos by time or file. When downloading the videos by time or file via web browser, users should enable browser plug-ins.

### 7.5.2 API Calling Flow

1. Check whether the device supports video download: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports downloading videos by time.

2. Download videos (plug-in-free):

   Get the capability of video download: `GET /ISAPI/ContentMgmt/download/capabilities>`.

   Download the videos: `GET /ISAPI/ContentMgmt/download?playbackURI=<playbackURI>&mediaID=<recordID>&downType=<downType>&encodeType=<encodeType>`.

   You can download videos by time or file via this protocol, and you can specify download method via the in the message. Details are as follows:

   - Download by file: the playbackURI uploads the start and end time, file name and size of the current file; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18T00:00:00Z;endtime=2016-07-18T00:25:04Z;name=00000000721000000&size=1065437356`.
   - Download by time: the playbackURI only uploads the start and end time; the device only parses time; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18T00:00:00Z;endtime=2016-07-18T00:25:04Z`.

   **Note**: The time in the URL should be in accordance with the ISO 8601 time format.

3. Download videos (with plug-in):

   Get the capability of video download: `GET /ISAPI/ContentMgmt/download/capabilities>`.

   Download the videos: `POST /ISAPI/ContentMgmt/download`.

   You can download videos by time or file via this protocol, and you can specify download method via the in the message. Details are as follows:

   - Download by file: the playbackURI uploads the start and end time, file name and size of the current video; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18 00:00:00&endtime=2016-07-18 00:25:04Z&name=00000000721000000&size=1065437356`.
   - Download by time: the playbackURI only uploads the start and end time; the device only parses time;

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`rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18 00:00:00&endtime=2016-07-18 00:25:04Z`.

**Note**: For website integrations, spaces in URL are not allowed, you can use &amp to escape the space.

# 8 PTZ

## 8.1 . PTZ Lock

### 8.1.1 Introduction to the Function

The client can control a network camera's PTZ lock. During the locked period, the PTZ linked with alarm events will not work, but manual rotation is available, until the client unlocks the PTZ. This function can be used for long-term monitoring of key locations to prevent the lens from being moved. When configuring or modifying intelligent applications that involve patrol, the PTZ should be locked first and then unlocked after the configuration is completed to achieve internal control of the intelligent application's operation, while the actual PTZ operation remains functional.

### 8.1.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `ManualPTZLockCap` indicates the support for manual `PTZ` control locking;
2. (Client) Control PTZ manually:

   Manually lock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLock?format=json`;

   Manually unlock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZUnlock?format=json`.
3. (Optional) The client configures the PTZ lock parameters to specify the lock duration:

   Capability: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `lockTime` indicates the maximum lock duration.

   Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.

   Get: `GET /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.
4. The client gets the remaining time for PTZ unlock: `GET /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLockCfg?format=json`.

## 8.2 Auxiliary Device Control

### 8.2.1 Introduction to the Function

The PTZ auxiliary devices help the device better display video content. The auxiliary devices includes lights, wipers, fans, and heaters. This function is used to control PTZ auxiliary devices.

### 8.2.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `PTZAuxType` indicates the support for PTZ auxiliary device control;
2. The client configures the PTZ auxiliary device parameters to control the auxiliary devices. It is recommended to first get all auxiliary device parameters before modifying parameter configurations:

   (Optional) Get all auxiliary device parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   (Optional) Get a single auxiliary device parameter: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`;

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(Optional) Configure auxiliary device parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

Configure a single auxiliary device parameter: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`.

**Notes:**

The `id` in the parameter configuration message corresponds to the `auxCtrlID` in the `url`, which is the auxiliary device index and starts from 1 by default.

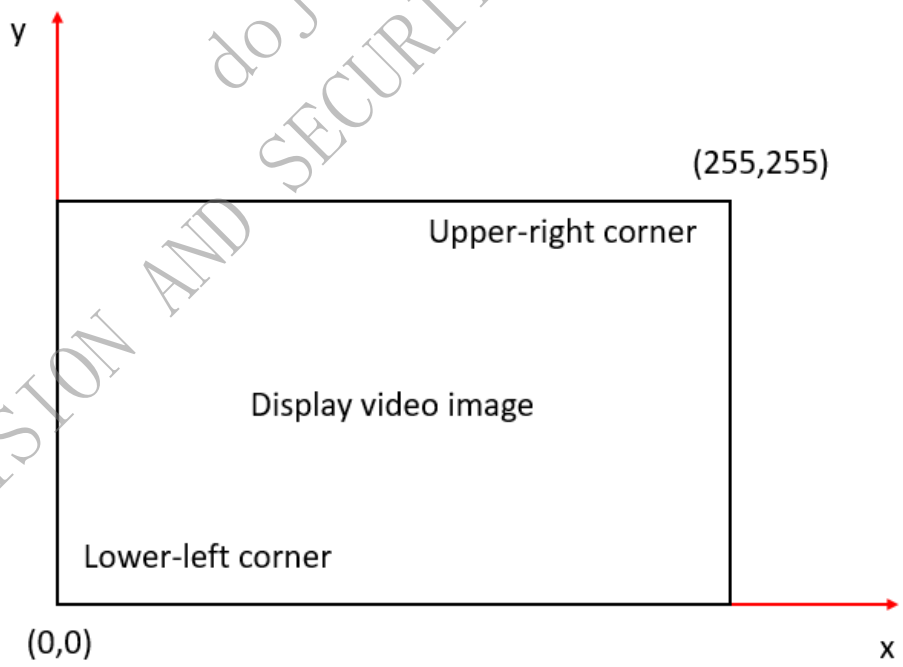
## 8.3 PTZ 3D Positioning Control

### 8.3.1 Introduction to the Function

Supports centering a point in the video image, and supports centering and zooming a rectangular area in the video image.

### 8.3.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPosition3D` indicates the support for 3D positioning control;
2. (Client) Send a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/position3D`:
    - The point in the video image is centered by the device, and the coordinates of `StartPoint` and `EndPoint` are the same;
    - The device zooms in and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is smaller than that of `EndPoint`;
    - The device zooms out and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is greater than that of `EndPoint`.




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## 8.4 PTZ Absolute Position

### 8.4.1 Introduction to the Function

The PTZ absolute position is unique and remains the same. The client can record certain PTZ absolute positions in actual application scenarios to repeatedly inspect at specified positions. For example, in a supermarket, record the PTZ

absolute position of the cash register and inspect the position periodically.

### 8.4.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. the returned `isSupportAbsoluteEx` indicates the support for high-precision absolute `PTZ` movement control;
2. (Client) Get the high-precision absolute `PTZ` movement control capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json`. The value range of control parameters will be returned;
3. The client specifies the PTZ to move to an absolute position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`. The `elevation` and `azimuth` represent the angles in the vertical and horizontal directions, `absoluteZoom` represents the zoom range, `focus` represents the focus range, and `absoluteRotate` represents the absolute rotation angle.
4. (Client) Get the PTZ absolute position: `GET /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`.

**Notes:**

- The zero coordinate position of `elevation` and `azimuth`, i.e., PT(0,0), can be specified by `PUT /ISAPI/PTZCtrl/channels/<channelID>/homeposition`.

## 8.5 PTZ Manual Control

### 8.5.1 Introduction to the Function

Adjust the horizontal and vertical positions of the PTZ, and control the zoom and focus of the lens at specified rotation/changing speeds. The client adjusts PTZ continuously through applying control commands, which is usually used for manual direction adjustment in remote live view.

### 8.5.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `ContinuousPanTiltSpace` in the message indicates the support for PTZ manual control;
2. The client sends a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/continuous`. The values of `pan`, `tilt`, `zoom`, and `rotate` represent the speed of rotation/change, and the characteristics (positive and negative) of the values represent the direction of rotation/change. If the control needs to be stopped, another command should be sent and values should be set as 0.

## 8.6 PTZ One-Touch Focus

### 8.6.1 Introduction to the Function

Help PTZ devices to focus. When defocus (blurry image) occurs , one-touch focus can make the image clear.

### 8.6.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsStart` indicates the support for one-touch focus;
2. (Client) Configure PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/start`;
3. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsReset` indicates the support for resetting focus parameters;
4. (Client) Reset the PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/reset`. The PTZ focus parameter setting will be restored.

## 8.7 Save PTZ Position

### 8.7.1 Introduction to the Function

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The client controls the device to save the current PTZ position information. After the PTZ leaves the current position, the client can control the PTZ to return to the saved position, but only one position can be saved for each PTZ.

### 8.7.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPTZSave` and `isSupportPTZSaveGet` indicates the support for keeping and getting `PTZ` parameters configurations;
2. (Client) Configure the device to save `PTZ` information: `PUT /ISAPI/PTZCtrl/channels/<channelID>/save?format=json`;
3. The client controls the PTZ to move to the saved position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/gotoPosition?format=json`. The PTZ will rotates to the saved position;
4. The client gets the saved PTZ position information: `GET /ISAPI/PTZCtrl/channels/<channelID>/save`. The PTZ position information will be returned and PTZ does not move.

# 9 Vehicle Recognition

## 9.1 Motor Vehicle Recognition

### 9.1.1 Introduction to the Function

This function is to detect whether there are vehicles appearing in the image for certain time periods. If a vehicle appears and the set recognition conditions are met, it will be captured, and its related information will be analyzed and uploaded as alarms. The integrator receives the event and analyzes the specific content as needed.

### 9.1.2. Device Event Upload

#### 9.1.2.1 Check Whether Device Supports Baseline Vehicle Recognition Event

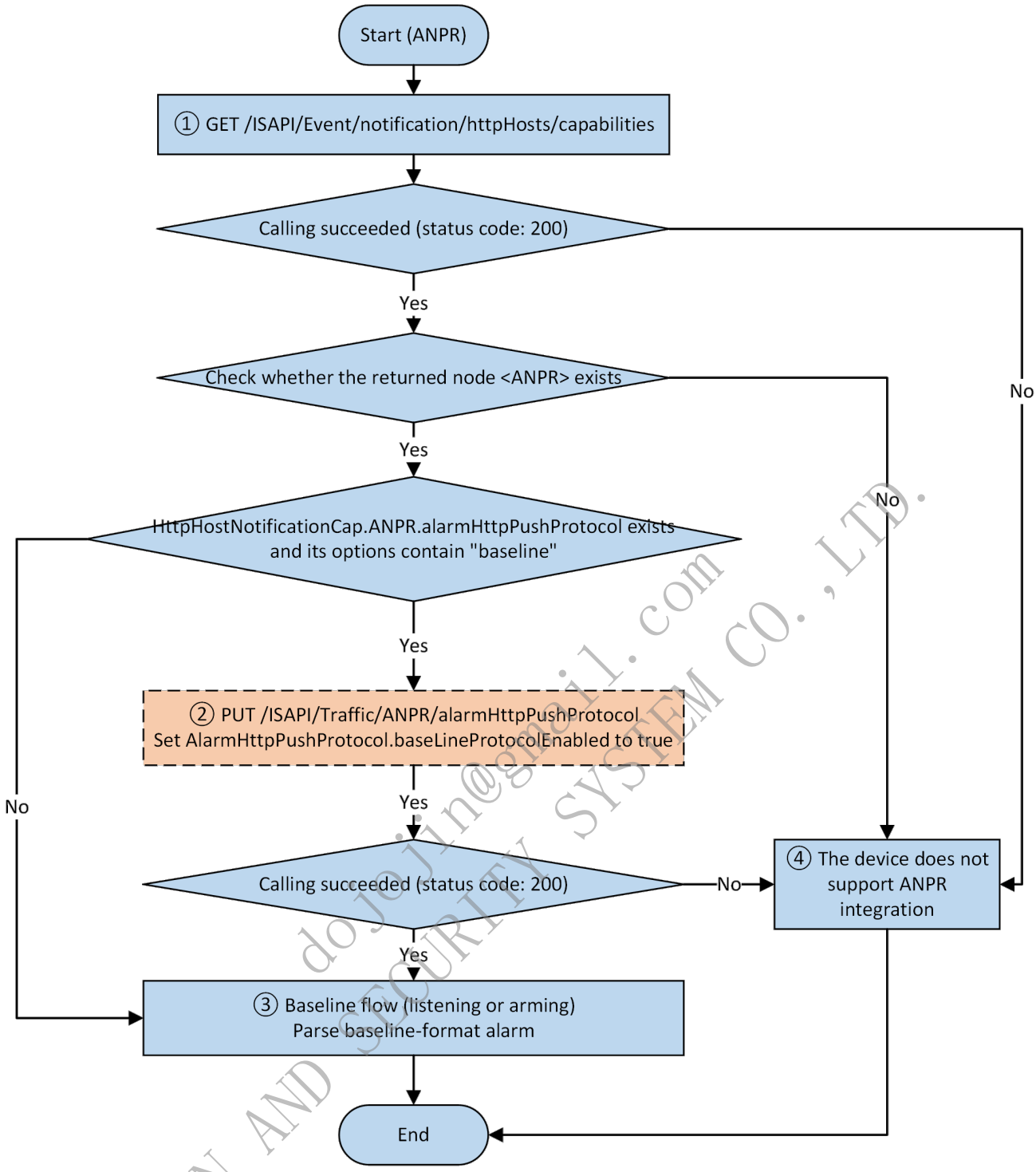
The integrator get the capability from device.

1. Get the system capability set via `GET /ISAPI/System/capabilities`: `true` is returned for `<ITCCap>.<isSupportVehicleDetection>` by device or `<ITCCap>.<isSupportHVTVehicleDetection>` is `true`.
2. Get the vehicle capture recognition capability via `GET /ISAPI/ITC/capabilities`: `true` is returned for `<isSupportVehicleDetection>` by device or `<isSupportHVTVehicleDetection>` is `true`.
3. Get the traffic service capability via `/ISAPI/Traffic/capabilities`: `<plateCap>` is returned by device.
4. Get the trigger mode capability set via `GET /ISAPI/ITC/TriggerMode/capabilities`: `<TriggerMode>` is returned.

When any one of the 4 conditions are met, it indicates the vehicle recognition event detection is supported. If none are met, the vehicle recognition event is not supported and please do not proceed to the following steps.

#### 9.1.2.2 Check Whether Device Supports Baseline Flow

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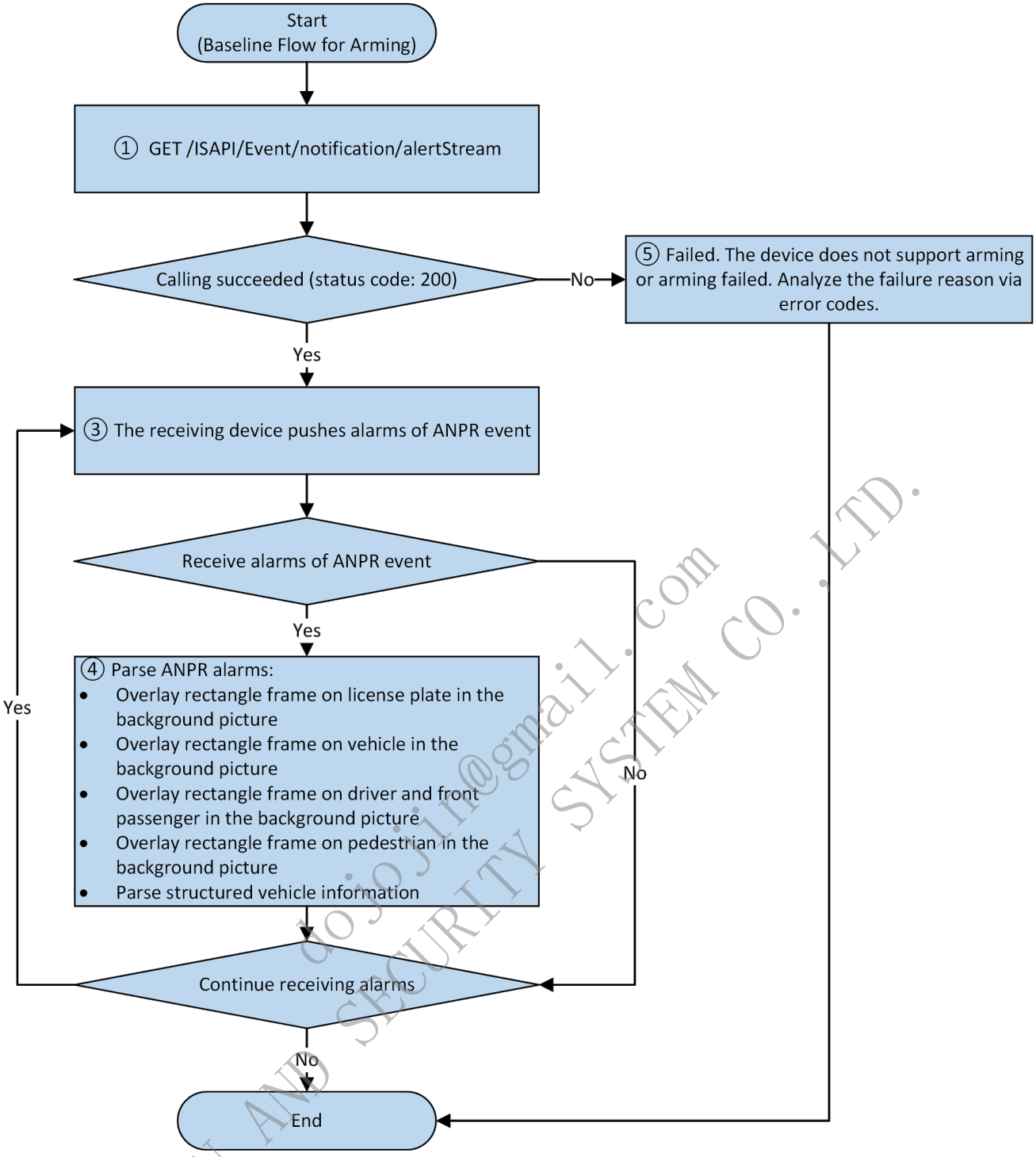



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1. (Client) Get the configuration capability of listening host via `GET /ISAPI/Event/notification/httpHosts/capabilities`.
2. (Client) Configure the alarm mode switch for device via `PUT /ISAPI/Traffic/ANPR/alarmHttpPushProtocol` and set `AlarmHttpPushProtocol.baseLineProtocolEnabled` to `true`.
3. Receive and parse alarms according to the baseline flow. See details in `4.3 Baseline Flow Event Upload` and `4.4 Baseline Flow Message Format and Example`.
4. The ANPR function integration is not supported by device.

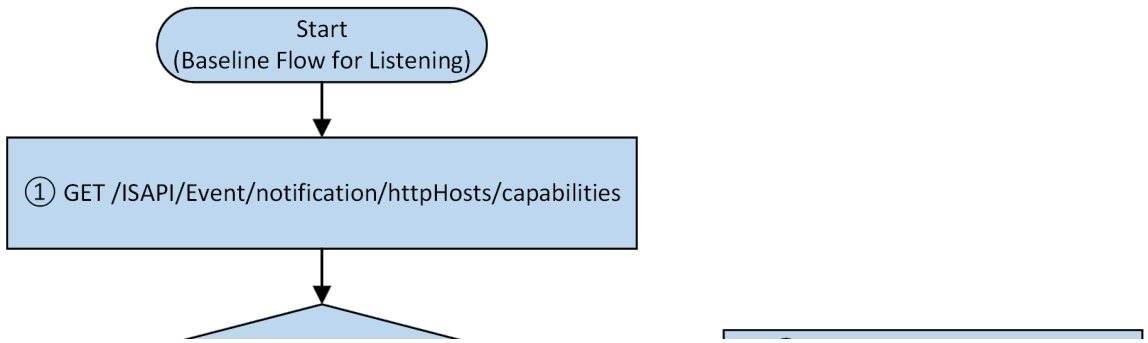
## 9.1.3 Baseline Flow Event Upload

### 9.1.3.1 Arming Flow

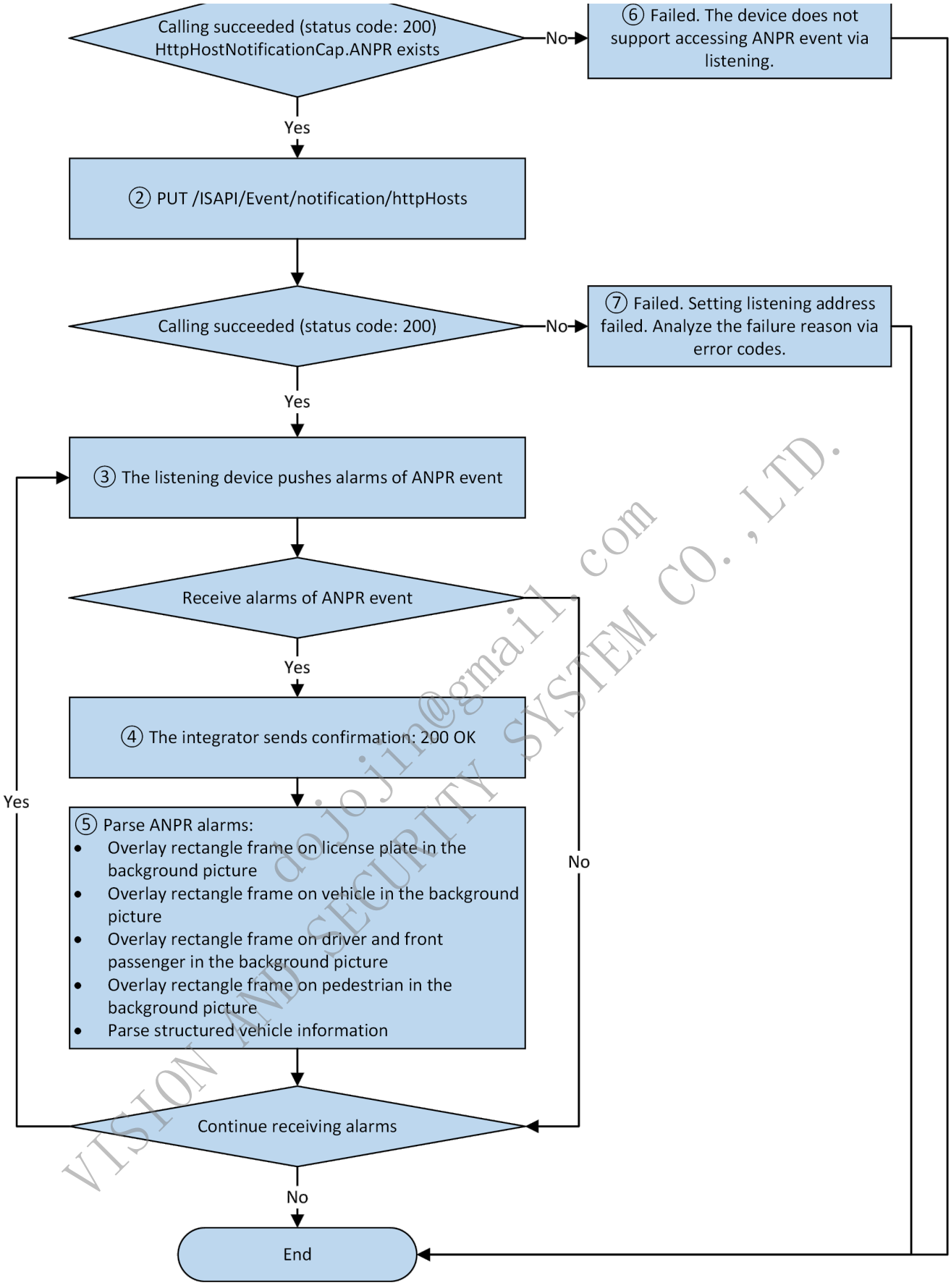



1. Arm the device.
2. Check whether arming succeeded.
3. Receive alarm notifications and check whether `eventType==ANPR`.
4. Parse ANPR alarm content.
5. Arming failed. Analyze the failure reason via error codes.

#### 9.1.3.2 Listening Flow




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1. Get the capability set of device HTTP listening host location.
2. Change the HTTP listening host location, IP address, port configuration to the IP address of the server for receiving alarm notifications.
3. Receive alarm notifications and check whether `eventType==ANPR`.
4. The integrator sends confirmation (200 OK) to the device and notify the device of the successful alarm receiving; if the integrator does not reply, the device will consider the alarm notification lost and upload it again.
5. Parse ANPR alarm content.

6. The device does not support accessing ANPR event via listening.
7. Setting listening address failed. Analyze the failure reason via error codes.

## 9.1.4 Baseline Flow Message Format and Example

### 9.1.4.1 Baseline Flow Event Upload Method Configuration

Set `baseLineProtocolEnabled` to `true`,

URI: `/ISAPI/Traffic/ANPR/alarmHttpPushProtocol`

Method: `GET, PUT`

Body:

```
<AlarmHttpPushProtocol version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <baseLineProtocolEnabled>true</baseLineProtocolEnabled>
</AlarmHttpPushProtocol>
```

### 9.1.2.2 Baseline Flow Event Upload Example

Reported by device.

```
POST /test HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: zh-CN
Content-Type: multipart/form-data; boundary=7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache

--7e13971310878
Content-Disposition: form-data; name="anpr.xml";filename="anpr.xml";
Content-Type: text/xml
Content-Length: 150

<EventNotificationAlert/>
-----------------------------7e13971310878
Content-Disposition: form-data; name="licensePlatePicture.jpg";filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 200

......JFIF.....`.`.....C.......... .
..
.............. $.' ",#..(7),01444.'9=82<.342...C. ....

.2!.!2222222222222222222222222222222222222222222222222222.....q.<.."..................................
.....................}........!1A..Qa."q.2....#B...R..$3br.
.....%&'()*456789:CDEFGHIJSTUVWXYZcdefghijstuvwxyz.....................................................................................................
...................w.......!1..AQ.aq."2...B.... #3R..br.
.$4.
--7e13971310878
Content-Disposition: form-data; name="detectionPicture.jpg";filename="detectionPicture.jpg";
Content-Type: image/jpeg
Content-Length: 9907

......JFIF.....`.`.....C.......... .
..
.............. $.' ",#..(7),01444.'9=82<.342...C. ....

.2!.!2222222222222222222222222222222222222222222222222222.....q.<.."..................................
.....................}........!1A..Qa."q.2....#B...R..$3br.
.....%&'()*456789:CDEFGHIJSTUVWXYZcdefghijstuvwxyz.....................................................................................................
...................w.......!1..AQ.aq."2...B.... #3R..br.
.$4.
--7e13971310878--
```

The receiver responds.

```
HTTP/1.1 200 OK
Connection: close
```

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**Note**: When there are multiple pictures of the same type, the following naming rule for will be applied: add suffix (e.g., "_1", "_2") after the extended name. For example, the names of three detection pictures can be detectionPicture_1.jpg, detectionPicture_2.jpg, and detectionPicture_3.jpg.

| Attribute Value | Attribute Type Character String |
|---|---|
| ANPR Alarm XML | anpr.xml |
| Detection Picture (Background Picture) | detectionPicture.jpg |
| License Plate Thumbnail | licensePlatePicture.jpg |
| Driver's Face Matting | pilotPicture.jpg |
| Passenger's Face Matting | copilotPicture.jpg |
| Composite Picture | compositePicture.jpg |
| License Plate Binary Picture | plateBinaryPicture.jpg |
| Non-Motor Vehicle Matting | nonMotorPicture.jpg |
| Pedestrian Picture | pedestrianDetectionPicture.jpg |
| Pedestrian Matting | pedestrianPicture.jpg |

# 10 Traffic Control

## 10.1 Traffic Enforcement of Vehicles

### 10.1.1 Introduction to the Function

#### 10.1.1.1 Smart Linkage System for Intelligent Transportation

**Application Scenario:** The smart linkage system of intelligent traffic consists of cameras and traffic speed domes, which support functions of remote monitoring via the network, video servers, and high definition, for traffic violation enforcement, traffic incident analysis, and traffic data statistics. Without people attending, the system can automatically capture and save key information for automatic enforcement, which helps save on labor costs, improve enforcement efficiency, and reduce disputes. The smart linkage system can be widely used in places that require large-scale high-definition monitoring, such as roads, airports, stations, docks, ports, tourist attractions, and streets.

**Enforcement Function:** The traffic enforcement event detection supports the function of violation enforcement, which can automatically detect and capture the evidence of events, including illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, and motor vehicle on non-motor vehicle lane, on the roads with no more than four lanes. The enforcement results will be uploaded as alarms in real time.

#### 10.1.1.2 Smart Linkage System for Parking Management

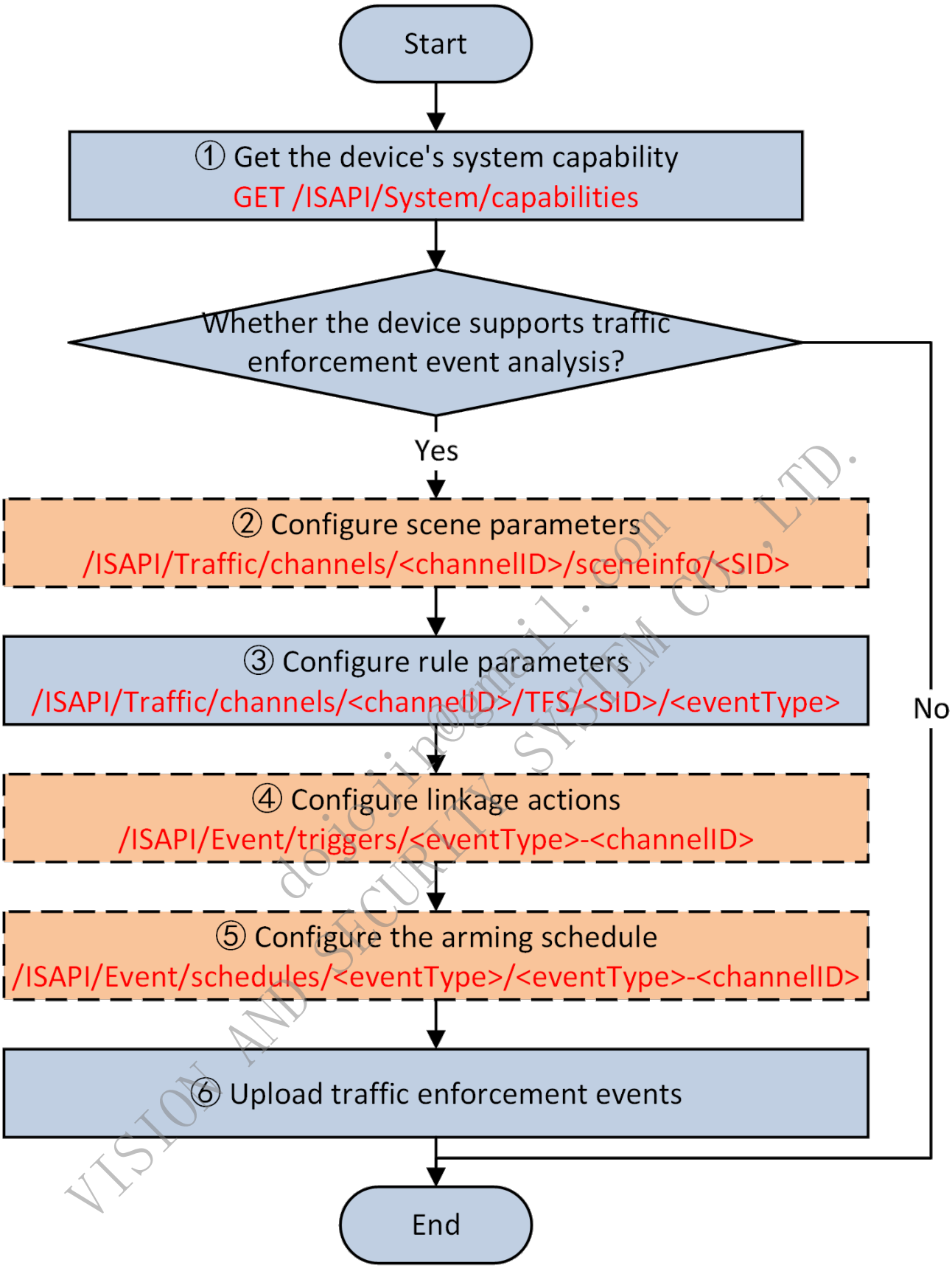
**Application Scenario:** The smart linkage system for parking management (hereinafter referred to as "the system") consists of parking speed domes and cameras (hereinafter referred to as "the devices"). The parking speed dome supports functions of remote monitoring via the network, video servers, and high definition. The system is easy to install and use without complex wiring. It focuses on the detection of the parallel parking event during enforcement. Without people attending, the system can automatically capture and save key information for automatic enforcement of the vehicle information in the parking space, which helps save on labor costs.

The devices are mainly used in places like the parking lot.

**Parallel Parking Detection:** The parallel parking detection can detect the parallel parking event of up to four parking spaces during enforcement and upload the vehicle information in the parking space in real time.

### 10.1.2 API Calling Flow

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1. **Get the device's system capability:**

   Get the device's system capability to check the traffic enforcement events supported by the device: `GET /ISAPI/System/capabilities`. The value of the node TFSEventSupport.opt can be illegalParking (illegal parking detection), wrongDirection (wrong-way driving detection), crossLane (driving on the lane line detection), laneChange (illegal lane change detection), vehicleExist (motor vehicle on non-motor vehicle lane detection), turnRound (illegal U-turning detection), parallelParking (parallel parking detection), notKeepDistance (not keeping vehicle distance detection), notSlowZebraCrossing (not slowing down at zebra crossing detection), overtakeRightSide (overtaking on the right detection), lowSpeed (driving in low speed detection), dragRacing

(street racing detection), changeLaneContinuously (continuous lane change detection), SSharpDriving (slalom driving detection), largeVehicleOccupyLine (lane occupation by large-sized vehicle detection), jamCrossLine (queue jumping detection), blackSmokeVehicle (vehicle emitting black smoke detection), or turnRightStop (large-sized vehicle not slowing down when turning right detection).

2. **Configure scenes:**

   Get the capability of configuring basic channel parameters to check whether the node `isEvidenceGettingSupport` exists and its value is true: `GET /ISAPI/Traffic/channels/<channelID>/capability`. If so, it indicates that the device supports configuring scene and rule parameters.

   Get the capability of configuring basic scene parameters of a specific channel: `GET /ISAPI/Traffic/channels/<channelID>/sceneinfo/capabilities`.

   Set the basic scene parameters for a specific channel: `PUT /ISAPI/Traffic/channels/<channelID>/sceneinfo/<SID>`. If you need to use the scene, the node SceneInfo.enabled should be set to true.

3. **Configure rules:**

   Get the capability of configuring the **event** rule parameters: `GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/XXX/capabilities`.

   Get the **event** rule parameters: `GET/ISAPI/Traffic/channels/<channelID>/TFS/<SID>/XXX`.

   Set the **event** rule parameters: `PUT/ISAPI/Traffic/channels/<channelID>/TFS/<SID>/XXX`.

   **Notes:**

   i. The value of `XXX` can be set to: illegalParking (illegal parking detection), wrongDirection (wrong-way driving detection), crossLane (driving on the lane line detection), laneChange (illegal lane change detection), vehicleExist (motor vehicle on non-motor vehicle lane detection), turnRound (illegal U-turning detection), parallelParking (parallel parking detection), notKeepDistance (not keeping vehicle distance detection), notSlowZebraCrossing (not slowing down at zebra crossing detection), overtakeRightSide (overtaking on the right detection), lowSpeed (driving in low speed detection), dragRacing (street racing detection), changeLaneContinuously (continuous lane change detection), SSharpDriving (slalom driving detection), largeVehicleOccupyLine (lane occupation by large-sized vehicle detection), jamCrossLine (queue jumping detection), blackSmokeVehicle (vehicle emitting black smoke detection), or turnRightStop (large-sized vehicle not slowing down when turning right detection).

   e.g.: Get the capability of configuring parameters of illegal parking detection: `GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking/capabilities`.

   Get or set parameters of illegal parking detection: `GET/PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking`.

   ii. For traffic speed domes, the rule parameters are required. For box cameras, the rule parameters are not required. The scene No. is 1 by default.

4. **Configure linkage actions:**

   Check whether the device supports configuring linkage actions of the illegal parking event by calling `GET /ISAPI/Event/triggersCap` and check whether the message `EventTriggersCap` contains the node `IllegalParkingTriggerCap`.

   Get the default or configured linkage action of the illegal parking event: `GET /ISAPI/Event/triggers/illegalParking-<channelID>`.

   Configure the linkage actions of the illegal parking event: `PUT /ISAPI/Event/triggers/illegalParking-<channelID>`.

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**Notes**: Only illegal parking detection supports linkage action configurations. You do not need to configure linkage actions for other events and they will be uploaded to the center by default.

5. **Configure arming schedules:**

   Check whether the device supports configuring the arming schedule for the illegal parking detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportillegalParking`.

   Configure the arming schedule for the illegal parking event: `GET/PUT /ISAPI/Event/schedules/illegalParkings/illegalParking-<channelID>`.

   Check whether the device supports configuring the arming schedule for the wrong-way driving detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportWrongDirection`.

   Configure the arming schedule for the wrong-way driving event: `GET/PUT /ISAPI/Event/schedules/wrongDirections/wrongDirection-<channelID>`.

   Check whether the device supports configuring the arming schedule for the illegal U-turning event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportTurnRound`.

   Configure the arming schedule for the illegal U-turning event: `GET/PUT /ISAPI/Event/schedules/turnRounds/turnRound-<channelID>`.

   Check whether the device supports configuring the arming schedule for the driving on the lane line event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportCrossLane`.

   Configure the arming schedule for the driving on the lane line event: `GET/PUT /ISAPI/Event/schedules/crossLanes/crossLane-<channelID>`.

   Check whether the device supports configuring the arming schedule for the illegal lane change detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportLaneChange`.

   Configure the arming schedule for the illegal lane change event: `GET/PUT /ISAPI/Event/schedules/laneChanges/laneChange-<channelID>`.

   Check whether the device supports configuring the arming schedule for the motor vehicle on non-motor vehicle lane event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportVehicleExist`.

   Configure the arming schedule for the motor vehicle on non-motor vehicle lane event: `GET/PUT /ISAPI/Event/schedules/vehicleExists/vehicleExist-<channelID>`.

   Check whether the device supports configuring the arming schedule for the parallel parking detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportParallelParking`.

   Configure the arming schedule for the parallel parking event: `GET/PUT /ISAPI/Event/schedules/parallelParkings/parallelParking-<channelID>`.

   **Notes:**

   i. You can determine whether to configure the arming schedule or not. If the arming schedule is not configured, it is all-day arming by default.

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ii. Only the illegal parking detection, wrong-way driving detection, illegal U-turning detection, driving on the lane line detection, illegal lane change detection, motor vehicle on non-motor vehicle lane detection, and parallel parking detection support arming schedule configuration. For other events, configuring the arming schedule is not supported and the default arming schedule is all-day arming.

6. **Upload the traffic enforcement events** (`"eventType":"TFS"`). In the event message, TFS.illegalTrafficEvent is the specific traffic violation event, TFS.VehicleInfo is the vehicle information of the traffic violation, and TFS.PlateInfo is the license plate information of the vehicle violation. The detectionPicturesNumber refers to the number of detected violation pictures. The detectionPictureTransType refers to the transmission method of the pictures.

### 10.1.3 Exception Handling

#### 10.1.3.1 Error Codes

| statusCode | statusString | subStatusCode | errorCode | errorMsg | Description | Rema |
|---|---|---|---|---|---|---|
| 6 | Invalid Content | intelligentTrafficMutexWithHighFrames | 0x60008014 | | Disable all functions of traffic incident detection, traffic violation enforcement, and traffic data statistics, or adjust the video frame rate to 50 or below. | |
| 6 | Invalid Content | intelligentTrafficMutexWithHighFramesEx | 0x60008018 | | Disable all functions of traffic incident detection, traffic violation enforcement, traffic data statistics, and vehicle detection, or adjust the video frame rate to 50 or below. | |

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## 10.2 Traffic Incidents of Vehicles

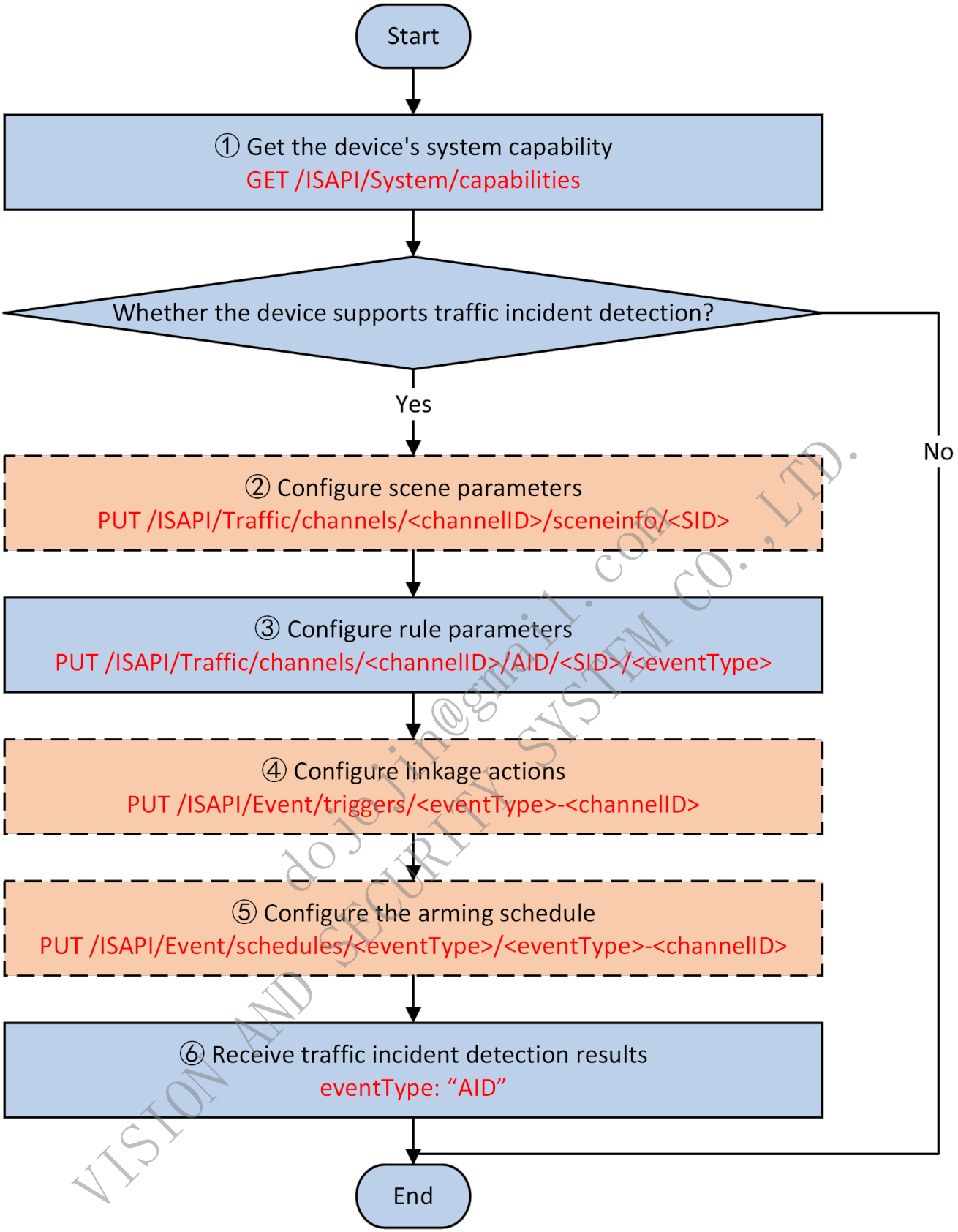
### 10.2.1 Introduction to the Function

**Application Scenario:** The smart linkage system of intelligent traffic consists of cameras and traffic speed domes, which support functions of remote monitoring via the network, video servers, and high definition, for traffic violation enforcement, traffic incident analysis, and traffic data statistics. Without people attending, the system can automatically capture and save key information for automatic enforcement, which helps save on labor costs, improve enforcement efficiency, and reduce disputes. The smart linkage system can be widely used in places that require large-scale high-definition monitoring, such as roads, airports, stations, docks, ports, tourist attractions, and streets.

**Intelligent Traffic Functions:** The AID function can detect traffic incidents that occurred on the roads with no more than four lanes and trigger alarms in real time. Supported incidents include thrown object detection, pedestrian detection, congestion detection, roadblock detection, traffic accident detection, fog detection, etc.

### 10.2.2 API Calling Flow

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1. **Get the device's system capability:**

   Get the device's system capability to check the traffic incidents supported by the device: `GET /ISAPI/System/capabilities`. The value of the node AIDEventSupport.opt can be abandonedObject (thrown object detection), pedestrian (pedestrian detection), congestion (congestion detection), roadBlock (roadblock detection), construction (construction detection), trafficAccident (traffic incident detection), fogDetection (fog detection), or obstacle (obstacle detection).

2. **Configure scenes:**

Get the capability of configuring basic channel parameters to check whether the node `isEvidenceGettingSupport` exists and its value is true: `GET /ISAPI/Traffic/channels/<channelID>/capability`. If so, it indicates that the device supports configuring scene and rule parameters.

Get the capability of configuring basic scene parameters of a specific channel: `GET /ISAPI/Traffic/channels/<channelID>/sceneinfo/capabilities`.

Set the basic scene parameters for a specific channel: `PUT /ISAPI/Traffic/channels/<channelID>/sceneinfo/<SID>`. If you need to use the scene, the node SceneInfo.enabled should be set to true.

3. **Configure rules:**

   Get the capability of configuring the **event** rule parameters: `GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/XXX/capabilities`.

   Get the **event** rule parameters: **`GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/XXX`**.

   Set the **event** rule parameters: `PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/XXX`.

   **Notes:**

   1. The value of `XXX` can be set to: abandonedObject (thrown object detection), pedestrian (pedestrian detection), congestion (congestion detection), roadBlock (roadblock detection), construction (construction detection), trafficAccident (traffic accident detection), fogDetection (fog deteciton), or obstacle (obstacle detection).

   e.g.: Get the capability of configuring parameters of thrown object detection: `GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/abandonedObject/capabilities`.

   Get or set parameters of thrown object detection: `GET/PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/abandonedObject`.

   2. For traffic speed domes, the rule parameters are required. For box cameras, the rule parameters are not required. The scene No. is 1 by default.

4. **Configure linkage actions:**

   Check whether the device supports configuring linkage actions for the thrown object event by calling `GET /ISAPI/Event/triggersCap` and checking whether the message `EventTriggersCap` contains the node `AbandonedObjectTriggerCap`.

   Configure linkage actions for the thrown object event: `GET/PUT /ISAPI/Event/triggers/abandonedObject-<channelID>`.

   Check whether the device supports configuring linkage actions for the pedestrian detection event by calling `GET /ISAPI/Event/triggersCap` and checking whether the message `EventTriggersCap` contains the node `PedestrianTriggerCap`.

   Configure linkage actions for the pedestrian detection: `GET/PUT /ISAPI/Event/triggers/pedestrian-<channelID>`.

   Check whether the device supports configuring linkage actions for the congestion event by calling `GET /ISAPI/Event/triggersCap` and checking whether the message `EventTriggersCap` contains the node `CongestionTriggerCap`.

   Configure linkage actions for the congestion event: `GET/PUT /ISAPI/Event/triggers/congestion-<channelID>`.

   Check whether the device supports configuring linkage actions for the roadblock detection event by calling `GET /ISAPI/Event/triggersCap` and checking whether the message `EventTriggersCap` contains the node `RoadBlockTriggerCap`.

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Configure linkage actions for the roadblock detection event: `GET/PUT /ISAPI/Event/triggers/roadBlock-<channelID>`.

Check whether the device supports configuring linkage actions for the construction detection event by calling `GET /ISAPI/Event/triggersCap` and checking whether the message `EventTriggersCap` contains the node `ConstructionTriggerCap`.

Configure linkage actions for the construction detection: `GET/PUT /ISAPI/Event/triggers/construction-<channelID>`.

Check whether the device supports configuring linkage actions for the traffic accident detection event by calling `GET /ISAPI/Event/triggersCap` and checking whether the message `EventTriggersCap` contains the node `TrafficAccidentTriggerCap`.

Configure linkage actions for the traffic accident detection: `GET/PUT /ISAPI/Event/triggers/trafficAccident-<channelID>`.

**Notes**:

1. You can determine whether to configure the linkage actions or not. If the linkage actions are not configured, the device will upload the event to the center by default.

2. The fog detection and obstacle detection do not support linkage action configuration. The device will upload them to the center by default.

5. **Configure arming schedules:**

   Check whether the device supports configuring the arming schedule for the thrown object event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportAbandonedObject`.

   Configure the arming schedule for the thrown object event: `GET/PUT /ISAPI/Event/schedules/abandonedObjects/abandonedObject-<channelID>`.

   Check whether the device supports configuring the arming schedule for the pedestrian detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportPedestrian`.

   Configure the arming schedule for the pedestrian detection: `GET/PUT /ISAPI/Event/schedules/pedestrians/pedestrian-<channelID>`.

   Check whether the device supports configuring the arming schedule for the congestion event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportCongestion`.

   Configure the arming schedule for the congestion event: `GET/PUT /ISAPI/Event/schedules/congestions/congestion-<channelID>`.

   Check whether the device supports configuring the arming schedule for the roadblock detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportRoadBlock`.

   Configure the arming schedule for the roadblock detection event: `GET/PUT /ISAPI/Event/schedules/roadBlocks/roadBlock-<channelID>`.

   Check whether the device supports configuring the arming schedule for the construction detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportConstruction`.

   Configure the arming schedule for the construction detection: `GET/PUT /ISAPI/Event/schedules/constructions/construction-<channelID>`.

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Check whether the device supports configuring the arming schedule for the traffic accident detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportTrafficAccident`.

Configure the arming schedule for the traffic incident detection: `GET/PUT /ISAPI/Event/schedules/trafficAccidents/trafficAccident-<channelID>`.

Check whether the device supports configuring the arming schedule for the fog detection by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportFogDetection`.

Configure the arming schedule for the fog detection: `GET/PUT /ISAPI/Event/schedules/fogDetection/fogDetection-<channelID>`.

**Notes:**

1. You can determine whether to configure the arming schedule or not. If the arming schedule is not configured, it is all-day arming by default.

2. The obstacle detection does not support arming schedule configuration. The default arming schedule of the device is all-day arming.

6. **Upload the traffic incident detection results** (`"eventType":"AID"`). In the event message, AID.illegalTrafficEvent is the specific traffic incident, AID.illegalTrafficEvent is the vehicle information, and AID.PlateInfo is the license plate information of the vehicle. The detectionPicturesNumber refers to the number of pictures in the alarm. The detectionPictureTransType refers to the transmission method of the pictures.

## 10.2.3 Exception Handling

#### 10.2.3.1 Error Codes

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| statusCode | statusString | subStatusCode | errorCode | errorMsg | Description | Rema |
|---|---|---|---|---|---|---|
| 6 | Invalid Content | intelligentTrafficMutexWithHighFrames | 0x60008014 | | Disable all functions of traffic incident detection, traffic violation enforcement, and traffic data statistics, or adjust the video frame rate to 50 or below. | |
| 6 | Invalid Content | intelligentTrafficMutexWithHighFramesEx | 0x60008018 | | Disable all functions of traffic incident detection, traffic violation enforcement, traffic data statistics, and vehicle detection, or adjust the video frame rate to 50 or below. | |

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# 11 Video Recognition

## 11.1 Professional Smart Abnormal Event Detection

### 11.1.1 Introduction to the Function

Professional Smart Abnormal Event Detection includes the following features:

- Line Crossing Detection
- Intrusion Detection
- Region Entrance Detection
- Region Exiting Detection
- Loitering Detection
- People Gathering Detection
- Rapid Movement Detection
- Parking Detection
- Unattended Baggage Detection
- Object Removal Detection

- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

Professional smart abnormal event detection is designed for specific scenarios: workplace activity detection and pedestrian activity detection.

Workplace activity detection includes but not limited to:

- Using a Mobile Phone
- Number of People Changes
- On/Off Duty
- Person Lingering

Pedestrian activity detection includes but not limited to:

- Running Detection
- People Gathering Detection
- Violent Motion Detection
- Person Falling Detection

You can configure workplace activity detection and pedestrian activity detection rules and alarm linkage actions to suit your needs.

Abnormal event detection can be categorized into two types: Basic smart abnormal event detection and professional smart abnormal event detection. Their differences are as follows:

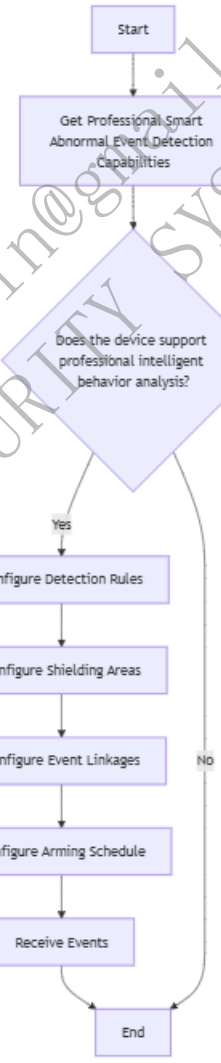
1. **Algorithms**: Traditional algorithms are used for basic smart abnormal event detection, while deep learning algorithms are used for professional smart abnormal event detection.
2. **Detection Effect**: Basic smart abnormal event detection can only analyze targets as people or vehicles. Professional smart abnormal event detection can not only distinguish between targets as people or vehicles but also identify specific behaviors, such as the number of people and human behaviors such as using a mobile phone, violent motion, and running.
3. **Supported Events**: Smart Basic Intelligent Abnormal Behavior Detection supports events such as:
    - Line Crossing Detection
    - Intrusion Detection
    - Region Entrance Detection
    - Region Exiting Detection
    - Loitering Detection
    - People Gathering Detection
    - Rapid Movement Detection
    - Parking Detection
    - Unattended Baggage Detection
    - Object Removal Detection

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Professional Smart Abnormal Event Detection supports all the events mentioned above and additional events including:

- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

### 11.1.2 API Calling Flow




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**Before taking the following steps, ensure that this smart detection has been enabled for devices.**

1. **Get Professional Smart Abnormal Event Detection Capabilities**
   - Call `GET /ISAPI/Intelligent/capabilities` to get professional smart abnormal event detetection capabilities and the field `isBehaviorSupport` is returned as `true` in the response message.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/capabilities` to get professional smart abnormal

event detetection capabilities for a specified channel and the field `isBehaviorSupport` is returned as `true` in the response message.

**Note**: `isBehaviorSupport`as `true` indicates support for professional smart abnormal event detetection. The fields of supported events are as follows:

- `isLineDetectionSupport` indicates support for **Line Crossing Detection**.
- `isFieldDetectionSupport` indicates support for **Intrusion Detection**.
- `isRegionEntranceSupport` indicates support for **Region Entrance Detection**.
- `isRegionExitingSupport` indicates support for **Region Exiting Detection**.
- `isLoiteringSupport` indicates support for **Loitering Detection**.
- `isGroupSupport` indicates support for **People Gathering Detection**.
- `isRapidMoveSupport` indicates support for **Rapid Movement Detection**.
- `isParkingSupport` indicates support for **Parking Detection**.
- `isUnattendedBaggageSupport` indicates support for **Unattended Baggage Detection**.
- `isAttendedBaggageSupport` indicates support for **Object Removal Detection**.
- `isRunningSupport` indicates support for **Running Detection**.
- `isPeopleNumChange` indicates support for **Abnormal Number of People Detection**.
- `isSpacingChangeSupport` indicates support for **Person Spacing Change Detection**.
- `isViolentMotionSupport` indicates support for **Violent Movement Detection**.
- `isLeavePositionSupport` indicates support for **On/Off Duty Detection**.
- `isFailDownSupport` indicates support for ** Falling Down Detection**.
- `isRetentionParamSupport` indicates support for **Lingering Detection**.
- `isSupportPersonDensityDetection` indicates support for **Person Density Detection**.
- `isVcaEventAdvReachHeightSupport` indicates support for **Climbing Detection**.
- `isSupportSituationAnalysis` indicates support for **Trend Analysis**.
- `isSupportToiletTarry` indicates support for **In-Toilet Overtime Detection**.
- `isSupportAudioAbnormal` indicates support for **Sudden Change of Sound Intensity Detection**.
- `isSupportStandUp` indicates support for **Student Standing Up Detection**.
- `isSupportGetUp` indicates support for **Getting Up from Bed Detection**.

2. **Configure Detection Rules**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/capabilities` to get professional smart abnormal event detection capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to get detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to configure detection rules.
   - Call `DELETE /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to delete detection rules.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to get all detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to configure all detection rules.

   **Note**:
   - The `eventType` field in the rules indicates the corresponding event type.
   - The `ruleType` field indicates the event support drawing a detection line or a polygon detection area. Some events (sudden change of sound intensity) do not support drawing a box or line, some events (line crossing, climbing) support drawing a line, and some events (except sudden change of sound intensity, line crossing, and climbing) support for polygon detection areas.

3. **(Optional) Configure Shielding Areas**

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- Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield/capabilities` to get professional smart area shielding capabilities.
- Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield` to get shielding areas for all events.
- Call `PUT /ISAPI/Intelligent/channels/<channelID>/Shield` to configure shielding areas for all events.

4. **(Optional) Configure Event Linkages**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to get event linkages.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to configure event linkages.
5. **(Optional) Configure Arming Schedule**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to get the schedule to disable the professional smart abnormal event detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to configure the schedule to disable the professional smart abnormal event detection rules.

   **Note**:
   - Up to eight time periods can be configured per day.
   - Each two time periods cannot overlap.
6. **Receive Events**
   - Line Crossing Detection: `"eventType":"linedetection"`
   - Intrusion Detection: `"eventType":"fielddetection"`
   - Region Entrance Detection: `"eventType":"regionEntrance"`
   - Region Exiting Detection: `"eventType":"regionExiting"`
   - Loitering Detection: `"eventType":"loitering"`
   - People Gathering Detection: `"eventType":"group"`
   - Rapid Movement Detection: `"eventType":"rapidMove"`
   - Parking Detection: `"eventType":"parking"`
   - Unattended Baggage Detection: `"eventType":"unattendedBaggage"`
   - Object Removal Detection: `"eventType":"attendedBaggage"`
   - Running Detection: `"eventType":"running"`
   - Abnormal Number of People Detection: `"eventType":"peopleNumChange"`
   - Person Spacing Change Detection: `"eventType":"spacingChange"`
   - Violent Movement Detection: `"eventType":"violentMotion"`
   - On/Off Duty Detection: `"eventType":"leavePosition"`
   - Falling Down Detection: `"eventType":"failDown"`
   - Lingering Detection: `"eventType":"retention"`
   - Person Density Event: `"eventType":"personDensityDetection"`
   - Climbing Detection: `"eventType":"advReachHeight"`
   - Trend Analysis: `"eventType":"situationAnalysis"`
   - In-Toilet Overtime Detection: `"eventType":"toiletTarry"`
   - Sudden Change of Sound Intensity Detection: `"eventType":"audioAbnormal"`
   - Student Standing Up Detection: `"eventType":"standUp"`
   - Getting Up from Bed Detection: `"eventType":"getUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`

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- Yard Lingering: `"eventType":"yardTarry"`
- Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
- Sitting Still Detection: `"eventType":"sitQuietly"`
- Yard Lingering: `"eventType":"yardTarry"`
- Using Mobile Phone Detection: `"eventType":"playCellphone"`

**Note**:

- The content of `targetAttrs` is in JSON format. When the event format is XML, use CDATA (Character Data) to include this data in the event and transmit it to the platform.
- CDATA (Character Data) is used to include large amounts of text data in XML documents. This text may contain special characters (such as `<`, `>`, `&`), which have special meanings in XML and typically need to be escaped. CDATA blocks allow these special characters to be included directly in the document without escaping, such as including JavaScript scripts or CSS styles in XML documents.

## 11.2 Upgrading the Deep Learning Model for Perimeter Intrusion Detection

### 11.2.1 Introduction to the Function

You can upgrade the deep learning model for perimeter intrusion detection. You can also upgrade both the local training model for NVRs and the training model for network camera channels.

### 11.2.2 API Calling Flow

1. Determine if the device supports upgrading the deep learning model for perimeter intrusion detection: Call `GET /ISAPI/Intelligent/capabilities`. If the field `isSupportUpgradeDepthPerimeterModel` is returned as `true` in the response message, this function is supported.
2. Get the deep learning model upgrade capabilities: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/capabilities?format=json`
3. Set the model version: `PUT /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/VersionOperation?format=json`
   - You cannot specify a version to upgrade. Only rolling back to the previous version and restoring to the factory model version are currently supported.
4. Get the model upgrade progress: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/process?format=json`
5. Set the model version for a channel: `PUT /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/VersionOperation?format=json`
6. Get the model upgrade progress for a channel: `GET /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/process?format=json`

# 12 Two-Way Audio

## 12.1 Two-Way Audio

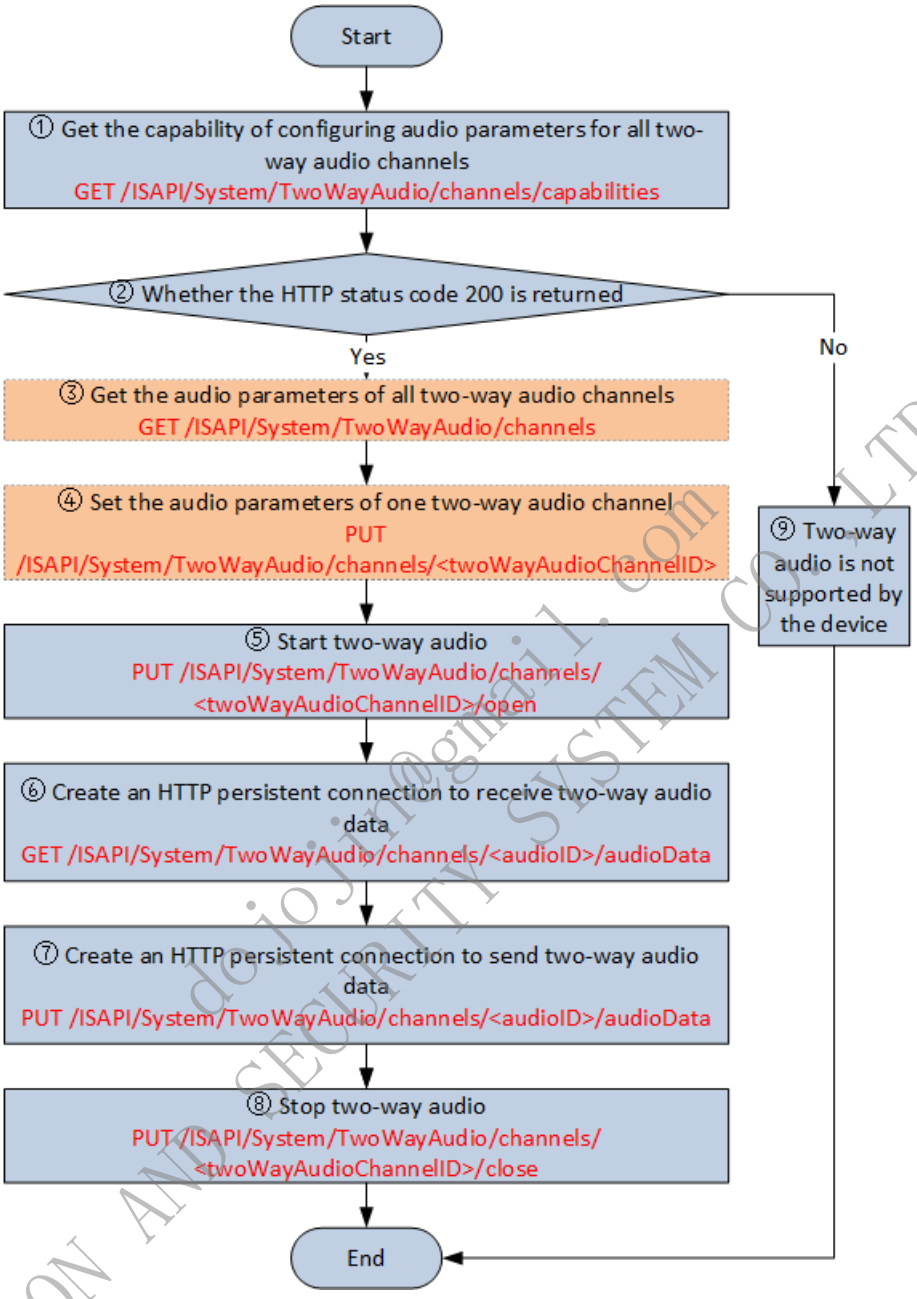
### 12.1.1 Introduction to the Function

This function is for the two-way audio between the client and device. When you call the API to implement the two-way audio, it requires the client to collect and encode the local audios and to decode the device audio data.

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Note: Before starting two-way audio, check whether the way of collecting device audio data, audio encoding format, volume, etc., is correct or not. The audio encoding format of the client should be consistent with the device audio encoding format.

### 12.1.2 API Calling Flow




1. Get the capability of configuring audio parameters for all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels/capabilities`. Check whether the device supports two-way audio. If the HTTP status code `200 OK` is returned, then the device supports two-way audio.
2. Get the audio parameters of all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels`. Get the No. of two-way audio channel for response and parsing `<id>`, audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, sampling rate `<audioSamplingRate>` and so on, to ensure the consistent of audio format during the two-way audio process.
3. Set the audio parameters of one two-way audio channel: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>`. This step is optional, and the `<audioID>` is the channel No. `<id>` responded and parsed in step 2.
4. Start two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/open`.
5. Create an HTTP persistent connection to receive the two-way audio data: `GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
6. Create an HTTP persistent connection to send the two-way audio data: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.

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7. When the two-way audio ends, the client disables the persistent connections in steps 6 and 7 and sends the message of stopping the two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/close`.

Note:

- Two-way audio over ISAPI supports digest authentication.
- It supports the two-way audio mode or sending only mode (the platform only call `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData` to send the audio data to devices).
- The `<audioID>` in step 3, 4, 5, 6, and 7 is the two-way audio channel No. which starts from 1. It can be parsed from the `<id>` in the device response message of the step 2.
- The audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, and sampling rate `<audioSamplingRate>` can be parsed from the device response message of the step 2 for audio playing and collection in the step 5 and 6.
- The step 6 and 7 use persistent connections. In the HTTP request headers, you don't need to set `Content-Legth`, but need to set the `Connection: keep-alive` and `Content-Type: application/octet-stream`.
- When the encoding format of audio data in the step 5 and 6 is AAC/MP2L2/MP3, the audio data length is variable, so the frame header should be lengthened by 4 bytes. When the encoding format is G.722.1/G.711alaw/G.711ulaw/G.726/G.729/G.729a/G.729b/PCM/ADPCM/Opus, the audio data length is fixed, there is no need to lengthen the frame header. For example, if the content length of a frame of MP3 audio data is 576 bytes and the header length is 4 bytes, the big end data is 0x00000240, the total length is 580 bytes.
- The damaged audio file in the AC3 format is usually incompatible when using. Currently, the audio algorithm library does not support this format.

Statistics of audio types with fixed lengths:

| Audio Type | Fixed Length | Frame Interval |
|---|---|---|
| G.722.1 | 80 | 40ms |
| G.711alaw | 160/320 | 20ms/40ms |
| G.711ulaw | 160/320 | 20ms/40ms |
| G.726 | 80 | 40ms |
| G.729 | 10 | 10ms |
| G.729a | 10 | 10ms |
| G.729b | 10 | 10ms |
| PCM | 1920 | None |
| ADPCM | 80 | 20ms |
| Opus | 32/64 | 20ms/40ms |

## 12.1.3 Exception Handling

#### 12.1.3.1 Error Codes

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| statusCode | statusString | subStatusCode | errorCode | errorMsg | Description | Re |
|---|---|---|---|---|---|---|
| 4 | Invalid Operation | twoWayAudioInProgressPleaseWait | 0x40002068 | Two-way audio in progress...Please wait. | Two-way audio in progress...Please wait. | Th op is all Tw au in pr Pl ag af tw au st |

### 12.1.4 Message Format and Example

#### 12.1.4.1 Receive Two-Way Audio Data

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

HTTP/1.1 200 OK
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the device.
```

#### 12.1.4.2 Send Two-Way Audio Data

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the client.
```

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# 13 Entrance & Exit Management and Control

## 13.1 Entrance & Exit Station Management

### 13.1.1 Function Introduction

In the parking fee management scenarios (e.g., parking lots), the entrance & exit station is used to configure charging rules, and to charge via cards/tickets.

### 13.1.2 API Calling Flow

**1. Check if the device supports vehicle card data management**

Get the total capabilities of entrance & exit station: `GET /ISAPI/Parking/TME/capabilities`. A success return indicates that the device supports card data management.

**2. Set entrance & exit station parameters**

Configure the basic rule parameters.

- Get the capabilities of entrance & exit station parameters: `GET`

`/ISAPI/Parking/channels/<channelID>/TMEConfiguration/capabilities`.

- Configure entrance & exit station parameters: `PUT /ISAPI/Parking/channels/<channelID>/TMEConfiguration`.
- Get entrance & exit station parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEConfiguration`.

(Optional) Configure the ticket printing format & content.

- Get the capabilities of ticket printing format & content: `GET /ISAPI/Parking/channels/<channelID>/paperPrintFormatInfo/capabilities`.
- Configure the ticket printing format & content: `PUT /ISAPI/Parking/channels/<channelID>/paperPrintFormatInfo`.
- Get the ticket printing format & content: `GET /ISAPI/Parking/channels/<channelID>/paperPrintFormatInfo`.

**3. Search for card and vehicle information of entrance/exit**

- Search for card information of entrance/exit: `POST /ISAPI/Parking/channels/<channelID>/search/TMECard`.
- Search for vehicle information of entrance/exit: `POST /ISAPI/Parking/channels/<channelID>/search/TMEVehicle`.

**4. Manage vehicle card information of entrance/exit**

Configure the vehicle card information of entrance/exit.

- Batch set multiple vehicle card information of entrance/exit: `PUT /ISAPI/Parking/channels/<channelID>/TMECardVehicle`.
- Batch get multiple vehicle card information of entrance/exit: `GET /ISAPI/Parking/channels/<channelID>/TMECardVehicle`.
- Set a single vehicle card information of entrance/exit: `PUT /ISAPI/Parking/channels/<channelID>/TMECardVehicle/<cardID>`. The `cardID` in URL specifies the vehicle card ID.
- Get a single vehicle card information of entrance/exit: `GET /ISAPI/Parking/channels/<channelID>/TMECardVehicle/<cardID>`. The `cardID` in URL specifies the vehicle card ID.
- Export the vehicle card information as binary data: `GET /ISAPI/Parking/channels/<channelID>/TMECardChargeData`.

**5. Set audio parameters of entrance & exit station**

- Set audio parameters: `PUT /ISAPI/Parking/channels/<channelID>/TMEVoiceConfiguration`.
- Get audio parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEVoiceConfiguration`.

**6. Top-up and payment**

The client sends account top-up information to the entrance & exit station to top up the account.

- Get capabilities of account top-up: `GET /ISAPI/Parking/channels/<channelID>/chargeAmount/capabilities`.
- Set account top-up parameters: `PUT /ISAPI/Parking/channels/<channelID>/chargeAmount`.

The client sends payment information to the entrance & exit station, which informs the vehicle owner of payment amount.

- Get capabilities of payment information: `GET /ISAPI/Parking/channels/<channelID>/paperChargeInfo/capabilities`.
- Set the payment information: `PUT /ISAPI/Parking/channels/<channelID>/paperChargeInfo`.

**7. Entrance & exit station status**

Get the entrance & exit station status: `GET /ISAPI/Parking/channels/<channelID>/TMESystemInfo`.

**8. Offline mode**

Configure offline mode parameters and charging rules in offline mode.

Offline mode parameters include whether to enable offline mode, whether to charge, whether to trigger an alarm, etc.

- Get the capability of offline mode parameters: `GET`

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`/ISAPI/Parking/channels/<channelID>/TMEOfflineConfiguration/capabilities`.

- Set offline mode parameters: `PUT /ISAPI/Parking/channels/<channelID>/TMEOfflineConfiguration`.
- Get offline mode parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEOfflineConfiguration`.

When the offline mode is enabled and charging is required, set the charging rule in offline mode.

- Get the capability of charging rules in offline mode: `GET /ISAPI/Parking/channels/<channelID>/TMEChargeRule/capabilities`.
- Batch set charging rules in offline mode: `PUT /ISAPI/Parking/channels/<channelID>/TMEChargeRule`.
- Batch get charging rules in offline mode: `GET /ISAPI/Parking/channels/<channelID>/TMEChargeRule`.
- Set a single charging rule in offline mode: `PUT /ISAPI/Parking/channels/<channelID>/TMEChargeRule/<ruleID>`. The `ruleID` specifies the charging rule ID.
- Get a single charging rule in offline mode: `GET /ISAPI/Parking/channels/<channelID>/TMEChargeRule/<ruleID>`. The `ruleID` specifies the charging rule ID.

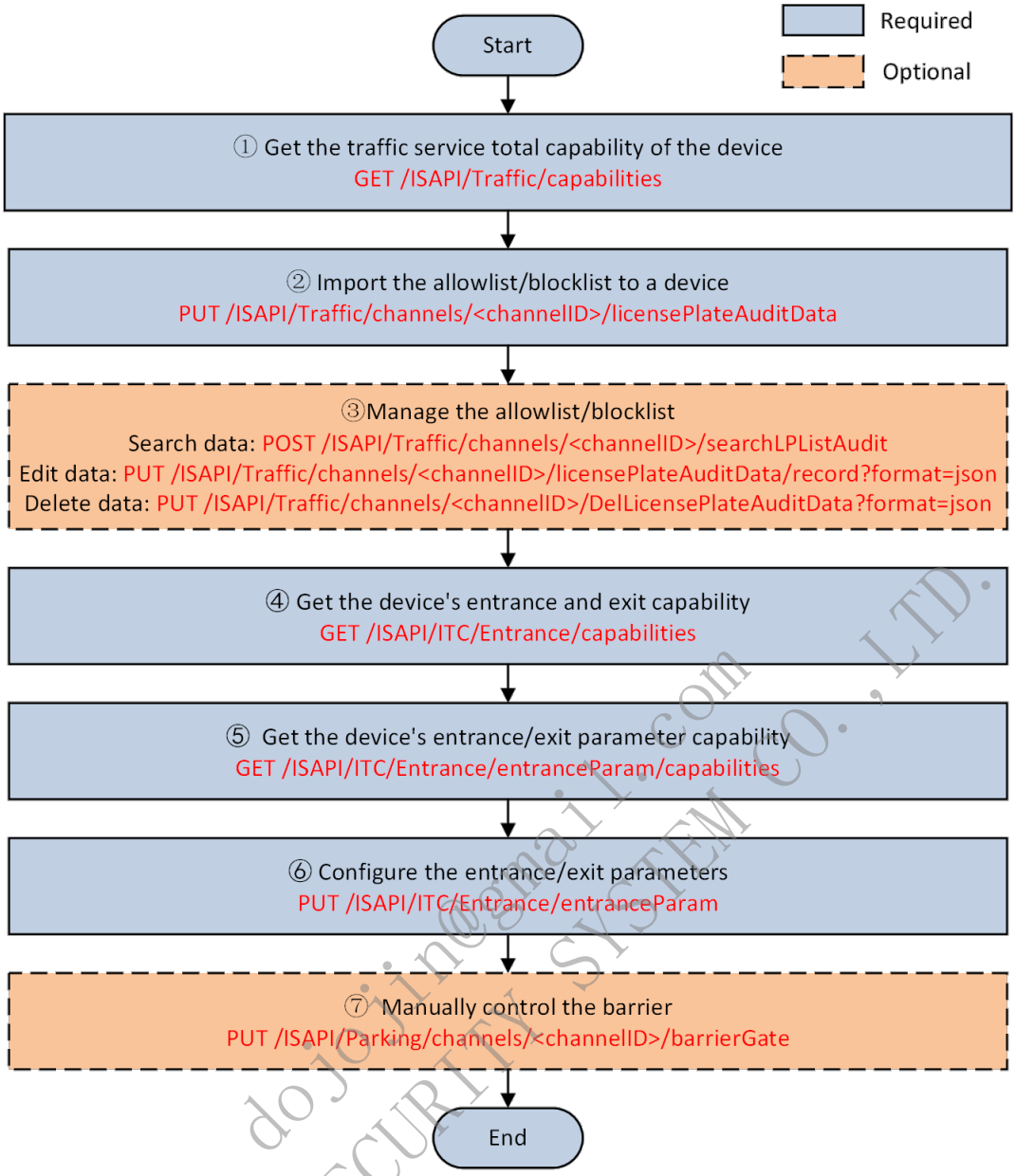
## 13.2 Entrance and Exit Barrier Control

### 13.2.1 Function Introduction

Devices use license plate recognition to compare captured license plate with the vehicle allowlist/blocklist, then automatically control the barrier based on the predefined rules. Manual control is also supported. Typical applications include entrances/exits, toll stations, parking lots, etc.

### 13.2.2 API Calling Flow

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**Steps**

1. Get the device's traffic service total capability: `GET /ISAPI/Traffic/capabilities`.

   Check the license plate recognition capabilities. Key capability fields are:

   - `$.TrafficCap.plateCap.isSupportPlateList` (whether the device supports importing/exporting the license plate blocklist and allowlist),
   - `$.TrafficCap.plateCap.isSupportLicenseModify` (whether the device supports adding or editing the license plate of imported allowlist/blocklist),
   - `$.TrafficCap.plateCap.isSupportLPAuditDataDelete` (whether the device supports deleting the imported allowlist/blocklist).

2. Import a vehicle allowlist/blocklist to the device: `PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData`.

   The default supported format is Excel. The sheet should include the following information: serial number, license plate number, group (0 for blocklist, 1 for allowlist), card number, license plate color, license plate type.

3. (Optional) Manage the vehicle allowlist/blocklist.

   - Search the data: `POST /ISAPI/Traffic/channels/<channelID>/searchLPListAudit`.
   - Add or edit the data: `PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData/record?format=json`.

- Delete the data: `PUT /ISAPI/Traffic/channels/<channelID>/DelLicensePlateAuditData?format=json`.

4. Get the device's entrance and exit capability: `GET /ISAPI/ITC/Entrance/capabilities`.

   Related capability fields are:

   - `$.EntranceCap.supportEntrance` (whether it supports entrance and exit control),
   - `$.EntranceCap.supportBarrierGateNum` (number of barriers).

5. Get the entrance and exit parameter capability of the device: `GET /ISAPI/ITC/Entrance/entranceParam/capabilities`.

6. Set the device's entrance and exit parameters: `PUT /ISAPI/ITC/Entrance/entranceParam`.

   The `$.EntranceParamList.EntranceParam.ctrlMod` field sets the barrier control mode:

   - 0-camera control (automatic),
   - 1-platform control (manual),
   - 2-both camera and platform are available.

7. (Optional) Manually control the barrier: `PUT /ISAPI/Parking/channels/<channelID>/barrierGate`.

# 14 API Reference

## 14.1 Device Management (General)

### 14.1.1 Exception Detection

#### 14.1.1.1 Get the linkage parameters of HDD error

**Request URL**

GET /ISAPI/Event/triggers/diskerror

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskerror
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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#### 14.1.1.2 Set the linkage parameters of HDD error

**Request URL**

PUT /ISAPI/Event/triggers/diskerror

**Query Parameter**

None

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskerror
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.1.3 Delete the linkage of HDD error

**Request URL**

DELETE /ISAPI/Event/triggers/diskerror

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.1.4 Get the linkage parameters of IP address conflict event

**Request URL**

GET /ISAPI/Event/triggers/ipconflict

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->ipconflict
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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### 14.1.1.5 Delete the linkage of IP address conflict event

**Request URL**

DELETE /ISAPI/Event/triggers/ipconflict

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.1.6 Set the linkage parameters of IP address conflict event

**Request URL**

PUT /ISAPI/Event/triggers/ipconflict

**Query Parameter**

None

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->ipconflict
  </id>
  <eventType>
    <!--req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.1.7 Get the linkage parameters of network disconnection

**Request URL**

GET /ISAPI/Event/triggers/nicbroken

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->nicbroken
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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### 14.1.1.8 Set the linkage parameters of network disconnection

**Request URL**

PUT /ISAPI/Event/triggers/nicbroken

**Query Parameter**

None

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->nicbroken
  </id>
  <eventType>
    <!--req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.1.9 Delete the linkage of network disconnection

**Request URL**

DELETE /ISAPI/Event/triggers/nicbroken

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.1.10 HDD error

**EventType:diskerror**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"diskerror" (HDD Error)-->diskerror
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:HDD Error-->diskerror
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <diskNo>
    <!--ro, opt, int, HDD number-->1
  </diskNo>
  <HDDList>
    <!--ro, opt, array, HDD list, subType:object-->
    <HDD>
      <!--ro, opt, object, HDD-->
      <id>
        <!--ro, req, int, index, desc:it starts from 1-->1
      </id>
      <diskNumber>
        <!--ro, req, int, HDD No.-->1
      </diskNumber>
    </HDD>
  </HDDList>
</EventNotificationAlert>
```

#### 14.1.1.11 Invalid access

#### EventType:illaccess

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event parameters, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the alarm device-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the alarm device-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the alarm device-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transfer protocol type, range:[0,64], desc:transfer protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZVIZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, No, of the channel that triggers an alarm, desc:No. of the video channel that triggers an alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"illaccess" (invalid access)-->illaccess
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for persistent events: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:illaccess-invalid access-->illaccess
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <username>
    <!--ro, opt, string, locked user name, range:[0,32], desc:the number of attempts by current user name has reached limit. The user name and IPv4 address may be locked at the same time-->test
  </username>
  <ipV4Address>
    <!--ro, opt, string, locked IPv4 address, desc:the number of attempts by current IPv4 address has reached limit. The user name and Ipv4 address may be locked at the same time-->test
  </ipV4Address>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:; “unlocked”, “locked”-->unlocked
  </lockStatus>
  <retryLoginTime>
    <!--ro, opt, int, remaining attempts, range:[0,65535], desc:it is valid when the value of lockStatus is “unlocked”-->0
  </retryLoginTime>
  <lockedTime>
    <!--ro, opt, int, locking duration (minutes), range:[0,65535], unit:秒, desc:it is valid when the value of lockStatus is “locked”-->0
  </lockedTime>
  <accessInfoList>
    <!--ro, opt, array, invalid access information list, subType:object-->
    <accessInfo>
      <!--ro, opt, object, invalid access information-->
      <ipV4Address>
        <!--ro, opt, string, Invalid access IP address-->test
      </ipV4Address>
      <dateTime>
        <!--ro, opt, datetime, invalid access time-->2004-05-03T17:30:08+08:00
      </dateTime>
    </accessInfo>
  </accessInfoList>
</EventNotificationAlert>
```

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### 14.1.1.12 Network disconnected

### EventType:nicbroken

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers an alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers an alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, communication protocol types, range:[0,64], desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome" (the value
should be "HTTP" when ISAPI protocol is transmitted via EZ protocol, and the value should be "EHome" when ISAPI protocol is transmitted via ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DevicePositionInfo>
    <!--ro, opt, object, device location, desc:the current location of the device when triggering the event, which is mainly for on-board device-->
    <divisionEW>
      <!--ro, opt, enum, eastern/western hemisphere, subType:string, desc:"E" (eastern hemisphere), "W" (western hemisphere)-->E
    </divisionEW>
    <longitude>
      <!--ro, opt, int, longitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </longitude>
    <divisionNS>
      <!--ro, opt, enum, southern/northern hemisphere, subType:string, desc:"S" (southern hemisphere), "N" (northern hemisphere)-->N
    </divisionNS>
    <latitude>
      <!--ro, opt, int, latitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </latitude>
    <direction>
      <!--ro, opt, int, vehicle direction, desc:actual direction (degree) (due north as 0 degree, calculated clockwise)-->100
    </direction>
    <speed>
      <!--ro, opt, int, speed, desc:cm/h-->100
    </speed>
    <height>
      <!--ro, opt, int, height, unit:cm, desc:cm-->1
    </height>
  </DevicePositionInfo>
</EventNotificationAlert>
```

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## 14.1.2 Event Subscription Management

### 14.1.2.1 Establish arming connection

**Request URL**

GET /ISAPI/Event/notification/alertStream?deployID=<deployID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| deployID | string | 0 (client-side arming), 1 (real-time arming). Defaults to the client-side arming when delpoyID does not exist. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event massage, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPV4 address of the device that triggers an alarm-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPV6 address of the device that triggers an alarm-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm, range:[0,65535]-->0
  </portNo>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address, range:[0,17]-->00-16-EA-AE-3C-40
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, range:[0,65535]-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, the uploading times of the same alarm, range:[0,65535]-->0
  </activePostCount>
  <eventType>
    <!--ro, req, string, heartbeat type-->test
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, range:[0,128]-->test
  </eventDescription>
</EventNotificationAlert>
```

### 14.1.2.2 Set alarm/event subscription parameters

**Request URL**

PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| subscribeEventID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, "all"-upload all alarms/events,"list"-upload specified alarm/event, subType:string, desc:"all"(upload all alarms/events), "list"(upload
specified alarm/event)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
```

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```
      <!--opt, object, uploading mode of specified alarm/event-->
      <type>
        <!--req, enum, alarm/event types, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD"(driving behavior detection) "GPSUpload", "HFPD"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(low frequency person detection), "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "LowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, subType:object-->
        <childDevID>
          <!--opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string-->1,2,3,4
      </audioChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4...", desc:event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
  </channels>
  <pictureURLType>
```

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```
    <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "localURL"(device local URL), "cloudStorageURL"(cloud storage URL)--
>cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:5 minute (default). If GUID is not reconnected in the internal, a new GUID
will be generated since the device starts a new arm period-->300
    </interval>
    <StatusSub>
      <!--opt, object, status description-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, subType:object-->
    <childDevID>
      <!--opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
```

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```
  </qosLevel>
  <audioChannels>
    <!--opt, string-->1,2,3,4
  </audioChannels>
</SubscribeEvent>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.1.2.3 Get alarm/event subscription parameters

**Request URL**

GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| subscribeEventID | string | When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), the (subscription ID) will be returned in message SubscribeEventResponse. When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), and if subscribeEventID is applied, it can be used to replace the in SubscribeEventResponse. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
</SubscribeEvent>
```

#### 14.1.2.4 Subscribe to events/alarms in arming mode

**Request URL**

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| deployID | enum | 0 (device local arming), 1 (real-time arming by platform). If the deployID is absent, it defaults to device local arming. |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm), "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRecogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID
capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status),
"certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)--
>mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)--
>cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get
all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
```

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```
      </minorType>
      <childDevIDList>
        <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
        <childDevID>
          <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string, arm by radar channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--opt, bool, upload by sub device, desc:upload by sub device-->true
      </childDeviceFormat>
      <unitChannels>
        <!--opt, string, arm by optical fiber channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--opt, object, status information-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <subscribeEventID>
    <!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message SubscribeEventResponse-->test
  </subscribeEventID>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
```

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```
    <!--opt, enum, arming level, subType:string, desc: high (default value), middle , low -->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
    <childDevID>
      <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
    </childDevID>
  </childDevIDList>
  <SubscribeISAPIMessage>
    <!--opt, object-->
    <EventTypeList>
      <!--opt, array, event list, subType:object, desc:event list-->
      <eventType>
        <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--req, array, event list, subType:object, range:[0,200]-->
      <Event>
        <!--opt, object, a single event-->
        <eventType>
          <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
  <radarChannels>
    <!--opt, string, batch arming by radar channel, desc:if this field is applied, radarChannels in Event will not take effect. To arm all radar channels,
this field should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
    <defaultLevel>
      <!--ro, req, enum, default qos level, subType:int, desc:default qos level-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, custom event qos level list, subType:object-->
      <event>
        <!--ro, opt, object, custom event qos level-->
        <eventType>
          <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
        </eventType>
        <level>
          <!--ro, req, enum, qos level of this event type, subType:int, desc:qos level of this event type-->1
        </level>
        <channels>
          <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the
channels listed in the array, and channels not listed use default qos level-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
  </audioChannels>
  <unitChannels>
    <!--opt, string, batch arming by optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm all fiber
channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </unitChannels>
  <retransmissionID>
    <!--opt, string, ANR arming ID, range:[1,64], desc:UUID (an unique ID generated by the client). If this field is applied during establishing the event
arming/subscription connection, the ANR of subscribed event will be enabled by default; when the network reconnected, the platform can apply the same UUID
to tell the device to re-upload saved event information during network disconnection period-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--opt, int, data transmission timeout for ANR, range:[1,168], unit:h, desc:data transmission timeout for ANR-->1
  </retransmissionTimeoutDuration>
  <subscribeWithoutPic>
    <!--opt, bool-->false
  </subscribeWithoutPic>
</SubscribeEvent>
```

## Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node
should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL).-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

### 14.1.2.5 Get the alarm/event subscription capability

**Request URL**

GET /ISAPI/Event/notification/subscribeEventCap

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat min="1" max="180">
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </heartbeat>
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to
channels according to channel list)-->list
  </channelMode>
  <eventMode opt="all,list,allAndChild">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to
specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
"IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"
(searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison),
"regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
        <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary" (binary), "localURL" (device local
URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
    </Event>
  </EventList>
  <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, attr:opt{req, string},def{req, string}, desc:"binary", "localURL"
(device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the
<pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of
uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <level opt="high,middle,low">
    <!--ro, opt, enum, arming level, subType:string, attr:opt{req, string}, desc:"high" (default value), "middle", "low"-->high
  </level>
</SubscribeEventCap>
```

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### 14.1.2.6 Cancel subscribing alarm/event

### Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| subscribeEventID | string | Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

### 14.1.2.7 Event subscription heartbeat

**EventType:heartBeat**

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, alarm message, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transmission communication protocol type, range:[0,64], desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type-->heartBeat
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->heartBeat
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
</EventNotificationAlert>
```

## 14.1.3 Online Upgrading Management

### 14.1.3.1 Get the online upgrade progress of device

**Request URL**

GET /ISAPI/System/onlineUpgrade/status

**Query Parameter**

None

**Request Message**

None

**Response Message**

dongjin@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "LanguageMismatch"
(language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade
version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "engineVersionMismatch" (mismatch of upgrade package engine
version)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <onlineUpdateDevInfoList>
    <!--ro, opt, object, online upgrade device information, desc:containing the online upgrade status of sub devices, sub modules, or sub channels-->
    <onlineUpdateDevInfo>
      <!--ro, opt, object, online upgrade device information-->
      <moduleType>
        <!--ro, req, enum, module type, subType:string, desc:"keypad", "detector", "cardReader" (card reader), "siren" (sounder), "relay", "transmitter"
(transmitter), "remoteCtrl" (remote control),"repeater" (repeater), "SSD", "extensionModule" (extension module)-->keypad
      </moduleType>
      <moduleID>
        <!--ro, req, int, module ID, range:[1,128], step:1-->1
      </moduleID>
      <model>
        <!--ro, opt, string, model, range:[1,32]-->test
      </model>
      <status>
        <!--ro, opt, enum, upgrade status, subType:string, desc:"allowance" (upgradeable, the upgrade command is not triggered), "ready" (waiting for
upgrade, the upgrade command has been triggered), "upgriding" (upgrading), "success" (upgraded), "failed" (upgrade failed)-->allowance
      </status>
      <percent>
        <!--ro, opt, int, current upgrade percentage, range:[0,100]-->1
      </percent>
      <updateTime>
        <!--ro, opt, datetime, upgrade completion time-->2004-05-03T17:30:08+08:00
      </updateTime>
    </onlineUpdateDevInfo>
  </onlineUpdateDevInfoList>
</OnlineUpgradeStatus>
```

### 14.1.3.2 Upload online upgrade status event

**EventType:onlineUpgradeStatus**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is an event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again. It can be regarded as the event triggering frequency*/
    "eventType":  "onlineUpgradeStatus",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Online Upgrade Status",
    /*ro, req, string, event description, desc:online upgrade status event*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:PUID. This field must be returned when ISAPI event information is transmitted via ISUP. It is the same as deviceID of URI (/ISAPI/System/Network/Ehome)*/
    "serialNumber":  "test",
    /*ro, opt, string, device short serial No., range:[1,9]*/
    "status":  "failure",
    /*ro, req, enum, upgrade status, subType:string, desc:"failure" (upgrade failed), "success" (upgrade succeeded), "upgrading"*/
    "failedReason":  "languageMismatch",
    /*ro, opt, enum, cause of failure, subType:string, desc:"languageMismatch" (language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error)*/
    "moduleType":  "cardReader",
    /*ro, opt, enum, device component type, subType:string, desc:"cardReader" (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen)*/
    "moduleID":  1,
    /*ro, opt, int, device component No., desc:it depends on the node moduleType*/
    "channel":  1,
    /*ro, opt, int, channel No., desc:it is used for IP channel (to upgrade digital channel managed by NVR)*/
    "taskID":  "test",
    /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?format=json*/
    "process":  1
    /*ro, opt, int, upgrade progress, range:[1,100], step:1, unit:%, dep:or,{$.status,eq,upgrading}, desc:progress percentage*/
}
```


## 14.1.4 Serial Port Management

### 14.1.4.1 Get serial port authentication service capability

**Request URL**

GET /ISAPI/System/Serial/authentication/capabilities?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
{
    "AuthenticationCap": {
    /*ro, opt, object, the serial port authentication service*/
        "enabled": {
        /*ro, req, object, whether to enable serial port authentication service*/
            "@opt": [true, false]
            /*ro, opt, array, whether to enable serial port authentication service, subType:bool*/
        },
        "serviceClosurePeriod": {
        /*ro, opt, object, service life, desc:unit: day it is required when enabled is "false"*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  30,
            /*ro, opt, int, the maximum value*/
            "@def":  7
            /*ro, opt, int, the default value*/
        }
    }
}
```

#### 14.1.4.2 Set parameters of serial authentication service

**Request URL**

PUT /ISAPI/System/Serial/authentication?format=json

**Query Parameter**

None

**Request Message**

```
{
    "Authentication": {
    /*opt, object*/
        "enabled":  false,
        /*req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*opt, int, service life, unit:d, desc:it is required when enabled is "false"*/
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

#### 14.1.4.3 Get parameters of serial authentication service

**Request URL**

GET /ISAPI/System/Serial/authentication?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "Authentication": {
    /*ro, opt, object*/
        "enabled":  false,
        /*ro, req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*ro, opt, int, service life, unit:d, desc:it is required when enabled is "false"”*/
    }
}
```

#### 14.1.4.4 Get the serial port capability of the device

**Request URL**

GET /ISAPI/System/Serial/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version{req, string, protocolVersion}-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt{req, string}-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
```

#### 14.1.4.5 Get the capability of a specific serial port

**Request URL**

GET /ISAPI/System/Serial/ports/<portID>/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| portID | string | -- |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO., LTD.
dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
  <rs485WorkMode
opt="Led,sensor,CaptureTrigger,transparent,GPS,ModbusRTU,supplementLight,audibleVisualAlarm,columnSpeaker,commModeUsing433MHz,signalLampControl">
    <!--ro, opt, string, working mode of RS-485 serial port, dep:and,{$.SerialPortList.SerialPort.serialPortType,eq,RS485}, attr:opt{req, string},
desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures. This node is valid only when
<serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
</SerialPort>
```

#### 14.1.4.6 Get the configuration parameters of a specified transparent transmission channel

**Request URL**

GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| portID | string | -- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<TransparentChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, transparent transmission channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the channel-->true
  </enabled>
  <serialPortID>
    <!--ro, req, string, serial port No.-->1
  </serialPortID>
</TransparentChannel>
```

#### 14.1.4.7 Set the parameters of a specific serial port supported by the device

**Request URL**

PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>

**Query Parameter**

VISION AND SECURITY SYSTEM CO., LTD. dojo_jin@gmail.com

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| portID | string | -- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<TransparentChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, serial port ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the serial port:-->true
  </enabled>
  <serialPortID>
    <!--req, string-->1
  </serialPortID>
</TransparentChannel>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.1.4.8 Close the transmission channel

**Request URL**

PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close?type=<type>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| portID | string | -- |
| channelID | string | -- |
| type | string | Data (purpose) type "general" (general data transparent transmission, the device sends the transparent transmission data via serial port, regardless of whether there is an external component connected to the serial port), "sensorCtrl" (sensor data transparent transmission control, when the device sends the transparent transmission data via serial port, it will also stop collecting sensor data, and compared with general type, the device will perform specific control and management on the sensors externally connected to the serial port) |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

#### 14.1.4.9 Open the transmission channel

**Request URL**

PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open?type=<type>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| portID | string | -- |
| channelID | string | Camera ID |
| type | string | Data (purpose) type. "general" (general data transparent transmission, the device sends the transparent transmission data via serial port, regardless of whether there is an external component connected to the serial port), "sensorCtrl" (sensor data transparent transmission control, when the device sends the transparent transmission data via serial port, it will also stop collecting sensor data, and compared with general type, the device will perform specific control and management on the sensors externally connected to the serial port) |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO., LTD.
dongjin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

#### 14.1.4.10 Set the parameters of a specific serial port supported by the device

**Request URL**

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| portID | string | -- |
| indexID | string | -- |
| childDevID | string | -- |
| deviceIndex | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"”-->half
  </duplexMode>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space"”-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware"”-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--opt, enum, working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures: "Led, CaptureTrigger".
This node is valid only when <serialPortType> is "RS485", subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial
port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is
"RS485"-->sensor
  </rs485WorkMode>
</SerialPort>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.1.4.11 Get the parameters of a specific port supported by the device

### Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| portID | string | -- |
| indexID | string | 0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information). |
| childDevID | string | -- |
| deviceIndex | string | For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--ro, opt, enum, working mode of RS-485 serial port, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
</SerialPort>
```

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### 14.1.4.12 Get the list of serial ports supported by the device

**Request URL**

GET /ISAPI/System/Serial/ports?permissionController=<indexID>&deviceIndex=<deviceIndex>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| indexID | string | 0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device. |
| deviceIndex | string | For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
    <rs485WorkMode>
      <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}-->sensor
    </rs485WorkMode>
  </SerialPort>
</SerialPortList>
```

## 14.1.5 System Maintenance

### 14.1.5.1 Get the storage capability of the device

**Request URL**

GET /ISAPI/ContentMgmt/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <isSupportSMARTTest>
    <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
  </isSupportSMARTTest>
  <isSupportMainAndSubRecord>
    <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
  </isSupportMainAndSubRecord>
  <isSupportLogConfig>
    <!--ro, opt, bool, whether the device supports log configuration, desc:related URI: /ISAPI/ContentMgmt/logConfig/capabilities-->true
  </isSupportLogConfig>
  <isSupportPictureInfo>
    <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
  </isSupportPictureInfo>
  <isSupportLogSearch>
    <!--ro, opt, bool, whether it supports log search-->true
  </isSupportLogSearch>
  <isSupportRecordSearch>
    <!--ro, opt, bool, whether it supports video search-->true
  </isSupportRecordSearch>
</RacmCap>
```

#### 14.1.5.2 Get the device status capability

**Request URL**

GET /ISAPI/ITC/ITCStatus/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ITCStatusCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isSupportfortifyLinkStatus>
    <!--ro, req, bool-->true
  </isSupportfortifyLinkStatus>
  <isSupportDetectorStatus>
    <!--ro, req, bool-->true
  </isSupportDetectorStatus>
  <relatedTriggerInputs>
    <!--ro, req, int, number of linked triggering inputs-->1
  </relatedTriggerInputs>
  <relatedSyncOutputs>
    <!--ro, req, int, number of linked synchronous outputs-->1
  </relatedSyncOutputs>
</ITCStatusCap>
```

#### 14.1.5.3 Get the live view status

**Request URL**

GET /ISAPI/ITC/ITCStatus/previewStatus

**Query Parameter**

None

**Request Message**

None

**Response Message**

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dojijin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<PreviewStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, live view status, attr:version{req, string, protocolVersion}-->
  <previewCount>
    <!--ro, req, int, number of channels for live view-->1
  </previewCount>
  <previewCfgList>
    <!--ro, opt, array, subType:object-->
    <previewCfg>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <previewIP>
        <!--ro, req, string, live view connection IP-->test
      </previewIP>
      <channelType>
        <!--ro, req, enum, channel type, subType:string, desc:"mainChannel", "subChannel", "noChannel"-->mainChannel
      </channelType>
      <frameRate>
        <!--ro, req, int, frame rate, desc:the value equals to the maximum frame rate *100. This node is set to 0 if there is no live view-->1
      </frameRate>
      <resolutionWidth>
        <!--ro, req, int, width of the video resolution-->1
      </resolutionWidth>
      <resolutionHeight>
        <!--ro, req, int, height of the video resolution-->1
      </resolutionHeight>
    </previewCfg>
  </previewCfgList>
  <snapResolutionWidth>
    <!--ro, req, int, width of the capture resolution-->1
  </snapResolutionWidth>
  <snapResolutionHeight>
    <!--ro, req, int, height of the capture resolution-->1
  </snapResolutionHeight>
  <triggerType>
    <!--ro, req, enum, trigger mode, subType:string, desc:trigger mode-->normal
  </triggerType>
</PreviewStatus>
```

#### 14.1.5.4 Get the system security capability

**Request URL**

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| userName | string | user name |
| deviceIndex | string | For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels. |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, dep:or, {$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the login password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>
  <isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
  </isSupportONVIFUserManagement>
  <isSupportConfigFileImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileImport>
  <isSupportConfigFileExport>
    <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileExport>
  <cfgFileSecretKeyLenLimit min="0" max="16">
    <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}-->1
  </cfgFileSecretKeyLenLimit>
  <isIrreversible>
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
  </isIrreversible>
  <salt>
    <!--ro, opt, string, the specific salt used by the user to log in-->test
  </salt>
  <isSupportGB35114Certificate>
    <!--ro, opt, bool, whether supports GB35114 protocol and CA/SIP authentication certificate-->true
  </isSupportGB35114Certificate>
  <isSupportAuthenticationMode opt="uKey,private">
    <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:“uKey”, “private”-->private
  </isSupportAuthenticationMode>
  <isSupportUserWithIssuerDN>
    <!--ro, opt, bool-->true
  </isSupportUserWithIssuerDN>
  <isSupportDeviceCertificatesManagement>
    <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not support this function-->true
  </isSupportDeviceCertificatesManagement>
  <SecurityAdvanced>
    <!--ro, opt, object, security advanced parameters-->
    <noOperationEnabled>
      <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
    </noOperationEnabled>
    <noOperationTime min="1" max="60" def="15">
      <!--ro, req, int, control timeout when there is no operation, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}-->15
    </noOperationTime>
    <passwordLimitLevel opt="middling,high">
      <!--ro, opt, enum, password strength limit level, subType:string, attr:opt{req, string}, desc:the configured password is required by the following limit _x000D_
“middling” (middle level), “high” (high level), middle level password contains 8 to 16 characters, including at least 2 types of the following characters: digits, lower case letters, upper case letters, and special characters; high level password contains 8 to 16 characters, including at least 3 types of the following characters: digits, lower case letters, upper case letters, and special characters-->middling
    </passwordLimitLevel>
  </SecurityAdvanced>
  <isSupportPasswordProtection>
    <!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET /ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
  </isSupportPasswordProtection>
</SecurityCap>
```

### 14.1.5.5 Get device system capability

### Request URL

GET /ISAPI/System/capabilities?type=<all>

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| all | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <isSupportPreview>
    <!--ro, opt, bool, whether the device supports live view-->true
  </isSupportPreview>
  <isSupportSystemMaintain>
    <!--ro, opt, bool, whether the device supports system maintenance, desc:for traffic devices, this node is required and must be true-->true
  </isSupportSystemMaintain>
  <isSupportReboot>
    <!--ro, opt, bool, whether the device supports rebooting, desc:for traffic devices, this node is required and must be true.-->true
  </isSupportReboot>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring to default settings, desc:for traffic devices, this node is required and must be true-->true
  </isSupportFactoryReset>
  <isSupportConfigurationData>
    <!--ro, opt, bool, whether the device supports importing and exporting the configuration files safely, desc:for traffic devices, this node is required
and must be true-->true
  </isSupportConfigurationData>
  <isSupportUpdatefirmware>
    <!--ro, opt, bool, whether the device supports upgrading, desc:for traffic devices, this node is required and must be true-->true
  </isSupportUpdatefirmware>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the device information, desc:for traffic devices, this node is required and must be true-->true
  </isSupportDeviceInfo>
  <isSupportTime>
    <!--ro, opt, bool, whether the device supports time configuration, desc:for traffic device, this node is required and must be true. For other devices,
the isSupportTimeCap is used to configure time-->true
  </isSupportTime>
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
      <isSupportPPPoE>
        <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
      </isSupportPPPoE>
      <isSupportBond>
        <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
      </isSupportBond>
      <isSupport802_1x>
        <!--ro, req, bool, whether the device supports 802.1x protocol-->true
      </isSupport802_1x>
      <isSupportNtp>
        <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
      </isSupportNtp>
      <isSupportFtp>
        <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
      </isSupportFtp>
      <isSupportUpnp>
        <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
      </isSupportUpnp>
      <isSupportDdns>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
      </isSupportDdns>
      <isSupportHttps>
        <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
      </isSupportHttps>
      <isSupport28181>
        <!--ro, opt, bool, N/A-->true
      </isSupport28181>
      <SnmpCap>
        <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
        <isSupport>
          <!--ro, req, bool, whether the device supports SNMP-->true
        </isSupport>
      </SnmpCap>
      <isSupportIPFilter>
        <!--ro, opt, bool, whether the device supports IP filtering-->true
      </isSupportIPFilter>
      <isSupportEZVIZ>
        <!--ro, opt, bool, whether the device supports EZ protocol-->true
      </isSupportEZVIZ>
      <isSupportEhome>
        <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
```

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```
        <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
      </isSupportEhome>
      <isSupportWirelessDial>
        <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
      </isSupportWirelessDial>
      <isSupportWifiProbe>
        <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
      </isSupportWifiProbe>
      <VerificationCodeModification>
        <!--ro, opt, object, device verification code configuration-->
        <verificationCodeType opt="normal,empty">
          <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
        </verificationCodeType>
      </VerificationCodeModification>
      <isSupportIntegrate>
        <!--ro, opt, bool, whether the device supports access protocol configuration-->true
      </isSupportIntegrate>
      <isSupportGetLinkSocketIP>
        <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
      </isSupportGetLinkSocketIP>
      <isSupportWebSocket>
        <!--ro, opt, bool, whether the device supports WebSocket-->true
      </isSupportWebSocket>
      <isSupportWebSocketS>
        <!--ro, opt, bool, whether the device supports WebSocketS-->true
      </isSupportWebSocketS>
      <isSupportEventDataOverWebSocket opt="true,false">
        <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
      </isSupportEventDataOverWebSocket>
      <isSupportPingDeny>
        <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
      </isSupportPingDeny>
      <isSupportClusterIntercom>
        <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is true or not be returned.-->true
      </isSupportClusterIntercom>
      <isSupportTSOverWebSocket>
        <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
      </isSupportTSOverWebSocket>
    </NetworkCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
      <rs485WorkMode opt="Led,CaptureTrigger">
        <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:Led (external LED screen, which is used for LED
display), CaptureTrigger (external device, which is used to trigger the transmission of the captured data)-->test
      </rs485WorkMode>
    </SerialCap>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportSubscribeIOTInfo>
      <!--ro, opt, bool, whether the device supports subscribing to the IoT information-->true
    </isSupportSubscribeIOTInfo>
  </SysCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
  </EventCap>
  <ITCCap>
    <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
    <isSupportVehicleDetection>
      <!--ro, opt, bool, whether the device supports vehicle detection-->true
    </isSupportVehicleDetection>
    <isSupportlicencePlateAuditData>
      <!--ro, opt, bool, whether the device supports importing or exporting the license plate blocklist and allowlist data-->true
    </isSupportlicencePlateAuditData>
    <isSupportSearchLPListAudit>
      <!--ro, opt, bool, whether the device supports searching for the license plate blocklist and allowlist-->true
    </isSupportSearchLPListAudit>
    <isSupportSyncSignalOutput>
      <!--ro, opt, bool, whether the device supports configuring output parameters-->true
    </isSupportSyncSignalOutput>
  </ITCCap>
  <TestCap>
    <!--ro, opt, object, test capability-->
    <isSupportEmailTest>
      <!--ro, opt, bool, whether the device supports email test-->true
    </isSupportEmailTest>
  </TestCap>
  <isSupportCloud>
    <!--ro, opt, bool, whether the device supports cloud storage-->true
  </isSupportCloud>
  <isSupportRTMP>
```

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```
    <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
  </isSupportRTMP>
  <VideoIntercomCap>
    <!--ro, opt, object, video intercom capability, desc:related URI: /ISAPI/VideoIntercom/capabilities-->
    <platform opt="R0,R1,S29,N10,N8,N4">
      <!--ro, opt, enum, platform type, subType:string, attr:opt{req, string}, desc:"R0", "R1"; it will be returned if the device supports-->R0
    </platform>
  </VideoIntercomCap>
  <isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
  </isSupportMixedTargetDetection>
  <isSupportDiscoveryMode>
    <!--ro, opt, bool, whether the device supports discovery mode, desc:related URI: /ISAPI/System/discoveryMode/capabilities-->true
  </isSupportDiscoveryMode>
  <isSupportDumpData>
    <!--ro, opt, bool, whether the device supports exporting the Dump file-->true
  </isSupportDumpData>
  <AIDEventSupport
opt="abandonedObject,pedestrian,congestion,roadBlock,construction,trafficAccident,fogDetection,wrongDirection,illegalParking,SSharpDriving,lowSpeed,dragRaci
ng,obstacle,vehNoYieldPedest,illegalMannedVeh,illegalMannedNonMotorVeh,umbrellaTentInstall,nonMotorVehOnVehLane,wearingNoHelmet,pedestRedLightRunning,pedest
OnNonMotorVehLane,pedestOnVehLane,conflagration,smoke,VehicleSpeedDrop,lowVisibility,animalsOccupy,targetSpeedDetection,emergencyLane,gasser,changeLaneConti
nuously">
    <!--ro, opt, string, supported AID event, attr:opt{opt, string}-->test
  </AIDEventSupport>
  <TFSEventSupport opt="illegalParking
,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,notSlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,chang
eLaneContinuously,SSharpDriving,largeVehicleOccupyLine,jamCrossLine,obstacle,blackSmokeVehicle,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,TFS
_nonZipperMerge,overSpeed,emergencyLane,gasser">
    <!--ro, opt, string, supported TFS, attr:opt{opt, string}-->test
  </TFSEventSupport>
  <isVehicleStatisticsSupport>
    <!--ro, opt, bool, whether the device supports configuring the parameters for traffic data statistics-->true
  </isVehicleStatisticsSupport>
  <isSupportVehicleFaceRecognition>
    <!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background: the traffic devices support "facial recognition"
(driver), however, the corresponding node is not added to the device to notify the integration of this capability-->true
  </isSupportVehicleFaceRecognition>
  <VQDEventSupport opt="luma,chroma,snow,streak,freeze,sigLose,clarity,jitter,block,flowers,noise,ghost,purple,ICR,protectiveFilm,blur,contrast">
    <!--ro, opt, string, VQD event list, attr:opt{opt, string}, desc:"Luma" (brightness exception); "Chroma" (image color cast); "Snow" (snow noise);
"Streak" (strip noise); "Freeze" (frozen screen); "SigLose" (signal loss); "Clarity" (resolution exception);         "JItter" (image flutter); "Block"
(video tampering); "Flowers" (blurred screen); "Noise" (image noise); "Ghost" (ghost image); "Purple" (purple fringing); "ICR" (ICP); "protectiveFilm"
(protective film unremoved)-->test
  </VQDEventSupport>
  <isSupportLCDScreen>
    <!--ro, opt, bool, whether the device supports LCD display screen-->true
  </isSupportLCDScreen>
  <isSupportImageCap>
    <!--ro, opt, bool, whether the device supports configuring image parameters-->true
  </isSupportImageCap>
  <isSupportEmmcInfo>
    <!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
  </isSupportEmmcInfo>
  <isSupportVehicleIllegalType opt="1,2...29">
    <!--ro, opt, string, the vehicle violation types supported by the device, attr:opt{opt, string}, desc:the content value is the same as the index value,
refer to NET_ITS_PLATE_RESULT.NET_DVR_VEHICLE_INFO.byIllegalType the capture camera is connected to the new platform, and the device supports "traffic
violation alarm"; however, the corresponding node is not added to the device to notify the integration of this capability; that is, the traffic violation
alarm capability lacks details. If later the platform needs to develop capability with details, the current device will be incompatible with the capability;
and detailed capability is hidden in the configuration of all trigger modes, which will influence overall access-->test
  </isSupportVehicleIllegalType>
  <isSupportRadar>
    <!--ro, opt, bool, whether the device supports radar functions, desc:related URI: /ISAPI/Radar/capabilities?format=json-->true
  </isSupportRadar>
  <isSupportISUPHttpPassthrough>
    <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
  </isSupportISUPHttpPassthrough>
  <isSupportDetectorMotionAlarm>
    <!--ro, opt, bool, whether the device supports the detector moving alarm, desc:eventType: detectorMotionAlarm-->true
  </isSupportDetectorMotionAlarm>
  <isSupportRadarVideoDetection>
    <!--ro, opt, bool, related URI: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
  </isSupportRadarVideoDetection>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
  </isSupportOnlineUpgradeTask>
  <OpenPlatformCap>
    <!--ro, opt, object, detailed HEOP capability supported by the device, desc:related URI: /ISAPI/Custom/OpenPlatform/capabilities-->
  </OpenPlatformCap>
  <isSupportDrivingDirectionAtIntersection>
    <!--ro, opt, bool, whether the device supports intersection driving direction detection, desc:eventType: drivingDirectionAtIntersection; this node is
not returned if not supported-->true
  </isSupportDrivingDirectionAtIntersection>
  <supportQuickConfigType opt="icr,dayNightGate,syncSignalOutput">
    <!--ro, opt, string, the list of supported functions of the quick configuration pop-up on the web, attr:opt{opt, string},
desc:the web quick configuration does not distinguish the configuration order, and it displays all supported function lists on the pop-up on the tab page
    icr refers to ICR switch configuration (related URL: /ISAPI/Image/channels/<channelID>/); icr
   dayNightGate refers to day and night mode switching (related URL: /ISAPI/Image/channels/<channelID>/dayNightGate);
    IOOutputs refers to synchronous signal output (related URL: /ISAPI/System/syncSignalOutput)-->syncSignalOutput
  </supportQuickConfigType>
  <isSupportUsersLoginNotification>
    <!--ro, opt, bool, whether the device supports parameters of notifications on user login, desc:/ISAPI/System/UsersLoginNotification?format=json-->true
  </isSupportUsersLoginNotification>
  <isSupportPowerIndicatorLightCtrlParams>
```

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```
    <!--ro, opt, bool, whether the device supports parameters of device battery indicator control, desc:GET/PUT /ISAPI/System/PowerIndicatorLightCtrlParams?
format=json-->true
  </isSupportPowerIndicatorLightCtrlParams>
  <isSupportExpresswayNetworkTollAccessParams>
    <!--ro, opt, bool, whether it supports protocol access parameters of network parking fee charging on highway, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
  </isSupportExpresswayNetworkTollAccessParams>
</DeviceCap>
```

#### 14.1.5.6 Export device configuration file

**Request URL**

GET /ISAPI/System/configurationData?secretkey=<secretkey>&mode=<mode>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| secretkey | string | The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing. |
| mode | enum | -- |

**Request Message**

None

**Response Message**

Binary Data

#### 14.1.5.7 Get the languages supported by the device

**Request URL**

GET /ISAPI/System/DeviceLanguage

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--ro, req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian",
"Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish",
"Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian,
"Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian", "Uzbek", "Kazak",
"Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
</DeviceLanguage>
```

#### 14.1.5.8 Set device language parameters

**Request URL**

PUT /ISAPI/System/DeviceLanguage

**Query Parameter**

None

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian",
"Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese",
"Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian" (Slovenian), "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish"-->SimChinese
  </language>
</DeviceLanguage>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.1.5.9 Get the capability of configuring the device language

### Request URL

GET /ISAPI/System/DeviceLanguage/capabilities

### Query Parameter

None

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device language configuration, attr:version{req, string, protocolVersion}-->
  <language
opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,Dutch,Portuguese,Spanish,Romanian,Turkish,J
apanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietnamese,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indon
esian,Ukrainian,EURSpanish,Uzbek,Kazak,Kirghiz,Farsi,Azerbaidzhan,Burmese,Mongolian,Anglicism,Estonian">
    <!--ro, req, enum, language, subType:string, attr:opt{req, string}, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese),
"English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish",
"Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic",
"Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian",
"Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
  <upgradeFirmWareEnabled>
    <!--ro, opt, bool, whether to enable upgrading the firmware-->true
  </upgradeFirmWareEnabled>
</DeviceLanguage>
```

## 14.1.5.10 Set restoring devices' default parameters

### Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| mode | enum | -- |
| devIndex | string | -- |
| loginPassword | string | -- |
| module | enum | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.1.5.11 Get the network service capability

**Request URL**

GET /ISAPI/System/Network/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <isSupport28181>
```

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```
      <!--ro, opt, bool, N/A-->true
    </isSupport28181>
    <SnmpCap>
      <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
      <isSupport>
        <!--ro, req, bool, whether the device supports SNMP-->true
      </isSupport>
    </SnmpCap>
    <isSupportIPFilter>
      <!--ro, opt, bool, whether the device supports IP filtering-->true
    </isSupportIPFilter>
    <isSupportEZVIZ>
      <!--ro, opt, bool, whether the device supports EZ protocol-->true
    </isSupportEZVIZ>
    <isSupportEhome>
      <!--ro, opt, bool, whether the device supports EHome (ISUP) protocol-->true
    </isSupportEhome>
    <isSupportWirelessDial>
      <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
    </isSupportWirelessDial>
    <isSupportWifiProbe>
      <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
    </isSupportWifiProbe>
    <VerificationCodeModification>
      <!--ro, opt, object, verification code editing-->
      <verificationCodeType opt="normal,empty">
        <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
      </verificationCodeType>
    </VerificationCodeModification>
    <isSupportIntegrate>
      <!--ro, opt, bool, whether the device supports access protocol configuration-->true
    </isSupportIntegrate>
    <isSupportWebSocket>
      <!--ro, opt, bool, whether the device supports WebSocket-->true
    </isSupportWebSocket>
    <isSupportWebSocketS>
      <!--ro, opt, bool, whether the device supports WebSocketS-->true
    </isSupportWebSocketS>
    <isSupportVideoImgDB>
      <!--ro, opt, bool, whether the device supports image and video library configuration-->true
    </isSupportVideoImgDB>
    <isSupportPrivatePlatom>
      <!--ro, opt, bool, whether the device supports third party platform configuration-->true
    </isSupportPrivatePlatom>
    <isSupportEventDataOverWebSocket opt="true,false">
      <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
    </isSupportEventDataOverWebSocket>
    <isSupportSip>
      <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
    </isSupportSip>
    <isSupportGantryETCServer>
      <!--ro, opt, bool, N/A, desc:N/A-->true
    </isSupportGantryETCServer>
    <isSupportDSTPServerCfg>
      <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
  format=json-->true
    </isSupportDSTPServerCfg>
    <isSupportDSTPProtocolsCap>
      <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
    </isSupportDSTPProtocolsCap>
    <isSupportTSOverWebSocket>
      <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
    </isSupportTSOverWebSocket>
  </NetworkCap>
```

#### 14.1.5.12 Set SSH parameters

**Request URL**

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>&controllerID=<controllerID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| readerID | string | -- |
| type | enum | -- |
| SOCChipID | string | -- |
| controllerID | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--wo, req, bool, whether to enable the function-->true
  </enabled>
  <port>
    <!--wo, opt, int-->22
  </port>
</SSH>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional
Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"
(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.1.5.13 Reboot device

**Request URL**

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>&controllerID=<controllerID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| devIndex | string | Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json. |
| module | enum | The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera). |
| loginPassword | string | -- |
| controllerID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

### 14.1.5.14 Get device status

### Request URL

GET /ISAPI/System/status?controllerID=<controllerID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| controllerID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device status, attr:version{opt, string, protocolVersion}-->
  <currentDeviceTime>
    <!--ro, opt, datetime, device current time-->1970-01-01T00:00:00+08:00
  </currentDeviceTime>
  <deviceUpTime>
    <!--ro, opt, int, device running time, unit:s-->1200
  </deviceUpTime>
  <TemperatureList>
    <!--ro, opt, array, temperature data list, subType:object-->
    <Temperature>
      <!--ro, opt, object, temperature data-->
      <tempSensorDescription>
        <!--ro, req, string, description-->test
      </tempSensorDescription>
      <temperature>
        <!--ro, req, float, temperature value-->1.000
      </temperature>
      <temperatureLowThreshold>
        <!--ro, opt, float, temperature low threshold-->1.000
      </temperatureLowThreshold>
      <temperatureHignThreshold>
        <!--ro, opt, float, temperature high threshold-->1.000
      </temperatureHignThreshold>
      <status>
        <!--ro, opt, enum, temperature status, subType:string, desc:"normal", "abnormal". Whether the current temperature is within the normal range, if
this field is absent, the device will not check the temperature status-->normal
      </status>
    </Temperature>
  </TemperatureList>
  <FanList>
    <!--ro, opt, array, fan data list, subType:object-->
    <Fan>
      <!--ro, opt, object, fan data-->
      <fanDescription>
```

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```
        <!--ro, req, string, description-->test
      </fanDescription>
      <speed>
        <!--ro, req, int, speed-->1
      </speed>
      <status>
        <!--ro, opt, enum, status, subType:string, desc:"normal", "busy", "abnormal"-->normal
      </status>
      <fanSpeedPercent>
        <!--ro, opt, int, fan speed percentage, range:[0,100]-->0
      </fanSpeedPercent>
      <fanRunMode>
        <!--ro, opt, enum, fan running mode, subType:string, desc:"manual", "temperatureControl"-->manual
      </fanRunMode>
      <fanSpeedGear>
        <!--ro, opt, int, fan speed gear, range:[1,4]-->0
      </fanSpeedGear>
      <fanLowThreshold>
        <!--ro, opt, int, fan usage low threshold rate, range:[0,100]-->0
      </fanLowThreshold>
      <fanHignThreshold>
        <!--ro, opt, int, fan usage high threshold rate, range:[0,100]-->0
      </fanHignThreshold>
    </Fan>
  </FanList>
  <PressureList>
    <!--ro, opt, array, pressure data list, subType:object-->
    <Pressure>
      <!--ro, opt, object, pressure data-->
      <pressureSensorDescription>
        <!--ro, req, string, description-->test
      </pressureSensorDescription>
      <pressure>
        <!--ro, req, int, pressure value-->1
      </pressure>
    </Pressure>
  </PressureList>
  <TamperList>
    <!--ro, opt, array, tampering sensor list, subType:object-->
    <Tamper>
      <!--ro, opt, object, tampering sensor-->
      <tamperSensorDescription>
        <!--ro, req, string, tampering sensor-->test
      </tamperSensorDescription>
      <tamper>
        <!--ro, req, bool, whether to enable-->true
      </tamper>
    </Tamper>
  </TamperList>
  <CPUList>
    <!--ro, opt, array, CPU data list, subType:object-->
    <CPU>
      <!--ro, opt, object, CPU data-->
      <cpuDescription>
        <!--ro, req, string, description-->test
      </cpuDescription>
      <cpuUtilization>
        <!--ro, req, int, CPU usage, range:[0,100], desc:CPU usage-->1
      </cpuUtilization>
      <cpuTemperature>
        <!--ro, opt, int, CPU temperature, range:[-40,80], unit:℃-->1
      </cpuTemperature>
      <status>
        <!--ro, opt, enum, status, subType:string, desc:"normal", "busy", "abnormal"-->normal
      </status>
      <CPULowThreshold>
        <!--ro, opt, int, CPU low threshold, range:[0,100]-->1
      </CPULowThreshold>
      <CPUHignThreshold>
        <!--ro, opt, int, CPU high threshold, range:[0,100]-->1
      </CPUHignThreshold>
      <cpuFiveSecondsUtilization>
        <!--ro, opt, int, CPU usage rate within 5 seconds, range:[0,100]-->1
      </cpuFiveSecondsUtilization>
      <cpuOneMinuteUtilization>
        <!--ro, opt, int, CPU usage rate within 1 minute, range:[0,100]-->1
      </cpuOneMinuteUtilization>
      <cpuFiveMinutessUtilization>
        <!--ro, opt, int, CPU usage rate within 5 minutes, range:[0,100]-->1
      </cpuFiveMinutessUtilization>
      <cpuHighestUtilizationProgressList>
        <!--ro, opt, array, progress list of the highest CPU usage rate, subType:object-->
        <cpuHighestUtilizationProgress>
          <!--ro, opt, object, progress of the highest CPU usage rate-->
          <progressName>
            <!--ro, opt, string, progress name, range:[1,32]-->test
          </progressName>
          <pID>
            <!--ro, opt, int, progress No., range:[1,32768], step:1-->1
          </pID>
          <progressCpuUtilization>
            <!--ro, opt, float, CPU usage rate of progress, range:[0.00,100.00], step:0.01-->0.01
          </progressCpuUtilization>
        </cpuHighestUtilizationProgress>
```

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```
      </cpuHighestUtilizationProgressList>
      <cpuHighestUtilizationThreadList>
        <!--ro, opt, array, thread list of the highest CPU usage rate, subType:object-->
        <cpuHighestUtilizationThread>
          <!--ro, opt, object, thread of the highest CPU usage rate-->
          <threadName>
            <!--ro, opt, string, thread name, range:[1,32]-->test
          </threadName>
          <tID>
            <!--ro, opt, int, thread No., range:[1,32768], step:1-->1
          </tID>
          <threadCpuUtilization>
            <!--ro, opt, float, CPU usage rate of progress, range:[0.00,100.00], step:0.01-->0.01
          </threadCpuUtilization>
        </cpuHighestUtilizationThread>
      </cpuHighestUtilizationThreadList>
      <cpuLoadList>
        <!--ro, opt, array, CPU load list, subType:object, desc:CPU average load amount in a time period-->
        <cpuLoad>
          <!--ro, opt, object, CPU load-->
          <cpuLoadTime>
            <!--ro, opt, int, CPU load time, range:[1,60], step:1, unit:min-->1
          </cpuLoadTime>
          <cpuAverageLoad>
            <!--ro, opt, float, CPU average load amount, range:[0.00,4.00], step:0.01-->0.01
          </cpuAverageLoad>
        </cpuLoad>
      </cpuLoadList>
      <highPrecisionCpuUtilizationList>
        <!--ro, opt, array, CPU usage rate list in high precision, subType:object-->
        <highPrecisionCpuUtilization>
          <!--ro, opt, object, CPU usage rate in high precision-->
          <cpuUtilization>
            <!--ro, opt, float, CPU usage rate, range:[0.00,100.00], step:0.01-->0.01
          </cpuUtilization>
        </highPrecisionCpuUtilization>
      </highPrecisionCpuUtilizationList>
      <highPrecisionCpuTemperature>
        <!--ro, opt, float, CPU temperature in high precision, range:[-40.0,80.0], step:0.1, unit:℃-->35.1
      </highPrecisionCpuTemperature>
      <cpuIndex>
        <!--ro, opt, string, CPU index-->1
      </cpuIndex>
    </CPU>
  </CPUList>
  <MemoryList>
    <!--ro, opt, array, memory list, subType:object-->
    <Memory>
      <!--ro, opt, object, memory-->
      <memoryDescription>
        <!--ro, req, string, description-->test
      </memoryDescription>
      <memoryUsage>
        <!--ro, req, float, used memory size, unit:MB-->1.0
      </memoryUsage>
      <memoryAvailable>
        <!--ro, req, float, available memory size, unit:MB-->1.0
      </memoryAvailable>
      <cacheSize>
        <!--ro, opt, float, Cache size, unit:MB-->1.0
      </cacheSize>
      <status>
        <!--ro, opt, enum, status, subType:string, desc:"normal", "busy", "abnormal"-->normal
      </status>
      <memoryUtilization>
        <!--ro, opt, int, memory usage rate, range:[0,100]-->1
      </memoryUtilization>
      <memoryUsageLowThreshold>
        <!--ro, opt, int, memory usage rate low threshold, range:[0,100]-->1
      </memoryUsageLowThreshold>
      <memoryUsageHignThreshold>
        <!--ro, opt, int, memory usage rate high threshold, range:[0,100]-->1
      </memoryUsageHignThreshold>
      <progressMemoryUsageList>
        <!--ro, opt, array, progress memory usage list, subType:object-->
        <progressMemoryUsage>
          <!--ro, opt, object, progress memory usage-->
          <progressName>
            <!--ro, opt, string, progress name, range:[1,32]-->test
          </progressName>
          <pID>
            <!--ro, opt, int, progress No., range:[1,32768], step:1-->1
          </pID>
          <memoryUsage>
            <!--ro, opt, float, used memory size, range:[0.00,1024.00], step:0.01, unit:MB-->1.00
          </memoryUsage>
        </progressMemoryUsage>
      </progressMemoryUsageList>
      <historyMemoryUsageList>
        <!--ro, opt, array, historical memory usage list, subType:object-->
        <historyMemoryUsage>
          <!--ro, opt, object, historical memory usage-->
          <memoryUsage>
```

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```
            <!--ro, opt, float, memory usage, range:[0.00,1024.00], step:0.01, unit:MB-->1.00
          </memoryUsage>
        </historyMemoryUsage>
      </historyMemoryUsageList>
    </Memory>
  </MemoryList>
  <NetPortStatusList>
    <!--ro, opt, array, network port status list, subType:object-->
    <NetPortStatus>
      <!--ro, opt, object, network port status-->
      <id>
        <!--ro, req, int, index-->1
      </id>
      <netPortDescription>
        <!--ro, req, enum, port description, subType:string, desc:“ctrl” (control port), “data1”, “data2”-->ctrl
      </netPortDescription>
      <linkStatus>
        <!--ro, opt, enum, connection status, subType:string, desc:“connected”, “disconnected”-->connected
      </linkStatus>
      <upSpeed>
        <!--ro, opt, int, upstream speed, unit:bps-->1
      </upSpeed>
      <downSpeed>
        <!--ro, opt, int, downstream speed, unit:bps-->1
      </downSpeed>
      <workSpeed>
        <!--ro, opt, int, NIC working rate, unit:bits/s-->1
      </workSpeed>
      <upStatus>
        <!--ro, opt, enum, upstream status, subType:string, desc:“normal”, “busy”, “abnormal”-->normal
      </upStatus>
      <downStatus>
        <!--ro, opt, enum, downstream status, subType:string, desc:“normal”, “busy”, “abnormal”-->normal
      </downStatus>
      <sendingAndReceivingRateList>
        <!--ro, opt, array, NIC sending/receiving data rate list, subType:object-->
        <sendingAndReceivingRate>
          <!--ro, opt, object, NIC sending/receiving data rate-->
          <sendingRate>
            <!--ro, opt, int, NIC sending data rate, range:[0,2147483647], step:1, unit:KB/s-->1
          </sendingRate>
          <receivingRate>
            <!--ro, opt, int, NIC receiving data rate, range:[0,2147483647], step:1, unit:KB/s-->1
          </receivingRate>
        </sendingAndReceivingRate>
      </sendingAndReceivingRateList>
    </NetPortStatus>
  </NetPortStatusList>
  <openFileHandles>
    <!--ro, opt, int, open file handle-->1
  </openFileHandles>
  <CameraList>
    <!--ro, opt, array, camera list, subType:object, desc:camera list-->
    <Camera>
      <!--ro, opt, object, camera-->
      <zoomReverseTimes>
        <!--ro, req, int, zoom reverse times-->0
      </zoomReverseTimes>
      <zoomTotalSteps>
        <!--ro, req, int, zoom total steps-->0
      </zoomTotalSteps>
      <focusReverseTimes>
        <!--ro, req, int, focus reverse times-->0
      </focusReverseTimes>
      <focusTotalSteps>
        <!--ro, req, int, focus total steps-->0
      </focusTotalSteps>
      <irisShiftTimes>
        <!--ro, req, int, iris shift times-->0
      </irisShiftTimes>
      <irisTotalSteps>
        <!--ro, req, int, iris total steps-->0
      </irisTotalSteps>
      <icrShiftTimes>
        <!--ro, req, int, ICR shift times-->0
      </icrShiftTimes>
      <icrTotalSteps>
        <!--ro, req, int, ICR total steps-->0
      </icrTotalSteps>
      <lensIntirTimes>
        <!--ro, req, int, lens Initialization times-->0
      </lensIntirTimes>
      <cameraRunTotalTime>
        <!--ro, req, int, camera total running time-->0
      </cameraRunTotalTime>
      <dayToNightChangeTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </dayToNightChangeTime>
      <nightToDayChangeTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </nightToDayChangeTime>
    </Camera>
  </CameraList>
  <DomeInfoList>
```

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```
  <!--ro, opt, array, PTZ information list, subType:object-->
  <DomeInfo>
    <!--ro, opt, object, PTZ information-->
    <domeRunTotalTime>
      <!--ro, opt, int, the combination of running time at -20° below/40° above/-20°- 40°-->1
    </domeRunTotalTime>
    <runTimeUnderNegativetwenty>
      <!--ro, opt, int, running time at -20° below-->1
    </runTimeUnderNegativetwenty>
    <runTimeBetweenNtwentyPforty>
      <!--ro, opt, int, running time at -20°- 40°-->1
    </runTimeBetweenNtwentyPforty>
    <runtimeOverPositiveforty>
      <!--ro, opt, int, running time at 40° above-->1
    </runtimeOverPositiveforty>
    <panTotalRounds>
      <!--ro, opt, int, pan rounds-->1
    </panTotalRounds>
    <tiltTotalRounds>
      <!--ro, opt, int, tilt rounds-->1
    </tiltTotalRounds>
    <heatState>
      <!--ro, opt, enum, heater status, subType:int, desc:0 (temperature control), 1 (remain open), 2 (remain closed)-->1
    </heatState>
    <fanState>
      <!--ro, opt, enum, fan status, subType:int, desc:0 (temperature control), 1 (remain open), 2 (remain closed)-->1
    </fanState>
    <panFrecRecord>
      <!--ro, opt, int, pan calling times, range:[0,2147483647]-->1
    </panFrecRecord>
    <tiltFrecRecord>
      <!--ro, opt, int, tilt calling times, range:[0,2147483647]-->1
    </tiltFrecRecord>
    <silpRingFrecRecord>
      <!--ro, opt, int, slip ring usage count, range:[0,2147483647]-->1
    </silpRingFrecRecord>
    <rotationRecord>
      <!--ro, opt, int, range:[0,2147483647]-->1
    </rotationRecord>
    <isPanLifeUrgent>
      <!--ro, opt, bool-->true
    </isPanLifeUrgent>
    <isTiltLifeUrgent>
      <!--ro, opt, bool-->true
    </isTiltLifeUrgent>
    <isRotationLifeUrgent>
      <!--ro, opt, bool-->true
    </isRotationLifeUrgent>
    <channelID>
      <!--ro, opt, int, range:[1,256]-->1
    </channelID>
  </DomeInfo>
</DomeInfoList>
<totalRebootCount>
  <!--ro, opt, int, device reboot times-->1
</totalRebootCount>
<deviceStatus>
  <!--ro, opt, enum, device running status, subType:string, desc:“normal”, “busy”, “abnormal”-->normal
</deviceStatus>
<dialStatus>
  <!--ro, opt, enum, dial-up status, subType:string, desc:“notDial” (no dialing), “dialing”, “finishDial” (dialing completed)-->notDial
</dialStatus>
<dialSignalStrength>
  <!--ro, opt, int, 4G signal intensity, range:[1,5], desc:refer to 4GSignalStrength-->1
</dialSignalStrength>
<batteryAllowance>
  <!--ro, opt, int, remaining battery, range:[0,100], desc:remaining battery, "0~100"%-->50
</batteryAllowance>
<HardDiskStatusList>
  <!--ro, opt, array, HDD Status list, subType:object-->
  <HardDiskStatus>
    <!--ro, opt, object, HDD Status-->
    <description>
      <!--ro, req, string, description-->test
    </description>
    <status>
      <!--ro, req, enum, status, subType:string, desc:“full”, “normal”, “abnormal”, “formatting”-->full
    </status>
    <hardDiskUsage>
      <!--ro, req, float, HDD usage space, range:[0.000,1048576.000], unit:MB-->0.000
    </hardDiskUsage>
    <hardDiskAvailable>
      <!--ro, req, float, HDD free Space, range:[0.000,1048576.000], unit:MB-->0.000
    </hardDiskAvailable>
    <DataList>
      <!--ro, opt, array, data list, subType:object, range:[1,6]-->
      <Data>
        <!--ro, opt, object, data-->
        <type>
          <!--ro, req, enum, data type, subType:string, desc:“free”, “picture”, “video”, “app”, “system”, “other”-->free
        </type>
        <percent>
          <!--ro, req, int, percentage, range:[0,100]-->1
```

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```
            </percent>
          </Data>
        </DataList>
        <hardDiskReadAndWriteSpeedList>
          <!--ro, opt, array, HDD reading and writing speed list, subType:object-->
          <hardDiskReadAndWriteSpeed>
            <!--ro, opt, object, HDD reading and writing speed-->
            <hardDiskReadSpeed>
              <!--ro, opt, int, HDD reading speed, range:[0,40000], step:1, unit:KB/s-->1
            </hardDiskReadSpeed>
            <hardDiskWriteSpeed>
              <!--ro, opt, int, HDD writing speed, range:[0,145000], step:1, unit:KB/s-->1
            </hardDiskWriteSpeed>
          </hardDiskReadAndWriteSpeed>
        </hardDiskReadAndWriteSpeedList>
      </HardDiskStatus>
    </HardDiskStatusList>
    <RadarStatusList>
      <!--ro, opt, array, radar status list, subType:object-->
      <RadarStatus>
        <!--ro, opt, object, radar status-->
        <description>
          <!--ro, req, string, description-->test
        </description>
        <status>
          <!--ro, req, enum, status, subType:string, desc:"normal", "busy", "abnormal"-->normal
        </status>
      </RadarStatus>
    </RadarStatusList>
    <BatteryList>
      <!--ro, opt, array, battery status list, subType:object-->
      <Battery>
        <!--ro, opt, object, battery status-->
        <id>
          <!--ro, req, int, index-->1
        </id>
        <state>
          <!--ro, opt, enum, battery status, subType:string, desc:"discharging", "charging", "idle"-->discharging
        </state>
        <percentage>
          <!--ro, opt, int, percentage, range:[0,100]-->50
        </percentage>
        <type>
          <!--ro, opt, enum, battery type, subType:int, desc:1 (storage battery), 0 (dry battery)-->0
        </type>
        <nominalVoltage>
          <!--ro, opt, int, nominal capacity, desc:unit: 10mAh-->1
        </nominalVoltage>
        <DP>
          <!--ro, opt, date, date of manufacture-->1970-01-01+08:00
        </DP>
        <remainingBattery>
          <!--ro, opt, int, remaining battery power, unit:mA-->10
        </remainingBattery>
        <totalVoltage>
          <!--ro, opt, int, total voltage, unit:mV-->30
        </totalVoltage>
        <cellVoltage>
          <!--ro, opt, int, single cell voltage, unit:mV-->30
        </cellVoltage>
        <current>
          <!--ro, opt, int, current, unit:mA-->30
        </current>
        <remainingDuration>
          <!--ro, opt, int, remaining battery usage time, range:[0,10000], unit:min,-->50
        </remainingDuration>
        <abnormalDescription>
          <!--ro, opt, string, range:[0,64]-->test
        </abnormalDescription>
        <isConnected>
          <!--ro, opt, bool-->true
        </isConnected>
        <temperature>
          <!--ro, opt, float, range:[-128.0,200.0], unit:℃-->25.0
        </temperature>
        <remainingBatteryCapacity>
          <!--ro, opt, int-->10
        </remainingBatteryCapacity>
        <totalgBatteryCapacity>
          <!--ro, opt, int-->10
        </totalgBatteryCapacity>
        <totalgChargingTimes>
          <!--ro, opt, int-->1
        </totalgChargingTimes>
        <batterylife>
          <!--ro, opt, object, battery life-->
          <year>
            <!--ro, opt, int, year-->2
          </year>
          <month>
            <!--ro, opt, int, month-->1
          </month>
          <day>
            <!--ro, opt, int, day-->1
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```
        <!--ro, opt, int, day-->1
      </day>
    </batterylife>
    <batteryHealth>
      <!--ro, opt, int, range:[0,100]-->80
    </batteryHealth>
    <isDetachableBattery>
      <!--ro, opt, bool-->false
    </isDetachableBattery>
  </Battery>
</BatteryList>
<AlarmInfoList>
  <!--ro, opt, array, alarm information list, subType:object, desc:add-->
  <AlarmInfo>
    <!--ro, opt, object, alarm information-->
    <channelId>
      <!--ro, req, int, channel No.-->1
    </channelId>
    <FaceAlarm>
      <!--ro, opt, object, face picture alarm-->
      <faceSnapRecvAlarm>
        <!--ro, req, int, alarm times of DSP received via face picture-->1
      </faceSnapRecvAlarm>
      <faceSnapSendAlarm>
        <!--ro, req, int, alarm times of face picture sent to the platform-->1
      </faceSnapSendAlarm>
    </FaceAlarm>
    <MixTargetAlarm>
      <!--ro, opt, object, multi-target-type detection alarm-->
      <mixTargetRecvAlarm>
        <!--ro, req, int, alarm times of DSP received via multi-target-type detection-->1
      </mixTargetRecvAlarm>
      <mixTargetSendAlarm>
        <!--ro, req, int, times of face picture alarm sent to the platform-->1
      </mixTargetSendAlarm>
    </MixTargetAlarm>
  </AlarmInfo>
</AlarmInfoList>
<AlertStreamServerList>
  <!--ro, opt, object, arming list-->
  <AlertStreamServer>
    <!--ro, opt, object, arming information-->
    <id>
      <!--ro, opt, int, index-->1
    </id>
    <protocolType>
      <!--ro, opt, enum, arming protocol type, subType:string, desc:"SDK", "ISAPI"-->SDK
    </protocolType>
    <ipAddress>
      <!--ro, opt, string-->172.6.64.7
    </ipAddress>
  </AlertStreamServer>
</AlertStreamServerList>
<USBStatusList>
  <!--ro, opt, array, USB status list, subType:object-->
  <USBStatus>
    <!--ro, opt, object, USB status-->
    <id>
      <!--ro, req, int, index-->1
    </id>
    <state>
      <!--ro, opt, enum, status, subType:string, desc:“connected”, “fullCapacity”, “deviceException”, “disconnected”-->disconnected
    </state>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when status is deviceException-->test
    </abnormalDescription>
  </USBStatus>
</USBStatusList>
<WifiStatusList>
  <!--ro, opt, array, Wi-Fi status list, subType:object-->
  <WifiStatus>
    <!--ro, opt, object, Wi-Fi status-->
    <id>
      <!--ro, req, int, index-->1
    </id>
    <state>
      <!--ro, opt, enum, status, subType:string, desc:“disabled”, “Wi-Fi”, “enable”, “disconnected”, “enableHotspot”-->disabled
    </state>
    <ssid>
      <!--ro, opt, string, SSID information-->test
    </ssid>
    <signalStrength>
      <!--ro, opt, int, signal strength, range:[1,5]-->1
    </signalStrength>
  </WifiStatus>
</WifiStatusList>
<dataLineCount>
  <!--ro, opt, int, number of data service channels-->1
</dataLineCount>
<commandLineCount>
  <!--ro, opt, int, number of command service channels-->1
</commandLineCount>
<switchNetFlowCount>
  <!--ro, opt, int, forwarding data stream, unit:MB-->1
```

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```
  </switchNetFlowCount>
  <resourceLoad>
    <!--ro, opt, int, decoding resources login progress, range:[0,100]-->1
  </resourceLoad>
  <networkLoad>
    <!--ro, opt, int, network connection progress, range:[0,100]-->1
  </networkLoad>
  <POEPowerList>
    <!--ro, opt, array, current device PoE power list, subType:object-->
    <POEPower>
      <!--ro, opt, object, device PoE power-->
      <usedPower>
        <!--ro, req, float, used PoE power, unit:W, desc:it is recommended to be accurate to 1 decimal place, unit: W-->1.0
      </usedPower>
      <fullPower>
        <!--ro, req, float, PoE total power, unit:W-->1.0
      </fullPower>
      <weekPOECrest>
        <!--ro, req, float, peak PoE power in a week, unit:W-->1.0
      </weekPOECrest>
    </POEPower>
  </POEPowerList>
  <ListenServerList>
    <!--ro, opt, array, get the user list for device listening, subType:object-->
    <ListenServer>
      <!--ro, opt, object, get the user for device listening-->
      <id>
        <!--ro, opt, int, index-->1
      </id>
      <protocolType>
        <!--ro, opt, enum, listening protocol type, subType:string, desc:"SDK", "ISAPI"-->SDK
      </protocolType>
      <ipAddress>
        <!--ro, opt, string, IP address of platform, server, NVR/CVR, and so on-->172.6.64.7
      </ipAddress>
    </ListenServer>
  </ListenServerList>
  <TECList>
    <!--ro, opt, array, TEC status list, subType:object, desc:TEC refers to Thermo Electric Cooler, when the temperature is too high, it can cool, and when
the temperature is too low, it can heat-->
    <TEC>
      <!--ro, opt, object, TEC status-->
      <id>
        <!--ro, opt, int, index-->0
      </id>
      <state>
        <!--ro, opt, enum, Work Status, subType:string, desc:"refrigerating", "heating", "notWorking", "abnormal"-->refrigerating
      </state>
      <temperature>
        <!--ro, req, float, TEC temperature, unit:℃-->0.000
      </temperature>
      <power>
        <!--ro, opt, float, unit:W-->45.2
      </power>
    </TEC>
  </TECList>
  <bootTime>
    <!--ro, opt, datetime, device startup time-->1970-01-01T00:00:00+08:00
  </bootTime>
  <lastCalibrationTime>
    <!--ro, opt, datetime, last time of device time synchronization-->1970-01-01T00:00:00+08:00
  </lastCalibrationTime>
  <lastCalibrationTimeDiff>
    <!--ro, opt, int, difference generated by the last  time synchronization, unit:s-->1200
  </lastCalibrationTimeDiff>
  <thermalMovementWorkingTime>
    <!--ro, opt, int, thermal camera module working time statistics, range:[0,99999], step:1, unit:h-->1200
  </thermalMovementWorkingTime>
  <tcpConnectionStatusList>
    <!--ro, opt, array, TCP connection status list, subType:object-->
    <tcpConnectionStatus>
      <!--ro, opt, object, TCP connection status, desc:number of TCP connection in different status-->
      <tcpStatusType>
        <!--ro, opt, enum, TCP connection type, subType:string, desc:"listen", "established", "closeWait", "timeWait"-->listen
      </tcpStatusType>
      <tcpConnectionNumber>
        <!--ro, opt, int, TCP connection amount, range:[0,65535], step:1-->1
      </tcpConnectionNumber>
    </tcpConnectionStatus>
  </tcpConnectionStatusList>
  <gpsStatus>
    <!--ro, opt, enum, GPS running status, subType:string, desc:"normal", "busy", "abnormal"-->normal
  </gpsStatus>
  <trafficLightStatus>
    <!--ro, opt, enum, traffic light signal detector connection status, subType:string, desc:"normal", "busy", "abnormal"-->normal
  </trafficLightStatus>
  <wifiProbeStatus>
    <!--ro, opt, enum, Wi-Fi probe connection status, subType:string, desc:"normal", "busy", "abnormal"-->normal
  </wifiProbeStatus>
  <scannerStatus>
    <!--ro, opt, object, scanning status-->
    <scannerLensStatus>
      <!--ro, opt, object, scanner status-->
```

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```
      <status>
        <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
      </status>
      <abnormalDescription>
        <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
      </abnormalDescription>
    </scannerLensStatus>
    <scannerFrequency>
      <!--ro, opt, int, scan frequency, range:[0,10000], unit:次/小时-->1
    </scannerFrequency>
  </scannerStatus>
  <keyboardStatus>
    <!--ro, opt, object, device keypad status-->
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
    </abnormalDescription>
  </keyboardStatus>
  <screenStatus>
    <!--ro, opt, object, device screen status-->
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
    </abnormalDescription>
  </screenStatus>
  <cameraLensStatus>
    <!--ro, opt, object, device camera status-->
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
    </abnormalDescription>
  </cameraLensStatus>
  <loudspeakerStatus>
    <!--ro, opt, object, device speaker status-->
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
    </abnormalDescription>
  </loudspeakerStatus>
  <earphoneStatus>
    <!--ro, opt, object, device handset status-->
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
    </abnormalDescription>
  </earphoneStatus>
  <pickUpStatus>
    <!--ro, opt, object, device pick-up status-->
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <abnormalDescription>
      <!--ro, opt, string, exception description, range:[0,64], desc:it is returned when the status is abnormal-->test
    </abnormalDescription>
  </pickUpStatus>
  <uploadTimeConsumingList>
    <!--ro, opt, object-->
    <uploadTimeConsuming>
      <!--ro, opt, object-->
      <uploadId>
        <!--ro, opt, int-->10
      </uploadId>
      <uploadIpaddr>
        <!--ro, opt, string, range:[0,40]-->test
      </uploadIpaddr>
      <uploadPort>
        <!--ro, opt, int-->65535
      </uploadPort>
      <uploadType>
        <!--ro, opt, string, range:[0,32]-->test
      </uploadType>
      <avgTime>
        <!--ro, opt, int, unit:s-->100
      </avgTime>
      <maxTime>
        <!--ro, opt, int, unit:s-->100
      </maxTime>
      <minTime>
        <!--ro, opt, int, unit:s-->100
      </minTime>
      <uploadSuccessNum>
        <!--ro, opt, int-->65535
      </uploadSuccessNum>
      <uploadTotalNum>
```

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```
      <!--ro, opt, int-->65535
    </uploadTotalNum>
    <uploadStartTime>
      <!--ro, opt, string, range:[0,20]-->test
    </uploadStartTime>
    <uploadEndTime>
      <!--ro, opt, string, range:[0,20]-->test
    </uploadEndTime>
    <uploadStatus>
      <!--ro, opt, string, range:[0,20]-->test
    </uploadStatus>
  </uploadTimeConsuming>
</uploadTimeConsumingList>
<SDCardStatusInfo>
  <!--ro, opt, object, SD card status information-->
  <videoRewritingTimes>
    <!--ro, opt, int, range:[0,2147483647]-->0
  </videoRewritingTimes>
  <snapshotRewritingTimes>
    <!--ro, opt, int, range:[0,2147483647]-->0
  </snapshotRewritingTimes>
</SDCardStatusInfo>
<powerSupplyStatusList>
  <!--ro, opt, array, subType:object, range:[1,3]-->
  <powerSupplyStatus>
    <!--ro, opt, object, power supply status-->
    <ID>
      <!--ro, req, int, index-->1
    </ID>
    <peripheralFlag>
      <!--ro, opt, bool-->true
    </peripheralFlag>
    <status>
      <!--ro, req, enum, status, subType:string-->normal
    </status>
    <currentPower>
      <!--ro, opt, float, range:[1,660], unit:W, dep:and,{$.DeviceStatus.powerSupplyStatusList[*].powerSupplyStatus.status,eq,on}-->1.0
    </currentPower>
    <maxPower>
      <!--ro, opt, float, range:[1,660], unit:W-->1.0
    </maxPower>
    <powerSupplyDetails>
      <!--ro, opt, string, range:[0,256]-->test
    </powerSupplyDetails>
  </powerSupplyStatus>
</powerSupplyStatusList>
<powerSupplyTotalMaxPower>
  <!--ro, opt, float, range:[1,1000], unit:W-->500.0
</powerSupplyTotalMaxPower>
<GPUList>
  <!--ro, opt, array, subType:object-->
  <GPU>
    <!--ro, opt, object-->
    <ID>
      <!--ro, req, int, Engine ID-->0
    </ID>
    <gpuTemperature>
      <!--ro, opt, int, CPU temperature, range:[-40,80], unit:℃-->1
    </gpuTemperature>
  </GPU>
</GPUList>
<solarPanelStatus>
  <!--ro, opt, object-->
  <voltage>
    <!--ro, opt, float, range:[0,99999], unit:V-->12.0
  </voltage>
  <current>
    <!--ro, opt, float, range:[0,99999], unit:A-->1.0
  </current>
  <powerGeneration>
    <!--ro, opt, float, range:[0,1000000], unit:kW·h-->1.0
  </powerGeneration>
</solarPanelStatus>
<currentDeviceTimeDST>
  <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
</currentDeviceTimeDST>
<loraModuleStatusList>
  <!--ro, opt, array, invalid material status, subType:object-->
  <loraModuleStatus>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int, sequence number-->1
    </id>
    <status>
      <!--ro, req, enum, status, subType:string, desc:"normal", "abnormal"-->normal
    </status>
    <loadStatus>
      <!--ro, opt, int, range:[0,16000], unit:%-->0
    </loadStatus>
    <totalLoad>
      <!--ro, opt, int, range:[0,16000], unit:%-->0
    </totalLoad>
  </loraModuleStatus>
```

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```
</loraModuleStatusList>
<dialStatusList>
  <!--ro, opt, array, subType:object-->
  <dialStatus>
    <!--ro, opt, object-->
    <networkStatus>
      <!--ro, req, enum, subType:string-->4G
    </networkStatus>
    <dialStatus>
      <!--ro, opt, enum, subType:string-->notDial
    </dialStatus>
    <dialSignalStrength>
      <!--ro, opt, int, range:[1,5]-->1
    </dialSignalStrength>
    <ICCID>
      <!--ro, opt, string-->test
    </ICCID>
    <band>
      <!--ro, opt, string, range:[0,100]-->N10
    </band>
    <PCI>
      <!--ro, opt, int, range:[0,4294967296]-->0
    </PCI>
    <cardType>
      <!--ro, opt, enum, subType:string-->maintenance
    </cardType>
    <businessCardStatus>
      <!--ro, opt, enum, subType:string, dep:and,{$.DeviceStatus.dialStatusList[*].cardType,eq,business}-->online
    </businessCardStatus>
    <usedTrafficValue>
      <!--ro, opt, float, range:[0.000,200.000], unit:GB-->0
    </usedTrafficValue>
  </dialStatus>
</dialStatusList>
<interfaceStatusList>
  <!--ro, opt, array, subType:object-->
  <interfaceStatus>
    <!--ro, opt, object-->
    <interfaceType>
      <!--ro, req, enum, subType:string-->HDMIIN
    </interfaceType>
    <interfaceTypeName>
      <!--ro, opt, string-->HDMIIN1
    </interfaceTypeName>
    <state>
      <!--ro, req, enum, subType:string-->connect
    </state>
  </interfaceStatus>
</interfaceStatusList>
<activatedTime>
  <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
</activatedTime>
<middlePlateMCUState>
  <!--ro, opt, enum, subType:string-->normal
</middlePlateMCUState>
<deviceInputVoltageStatus>
  <!--ro, opt, enum, subType:string-->normal
</deviceInputVoltageStatus>
<deviceInputVoltage>
  <!--ro, opt, float, range:[0,220], dep:and,{$.DeviceStatus.deviceInputVoltageStatus,eq,normal}-->1.0
</deviceInputVoltage>
<encodingCardList>
  <!--ro, opt, array, subType:object-->
  <encodingCard>
    <!--ro, opt, object-->
    <slotID>
      <!--ro, req, int-->1
    </slotID>
    <status>
      <!--ro, req, enum, subType:string-->normal
    </status>
    <abnormalChips>
      <!--ro, opt, int, dep:and,{$.DeviceStatus.encodingCardList[*].encodingCard.status,eq,abnormal}-->4
    </abnormalChips>
  </encodingCard>
</encodingCardList>
<SSDCardList>
  <!--ro, opt, array, subType:object-->
  <SSDCard>
    <!--ro, opt, object-->
    <SSDCardID>
      <!--ro, opt, int-->1
    </SSDCardID>
    <status>
      <!--ro, req, enum, subType:string-->normal
    </status>
    <slotID>
      <!--ro, opt, int-->1
    </slotID>
    <type>
      <!--ro, opt, string-->test
    </type>
    <serialNumber>
      <!--ro, opt, string-->test
```

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```
        </serialNumber>
        <firmwareVersion>
          <!--ro, opt, string-->test
        </firmwareVersion>
        <residualLife>
          <!--ro, opt, int, unit:%-->90
        </residualLife>
        <SSDCardSilkScreen>
          <!--ro, opt, enum, subType:string-->JP12
        </SSDCardSilkScreen>
      </SSDCard>
    </SSDCardList>
    <SDCardInfo>
      <!--ro, opt, object-->
      <memoryCapacity>
        <!--ro, opt, float, range:[0.00,1024.00], unit:MB-->100.00
      </memoryCapacity>
    </SDCardInfo>
    <routeConnectStatus>
      <!--ro, opt, enum, subType:string-->online
    </routeConnectStatus>
    <controlStatusInfo>
      <!--ro, opt, object-->
      <dualControlStatus>
        <!--ro, opt, enum, subType:string-->normal
      </dualControlStatus>
      <mismatchedTypeList>
        <!--ro, opt, array, subType:object, range:[1,6], dep:and,{$.DeviceStatus.controlStatusInfo.dualControlStatus,eq,mismatched}-->
        <mismatchedType>
          <!--ro, req, enum, subType:string-->CPU
        </mismatchedType>
      </mismatchedTypeList>
      <controlStatusList>
        <!--ro, opt, array, subType:object, range:[1,2]-->
        <controlStatus>
          <!--ro, req, object, dep:and,{$.DeviceStatus.controlStatusInfo.dualControlStatus,eq,abnormal}-->
          <controlID>
            <!--ro, req, string, range:[1,64]-->1
          </controlID>
          <slot>
            <!--ro, req, enum, subType:string-->left
          </slot>
          <statusInfoList>
            <!--ro, opt, array, subType:object, range:[1,4]-->
            <statusInfo>
              <!--ro, opt, object-->
              <status>
                <!--ro, opt, enum, subType:string-->offline
              </status>
              <abnormalDetails>
                <!--ro, opt, string-->test
              </abnormalDetails>
            </statusInfo>
          </statusInfoList>
        </controlStatus>
      </controlStatusList>
    </controlStatusInfo>
    <sonarConnectStatus>
      <!--ro, opt, enum, subType:string-->normal
    </sonarConnectStatus>
    <humidityList>
      <!--ro, opt, array, subType:object, range:[1,1]-->
      <humidity>
        <!--ro, opt, object-->
        <description>
          <!--ro, req, string, range:[1,64]-->湿度-1
        </description>
        <humidity>
          <!--ro, req, float, range:[0,100], unit:%-->45.0
        </humidity>
      </humidity>
    </humidityList>
  </DeviceStatus>
```

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#### 14.1.5.15 Get the video analysis task capability

**Request URL**

GET /ISAPI/System/Video/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->0
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of local menus that can be displayed-->0
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended analog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, enum, statistics type, subType:string, desc:statistics type-->human
  </countingType>
  <isSupportPicture>
    <!--ro, opt, bool, whether the device supports OSD picture overlay, desc:related URI: /ISAPI/System/Video/inputs/channels/<channelID>/image/picture--
>true
  </isSupportPicture>
  <isSupportPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view-->true
  </isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--ro, opt, bool, whether the device supports searching for the recording status-->true
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--ro, opt, bool, whether the device supports privacy mask-->true
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
  </isSupportBinocularPreviewSwitch>
  <isSupportCalibCheck>
    <!--ro, opt, bool, whether the device supports calibration verification-->true
  </isSupportCalibCheck>
  <isSupportPIP>
    <!--ro, opt, bool-->true
  </isSupportPIP>
  <isSupportFocusVideoMode>
    <!--ro, opt, bool-->true
  </isSupportFocusVideoMode>
  <isSupportExternalChannel>
    <!--ro, opt, bool, whether the device supports extended analog channels-->true
  </isSupportExternalChannel>
  <isSupportMultiChannelCounting>
    <!--ro, opt, bool-->true
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--ro, opt, bool, whether the device supports people counting ANR-->true
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--ro, opt, bool, whether the device supports heat map data ANR-->true
  </isSupportHeatmapCollection>
  <channelFlexible opt="name,enable,online,linknum">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->name
  </channelFlexible>
  <isSupportOutputsResource>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/resource?format=json, desc:related URI: /ISAPI/System/Video/outputs/resource?format=json--
>true
  </isSupportOutputsResource>
  <OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->GBK
  </OSDLanguage>
  <isSupportMixedChannel>
    <!--ro, opt, bool-->true
  </isSupportMixedChannel>
  <isSupportMixedChannelStatus>
    <!--ro, opt, bool-->true
  </isSupportMixedChannelStatus>
  <isSupportOutputCourseware>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/courseware/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/courseware/capabilities?format=json-->true
  </isSupportOutputCourseware>
  <isSupportVideoInputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/inputs/mode/capabilities, desc:related URI: /ISAPI/System/Video/inputs/mode/capabilities-->true
  </isSupportVideoInputMode>
  <isSupportVideoOutputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/mode/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/mode/capabilities?format=json-->true
  </isSupportVideoOutputMode>
  <PreviewMode>
    <!--ro, opt, object-->
    <PIPType>
```

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```
        <!--ro, opt, object-->
        <mainScreenChannelID>
          <!--ro, opt, int-->0
        </mainScreenChannelID>
        <subScreenChannelID>
          <!--ro, opt, int-->0
        </subScreenChannelID>
      </PIPType>
    </PreviewMode>
    <isSupportViewTag>
      <!--ro, opt, bool-->true
    </isSupportViewTag>
    <isSupportMenuStatus>
      <!--ro, opt, bool-->true
    </isSupportMenuStatus>
  </VideoCap>
```

## 14.1.6 Upgrade Management

### 14.1.6.1 Upgrade devices

**Request URL**

POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpdateAuxDevice=<isUpdateAuxDevice>&confirmEnabled=<confirmEnabled>&upgradeToken=<upgradeToken>

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| type | enum | device type 1.alarm device: "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder), "sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "repeater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob), "subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR module), "networkInputAndOutputModule" (network input and output module via network TAP) 2. access control device: "cardReader" (485 card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm program of card reader), "bluetoothModule" (bluetooth module), "MCU"-MCU upgrade, "temperatureModule" (temperature measurement module), "faceModule" (face module), "touchScreenModule" (touch screen module), "netReader" (network card reader), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "QRCodeModule" (QR code module) 3. intelligent cabinet: "lockControlBoard" (lock control board) 4. "ledReceiverCard"-receiving card; 5. "subsystem"-sub system; 6. radar-assisted device: "warningScreenFirmware"-firmware upgrade of pre-alarm screen, "warningScreenFont"-font library upgrade of pre-alarm screen, "warningScreenVoice"-audio upgrade of pre-alarm screen; 7. "electricGenie"-electric Genie |
| moduleAddress | string | module address is an exclusive node for Ax Hybrid example: keypad: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; relay: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; sounder: GET /ISAPI/SecurityCP/Configuration/wirelessSiren?format=json; output module: GET /ISAPI/SecurityCP/Configuration/outputModules?format=json; extension module: GET /ISAPI/SecurityCP/Configuration/extensionModule?format=json; zone: GET /ISAPI/SecurityCP/Configuration/zones?format=json |
| indexID | string | device No. 1. AX Hybrid Pro and wireless security control upgrade by ID; 2. access control devices upgrade by ID; 3. intelligent cabinet upgrade by IS |
| devIndex | string | -- |
| isUpdateAuxDevice | string | -- |
| confirmEnabled | string | -- |
| upgradeToken | string | -- |

**Request Message**

Binary Data

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:it describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information, range:[0,1024], desc:this node is used for debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.1.6.2 Get the device upgrade progress

**Request URL**

GET /ISAPI/System/upgradeStatus?format=json&type=<type>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| type | string | -- |

**Request Message**

None

**Response Message**

```
{
    "requestURL":  "/ISAPI/Streaming/channels/1",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, req, string, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
    "errorMsg":  "ok",
    /*ro, opt, string, This field is required when the value of statusCode is not 1. Detailed error description of a certain parameter can be provided*/
    "upgrading":  "TRUE",
    /*ro, opt, string, whether the device is upgrading: “TRUE" (upgrading), "FALSE" (not in upgrading)*/
    "percent":  22,
    /*ro, opt, int, upgrade progress (% complete)*/
    "idList": [
    /*ro, opt, array, ID list, subType:object*/
        {
            "id":  "test",
            /*ro, req, string, analysis unit ID*/
            "percent":  22,
            /*ro, opt, int, upgrade progress (% complete)*/
            "status":  "test"
            /*ro, opt, string, "backingUp" (backing up upgrade)*/
        }
    ]
}
```

#### 14.1.6.3 Get the device upgrading status and progress

**Request URL**

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| type | string | "usbmic" (array microphone module) |
| childDevID | string | -- |
| id | string | The id indicates the peripheral No. to be upgraded. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

## 14.1.7 Time Management

### 14.1.7.1 Get device time synchronization management parameters

**Request URL**

GET /ISAPI/System/time

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

#### 14.1.7.2 Set device time synchronization management parameters

**Request URL**

PUT /ISAPI/System/time

**Query Parameter**

None

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

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### 14.1.7.3 Get the capability of device time synchronization management

#### Request URL

GET /ISAPI/System/time/capabilities

#### Query Parameter

None

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time
synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or
no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is
configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-
MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports
configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI:
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

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```
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


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#### 14.1.7.4 Set device local time

**Request URL**

PUT /ISAPI/System/time/localTime

**Query Parameter**

None

**Request Message**

Binary Data

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

### 14.1.7.5 Set parameters of all NTP servers

### Request URL

PUT /ISAPI/System/time/ntpServers

### Query Parameter

None

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

### Response Message

dongjin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.1.7.6 Get the parameters of a specific NTP (Network Time Protocol) server

**Request URL**

GET /ISAPI/System/time/ntpServers

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

#### 14.1.7.7 Set the parameters of a NTP server

**Request URL**

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| NTPServerID | string | -- |

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### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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## 14.1.7.8 Get the parameters of a NTP server

### Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| NTPServerID | string | NTP server No. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

#### 14.1.7.9 Get the configuration capability of a specific NTP server

**Request URL**

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| NTPServerID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)--
>hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval,
unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

### 14.1.7.10 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

**Request URL**

GET /ISAPI/System/time/ntpServers/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

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dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```

### 14.1.7.11 Execute a test to check if the NTP server is available

#### Request URL

POST /ISAPI/System/time/ntpServers/test

#### Query Parameter

None

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPTestDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--req, enum, "ipaddress,hostname", subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--req, int, port No.-->1
  </portNo>
</NTPTestDescription>
```

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPTestResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <errorDescription>
    <!--ro, req, string, error description-->test
  </errorDescription>
</NTPTestResult>
```

### 14.1.7.12 Set the device time type

#### Request URL

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PUT /ISAPI/System/time/timeType?format=json

**Query Parameter**

None

**Request Message**

```
{
    "TimeType": {
    /*req, object, time type information, desc:time type information*/
        "type":  "UTC",
        /*req, enum, time type, subType:string, desc:"UTC, local "*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*opt, enum, subType:string*/
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

#### 14.1.7.13 Get the device time type

**Request URL**

GET /ISAPI/System/time/timeType?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
{
    "TimeType": {
    /*ro, req, object*/
        "type":  "UTC",
        /*ro, req, enum, value: local,UTC, subType:string, desc:"UTC,local"*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*ro, opt, enum, subType:string*/
    }
}
```

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# 14.2 Video (General)

## 14.2.1 Wide Dynamic Range / Backlight Compensation / Highlight Compensation

### 14.2.1.1 Set the WDR (Wide Dynamic Range) parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/hdr

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode>
    <!--req, enum, WDR (Wide Dynamic Range) mode, subType:string, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto"-->Sdr
  </hdrMode>
  <HdrSwitch>
    <!--opt, object-->
    <hdrSwitchMode>
      <!--req, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness. <timeSwitch> nad <lightSwitch>
are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--req, object, enable WDR by time-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, start time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, start time (minute)-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel>
    <!--opt, enum, WDR level, subType:string, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel>
    <!--opt, int-->1
  </SdrLevel>
</Hdr>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.1.2 Get the WDR (Wide Dynamic Range) parameters of a specific channel

### Request URL

GET /ISAPI/Image/channels/<channelID>/hdr

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

### Request Message

None

### Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojijin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode>
    <!--ro, req, enum, WDR mode, subType:string, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto".-->Sdr
  </hdrMode>
  <HdrSwitch>
    <!--ro, opt, object-->
    <hdrSwitchMode>
      <!--ro, req, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness. <timeSwitch> nad
<lightSwitch> are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--ro, req, object, enable WDR by time-->
      <startHour>
        <!--ro, opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--ro, opt, int, start time (hour)-->1
      </endHour>
      <endMinute>
        <!--ro, opt, int, start time (minute)-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel>
    <!--ro, opt, enum, WDR level, subType:string, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel>
    <!--ro, opt, int-->1
  </SdrLevel>
</Hdr>
```

#### 14.2.1.3 Get the WDR (Wide Dynamic Range) capability of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/hdr/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO., LTD.
dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode opt="close,realHdr,digitalHdr,Sdr,BLC">
    <!--ro, req, enum, WDR mode, subType:string, attr:opt{req, string}, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto".--
>Sdr
  </hdrMode>
  <HdrSwitch>
    <!--ro, opt, object-->
    <hdrSwitchMode opt="open,timeCtrl,lightCtrl">
      <!--ro, req, enum, switch mode, subType:string, attr:opt{req, string}, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness.
<timeSwitch> nad <lightSwitch> are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--ro, req, object, enable WDR by time-->
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->1
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->1
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->1
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel opt="level1,level2,level3,level4,level5,level6,level7">
    <!--ro, opt, enum, WDR level, subType:string, attr:opt{req, string}, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel min="0" max="100" def="50">
    <!--ro, opt, int, dep:and,{$.Hdr.hdrMode,eq,Sdr}, attr:min{req, int},max{req, int},def{req, int}-->1
  </SdrLevel>
</Hdr>
```

#### 14.2.1.4 Get the halo inhibition parameters of the traffic lights of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/lightHal

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<lightHal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, halo inhibition parameters of the traffic lights, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <halLevel>
    <!--ro, req, int, halo inhibition intensity, range:[0,100]-->0
  </halLevel>
  <halRange>
    <!--ro, req, int, halo inhibition range, range:[0,2064]-->0
  </halRange>
</lightHal>
```

#### 14.2.1.5 Set the halo inhibition parameters of the traffic lights of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/lightHal

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<lightHal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, halo inhibition parameters of the traffic lights, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <halLevel>
    <!--req, int, halo inhibition intensity, range:[0,100]-->0
  </halLevel>
  <halRange>
    <!--req, int, halo inhibition range, range:[0,2064]-->0
  </halRange>
</lightHal>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.2.1.6 Set the wide dynamic range (WDR) parameters in auto mode of a specified channel

### Request URL

PUT /ISAPI/Image/channels/<channelID>/WDR

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WDR xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->auto
  </mode>
  <WDRLevel>
    <!--opt, int, range:[1,100]-->50
  </WDRLevel>
  <WDRContrastLevel>
    <!--opt, int-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--opt, int-->1
  </WDRLevel1>
</WDR>
```

### Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.1.7 Get the WDR (Wide Dynamic Range) parameters in auto mode of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/WDR

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Video channel No. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<WDR xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, WDR mode, subType:string, desc:WDR mode-->auto
  </mode>
  <WDRLevel>
    <!--ro, opt, int, WDR level, range:[0,100], desc:WDR level-->50
  </WDRLevel>
  <WDRContrastLevel>
    <!--ro, opt, int-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--ro, opt, int-->1
  </WDRLevel1>
  <timeMode>
    <!--ro, opt, object, dep:and,{$.WDR.mode,eq,time}-->
    <timeRangeList>
      <!--ro, req, array, subType:object, range:[0,1]-->
      <timeRange>
        <!--ro, req, object-->
        <startTime>
          <!--ro, req, time-->08:00:00
        </startTime>
        <endTime>
          <!--ro, req, time-->18:00:00
        </endTime>
      </timeRange>
    </timeRangeList>
  </timeMode>
  <WDRStreakInhibition>
    <!--ro, opt, bool-->false
  </WDRStreakInhibition>
  <WDRCloseToOpenFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDRCloseToOpenFilterLevel>
  <WDROpenToCloseFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDROpenToCloseFilterLevel>
</WDR>
```

### 14.2.2 Exposure Parameters

#### 14.2.2.1 Get the configuration parameters of the low illumination electronic shutter of a specified channel

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**Request URL**

GET /ISAPI/Image/channels/<channelID>/DSS

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<DSS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, low illumination electronic shutter, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <DSSLevel>
    <!--ro, opt, enum, low illumination electronic shutter level in exposure, subType:string,
desc:"*1.25,*1.5,*2,*3*4,*6,*8,*12,*16,*24,*32,*48,*64,*96,*128,*256,*512,auto". "*2" indicates that the speed of the low illumination electronic shutter is
2 times of the normal shutter speed-->*1.25
  </DSSLevel>
  <mode>
    <!--ro, opt, enum, mode, subType:string, desc:"frame" (frame mode, it indicates that node DSSLevel is valid), "level" (level mode, it indicates that
slowShutterLevel is valid). If this node does not exist, the default mode is frame mode-->frame
  </mode>
  <slowShutterLevel>
    <!--ro, req, int, shutter level, range:[1,6]-->1
  </slowShutterLevel>
</DSS>
```

### 14.2.2.2 Set the configuration parameters of the low illumination electronic shutter for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/DSS

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<DSS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, low illumination electronic shutter, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <DSSLevel>
    <!--opt, enum, low illumination level, subType:string, desc:low illumination level-->*1.25
  </DSSLevel>
  <mode>
    <!--opt, enum, mode, subType:string, desc:"frame" (frame mode, it indicates that node DSSLevel is valid), "level" (level mode, it indicates that
slowShutterLevel is valid). If this node does not exist, the default mode is frame mode-->frame
  </mode>
  <slowShutterLevel>
    <!--req, int, slow shutter level in exposure, range:[1,6]-->1
  </slowShutterLevel>
</DSS>
```

**Response Message**

VISION AND SECURITY SYSTEM CO., LTD. dojo jin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.2.3 Set the exposure mode for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/exposure

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--req, enum, exposure type, subType:string, desc:exposure type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--opt, int, automatic aperture level-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <Type>
      <!--opt, enum, "AUTO", "MANUAL", subType:string, desc:"AUTO", "MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--opt, int, focal length-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
  </OverexposeSuppress>
  <pIris>
    <!--opt, object, aperture-->
    <pIrisType>
      <!--opt, enum, aperture type, subType:string, desc:aperture type-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--opt, int, aperture level-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--opt, object, face exposure-->
    <enabled>
      <!--opt, bool, whether to enable face exposure-->true
    </enabled>
    <sensitivity>
      <!--opt, int, sensitivity of face exposure-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is available when ExposureType is set to "pIris-General"-->
    <irisLevel>
      <!--opt, int, aperture level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
</Exposure>
```

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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#### 14.2.2.4 Get the exposure mode of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/exposure?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Camera ID |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--ro, req, enum, lens type, subType:string, desc:lens type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--ro, opt, int-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <Type>
      <!--ro, opt, enum, this node depends on <enabled>,"AUTO,MANUAL", subType:string, desc:this node depends on <enabled>,"AUTO,MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
    <hightLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </hightLightDistanceLevel>
    <lowLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </lowLightDistanceLevel>
  </OverexposeSuppress>
  <pIris>
    <!--ro, opt, object, this node depends on <ExposureType>,"AUTO,MANUAL"-->
    <pIrisType>
      <!--ro, opt, enum, subType:string-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--ro, opt, int, this node depends on <pIrisType>-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--ro, opt, object, face exposure-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivity>
      <!--ro, opt, int, sensitivity-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--ro, opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is valid when <ExposureType> is set to "pIris-General"-->
    <irisLevel>
      <!--ro, opt, int, iris level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
  <combinationParams>
    <!--ro, opt, int, range:[0,100]-->50
  </combinationParams>
  <opticalFilter>
    <!--ro, opt, enum, subType:string-->none
  </opticalFilter>
  <lowGhostingExposure>
    <!--ro, opt, object-->
    <lowGhostingShutterLevel>
      <!--ro, opt, enum, subType:string-->1/100
    </lowGhostingShutterLevel>
  </lowGhostingExposure>
  <exposureCompensateLevel>
    <!--ro, opt, int, range:[-100,100]-->0
  </exposureCompensateLevel>
  <exposureControl>
    <!--ro, opt, enum, subType:string-->center
  </exposureControl>
</Exposure>
```

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dojojin@gmail.com

#### 14.2.2.5 Set the iris parameters of a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/iris

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Iris xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <IrisLevel>
    <!--req, int-->1
  </IrisLevel>
  <irisSpeed>
    <!--opt, int-->1
  </irisSpeed>
</Iris>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.2.6 Get the iris parameters of a specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/iris

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Iris xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <IrisLevel>
    <!--ro, req, int-->1
  </IrisLevel>
  <irisSpeed>
    <!--ro, opt, int-->1
  </irisSpeed>
</Iris>
```

#### 14.2.2.7 Set the multi-shutter parameters for a specified channel

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/multishut

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMultishut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <doubleShutEnable>
    <!--req, bool, whether to enable dual-shutter-->true
  </doubleShutEnable>
  <codecType>
    <!--req, enum, type, subType:string, desc:type-->all_ia
  </codecType>
  <threeShutEnable>
    <!--opt, bool, whether to enable three-shutter-->true
  </threeShutEnable>
  <doubleVideoEnable>
    <!--opt, bool, whether to enable dual-video-->true
  </doubleVideoEnable>
</ImageMultishut>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.2.8 Get the multi-shutter capability of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/multishut/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMultishut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <doubleShutEnable>
    <!--ro, req, bool, whether to enable dual-shutter-->true
  </doubleShutEnable>
  <codecType>
    <!--ro, req, enum, "all_ia" (algorithm stream), "all_record" (record stream), "all_threeShut", subType:string, desc:"all_ia" (algorithm stream),
"all_record" (record stream), "all_threeShut"-->all_ia
  </codecType>
  <threeShutEnable>
    <!--ro, opt, bool-->true
  </threeShutEnable>
  <doubleVideoEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </doubleVideoEnable>
  <multiCapture>
    <!--ro, opt, object-->
    <iaCaptureEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </iaCaptureEnabled>
    <recordCaptureEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </recordCaptureEnabled>
    <captureCombineType opt="none,iaAndRecord">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
    </captureCombineType>
  </multiCapture>
</ImageMultishut>
```

#### 14.2.2.9 Get the multi-shutter parameters of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/multishut?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD.
dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMultishut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <doubleShutEnable>
    <!--ro, req, bool, whether to enable dual-shutter-->true
  </doubleShutEnable>
  <codecType>
    <!--ro, req, enum, type, subType:string, desc:type-->all_ia
  </codecType>
  <threeShutEnable>
    <!--ro, opt, bool-->true
  </threeShutEnable>
  <doubleVideoEnable>
    <!--ro, opt, bool-->true
  </doubleVideoEnable>
  <multiCapture>
    <!--ro, opt, object-->
    <iaCaptureEnabled>
      <!--ro, opt, bool-->true
    </iaCaptureEnabled>
    <recordCaptureEnabled>
      <!--ro, opt, bool-->true
    </recordCaptureEnabled>
    <captureCombineType>
      <!--ro, opt, enum, subType:string-->none
    </captureCombineType>
  </multiCapture>
</ImageMultishut>
```

### 14.2.2.10 Set the regional exposure parameters of a specified channel

#### Request URL

PUT /ISAPI/Image/channels/<channelID>/regionalExposure

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionalExposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <StartPoint>
    <!--opt, object-->
    <positionX>
      <!--req, int, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, range:[0,1000]-->0
    </positionY>
  </StartPoint>
  <EndPoint>
    <!--opt, object-->
    <positionX>
      <!--req, int, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, range:[0,1000]-->0
    </positionY>
  </EndPoint>
  <enabled>
    <!--opt, bool-->false
  </enabled>
  <showRuleEnabled>
    <!--opt, bool-->false
  </showRuleEnabled>
</RegionalExposure>
```

#### Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.2.11 Set the exposure time parameters of a specific channel

#### Request URL

PUT /ISAPI/Image/channels/<channelID>/shutter

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shutter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <ShutterLevel>
    <!--opt, enum, exposure level, subType:string, desc:"1/1", "1/2", "1/3", "1/6", "1/12", "1/25", "1/50", "1/75", "1/100", "1/120", "1/125", "1/150",
"1/175", this node is valid when exposure type is "shutter first"-->1/1
  </ShutterLevel>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleShutterLevel>
    <!--opt, int-->1
  </scheduleShutterLevel>
  <ScheduleList>
    <!--opt, array, subType:object-->
    <Schedule>
      <!--opt, object-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, end time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, end time (minute)-->1
      </endMinute>
    </Schedule>
  </ScheduleList>
  <redShutterLevel>
    <!--opt, int, range:[0,20000]-->1
  </redShutterLevel>
</Shutter>
```

#### Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.2.12 Set the lens aperture parameters for a specified video input channel

**Request URL**

PUT /ISAPI/System/Video/inputs/channels/<channelID>/iris

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<IrisData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, aperture, attr:version{req, string, protocolVersion}-->
  <iris>
    <!--opt, int, aperture adjusting speed, range:[-100,100], desc:aperture adjusting speed-->1
  </iris>
</IrisData>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.3 Focus Parameters

#### 14.2.3.1 Get the focus parameters of the specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/focusConfiguration

**Query Parameter**

dojo.jin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, focus configuration, attr:version{req, string, protocolVersion}-->
  <focusStyle>
    <!--ro, req, enum, focus mode, subType:string, desc:"AUTO" (automatic focus), "MANUAL" (manual focus), "SEMIAUTOMATIC" (semi-automatic focus)-->AUTO
  </focusStyle>
  <focusLimited>
    <!--ro, opt, enum, focus limits, subType:int, desc:focus limits-->10
  </focusLimited>
  <focusPosition>
    <!--ro, opt, int, focus point position-->1
  </focusPosition>
  <focusSpeed>
    <!--ro, opt, int, focus speed-->1
  </focusSpeed>
  <focusSensitivity>
    <!--ro, opt, int, focus sensitivity-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled>
    <!--ro, opt, bool, whether to enable temperature change adaption-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--ro, opt, int, relative focus point position-->1
  </relativeFocusPos>
  <lensType>
    <!--ro, opt, enum, lens type, subType:string, desc:lens type-->1140
  </lensType>
  <highTemperaturePriority>
    <!--ro, opt, bool, whether to enable high temperature priority mode-->true
  </highTemperaturePriority>
  <focusRange>
    <!--ro, opt, enum, focus adjusting range, subType:string, desc:focus adjusting range-->max
  </focusRange>
  <focusDistanceMode>
    <!--ro, opt, enum, focus distance mode, subType:string, desc:focus distance mode-->compatible
  </focusDistanceMode>
  <zoomFocusFollowEnabled>
    <!--ro, opt, bool-->true
  </zoomFocusFollowEnabled>
  <focusRegion>
    <!--ro, opt, enum, subType:string, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}-->center
  </focusRegion>
  <focusScene>
    <!--ro, opt, enum, subType:string-->normal
  </focusScene>
  <timeFocusEnabled>
    <!--ro, opt, bool, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}-->true
  </timeFocusEnabled>
  <timeFocus>
    <!--ro, opt, int, range:[1,1440], unit:min, dep:and,{$.FocusConfiguration.timeFocusEnabled,eq,true}-->1
  </timeFocus>
  <focusDistance>
    <!--ro, opt, float, range:[0.30,99.99], step:0.01, unit:m, dep:and,{$.FocusConfiguration.focusStyle,eq,distanceCalculate}-->0.35
  </focusDistance>
  <focusStrategy>
    <!--ro, opt, enum, subType:string-->DTOF
  </focusStrategy>
</FocusConfiguration>
```

VISION AND SECURITY SYSTEM CO.,LTD.
dojojin@gmail.com

### 14.2.3.2 Set the focus parameters for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/focusConfiguration

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, focus configuration, attr:version{req, string, protocolVersion}-->
  <focusStyle>
    <!--req, enum, focus mode, subType:string, desc:"auto" (automatic focus), "manual" (manual focus), "semiautomatic" (semiautomatic focus)-->AUTO
  </focusStyle>
  <focusLimited>
    <!--opt, enum, subType:int-->10
  </focusLimited>
  <focusPosition>
    <!--opt, int, focus point position-->1
  </focusPosition>
  <focusSpeed>
    <!--opt, int, focus speed-->1
  </focusSpeed>
  <focusSensitivity>
    <!--opt, int, focus sensitivity-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled>
    <!--opt, bool, whether to enable temperature change adaption-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--opt, int, relative focus point position-->1
  </relativeFocusPos>
  <lensType>
    <!--opt, enum, lens type, subType:string, desc:"1140", "1050"-->1140
  </lensType>
  <highTemperaturePriority>
    <!--opt, bool, whether to enable high temperature priority mode-->true
  </highTemperaturePriority>
  <focusRange>
    <!--opt, enum, focus adjustment range, subType:string, desc:"min" (the minimum), "medium" (the medium), "max" (the maximum)-->max
  </focusRange>
  <focusDistanceMode>
    <!--opt, enum, focus distance mode, subType:string, desc:focus distance mode-->compatible
  </focusDistanceMode>
</FocusConfiguration>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.3.3 Get the image capability of a specified channel

### Request URL

GET /ISAPI/Image/channels/<channelID>/imageCap

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

### Request Message

None

### Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojijin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isSupportRegionalExposure>
    <!--ro, opt, bool-->true
  </isSupportRegionalExposure>
  <isSupportRegionalFocus>
    <!--ro, opt, bool-->true
  </isSupportRegionalFocus>
  <isSupportManualBackGroundCorrect>
    <!--ro, opt, bool-->true
  </isSupportManualBackGroundCorrect>
  <isSupportManualShutterCorrect>
    <!--ro, opt, bool-->true
  </isSupportManualShutterCorrect>
  <RegionalExposure>
    <!--ro, opt, object-->
    <enabled opt="true,false" def="false">
      <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
    </enabled>
    <showRlueEnabled opt="true,false" def="false">
      <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
    </showRlueEnabled>
  </RegionalExposure>
</ImageCap>
```

#### 14.2.3.4 Set the regional focus parameters for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/regionalFocus

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionalFocus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, regional focus parameters, attr:version{req, string, protocolVersion}-->
  <StartPoint>
    <!--opt, object, start point, desc:the origin is the upper-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
  </StartPoint>
  <EndPoint>
    <!--opt, object, end point, desc:the origin is the upper-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
  </EndPoint>
</RegionalFocus>
```

**Response Message**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.3.5 Set the focus parameters for a specified video input channel

**Request URL**

PUT /ISAPI/System/Video/inputs/channels/<channelID>/focus

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<FocusData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of focus control, attr:version{req, string, protocolVersion}-->
  <focus>
    <!--req, int, focus control, range:[-100,100], desc:focus control-->0
  </focus>
</FocusData>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.4 Image Adjustment

#### 14.2.4.1 Set the gamma correction parameters for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/gammaCorrection

**Query Parameter**

dojo jit@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<gammaCorrection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, gamma correction parameters, attr:version{req, string, protocolVersion}-->
  <gammaCorrectionEnabled>
    <!--opt, bool, whether to enable gamma correction-->true
  </gammaCorrectionEnabled>
  <gammaCorrectionLevel>
    <!--opt, int, gamma correction level-->1
  </gammaCorrectionLevel>
</gammaCorrection>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.4.2 Get the gamma correction parameters of a specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/gammaCorrection?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<gammaCorrection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <gammaCorrectionEnabled>
    <!--ro, opt, bool-->true
  </gammaCorrectionEnabled>
  <gammaCorrectionLevel>
    <!--ro, opt, int-->1
  </gammaCorrectionLevel>
</gammaCorrection>
```

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#### 14.2.4.3 Set the parameters for visual and thermal image fusion

**Request URL**

PUT /ISAPI/Image/dualLightFusion?format=json&channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
{
    "enabled": true,
    /*opt, bool, whether to enable visual and thermal image fusion, desc:when the illumination is low, the visual and thermal image fusion can improve the
live view effect*/
    "switchingThreshold": 2
    /*opt, int, range:[0,7], step:1*/
}
```

**Response Message**

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

### 14.2.5 Image Adjustment

#### 14.2.5.1 Set image parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <DSS>
    <!--opt, object, DSS-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <mode>
      <!--opt, enum, subType:string-->frame
    </mode>
    <slowShutterLevel>
      <!--opt, int, range:[1,6]-->1
    </slowShutterLevel>
  </DSS>
  <WhiteBalance>
    <!--opt, object, white balance-->
    <WhiteBalanceStyle>
      <!--opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
```

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```
      <whiteBalanceLevel>
        <!--opt, int, range:[0,100]-->1
      </whiteBalanceLevel>
    </WhiteBalance>
    <Exposure>
      <!--opt, object-->
      <ExposureType>
        <!--req, enum, subType:string-->auto
      </ExposureType>
      <pIris>
        <!--opt, object, aperture-->
        <pIrisType>
          <!--opt, enum, subType:string-->MANUAL
        </pIrisType>
        <IrisLevel>
          <!--opt, int, range:[1,100]-->1
        </IrisLevel>
      </pIris>
    </Exposure>
    <Sharpness>
      <!--opt, object-->
      <SharpnessLevel>
        <!--req, int, range:[0,100]-->1
      </SharpnessLevel>
    </Sharpness>
    <gammaCorrection>
      <!--opt, object-->
      <gammaCorrectionEnabled>
        <!--opt, bool-->true
      </gammaCorrectionEnabled>
      <gammaCorrectionLevel>
        <!--opt, int, range:[0,100]-->1
      </gammaCorrectionLevel>
    </gammaCorrection>
    <Shutter>
      <!--opt, object-->
      <ShutterLevel>
        <!--opt, enum, subType:string-->1/1
      </ShutterLevel>
    </Shutter>
    <Gain>
      <!--opt, object, gain-->
      <GainLevel>
        <!--opt, string, range:[0,10]-->test
      </GainLevel>
    </Gain>
    <Color>
      <!--opt, object-->
      <brightnessLevel>
        <!--opt, int, brightness, range:[0,100]-->0
      </brightnessLevel>
      <contrastLevel>
        <!--opt, int, contrast, range:[0,100]-->0
      </contrastLevel>
      <saturationLevel>
        <!--opt, int, saturation, range:[0,100]-->0
      </saturationLevel>
    </Color>
    <lightHal>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <halLevel>
        <!--req, int, range:[0,100]-->0
      </halLevel>
      <halRange>
        <!--req, int, range:[0,2064]-->0
      </halRange>
    </lightHal>
  </ImageChannel>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.5.2 Set the brightness enhancement parameters for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/brightEnhance

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, brightness enhancement parameters, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--req, bool, whether to enable-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.5.3 Get the brightness enhancement parameters of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/brightEnhance?parameterType=<parameterType>

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--ro, req, bool, whether to enable brightness enhancement-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--ro, req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

### 14.2.5.4 Get image configuration capability of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Video channel No. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, temperature range capability, see XML_Cap_tempRange for details, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <FocusConfiguration>
    <!--ro, opt, object-->
    <focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC,distanceCalculate">
      <!--ro, req, string, attr:opt{req, string}-->test
    </focusStyle>
    <focusStatus opt="success,failed,working">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->success
    </focusStatus>
  </FocusConfiguration>
  <DSS>
    <!--ro, opt, object, DSS-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <mode opt="frame,level">
      <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->frame
    </mode>
    <slowShutterLevel min="1" max="6">
      <!--ro, opt, int, range:[1,6], attr:min{req, int},max{req, int}-->1
    </slowShutterLevel>
  </DSS>
  <WhiteBalance>
    <!--ro, opt, object, white balance-->
```

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```
    <WhiteBalanceStyle opt="manual、auto1、auto2、locked、outdoor、indoor、daylightLamp、sodiumlight、autotrace、onece、autooutdoor、autosodiumlight、
mercurylight、auto、incandescentlight、warmlight、naturallight,customColorTemperature,highTemperature">
      <!--ro, req, string, attr:opt{req, string}-->test
    </WhiteBalanceStyle>
    <WhiteBalanceRed min="-127" max="127">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue min="-127" max="127">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </WhiteBalanceBlue>
    <whiteBalanceLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </whiteBalanceLevel>
  </WhiteBalance>
  <Exposure>
    <!--ro, opt, object-->
    <ExposureType opt="auto,IrisFirst,ShutterFirst,gainFirst,manual,pIris,T5280-PQ1,T5289-PQ1,T1140-PQ1,T2712-PQ1,HV1250P-MPIR,pc,pIris-General,HZ1140P-
8MPIRA,HZ1135P-8MPSGA,MV1550P-12MPIR,MZ1545P-12MP,HZ0618P-4MPSG,MZ1050P-12MP,SF5014M-12MP,SF3018M-10MP,MZ1140M,MZ2514M-10MP,MZ7018M-8MP,MZ2516M-
10MPIR,indoor50HZ,indoor60HZ,lowGhosting,customize">
      <!--ro, req, string, attr:opt{req, string}-->test
    </ExposureType>
    <pIris>
      <!--ro, opt, object, aperture-->
      <pIrisType opt="AUTO,MANUAL">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->MANUAL
      </pIrisType>
      <IrisLevel min="1" max="100">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </IrisLevel>
    </pIris>
  </Exposure>
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel min="0" max="255">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </SharpnessLevel>
  </Sharpness>
  <gammaCorrection>
    <!--ro, opt, object-->
    <gammaCorrectionEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </gammaCorrectionEnabled>
    <gammaCorrectionLevel min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </gammaCorrectionLevel>
  </gammaCorrection>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1
/10000,1/225,1/300,1/400,,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </ShutterLevel>
  </Shutter>
  <Gain>
    <!--ro, opt, object, gain-->
    <GainLevel min="0" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--ro, opt, object, image adjustment capability-->
    <brightnessLevel min="0" max="255">
      <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->0
    </brightnessLevel>
    <contrastLevel min="0" max="255">
      <!--ro, opt, int, contrast, attr:min{req, int},max{req, int}-->0
    </contrastLevel>
    <saturationLevel min="0" max="255">
      <!--ro, opt, int, saturation, attr:min{req, int},max{req, int}-->0
    </saturationLevel>
    <nightMode opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </nightMode>
  </Color>
  <DayNightGate>
    <!--ro, opt, object-->
    <brightnessThreld min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </brightnessThreld>
    <nightModeCtrl>
      <!--ro, opt, object-->
      <nightModeCtrlMode opt="manual,time,auto">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->manual
      </nightModeCtrlMode>
      <manualMode>
        <!--ro, opt, object-->
        <manualModeVal opt="day,night">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
        </manualModeVal>
      </manualMode>
      <timeMode>
        <!--ro, opt, object-->
        <scheduleType opt="day,range">
```

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```
        <scheduleType opt="day,range">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
        </scheduleType>
        <switchList>
          <!--ro, opt, array, subType:object-->
          <switch>
            <!--ro, opt, object-->
            <nightModeTmeId min="0" max="1">
              <!--ro, opt, int, range:[0,1], attr:min{req, int},max{req, int}-->1
            </nightModeTmeId>
            <nightModeStartHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </nightModeStartHour>
            <nightModeStartMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </nightModeStartMinute>
            <nightModeEndHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </nightModeEndHour>
            <nightModeEndMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </nightModeEndMinute>
          </switch>
        </switchList>
        <timeScheduleList size="2">
          <!--ro, opt, object, attr:size{req, int}-->
          <timeSchedule>
            <!--ro, opt, object-->
            <triggerData>
              <!--ro, opt, object-->
              <startTime>
                <!--ro, opt, object-->
                <monthOfYear min="1" max="12">
                  <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->1
                </monthOfYear>
              </startTime>
              <endTime>
                <!--ro, opt, object-->
                <monthOfYear min="1" max="12">
                  <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->12
                </monthOfYear>
              </endTime>
            </triggerData>
            <timeSwitchList size="2">
              <!--ro, opt, object, attr:size{req, int}-->
              <timeSwitch>
                <!--ro, opt, object-->
                <timeID min="0" max="1">
                  <!--ro, req, int, range:[0,1], attr:min{req, int},max{req, int}-->1
                </timeID>
                <startHour min="0" max="23">
                  <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
                </startHour>
                <startMinute min="0" max="59">
                  <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
                </startMinute>
                <endHour min="0" max="23">
                  <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
                </endHour>
                <endMinute min="0" max="59">
                  <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
                </endMinute>
              </timeSwitch>
            </timeSwitchList>
          </timeSchedule>
        </timeScheduleList>
      </timeMode>
      <autoMode>
        <!--ro, opt, object-->
        <autoSwtichThresholdVal min="0" max="100">
          <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
        </autoSwtichThresholdVal>
        <autoNightSwtichThresholdVal min="0" max="100">
          <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
        </autoNightSwtichThresholdVal>
        <autoDayToNightDelayTime min="0" max="120">
          <!--ro, opt, int, range:[0,120], attr:min{req, int},max{req, int}-->1
        </autoDayToNightDelayTime>
        <autoNightToDayDelayTime min="0" max="120">
          <!--ro, opt, int, range:[0,120], attr:min{req, int},max{req, int}-->1
        </autoNightToDayDelayTime>
      </autoMode>
    </nightModeCtrl>
  </DayNightGate>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType opt="auto,day,night,schedule,eventTrigger,videoAuto,IrcutFilterType,darkFighterX,darkFighterXAuto,darkFighterXSchedule,colorVuAuto">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
    </IrcutFilterType>
  </IrcutFilter>
  <isSupportMovingTargetShutterParams>
    <!--ro, opt, bool-->true
  </isSupportMovingTargetShutterParams>
  <NoiseReduceExt>
```

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```
    <!--ro, opt, object-->
    <mode opt="close,general,trailingNoiseBalance,advanced">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel min="0" max="255">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
      </generalLevel>
    </GeneralMode>
    <AdvancedMode>
      <!--ro, opt, object-->
      <spatialLevel min="0" max="100">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
      </spatialLevel>
      <temporalLevel min="0" max="100">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
      </temporalLevel>
    </AdvancedMode>
  </NoiseReduceExt>
  <isSupportJPEGParam>
    <!--ro, opt, bool-->true
  </isSupportJPEGParam>
  <isSupportDarkEnhance>
    <!--ro, opt, bool-->true
  </isSupportDarkEnhance>
  <isSupportIcr>
    <!--ro, opt, bool-->true
  </isSupportIcr>
  <isSupportMultishut>
    <!--ro, opt, bool-->true
  </isSupportMultishut>
  <isSupportGrayRange>
    <!--ro, opt, bool-->true
  </isSupportGrayRange>
  <isSupportCapImage>
    <!--ro, opt, bool-->true
  </isSupportCapImage>
  <isSupportPlateBright>
    <!--ro, opt, bool-->true
  </isSupportPlateBright>
  <NoiseReduce2D>
    <!--ro, opt, object-->
    <noiseReduce2DEnable>
      <!--ro, opt, bool-->true
    </noiseReduce2DEnable>
    <noiseReduce2DLevel min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </noiseReduce2DLevel>
  </NoiseReduce2D>
  <BrightEnhance>
    <!--ro, opt, object-->
    <brightEnhanceLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </brightEnhanceLevel>
  </BrightEnhance>
</ImageChannel>
```

#### 14.2.5.5 Set the image adjustment parameters in auto mode of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/color

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted--
>true
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.5.6 Get the image adjustment parameters in auto mode of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/color?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```

### 14.2.5.7 Set the gray scale range parameters of a specific channel

#### Request URL

PUT /ISAPI/Image/channels/<channelID>/grayRange

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GrayRange xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, gray scale range, attr:version{req, string, protocolVersion}-->
  <grayValueType>
    <!--req, enum, gray scale value, subType:string, desc:"hightContrast"-high contrast with less details in shadows, the value ranges from 0 to 255;
"moreDetail"-more detail in shadows but low contrast, the value ranges from 16 to 235. "hightContrast" (high contrast with less details in shadows),
"moreDetail" (more detail in shadows but low contrast)-->hightContrast
  </grayValueType>
</GrayRange>
```

#### Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dejojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.5.8 Get the grayscale range capability of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/grayRange/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<GrayRange xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <grayValueType opt="hightContrast,moreDetail">
    <!--ro, req, string, grayscale value, attr:opt{req, string}, desc:"hightContrast"-high contrast with less details in shadows, the value ranges from 0 to
255; "moreDetail"-more detail in shadows but low contrast,-->test
  </grayValueType>
</GrayRange>
```

#### 14.2.5.9 Get the grayscale range parameters of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/grayRange?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD. dongjin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<GrayRange xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <grayValueType>
    <!--ro, req, enum, grayscale value, subType:string, desc:"hightContrast"-high contrast with less details in shadows, the value ranges from 0 to 255;
"moreDetail"-more detail in shadows but low contrast, the value ranges from 16 to 235. "hightContrast" (high contrast with less details in shadows),
"moreDetail" (more detail in shadows but low contrast)-->hightContrast
  </grayValueType>
</GrayRange>
```

### 14.2.5.10 Get day/night auto switch parameters of a specific channel

#### Request URL

GET /ISAPI/Image/channels/<channelID>/ircutFilter

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--ro, opt, enum, day/night auto switch mode, subType:string, desc:"auto, day, night, schedule, eventTrigger, darkFighterX, darkFighterXAuto,
darkFighterXSchedule"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--ro, opt, int, level of switching day to night, range:[0,7]-->4
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--ro, opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--ro, opt, int, level of switching night to day, range:[0,7]-->4
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--ro, opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--ro, opt, object-->
    <scheduleType>
      <!--ro, req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--ro, req, object, time period on schedule-->
      <beginTime>
        <!--ro, req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--ro, opt, object-->
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:event type-->IO
    </eventType>
    <IrcutFilterAction>
      <!--ro, req, enum, "day, night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
  <dayNightStatus>
    <!--ro, opt, enum, subType:string-->day
  </dayNightStatus>
</IrcutFilter>
```

### 14.2.5.11 Set day/night auto switch parameters of a specific channel

#### Request URL

VISION AND SECURITY SYSTEM CO.,LTD.
dojojin@gmail.com

PUT /ISAPI/Image/channels/<channelID>/ircutFilter

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, IrcutFilter, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--opt, enum, IrcutFilter type, subType:string, desc:"auto", "day", "night", "schedule" , "eventTrigger", "videoAuto", "darkFighterX",
"darkFighterXAuto", "darkFighterXSchedule", "colorVuAuto"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, enum, level of switching day to night, subType:string, desc:"low", "normal", "high"-->low
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, enum, level of switching night to day, subType:string, desc:"low", "normal", "high"-->low
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--opt, object, schedule-->
    <scheduleType>
      <!--req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--req, object, time period on schedule-->
      <beginTime>
        <!--req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--opt, object, event trigger-->
    <eventType>
      <!--req, enum, event type, subType:string, desc:"IO", "VMD"-->IO
    </eventType>
    <IrcutFilterAction>
      <!--req, enum, "day,night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
</IrcutFilter>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

### 14.2.5.12 Get the contrast enhancement parameters of a specific channel

### Request URL

GET /ISAPI/Image/channels/<channelID>/lse

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--ro, opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

## 14.2.5.13 Set the contrast enhancement parameters of a specific channel

### Request URL

PUT /ISAPI/Image/channels/<channelID>/lse

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--opt, enum, LSE switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

### Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.5.14 Get the contrast enhancement capability of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/lse/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, req, int, contrast enhancement level, range:[0,100]-->50
  </lseLevel>
  <lseHaloLevel>
    <!--ro, req, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, opt, int, brightness level, range:[0,100], dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

#### 14.2.5.15 Set the license plate brightness compensation parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/plateBright

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

VISION AND SECURITY SYSTEM CO.,LTD. doujijin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateBright xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, license plate brightness compensation, attr:version{req, string, protocolVersion}-->
  <plateBrightEnabled>
    <!--req, bool, whether to enable license plate brightness compensation-->true
  </plateBrightEnabled>
  <plateBrightSensitivity>
    <!--opt, int, license plate brightness compensation sensitivity, range:[1,100]-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--req, bool, whether to enable supplement light correction factor-->true
  </correctFactorEnabled>
  <correctFactor>
    <!--opt, int, supplement light correction factor, range:[0,100]-->1
  </correctFactor>
</PlateBright>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.2.5.16 Get the license plate brightness compensation capability of a specific channel

### Request URL

GET /ISAPI/Image/channels/<channelID>/plateBright/capabilities

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateBright xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate brightness compensation parameters, attr:version{req, string, protocolVersion}-->
  <plateBrightEnabled>
    <!--ro, req, bool, whether to enable license plate brightness compensation-->true
  </plateBrightEnabled>
  <plateBrightSensitivity min="0" max="100">
    <!--ro, opt, int, license plate brightness compensation sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--ro, opt, bool, whether to enable supplement light correction factor-->true
  </correctFactorEnabled>
  <correctFactor min="0" max="100">
    <!--ro, opt, int, supplement light correction factor, range:[0,100], attr:min{req, int},max{req, int}-->1
  </correctFactor>
</PlateBright>
```

## 14.2.5.17 Get the license plate brightness compensation parameters of a specific channel

### Request URL

dojojin@gmail.com VISION AND SECURITY SYSTEM CO. LTD.

GET /ISAPI/Image/channels/<channelID>/plateBright?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateBright xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate brightness compensation parameters, attr:version{req, string, protocolVersion}-->
  <plateBrightEnabled>
    <!--ro, req, bool, whether to enable license plate brightness compensation-->true
  </plateBrightEnabled>
  <plateBrightSensitivity>
    <!--ro, opt, int, license plate brightness compensation sensitivity, range:[0,100]-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--ro, opt, bool, whether to enable supplement light correction factor-->true
  </correctFactorEnabled>
  <correctFactor>
    <!--ro, opt, int, supplement light correction factor, range:[0,100]-->1
  </correctFactor>
</PlateBright>
```

### 14.2.5.18 Restore the image parameters of a specific channel to default settings

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/restore

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel ID |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD.
doc.jojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

### 14.2.5.19 Set the sharpness control parameters of a specific channel

#### Request URL

PUT /ISAPI/Image/channels/<channelID>/sharpness

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.5.20 Get the sharpness control parameters of a specific channel

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

**Request URL**

GET /ISAPI/Image/channels/<channelID>/sharpness?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

### 14.2.5.21 Get the image processing parameters of a specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Video channel No. |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <DSS>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <mode>
      <!--ro, opt, enum, subType:string-->frame
    </mode>
    <slowShutterLevel>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
      <!--ro, opt, int, range:[1,6]-->1
    </slowShutterLevel>
  </DSS>
  <WhiteBalance>
    <!--ro, opt, object-->
    <WhiteBalanceStyle>
      <!--ro, opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceBlue>
    <whiteBalanceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </whiteBalanceLevel>
  </WhiteBalance>
  <Exposure>
    <!--ro, opt, object-->
    <ExposureType>
      <!--ro, req, enum, subType:string-->auto
    </ExposureType>
    <pIris>
      <!--ro, opt, object-->
      <pIrisType>
        <!--ro, opt, enum, subType:string-->MANUAL
      </pIrisType>
      <IrisLevel>
        <!--ro, opt, int, range:[1,100]-->1
      </IrisLevel>
    </pIris>
  </Exposure>
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel>
      <!--ro, req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <gammaCorrection>
    <!--ro, opt, object-->
    <gammaCorrectionEnabled>
      <!--ro, opt, bool-->true
    </gammaCorrectionEnabled>
    <gammaCorrectionLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </gammaCorrectionLevel>
  </gammaCorrection>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel>
      <!--ro, opt, enum, subType:string-->1/1
    </ShutterLevel>
  </Shutter>
  <Gain>
    <!--ro, opt, object-->
    <GainLevel>
      <!--ro, opt, string, range:[0,10]-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </saturationLevel>
    <nightMode>
      <!--ro, opt, bool-->true
    </nightMode>
  </Color>
  <DayNightGate>
    <!--ro, opt, object-->
    <brightnessThreld>
      <!--ro, opt, int-->1
    </brightnessThreld>
    <nightModeCtrl>
      <!--ro, opt, object-->
      <nightModeCtrlMode>
        <!--ro, opt, enum, subType:string-->manual
      </nightModeCtrlMode>
      <manualMode>
        <!--ro, opt, object-->
        <manualModeVal>
          <!--ro, opt, enum, subType:string-->day
        </manualModeVal>
      </manualMode>
      <timeMode>
        <!--ro, opt, object-->
```

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```
        <scheduleType>
          <!--ro, opt, enum, subType:string-->day
        </scheduleType>
        <switchList>
          <!--ro, opt, array, subType:object-->
          <switch>
            <!--ro, opt, object-->
            <nightModeStartHour>
              <!--ro, opt, int, range:[0,23]-->1
            </nightModeStartHour>
            <nightModeStartMinute>
              <!--ro, opt, int, range:[0,59]-->1
            </nightModeStartMinute>
            <nightModeEndHour>
              <!--ro, opt, int, range:[0,23]-->1
            </nightModeEndHour>
            <nightModeEndMinute>
              <!--ro, opt, int, range:[0,59]-->1
            </nightModeEndMinute>
          </switch>
        </switchList>
        <timeScheduleList>
          <!--ro, opt, array, subType:object, range:[0,2]-->
          <timeSchedule>
            <!--ro, opt, object-->
            <triggerData>
              <!--ro, opt, object-->
              <startTime>
                <!--ro, opt, object-->
                <monthOfYear>
                  <!--ro, opt, int, range:[1,12]-->0
                </monthOfYear>
              </startTime>
              <endTime>
                <!--ro, opt, object-->
                <monthOfYear>
                  <!--ro, opt, int, range:[1,12]-->0
                </monthOfYear>
              </endTime>
            </triggerData>
            <timeSwitchList>
              <!--ro, opt, array, subType:object, range:[0,1]-->
              <timeSwitch>
                <!--ro, opt, object-->
                <startHour>
                  <!--ro, opt, int, range:[0,23]-->1
                </startHour>
                <startMinute>
                  <!--ro, opt, int, range:[0,59]-->1
                </startMinute>
                <endHour>
                  <!--ro, opt, int, range:[0,23]-->1
                </endHour>
                <endMinute>
                  <!--ro, opt, int, range:[0,59]-->1
                </endMinute>
              </timeSwitch>
            </timeSwitchList>
          </timeSchedule>
        </timeScheduleList>
      </timeMode>
      <autoMode>
        <!--ro, opt, object-->
        <autoSwtichThresholdVal>
          <!--ro, opt, int, range:[1,100]-->1
        </autoSwtichThresholdVal>
      </autoMode>
    </nightModeCtrl>
  </DayNightGate>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType>
      <!--ro, opt, enum, subType:string-->auto
    </IrcutFilterType>
    <Schedule>
      <!--ro, opt, object-->
      <scheduleType>
        <!--ro, req, enum, subType:string-->day
      </scheduleType>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </IrcutFilter>
  <lightHal>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
```

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```
      </enabled>
      <halLevel>
        <!--ro, req, int, range:[0,100]-->0
      </halLevel>
      <halRange>
        <!--ro, req, int, range:[0,2064]-->0
      </halRange>
    </lightHal>
    <NoiseReduce2D>
      <!--ro, opt, object-->
      <noiseReduce2DEnable>
        <!--ro, req, bool-->true
      </noiseReduce2DEnable>
      <noiseReduce2DLevel>
        <!--ro, opt, int, range:[0,255]-->1
      </noiseReduce2DLevel>
    </NoiseReduce2D>
    <BrightEnhance>
      <!--ro, opt, object-->
      <brightEnhanceEnabled>
        <!--ro, req, bool-->true
      </brightEnhanceEnabled>
      <brightEnhanceLevel>
        <!--ro, req, int, range:[0,100]-->1
      </brightEnhanceLevel>
    </BrightEnhance>
    <NoiseReduceExt>
      <!--ro, opt, object-->
      <mode>
        <!--ro, req, enum, subType:string-->close
      </mode>
      <GeneralMode>
        <!--ro, opt, object-->
        <generalLevel>
          <!--ro, opt, int, range:[0,255]-->1
        </generalLevel>
      </GeneralMode>
      <AdvancedMode>
        <!--ro, opt, object-->
        <spatialLevel>
          <!--ro, opt, int, range:[0,100]-->1
        </spatialLevel>
        <temporalLevel>
          <!--ro, opt, int, range:[0,100]-->1
        </temporalLevel>
      </AdvancedMode>
    </NoiseReduceExt>
    <PlateBright>
      <!--ro, opt, object-->
      <plateBrightEnabled>
        <!--ro, req, bool-->true
      </plateBrightEnabled>
      <plateBrightSensitivity>
        <!--ro, opt, int, range:[1,100]-->1
      </plateBrightSensitivity>
    </PlateBright>
    <ImageMultishut>
      <!--ro, opt, object-->
      <doubleShutEnable>
        <!--ro, req, bool-->true
      </doubleShutEnable>
      <codecType>
        <!--ro, req, enum, subType:string-->all_ia
      </codecType>
    </ImageMultishut>
    <JPEGParam>
      <!--ro, opt, object-->
      <JPEGSize>
        <!--ro, opt, int, range:[64,8196], unit:KB-->64
      </JPEGSize>
      <MergeJPEGSize>
        <!--ro, opt, int, range:[64,8196], unit:KB-->64
      </MergeJPEGSize>
      <EXIFInformationEnabled>
        <!--ro, opt, bool-->true
      </EXIFInformationEnabled>
    </JPEGParam>
    <DarkEnhance>
      <!--ro, opt, object-->
      <darkEnhanceEnabled>
        <!--ro, req, bool-->true
      </darkEnhanceEnabled>
      <darkEnhanceLevel>
        <!--ro, opt, int, range:[1,100]-->1
      </darkEnhanceLevel>
    </DarkEnhance>
    <GrayRange>
      <!--ro, opt, object-->
      <grayValueType>
        <!--ro, req, enum, subType:string-->hightContrast
      </grayValueType>
    </GrayRange>
  </ImageChannel>
```

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#### 14.2.5.22 Get the image mode parameters of all channels

**Request URL**

GET /ISAPI/Image/channels/imageModes

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageModeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, image mode configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <ImageMode>
    <!--ro, opt, object, image mode-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:"standard", "indoor", "outdoor", "dimLight", "bright", "soft", "vivid"-->standard
    </type>
    <recommendation>
      <!--ro, req, object, recommended configurations-->
      <brightnessLevel>
        <!--ro, opt, int, brightness, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, contrast, range:[0,100]-->1
      </contrastLevel>
      <sharpnessLevel>
        <!--ro, opt, int, sharpness, range:[0,100]-->1
      </sharpnessLevel>
      <saturationLevel>
        <!--ro, opt, int, saturation, range:[0,100]-->1
      </saturationLevel>
      <hueLevel>
        <!--ro, opt, int, hue, range:[0,100]-->1
      </hueLevel>
      <deNoiseLevel>
        <!--ro, opt, int, noise reduction, range:[0,100]-->1
      </deNoiseLevel>
    </recommendation>
  </ImageMode>
</ImageModeList>
```

## 14.2.6 Image Enhancement

#### 14.2.6.1 Set the dark space enhancement parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/DarkEnhance

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<DarkEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, dark space enhancement, attr:version{req, string, protocolVersion}-->
  <darkEnhanceEnabled>
    <!--req, bool, whether to enable dark space enhancement-->true
  </darkEnhanceEnabled>
  <darkEnhanceLevel>
    <!--opt, int, dark space enhancement level, range:[1,100]-->1
  </darkEnhanceLevel>
</DarkEnhance>
```

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.6.2 Get the dark space enhancement parameters of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/DarkEnhance

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<DarkEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, dark space enhancement, attr:version{req, string, protocolVersion}-->
  <darkEnhanceEnabled>
    <!--ro, req, bool, whether to enable dark space enhancement-->true
  </darkEnhanceEnabled>
  <darkEnhanceLevel>
    <!--ro, opt, int, dark space enhancement level, range:[1,100]-->1
  </darkEnhanceLevel>
</DarkEnhance>
```

#### 14.2.6.3 Get the electronic defogging parameters of a specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/defog

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Defog xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</Defog>
```

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#### 14.2.6.4 Set the electronic defogging parameters for a specified channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/defog

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Defog xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, defog configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
</Defog>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.6.5 Get defog mode parameters of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/dehaze

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<Dehaze xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, "open,close,auto", attr:version{req, string, protocolVersion}-->
  <DehazeMode>
    <!--ro, opt, enum, defog mode, subType:string, desc:"open", "close", "auto"-->open
  </DehazeMode>
  <DehazeLevel>
    <!--ro, opt, int, defog level-->1
  </DehazeLevel>
  <dehazeIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.Dehaze.DehazeMode,eq,advancedMode}-->0
  </dehazeIntensity>
  <smoothIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.Dehaze.DehazeMode,eq,advancedMode}-->0
  </smoothIntensity>
</Dehaze>
```

#### 14.2.6.6 Set defog mode parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/dehaze

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Dehaze xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, "open,close,auto", attr:version{req, string, protocolVersion}-->
  <DehazeMode>
    <!--opt, enum, defog mode, subType:string, desc:"open", "close", "auto"-->open
  </DehazeMode>
  <DehazeLevel>
    <!--opt, int, defog level-->1
  </DehazeLevel>
</Dehaze>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.6.7 Set the picture gain parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/gain

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

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### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Gain xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, gain, attr:version{req, string, protocolVersion}-->
  <GainLevel>
    <!--opt, int, gain level-->1
  </GainLevel>
  <GainWindow>
    <!--opt, object-->
    <RegionCoordinatesList>
      <!--opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleGainLevel>
    <!--opt, int, dep:and,{$.Gain.scheduleEnabled,eq,true}-->1
  </scheduleGainLevel>
  <Schedule>
    <!--opt, object, dep:and,{$.Gain.scheduleEnabled,eq,true}-->
    <startHour min="0" max="23">
      <!--opt, int, start time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </startHour>
    <startMinute min="0" max="59">
      <!--opt, int, start time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </startMinute>
    <endHour min="0" max="23">
      <!--opt, int, end time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </endHour>
    <endMinute min="0" max="59">
      <!--opt, int, end time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </endMinute>
  </Schedule>
  <redGainLevel>
    <!--opt, int, range:[0,100]-->1
  </redGainLevel>
</Gain>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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### 14.2.6.8 Get the picture gain parameters of a specific channel

### Request URL

GET /ISAPI/Image/channels/<channelID>/gain?parameterType=<parameterType>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Gain xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, gain, attr:version{req, string, protocolVersion}-->
  <GainLevel>
    <!--ro, opt, int, gain level-->1
  </GainLevel>
  <GainWindow>
    <!--ro, opt, object-->
    <RegionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
  <scheduleEnabled>
    <!--ro, opt, bool-->true
  </scheduleEnabled>
  <scheduleGainLevel>
    <!--ro, opt, int, dep:and,{$.Gain.scheduleEnabled,eq,true}-->1
  </scheduleGainLevel>
  <Schedule>
    <!--ro, opt, object, time range, dep:and,{$.Gain.scheduleEnabled,eq,true}-->
    <startHour min="0" max="23">
      <!--ro, opt, int, start time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </startHour>
    <startMinute min="0" max="59">
      <!--ro, opt, int, start time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </startMinute>
    <endHour min="0" max="23">
      <!--ro, opt, int, end time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </endHour>
    <endMinute min="0" max="59">
      <!--ro, opt, int, end time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </endMinute>
  </Schedule>
  <redGainLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </redGainLevel>
  <minExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </minExposureGainLevel>
  <maxExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </maxExposureGainLevel>
</Gain>
```

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### 14.2.6.9 Set the DNR (Digital Noise Reduction) parameters in auto mode of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/noiseReduce

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--opt, object, mode-->
    <generalLevel>
      <!--req, int, level-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, advanced mode-->
    <FrameNoiseReduceLevel>
      <!--req, int-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int-->1
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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dongjin@gmail.com

## 14.2.6.10 Get the DNR (Digital Noise Reduction) parameters in auto mode of a specific channel

### Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, this node is valid only when <mode> is "general"-->
    <generalLevel>
      <!--ro, req, int-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, advanced mode-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int-->1
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

### 14.2.6.11 Set the 2D noise reduction parameters of a specific channel

#### Request URL

PUT /ISAPI/Image/channels/<channelID>/NoiseReduce2D

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce2D xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, 2D noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <noiseReduce2DEnable>
    <!--req, bool, whether to enable 2D noise reduction-->true
  </noiseReduce2DEnable>
  <noiseReduce2DLevel min="0" max="100">
    <!--opt, int, 2D noise reduction level, attr:min{req, int},max{req, int}-->1
  </noiseReduce2DLevel>
</NoiseReduce2D>
```

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.6.12 Get 2D noise reduction parameters of a specific channel

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

**Request URL**

GET /ISAPI/Image/channels/<channelID>/NoiseReduce2D?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce2D xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <noiseReduce2DEnable>
    <!--ro, req, bool-->true
  </noiseReduce2DEnable>
  <noiseReduce2DLevel min="0" max="255">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </noiseReduce2DLevel>
</NoiseReduce2D>
```

### 14.2.6.13 Set the 3D noise reduction parameters of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/noiseReduceExt

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduceExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--opt, object, general mode-->
    <generalLevel>
      <!--opt, int, general level, range:[0,100]-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, advanced mode-->
    <spatialLevel>
      <!--opt, int, spatial level, range:[0,100]-->1
    </spatialLevel>
    <temporalLevel>
      <!--opt, int, level, range:[0,100]-->1
    </temporalLevel>
  </AdvancedMode>
</NoiseReduceExt>
```

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.6.14 Get 3D noise reduction parameters of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/noiseReduceExt?parameterType=<parameterType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduceExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, general mode-->
    <generalLevel>
      <!--ro, opt, int, general level, range:[0,100]-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, advanced mode-->
    <spatialLevel>
      <!--ro, opt, int, spatial level, range:[0,100]-->1
    </spatialLevel>
    <temporalLevel>
      <!--ro, opt, int, temporal level, range:[0,100]-->1
    </temporalLevel>
  </AdvancedMode>
  <trailingNoiseBalance>
    <!--ro, opt, object-->
    <level>
      <!--ro, opt, int, range:[0,100]-->1
    </level>
  </trailingNoiseBalance>
</NoiseReduceExt>
```

## 14.2.7 Image Light Supplement

#### 14.2.7.1 Set the day/night mode switching parameters for a specified channel

**Request URL**

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PUT /ISAPI/Image/channels/<channelID>/dayNightGate

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DayNightGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, day/night mode switching parameters, attr:version{req, string, protocolVersion}-->
  <brightnessThreld>
    <!--opt, int, brightness-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--opt, object, day/night mode control-->
    <nightModeCtrlMode>
      <!--opt, enum, day/night switching mode, subType:string, desc:"manual" (manual switch), "time" (timed switch), "auto" (auto switch)-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--opt, object, manual switch-->
      <manualModeVal>
        <!--opt, enum, manual switching mode, subType:string, desc:"day", "night"-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--opt, object, timed switch-->
      <switchList>
        <!--opt, array, subType:object-->
        <switch>
          <!--opt, object-->
          <nightModeTmeId>
            <!--opt, int-->1
          </nightModeTmeId>
          <nightModeStartHour>
            <!--opt, int, range:[0,23]-->1
          </nightModeStartHour>
          <nightModeStartMinute>
            <!--opt, int, range:[0,59]-->1
          </nightModeStartMinute>
          <nightModeEndHour>
            <!--opt, int, range:[0,23]-->1
          </nightModeEndHour>
          <nightModeEndMinute>
            <!--opt, int, range:[0,59]-->1
          </nightModeEndMinute>
        </switch>
      </switchList>
    </timeMode>
    <autoMode>
      <!--opt, object, auto switch-->
      <autoSwtichThresholdVal>
        <!--opt, int, threshold for switching, range:[1,100]-->1
      </autoSwtichThresholdVal>
    </autoMode>
  </nightModeCtrl>
</DayNightGate>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.7.2 Get the day/night switch parameters of the specific channel

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

### Request URL

GET /ISAPI/Image/channels/<channelID>/dayNightGate

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DayNightGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, day/night switch, attr:version{req, string, protocolVersion}-->
  <brightnessThreld>
    <!--ro, opt, int, brightness-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--ro, opt, object, day/night mode switching control-->
    <nightModeCtrlMode>
      <!--ro, opt, enum, day/night switch mode, subType:string, desc:"manual" (manual switch mode), "time" (timed switch mode), "auto" (automatic switch
mode)-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--ro, opt, object, manual switch-->
      <manualModeVal>
        <!--ro, opt, enum, "day", "night", subType:string, desc:"day", "night"-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--ro, opt, object, timed switch-->
      <scheduleType>
        <!--ro, opt, enum, subType:string-->day
      </scheduleType>
      <switchList>
        <!--ro, opt, array, subType:object-->
        <switch>
          <!--ro, opt, object, timed switch settings-->
          <nightModeTmeId>
            <!--ro, opt, int, night mode ID-->1
          </nightModeTmeId>
          <nightModeStartHour>
            <!--ro, opt, int, night mode starting hour, range:[0,23]-->1
          </nightModeStartHour>
          <nightModeStartMinute>
            <!--ro, opt, int, night mode starting minute, range:[0,59]-->1
          </nightModeStartMinute>
          <nightModeEndHour>
            <!--ro, opt, int, night mode ending hour, range:[0,23]-->1
          </nightModeEndHour>
          <nightModeEndMinute>
            <!--ro, opt, int, night mode ending minute, range:[0,59]-->1
          </nightModeEndMinute>
        </switch>
      </switchList>
      <timeScheduleList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <timeSchedule>
          <!--ro, opt, object-->
          <triggerData>
            <!--ro, opt, object-->
            <startTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->0
              </monthOfYear>
            </startTime>
            <endTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->0
              </monthOfYear>
            </endTime>
          </triggerData>
          <timeSwitchList>
            <!--ro, opt, array, subType:object, range:[0,1]-->
            <timeSwitch>
              <!--ro, opt, object-->
              <startHour>
                <!--ro, opt, int, range:[0,23]-->1
              </startHour>
              <startMinute>
                <!--ro, opt, int, range:[0,59]-->1
```

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```
            </startMinute>
            <endHour>
              <!--ro, opt, int, range:[0,23]-->1
            </endHour>
            <endMinute>
              <!--ro, opt, int, range:[0,59]-->1
            </endMinute>
          </timeSwitch>
        </timeSwitchList>
      </timeSchedule>
    </timeScheduleList>
    <dayNightStatus>
      <!--ro, opt, enum, subType:string-->day
    </dayNightStatus>
  </timeMode>
  <autoMode>
    <!--ro, opt, object, automatic switch-->
    <autoSwtichThresholdVal>
      <!--ro, opt, int, threshold for automatic switch, range:[0,100]-->1
    </autoSwtichThresholdVal>
    <autoNightSwtichThresholdVal>
      <!--ro, opt, int, range:[0,100]-->1
    </autoNightSwtichThresholdVal>
    <autoDayToNightDelayTime>
      <!--ro, opt, int, range:[0,120]-->1
    </autoDayToNightDelayTime>
    <autoNightToDayDelayTime>
      <!--ro, opt, int, range:[0,120]-->1
    </autoNightToDayDelayTime>
    <ICRChange>
      <!--ro, opt, bool-->false
    </ICRChange>
    <lightChange>
      <!--ro, opt, bool-->false
    </lightChange>
    <WDRChange>
      <!--ro, opt, bool-->false
    </WDRChange>
    <dayNightStatus>
      <!--ro, opt, enum, subType:string-->day
    </dayNightStatus>
  </autoMode>
 </nightModeCtrl>
</DayNightGate>
```

#### 14.2.7.3 Set the IR-cut filter parameters for cameras of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/icr

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ICR configuration parameters, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--opt, object, IR-cut filter-->
    <ICRCtrlMode>
      <!--req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-manually
switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ICRCtrlMode>
  </ICRCtrl>
</ImageIcrE>
```

**Response Message**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.7.4 Get the IR-cut filter parameters for cameras of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/icr

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ICR parameters configuration, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--ro, opt, object, IR-cut filter-->
    <ICRCtrlMode>
      <!--ro, req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-
manually switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ICRCtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,manual}-->
      <ManualPresetVal>
        <!--ro, req, enum, manually switch, subType:string, desc:"day", "night"-->day
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,time}-->
      <scheduleType>
        <!--ro, opt, enum, subType:string-->day
      </scheduleType>
      <SwitchList>
        <!--ro, opt, array, schedule list for switching, subType:object, range:[0,2], dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,day}, desc:up to
two time period control can be supported. For example: time period is from 6:00  to 18:00,the preset ID is 1-->
        <Preset>
          <!--ro, opt, object, preset-->
          <PresetId>
            <!--ro, req, int, preset ID-->1
          </PresetId>
          <PresetVal>
            <!--ro, opt, enum, "day","night", subType:string, desc:"day", "night"-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId>
            <!--ro, req, int, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--ro, opt, int, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
```

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```
            <endMinute>
              <!--ro, opt, int, range:[0,59]-->1
            </endMinute>
            <PresetVal>
              <!--ro, opt, enum, subType:string-->day
            </PresetVal>
          </TimeSwitch>
        </SwitchList>
        <dayNightStatus>
          <!--ro, opt, enum, subType:string-->day
        </dayNightStatus>
        <timeScheduleList>
          <!--ro, opt, array, subType:object, range:[0,2], dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,range}-->
          <timeSchedule>
            <!--ro, opt, object-->
            <triggerData>
              <!--ro, opt, object-->
              <startTime>
                <!--ro, opt, object-->
                <monthOfYear>
                  <!--ro, opt, int, range:[1,12]-->1
                </monthOfYear>
              </startTime>
              <endTime>
                <!--ro, opt, object-->
                <monthOfYear>
                  <!--ro, opt, int, range:[1,12]-->1
                </monthOfYear>
              </endTime>
            </triggerData>
            <timeSwitchList>
              <!--ro, opt, array, subType:object, range:[0,2]-->
              <timeSwitch>
                <!--ro, opt, object-->
                <timeID>
                  <!--ro, req, int, range:[0,1]-->1
                </timeID>
                <startHour>
                  <!--ro, opt, int, range:[0,23]-->1
                </startHour>
                <startMinute>
                  <!--ro, opt, int, range:[0,59]-->1
                </startMinute>
                <endHour>
                  <!--ro, opt, int, range:[0,23]-->1
                </endHour>
                <endMinute>
                  <!--ro, opt, int, range:[0,59]-->1
                </endMinute>
                <dayNightVal>
                  <!--ro, opt, enum, subType:string-->day
                </dayNightVal>
              </timeSwitch>
            </timeSwitchList>
          </timeSchedule>
        </timeScheduleList>
      </TimeMode>
      <AutoMode>
        <!--ro, opt, object, automatically switch according to the light change, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,auto}-->
        <DayNightFilterTh>
          <!--ro, req, int, day/night auto switch sensitivity, range:[0,100]-->1
        </DayNightFilterTh>
        <ICRAutoSwitch min="0" max="100">
          <!--ro, opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
        </ICRAutoSwitch>
        <dayNightStatus>
          <!--ro, opt, enum, subType:string-->day
        </dayNightStatus>
      </AutoMode>
    </ICRCtrl>
    <ECtrl>
      <!--ro, opt, object, -E device,halo inhibition filter,which can be used to inhibit the halo-->
      <ECtrlMode>
        <!--ro, req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-
manually switch, "time"-switch by schedule, "auto"-automatically switch-->close
      </ECtrlMode>
      <ManualMode>
        <!--ro, opt, object, manually switch, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,manual}-->
        <ManualPresetVal>
          <!--ro, req, int, manually switch, range:[0,100]-->1
        </ManualPresetVal>
      </ManualMode>
      <TimeMode>
        <!--ro, opt, object, switch by schedule, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,time}-->
        <SwitchList>
          <!--ro, opt, array, schedule list for switching, subType:object, range:[0,2], desc:up to two time period control can be supported. For example: time
period is from 6:00  to 18:00,the preset ID is 1-->
          <Preset>
            <!--ro, opt, object, preset-->
            <PresetId>
              <!--ro, req, int, preset ID-->1
            </PresetId>
            <PresetVal>
              <!--ro, opt, enum, "day", "night", subType:string, desc:"day", "night"-->day
```

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dojojin@gmail.com

```
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId>
            <!--ro, req, int, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--ro, opt, int, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </endMinute>
          <PresetVal>
            <!--ro, opt, enum, subType:string-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
    </TimeMode>
    <AutoMode>
      <!--ro, opt, object, automatically switch according to the light change, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,auto}-->
      <DayNightFilterTh>
        <!--ro, req, int, day/night auto switch sensitivity, range:[0,100]-->1
      </DayNightFilterTh>
      <ExceptionCatchTh>
        <!--ro, req, int, exception detection sensitivity, range:[0,100]-->1
      </ExceptionCatchTh>
    </AutoMode>
  </ECtrl>
</ImageIcrE>
```

#### 14.2.7.5 Get the IR-cut filter capability for cameras of a specific channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/icr/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ICR parameters configuration, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--ro, opt, object, IR-cut filter-->
    <ICRCtrlMode opt="close,manual,time,auto">
      <!--ro, req, string, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, attr:opt{req, string}-->test
    </ICRCtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch-->
      <ManualPresetVal opt="day,night">
        <!--ro, req, string, manually switch, attr:opt{req, string}-->test
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule-->
      <scheduleType opt="day,range">
        <!--ro, opt, string, attr:opt{req, string}-->day
      </scheduleType>
      <SwitchList size="2">
        <!--ro, opt, array, subType:object, dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,day}, attr:size{req, int}-->
        <Preset>
          <!--ro, req, object-->
          <PresetId min="0" max="1">
            <!--ro, req, int, attr:min{req, int},max{req, int}-->1
          </PresetId>
          <PresetVal opt="day,night">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
```

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```
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId min="0" max="1">
            <!--ro, req, int, attr:min{req, int},max{req, int}-->1
          </timeId>
          <startHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </startHour>
          <startMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </startMinute>
          <endHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </endHour>
          <endMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </endMinute>
          <PresetVal opt="day,night">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
      <timeScheduleList size="2">
        <!--ro, opt, object, dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,range}, attr:size{req, int}-->
        <timeSchedule>
          <!--ro, opt, object-->
          <triggerData>
            <!--ro, opt, object-->
            <startTime>
              <!--ro, opt, object-->
              <monthOfYear min="1" max="12">
                <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->0
              </monthOfYear>
            </startTime>
            <endTime>
              <!--ro, opt, object-->
              <monthOfYear min="1" max="12">
                <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->0
              </monthOfYear>
            </endTime>
          </triggerData>
          <timeSwitchList size="2">
            <!--ro, opt, object, attr:size{req, int}-->
            <timeSwitch>
              <!--ro, opt, object-->
              <timeID min="0" max="1">
                <!--ro, req, int, range:[0,1], attr:min{req, int},max{req, int}-->1
              </timeID>
              <startHour min="0" max="23">
                <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
              </startHour>
              <startMinute min="0" max="59">
                <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
              </startMinute>
              <endHour min="0" max="23">
                <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
              </endHour>
              <endMinute min="0" max="59">
                <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
              </endMinute>
              <dayNightVal opt="day,night">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
              </dayNightVal>
            </timeSwitch>
          </timeSwitchList>
        </timeSchedule>
      </timeScheduleList>
    </TimeMode>
    <AutoMode>
      <!--ro, opt, object, automatically switch according to the light change-->
      <DayNightFilterTh min="0" max="100">
        <!--ro, req, int, day/night auto switch sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
      </DayNightFilterTh>
      <ICRAutoSwitch min="0" max="100">
        <!--ro, opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
      </ICRAutoSwitch>
    </AutoMode>
  </ICRCtrl>
  <ECtrl>
    <!--ro, opt, object, -E device,halo inhibition filter,which can be used to inhibit the halo-->
    <ECtrlMode opt="close,manual,time,auto">
      <!--ro, req, string, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, attr:opt{req, string}-->test
    </ECtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch-->
      <ManualPresetVal min="0" max="100">
        <!--ro, req, int, manually switch, range:[0,100], attr:min{req, int},max{req, int}-->1
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule-->
      <SwitchList size="2">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <Preset>
```

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```
            <!--ro, req, object-->
            <PresetId min="0" max="1">
              <!--ro, req, int, attr:min{req, int},max{req, int}-->1
            </PresetId>
            <PresetVal opt="day,night">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
            </PresetVal>
          </Preset>
          <TimeSwitch>
            <!--ro, opt, object-->
            <timeId min="0" max="1">
              <!--ro, req, int, attr:min{req, int},max{req, int}-->1
            </timeId>
            <startHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </startHour>
            <startMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </startMinute>
            <endHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </endHour>
            <endMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </endMinute>
            <PresetVal opt="day,night">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
            </PresetVal>
          </TimeSwitch>
        </SwitchList>
      </TimeMode>
      <AutoMode>
        <!--ro, opt, object, automatically switch according to the light change-->
        <DayNightFilterTh min="0" max="100">
          <!--ro, req, int, day/night auto switch sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
        </DayNightFilterTh>
        <ExceptionCatchTh min="0" max="100">
          <!--ro, req, int, "0-100", range:[0,100], attr:min{req, int},max{req, int}-->1
        </ExceptionCatchTh>
      </AutoMode>
    </ECtrl>
  </ImageIcrE>
```

#### 14.2.7.6 Set the mounting scenario mode of a camera

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/mountingScenario

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","LowIllumination"-Low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor“, “outdoor”, “day”, “night”, “morning”, “nightfall”, “mode1”, “mode2”, “mode3”, “mode4”, “street”, “LowIllumination”-Low illumination, “custom1”, “custom2”, “normal”, “road”, “faceSnap”-face picture capture, “highway”, “road2”, “backlight”-back light, “frontlight”-front light, “facenight”, “fastRoadAndLight”, “fastRoadNoLihgt”, “normalRoadAndLight”, “normalRoadNoLight”, “cityRoadAndBright”-->indoor
  </mode>
</MountingScenario>
```

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.7.7 Get the capability of configurations of mounting scenario mode

#### Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario/capabilities

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR,sportSnap,dustMoisture,rampWay">
    <!--ro, req, enum, Mode, subType:string, attr:opt{req, string}, desc:"lowIllumination"-Low illumination, "backlight"-back light, "frontlight"-front
light, "faceSnap"-face picture capture-->indoor
  </mode>
  <isSupportRecommendation>
    <!--ro, opt, bool-->true
  </isSupportRecommendation>
  <modeNameList size="2">
    <!--ro, opt, object, attr:size{req, int}-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

### 14.2.7.8 Get the mounting scenario mode of a camera

#### Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario?parameterType=<parameterType>

#### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| parameterType | string | Parameter type. When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured, it will be the value of the parameter configured on the device. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-
Low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor”, “outdoor”, “day”, “night”,
“morning”, “nightfall”, “mode1”, “mode2”,“ mode3”, “mode4”, “highway”, “road2”, “faceSnap”, “backlight”, “frontlight”, “lowIllumination”-low illumination,
“backlight”-back light, “frontlight”-front light, “faceSnap”-face picture capture-->indoor
  </mode>
  <modeNameList>
    <!--ro, opt, array, subType:object-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode>
        <!--ro, req, enum, subType:string-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

## 14.2.8 Manual Capture

### 14.2.8.1 Get the manual capture result information of a specified channel

**Request URL**

GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>&maxSnapResolution=<maxSnapResolution>&livingBodyDetect=<livingBodyDetect>

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| trackStreamID | string | Channel No. * 100 + stream type (-1: main stream, -2: sub-stream). |
| snapShotImageType | string | Image formats: "bayer" (image not processed by ISP, which can be used to test whether the image clarity of the factory equipment meets the standard), "JPEGWITHRULE" (this type of image is specially for HIKMICRO devices, indicating a JPEG image captured with temperature rule information overlaid), "JPEGWITHOUTRULE" (when this type is used by HIKMICRO devices, it indicates a JPEG image captured without temperature rule information overlaid), "JPEG" (when this type is used by HIKMICRO devices, whether the captured JPEG image includes rule information depends on the enabling status of overlay temperature information on stream function. If it is enabled, the image returned will have rule information overlaid; otherwise, it will not. If the HIKMICRO device calls the URL without using this parameter, the logic for the returned JPEG image is consistent with the above JPEG type, depending on the enabling status of overlay temperature information on stream function). |
| videoResolutionWidth | string | Resolution width, and if not configured, it is 704 by default (D1 resolution width). |
| videoResolutionHeight | string | Resolution height, and if not configured, it is 576 by default (D1 resolution height). |
| imageQuality | string | Image quality: "best", "better", "general" |
| x | string | x-coordinate of the specified area, the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen |
| y | string | y-coordinate of the specified area, the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen |
| hight | string | Height of the specified area. The value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen. |
| width | string | Width of the specified area. The value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen. |
| maxSnapResolution | enum | Capture images using the maximum image capture resolution. For certain devices, such as low-power devices, the actual maximum image capture resolution differs from the video encoding resolution. During certain intelligent detections, images are captured at the maximum image capture resolution. If not configured, the default value is 1. |
| livingBodyDetect | enum | Whether to perform face anti-spoofing detection when capturing images. |

**Request Message**

None

**Response Message**

Binary Data

## 14.2.9 Video OSD Information Overlay

### 14.2.9.1 Get the OSD parameters of a specified channel

**Request URL**

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, screen size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, whether it is flickering-->true
    </flashing>
    <backgroundColor>
      <!--ro, opt, object, background color-->
      <enabled>
        <!--ro, req, bool, whether to enable the background color-->true
      </enabled>
      <color>
        <!--ro, opt, string, background color RGB, desc:hexadecimal digit-->000000
      </color>
      <transparency>
        <!--ro, opt, int, background color transparency, range:[0,100]-->50
      </transparency>
    </backgroundColor>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--ro, opt, object, text content-->
      <id>
        <!--ro, req, string, text No.-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--ro, req, string, overlay text-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool, whether the overlaid text remains the previous settings after device rebooting, desc:if the value is true or the node is not
transmitted, it indicates to write OSD to flash, and it is not suitable for frequent or fast callings. If the value is false, it indicates not to writer OSD
to flash and the OSD settings will not restore-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object, channel time information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <displayWeek>
      <!--ro, opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object, channel name overlay-->
```

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```
    <!--ro, opt, object, channel name overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--ro, opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--ro, opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <alignment>
    <!--ro, opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
  </alignment>
  <publicSecurity>
    <!--ro, opt, bool-->true
  </publicSecurity>
  <boundary>
    <!--ro, opt, int, the minimum boundary, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->1
  </boundary>
  <upDownboundary>
    <!--ro, opt, int, range:[0,2]-->1
  </upDownboundary>
  <PackingSpaceRecognitionOverlay>
    <!--ro, opt, object, display of parking space detection result-->
    <isParkedEnabled>
      <!--ro, opt, bool, whether to display the parking space status (occupied or free)-->true
    </isParkedEnabled>
  </PackingSpaceRecognitionOverlay>
  <VisibilityDisplay>
    <!--ro, opt, object, visibility information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable. If it is enabled, the visibility information will be overlaid-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </VisibilityDisplay>
</VideoOverlay>
```

#### 14.2.9.2 Set OSD parameters of a specific video input channel

**Request URL**

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--req, object, screen size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt, object, attributes-->
  </attribute>
  <TextOverlayList>
    <!--opt, array, text content list, subType:object-->
```

VISION AND SECURITY SYSTEM CO., LTD. dajiajin@gmail.com

```
    <!--opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--opt, object, text content-->
      <id>
        <!--req, string, text No.-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--req, string, overlay text-->test
      </displayText>
      <isPersistentText>
        <!--opt, bool, whether the overlaid text remains the previous settings after device rebooting, desc:if the value is true or the node is not
transmitted, it indicates to write OSD to flash, and it is not suitable for frequent or fast callings. If the value is false, it indicates not to writer OSD
to flash and the OSD settings will not restore-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--opt, object, channel time information overlay-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <displayWeek>
      <!--opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--opt, object, channel name overlay-->
    <enabled>
      <!--req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <alignment>
    <!--opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
  </alignment>
  <publicSecurity>
    <!--opt, bool-->true
  </publicSecurity>
  <boundary>
    <!--opt, int, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->000000
  </boundary>
  <upDownboundary>
    <!--opt, int, range:[0,2]-->1
  </upDownboundary>
</VideoOverlay>
```

## Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, it is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

### 14.2.9.3 Get the OSD capability of a specific video input channel

#### Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_BatteryPowerOverlay, attr:version{req, string, protocolVersion}-->
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent opt="true,false">
      <!--ro, req, bool, whether it is transparent, attr:opt{req, string}-->true
    </transparent>
    <flashing opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </flashing>
    <backgroundColor>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <color>
        <!--ro, opt, string-->000000
      </color>
      <transparency min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->50
      </transparency>
    </backgroundColor>
  </attribute>
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <TextOverlay>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, listening host ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayText min="0" max="64">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </displayText>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

```
        <isPersistentText>
          <!--ro, opt, bool-->true
        </isPersistentText>
      </TextOverlay>
    </TextOverlayList>
    <DateTimeOverlay>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayWeek opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </displayWeek>
    </DateTimeOverlay>
    <channelNameOverlay>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <positionX>
        <!--ro, opt, int, range:[0,1920]-->1
      </positionX>
      <positionY>
        <!--ro, opt, int, range:[0,1920]-->1
      </positionY>
      <name min="0" max="128">
        <!--ro, opt, string, range:[0,128], attr:min{req, int},max{req, int}-->Camera 01
      </name>
    </channelNameOverlay>
    <fontSize opt="16*16,32*32,48*48,64*64,80*80,96*96,128*128,adaptive">
      <!--ro, opt, string, font size, attr:opt{req, string}-->adaptive
    </fontSize>
    <frontColorMode opt="auto,customize,outline" def="auto">
      <!--ro, opt, string, font color mode, attr:opt{req, string},def{opt, string}-->outline
    </frontColorMode>
    <frontColor>
      <!--ro, opt, string, font color, desc:font color-->000000
    </frontColor>
    <alignment opt="customize,alignRight,alignLeft,GB,GB2025,allRight,allLeft,adsorption">
      <!--ro, opt, string, alignment mode, attr:opt{req, string}-->customize
    </alignment>
    <lowRightTextNumber min="1" max="6">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </lowRightTextNumber>
    <lowLeftTextNumber min="7" max="8">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->7
    </lowLeftTextNumber>
    <boundary min="0" max="2">
      <!--ro, opt, int, dep:and,{$.VideoOverlay.alignment,eq,GB}, attr:min{req, int},max{req, int}-->1
    </boundary>
    <upDownboundary min="0" max="5">
      <!--ro, opt, int, attr:min{opt, int},max{opt, int}-->1
    </upDownboundary>
  </VideoOverlay>
```

#### 14.2.9.4 Get the channel name overlay parameters of a specified video input channel

**Request URL**

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, channel name overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--ro, opt, string, channel name, range:[0,128]-->Camera 01
  </name>
  <OverlayRegion>
    <!--ro, opt, object, coordinates of overlaid string frame (the origin is the is the upper-left corner of the screen)-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

### 14.2.9.5 Set the channel name overlay parameters for a specified video input channel

#### Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--opt, string, channel name, range:[0,32]-->Camera 01
  </name>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

#### Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.9.6 Set the parameters of date and time information overlay for a specified video input channel

#### Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, configurations of date and time information overlay of video input channels, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable date and time information overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, time format, subType:string, desc:"12hour" (12-hour system), "24hour" (24-hour system)-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--opt, bool, whether to display date-->true
  </displayDate>
  <displayTime>
    <!--opt, bool, whether to display time-->true
  </displayTime>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

#### Response Message

dojo jin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.9.7 Get the parameters of date and time information overlay of a specific channel

#### Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable date and time overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlay area, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlay area, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY"--
>YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--ro, opt, bool-->true
  </displayDate>
  <displayTime>
    <!--ro, opt, bool-->true
  </displayTime>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

### 14.2.9.8 Set the text overlay parameters of a specific video input channel

#### Request URL

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VISION AND SECURITY SYSTEM CO.,LTD.

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_TextOverlay, subType:object, attr:version{opt, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, text No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate-->1
    </positionY>
    <displayText>
      <!--req, string, overlay character-->test
    </displayText>
    <OverlayRegion>
      <!--opt, object-->
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

## 14.2.9.9 Get parameters of text information overlay of a specified video input channel

### Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, text information overlay, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object, text overlay configuration-->
    <id>
      <!--ro, req, string, text No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the
screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the
screen-->1
    </positionY>
    <displayText>
      <!--ro, req, string, overlaid text-->test
    </displayText>
    <OverlayRegion>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

## 14.2.9.10 Add a text to overlay on the video for a specific video input channel

### Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, text overlay-->test
  </displayText>
</TextOverlay>
```

### Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

### 14.2.9.11 Set the parameters of a specific piece of text overlay information of a specific video input channel

#### Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| textID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, text No.-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlay character-->test
  </displayText>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

#### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.2.9.12 Get the text overlay parameters of a specified video input channel

**Request URL**

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| textID | string | Text overlay ID |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay configuration, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, text No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-
->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-
->1
  </positionY>
  <displayText>
    <!--ro, req, string, overlaid text-->test
  </displayText>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

## 14.2.10 Video Privacy Mask

#### 14.2.10.1 Get the privacy mask configuration capability of a specified channel

**Request URL**

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/privacyMaskCap

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### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the privacy mask configuration capability, attr:version{req, string, protocolVersion}-->
  <maskTypeDscriptor opt="gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic,black,matchman,groundGlass,custom">
    <!--ro, opt, enum, mask type, subType:string, attr:opt{req, string}, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-
transparent", "mosaic", "black"-->gray
  </maskTypeDscriptor>
  <privacymaskName min="0" max="16">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </privacymaskName>
  <regionNum min="0" max="8">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->8
  </regionNum>
  <minZoomdoorlimit>
    <!--ro, opt, int, the minimum zooming in limit-->1
  </minZoomdoorlimit>
  <maxZoomdoorlimit>
    <!--ro, opt, int, the maximum zooming in limit-->1
  </maxZoomdoorlimit>
  <videoPrivacyType opt="privacyMask,privacyCover">
    <!--ro, opt, enum, privacy type, subType:string, attr:opt{req, string}, desc:"privacyMask"-privacy mask, "privacyCover"-video tampering-->privacyMask
  </videoPrivacyType>
  <supportRegionType>
    <!--ro, opt, enum, the default type of area drawing, subType:string, desc:"rectangle". If this node is not supported, the area is rectangular for
privacy mask, and is polygon for video tampering by default-->rectangle
  </supportRegionType>
  <mediaObjectEncryption>
    <!--ro, opt, object, encryption of the private target in media data, desc:if it is enabled, the main steam of corresponding area will be covered, but
the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted area if this node is not returned, it indicates that device does not support this function-->
    <enabled opt="true,false" def="false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string},def{req, string}-->true
    </enabled>
    <maskType opt="mosaic" def="mosaic">
      <!--ro, opt, string, mask type, attr:opt{req, string},def{req, string}, desc:"mosaic"-->mosaic
    </maskType>
  </mediaObjectEncryption>
  <polygonCoordinatesNum>
    <!--ro, opt, int, dep:and,{$.PrivacyMaskCap.supportRegionType,eq,polygon}-->1
  </polygonCoordinatesNum>
  <regionRangeType opt="fullScreen,region">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->region
  </regionRangeType>
  <privacyMaskRegionListNum>
    <!--ro, opt, int, range:[0,8]-->4
  </privacyMaskRegionListNum>
  <switch opt="local,remote">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->local
  </switch>
  <isSupportUpdateBackgroundPicture>
    <!--ro, opt, bool-->true
  </isSupportUpdateBackgroundPicture>
  <isSupportHumanBodyPrivacyMaskParams>
    <!--ro, opt, bool-->true
  </isSupportHumanBodyPrivacyMaskParams>
  <isSupportHeadShoulderPrivacyMaskParams>
    <!--ro, opt, bool-->true
  </isSupportHeadShoulderPrivacyMaskParams>
  <isSupportPlatePrivacyMaskParams>
    <!--ro, opt, bool-->true
  </isSupportPlatePrivacyMaskParams>
  <isSupportImportPrivacyMaskPicture>
    <!--ro, opt, bool-->true
  </isSupportImportPrivacyMaskPicture>
</PrivacyMaskCap>
```

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### 14.2.10.2 Get the parameters of all privacy mask areas of a specified channel

### Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--ro, opt, object, configurations of a privacy mask area-->
    <id>
      <!--ro, req, int, area ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--ro, opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mediaObjectEncryption>
      <!--ro, opt, object, encryption of the private target in media data, desc:it is disabled when this field is not returned. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
      </maskType>
    </mediaObjectEncryption>
    <regionRangeType>
      <!--ro, opt, enum, subType:string-->region
    </regionRangeType>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

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### 14.2.10.3 Set the parameters of all privacy mask areas of a specified channel

**Request URL**

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--opt, object, configurations of a privacy mask area-->
    <id>
      <!--req, int, area ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--req, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mediaObjectEncryption>
      <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
      </maskType>
    </mediaObjectEncryption>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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### 14.2.10.4 Get the privacy mask parameters of a specific channel

### Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| switchType | string | Analysis unit switch: "local" (analysis by local device, e.g., executed locally by NVR), "remote" (the device transmits data to sub-device for analysis, e.g., from the NVR to the network camera). |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--ro, opt, array, list of video tampering region(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--ro, opt, object, configurations of a privacy mask region-->
      <id>
        <!--ro, req, int, region ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the region-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--ro, opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--ro, opt, object, encryption of the private target in media data, desc:If it is enabled, the main steam of corresponding region will be covered,
but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted region.-->
        <enabled>
          <!--ro, req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
      <regionRangeType>
        <!--ro, opt, enum, subType:string-->region
      </regionRangeType>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--ro, opt, enum, region type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

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### 14.2.10.5 Set the privacy mask parameters of a specified channel

#### Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| switchType | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--opt, array, list of video privacy mask area(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--opt, object, configurations of a privacy mask area-->
      <id>
        <!--req, int, area ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--opt, int, zooming value limit range, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
        <enabled>
          <!--req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

#### Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.2.11 Privacy Mask

### 14.2.11.1 Video tampering alarm

**EventType:shelteralarm**

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event details, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 Address-->10.10.10.10
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->ABCD:EF01:2345:6789:ABCD:EF01:2345:6789
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP, HTTPS, EHome". The value should be "HTTP" when ISAPI protocol
is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP.-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type: shelteralarm, subType:string, desc:event type: shelteralarm-->shelteralarm
  </eventType>
  <eventState>
    <!--ro, req, enum, event status (for persistent event), subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID-->test
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, req, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <Image>
    <!--ro, opt, object, event related pictures-->
    <resourcesContentType>
      <!--ro, opt, enum, resource transmission type, subType:string, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the
returned alarm message contains additional resource (e.g., pictures), this field should be returned.-->binary
    </resourcesContentType>
    <resourcesContent>
      <!--ro, opt, string, resource ID, range:[1,128], desc:when the returned alarm message contains additional resources (e.g., pictures), this field
should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of
resourcesContentType is "url", its value is the resource URL-->image
    </resourcesContent>
  </Image>
</EventNotificationAlert>
```

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| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| EventNotificationAlert | [Message content] | application/xml | -- | -- | -- |
| image | [Binary picture data] | image/jpeg | image | image.jpg | -- |

**Note:** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

## 14.2.12 Image White Balance

### 14.2.12.1 Set the white balance parameters of s specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/whiteBalance

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--opt, enum, white balance type, subType:string, desc:"auto", "atuo1", "auto2" "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight",
"mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceBlue>
</WhiteBalance>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.2.12.2 Get the white balance parameters of s specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/whiteBalance?parameterType=<parameterType>

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**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| parameterType | string | When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--ro, opt, enum, "auto,manual,indoor,outdoor,autotrace,onece,sodiumlight,mercurylight,auto0,atuo1,fluorescent,natural,warm,incandescent",
subType:string, desc:"auto", "auto1", "atuo2", "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight", "mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceBlue>
  <whiteBalanceLevel>
    <!--ro, opt, int, range:[0,100], dep:or,{$.WhiteBalance.WhiteBalanceStyle,eq,auto},{$.WhiteBalance.WhiteBalanceStyle,eq,auto1},
{$.WhiteBalance.WhiteBalanceStyle,eq,auto2}-->1
  </whiteBalanceLevel>
  <customColorTemperatureLevel>
    <!--ro, opt, int, range:[1800,10000]-->1
  </customColorTemperatureLevel>
  <WhiteBalanceRedFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceRedFineTuning>
  <WhiteBalanceBlueFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceBlueFineTuning>
  <autoMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,auto}-->video
  </autoMode>
  <manualMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,manual}-->colorFineTuning
  </manualMode>
</WhiteBalance>
```

# 14.3 Video Management

## 14.3.1 Scheduled Recording

### 14.3.1.1 Add a recording schedule

**Request URL**

POST /ISAPI/ContentMgmt/record/tracks

**Query Parameter**

None

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search all recording schedules, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream
of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should
be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--req, int, channel No.-->1
  </Channel>
```

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```
  </Channel>
  <Enable>
    <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20
fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--opt, int, recording size-->1
  </Size>
  <Duration min="0" max="750">
    <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours; 0
indicates the video is valid until it is recurrently overwritten-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
    <!--req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture
alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures),
"AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement"
(temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm),
"zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to
vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed
detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--opt, bool, whether to support recurrently overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--opt, object, video source description-->
    <SrcGUID>
      <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers
to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream
of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcUrlMethods>
      <!--req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--req, string, login source password-->admin:admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--opt, object, scheduled start time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--opt, object, scheduled end time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--opt, time, time-->08:00:00
            </TimeOfDay>
```

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```
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--req, string, Description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
            </ActionRecordingMode>
            <PreRecordTimeSeconds>
              <!--opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds>
              <!--opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
            </PostRecordTimeSeconds>
          </Actions>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--opt, object, custom extension-->
      <CustomExtensionName>
        <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <SaveAudio>
        <!--opt, bool, whether to enable audio recording-->true
      </SaveAudio>
      <PreRecordTimeSeconds>
        <!--opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
      <HolidaySchedule>
        <!--opt, object, holiday schedule-->
        <ScheduleBlock>
          <!--opt, object, schedule module-->
          <ScheduleBlockGUID>
            <!--req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
          </ScheduleBlockType>
        </ScheduleBlock>
      </HolidaySchedule>
    </CustomExtension>
  </CustomExtensionList>
  <IntelligentRecord>
    <!--opt, bool, whether to enable VCA recording function-->false
  </IntelligentRecord>
  <delayTime>
    <!--opt, int, capture delay time, unit:s-->1
  </delayTime>
</Track>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.3.1.2 Get all parameters of a recording schedule

**Request URL**

GET /ISAPI/ContentMgmt/record/tracks

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TrackList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, get all parameters of a recording schedule, subType:object, attr:version{req, string, protocolVersion}-->
  <Track>
    <!--ro, opt, object, recording schedule configuration-->
    <id>
      <!--ro, opt, int, channel number corresponding to stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main
stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type
should be subject to that of SrcUrl-->201
    </id>
    <Channel>
      <!--ro, req, int, channel No.-->1
    </Channel>
    <Enable>
      <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
    </Enable>
    <Description>
      <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-
SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
    </Description>
    <TrackGUID>
      <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
    </TrackGUID>
    <Size>
      <!--ro, opt, int, recording size-->1
    </Size>
    <Duration min="0" max="750">
      <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15
hours; 0 indicates the video is valid until it is recurrently overwritten-->P10DT15H
    </Duration>
    <DefaultRecordingMode>
      <!--ro, req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM"
(capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording),
"temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by
default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm"
(panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from
vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp"
(get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
    </DefaultRecordingMode>
    <LoopEnable>
      <!--ro, opt, bool, whether to support recurrently overwriting-->true
    </LoopEnable>
    <SrcDescriptor>
      <!--ro, opt, object, video source description-->
      <SrcGUID>
```

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```
        <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
      </SrcGUID>
      <SrcChannel>
        <!--ro, req, int, source channel-->1
      </SrcChannel>
      <StreamHint>
        <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
      </StreamHint>
      <SrcDriver>
        <!--ro, req, string, stream executable driver name-->RTP/RTSP
      </SrcDriver>
      <SrcType>
        <!--ro, opt, string, source type-->mp4 video
      </SrcType>
      <SrcUrl>
        <!--ro, req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID>
refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main
stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
      </SrcUrl>
      <SrcUrlMethods>
        <!--ro, req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
      </SrcUrlMethods>
      <SrcLogin>
        <!--ro, req, string, login source password-->admin:admin
      </SrcLogin>
    </SrcDescriptor>
    <TrackSchedule>
      <!--ro, opt, object, recording schedule-->
      <ScheduleBlockList>
        <!--ro, opt, array, recording schedule list, subType:object-->
        <ScheduleBlock>
          <!--ro, opt, object, recording schedule module-->
          <ScheduleBlockGUID>
            <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
          </ScheduleBlockType>
          <ScheduleAction>
            <!--ro, opt, array, schedule, subType:object, desc:N/A-->
            <id>
              <!--ro, req, int, ID-->1
            </id>
            <ScheduleActionStartTime>
              <!--ro, opt, object, scheduled start time-->
              <DayOfWeek>
                <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
              </DayOfWeek>
              <TimeOfDay>
                <!--ro, opt, time, time-->08:00:00
              </TimeOfDay>
            </ScheduleActionStartTime>
            <ScheduleActionEndTime>
              <!--ro, opt, object, scheduled end time-->
              <DayOfWeek>
                <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
              </DayOfWeek>
              <TimeOfDay>
                <!--ro, opt, time, time-->08:00:00
              </TimeOfDay>
            </ScheduleActionEndTime>
            <ScheduleDSTEnable>
              <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
            </ScheduleDSTEnable>
            <Description>
              <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
            </Description>
            <Actions>
              <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
              <Record>
                <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </Record>
              <Log>
                <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </Log>
              <SaveImg>
                <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </SaveImg>
              <ActionRecordingMode>
                <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "Linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
              </ActionRecordingMode>
              <PreRecordTimeSeconds>
                <!--ro, opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
              </PreRecordTimeSeconds>
              <PostRecordTimeSeconds>
                <!--ro, opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
              </PostRecordTimeSeconds>
            </Actions>
```

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```
          </ScheduleAction>
        </ScheduleBlock>
      </ScheduleBlockList>
    </TrackSchedule>
    <CustomExtensionList>
      <!--ro, opt, array, custom extension list, subType:object-->
      <CustomExtension>
        <!--ro, opt, object, custom extension-->
        <CustomExtensionName>
          <!--ro, opt, string, custom extension name-->www.xxxx.com/RaCM/trackExt/ver10
        </CustomExtensionName>
        <enableSchedule>
          <!--ro, opt, bool, whether to enable recording schedule configuration-->true
        </enableSchedule>
        <SaveAudio>
          <!--ro, opt, bool, whether to enable audio recording-->true
        </SaveAudio>
        <PreRecordTimeSeconds>
          <!--ro, opt, int, pre-record, unit:s-->1
        </PreRecordTimeSeconds>
        <PostRecordTimeSeconds>
          <!--ro, opt, int, post-record, unit:s-->1
        </PostRecordTimeSeconds>
        <HolidaySchedule>
          <!--ro, opt, object, holiday schedule-->
          <ScheduleBlock>
            <!--ro, opt, object, schedule module-->
            <ScheduleBlockGUID>
              <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
            </ScheduleBlockGUID>
            <ScheduleBlockType>
              <!--ro, opt, string, schedule module type-->www.xxxx.com/racm/schedule/ver10
            </ScheduleBlockType>
          </ScheduleBlock>
        </HolidaySchedule>
      </CustomExtension>
    </CustomExtensionList>
    <IntelligentRecord>
      <!--ro, opt, bool, whether to enable VCA recording function-->false
    </IntelligentRecord>
    <delayTime>
      <!--ro, opt, int, capture delay time, unit:s-->1
    </delayTime>
    <durationEnabled>
      <!--ro, opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time
via the node "Duration" when it is supported-->true
    </durationEnabled>
    <redundancyRec>
      <!--ro, opt, bool, whether it is redundant recording, desc:it is false when this node does not exist-->false
    </redundancyRec>
    <passbackRecord>
      <!--ro, opt, bool, whether to copy back recording, desc:it is false when this node does not exist-->false
    </passbackRecord>
    <lockDuration>
      <!--ro, opt, int, recording locking duration, unit:h, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording
is longer than the locked part, it will not be locked-->0
    </lockDuration>
    <recordBackup>
      <!--ro, opt, bool, whether to enable video archiving, desc:it is false when this node does not exist-->false
    </recordBackup>
    <SVCLevel>
      <!--ro, opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted or this node does not exist, 1-half extracted, 2-three quarters
extracted-->0
    </SVCLevel>
    <recordManage>
      <!--ro, opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording
schedule is still in use, e.g., motion detection and copy-back); it is false when this node does not exist-->false
    </recordManage>
    <extraSaveAudio>
      <!--ro, opt, bool, whether to enable separate audio storage, desc:it is false when this node does not exist-->false
    </extraSaveAudio>
  </Track>
</TrackList>
```

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#### 14.3.1.3 Set all recording schedules

**Request URL**

PUT /ISAPI/ContentMgmt/record/tracks

**Query Parameter**

None

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TrackList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```
<!--req, array, all recording schedules parameters, subType:object, attr:version{req, string, protocolVersion}-->
<Track>
  <!--opt, object, recording schedule configuration-->
  <id>
    <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--opt, int, video size-->1
  </Size>
  <Duration min="0" max="750">
    <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours; 0 indicates the video is valid until it is recurrently overwritten-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
    <!--req, enum, default recording type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking), "parallelParkingContinuousRecord" (from vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--opt, bool, whether to support recurrently overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--opt, object, video source description-->
    <SrcGUID>
      <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcUrlMethods>
      <!--req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--req, string, login source password-->admin:admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--opt, object, scheduled start time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
```

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```
            <!--opt, time, time-->08:00:00
          </TimeOfDay>
        </ScheduleActionStartTime>
        <ScheduleActionEndTime>
          <!--opt, object, scheduled end time-->
          <DayOfWeek>
            <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
          </DayOfWeek>
          <TimeOfDay>
            <!--opt, time, time-->08:00:00
          </TimeOfDay>
        </ScheduleActionEndTime>
        <ScheduleDSTEnable>
          <!--req, bool, whether to enable Daylight Saving Time (DST)-->true
        </ScheduleDSTEnable>
        <Description>
          <!--req, string, description-->PreMorning (Midnight to 8am,local time)
        </Description>
        <Actions>
          <!--opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
          <Record>
            <!--opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
          </Record>
          <Log>
            <!--opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
          </Log>
          <SaveImg>
            <!--opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
          </SaveImg>
          <ActionRecordingMode>
            <!--req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
          </ActionRecordingMode>
          <PreRecordTimeSeconds>
            <!--opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
          </PreRecordTimeSeconds>
          <PostRecordTimeSeconds>
            <!--opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
          </PostRecordTimeSeconds>
        </Actions>
      </ScheduleAction>
    </ScheduleBlock>
  </ScheduleBlockList>
</TrackSchedule>
<CustomExtensionList>
  <!--opt, array, custom extension list, subType:object-->
  <CustomExtension>
    <!--opt, object, custom extension-->
    <CustomExtensionName>
      <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
    </CustomExtensionName>
    <enableSchedule>
      <!--opt, bool, whether to enable recording schedule configuration-->true
    </enableSchedule>
    <SaveAudio>
      <!--opt, bool, whether to enable audio recording-->true
    </SaveAudio>
    <PreRecordTimeSeconds>
      <!--opt, int, pre-record, unit:s-->1
    </PreRecordTimeSeconds>
    <PostRecordTimeSeconds>
      <!--opt, int, post-record, unit:s-->1
    </PostRecordTimeSeconds>
    <HolidaySchedule>
      <!--opt, object, holiday schedule-->
      <ScheduleBlock>
        <!--opt, object, schedule module-->
        <ScheduleBlockGUID>
          <!--req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
        </ScheduleBlockType>
      </ScheduleBlock>
    </HolidaySchedule>
  </CustomExtension>
</CustomExtensionList>
<IntelligentRecord>
  <!--opt, bool, whether to enable VCA recording function-->false
</IntelligentRecord>
<delayTime>
  <!--opt, int, capture delay time, unit:s-->1
</delayTime>
<durationEnabled>
  <!--opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via
the node Duration when it is supported-->true
</durationEnabled>
<redundancyRec>
  <!--opt, bool, whether it is redundant recording, desc:it is false when this node does not exist-->false
```

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```
      <!--opt, bool, whether it is redundant recording, desc:it is false when this node does not exist-->false
    </redundancyRec>
    <passbackRecord>
      <!--opt, bool, whether to copy back recording, desc:it is false when this node does not exist-->false
    </passbackRecord>
    <lockDuration>
      <!--opt, int, recording locking duration, unit:h, desc:0-unlocked or this Field does not exist, 0xffff-locked permanently; if the total recording is longer than the locked part, it will not be locked-->0
    </lockDuration>
    <recordBackup>
      <!--opt, bool, whether to enable video archiving, desc:it is false when this node does not exist-->false
    </recordBackup>
    <SVCLevel>
      <!--opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted or this node does not exist, 1-half extracted, 2-three quarters extracted-->0
    </SVCLevel>
    <recordManage>
      <!--opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording schedule is still in use, e.g., motion detection and copy-back); it is false when this node does not exist-->false
    </recordManage>
    <extraSaveAudio>
      <!--opt, bool, whether to enable separate audio storage, desc:it is false when this node does not exist-->false
    </extraSaveAudio>
  </Track>
</TrackList>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

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## 14.3.1.4 Get the configuration capability of the recording schedule

### Request URL

GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/capabilities

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| trackStreamID | string | Channel number corresponding to the stream type, for example, 101 indicates main stream of channel 1. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of the recording schedule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 1, 202 refers to the sub-stream of channel 2, and so on-->201
  </id>
  <Channel>
    <!--ro, req, int, channel No.-->1
  </Channel>
```

```
  <Enable>
    <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20
fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--ro, opt, int, recording size-->1
  </Size>
  <Duration min="0" max="750">
    <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15
hours-->P10DT15H
  </Duration>
  <DefaultRecordingMode
opt="CMR,MOTION,ALARM,EDR,ALARMANDMOTION,AllEvent,POS,PECCANCY,SMART,temperatureIntervalMeasurement,methaneConcentrationException,thermalVehicleDetection,fi
shingShipDetection,consultation,panicAlarm,zoneAlarm,parallelParking,parallelParkingContinuousRecord,linedetection,toiletTarry,getUp,retention,failDown">
    <!--ro, req, enum, default recording type, subType:string, attr:opt{req, string}, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection
pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and
motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording),
"temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by
default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm"
(panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from
vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp"
(get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--ro, opt, bool, whether to support recurrent overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--ro, opt, object, video source description-->
    <SrcGUID>
      <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--ro, req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--ro, req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--ro, opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl
opt="rtsp://localhost/PSIA/Streaming/channels/101,rtsp://localhost/PSIA/Streaming/channels/102,rtsp://localhost/PSIA/Streaming/channels/103,rtsp://localhost
/PSIA/Streaming/channels/107">
      <!--ro, req, string, source URL, attr:opt{req, string}-->rtsp://10.3.2.26/mpeg4/media.amp
    </SrcUrl>
    <SrcUrlMethods>
      <!--ro, req, string, methods supported by the source, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--ro, req, string, login source password, desc:user name for source channel RTSP authentication returned fby device; encryption required for
sensitive information-->admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--ro, opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--ro, opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--ro, opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--ro, opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--ro, req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--ro, opt, object, scheduled start time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--ro, opt, object, scheduled end time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
```

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```
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "Command" (recording by commands), "SMART" (smart recording), "AllEvent" (all events), "temperatureIntervalMeasurement" (temperature range detection), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
            </ActionRecordingMode>
            <PreRecordTimeSeconds min="0" max="10" def="5">
              <!--ro, opt, int, pre-record duration, unit:s, attr:min{req, int},max{req, int},def{req, int}, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds min="30" max="600" def="30">
              <!--ro, opt, int, post-record duration, unit:s, attr:min{req, int},max{req, int},def{req, int}, desc:from 30 to 600 seconds; default: 30 seconds-->30
            </PostRecordTimeSeconds>
          </Actions>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--ro, opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--ro, opt, object, custom extension-->
      <CustomExtensionName>
        <!--ro, opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--ro, opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <SaveAudio>
        <!--ro, opt, bool, whether to enable audio recording-->true
      </SaveAudio>
      <PreRecordTimeSeconds>
        <!--ro, opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--ro, opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
      <HolidaySchedule>
        <!--ro, opt, object, holiday schedule-->
        <ScheduleBlock>
          <!--ro, opt, object, schedule module-->
          <ScheduleBlockGUID>
            <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--ro, opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
          </ScheduleBlockType>
        </ScheduleBlock>
      </HolidaySchedule>
    </CustomExtension>
  </CustomExtensionList>
  <IntelligentRecord>
    <!--ro, opt, enum, whether to enable VCA recording, subType:string, desc:0-no, 1-yes.-->0
  </IntelligentRecord>
  <delayTime opt="0,3,4,5,10,30,60,120,300">
    <!--ro, opt, int, capture delay time, unit:s, attr:opt{req, string}-->1
  </delayTime>
  <durationEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable video expiry date, attr:opt{req, string},
desc:1. front-end device: this node is not returned when this function is not supported; the video expiry date will be set by the node <Duration> when this function is supported and <Duration> is not 0
2. back-end device: the video expiry date will be set by the node <Duration> when this node is not returned; this node is "false" when <Duration> is 0-->true
  </durationEnabled>
  <redundancyRec opt="true,false">
    <!--ro, opt, bool, whether it is redundant recording, attr:opt{req, string}-->false
  </redundancyRec>
  <passbackRecord opt="true,false">
    <!--ro, opt, bool, whether to copy back recording, attr:opt{req, string}-->false
```

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```
  </passbackRecord>
  <lockDuration min="0" max="65535">
    <!--ro, opt, int, recording locking duration, unit:h, attr:min{req, int},max{req, int}, desc:0-unlocked or this node does not exist, 0xffff-locked
permanently; if the total recording is longer than the locked part, it will not be locked-->0
  </lockDuration>
  <recordBackup opt="true,false">
    <!--ro, opt, bool, whether to enable video archiving, attr:opt{req, string}-->false
  </recordBackup>
  <SVCLevel opt="0,1,2,3">
    <!--ro, opt, enum, SVC frame extracting type, subType:int, attr:opt{req, string}, desc:0-not extracted, 1-half extracted, 2-three quarters extracted-->0
  </SVCLevel>
  <recordManage opt="true,false">
    <!--ro, opt, bool, whether to enable video scheduling, attr:opt{req, string}, desc:enabled-scheduled recording; disabled-no scheduled recording but the
recording schedule still in use for, e.g., motion detection and copy-back-->false
  </recordManage>
  <extraSaveAudio opt="true,false">
    <!--ro, opt, bool, whether to enable separate audio storage, attr:opt{req, string}-->false
  </extraSaveAudio>
</Track>
```

#### 14.3.1.5 Get the recording schedule parameters of a specified channel

**Request URL**

GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| trackStreamID | string | Channel number corresponding to the stream type, for example, 101 indicates main stream of channel 1. |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of a recording schedule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main
stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type
should be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--ro, req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20
fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--ro, opt, int, video size-->1
  </Size>
  <Duration min="0" max="750">
    <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15
hours-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
    <!--ro, req, enum, default recording type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM"
(capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "ALLEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording),
"temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by
default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm"
(panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from
vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp"
(get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--ro, opt, bool, whether to support recurrent overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--ro, opt, object, video source description-->
    <SrcGUID>
```

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```
      <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--ro, req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--ro, req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--ro, opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--ro, req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID>
refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main
stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcUrlMethods>
      <!--ro, req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--ro, req, string, login source password, desc:user name for source channel RTSP authentication returned by device; encryption required for
sensitive information-->admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--ro, opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--ro, opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--ro, opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--ro, opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--ro, req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--ro, opt, object, scheduled start time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--ro, opt, object, scheduled end time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
            </ActionRecordingMode>
            <PreRecordTimeSeconds>
              <!--ro, opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds>
              <!--ro, opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
            </PostRecordTimeSeconds>
```

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```
            </Actions>
          </ScheduleAction>
        </ScheduleBlock>
      </ScheduleBlockList>
    </TrackSchedule>
    <CustomExtensionList>
      <!--ro, opt, array, custom extension list, subType:object-->
      <CustomExtension>
        <!--ro, opt, object, custom extension-->
        <CustomExtensionName>
          <!--ro, opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
        </CustomExtensionName>
        <enableSchedule>
          <!--ro, opt, bool, whether to enable recording schedule configuration-->true
        </enableSchedule>
        <SaveAudio>
          <!--ro, opt, bool, whether to enable audio recording-->true
        </SaveAudio>
        <PreRecordTimeSeconds>
          <!--ro, opt, int, pre-record, unit:s-->1
        </PreRecordTimeSeconds>
        <PostRecordTimeSeconds>
          <!--ro, opt, int, post-record, unit:s-->1
        </PostRecordTimeSeconds>
        <HolidaySchedule>
          <!--ro, opt, object, holiday schedule-->
          <ScheduleBlock>
            <!--ro, opt, object, schedule module-->
            <ScheduleBlockGUID>
              <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
            </ScheduleBlockGUID>
            <ScheduleBlockType>
              <!--ro, opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
            </ScheduleBlockType>
          </ScheduleBlock>
        </HolidaySchedule>
      </CustomExtension>
    </CustomExtensionList>
    <IntelligentRecord>
      <!--ro, opt, bool, whether to enable VCA recording-->false
    </IntelligentRecord>
    <delayTime>
      <!--ro, opt, int, capture delay time, unit:s-->1
    </delayTime>
    <durationEnabled>
      <!--ro, opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via
the node Duration when it is supported-->true
    </durationEnabled>
    <redundancyRec>
      <!--ro, opt, bool, whether it is redundant recording-->false
    </redundancyRec>
    <passbackRecord>
      <!--ro, opt, bool, whether to copy back recording-->false
    </passbackRecord>
    <lockDuration>
      <!--ro, opt, int, recording locking duration, unit:h, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording is
longer than the locked part, it will not be locked-->0
    </lockDuration>
    <recordBackup>
      <!--ro, opt, bool, whether to enable video archiving-->false
    </recordBackup>
    <SVCLevel>
      <!--ro, opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted, 1-half extracted, 2-three quarters extracted-->0
    </SVCLevel>
    <recordManage>
      <!--ro, opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording schedule
is still in use, e.g., motion detection and copy-back)-->false
    </recordManage>
    <extraSaveAudio>
      <!--ro, opt, bool, whether to enable separate audio storage-->false
    </extraSaveAudio>
  </Track>
```

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#### 14.3.1.6 Set the recording schedule for a specified channel

**Request URL**

PUT /ISAPI/ContentMgmt/record/tracks/<trackStreamID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| trackStreamID | string | Channel number corresponding to the stream type, for example, 101 indicates main stream of channel 1. |

## Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, get the parameters of a recording schedule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--opt, int, recording size-->1
  </Size>
  <Duration min="0" max="750">
    <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
    <!--req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--opt, bool, whether to support recurrently overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--opt, object, video source description-->
    <SrcGUID>
      <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcUrlMethods>
      <!--req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--req, string, login source password, desc:user name for source channel RTSP authentication; encryption required for sensitive information-->admin:admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
```

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```
            <!--opt, object, scheduled start time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--opt, object, scheduled end time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--req, string, Description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
            </ActionRecordingMode>
            <PreRecordTimeSeconds>
              <!--opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds>
              <!--opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
            </PostRecordTimeSeconds>
          </Actions>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--opt, object, custom extension-->
      <CustomExtensionName>
        <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <SaveAudio>
        <!--opt, bool, whether to enable audio recording-->true
      </SaveAudio>
      <PreRecordTimeSeconds>
        <!--opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
      <HolidaySchedule>
        <!--opt, object, holiday schedule-->
        <ScheduleBlock>
          <!--opt, object, schedule module-->
          <ScheduleBlockGUID>
            <!--req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
          </ScheduleBlockType>
        </ScheduleBlock>
      </HolidaySchedule>
    </CustomExtension>
  </CustomExtensionList>
  <IntelligentRecord>
    <!--opt, bool, whether to enable VCA recording function-->false
  </IntelligentRecord>
  <delayTime>
    <!--opt, int, capture delay time, unit:s-->1
  </delayTime>
```

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```
  <durationEnabled>
    <!--opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via the node Duration when it is supported-->true
  </durationEnabled>
  <redundancyRec>
    <!--opt, bool, whether it is redundant recording-->false
  </redundancyRec>
  <passbackRecord>
    <!--opt, bool, whether to copy back recording-->false
  </passbackRecord>
  <lockDuration>
    <!--opt, int, recording locking duration, unit:h, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording is longer than the locked part, it will not be locked-->0
  </lockDuration>
  <recordBackup>
    <!--opt, bool, whether to enable video archiving-->false
  </recordBackup>
  <SVCLevel>
    <!--opt, enum, SVC frame extracting type, subType:int, desc:0 (not extracted), 1 (half extracted), 2 (three quarters extracted), 3 (a quarter extracted)-->0
  </SVCLevel>
  <recordManage>
    <!--opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording schedule is still in use, e.g., motion detection and copy-back)-->false
  </recordManage>
  <extraSaveAudio>
    <!--opt, bool, whether to enable separate audio storage-->false
  </extraSaveAudio>
</Track>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

## 14.3.2 Video Download

### 14.3.2.1 Get the capability set of downloading videos

**Request URL**

GET /ISAPI/ContentMgmt/download/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<DownloadAbility xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the capability of downloading, attr:version{req, string, protocolVersion}-->
  <isSupportDownloadbyTime>
    <!--ro, opt, bool, whether it supports downloading by time-->true
  </isSupportDownloadbyTime>
  <isSupportDownloadbyFileName>
    <!--ro, opt, bool, whether it supports downloading by file name-->true
  </isSupportDownloadbyFileName>
  <isSupportDownloadCourseFile>
    <!--ro, opt, bool, whether it supports downloading class files-->true
  </isSupportDownloadCourseFile>
  <isSupportDownloadToUSB>
    <!--ro, opt, bool, whether it supports exporting files to devices via USB-->true
  </isSupportDownloadToUSB>
</DownloadAbility>
```

#### 14.3.2.2 Download video files

**Request URL**

GET /ISAPI/ContentMgmt/download?playbackURI=<playbackURI>&userName=<userName>&password=<password>&onlyVerification=<onlyVerification>&mediaID=<recordID>&startTime=<time1>&endTime=<time2>&downType=<downType>&encodeType=<encodeType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| playbackURI | string | Video file URL, format: rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18T00:00:00Z&endtime=2016-07-18T00:25:04Z&name=00000000721000000&size=1065437356&timeType=STD. Download by file: the playbackURI is configured with the start and end times of the current file, the file name, and the file size.Download by time: the playbackURI is configured with the start and end times for the time-based download, without uploading the file name and size. The device only parses the time and downloads based on the specified time. A video file will be returned |
| userName | string | Double verification user name. |
| password | string | Double verification password. |
| onlyVerification | string | If onlyVerification=true is contsined in the URL, it indicates that the current protocol is only used for authentication and not for downloading. |
| recordID | string | Video file ID. If this parameter is does not exist, the HTTP 400 error will be returned by the collection station device, with ISAPI error code 4 (invalid operation). |
| time1 | string | Start time. When mediaID= exists, this parameter is valid, indicating the download starts from this timestamp within the video file, for example, the collection station device can download segment by specifying mediaID file. |
| time2 | string | End time. When mediaID= exists, this parameter is valid, indicating the download ends at this timestamp within the video file, for example, the collection station device can download segment by specifying mediaID file. |
| downType | string | 1 (thumbnail), 2 (file). If this parameter does not exist, the default is to download the file. |
| encodeType | string | Encapsulation format: "ps", "mp4" (default). |

**Request Message**

None

**Response Message**

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Binary Data

## 14.3.3 Video Search

### 14.3.3.1 Get the capability of searching for video and picture files

**Request URL**

GET /ISAPI/ContentMgmt/search/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <timeSpanList>
    <!--ro, opt, array, list of time periods, subType:object-->
    <timeSpan>
      <!--ro, opt, object, time period-->
      <laneNumber min="1" max="99">
        <!--ro, opt, int, lane No., range:[1,99], attr:min{req, int},max{req, int}-->1
      </laneNumber>
      <carType opt="all,unknow,coach,truck,car,minibus,smalltruck,bigcar,smallcar,human,tumbrel,trike,suv,midcar">
        <!--ro, opt, enum, vehicle type, subType:string, attr:opt{req, string}, desc:"coach", "truck", "car", "minibus", "smalltruck", "bigcar", "smallcar",
"human", "tumbrel", "trike"; "suv"; "midcar", "all", "unknown"-->coach
      </carType>
      <illegalType
opt="all,normal,lowSpeed,highSpeed,reverse,redLight,driveLine,direction,intersectionCongest,nonDriveWay,changeLane,wrongLine,badSign,intersectionStop,greenL
ightStop,overYellow,illegalParking,unTurn,emergencyLane,banRight,banLeft,failSafeBelt,pedRedLight,failYieldPed,overStop,bigCarOccupy,leftNotYieldStraight,ri
ghtNotYieldLeft,uTurnNotYieldGoingStaight,leftSideDrivingAfterLeftTurn,gasser,threeWheelPassenger,uphone,congestion,accident,runGreeLight,nonMotorExist,fail
Helmet,smoke,playPhone,failSafeBeltVice,roadBlock,construction,notKeepDistance,overtakeRightSide,dragRacing,changeLaneContinuously,SSharpDriving,abandonedOb
ject,pedestrian,notSlowZebraCrossing,overThermal,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,smog,conflagration,animalsOccupy,irregularSafeBel
tVic,notYieldToRightRoad,rampNotYieldToMainRoad,enterNotYieldToInRoundabout,TruckOccupyOvertakingLane,noAlterTraffic,banStraight,followClose">
        <!--ro, opt, enum, violation type, subType:string, attr:opt{req, string}, desc:"all", "normal", "lowSpeed", "highSpeed", "reverse", "redLight",
"driveLine", "direction", "intersectionCongest", "nonDriveWay", "changeLane", "wrongLine", "badSign", "intersectionStop", "greenLightStop", "overYello",
"illegalParking", "unTurn", "emergencyLane", "banRight", "banLeft", "failSafeBelt", "pedRedLight", "failYieldPed", "overStop", "bigCarOccupy",
"leftNotYieldStraight, "rightNotYieldLeft", "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "gasser", "threeWheelPassenger", "uphone",
"congestion", "accident", "runGreeLight", "nonMotorExist", "failHelmet", "smoke", "playPhone", "failSafeBeltVice", "roadBlock", "construction",
"notKeepDistance", "overtakeRightSide", "dragRacing", "changeLaneContinuously", "SSharpDriving", "abandonedObject", "pedestrian", "notSlowZebraCrossing",
"overThermal", "turnRightStop", "occupyDedicatedLane", "notDriveInDedicatedLane", "smog", "conflagration", "animalsOccupy"-->all
      </illegalType>
      <startTime>
        <!--ro, req, datetime, start time-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time-->1970-01-01T00:00:00+08:00
      </endTime>
    </timeSpan>
  </timeSpanList>
</CMSearchDescription>
```

### 14.3.3.2 Search for video and picture files

**Request URL**

POST /ISAPI/ContentMgmt/search?timeType=<timeType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| timeType | enum | 1. If "isSupportTimeTypeSTD" is true in GET /ISAPI/System/capabilities, it indicates that the device supports parsing the optional parameter timeType=STD in the URL.2. If the parameter timeType=STD is included in the URL, time-related parameters (e.g., "datetime") is processed in ISO8601 time format.3. If the parameter timeType=STD is not included in the URL, time-related parameters is processed according to the original time format set by the device. |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searches, the search history will be saved in the memory to speed up next search-->test
  </searchID>
  <trackIDList>
    <!--opt, array, channel list, subType:object-->
    <trackID>
      <!--opt, int, channel stream ID, desc:"XX01" (main stream videos of XX channel), "XX03" (search for XX channel pictures), 1 by default-->101
    </trackID>
  </trackIDList>
  <timeSpanList>
    <!--opt, array, list of time periods, subType:object-->
    <timeSpan>
      <!--opt, object, time period-->
      <startTime>
        <!--req, datetime, start time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:00:00Z
      </startTime>
      <endTime>
        <!--req, datetime, end time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T01:00:00Z
      </endTime>
      <laneNumber>
        <!--opt, int, lane No.-->1
      </laneNumber>
      <carType>
        <!--opt, enum, vehicle type, subType:string, desc:"coach", "truck", "car", "minibus", "smalltruck" (small-sized truck), "bigcar" (large-sized vehicle), "smallcar" (small-sized vehicle), "human", "tumbrel" (two wheeler), "trike" (tricycle), "suv", "midcar" (middle-sized bus), "all", "unknown"-->coach
      </carType>
      <illegalType>
        <!--opt, enum, violation type, subType:string, desc:"all", "normal", "lowSpeed" (driving in low speed), "highSpeed" (driving in high speed), "reverse" (wrong-way driving), "redLight" (red light running), "driveLine" (driving on the lane line), "direction" (driving against direction guidance), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "changeLane" (illegal lane change), "wrongLine" (occupy dedicated lane), "badSign" (prohibition violation), "intersectionStop" (stop vehicle at intersection), "greenLightStop (stop vehicle when green light), "overYello" (over yellow line), "illegalParking" (illegal parking), "unTurn" (illegal U-turn), "emergencyLane" (occupy emergency lane), "banRight" (no right turn), "banLeft" (no left turn), "failSafeBelt" (seatbelt unbuckled), "pedRedLight" (pedestrian red light running), "failYieldPed" (not yield to pedestrian), "overStop" (over stop line), "bigCarOccupy" (lane occupation by large-sized vehicle), "leftNoYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStraight", "leftSideDrivingAfterLeftTurn", "gasser", "threeWheel", "Passenger", "uphone", "congestion", "accident", "runGreeLight", "nonMotorExist", "failHelmet", "smoke", "playPhone", "failSafeBeltVice", "roadBlock", "construction", "notKeepDistance", "overtakeRightSide", "dragRacing", "changeLaneContinuously", "SSharpDriving", "abandonedObject", "pedestrian", "notSlowZebraCrossing", "overThermal", "turnRightStop", "occupyDedicatedLane", "notDriveInDedicatedLane", "smog", "conflagration", "animalsOccupy", "irregularSafeBeltVic", "notYieldToRightRoad", "rampNotYieldToMainRoad", "enterNotYieldToInRoundabout", "TruckOccupyOvertakingLane", "noAlterTraffic", "banStraight", "followClose"-->all
      </illegalType>
      <vehicleTypeByWeight>
        <!--opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1 (7 and fewer seats passenger vehicle / 2 tons or fewer truck), 2 (8 to 19 seats passenger vehicle / 2 to 5 tons truck), 3 (20 to 39 seats passenger vehicle/5 to 10 tons truck), 4 (40 and more seats passenger vehicle/10 to 15 tons truck), 5 (15 and more tons truck)-->4
      </vehicleTypeByWeight>
      <mediaID>
        <!--opt, int, file ID, desc:file ID (media ID), which is used to return video file information after segmented by startTime and endTime-->0
      </mediaID>
      <listType>
        <!--ro, opt, enum, name list type, subType:string, desc:"black" (blocklist). If this node is empty, it indicates that the name list type is not limited, corresponding to the listType in ANPR-->black
      </listType>
    </timeSpan>
  </timeSpanList>
  <sourceIDList>
    <!--opt, array, stream ID list, subType:object-->
    <sourceID>
      <!--req, string, GUID, range:[0,64]-->test
    </sourceID>
  </sourceIDList>
  <searchTypeList>
    <!--opt, array, search type list, subType:object, range:[0,20]-->
    <searchKeyword>
      <!--opt, string, keyword-->test
    </searchKeyword>
    <keywordType>
      <!--opt, enum, keyword type, subType:string, desc:“policeCode” (police No.), “shortSerialNumber” (device serial No.); when searching for videos stored in dock stations via the platform, videos can be filtered according to the serial No. of body camera; if the dock station does not send this parameter, the policeCode will be used by default-->shortSerialNumber
    </keywordType>
  </searchTypeList>
  <contentTypeList>
    <!--opt, array, file type list for search, subType:object-->
    <contentType>
      <!--opt, enum, file type for search, subType:string, desc:"video", "audio", "metadata”, "text", "mixed", "other". Note: "metadata" is used when searching for pictures-->video
    </contentType>
  </contentTypeList>
  <uploadState>
    <!--opt, enum, uploading status, subType:string, desc:"all", "uploaded", "nonUploaded” (not uploaded), "uploadFailed” (uploading failed)-->all
  </uploadState>
  <lockState>
    <!--opt, enum, locking status, subType:string, desc:"Lock” (Locked), "unlock” (unlocked), "all"-->lock
  </lockState>
  <maxResults>
    <!--opt, int, the maximum number of results returned for one time, range:[1,500]-->500
```

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```
  </maxResults>
  <searchResultPostion>
    <!--opt, int, start position for search, desc:the node of body camera is in front of <maxResults>-->1
  </searchResultPostion>
  <metadataList>
    <!--req, array, search conditions, subType:object, desc:only supports a single description list-->
    <metadataDescriptor>
      <!--req, enum, video or file type for search, subType:string, desc:1. the value of CMSearchDescription->contentTypeList->contentType indicates the type (video or picture) for search; 2. the video and picture file type supported by the device will be returned by the <recordSearchType> node and the <pictureSearchType> node in the RacmCap message returned by calling /ISAPI/ContentMgmt/capabilities; 3. transmit the data in the format of "recordType.meta..com/[type]" to search for videos or pictures; if /[type] is not contained, it indicates that all types of video files will be searched; if the [type] is set to "allPic", it indicates that all types of picture files will be searched; the [type] refers to the followings: "imageCapture" (scheduled capture by channel), "pothole", "crack", "roadMaint" (road maintenance), "scheduledRecordUpload" (scheduled recording by channel), "ReID" (target pattern detection), "thermalVehicleDetection" (thermal-based vehicle detection), "objectsThrownDetection", "methaneConcentrationException", "methaneLightIntensityException" (hazardous gas detection), "bannerDetection", "waterQualityDetection", "waterLevelDetection", "parallelParking", "waterLevelAndRainfall" (water level and rainfall), "AIOpenPlatform" (events on AI open platform), "fishingShipDetection", "variableDataEvent", "VOCsGasDetection", "angleDeviationDetection" (tilt detection), "pir", "shipChannelAbnormal" (ship deviation), "radarVideoDetection" (radar-assisted target detection), "AIDfogDetection", "consultation", "panicAlarm", "zoneAlarm", "smokeAndFireDetection", "smokeDetection", "facedetection", "FieldDetection" (intrusion detection), "scenechangedetection", "LineDetection" (line crossing detection), "regionEntrance", "regionExiting", "Loitering", "group" (people gathering), "rapidMove" (fast moving), "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapShot" (manual capture), "playSnapShot" (capture in playback), "safetyHelmet" (hard hat detection), "personQueueCounting" (people queuing-up detection), "personQueueTime" (waiting time in queue detection), "violentMotion", "leavePosition" (absent detection), "advReachHeight" (climing detection), "peopleNumChange" (people count change), "mixedTargetDetection" (multi-target-type detection), "illegalParking" (parking violation), "pedestrian" (pedestrian detection), "construction", "roadblock", "evidence" (manual evidence capture), "crosslane" (driving on the lane line), "vehicleexist" (motor vehicle on non-motor vehicle lane), "Lanechange", "wrongdirection" (wron-way driving), "congestion", "turnround", "abandonedObject", "trafficAccident", "fogDetection", "sensorAlarm", "parallelParkingContinuousRecord" (contimuous recording for parallel parking), "GasTankDetection", "swipePatrol" (patrolling by card), "alarmResult" (face picture comparison), "UVPhotonDetection", "DevPowerStatusReport" (device power status uploading), "playCellphone" (using mobile phone detection), "HFPD", "audioAnalysis", "PeopleFallingDownDetection", "PeopleSedentary", "RadarFallenObjectsOnTracks", "VegetationSpectrumDetection", "personDensityDetection", "ChannelClearEvent", "channelOccupy", ChannelPassingEvent", and "FallenObjectsOnTracksEvent"-->recordType.meta..com/imageCapture
    </metadataDescriptor>
    <SearchProperity>
      <!--opt, object, search by features-->
      <plateSearchMask>
        <!--opt, enum, license plate-->test
      </plateSearchMask>
      <stateOrProvince>
        <!--opt, int, province, desc:"0xffff" (undefined)-->0xffff
      </stateOrProvince>
      <country>
        <!--opt, enum, country/region, subType:string, desc:0: Algorithm library does not support the national identification card; 1: CZ - Czech Republic; 2: FRA - France; 3: DE - Germany; 4: E - Spain; 5: IT - Italy; 6: NL - Netherlands; 7: PL - Poland; 8: SVK - Slovakia; 9: BY - Belorussia; 10: MDA - Moldova; 11: RU - Russia; 12: UA - Ukraine; 169: Canada; 172: US; 227: Australia; 0xff: All-->0
      </country>
      <subType>
        <!--opt, enum, sub types, subType:string, desc:only supported by vehicleDetection and HVTVehicleDetection. "all", "motorVehicle", "nonMotorVehicle", "pedestrian"-->all
      </subType>
      <capTypeDescriptor>
        <!--opt, enum, types of captured pictures, subType:string, desc:"event", "evidence". Events include parkEvent, crosslane, vehicleexist, lanechange, wrongdirection, congestion, turnaround, edfManual. Evidences include parkEvent, crosslane, vehicleexist, lanechange, wrongdirection, and anyType-->anyType
      </capTypeDescriptor>
      <nation>
        <!--opt, enum, area, subType:string, desc:EU: Europe Region; ER: Middle East; AP: Asia Pacific; AfricaAndAmerica: Africa and America-->EU
      </nation>
      <vehicleType>
        <!--opt, enum, vehicle type, subType:string, desc:"unknown", "bus", "truck", "vehicle", "van", "buggy", "SUVMPV": SUV/MPV, "mediumBus"-->bus
      </vehicleType>
      <absoulteParkingNum>
        <!--opt, string, parking space No., desc:the parking space No. is the same as the number on the actual ground surface, which can be captured by the camera-->test
      </absoulteParkingNum>
    </SearchProperity>
  </metadataList>
  <cardNumber>
    <!--opt, string, card No., range:[0,32], desc:if this node is empty or does not exist, it indicates that the card No. is not restricted-->test
  </cardNumber>
  <drawFrame>
    <!--opt, bool, whether to enable frame extraction, desc:the value of this node is false by default-->false
  </drawFrame>
  <AlarmInNoList>
    <!--opt, array, triggered alarm input channel list, subType:object-->
    <alarmInNo>
      <!--req, int, triggered alarm input channel No.-->0
    </alarmInNo>
  </AlarmInNoList>
  <ruleID>
    <!--opt, string, rule ID, desc:if this node is not configured, the search result will not be restricted by any rule-->test
  </ruleID>
  <roomIndex>
    <!--opt, int, interrogation room No., desc:it starts from 1-->0
  </roomIndex>
  <StreamIDParam>
    <!--opt, object, stream ID parameters-->
    <streamID>
      <!--req, string, stream ID, range:[0,72]-->test
    </streamID>
    <channel>
      <!--opt, int, channel No., range:[1,256]-->1
    </channel>
    <cmdType>
      <!--opt, int, trigger type, range:[0,255], desc:connect NVR with cloud storage to use it-->0
    </cmdType>
    <backupVolumeNum>
      <!--opt, int, archive No., range:[0,255], desc:applicable to CVR-->0
```

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```
    </backupVolumeNum>
    <archiveLabel>
      <!--opt, string, archive No.-->test
    </archiveLabel>
  </StreamIDParam>
  <PosAlarm>
    <!--opt, object, pos video-->
    <caseSensitive>
      <!--req, bool, whether it is case-sensitive-->true
    </caseSensitive>
    <combinateMode>
      <!--req, enum, keyword combination, subType:string, desc:"or", "and"-->or
    </combinateMode>
    <KeyWordList>
      <!--opt, array, keyword list, subType:object, desc:supports up to three key words-->
      <keyWord>
        <!--req, string, keyword, range:[0,128]-->test
      </keyWord>
    </KeyWordList>
  </PosAlarm>
  <TrialParam>
    <!--opt, object, search conditions of trail information-->
    <caseNo>
      <!--opt, string, case No., range:[0,56]-->test
    </caseNo>
    <caseName>
      <!--opt, string, case name, range:[0,56]-->test
    </caseName>
    <litigant1>
      <!--opt, string, party 1, range:[0,32]-->test
    </litigant1>
    <litigant2>
      <!--opt, string, party 2, range:[0,32]-->test
    </litigant2>
    <chiefJudge>
      <!--opt, string, jury, range:[0,32]-->test
    </chiefJudge>
    <caseType>
      <!--opt, enum, case type, subType:int, desc:0 (all), 1 (penal case), 2 (civil case)-->0
    </caseType>
  </TrialParam>
  <ConferenceParam>
    <!--opt, object, video conference parameters-->
    <conferenceID>
      <!--opt, string, conference ID, range:[1,64]-->test
    </conferenceID>
  </ConferenceParam>
  <ATMParam>
    <!--opt, object, ATM search-->
    <transactionType>
      <!--opt, enum, transaction type, subType:int, desc:0 (all), 1(search), 2(withdrawal), 3 (deposit), 4(change password), 5(transfer), 6(search with no
card), 7(deposit with no card), 8 (cash swallowed), 9(card swallowed) 10(undefined)-->0
    </transactionType>
    <transationAmount>
      <!--opt, int, amount, step:1, desc:unit: yuan-->0
    </transationAmount>
  </ATMParam>
  <labelSearch>
    <!--opt, object, search by tag type list of video file, desc:if the node is not returned, it means search for all videos. Related URI:
/ISAPI/ContentMgmt/record/control/remark?format=json&security=<security>&iv=<iv>-->
    <labelSearchType>
      <!--opt, enum, type of searching by video file tag, subType:string, desc:"recordLabelTypeList", "allLabel", "noLabel"-->recordLabelTypeList
    </labelSearchType>
    <labelTypeList>
      <!--opt, object, by tag type list of video file, dep:and,{$.CMSearchDescription.recordLabelSearch.recordLabelSearchType,eq,recordLabelTypeList}-->
      <labelTypes>
        <!--opt, object, tag type of video file-->
        <labelType>
          <!--opt, enum, tag type of video file, subType:string, desc:"importance" (important), "unimportance" (not important)-->importance
        </labelType>
      </labelTypes>
    </labelTypeList>
  </labelSearch>
  <dockStationAudioFile>
    <!--opt, object, dock station audio file-->
    <analysisState>
      <!--ro, opt, enum, analysis status, subType:string, desc:"complete" (completed), "running", "notStarted" (not started)-->complete
    </analysisState>
    <keyWordList>
      <!--opt, array, dock station audio file label, subType:object, range:[0,100]-->
      <keyWord>
        <!--opt, string, label, range:[0,256]-->test
      </keyWord>
    </keyWordList>
  </dockStationAudioFile>
  <objectsThrownDetectionSearch>
    <!--opt, object, object thrown from building search condition-->
    <alarmLevel>
      <!--opt, int, alarm priority, range:[1,6], desc:1 (low), 2 (medium), 3 (high), 4 (low and medium), 5 (low and high), 6 (medium and high). If this node
is empty, alarms of all levels will be displayed-->3
    </alarmLevel>
    <lowFloorID>
      <!--opt, int, low floor, range:[-100,100]-->0
```

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```
    </lowFloorID>
    <highFloorID>
      <!--opt, int, high floor, range:[-100,100]-->0
    </highFloorID>
  </objectsThrownDetectionSearch>
  <alarmResultSearch>
    <!--opt, object, face picture comparison search condition-->
    <faceContrastType>
      <!--req, enum, face picture comparison type, subType:string, desc:"all", "faceContrast" (face picture comparison succeeded), "whiteFaceContrast"
(allowlist face picture comparison), "faceContrastFailure" (face picture comparison failed)-->faceContrast
    </faceContrastType>
  </alarmResultSearch>
  <radarID>
    <!--opt, int, radar channel No.-->1
  </radarID>
  <synthesisType>
    <!--opt, enum, subType:string-->all
  </synthesisType>
</CMSearchDescription>
```

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searches, the search history will be saved in the memory to speed up next search-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search status, subType:string, desc:true (succeeded), false (failed)-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK", "NO MATCHES", "MORE"-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of returned records-->500
  </numOfMatches>
  <matchList>
    <!--ro, opt, array, result list, subType:object-->
    <searchMatchItem>
      <!--ro, opt, object, results-->
      <sourceID>
        <!--ro, req, string, source ID, desc:source ID-->test
      </sourceID>
      <trackID>
        <!--ro, opt, int, channel stream ID, desc:"XX01" (main stream videos of XX channel), "XX03" (search for XX channel pictures), 1 by default-->101
      </trackID>
      <timeSpan>
        <!--ro, opt, object, time range-->
        <startTime>
          <!--ro, req, datetime, video start time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:10:00Z
        </startTime>
        <endTime>
          <!--ro, req, datetime, video end time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:20:00Z
        </endTime>
      </timeSpan>
      <mediaSegmentDescriptor>
        <!--ro, opt, object, data description-->
        <contentType>
          <!--ro, req, string, file type, desc:corresponding to conditions-->test
        </contentType>
        <codecType>
          <!--ro, req, string, video encoding type-->test
        </codecType>
        <rateType>
          <!--ro, opt, string, video bit rate-->test
        </rateType>
        <playbackURI>
          <!--ro, opt, string, video URL,
desc:"name=" is followed by the video file name, and "size=" is followed by the size of the video file. The "starttime" and "endtime" are UTC time, but as
the time returned by front-end devices is UTC time, while the time returned by DVR/NVR devices is the devices' local time, there are different mechanisms to
deal with the time:
1. For DVR/NVR devices that return the local time:
    1) When the request URL does not include "timeType=STD" (/ISAPI/ContentMgmt/search), then the returned playbackURI will not include "timeType=STD". The
"starttime" and "endtime" will refer to the devices' local time.
    2) When the request URL includes "timeType=STD" (/ISAPI/ContentMgmt/search&timeType=STD), and the devices support parsing "timeType"
("isSupportTimeTypeSTD" is true in GET /ISAPI/System/capabilities), then the returned playbackURI will include "timeType=STD". The "starttime" and "endtime"
will refer to the UTC time in ISO 8601 format.
2. For IPC devices that return UTC time, whether or not the request URL includes "timeType=STD", the devices do not need to parse "timeType". The returned
playbackURI will not include "timeType=STD", and the "starttime" and "endtime" will refer to the UTC time in ISO 8601 format.
3. To play back the video files, synchronize the playbackURI to the video playback API: /ISAPI/Streaming/tracks/<trackStreamID>?starttime=
<startTime>&endtime=<endTimeime>&name=<fileName>&size=<fileSize>&timeType=<timeType>.-->rtsp://10.65.130.168/Streaming/tracks/201/?
starttime=20190213T091134Z&amp;endtime=20190213T092116Z&amp;name=01010000093000100&amp;size=732335288;timeType=STD
        </playbackURI>
        <lockStatus>
          <!--ro, opt, enum, whether video file is locked, subType:string, desc:"lock", "unlock"-->lock
        </lockStatus>
        <remainLockTime>
          <!--ro, opt, int, remaining lock time, unit:s, desc:only returned when "lockStatus" is "lock". If 0 is returned, it indicates that it is locked
permanently-->1
```

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```
permanently -->1
        </remainLockTime>
        <name>
          <!--ro, opt, string, file name-->test
        </name>
        <size>
          <!--ro, opt, int, file size-->0
        </size>
        <mediaID>
          <!--ro, opt, int, file ID, desc:unique identifier of file-->0
        </mediaID>
        <remark>
          <!--ro, opt, string, file remarks-->test
        </remark>
        <GPSInfo>
          <!--ro, opt, string, file GPS, desc:required, if no GPS information, a blank is returned-->test
        </GPSInfo>
        <recorderCode>
          <!--ro, opt, string, video recorder No.-->test
        </recorderCode>
        <uploadState>
          <!--ro, opt, enum, upload status, subType:string, desc:"uploaded", "nonUploaded", "uploadFailed", "uploading"-->uploaded
        </uploadState>
        <uploadTime>
          <!--ro, opt, datetime, video uploading time, desc:ISO8601 time (TD format, local time and time difference)-->1970-01-01T00:00:00+08:00
        </uploadTime>
        <policeCode>
          <!--ro, opt, string, police No., desc:sensitive information is encrypted-->test
        </policeCode>
        <policeName>
          <!--ro, opt, string, police name, desc:sensitive information is encrypted-->test
        </policeName>
        <shootingTime>
          <!--ro, opt, datetime, time, desc:ISO8601 time (TD format, local time and time difference)-->1970-01-01T00:00:00+08:00
        </shootingTime>
        <FileTime>
          <!--ro, opt, int, the file is too long, unit:min-->1
        </FileTime>
        <remarks>
          <!--ro, opt, string, file remarks-->test
        </remarks>
        <cloudStorageType>
          <!--ro, opt, enum, storage server type, subType:string, desc:0 (cloud storage), 1 (CVR), 2 (pstor)-->0
        </cloudStorageType>
        <fileUrl>
          <!--ro, opt, string, file URL-->test
        </fileUrl>
        <thumbnailUrl>
          <!--ro, opt, string, thumbnail URL of picture or video-->test
        </thumbnailUrl>
        <encryption>
          <!--ro, opt, object, encryption-->
          <enabled>
            <!--ro, opt, bool, whether to encrypt the file-->true
          </enabled>
          <encryptionkey>
            <!--ro, opt, string, encryption key, desc:sensitive information is encrypted-->test
          </encryptionkey>
          <encryptionLevel>
            <!--ro, opt, enum, encryption level, subType:int, desc:1 (AES128), 2 (AES256)-->1
          </encryptionLevel>
        </encryption>
        <videoTargetTypeList>
          <!--ro, opt, object, list of target types, desc:this node is only applicable for HEOP protocol of intercom products. Several target types are
included in the time range. Every type is for a stream. Playback is according to video of corresponding time-->
          <videoTargetType>
            <!--ro, opt, enum, list of target types, subType:string, desc:this node is only applicable for HEOP protocol of intercom products. Several
target types are included in the time range. Every type is for a stream. Playback is according to video of corresponding time. "local", "peer"-->local
          </videoTargetType>
        </videoTargetTypeList>
        <TargetRegionList size="8">
          <!--ro, opt, array, search for the target region list of the picture, subType:object, attr:size{req, int}, desc:related URL:
/ISAPI/ContentMgmt/capabilities  targetRegionEventPicSearch-->
          <TargetRegion>
            <!--ro, opt, object, target area-->
            <regionID>
              <!--ro, req, string, detection area ID-->1
            </regionID>
            <RegionCoordinatesList>
              <!--ro, req, array, list of point coordinates, subType:object, range:[3,10], desc:rectangle-->
              <RegionCoordinates>
                <!--ro, opt, object, point coordinates, desc:the origin is the lower-left corner of the screen-->
                <positionX>
                  <!--ro, req, float, X-coordinate, range:[0,1]-->1
                </positionX>
                <positionY>
                  <!--ro, req, float, Y-coordinate, range:[0,1]-->1
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </TargetRegion>
        </TargetRegionList>
        <objectsThrownDetectionResult>
          <!--ro, opt, object, object thrown from building search result-->
```

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```
          <alarmLevel>
            <!--ro, opt, int, alarm level, desc:it is categorized into three levels according to the target track and the actual event credibility. The higher the level, the greater the credibility of the alarm being triggered by an actual object thrown from building event.-->0
          </alarmLevel>
          <floorID>
            <!--ro, opt, int, floor No., range:[-100,100]-->0
          </floorID>
        </objectsThrownDetectionResult>
        <synthesisType>
          <!--ro, opt, enum, subType:string-->auto
        </synthesisType>
        <bodyCameraShortSN>
          <!--ro, opt, string, range:[0,9]-->P20151543
        </bodyCameraShortSN>
        <shipsDetection>
          <!--ro, opt, object-->
          <shipName>
            <!--ro, opt, string, range:[1,64]-->浙嘉兴资03181
          </shipName>
          <shipsSpeed>
            <!--ro, opt, float, range:[0.0,300.0], unit:km/h-->23.4
          </shipsSpeed>
          <shipsLength>
            <!--ro, opt, float, range:[0.000,500.000], unit:m-->50.000
          </shipsLength>
          <shipLocation>
            <!--ro, opt, object-->
            <longitudeType>
              <!--ro, opt, enum, subType:string-->E
            </longitudeType>
            <latitudeType>
              <!--ro, opt, enum, subType:string-->S
            </latitudeType>
            <longitude>
              <!--ro, opt, object-->
              <degree>
                <!--ro, opt, int, range:[0,180]-->0
              </degree>
              <minute>
                <!--ro, opt, int, range:[0,60]-->0
              </minute>
              <sec>
                <!--ro, opt, float, range:[0,60.000000]-->0.000
              </sec>
            </longitude>
            <latitude>
              <!--ro, opt, object-->
              <degree>
                <!--ro, opt, int, range:[0,180]-->0
              </degree>
              <minute>
                <!--ro, opt, int, range:[0,60]-->0
              </minute>
              <sec>
                <!--ro, opt, float, range:[0,60.000000]-->0.000
              </sec>
            </latitude>
          </shipLocation>
          <abnormalType>
            <!--ro, opt, enum, subType:int-->shipCourseDeviation
          </abnormalType>
          <shipChannelID>
            <!--ro, opt, int, range:[1,4]-->4
          </shipChannelID>
        </shipsDetection>
      </mediaSegmentDescriptor>
      <metadataMatches>
        <!--ro, opt, object, metadata match information-->
        <metadataDescriptor>
          <!--ro, req, string, metadata description, desc:eg: recordType.meta..com/MOTION. The supported video file type by a specific device can be returned by opt in RacmCap->recordSearchType of /ISAPI/ContentMgmt/capabilities-->test
        </metadataDescriptor>
      </metadataMatches>
      <face>
        <!--ro, opt, object, face search result-->
        <name>
          <!--ro, opt, string, name-->test
        </name>
        <gender>
          <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
        </gender>
        <certificateType>
          <!--ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (the other certificate)-->officerID
        </certificateType>
        <certificateNumber>
          <!--ro, opt, string, certificate No.-->test
        </certificateNumber>
        <currTemperature>
          <!--ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃-->36.5
        </currTemperature>
        <mask>
          <!--ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown", "no", "yes"-->unknown
        </mask>
```

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```
        <mask>
        <phoneNumber>
          <!--ro, opt, string, phone number, range:[1,64], desc:it is valid when event type is face picture comparison-->test
        </phoneNumber>
        <FDID>
          <!--ro, opt, string, face picture library ID, range:[1,64], desc:it is valid when event type is face picture comparison-->test
        </FDID>
        <FDLibName>
          <!--ro, opt, string, face picture library name, range:[1,64], desc:it is valid when event type is face picture comparison-->test
        </FDLibName>
        <FDPicURL>
          <!--ro, opt, string, face picture library picture URL, desc:it is valid when event type is face picture comparison-->test
        </FDPicURL>
        <similarity>
          <!--ro, opt, float, comparison similarity, range:[0,1], step:0.001, desc:it is valid when event type is face picture comparison-->test
        </similarity>
      </face>
      <AlarmInNoList>
        <!--ro, opt, array, triggered alarm input channel list, subType:object-->
        <alarmInNo>
          <!--ro, req, int, triggered alarm input channel No.-->0
        </alarmInNo>
      </AlarmInNoList>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:if this node is not returned, the search result is not restricted by any rule-->test
      </ruleID>
      <roomIndex>
        <!--ro, opt, int, interrogation room No., desc:it starts from 1-->0
      </roomIndex>
      <StreamIDParam>
        <!--ro, opt, object, stream ID parameters-->
        <streamID>
          <!--ro, req, string, stream ID, range:[0,72]-->test
        </streamID>
        <cmdType>
          <!--ro, opt, int, trigger type, range:[0,255], desc:connect NVR with cloud storage to use it-->0
        </cmdType>
        <backupVolumeNum>
          <!--ro, opt, int, archive No., range:[0,255], desc:applicable to CVR-->0
        </backupVolumeNum>
        <archiveLabel>
          <!--ro, opt, string, archive No.-->test
        </archiveLabel>
      </StreamIDParam>
      <PosAlarm>
        <!--ro, opt, object, pos video-->
        <caseSensitive>
          <!--ro, req, bool, whether it is case-sensitive-->true
        </caseSensitive>
        <combinateMode>
          <!--ro, req, enum, keyword combination, subType:string, desc:"or", "and"-->or
        </combinateMode>
        <KeyWordList>
          <!--ro, opt, array, keyword list, subType:object, desc:supports a maximum of three words-->
          <keyWord>
            <!--ro, req, string, keyword, range:[0,128]-->test
          </keyWord>
        </KeyWordList>
      </PosAlarm>
      <TrialParam>
        <!--ro, opt, object, search conditions of trail information-->
        <caseNo>
          <!--ro, opt, string, case No., range:[0,56]-->test
        </caseNo>
        <caseName>
          <!--ro, opt, string, case name, range:[0,56]-->test
        </caseName>
        <litigant1>
          <!--ro, opt, string, party 1, range:[0,32]-->test
        </litigant1>
        <litigant2>
          <!--ro, opt, string, party 2, range:[0,32]-->test
        </litigant2>
        <chiefJudge>
          <!--ro, opt, string, jury, range:[0,32]-->test
        </chiefJudge>
        <caseType>
          <!--ro, opt, enum, case type, subType:int, desc:0 (all), 1 (penal case), 2 (civil case)-->0
        </caseType>
      </TrialParam>
      <plateNo>
        <!--ro, opt, string, license plate number, range:[1,64], desc:noPlate (vehicle without license plate), unknown (no license plate recognized),
XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string)-->蓝浙AX2P12
      </plateNo>
      <ConferenceParam>
        <!--ro, opt, object, video conference parameters-->
        <conferenceName>
          <!--ro, opt, string, conference name, range:[1,64], desc:from the "conferenceName" of POST /ISAPI/VCS/conferences-->test
        </conferenceName>
      </ConferenceParam>
      <ATMParam>
        <!--ro, opt, object, ATM search-->
        <transactionType>
          <!--ro, opt, enum, transaction type, subType:int, desc:0 (all), 1(search), 2(withdrawal), 3 (deposit), 4 (change password), 5(transfer), 6 (search
```

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```
with no card), 7 (deposit with no card), 8 (cash swallowed), 9 (card swallowed) 10 (undefined)-->0
        </transactionType>
        <transationAmount>
          <!--ro, opt, int, amount, step:1, desc:unit: yuan-->0
        </transationAmount>
      </ATMParam>
      <labelTypeList>
        <!--ro, opt, object, tag type list of video file, desc:URL for configuring video tags: /ISAPI/ContentMgmt/record/control/remark?
format=json&security=<security>&iv=<iv>-->
        <labelTypes>
          <!--ro, opt, object, tag type of video file-->
          <labelType>
            <!--ro, opt, enum, tag type of video file, subType:string, desc:"importance", "unimportance"-->importance
          </labelType>
        </labelTypes>
      </labelTypeList>
      <isInFalseAlarmMaterialLib>
        <!--ro, opt, bool, whether imported to false alarm library, desc:not imported if this node is not returned or false is returned-->true
      </isInFalseAlarmMaterialLib>
      <vegetationSpectrumDetection>
        <!--ro, opt, object-->
        <vegetationIndex>
          <!--ro, opt, enum, subType:string-->NDVI
        </vegetationIndex>
        <vegetationIndexInitData>
          <!--ro, opt, enum, subType:string-->NDVI
        </vegetationIndexInitData>
        <reflectivityType>
          <!--ro, opt, enum, subType:string-->blue
        </reflectivityType>
        <isVisibleImage>
          <!--ro, opt, bool-->true
        </isVisibleImage>
      </vegetationSpectrumDetection>
    </searchMatchItem>
    <dockStationAudioFile>
      <!--ro, opt, object, dock station audio file-->
      <analysisState>
        <!--ro, opt, enum, analysis status, subType:string, desc:"complete" (completed), "running", "notStarted" (not started)-->complete
      </analysisState>
      <keyWordList>
        <!--ro, opt, array, dock station audio file label, subType:object, range:[0,100]-->
        <keyWord>
          <!--ro, opt, string, label, range:[0,256]-->test
        </keyWord>
      </keyWordList>
    </dockStationAudioFile>
    <pictureType>
      <!--ro, opt, enum, subType:string-->licensePlatePicture
    </pictureType>
  </matchList>
  <totalMatches>
    <!--ro, opt, int, total number of matched results-->12
  </totalMatches>
</CMSearchResult>
```

#### 14.3.3.3 Get the capability of searching for recording start and end time

**Request URL**

GET /ISAPI/ContentMgmt/time/search/capabilities?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
{
    "TimeSearchCond": {
    /*ro, opt, object*/
        "@size":  1024,
        /*ro, req, int, the maximum number of search conditions that can be set*/
        "channelID": {
        /*ro, req, object, channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the maximum value*/
        },
        "streamType": {
        /*ro, opt, object, stream type*/
            "@opt": ["main", "sub", "adaptive"]
            /*ro, req, array, options, subType:string, desc:“main”, “sub”, “adaptive”*/
        }
    },
    "TimeSearchResult": {
    /*ro, opt, object*/
        "@size":  1024,
        /*ro, req, int, the maximum number of search results that can be returned*/
        "channelID": {
        /*ro, req, object, channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the maximum value*/
        },
        "streamType": {
        /*ro, opt, object, stream type*/
            "@opt": ["main", "sub", "adaptive"]
            /*ro, req, array, options, subType:string, desc:“main”, “sub”, “adaptive”*/
        },
        "hasRecord": {
        /*ro, req, object, whether the video exists (startTime and endTime are valid only when the value of hasRecord is "true")*/
            "@opt": [true, false]
            /*ro, req, array, options, subType:bool*/
        },
        "startTime": {
        /*ro, opt, object, recording start time (e.g.,2010-01-01T00:00:00+08:00, this field is required when the value of hasRecord is "true")*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  64
            /*ro, req, int, the maximum value*/
        },
        "endTime": {
        /*ro, opt, object, recording end time (e.g.,2010-01-01T00:00:00+08:00, this field is required when the value of hasRecord is "true")*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  64
            /*ro, req, int, the maximum value*/
        }
    }
}
```

# 14.4 PTZ Control

## 14.4.1 PTZ Manual Control

### 14.4.1.1 Get the pan/tilt speed ratio of a specified channel

**Request URL**

GET /ISAPI/Image/channels/<channelID>/proportionalpan

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<proportionalpan xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</proportionalpan>
```

#### 14.4.1.2 Set the pan/tilt speed ratio of a specific channel

**Request URL**

PUT /ISAPI/Image/channels/<channelID>/proportionalpan

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<proportionalpan xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, it is true by default, desc:it is true by default-->true
  </enabled>
</proportionalpan>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.4.1.3 Control PTZ to pan and tilt

**Request URL**

PUT /ISAPI/PTZCtrl/channels/<channelID>/continuous?type=<type>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| type | enum | Module type. 1. Determine whether the device supports the optional parameter "type" by checking if /ISAPI/PTZCtrl/channels//capabilities returns continuousPanTiltType; 2. PTZLaser (laser PTZ), which indicates the control of the laser PTZ linked with the specified channel's PTZ camera |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, panning, tilting, and zooming positive direction, attr:version{req, string, protocolVersion}-->
  <pan>
    <!--opt, int, panning positive direction, range:[-100,100], desc:panning positive direction-->60
  </pan>
  <tilt>
    <!--opt, int, tilting positive direction, range:[-100,100], desc:tilting positive direction-->60
  </tilt>
  <zoom>
    <!--opt, int, zoom value, range:[-100,100], desc:the value represents the zoom speed. Positive value represents zooming in, and the larger the value,
the faster the speed. Negative value represents zooming out, and the smaller the value, the faster the speed. Value 0 represents stopping rotating-->60
  </zoom>
  <rotate>
    <!--opt, int, lens rotation value, range:[-100,100], desc:the value represents the rotation speed. Positive value represents clockwise rotation, and the
speed rises with the larger value. Negative value represents counterclockwise rotation, and the speed rises with the smaller value. Value 0 represents
stopping rotating-->60
  </rotate>
</PTZData>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:detailed information of custom error returned by device applications, used
for fast debugging-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

# 14.5 Vehicle Recognition

## 14.5.1 Motor Vehicle Detection

### 14.5.1.1 Get the total capability of vehicle capture and recognition

**Request URL**

GET /ISAPI/ITC/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<ITCCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isSupportITCStatus>
    <!--ro, opt, bool-->true
  </isSupportITCStatus>
  <isSupportIllegalDictionary>
    <!--ro, opt, bool-->true
  </isSupportIllegalDictionary>
  <isSupportITCSetUp>
    <!--ro, opt, bool-->true
  </isSupportITCSetUp>
  <isSupportTrafficParam>
    <!--ro, opt, bool-->true
  </isSupportTrafficParam>
  <isSupportManualCap>
    <!--ro, opt, bool-->true
  </isSupportManualCap>
  <isSupportIllegalUploadPic>
    <!--ro, opt, bool-->true
  </isSupportIllegalUploadPic>
  <isSupportContinueCap>
    <!--ro, opt, bool-->true
  </isSupportContinueCap>
  <isSupportWiper>
    <!--ro, opt, bool-->true
  </isSupportWiper>
  <isSupportPlateRecognitionParam>
    <!--ro, opt, bool-->true
  </isSupportPlateRecognitionParam>
  <isSupportSyncSignalOutput>
    <!--ro, opt, bool-->true
  </isSupportSyncSignalOutput>
  <isSupportImageMerge>
    <!--ro, opt, bool-->true
  </isSupportImageMerge>
  <isSupportCabinetParam>
    <!--ro, opt, bool-->true
  </isSupportCabinetParam>
  <isSupportCarFeatureParam>
    <!--ro, opt, bool-->true
  </isSupportCarFeatureParam>
  <isSupportIllegalCodeData>
    <!--ro, opt, bool-->true
  </isSupportIllegalCodeData>
  <isSupportSnapshot>
    <!--ro, opt, bool-->true
  </isSupportSnapshot>
  <isSupportSyncPower>
    <!--ro, opt, bool-->true
  </isSupportSyncPower>
  <isSupportNetStorage>
    <!--ro, opt, bool-->true
  </isSupportNetStorage>
  <isSupportLightCorrect>
    <!--ro, opt, bool-->true
  </isSupportLightCorrect>
  <isSupportEntranceShowCap>
    <!--ro, opt, bool-->true
  </isSupportEntranceShowCap>
  <isSupportAlgorithmsState>
    <!--ro, opt, bool-->true
  </isSupportAlgorithmsState>
  <isSupportAlgorithmsVersion>
    <!--ro, opt, bool-->true
  </isSupportAlgorithmsVersion>
  <isSupportVehicleDetection>
    <!--ro, opt, bool-->true
  </isSupportVehicleDetection>
  <isSupportDifferIllegalPic>
    <!--ro, opt, bool-->true
  </isSupportDifferIllegalPic>
  <LSTSCap>
    <!--ro, opt, object-->
    <isSupportLiveReceivePicture>
      <!--ro, opt, bool-->true
    </isSupportLiveReceivePicture>
    <isSupportTrafficLightStatus>
      <!--ro, opt, bool-->true
    </isSupportTrafficLightStatus>
    <isSupportTrafficStatistics>
      <!--ro, opt, bool-->true
    </isSupportTrafficStatistics>
  </LSTSCap>
</ITCCap>
```

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#### 14.5.1.2 Get the vehicle feature parameters capability

**Request URL**

GET /ISAPI/ITC/carFeatureParam/capabilities?channelID=<channelID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | optional, for specifying channel No. for multi-channel devices; when this node is not returned, it is channel 1 by default |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CarFeatureParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle feature parameters, attr:version{req, string, protocolVersion}-->
  <isSupportCarColor>
    <!--ro, opt, bool, whether to support vehicle color recognition-->true
  </isSupportCarColor>
  <isSupportCarLogo>
    <!--ro, opt, bool, whether to support vehicle logo recognition-->true
  </isSupportCarLogo>
  <isSupportSafeBelt>
    <!--ro, opt, bool, whether to support safety belt detection, desc:whether to support safety belt detection-->true
  </isSupportSafeBelt>
  <isSupportSunVisor>
    <!--ro, opt, bool, whether to support sun visor detection-->true
  </isSupportSunVisor>
  <isSupportHighPollutVehicle>
    <!--ro, opt, bool, whether to support yellow-label vehicle detection-->true
  </isSupportHighPollutVehicle>
  <isSupportDangerousVehicle>
    <!--ro, opt, bool, whether to support dangerous goods transport vehicle detection 1-->true
  </isSupportDangerousVehicle>
  <isSupportVehicleSubBrand>
    <!--ro, opt, bool, whether to support  vehicle sub-brand detection-->true
  </isSupportVehicleSubBrand>
  <isSupportEnvprosign>
    <!--ro, opt, bool, whether to support dangerous goods transport vehicle detection 2-->true
  </isSupportEnvprosign>
  <isSupportFaceDetect>
    <!--ro, req, bool, whether to enable face matting-->true
  </isSupportFaceDetect>
  <isSupportUphone>
    <!--ro, opt, bool, whether to support phone call detection-->true
  </isSupportUphone>
  <isSupportPendant>
    <!--ro, opt, bool, whether to support window hangings detection-->true
  </isSupportPendant>
  <isSupportPilotFace>
    <!--ro, req, bool, whether to support driver's face matting-->true
  </isSupportPilotFace>
  <isSupportCopilotFace>
    <!--ro, req, bool, whether to support front passenger's face picture matting-->true
  </isSupportCopilotFace>
  <isSupportNonmotorface>
    <!--ro, opt, bool, whether to support non-motor vehicle driver's face matting-->true
  </isSupportNonmotorface>
  <faceImgScale min="1" max="10">
    <!--ro, req, int, close-up face picture zooming ratio, range:[1,10], attr:min{req, int},max{req, int}-->1
  </faceImgScale>
  <faceImgRatio min="1" max="3">
    <!--ro, req, int, close-up face picture matting ratio, range:[1,3], attr:min{req, int},max{req, int}, desc:close-up face picture matting ratio-->1
  </faceImgRatio>
  <FaceImgInfoEx>
    <!--ro, opt, object, extended information of face matting-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable face matting types of close-up face picture, attr:opt{req, string}, desc:the node faceImgScale will be invalid
when this function is enabled-->false
    </enabled>
    <faceImgType opt="custom,IDCard">
      <!--ro, opt, enum, close-up face picture type, subType:string, attr:opt{req, string}, desc:"custom", "IDCard" (ID picture)-->custom
    </faceImgType>
    <faceCloseupRatio>
      <!--ro, opt, object, target capture size, desc:this node is valid when faceImgType is "custom"-->
      <upRatio min="0" max="100">
        <!--ro, req, int, enlarge from top, attr:min{req, int},max{req, int}-->20
      </upRatio>
      <downRatio min="0" max="100">
        <!--ro, req, int, enlarge from bottom, attr:min{req, int},max{req, int}-->20
      </downRatio>
      <leftRatio min="0" max="100">
        <!--ro, req, int, enlarge from left, attr:min{req, int},max{req, int}-->20
      </leftRatio>
      <rightRatio min="0" max="100">
```

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```
        <!--ro, req, int, enlarge from right, attr:min{req, int},max{req, int}-->20
      </rightRatio>
    </faceCloseupRatio>
  </FaceImgInfoEx>
  <faceImgOutput opt="none,overPic,alarmUpload,overPicAndAlarmUpload">
    <!--ro, opt, enum, close-up face picture output mode, subType:string, attr:opt{req, string}, desc:"none" (no output), "overPic" (overlay on scene
picture), "alarmUpload" (upload arm), "overPicAndAlarmUpload" (overlay on scene picture and upload arm)-->none
  </faceImgOutput>
  <isSupportDrawFaceRect>
    <!--ro, opt, bool, whether to overlay frame on face matting-->true
  </isSupportDrawFaceRect>
  <isSupportFaceImgBrightenEnhence>
    <!--ro, opt, bool, whether to support close-up face picture contrast enhancement-->true
  </isSupportFaceImgBrightenEnhence>
  <faceImgBrightenEnhenceLevel min="0" max="100">
    <!--ro, req, int, contrast enhancement level, range:[0,100], attr:min{req, int},max{req, int}-->0
  </faceImgBrightenEnhenceLevel>
  <tissueBoxEnable>
    <!--ro, opt, bool, whether to support enabling tissue box detection-->true
  </tissueBoxEnable>
  <frontChildEnable>
    <!--ro, opt, bool, whether to support enabling baby in arm detection-->true
  </frontChildEnable>
  <labelEnable>
    <!--ro, opt, bool, whether to support enabling label detection-->true
  </labelEnable>
  <decorationEnable>
    <!--ro, opt, bool, whether to support enabling decoration detection-->true
  </decorationEnable>
  <safeBeltSensitivity min="1" max="100">
    <!--ro, opt, int, safety belt detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:this node is valid when safeBeltEnabled is
"true"-->1
  </safeBeltSensitivity>
  <uphoneSensitivity min="1" max="100">
    <!--ro, opt, int, phone call detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:this node is valid when uphoneEnabled is
"true"-->1
  </uphoneSensitivity>
  <pendantSensitivity min="1" max="100">
    <!--ro, opt, int, window hangings detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:this node is valid when pendantEnabled is
"true"-->1
  </pendantSensitivity>
  <isSupportSmoking>
    <!--ro, opt, bool, whether to support smoking detection-->true
  </isSupportSmoking>
  <isSupportHelmet>
    <!--ro, opt, bool, whether to support helmet detection-->true
  </isSupportHelmet>
  <isSupportTwoWheelVehicle>
    <!--ro, opt, bool, whether to support manned two wheeler detection-->true
  </isSupportTwoWheelVehicle>
  <targetCutout>
    <!--ro, opt, object, target face matting-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable target face matting, attr:opt{req, string}-->true
    </enabled>
    <size>
      <!--ro, req, enum, target matting ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
    </size>
    <zoom>
      <!--ro, req, enum, zooming in/out ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
    </zoom>
  </targetCutout>
  <playMobilePhone>
    <!--ro, opt, bool, whether to enable driver using mobile phone detection-->true
  </playMobilePhone>
  <isSupportShed>
    <!--ro, opt, bool, whether to enable non-motor vehicle umbrella tent detection-->true
  </isSupportShed>
  <separatData>
    <!--ro, opt, bool, upload license plate No. and picture information separately and for multiple times-->true
  </separatData>
  <rudder opt="left,right">
    <!--ro, opt, enum, left/right rudder, subType:string, attr:opt{req, string}, desc:"left" (by default), "right"-->left
  </rudder>
  <helmetSensitivity>
    <!--ro, opt, int, helmet detection sensitivity, range:[1,100]-->0
  </helmetSensitivity>
  <helmetDaySensitivity min="1" max="100">
    <!--ro, req, int, Sensitivity for Not Wearing Helmet at Daytime, attr:min{req, int},max{req, int}-->0
  </helmetDaySensitivity>
  <helmetNightSensitivity min="1" max="100">
    <!--ro, req, int, Sensitivity for Not Wearing Helmet at Night, attr:min{req, int},max{req, int}-->0
  </helmetNightSensitivity>
  <shedSensitivity min="1" max="100">
    <!--ro, opt, int, non-motor vehicle umbrella tent detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->0
  </shedSensitivity>
  <twoWheelVehicleSensitivity min="1" max="100">
    <!--ro, opt, int, manned two wheeler detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->0
  </twoWheelVehicleSensitivity>
  <isSupportFocedCopilotFace>
    <!--ro, opt, bool, whether to mandate face matting for front passenger-->true
  </isSupportFocedCopilotFace>
  <smokingSensitivity min="0" max="100">
```

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```
    <!--ro, opt, int, smoking detection sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
  </smokingSensitivity>
  <CarWindowFeature>
    <!--ro, opt, object, vehicle window feature detection-->
    <tempPlateEnable opt="true,false">
      <!--ro, opt, bool, whether to enable detection of temporary license plate on the window, attr:opt{req, string}-->true
    </tempPlateEnable>
    <tempPlateSensitivity min="0" max="100">
      <!--ro, opt, int, detection sensitivity of temporary license plate on the window, range:[0,100], attr:min{req, int},max{req, int}-->1
    </tempPlateSensitivity>
    <passCardEnable opt="true,false">
      <!--ro, opt, bool, whether to enable detection of entry & exit pass on the window, attr:opt{req, string}-->true
    </passCardEnable>
    <passCardSensitivity min="0" max="100">
      <!--ro, opt, int, detection sensitivity of entry & exit pass on the window, range:[0,100], attr:min{req, int},max{req, int}-->1
    </passCardSensitivity>
    <carCardEnable opt="true,false">
      <!--ro, opt, bool, whether to enable detection of cards attached to the window, attr:opt{req, string}-->true
    </carCardEnable>
    <carCardSensitivity min="0" max="100">
      <!--ro, opt, int, detection sensitivity of cards attached to the window, range:[0,100], attr:min{req, int},max{req, int}-->1
    </carCardSensitivity>
  </CarWindowFeature>
  <CarBodyFeature>
    <!--ro, opt, object, bodywork feature detection-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable bodywork feature detection, attr:opt{req, string}-->true
    </enabled>
  </CarBodyFeature>
  <vehicleUseEnable opt="true,false">
    <!--ro, opt, bool, whether to enable vehicle purpose detection, attr:opt{req, string}-->true
  </vehicleUseEnable>
  <helmetCheckMode opt="checkOutFirst,checkAccurateFirst,custorm">
    <!--ro, opt, enum, helmet detection mode, subType:string, dep:and,{$.CarFeatureParam.isSupportHelmet,eq,true}, attr:opt{req, string},
desc:1. "checkOutFirst", "checkAccurateFirst"; it is "checkAccurateFirst" by default when this node is not returned
2. during detection of not wearing helmet: "checkOutFirst" is for higher detection rate but comes with higher false alarm rate which requires manual
verification; "checkAccurateFirst" is for high-quality detection but comes with higher under-detection rate-->checkOutFirst
  </helmetCheckMode>
  <reflectiveStripeEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable reflective stripe detection, attr:opt{req, string}-->true
  </reflectiveStripeEnabled>
</CarFeatureParam>
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| CarFeatureParam | [Message content] | application/xml | -- | -- | -- |

**Notes:**

newly added reflectiveStripeEnabled as reflective stripe detection for vehicle feature parameters

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

### 14.5.1.3 Get the vehicle feature configuration parameters

**Request URL**

GET /ISAPI/ITC/carFeatureParam?channelID=<channelID>

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | optional, for specifying channel No. for multi-channel devices; when this node is not returned, it is channel 1 by default |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CarFeatureParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle feature parameters, attr:version{req, string, protocolVersion}-->
  <colorEnabled>
    <!--ro, req, bool, vehicle color recognition-->true
  </colorEnabled>
  <carLogoEnabled>
    <!--ro, req, bool, vehicle log recognition-->true
  </carLogoEnabled>
  <sunVisorEnabled>
    <!--ro, req, bool, sun visor detection-->true
  </sunVisorEnabled>
  <faceDetectEnabled>
    <!--ro, req, bool, face picture matting-->true
  </faceDetectEnabled>
  <uphoneEnabled>
    <!--ro, opt, bool, phone call detection-->true
  </uphoneEnabled>
  <pendantEnabled>
    <!--ro, opt, bool, window hangings detection-->true
  </pendantEnabled>
  <tissueBoxEnable>
    <!--ro, opt, bool, tissue box detection-->true
  </tissueBoxEnable>
  <frontChildEnable>
    <!--ro, opt, bool, baby in arm detection-->true
  </frontChildEnable>
  <labelEnable>
    <!--ro, opt, bool, label detection-->true
  </labelEnable>
  <uphoneSensitivity>
    <!--ro, opt, int, phone call detection sensitivity, range:[1,100]-->1
  </uphoneSensitivity>
  <pendantSensitivity>
    <!--ro, opt, int, window hangings detection sensitivity, range:[1,100]-->1
  </pendantSensitivity>
  <decorationEnable>
    <!--ro, opt, bool, decoration detection-->true
  </decorationEnable>
  <OutputCtrl>
    <!--ro, opt, object, face matting-->
    <hostFaceEnabled>
      <!--ro, req, bool, driver's face matting-->true
    </hostFaceEnabled>
    <viceFaceEnabled>
      <!--ro, req, bool, front passenger's face matting-->true
    </viceFaceEnabled>
    <nonMotorFaceEnabled>
      <!--ro, req, bool, non-motor vehicle driver's face matting-->true
    </nonMotorFaceEnabled>
    <faceImgScale>
      <!--ro, req, int, close-up face picture zooming ratio, range:[1,10]-->1
    </faceImgScale>
    <faceImgRatio>
      <!--ro, req, enum, close-up face picture cutting ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
    </faceImgRatio>
    <faceImgOutputMode>
      <!--ro, req, enum, close-up face picture output mode, subType:string, desc:"none" (no output), "overPic" (overlay on scene picture), "alarmUpload"
(upload arm), "overPicAndAlarmUpload" (overlay on scene picture and upload arm)-->none
    </faceImgOutputMode>
    <DrawFaceRectEnabled>
      <!--ro, req, bool, whether to overlay frame on face matting-->true
    </DrawFaceRectEnabled>
    <faceImgBrightenEnhenceEnabled>
      <!--ro, req, bool, close-up face picture contrast enhancement-->true
    </faceImgBrightenEnhenceEnabled>
    <faceImgBrightenEnhenceLevel>
      <!--ro, req, int, contrast enhancement level, range:[0,100]-->0
    </faceImgBrightenEnhenceLevel>
    <faceCopyIllegal>
      <!--ro, opt, bool, whether need enable face matting for violation picture-->true
    </faceCopyIllegal>
    <forcedViceFaceEnabled>
      <!--ro, opt, bool, whether to mandate face matting for front passenger-->true
    </forcedViceFaceEnabled>
  </OutputCtrl>
```

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```
<smokingEnabled>
  <!--ro, req, bool, smoking detection-->true
</smokingEnabled>
<helmetEnabled>
  <!--ro, opt, bool, helmet detection-->true
</helmetEnabled>
<twoWheelVehicleEnabled>
  <!--ro, opt, bool, manned two wheeler detection-->true
</twoWheelVehicleEnabled>
<playMobilePhone>
  <!--ro, opt, bool, whether to enable driver using mobile phone detection-->true
</playMobilePhone>
<isSupportShed>
  <!--ro, opt, bool, whether to enable non-motor vehicle umbrella tent detection-->true
</isSupportShed>
<separatData>
  <!--ro, opt, bool, upload license plate No. and picture information separately and for multiple times-->true
</separatData>
<rudder>
  <!--ro, opt, enum, left/right rudder, subType:string, desc:"left", "right"-->left
</rudder>
<helmetSensitivity>
  <!--ro, opt, int, helmet detection sensitivity, range:[1,100]-->0
</helmetSensitivity>
<helmetDaySensitivity>
  <!--ro, req, int, sensitivity for detecting not wearing helmet at daytime-->0
</helmetDaySensitivity>
<helmetNightSensitivity>
  <!--ro, req, int, sensitivity for detecting not wearing helmet at night-->0
</helmetNightSensitivity>
<shedSensitivity>
  <!--ro, opt, int, non-motor vehicle umbrella tent detection sensitivity, range:[1,100]-->0
</shedSensitivity>
<twoWheelVehicleSensitivity>
  <!--ro, opt, int, manned two wheeler detection sensitivity, range:[1,100]-->0
</twoWheelVehicleSensitivity>
<smokingSensitivity>
  <!--ro, opt, int, smoking detection sensitivity, range:[0,100]-->0
</smokingSensitivity>
<CarWindowFeature>
  <!--ro, opt, object, vehicle window feature detection-->
  <tempPlateEnable>
    <!--ro, opt, bool, whether to enable detection of temporary license plate on the window, desc:detect whether there is temporary license plate on the
window-->true
  </tempPlateEnable>
  <tempPlateSensitivity>
    <!--ro, opt, int, detection sensitivity of temporary license plate on the window, range:[0,100]-->1
  </tempPlateSensitivity>
  <passCardEnable>
    <!--ro, opt, bool, whether to enable detection of entry & exit pass on the window, desc:whether there is entry & exit pass on the window-->true
  </passCardEnable>
  <passCardSensitivity>
    <!--ro, opt, int, detection sensitivity of entry & exit pass on the window, range:[0,100]-->1
  </passCardSensitivity>
  <carCardEnable>
    <!--ro, opt, bool, whether to enable detection of cards attached to the window, desc:whether there is any card (business cards, flyers, etc) attached
to the window-->true
  </carCardEnable>
  <carCardSensitivity>
    <!--ro, opt, int, detection sensitivity of cards attached to the window, range:[0,100]-->1
  </carCardSensitivity>
</CarWindowFeature>
<CarBodyFeature>
  <!--ro, opt, object, bodywork feature detection-->
  <enabled>
    <!--ro, opt, bool, whether to enable bodywork feature detection, desc:whether the vehicle is with spare tire, with sunroof, with roof rack, painted,
people sticking out of the sunroof, covered (for dump truck)-->true
  </enabled>
</CarBodyFeature>
<vehicleUseEnable>
  <!--ro, opt, bool, whether to enable vehicle purpose detection, desc:recognize bus, school bus, taxi, and ambulance-->true
</vehicleUseEnable>
<helmetCheckMode>
  <!--ro, opt, enum, helmet detection mode, subType:string, dep:and,{$.CarFeatureParam.helmetEnabled,eq,true},
desc:1. "checkOutFirst", "checkAccurateFirst"; it is "checkAccurateFirst" by default when this node is not returned
2. during detection of not wearing helmet: "checkOutFirst" is for higher detection rate but comes with higher false alarm rate which requires manual
verification; "checkAccurateFirst" is for high-quality detection but comes with higher under-detection rate-->checkOutFirst
</helmetCheckMode>
<reflectiveStripeEnabled>
  <!--ro, opt, bool, whether to enable reflective stripe detection-->true
</reflectiveStripeEnabled>
</CarFeatureParam>
```

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#### 14.5.1.4 Set the vehicle feature configuration parameters

**Request URL**

PUT /ISAPI/ITC/carFeatureParam?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | optional, for specifying channel No. for multi-channel devices; when this node is not returned, it is channel 1 by default |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CarFeatureParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle feature parameters, attr:version{req, string, protocolVersion}-->
  <colorEnabled>
    <!--req, bool, whether to enable vehicle color recognition-->true
  </colorEnabled>
  <carLogoEnabled>
    <!--req, bool, whether to enable vehicle log recognition-->true
  </carLogoEnabled>
  <pilotSafebeltEnabled>
    <!--req, bool, whether to enable safety belt detection-->true
  </pilotSafebeltEnabled>
  <sunVisorEnabled>
    <!--req, bool, Sun Visor Detection-->true
  </sunVisorEnabled>
  <envprosignEnabled>
    <!--req, bool, Dangerous Goods Transport Vehicle Detection-->true
  </envprosignEnabled>
  <dangmarkEnabled>
    <!--req, bool, whether to enable dangerous goods transport vehicle detection-->true
  </dangmarkEnabled>
  <faceDetectEnabled>
    <!--req, bool, Face Picture Matting-->true
  </faceDetectEnabled>
  <uphoneEnabled>
    <!--opt, bool, Phone Call Detection-->true
  </uphoneEnabled>
  <pendantEnabled>
    <!--opt, bool, whether to enable window hangings detection-->true
  </pendantEnabled>
  <tissueBoxEnable>
    <!--opt, bool, Tissue Box Detection-->true
  </tissueBoxEnable>
  <frontChildEnable>
    <!--opt, bool, Baby in Arm Detection-->true
  </frontChildEnable>
  <labelEnable>
    <!--opt, bool, Label Detection-->true
  </labelEnable>
  <safebeltSensitivity>
    <!--opt, int, safety belt detection sensitivity, range:[1,100]-->1
  </safebeltSensitivity>
  <uphoneSensitivity>
    <!--opt, int, Phone Call Detection, range:[1,100]-->1
  </uphoneSensitivity>
  <pendantSensitivity>
    <!--opt, int, window hangings detection sensitivity, range:[1,100]-->1
  </pendantSensitivity>
  <decorationEnable>
    <!--opt, bool, Decoration Detection-->true
  </decorationEnable>
  <OutputCtrl>
    <!--opt, object, Face Picture Matting-->
    <hostFaceEnabled>
      <!--req, bool, Driver's Face Picture Matting-->true
    </hostFaceEnabled>
    <viceFaceEnabled>
      <!--req, bool, Passenger's Face Picture Matting-->true
    </viceFaceEnabled>
    <nonMotorFaceEnabled>
      <!--req, bool, Pedestrian's and Non-Motor Vehicle User's Face Picture Matting-->true
    </nonMotorFaceEnabled>
    <faceImgScale>
      <!--req, int, face close-up picture cutting ratio, range:[1,10]-->1
    </faceImgScale>
    <faceImgRatio>
      <!--req, enum, Face Close-up Picture Cutting Ratio, subType:int, desc:face close-up picture zooming ratio,which ranges from 1 to 3-->1
    </faceImgRatio>
    <FaceImgInfoEx>
      <!--ro, opt, object, extended information of face matting-->
      <enabled>
        <!--ro, req, bool, whether to enable face matting types of close-up face picture, desc:the node faceImgScale will be invalid when this function is
enabled-->false
      </enabled>
      <faceImgType>
        <!--ro, req, enum, close-up face picture type, subType:string, desc:"custom", "IDCard" (ID picture)-->IDCard
      </faceImgType>
      <faceCloseupRatio>
        <!--ro, opt, object, target capture size, desc:this node is valid when faceImgType is "custom"-->
```

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```
          <upRatio>
            <!--ro, req, int, enlarge from top, range:[0,100]-->20
          </upRatio>
          <downRatio>
            <!--ro, req, int, enlarge from bottom, range:[0,100]-->20
          </downRatio>
          <leftRatio>
            <!--ro, req, int, enlarge from left, range:[0,100]-->20
          </leftRatio>
          <rightRatio>
            <!--ro, req, int, enlarge from right, range:[0,100]-->20
          </rightRatio>
        </faceCloseupRatio>
      </FaceImgInfoEx>
      <faceImgOutputMode>
        <!--req, enum, close-up face picture output mode, subType:string, desc:"none" (no output), "overPic" (overlay on scene picture), "alarmUpload" (upload
arm), "overPicAndAlarmUpload" (overlay on scene picture and upload arm)-->none
      </faceImgOutputMode>
      <DrawFaceRectEnabled>
        <!--req, bool, whether to overlay frame on face matting-->true
      </DrawFaceRectEnabled>
      <faceImgBrightenEnhenceEnabled>
        <!--req, bool, close-up face picture contrast enhancement-->true
      </faceImgBrightenEnhenceEnabled>
      <faceImgBrightenEnhenceLevel>
        <!--req, int, contrast enhancement level, range:[0,100]-->0
      </faceImgBrightenEnhenceLevel>
      <faceCopyIllegal>
        <!--opt, bool, whether need enable face matting for violation picture-->true
      </faceCopyIllegal>
      <forcedViceFaceEnabled>
        <!--opt, bool, whether to mandate face matting for front passenger-->true
      </forcedViceFaceEnabled>
    </OutputCtrl>
    <smokingEnabled>
      <!--req, bool, smoking detection-->true
    </smokingEnabled>
    <perfumeBoxEnabled>
      <!--opt, bool, perfume box detection-->true
    </perfumeBoxEnabled>
    <helmetEnabled>
      <!--opt, bool, helmet detection-->true
    </helmetEnabled>
    <twoWheelVehicleEnabled>
      <!--opt, bool, manned two wheeler detection-->true
    </twoWheelVehicleEnabled>
    <threeWheelVehicleEnabled>
      <!--opt, bool, manned tricycle detection-->true
    </threeWheelVehicleEnabled>
    <targetCutout>
      <!--opt, object, target face matting-->
      <enabled>
        <!--opt, bool, whether to enable target face matting-->true
      </enabled>
      <size>
        <!--req, enum, target matting ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
      </size>
      <zoom>
        <!--req, enum, zooming in/out ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
      </zoom>
    </targetCutout>
    <playMobilePhone>
      <!--opt, bool, whether to enable driver using mobile phone detection-->true
    </playMobilePhone>
    <isSupportShed>
      <!--opt, bool, whether to enable non-motor vehicle umbrella tent detection-->true
    </isSupportShed>
    <separatData>
      <!--opt, bool, upload license plate No. and picture information separately and for multiple times-->true
    </separatData>
    <rudder>
      <!--opt, enum, left/right rudder, subType:string, desc:"left", "right"-->left
    </rudder>
    <helmetSensitivity>
      <!--opt, int, helmet detection sensitivity, range:[1,100]-->0
    </helmetSensitivity>
    <helmetDaySensitivity>
      <!--ro, req, int, Sensitivity for Not Wearing Helmet at Daytime-->0
    </helmetDaySensitivity>
    <helmetNightSensitivity>
      <!--ro, req, int, Sensitivity for Not Wearing Helmet at Night-->0
    </helmetNightSensitivity>
    <shedSensitivity>
      <!--opt, int, non-motor vehicle umbrella tent detection sensitivity, range:[1,100]-->0
    </shedSensitivity>
    <twoWheelVehicleSensitivity>
      <!--opt, int, manned two wheeler detection sensitivity, range:[1,100]-->0
    </twoWheelVehicleSensitivity>
    <smokingSensitivity>
      <!--opt, int, smoking detection sensitivity, range:[0,100]-->0
    </smokingSensitivity>
    <CarWindowFeature>
      <!--opt, object, vehicle window feature detection-->
      <tempPlateEnable>
```

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```
      <!--opt, bool, whether to enable detection of temporary license plate on the window, desc:detect whether there is temporary license plate on the window-->true
    </tempPlateEnable>
    <tempPlateSensitivity>
      <!--opt, int, detection sensitivity of temporary license plate on the window, range:[0,100]-->1
    </tempPlateSensitivity>
    <passCardEnable>
      <!--opt, bool, whether to enable detection of entry & exit pass on the window, desc:whether there is entry & exit pass on the window-->true
    </passCardEnable>
    <passCardSensitivity>
      <!--opt, int, detection sensitivity of entry & exit pass on the window, range:[0,100]-->1
    </passCardSensitivity>
    <carCardEnable>
      <!--opt, bool, whether to enable detection of cards attached to the window, desc:whether there is any card (business cards, flyers, etc.) attached to the window-->true
    </carCardEnable>
    <carCardSensitivity>
      <!--opt, int, detection sensitivity of cards attached to the window, range:[0,100]-->1
    </carCardSensitivity>
  </CarWindowFeature>
  <CarBodyFeature>
    <!--opt, object, bodywork feature detection-->
    <enabled>
      <!--opt, bool, whether to enable bodywork feature detection, desc:whether the vehicle is with spare tire, with sunroof, with roof rack, painted, people sticking out of the sunroof, covered (for dump truck)-->true
    </enabled>
  </CarBodyFeature>
  <vehicleUseEnable>
    <!--opt, bool, whether to enable vehicle purpose detection, desc:recognize bus, school bus, taxi, and ambulance-->true
  </vehicleUseEnable>
  <channelId>
    <!--ro, opt, int, channel No., range:[1,5]-->1
  </channelId>
  <helmetCheckMode>
    <!--opt, enum, helmet detection mode, subType:string, dep:and,{$.CarFeatureParam.helmetEnabled,eq,true}, desc:1. "checkOutFirst", "checkAccurateFirst"; it is "checkAccurateFirst" by default when this node is not returned
2. during detection of not wearing helmet: "checkOutFirst" is for higher detection rate but comes with higher false alarm rate which requires manual verification; "checkAccurateFirst" is for high-quality detection but comes with higher under-detection rate-->checkOutFirst
  </helmetCheckMode>
  <reflectiveStripeEnabled>
    <!--ro, opt, bool, whether to enable reflective stripe detection-->true
  </reflectiveStripeEnabled>
  <illegalMannedNonMotorVehicle>
    <!--opt, object-->
    <peopleNumThreshold>
      <!--opt, int, range:[2,3]-->2
    </peopleNumThreshold>
  </illegalMannedNonMotorVehicle>
</CarFeatureParam>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.5.1.5 Get the capability of arming parameters for traffic violations capture

### Request URL

GET /ISAPI/ITC/illegalSchedules/capabilities?format=json&channelID=<channelID>

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
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```
{
    "eventPriorityEnable": {
    /*ro, opt, object, whether to enable violation priority*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "IllegalScheduleListCap": {
    /*ro, req, object, list of arming parameters for traffic violations capture, desc:this node is only available for ANPR violation information, related node is illegalInfo*/
        "@size":  1,
        /*ro, req, int, the maximum number supported in a single capture*/
        "Schedule": {
        /*ro, opt, object, arming parameters*/
            "eventType":  "post,driveLine",
            /*ro, req, string, event type, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock", "construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing), "changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm), "notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop", "animalsOccupy", "suonaDet" (whistle), "roarDet"*/
            "noPlateCaptureEnable": {
            /*ro, opt, object, whether to enable no plate vehicle capture*/
                "@opt": [true, false]
                /*ro, opt, array, options, subType:bool*/
            },
            "eventPriority": {
            /*ro, opt, object, event capture priority, desc:value range 0 to 10; the higher the value, the higher the priority; when the value is 0, this node will not have priority*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  10
                /*ro, req, int, the maximum value*/
            },
            "TimeBlockList": {
            /*ro, opt, object, list of time periods*/
                "@size":  7,
                /*ro, req, int, the maximum number supported in a single capture*/
                "TimeBlock": {
                /*ro, opt, object, time period*/
                    "dayOfWeek": {
                    /*ro, req, object, day of a week, desc:Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  7
                        /*ro, req, int, the maximum value*/
                    },
                    "TimeRangeList": {
                    /*ro, opt, object, list of upload intervals*/
                        "@size":  4,
                        /*ro, req, int, the maximum number supported in a single capture*/
                        "TimeRange": {
                        /*ro, opt, object, time interval*/
                            "beginTime":  "00:00:00+08:00",
                            /*ro, req, time, start time*/
                            "endTime":  "00:00:00+08:00"
                            /*ro, req, time, end time*/
                        }
                    }
                }
            },
            "RelatedLanList": {
            /*ro, opt, object, list of linked lane information*/
                "@size":  6,
                /*ro, opt, int, size*/
                "RelatedLan": {
                /*ro, opt, object, linked lane information*/
                    "lanID": {
                    /*ro, opt, object, whether to enable linked lane No.*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  6
                        /*ro, req, int, the maximum value*/
                    }
                }
            }
        }
    }
}
```

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### 14.5.1.6 Get the arming parameters for traffic violation capture

#### Request URL

GET /ISAPI/ITC/illegalSchedules?format=json&channelID=<channelID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

None

#### Response Message

```
{
    "eventPriorityEnable":  true,
    /*ro, req, bool, whether to enable violation priority*/
    "IllegalScheduleList": [
    /*ro, req, array, arming time of traffic violation capture, subType:object, desc:this node is only available for ANPR violation information, which is related to the node illegalInfo*/
        {
            "Schedule": {
            /*ro, opt, object, arming schedules*/
                "eventType":  "post",
                /*ro, req, enum, event type, subType:string, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock", "construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing), "changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm), "notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop", "animalsOccupy", "suonaDet" (whistle), "roarDet"*/
                "noPlateCaptureEnable":  true,
                /*ro, opt, bool, whether to enable no plate vehicle capture*/
                "eventPriority":  0,
                /*ro, opt, int, event capture priority, range:[0,10], desc:value ranges from 0 to 10; the higher the value, the higher the priority; when the value is 0, this event will not have priority*/
                "TimeBlockList": [
                /*ro, opt, array, list of time periods, subType:object, range:[0,7]*/
                    {
                        "TimeBlock": {
                        /*ro, opt, object, time period*/
                            "dayOfWeek":  1,
                            /*ro, req, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
                            "TimeRangeList": [
                            /*ro, opt, array, list of time intervals, subType:object, range:[0,4], desc:4 time periods can be configured for a day*/
                                {
                                    "TimeRange": {
                                    /*ro, opt, object, time interval*/
                                        "beginTime":  "00:00:00+08:00",
                                        /*ro, req, time, start time*/
                                        "endTime":  "00:00:00+08:00"
                                        /*ro, req, time, end time*/
                                    }
                                }
                            ]
                        }
                    }
                ],
                "RelatedLanList": [
                /*ro, opt, array, list of linked lane information, subType:object, range:[0,6]*/
                    {
                        "RelatedLan": {
                        /*ro, opt, object, linked lane information*/
                            "lanID":  1
                            /*ro, opt, int, whether to enable linked lane No.*/
                        }
                    }
                ]
            }
        }
    ]
}
```

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### 14.5.1.7 Set the arming parameters for traffic violations capture

#### Request URL

PUT /ISAPI/ITC/illegalSchedules?format=json&channelID=<channelID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
{
    "eventPriorityEnable":  true,
    /*req, bool, whether to enable violation priority*/
    "IllegalScheduleList": [
    /*req, array, arming time of traffic violations capture, subType:object, desc:this node is only available for ANPR violation information, related node
is illegalInfo*/
        {
            "Schedule": {
            /*opt, object, arming schedules*/
                "eventType":  "post",
                /*req, enum, event type, subType:string, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving),
"banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt",
"uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-
motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on
non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction
guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion),
"leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight",
"leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red
light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock",
"construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing),
"changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm),
"notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield
at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to
vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when
turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop",
"animalsOccupy", "suonaDet" (whistle), "roarDet"*/
                "noPlateCaptureEnable":  true,
                /*opt, bool, whether to enable no plate vehicle capture*/
                "eventPriority":  0,
                /*opt, int, event capture priority, range:[0,10], desc:value range 0 to 10; the higher the value, the higher the priority; when the value is
0, this node will not have priority*/
                "TimeBlockList": [
                /*opt, array, list of time periods, subType:object, range:[0,7]*/
                    {
                        "TimeBlock": {
                        /*opt, object, time period*/
                            "dayOfWeek":  1,
                            /*req, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7
(Sunday)*/
                            "TimeRangeList": [
                            /*opt, array, list of time intervals, subType:object, range:[0,4], desc:4 time periods can be configured for a day*/
                                {
                                    "TimeRange": {
                                    /*opt, object, time interval*/
                                        "beginTime":  "00:00:00+08:00",
                                        /*req, time, start time*/
                                        "endTime":  "00:00:00+08:00"
                                        /*req, time, end time*/
                                    }
                                }
                            ]
                        }
                    }
                ],
                "RelatedLanList": [
                /*opt, array, list of linked lane information, subType:object, range:[0,6]*/
                    {
                        "RelatedLan": {
                        /*opt, object, linked lane information*/
                            "lanID":  1
                            /*opt, int, whether to enable linked lane No.*/
                        }
                    }
                ]
            }
        }
    ]
}
```

#### Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

#### 14.5.1.8 Set the composite picture parameters

**Request URL**

PUT /ISAPI/ITC/imageMerge

**Query Parameter**

None

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMerge xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <isMerge>
    <!--req, bool-->true
  </isMerge>
  <oneMergeType>
    <!--req, enum, subType:int-->101
  </oneMergeType>
  <twoMergeType>
    <!--req, enum, subType:int-->201
  </twoMergeType>
  <threeMergeType>
    <!--req, enum, subType:int-->301
  </threeMergeType>
  <fourMergeType>
    <!--req, enum, subType:int-->401
  </fourMergeType>
  <closeupScale min="0" max="100">
    <!--opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </closeupScale>
  <closeupIndex min="0" max="5">
    <!--opt, int, range:[0,5], attr:min{req, int},max{req, int}-->0
  </closeupIndex>
  <positionOffset>
    <!--opt, int, range:[0,1535]-->50
  </positionOffset>
  <featureEnable>
    <!--opt, bool-->true
  </featureEnable>
</ImageMerge>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.5.1.9 Get ANPR configuration capability

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### Request URL

GET /ISAPI/ITC/plateRecognitionParam/capabilities?channelID=<channelID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelPlateRecognitionParam is "true" returned from GET /ISAPI/ITC/capabilities). |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognitionParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ANPR parameters, attr:version{req, string, protocolVersion}-->
  <isSupportdefaultCHN>
    <!--ro, opt, string, province name-->test
  </isSupportdefaultCHN>
  <isSupportfrontPlateReco>
    <!--ro, opt, bool, whether to support front license plate recognition-->true
  </isSupportfrontPlateReco>
  <isSupportrearPlateReco>
    <!--ro, opt, bool, whether to support rear license plate recognition-->true
  </isSupportrearPlateReco>
  <isSupportsmallPlateReco>
    <!--ro, opt, bool, whether to support small license plate recognition-->true
  </isSupportsmallPlateReco>
  <isSupportlargePlateReco>
    <!--ro, opt, bool, whether to support large license plate recognition-->true
  </isSupportlargePlateReco>
  <isSupportfarmVehicle>
    <!--ro, opt, bool, whether to support agricultural vehicle recognition-->true
  </isSupportfarmVehicle>
  <isSupportmotor>
    <!--ro, opt, bool, whether to support motorcycle recognition-->true
  </isSupportmotor>
  <isSupportfuzzyDisc>
    <!--ro, opt, bool, whether to support fuzzy recognition-->true
  </isSupportfuzzyDisc>
  <isSupportmicroPlateReg>
    <!--ro, opt, bool, whether to support tiny license plate recognition-->true
  </isSupportmicroPlateReg>
  <isSupportcAPlateRegEnabled>
    <!--ro, opt, bool, whether to support civil aviation license plate recognition-->true
  </isSupportcAPlateRegEnabled>
  <isSupporttiltPlateRegEnabled>
    <!--ro, opt, bool, whether to support tilted license plate recognition-->true
  </isSupporttiltPlateRegEnabled>
  <isSupportsuperPlateRecoEnabled>
    <!--ro, opt, bool, whether to support oversized license plate recognition-->true
  </isSupportsuperPlateRecoEnabled>
  <isSupportEmbassyPlateRecoEnabled>
    <!--ro, opt, bool, whether to support embassy license plate recognition-->true
  </isSupportEmbassyPlateRecoEnabled>
  <isSupportfrontAndRearReg>
    <!--ro, opt, bool, whether to enable front license plate recognition and rear license plate recognition-->true
  </isSupportfrontAndRearReg>
  <nation opt="EU,ER,EUandCIS,ME,APAC,AFandAM,All">
    <!--ro, opt, enum, country/region, subType:string, attr:opt{req, string}, desc:"ER" (Russian-speaking region), "EU" (Europe), "ME" (Middle East), "APAC"
(Asia-Pacific region), "AFandAM" (Africa and America), "All" (all regions), "EUandCIS" (Europe and Russia)-->ER
  </nation>
  <countryIndex opt="1,2,3,4,5,6,7,8,12,14,17,18,19,20,23,39,44,55">
    <!--ro, opt, int, country/region index, attr:opt{req, string}-->0
  </countryIndex>
  <nonMotorPlateCityRecoEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable belonged city recognition of non-motor vehicle license plate, attr:opt{req, string}-->true
  </nonMotorPlateCityRecoEnabled>
  <city>
    <!--ro, opt, string, belonged city of license plate-->test
  </city>
  <fakePlateFilterEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->false
  </fakePlateFilterEnabled>
  <citySeparatorEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </citySeparatorEnabled>
  <fakePlateDetectionUploadEnabled opt="true,false">
    <!--ro, opt, bool, dep:and,{$.PlateRecognitionParam.fakePlateFilterEnabled,eq,true}, attr:opt{req, string}-->false
  </fakePlateDetectionUploadEnabled>
```

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```
<provinceThresholds min="0" max="100">
  <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
</provinceThresholds>
<secondLicensePlateRegEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</secondLicensePlateRegEnabled>
<yellowGreenPlateRecoRule opt="yellow,green,yellowGreen,yellowGreenBicolor">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->yellow
</yellowGreenPlateRecoRule>
<nonStandardPlateIllegalHandleRules opt="none,filter,wipePlate">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
</nonStandardPlateIllegalHandleRules>
<nonStandardPlatePostHandleRules opt="none,filter,wipePlate">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
</nonStandardPlatePostHandleRules>
<nonStandardPlateThresholds min="0" max="100">
  <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
</nonStandardPlateThresholds>
<nonStandardPlateMinCorrectFactor min="0" max="100">
  <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
</nonStandardPlateMinCorrectFactor>
<nonStandardPlateMaxCorrectFactor min="0" max="100">
  <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
</nonStandardPlateMaxCorrectFactor>
<nonStandardVehicleCorrectFactor min="0" max="100">
  <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
</nonStandardVehicleCorrectFactor>
<nonStandardPlateRegEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</nonStandardPlateRegEnabled>
<threePlateRecoEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</threePlateRecoEnabled>
<newEnergyGreenPlateRecoRule opt="green,newEnergyGreen,gradationGreen">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->green
</newEnergyGreenPlateRecoRule>
<customPlateRule>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <customPlateColorEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </customPlateColorEnabled>
  <customPlateColor opt="black,blue,green,white,yellow">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->black
  </customPlateColor>
  <wildcardRuleList size="10">
    <!--ro, opt, object, attr:size{req, int}-->
    <wildcardRule>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <wildcard min="0" max="16">
        <!--ro, opt, string, range:[0,16], attr:min{req, int},max{req, int}-->蓝园A?!!**
      </wildcard>
    </wildcardRule>
  </wildcardRuleList>
</customPlateRule>
<plateMappingRule>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <plateMappingList size="64">
    <!--ro, opt, array, subType:object, range:[0,64], attr:size{req, int}-->
    <plateMapping>
      <!--ro, opt, object-->
      <detectPlateNo min="0" max="16">
        <!--ro, opt, string, range:[0,16], attr:min{req, int},max{req, int}-->浙A12348
      </detectPlateNo>
      <replacePlateNo min="0" max="16">
        <!--ro, opt, string, range:[0,16], attr:min{req, int},max{req, int}-->浙A1234B
      </replacePlateNo>
    </plateMapping>
  </plateMappingList>
</plateMappingRule>
<primaryAndSecondaryPlate>
  <!--ro, opt, array, subType:object-->
  <enabled>
    <!--ro, opt, array, subType:object-->
    <mainlandPlateEnable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </mainlandPlateEnable>
    <hongKongPlateEnable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </hongKongPlateEnable>
    <maCaoPlateEnable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </maCaoPlateEnable>
  </enabled>
  <plateRegionType>
```

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```
      <!--ro, opt, array, subType:object-->
      <mainAreaOption opt="mainland,hongKong,maCao">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->mainland
      </mainAreaOption>
      <secondAreaOption opt="mainland,hongKong,maCao">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->hongKong
      </secondAreaOption>
    </plateRegionType>
  </primaryAndSecondaryPlate>
</PlateRecognitionParam>
```

### 14.5.1.10 Set ANPR parameters

#### Request URL

PUT /ISAPI/ITC/plateRecognitionParam?channelID=<channelID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognitionParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ANPR parameters, attr:version{req, string, protocolVersion}-->
  <defaultCHN>
    <!--req, string, province/state abbreviation-->test
  </defaultCHN>
  <frontPlateRecoEnabled>
    <!--opt, bool, front license plate recognition, desc:front license plate recognition-->true
  </frontPlateRecoEnabled>
  <rearPlateRecoEnabled>
    <!--opt, bool, rear license plate recognition-->true
  </rearPlateRecoEnabled>
  <smallPlateRecoEnabled>
    <!--opt, bool, small license plate recognition-->true
  </smallPlateRecoEnabled>
  <largePlateRecoEnabled>
    <!--opt, bool, large license plate recognition-->true
  </largePlateRecoEnabled>
  <farmVehicleEnabled>
    <!--opt, bool, agricultural vehicle recognition-->true
  </farmVehicleEnabled>
  <motorEnabled>
    <!--opt, bool, motorcycle recognition-->true
  </motorEnabled>
  <fuzzyDiscEnabled>
    <!--opt, bool, fuzzy recognition-->true
  </fuzzyDiscEnabled>
  <microPlateRegEnabled>
    <!--opt, bool, tiny license plate recognition-->true
  </microPlateRegEnabled>
  <cAPlateRegEnabled>
    <!--opt, bool, civil aviation license plate recognition-->true
  </cAPlateRegEnabled>
  <tiltPlateRegEnabled>
    <!--opt, bool, tilted license plate recognition-->true
  </tiltPlateRegEnabled>
  <superPlateRecoEnabled>
    <!--opt, bool, oversized license plate recognition-->true
  </superPlateRecoEnabled>
  <embassyPlateRecoEnabled>
    <!--opt, bool, embassy license plate recognition-->true
  </embassyPlateRecoEnabled>
  <nation>
    <!--opt, enum, country/region, subType:string, desc:"ER" (Russian-speaking region), "EU" (Europe), , "EUandCIS" (Europe and Russia), "ME" (Middle East),
"APAC" (Asia-Pacific region), "AFandAM" (Africa and America), "All" (all regions)-->ER
  </nation>
  <countryIndex>
    <!--opt, int, country/region index, desc:refer to the <CRIndex> data field-->0
  </countryIndex>
  <nonMotorPlateCityRecoEnabled>
    <!--opt, bool, whether to enable belonged city recognition of non-motor vehicle license plate-->true
  </nonMotorPlateCityRecoEnabled>
  <city>
    <!--opt, string, License Plate Belonged City-->test
  </city>
  <fakePlateFilterEnabled>
    <!--opt, bool, filter fake license plate-->false
  </fakePlateFilterEnabled>
</PlateRecognitionParam>
```

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### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

### 14.5.1.11 Get ANPR parameters

### Request URL

GET /ISAPI/ITC/plateRecognitionParam?channelID=<channelID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelPlateRecognitionParam is "true" returned from GET /ISAPI/ITC/capabilities). |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognitionParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ANPR parameters, attr:version{req, string, protocolVersion}-->
  <defaultCHN>
    <!--ro, opt, string, province name-->test
  </defaultCHN>
  <frontPlateRecoEnabled>
    <!--ro, opt, bool, front license plate recognition, desc:license plate recognition principles: when <isSupportfrontAndRearReg> is true and both front
and rear license plate recognition are supported, "true" is returned and the front license plate recognition parameters will be modified by default; when
<isSupportfrontAndRearReg> is false and either front or rear license plate recognition is supported, "false" is returned-->true
  </frontPlateRecoEnabled>
  <rearPlateRecoEnabled>
    <!--ro, opt, bool, rear license plate recognition-->true
  </rearPlateRecoEnabled>
  <smallPlateRecoEnabled>
    <!--ro, opt, bool, small license plate recognition-->true
  </smallPlateRecoEnabled>
  <largePlateRecoEnabled>
    <!--ro, opt, bool, large license plate recognition-->true
  </largePlateRecoEnabled>
  <farmVehicleEnabled>
    <!--ro, opt, bool, agricultural vehicle recognition-->true
  </farmVehicleEnabled>
  <motorEnabled>
    <!--ro, opt, bool, motorcycle recognition-->true
  </motorEnabled>
  <fuzzyDiscEnabled>
    <!--ro, opt, bool, fuzzy recognition-->true
  </fuzzyDiscEnabled>
  <microPlateRegEnabled>
    <!--ro, opt, bool, tiny license plate recognition-->true
  </microPlateRegEnabled>
  <cAPlateRegEnabled>
    <!--ro, opt, bool, civil aviation license plate recognition-->true
  </cAPlateRegEnabled>
```

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```
<tiltPlateRegEnabled>
  <!--ro, opt, bool, tilted license plate recognition-->true
</tiltPlateRegEnabled>
<superPlateRecoEnabled>
  <!--ro, opt, bool, large license plate recognition-->true
</superPlateRecoEnabled>
<embassyPlateRecoEnabled>
  <!--ro, opt, bool, embassy license plate recognition-->true
</embassyPlateRecoEnabled>
<nation>
  <!--ro, opt, enum, country/region, subType:string, desc:"ER" (Russian-speaking region), "EU" (Europe), , "EUandCIS" (Europe and Russia), "ME" (Middle East), "APAC" (Asia-Pacific region), "AFandAM" (Africa and America), "All" (all regions)-->ER
</nation>
<countryIndex>
  <!--ro, opt, int, country/region index, desc:refer to the <CRIndex> data field-->0
</countryIndex>
<nonMotorPlateCityRecoEnabled>
  <!--ro, opt, bool, whether to enable belonged city recognition of non-motor vehicle license plate-->true
</nonMotorPlateCityRecoEnabled>
<city>
  <!--ro, opt, string, license plate belonged city-->test
</city>
<fakePlateFilterEnabled>
  <!--ro, opt, bool, filter fake license plate, desc:filter fake license plate-->false
</fakePlateFilterEnabled>
<citySeparatorEnabled>
  <!--ro, opt, bool-->true
</citySeparatorEnabled>
<fakePlateDetectionUploadEnabled>
  <!--ro, opt, bool, dep:and,{$.PlateRecognitionParam.fakePlateFilterEnabled,eq,true}-->false
</fakePlateDetectionUploadEnabled>
<provinceThresholds>
  <!--ro, opt, int, range:[0,100]-->0
</provinceThresholds>
<secondLicensePlateRegEnabled>
  <!--ro, opt, bool-->true
</secondLicensePlateRegEnabled>
<yellowGreenPlateRecoRule>
  <!--ro, opt, enum, subType:string-->yellow
</yellowGreenPlateRecoRule>
<nonStandardPlateIllegalHandleRules>
  <!--ro, opt, enum, subType:string-->none
</nonStandardPlateIllegalHandleRules>
<nonStandardPlatePostHandleRules>
  <!--ro, opt, enum, subType:string-->none
</nonStandardPlatePostHandleRules>
<nonStandardPlateThresholds>
  <!--ro, opt, int, range:[0,100]-->0
</nonStandardPlateThresholds>
<nonStandardPlateMinCorrectFactor>
  <!--ro, opt, int, range:[0,100]-->0
</nonStandardPlateMinCorrectFactor>
<nonStandardPlateMaxCorrectFactor>
  <!--ro, opt, int, range:[0,100]-->0
</nonStandardPlateMaxCorrectFactor>
<nonStandardVehicleCorrectFactor>
  <!--ro, opt, int, range:[0,100]-->0
</nonStandardVehicleCorrectFactor>
<nonStandardPlateRegEnabled>
  <!--ro, opt, bool-->true
</nonStandardPlateRegEnabled>
<threePlateRecoEnabled>
  <!--ro, opt, bool-->true
</threePlateRecoEnabled>
<newEnergyGreenPlateRecoRule>
  <!--ro, opt, enum, subType:string-->green
</newEnergyGreenPlateRecoRule>
<customPlateRule>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <customPlateColorEnabled>
    <!--ro, opt, bool-->true
  </customPlateColorEnabled>
  <customPlateColor>
    <!--ro, opt, enum, subType:string-->black
  </customPlateColor>
  <wildcardRuleList>
    <!--ro, opt, array, subType:object, range:[0,10]-->
    <wildcardRule>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <wildcard>
        <!--ro, opt, string, range:[0,16]-->蓝园A?!!**
      </wildcard>
    </wildcardRule>
  </wildcardRuleList>
</customPlateRule>
<plateMappingRule>
  <!--ro, opt, object-->
  <enabled>
```

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```
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <plateMappingList>
        <!--ro, opt, array, subType:object, range:[0,64]-->
        <plateMapping>
          <!--ro, opt, object-->
          <detectPlateNo>
            <!--ro, opt, string, range:[0,16]-->浙A12348
          </detectPlateNo>
          <replacePlateNo>
            <!--ro, opt, string, range:[0,16]-->浙A1234B
          </replacePlateNo>
        </plateMapping>
      </plateMappingList>
    </plateMappingRule>
    <primaryAndSecondaryPlate>
      <!--ro, opt, array, subType:object-->
      <enabled>
        <!--ro, opt, array, subType:object-->
        <mainlandPlateEnable>
          <!--ro, opt, bool-->true
        </mainlandPlateEnable>
        <hongKongPlateEnable>
          <!--ro, opt, bool-->true
        </hongKongPlateEnable>
        <maCaoPlateEnable>
          <!--ro, opt, bool-->true
        </maCaoPlateEnable>
      </enabled>
      <plateRegionType>
        <!--ro, opt, array, subType:object-->
        <mainAreaOption>
          <!--ro, opt, enum, subType:string-->mainland
        </mainAreaOption>
        <secondAreaOption>
          <!--ro, opt, enum, subType:string-->hongKong
        </secondAreaOption>
      </plateRegionType>
    </primaryAndSecondaryPlate>
  </PlateRecognitionParam>
```

#### 14.5.1.12 Get trigger mode capability of traffic cameras

**Request URL**

GET /ISAPI/ITC/TriggerMode/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TriggerMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, trigger mode, attr:version{req, string, protocolVersion}-->
  <usedTriggerMode
opt="postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postHVT,postMPR,epoliceRS485,postEpoliceRS485,videoEpolice,TPS,videoMonitor,postNoComityPed,pos
tRedLightPed,postSideRadarSpeed,parkingIncident">
    <!--ro, req, enum, trigger mode supported by the device, different devices support different trigger modes, and a device may support multiple trigger
modes, subType:string, attr:opt{req, string}, desc:"postMobile", "postDetect", "TPS", "trafficIncident", "trafficIncidentTPS", "roadWarning", "postIOSpeed",
"postSingleIO", "postRS485", "postRadar", "postVTCoil", "postHVT", "postMPR", "epoliceRS485", "postEpoliceRS485", "videoEpolice", "videoMonitor",
"postNoComityPed", "postRedLightPed", "postSideRadarSpeed", "parkingIncident", "vehicleDetect", "boxNumDetection", "parkSpeedDetection"-->postMobile
  </usedTriggerMode>
  <isSupportChannelConfig>
    <!--ro, opt, bool, whether it supports channel configuration, desc:related URL: /ISAPI/ITC/channels/<ID>/TriggerMode/capabilities-->true
  </isSupportChannelConfig>
  <isSupportChannelSwitchStatus>
    <!--ro, opt, bool, whether it supports switching trigger mode status by channel No., desc:true (the optional node <channelID> in
/ISAPI/ITC/TriggerMode/switchStatus?format=json&channelID=<channelID> is supported)-->true
  </isSupportChannelSwitchStatus>
</TriggerMode>
```

#### 14.5.1.13 Get parameters of intersection violation system vehicle detector

**Request URL**

GET /ISAPI/ITC/TriggerMode/epoliceRS485

**Query Parameter**

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None

**Request Message**

None

**Response Message**

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <trafficLightSource>
    <!--ro, req, enum, traffic light source, subType:int, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <EpoliceRS485LaneList>
    <!--ro, opt, array, lane list, subType:object-->
    <EpoliceRS485Lane>
      <!--ro, opt, object, lane-->
      <laneId>
        <!--ro, req, int, lane ID, range:[1,4]-->0
      </laneId>
      <laneDirectionType>
        <!--ro, req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--ro, req, int, linked Lane No., range:[1,16]-->0
      </relatedDriveWay>
      <redLightDriveWay>
        <!--ro, req, int, red light lane No., range:[1,16]-->0
      </redLightDriveWay>
      <yellowLightDriveWay>
        <!--ro, req, int, yellow light lane No., range:[1,16]-->0
      </yellowLightDriveWay>
      <OSDDriveWay>
        <!--ro, req, int, overlay lane No., range:[1,99]-->0
      </OSDDriveWay>
      <distance>
        <!--ro, req, int, coil distance, range:[0,2000], unit:cm-->0
      </distance>
      <recordEnable>
        <!--ro, req, bool, whether to enable recording-->true
      </recordEnable>
      <recordType>
        <!--ro, req, enum, recording type, subType:string, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
      </recordType>
      <preRecordTime>
        <!--ro, req, int, pre-record duration, range:[0,100]-->0
      </preRecordTime>
      <recordDelayTime>
        <!--ro, req, int, post-record duration, range:[0,100]-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--ro, req, int, timeout duration, range:[0,100]-->0
      </recordTimeOut>
      <serialProtocol>
        <!--ro, req, enum, serial port protocol, subType:int, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
      </serialProtocol>
      <runRedLightCaptureLogic>
        <!--ro, req, enum, red light running capture logic, subType:string, desc:"singelPro" (single_enter 1_exit 1_delay 1_1), "singlePro2" (singelPro),
"doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit 1_delay 2_1), "doublePro4"
(double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter 1_exit 2_delay 2_2),
"doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
      </runRedLightCaptureLogic>
      <oppositeDirectionCaptureLogic>
        <!--ro, req, enum, Wrong-Way Driving Capture Logic, subType:string, desc:"noCap" (do not capture), "reversePro1" (double/three_enter_enter_delay),
"reversePro2" (double/three_enter_delay)-->noCap
      </oppositeDirectionCaptureLogic>
      <intervalType>
        <!--ro, req, enum, Burst Interval Type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
      </intervalType>
      <interval>
        <!--ro, req, int, interval, range:[0,2000]-->0
      </interval>
      <IOOutList>
        <!--ro, opt, array, list of I/O output ports, subType:object-->
        <IOOut>
          <!--ro, req, object, I/O output port-->
          <id>
            <!--ro, req, int, linked I/O output No., range:[1,10]-->0
          </id>
          <enabled>
            <!--ro, req, bool, whether to link the I/O output-->true
          </enabled>
        </IOOut>
      </IOOutList>
      <flashMode>
        <!--ro, req, enum, flashing mode, subType:string, desc:"together" (simultaneously), "alternation" (alternatively)-->together
      </flashMode>
    </EpoliceRS485Lane>
  </EpoliceRS485LaneList>
</EpoliceRS485>
```

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#### 14.5.1.14 Set parameters of intersection violation system vehicle detector

**Request URL**

PUT /ISAPI/ITC/TriggerMode/epoliceRS485

**Query Parameter**

None

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <trafficLightSource>
    <!--req, enum, traffic light source, subType:int, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <EpoliceRS485LaneList>
    <!--opt, array, lane list, subType:object-->
    <EpoliceRS485Lane>
      <!--opt, object, lane-->
      <laneId>
        <!--req, int, lane ID, range:[1,4]-->0
      </laneId>
      <laneDirectionType>
        <!--req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to
south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--req, int, linked Lane No., range:[1,16]-->0
      </relatedDriveWay>
      <redLightDriveWay>
        <!--req, int, red light lane No., range:[1,16]-->0
      </redLightDriveWay>
      <yellowLightDriveWay>
        <!--req, int, yellow light lane No., range:[1,16]-->0
      </yellowLightDriveWay>
      <OSDDriveWay>
        <!--req, int, Overlay Lane No., range:[1,99]-->0
      </OSDDriveWay>
      <distance>
        <!--req, int, coil distance, range:[0,2000], unit:cm-->0
      </distance>
      <recordEnable>
        <!--req, bool, whether to enable recording-->true
      </recordEnable>
      <recordType>
        <!--req, enum, recording type, subType:string, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
      </recordType>
      <preRecordTime>
        <!--req, int, pre-record duration, range:[0,100]-->0
      </preRecordTime>
      <recordDelayTime>
        <!--req, int, post-record, range:[0,100]-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--req, int, timeout duration, range:[0,100]-->0
      </recordTimeOut>
      <serialProtocol>
        <!--req, enum, serial port protocol, subType:int, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
      </serialProtocol>
      <runRedLightCaptureLogic>
        <!--req, enum, Red Light Running Capture Logic, subType:string, desc:"singelPro" (single_enter 1_exit 1_delay 1_1), "singlePro2" (singelPro),
"doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit 1_delay 2_1), "doublePro4"
(double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter 1_exit 2_delay 2_2),
"doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
      </runRedLightCaptureLogic>
      <oppositeDirectionCaptureLogic>
        <!--req, enum, Wrong-Way Driving Capture Logic, subType:string, desc:"noCap" (do not capture), "reversePro1" (double/three_enter_enter_delay),
"reversePro2" (double/three_enter_delay)-->noCap
      </oppositeDirectionCaptureLogic>
      <intervalType>
        <!--req, enum, Burst Interval Type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
      </intervalType>
      <interval>
        <!--req, int, interval, range:[0,2000]-->0
      </interval>
      <IOOutList>
        <!--opt, array, list of I/O output ports, subType:object-->
        <IOOut>
          <!--req, object, I/O output port-->
          <id>
            <!--req, int, linked I/O output No., range:[1,10]-->0
          </id>
          <enabled>
            <!--req, bool, whether to link the I/O output-->true
          </enabled>
        </IOOut>
      </IOOutList>
      <flashMode>
        <!--req, enum, flashing mode, subType:string, desc:"together" (simultaneously), "alternation" (alternatively)-->together
      </flashMode>
    </EpoliceRS485Lane>
  </EpoliceRS485LaneList>
</EpoliceRS485>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.5.1.15 Get the parameters capability of intersection violation system vehicle detector

**Request URL**

GET /ISAPI/ITC/TriggerMode/epoliceRS485/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="6">
    <!--ro, req, int, number of supported lanes, range:[1,6], attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <trafficLightSource opt="0,1">
    <!--ro, req, enum, traffic light source, subType:int, attr:opt{req, string}, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <relatedDriveWay min="1" max="16">
    <!--ro, req, int, linked lane No., range:[1,16], attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
  <redLightDriveWay min="1" max="16">
    <!--ro, req, int, red light lane No., range:[1,16], attr:min{req, int},max{req, int}-->0
  </redLightDriveWay>
  <yellowLightDriveWay min="1" max="16">
    <!--ro, req, int, yellow light lane No., range:[1,16], attr:min{req, int},max{req, int}-->0
  </yellowLightDriveWay>
  <OSDDriveWay min="1" max="99">
    <!--ro, req, int, overlay lane No., range:[1,99], attr:min{req, int},max{req, int}-->0
  </OSDDriveWay>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, lane direction, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to
north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to
northeast)-->0
  </laneDirectionType>
  <distance min="0" max="2000">
    <!--ro, req, int, coil distance, range:[0,200], attr:min{req, int},max{req, int}-->0
  </distance>
  <recordEnable>
    <!--ro, req, bool, whether to enable recording-->true
  </recordEnable>
  <recordType opt="delayRecord,preRecord">
    <!--ro, req, enum, recording type, subType:string, attr:opt{req, string}, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
  </recordType>
  <preRecordTime min="0" max="100">
    <!--ro, req, int, pre-record, range:[0,100], attr:min{req, int},max{req, int}-->0
  </preRecordTime>
  <recordDelayTime min="0" max="100">
    <!--ro, req, int, post-record, range:[0,100], attr:min{req, int},max{req, int}-->0
  </recordDelayTime>
  <recordTimeOut min="0" max="100">
    <!--ro, req, int, timeout duration, range:[0,100], attr:min{req, int},max{req, int}-->0
  </recordTimeOut>
  <serialProtocol opt="0,1,2">
    <!--ro, req, enum, serial port protocol, subType:int, attr:opt{req, string}, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
  </serialProtocol>
  <runRedLightCaptureLogic opt="singelPro,doublePro1,doublePro2,doublePro3,doublePro4,doublePro5,doublePro6,doublePro7,doublePro8,singlePro2,other">
    <!--ro, req, enum, red light running capture logic, subType:string, attr:opt{req, string}, desc:"singelPro" (single_enter 1_exit 1_delay 1_1),
"singlePro2" (singelPro), "doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit
1_delay 2_1), "doublePro4" (double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter
1_exit 2_delay 2_2), "doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
  </runRedLightCaptureLogic>
  <oppositeDirectionCaptureLogic opt="noCap,reversePro1,reversePro2">
    <!--ro, req, enum, Wrong-Way Driving Capture Logic, subType:string, attr:opt{req, string}, desc:"noCap" (do not capture), "reversePro1"
(double/three_enter_enter_delay), "reversePro2" (double/three_enter_delay)-->noCap
  </oppositeDirectionCaptureLogic>
  <intervalType opt="time,distance">
    <!--ro, req, enum, Burst Interval Type, subType:string, attr:opt{req, string}, desc:"time" (by time), "distance" (by distance)-->time
  </intervalType>
  <interval>
    <!--ro, req, int, interval, range:[0,2000]-->0
  </interval>
  <IOOut min="1" max="7">
    <!--ro, req, int, number of I/O outputs, range:[1,7], attr:min{req, int},max{req, int}-->0
  </IOOut>
  <flashMode opt="together,alternation">
    <!--ro, req, enum, flashing mode, subType:string, attr:opt{req, string}, desc:"together" (simultaneously), "alternation" (alternatively)-->together
  </flashMode>
</EpoliceRS485>
```

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### 14.5.1.16 Get the recommended configurations of intersection violation system vehicle detector

**Request URL**

GET /ISAPI/ITC/TriggerMode/epoliceRS485/recommendation

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <trafficLightSource>
    <!--ro, req, enum, traffic light source, subType:int, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <EpoliceRS485LaneList>
    <!--ro, opt, array, lane list, subType:object-->
    <EpoliceRS485Lane>
      <!--ro, opt, object, lane-->
      <laneId>
        <!--ro, req, int, lane ID, range:[1,4]-->0
      </laneId>
      <laneDirectionType>
        <!--ro, req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--ro, req, int, linked Lane No., range:[1,16]-->0
      </relatedDriveWay>
      <redLightDriveWay>
        <!--ro, req, int, red light lane No., range:[1,16]-->0
      </redLightDriveWay>
      <yellowLightDriveWay>
        <!--ro, req, int, yellow light lane No., range:[1,16]-->0
      </yellowLightDriveWay>
      <OSDDriveWay>
        <!--ro, req, int, overlay lane No., range:[1,99]-->0
      </OSDDriveWay>
      <distance>
        <!--ro, req, int, coil distance, range:[0,2000], unit:cm-->0
      </distance>
      <recordEnable>
        <!--ro, req, bool, whether to enable recording-->true
      </recordEnable>
      <recordType>
        <!--ro, req, enum, recording type, subType:string, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
      </recordType>
      <preRecordTime>
        <!--ro, req, int, pre-record duration, range:[0,100]-->0
      </preRecordTime>
      <recordDelayTime>
        <!--ro, req, int, post-record, range:[0,100]-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--ro, req, int, timeout duration, range:[0,100]-->0
      </recordTimeOut>
      <serialProtocol>
        <!--ro, req, enum, serial port protocol, subType:int, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
      </serialProtocol>
      <runRedLightCaptureLogic>
        <!--ro, req, enum, Red Light Running Capture Logic, subType:string, desc:"singelPro" (single_enter 1_exit 1_delay 1_1), "singlePro2" (singelPro),
"doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit 1_delay 2_1), "doublePro4"
(double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter 1_exit 2_delay 2_2),
"doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
      </runRedLightCaptureLogic>
      <oppositeDirectionCaptureLogic>
        <!--ro, req, enum, Wrong-Way Driving Capture Logic, subType:string, desc:"noCap" (do not capture), "reversePro1" (double/three_enter_enter_delay),
"reversePro2" (double/three_enter_delay)-->noCap
      </oppositeDirectionCaptureLogic>
      <intervalType>
        <!--ro, req, enum, burst interval type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
      </intervalType>
      <interval>
        <!--ro, req, int, interval, range:[0,2000]-->0
      </interval>
      <IOOutList>
        <!--ro, opt, array, list of I/O output ports, subType:object-->
        <IOOut>
          <!--ro, req, object, I/O output port-->
          <id>
            <!--ro, req, int, linked I/O output No., range:[1,10]-->0
          </id>
          <enabled>
            <!--ro, req, bool, whether to link the I/O output-->true
          </enabled>
        </IOOut>
      </IOOutList>
      <flashMode>
        <!--ro, req, enum, flashing mode, subType:string, desc:"together" (simultaneously), "alternation" (alternatively)-->together
      </flashMode>
    </EpoliceRS485Lane>
  </EpoliceRS485LaneList>
</EpoliceRS485>
```

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#### 14.5.1.17 Get the status of video detection traffic lights

**Request URL**

GET /ISAPI/ITC/TriggerMode/lightStatus?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelLightStatus is "true" returned from /ISAPI/ITC/TriggerMode/capabilities). |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<VepLightStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, status of video detection traffic lights, attr:version{req, string, protocolVersion}-->
  <lightStatus opt="normal,abnomal">
    <!--ro, req, enum, traffic light status, subType:string, attr:opt{req, string}, desc:"normal", "abnormal"-->normal
  </lightStatus>
</VepLightStatus>
```

#### 14.5.1.18 Get the license plate recognition configuration capability of a specific trigger mode

**Request URL**

GET /ISAPI/ITC/TriggerMode/plateRecognition/<triggerMode>/capabilities?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| triggerMode | enum | -- |
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelPlateRecognition is "true" returned from /ISAPI/ITC/TriggerMode/capabilities). |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the license plate recognition parameters of a specific trigger mode, attr:version{req, string, protocolVersion}-->
  <triggerMode opt="postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postHVT,postMpr,epoliceRS485,postEpoliceRS485,videoEpolice,postBOT">
    <!--ro, req, enum, trigger modes supported by device, subType:string, attr:opt{req, string}, desc:"postiospeed" (checkpoint i/o speed detection),
"postsingleio" (checkpoint single i/o trigger), "postrs485" (checkpoint vehicle detector), "postradar" (checkpoint radar), "postvtcoil" (checkpoint virtual
coil), "posthvt" (checkpoint multi-target-type detection), "postmpr" (entrance and exit video trigger), "epolicers485" (intersection violation system
vehicle detector), "postepolicers485" (checkpoint intersection violation system vehicle detector), "videoepolice" (video intersection violation system),
"postbot" (bus lane occupation)-->postIOSpeed
  </triggerMode>
  <backupType opt="0,1">
    <!--ro, req, enum, backup mode type, subType:int, attr:opt{req, string}, desc:0-backup mode, 1-normal configuration mode. When this node is 0, the
current trigger mode is checkpoint vehicle detector; when this node is 1, the current trigger mode is checkpoint virtual coil-->0
  </backupType>
  <plateRecogEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable license plate recognition function of this region, attr:opt{req, string}-->true
  </plateRecogEnabled>
  <regionMode opt="rectangle,polygon">
    <!--ro, req, enum, area type, subType:string, attr:opt{req, string}, desc:"rectangle", "polygon"-->rectangle
  </regionMode>
  <plateRegionCount min="1" max="6">
    <!--ro, req, int, total number of linked license plate recognition regions, range:[1,6], attr:min{req, int},max{req, int}-->0
  </plateRegionCount>
</PlateRecognition>
```

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#### 14.5.1.19 Get the license plate recognition parameters of all trigger modes

**Request URL**

GET /ISAPI/ITC/TriggerMode/plateRecognition?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelPlateRecognition is "true" returned from /ISAPI/ITC/TriggerMode/capabilities). |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognitionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, see details in the message of XML_PlateRecognition, subType:object, attr:version{req, string, protocolVersion}-->
  <PlateRecognition>
    <!--ro, opt, object-->
    <triggerMode>
      <!--ro, req, enum, subType:string-->postIOSpeed
    </triggerMode>
    <backupType>
      <!--ro, req, enum, subType:int-->0
    </backupType>
    <plateRecogEnabled>
      <!--ro, opt, bool-->true
    </plateRecogEnabled>
    <regionMode>
      <!--ro, req, enum, region type, subType:string, desc:region type-->rectangle
    </regionMode>
    <plateRegionCount>
      <!--ro, req, int, range:[1,6]-->0
    </plateRegionCount>
    <PlateRegionList>
      <!--ro, req, array, subType:object-->
      <PlateRegion>
        <!--ro, req, object-->
        <regionId>
          <!--ro, req, int, region ID, range:[1,6]-->0
        </regionId>
        <RegionCoordinatesList>
          <!--ro, req, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, req, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </PlateRegion>
    </PlateRegionList>
  </PlateRecognition>
</PlateRecognitionList>
```

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#### 14.5.1.20 Get parameters of checkpoint intersection violation system vehicle detector

**Request URL**

GET /ISAPI/ITC/TriggerMode/postEpoliceRS485

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
</PostRS485>
```

### 14.5.1.21 Set parameters of checkpoint intersection violation system vehicle detector

#### Request URL

PUT /ISAPI/ITC/TriggerMode/postEpoliceRS485

#### Query Parameter

None

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
</PostRS485>
```

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

### 14.5.1.22 Get the parameters capability of checkpoint intersection violation system vehicle detector

#### Request URL

GET /ISAPI/ITC/TriggerMode/postEpoliceRS485/capabilities

#### Query Parameter

None

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="6">
    <!--ro, req, int, number of supported lanes, attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <maxSwitchTime min="5" max="255">
    <!--ro, req, int, fault-tolerant time, attr:min{req, int},max{req, int}-->0
  </maxSwitchTime>
  <backupMode opt="0,1,2">
    <!--ro, req, enum, Backup Mode, subType:int, attr:opt{req, string}, desc:"0" (none), "1" (trigger of checkpoint virtual coil) Pattern-->0
  </backupMode>
  <relatedDriveWay min="1" max="99">
    <!--ro, req, int, linked lane No., attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
```

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```
    <OSDDriveWay min="1" max="99">
      <!--ro, req, int, lane No. to be linked, attr:min{req, int},max{req, int}-->0
    </OSDDriveWay>
    <laneType opt="none,highway,urbanFastWay,other">
      <!--ro, req, enum, lane attribute, subType:string, attr:opt{req, string}, desc:"none"-not configured, "highway"-high way, "urbanFastWay"-urban
expressway, "other"-->none
    </laneType>
    <laneUsage opt="carriageWay,emergency">
      <!--ro, req, enum, lane purpose, subType:string, attr:opt{req, string}, desc:"carriageWay" (normal lane), "emergency" (emergency lane)-->carriageWay
    </laneUsage>
    <distance min="0" max="20000">
      <!--ro, req, int, coil distance, attr:min{req, int},max{req, int}-->0
    </distance>
    <delayTime min="0" max="2000">
      <!--ro, req, int, trigger delay time, attr:min{req, int},max{req, int}-->0
    </delayTime>
    <delayDistance min="0" max="255">
      <!--ro, req, int, trigger delay distance, attr:min{req, int},max{req, int}-->0
    </delayDistance>
    <speedCapEnabled>
      <!--ro, req, bool, overspeed capture-->true
    </speedCapEnabled>
    <lowSpeedCapEnable>
      <!--ro, req, bool, low speed capture-->true
    </lowSpeedCapEnable>
    <emergencyCapEnable>
      <!--ro, req, bool, occupying emergency lane capture-->true
    </emergencyCapEnable>
    <signSpeed min="0" max="255">
      <!--ro, req, int, marked speed limit, attr:min{req, int},max{req, int}-->0
    </signSpeed>
    <speedLimit min="0" max="255">
      <!--ro, req, int, speed limit, attr:min{req, int},max{req, int}-->0
    </speedLimit>
    <bigCarSignSpeed min="0" max="255">
      <!--ro, req, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarSignSpeed>
    <bigCarLowSpeedLimit min="0" max="255">
      <!--ro, req, int, low speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarLowSpeedLimit>
    <lowSpeedLimit min="0" max="60">
      <!--ro, req, int, low speed limit for small-sized vehicle, attr:min{req, int},max{req, int}-->0
    </lowSpeedLimit>
    <bigCarSpeedLimit min="0" max="60">
      <!--ro, req, int, speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarSpeedLimit>
    <carHighSpeed min="0" max="255">
      <!--ro, req, int, abnormal overspeed, attr:min{req, int},max{req, int}-->0
    </carHighSpeed>
    <carLowSpeed min="0" max="255">
      <!--ro, req, int, abnormal low speed, attr:min{req, int},max{req, int}-->0
    </carLowSpeed>
    <bigCarHighSpeed min="0" max="255">
      <!--ro, req, int, abnormal overspeed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed min="0" max="255">
      <!--ro, req, int, abnormal low speed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarLowSpeed>
    <snapTimes min="0" max="5">
      <!--ro, req, int, continuous capture times, attr:min{req, int},max{req, int}-->0
    </snapTimes>
    <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
      <!--ro, req, enum, types of lane directions, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from
south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from
southwest to northeast)-->0
    </laneDirectionType>
    <intervalType opt="time,distance">
      <!--ro, req, enum, Burst Interval Type, subType:string, attr:opt{req, string}, desc:"time" (by time), "distance" (by distance)-->time
    </intervalType>
    <IOOut min="1" max="7">
      <!--ro, req, int, I/O outputs, attr:min{req, int},max{req, int}-->0
    </IOOut>
    <flashMode opt="together,alternation">
      <!--ro, req, enum, flashing mode, subType:string, attr:opt{req, string}, desc:"together" (simultaneously), "alternation" (alternatively)-->together
    </flashMode>
    <emergencyTimeSwitchList size="4">
      <!--ro, req, array, time period for emergency lanes, subType:object, attr:size{req, int}-->
      <emergencyTimeSwitch>
        <!--ro, req, object, emergency lane time-->
        <timeId min="1" max="4">
          <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
        </timeId>
        <startHour min="0" max="23">
          <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
        </startHour>
        <startMinute min="0" max="59">
          <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
        </startMinute>
        <endHour min="0" max="23">
          <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
        </endHour>
        <endMinute min="0" max="59">
          <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
```

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```
        </endMinute>
      </emergencyTimeSwitch>
    </emergencyTimeSwitchList>
    <noSupportPlateRegion>
      <!--ro, opt, bool, areas where ANPR is not supported-->true
    </noSupportPlateRegion>
    <LaneLine>
      <!--ro, opt, object, lane line-->
      <lineName min="0" max="64">
        <!--ro, req, string, line name, attr:min{req, int},max{req, int}-->test
      </lineName>
      <RegionCoordinatesList num="2">
        <!--ro, opt, array, region coordinate list, subType:object, range:[0,2], attr:num{opt, object}-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LaneLine>
    <LaneRightBoundaryLine>
      <!--ro, req, object, right border line of lane-->
      <lineName>
        <!--ro, req, enum, line name, subType:string, desc:"laneRightBoundaryLine" (right border line of lane)-->laneRightBoundaryLine
      </lineName>
      <lineType>
        <!--ro, req, enum, line type, subType:string, desc:"unknown" (white line), "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
      </lineType>
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </LaneRightBoundaryLine>
    <safeBelt>
      <!--ro, opt, bool, whether to enable seatbelt detection-->true
    </safeBelt>
    <safeBeltCapNo min="1" max="3">
      <!--ro, opt, int, number of captured pictures in which the driver does not wear the seatbelt, attr:min{req, int},max{req, int}-->0
    </safeBeltCapNo>
    <uphone>
      <!--ro, opt, bool, whether to enable phone call detection-->true
    </uphone>
    <uphoneCapNo min="1" max="3">
      <!--ro, opt, int, number of captured pictures in which the driver is making call, attr:min{req, int},max{req, int}-->0
    </uphoneCapNo>
  </PostRS485>
```

#### 14.5.1.23 Get the recommended configurations of checkpoint intersection violation system vehicle detector

**Request URL**

GET /ISAPI/ITC/TriggerMode/postEpoliceRS485/recommendation

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
</PostRS485>
```

#### 14.5.1.24 Get statuses of switching trigger mode

**Request URL**

GET /ISAPI/ITC/TriggerMode/switchStatus?format=json&channelID=<channelID>

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | channel No. (this optional parameter is supported when isSupportChannelSwitchStatus is "true" returned from GET /ISAPI/ITC/TriggerMode/capabilities) |

**Request Message**

None

**Response Message**

```
{
    "switchStatus":  "completed"
    /*ro, req, enum, status, subType:string, desc:"completed", "executing"*/
}
```

#### 14.5.1.25 Get the trigger mode configuration capability of traffic incidents

**Request URL**

GET /ISAPI/ITC/TriggerMode/trafficIncident/capabilities?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
{
    "TrafficIncidentCap": {
    /*ro, opt, object, configuration capability of traffic incident detection*/
        "enabled": [true, false],
        /*ro, req, array, whether to enable traffic incident detection, subType:bool*/
        "sceneMode":  "highway,urban,tunnel",
        /*ro, opt, string, scene mode, desc:"highway", "urban" (urban road), "tunnel"*/
        "isSupportChannelTrafficIncident":  false
        /*ro, opt, bool, whether it support configuring parameters of traffic incident detection, desc:if this node is true, it indicates that channelID can be set in /ISAPI/ITC/TriggerMode/trafficIncident?format=json&channelID=<channelID> and /ISAPI/ITC/TriggerMode/trafficIncident/defaultParam?format=json&channelID=<channelID>*/
    }
}
```

#### 14.5.1.26 Set the trigger mode parameters of traffic incidents

**Request URL**

PUT /ISAPI/ITC/TriggerMode/trafficIncident?format=json&channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | channel No. (this optional parameter is supported when isSupportChannelTrafficIncident is "true" returned from GET /ISAPI/ITC/TriggerMode/trafficIncident/capabilities?format=json) |

**Request Message**

```
{
    "TrafficIncident": {
    /*opt, object, parameters of traffic incident detection*/
        "enabled":  true,
        /*req, bool, whether to enable the traffic incident trigger mode*/
        "sceneMode":  "highway"
        /*opt, enum, scene mode, subType:string, desc:"highway", "urban" (urban road), "tunnel"*/
    }
}
```

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**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:error code*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

### 14.5.1.27 Get the trigger mode parameters of traffic incidents

**Request URL**

GET /ISAPI/ITC/TriggerMode/trafficIncident?format=json&channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | channel No. (this optional parameter is supported when isSupportChannelTrafficIncident is "true" returned from GET /ISAPI/ITC/TriggerMode/trafficIncident/capabilities?format=json) |

**Request Message**

None

**Response Message**

```
{
    "TrafficIncident": {
    /*ro, opt, object, parameters of traffic incident detection*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "sceneMode":  "highway"
        /*ro, opt, enum, scene mode, subType:string, desc:"highway", "urban" (urban road), "tunnel"*/
    }
}
```

### 14.5.1.28 Get the capability of video intersection violation parameters

**Request URL**

GET /ISAPI/ITC/TriggerMode/videoEpolice/capabilities?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelVideoEpolice is "true" returned from /ISAPI/ITC/TriggerMode/capabilities). |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoEpolice xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, video intersection violation parameters, attr:version{req, string, protocolVersion}-->
  <intervalType opt="time,distance">
    <!--ro, req, enum, burst interval type, subType:string, attr:opt{req, string}, desc:"time" (by time), "distance" (by distance)-->time
  </intervalType>
  <interval min="1" max="1000">
    <!--ro, req, int, time interval, range:[1,1000], attr:min{req, int},max{req, int}-->0
  </interval>
  <post>
    <!--ro, req, bool, whether to enable checkpoint detection-->true
```

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```
</post>
<postCapNo opt="1,3">
  <!--ro, req, int, captured pictures of checkpoint, range:[1,3], attr:opt{req, string}-->1
</postCapNo>
<driveLine>
  <!--ro, req, bool, whether to enable driving on lane line detection-->true
</driveLine>
<drivelineCapNo opt="2,3">
  <!--ro, req, int, capture times, range:[2,3], attr:opt{req, string}-->0
</drivelineCapNo>
<driveLineSensitivity min="0" max="100">
  <!--ro, req, int, over line sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
</driveLineSensitivity>
<reverse>
  <!--ro, req, bool, whether to enable wrong-way driving detection-->true
</reverse>
<reverseCapNo opt="2,3">
  <!--ro, req, int, number of captured pictures, range:[2,3], attr:opt{req, string}-->0
</reverseCapNo>
<redLight>
  <!--ro, req, bool, whether to enable red light running detection-->true
</redLight>
<nonMotorRedLightEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</nonMotorRedLightEnabled>
<direction>
  <!--ro, req, bool, whether to enable detection of driving in wrong lane at intersection-->true
</direction>
<intersectionCongest>
  <!--ro, req, bool, whether to enable intersection congestion detection-->true
</intersectionCongest>
<nonDriveWay>
  <!--ro, opt, bool, whether to enable detection of motor vehicle on non-motor vehicle lane-->true
</nonDriveWay>
<nonDriveWayCapNo opt="2,3">
  <!--ro, opt, int, capture times, range:[2,3], attr:opt{req, string}-->0
</nonDriveWayCapNo>
<changeLane>
  <!--ro, req, bool, whether to enable illegal lane change detection-->true
</changeLane>
<changeLaneCapNo opt="2,3">
  <!--ro, req, int, number of captured pictures, range:[2,3], attr:opt{req, string}-->0
</changeLaneCapNo>
<changeLaneFirstImageReserveEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</changeLaneFirstImageReserveEnabled>
<banSign>
  <!--ro, opt, bool, whether to enable ban breaking detection-->true
</banSign>
<banSignCapNo opt="2,3">
  <!--ro, opt, int, number of captured pictures, range:[2,3], attr:opt{req, string}-->0
</banSignCapNo>
<notCatchingReverseVehiclesEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</notCatchingReverseVehiclesEnabled>
<intersectionStop>
  <!--ro, opt, object, stop vehicle on intersection-->
  <enable opt="true,false">
    <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
  </enable>
  <lastTime min="10" max="180">
    <!--ro, req, int, congestion duration, unit: second, range:[0,180], attr:min{req, int},max{req, int}-->0
  </lastTime>
  <sensitivity min="1" max="100">
    <!--ro, opt, int, triggering sensitivity, which is between 0 to 100, range:[1,100], attr:min{req, int},max{req, int}-->0
  </sensitivity>
  <bigCarRatio min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->0
  </bigCarRatio>
  <smallCarRatio min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->0
  </smallCarRatio>
  <capNo opt="3">
    <!--ro, req, int, number of captures, attr:opt{req, string}-->3
  </capNo>
</intersectionStop>
<greenLightStop>
  <!--ro, opt, bool, whether to enable detection of parking at green light-->true
</greenLightStop>
<overSpeed>
  <!--ro, req, bool, whether to enable overspeed detection-->true
</overSpeed>
<overSpeedCapNo opt="2,3">
  <!--ro, req, int, number of captured pictures in which the overspeed is detected, range:[2,3], attr:opt{req, string}-->0
</overSpeedCapNo>
<detectMode>
  <!--ro, opt, enum, detection mode, subType:string, desc:"video", "coilLoop" (video and coil)-->video
</detectMode>
<illegalTurn>
  <!--ro, req, bool, whether to enable illegal U-turning detection-->true
</illegalTurn>
<reverseLaneCheckpointFilteringEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
```

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```
    </reverseLaneCheckpointFilteringEnabled>
    <TrafficLight opt="IO,RS485,videoDetect,network,autoDetectTrafficLight">
      <!--ro, req, object, traffic lights, attr:opt{req, string}-->
      <source>
        <!--ro, req, enum, source, subType:string, desc:"IO", "RS485", "videoDetect" (video detection), "network"-->IO
      </source>
      <RS485LightList>
        <!--ro, opt, array, list of 485 indicator lights, subType:object-->
        <RS485Light>
          <!--ro, opt, object, 485 indicator light-->
          <lightId>
            <!--ro, req, int, "1-3", range:[1,6]-->0
          </lightId>
          <lightType>
            <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:guiding direction of traffic light: "left,straight,right"-->left
          </lightType>
          <relatedLightChan>
            <!--ro, req, int, linked channel of red light, range:[0,16]-->0
          </relatedLightChan>
          <yellowLightChan>
            <!--ro, req, int, linked channel of yellow light, range:[0,16]-->0
          </yellowLightChan>
        </RS485Light>
      </RS485LightList>
      <NetworkLightList>
        <!--ro, opt, array, list of network indicator lights, subType:object-->
        <NetworkLight>
          <!--ro, opt, object, network indicator light-->
          <lightId>
            <!--ro, req, int, light ID, range:[1,6]-->0
          </lightId>
          <lightType>
            <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:guiding direction of traffic light: "left,straight,right"-->left
          </lightType>
          <relatedLightChan>
            <!--ro, req, int, linked channel of red light, range:[0,16]-->0
          </relatedLightChan>
          <yellowLightChan>
            <!--ro, req, int, linked channel of yellow light, range:[0,16]-->0
          </yellowLightChan>
        </NetworkLight>
      </NetworkLightList>
    </TrafficLight>
    <relatedLaneCount min="1" max="6">
      <!--ro, req, int, number of supported lanes, range:[1,6], attr:min{req, int},max{req, int}-->0
    </relatedLaneCount>
    <relatedDriveWay min="1" max="99">
      <!--ro, req, int, linked lane No., range:[1,99], attr:min{req, int},max{req, int}-->0
    </relatedDriveWay>
    <recordEnable>
      <!--ro, req, bool, whether to enable recording when traffic violation happened-->true
    </recordEnable>
    <recordType opt="delayRecord,preRecord">
      <!--ro, req, enum, recording type, subType:string, attr:opt{req, string}, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
    </recordType>
    <preRecordTime min="0" max="100">
      <!--ro, req, int, pre-record time, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
    </preRecordTime>
    <recordDelayTime min="0" max="100">
      <!--ro, req, int, post-record time, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
    </recordDelayTime>
    <recordTimeOut min="0" max="100">
      <!--ro, req, int, recording timeout, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
    </recordTimeOut>
    <signSpeed min="0" max="255">
      <!--ro, opt, int, marked speed limit, range:[0,255], unit:km/h, attr:min{req, int},max{req, int}-->0
    </signSpeed>
    <speedLimit min="0" max="255">
      <!--ro, opt, int, speed limit, range:[0,255], unit:km/h, attr:min{req, int},max{req, int}-->0
    </speedLimit>
    <usageType opt="carriageWay,bus,slow,nonmotor,reverse,emergency,banNonMotor,ramp,fast,special">
      <!--ro, req, enum, lane usage, subType:string, attr:opt{req, string}, desc:"carriageWay" (roadway), "bus" (bus lane), "slow" (slow lane), "nonmotor"
(non-motor lane), "reverse" (contraflow lane), "emergency" (emergency lane), "banNonMotor" (non-motor vehicle vehicles prohibited lane), "ramp"--
>carriageWay
    </usageType>
    <LaneUsageList size="5">
      <!--ro, opt, object, list of lane purposes, attr:size{req, int}, desc:list of lane purposes-->
      <laneUsage opt="banTrucks,bus,banLeft,banRight,motor,banStraight,variableLane">
        <!--ro, req, enum, lane purpose, subType:string, attr:opt{req, string}, desc:lane usage type: "bus"-bus lane,"banTrucks"-truck prohibited
lane,"banLeft"-left turn prohibited lane,"banRight"-right turn prohibited lane,"motor"-motorcycle lane-->banTrucks
      </laneUsage>
    </LaneUsageList>
    <variableLaneTimeSwitchList size="4">
      <!--ro, opt, object, time period for changeable lanes, attr:size{req, int}-->
      <variableLaneTimeSwitch>
        <!--ro, req, object, time for changeable lanes-->
        <timeId min="1" max="4">
          <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
        </timeId>
        <startHour min="0" max="23">
          <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
        </startHour>
        <startMinute min="0" max="59">
          <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
```

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```
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
      </endMinute>
      <laneDirection opt="unkown,left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight,leftWait,straightWait,strightWaitRight">
        <!--ro, req, enum, driving direction, subType:string, attr:opt{req, string}, desc:"unknown", "left", "straight", "leftStraight", "right",
"leftRight", "rightStraight", "leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->leftRightStraight
      </laneDirection>
    </variableLaneTimeSwitch>
  </variableLaneTimeSwitchList>
  <variableLaneTimeSwitchType opt="day,week">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->week
  </variableLaneTimeSwitchType>
  <variableLaneWeekSwitchList size="7">
    <!--ro, opt, object, attr:size{req, int}-->
    <variableLaneWeekSwitch>
      <!--ro, req, object-->
      <dayOfWeek opt="1,2,3,4,5,6,7">
        <!--ro, req, enum, subType:int, attr:opt{req, string}-->1
      </dayOfWeek>
      <timeRangeList>
        <!--ro, req, object-->
        <timeRange>
          <!--ro, req, object-->
          <ID min="1" max="4">
            <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->1
          </ID>
          <startHour min="0" max="23">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
          </startHour>
          <startMinute min="0" max="59">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
          </startMinute>
          <endHour min="0" max="23">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
          </endHour>
          <endMinute min="0" max="59">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
          </endMinute>
          <laneDirection opt="unkown,left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight,leftWait,straightWait,strightWaitRight">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
          </laneDirection>
        </timeRange>
      </timeRangeList>
    </variableLaneWeekSwitch>
  </variableLaneWeekSwitchList>
  <directionType opt="unkown,left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight,leftWait,straightWait,strightWaitRight">
    <!--ro, req, enum, driving direction, subType:string, attr:opt{req, string}, desc:direction type,multiple options can be supported at one time--
>leftRightStraight
  </directionType>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, lane direction type, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south
to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to
northeast)-->0
  </laneDirectionType>
  <stopLineDistance min="0" max="300">
    <!--ro, req, int, distance between the vehicle and the stop line in the second captured picture of red light running violation, range:[0,300], unit:m,
attr:min{req, int},max{req, int}, desc:distance between the vehicle and the stop line in the second captured picture of red light running violation-->0
  </stopLineDistance>
  <loopSensitivity min="1" max="100">
    <!--ro, opt, int, loop sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
  </loopSensitivity>
  <capPositionOffset min="0" max="300">
    <!--ro, req, int, second captured picture deviation, range:[0,300], attr:min{req, int},max{req, int}, desc:unit: pixel-->50
  </capPositionOffset>
  <videoDetectLightId min="1" max="12">
    <!--ro, req, int, area ID of video detection traffic light, range:[1,12], attr:min{req, int},max{req, int}-->1
  </videoDetectLightId>
  <lightNum min="0" max="12">
    <!--ro, req, int, number of lights, range:[1,3], attr:min{req, int},max{req, int}-->1
  </lightNum>
  <straightLight>
    <!--ro, req, bool, straight signal-->true
  </straightLight>
  <leftLight>
    <!--ro, req, bool, left turn signal-->true
  </leftLight>
  <rightLight>
    <!--ro, req, bool, right turn signal-->true
  </rightLight>
  <greenLight>
    <!--ro, req, bool, green light-->true
  </greenLight>
  <yellowLight>
    <!--ro, req, bool, yellow light-->true
  </yellowLight>
  <yellowLightTime min="0" max="10000">
    <!--ro, req, int, yellow time, range:[0,10000], unit:ms, attr:min{req, int},max{req, int}-->0
  </yellowLightTime>
  <beginTime opt="1, 2, 3">
```

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```
      <!--ro, req, enum, recording start time, subType:int, attr:opt{req, string}, desc:recording start time-->1
    </beginTime>
    <leftNotYieldStraight>
      <!--ro, opt, object, detection of left turn not yield to straight-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of left turn not yield to straight driving, attr:opt{req, string}-->true
      </enable>
      <capNo opt="3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
    </leftNotYieldStraight>
    <rightNotYieldLeft>
      <!--ro, opt, object, detection of right turn not yield to left turn-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of right turn not yield to left turn, attr:opt{req, string}-->true
      </enable>
      <capNo opt="3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
    </rightNotYieldLeft>
    <uTurnNotYieldGoingStaight>
      <!--ro, opt, object, detection of U-turn not yield to straight driving-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of U-turn not yield to straight driving, attr:opt{req, string}-->true
      </enable>
      <capNo opt="3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
    </uTurnNotYieldGoingStaight>
    <leftSideDrivingAfterLeftTurn>
      <!--ro, opt, object, turn left in short turn radius-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of left turn not yield to straight driving, attr:opt{req, string}-->true
      </enable>
      <capNo opt="3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
    </leftSideDrivingAfterLeftTurn>
    <failYieldPed>
      <!--ro, opt, object, failing to yield for pedestrians-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of left turn not yield to straight driving, attr:opt{req, string}-->true
      </enable>
      <cameraPosition opt="crossing,roadSection">
        <!--ro, req, enum, camera position, subType:string, attr:opt{req, string}, desc:Yellow Time-->crossing
      </cameraPosition>
      <noDisplacementPedEnabled opt="true,false">
        <!--ro, req, bool, whether to enable no displacement capture of pedestrian, attr:opt{req, string}-->true
      </noDisplacementPedEnabled>
      <failYieldPedLineList>
        <!--ro, opt, array, list of failing to yield for pedestrians lines, subType:object-->
        <failYieldPedLine>
          <!--ro, opt, object, failing to yield for pedestrians line-->
          <lineId opt="1,2">
            <!--ro, req, int, line ID, range:[1,2], attr:opt{req, string}-->0
          </lineId>
          <lineName>
            <!--ro, req, string, line name, range:[0,64]-->test
          </lineName>
          <RegionCoordinatesList size="2">
            <!--ro, opt, array, subType:object, range:[0,2], attr:size{opt, int}-->
            <RegionCoordinates>
              <!--ro, opt, object-->
              <positionX>
                <!--ro, req, int, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--ro, req, int, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </failYieldPedLine>
      </failYieldPedLineList>
      <failYieldPedSensitivity min="1" max="100">
        <!--ro, req, int, not yielding to pedestrian sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
      </failYieldPedSensitivity>
      <failYieldnonMotor opt="true,false">
        <!--ro, opt, bool, whether to enable capturing not yielding to pedestrian, attr:opt{req, string}-->true
      </failYieldnonMotor>
      <failYieldNum min="1" max="7">
        <!--ro, opt, int, number of pedestrian not yielded to, attr:min{req, int},max{req, int}-->1
      </failYieldNum>
      <failYieldAreaType opt="rectangle,polygon">
        <!--ro, opt, enum, detection area of failing to yield for pedestrians, subType:string, attr:opt{req, string}, desc:"rectangle", "polygon"-->rectangle
      </failYieldAreaType>
      <failYieldDistanceMode opt="car,lane">
        <!--ro, opt, enum, distance mode of failing to yield for pedestrians, subType:string, attr:opt{req, string}, desc:"car" (distance from pedestrian to
vehicle center), "lane" (distance from pedestrian to lane center)-->car
      </failYieldDistanceMode>
      <failYieldSnapPosition min="0" max="100">
        <!--ro, opt, int, distance range for capturing failing to yield for pedestrians, attr:min{req, int},max{req, int}-->20
      </failYieldSnapPosition>
      <capNo opt="3">
```

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    <!--ro, req, int, number of captures, attr:opt{req, string}-->3
  </capNo>
</failYieldPed>
<illegalParking>
  <!--ro, opt, object, illegal parking detection-->
  <enable opt="true,false">
    <!--ro, req, bool, whether to enable illegal parking detection, attr:opt{req, string}-->true
  </enable>
  <sensitivity min="0" max="100">
    <!--ro, req, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
  </sensitivity>
  <lastTime min="1" max="180">
    <!--ro, req, int, parking time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </lastTime>
  <illegalParkingLineList>
    <!--ro, opt, array, subType:object, range:[0,2]-->
    <illegalParkingLine>
      <!--ro, opt, object-->
      <lineId opt="1,2,3">
        <!--ro, req, int, attr:opt{req, string}-->0
      </lineId>
      <lineName>
        <!--ro, req, string, range:[0,64]-->test
      </lineName>
      <RegionCoordinatesList num="2">
        <!--ro, opt, array, subType:object, range:[0,2], attr:num{opt, int}-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </illegalParkingLine>
  </illegalParkingLineList>
  <capNo opt="3">
    <!--ro, req, int, number of captures, attr:opt{req, string}-->3
  </capNo>
</illegalParking>
<gasser>
  <!--ro, opt, object, queue jumping-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable queue jumping detection, attr:opt{req, string}-->true
  </enabled>
  <sensitivity min="0" max="100">
    <!--ro, req, int, detection sensitivity of queue jumping, range:[1,100], attr:min{req, int},max{req, int}-->1
  </sensitivity>
  <congestionThreshold min="1" max="100">
    <!--ro, req, int, congestion threshold, range:[1,100], attr:min{req, int},max{req, int}-->1
  </congestionThreshold>
  <capInterval min="0" max="6000">
    <!--ro, req, int, second and third capture detection, range:[0,6000], unit:ms, attr:min{req, int},max{req, int}-->0
  </capInterval>
  <capNo opt="3">
    <!--ro, req, int, number of captures, attr:opt{req, string}-->3
  </capNo>
</gasser>
<congestion>
  <!--ro, opt, object, congestion-->
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable congestion detection, attr:opt{req, string}-->true
  </enabled>
  <interval min="1" max="10">
    <!--ro, req, int, alarm interval, range:[1,10], unit:min, attr:min{req, int},max{req, int}-->1
  </interval>
  <sensitivity min="1" max="100">
    <!--ro, req, int, congestion sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
  </sensitivity>
  <lastTime min="1" max="60">
    <!--ro, req, int, congestion duration, range:[1,60], unit:s, attr:min{req, int},max{req, int}-->1
  </lastTime>
  <capNo opt="3">
    <!--ro, req, int, number of captures, attr:opt{req, string}-->3
  </capNo>
</congestion>
<accident>
  <!--ro, opt, object, accident detection-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable accident detection, attr:opt{req, string}-->true
  </enabled>
  <lastTime min="1" max="180">
    <!--ro, req, int, event time limit, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </lastTime>
  <interval min="1" max="60">
    <!--ro, req, int, repeated capture interval, range:[1,60], unit:min, attr:min{req, int},max{req, int}-->1
  </interval>
  <capNo opt="3">
    <!--ro, req, int, number of captures, attr:opt{req, string}-->3
  </capNo>
</accident>
<runGreenLight opt="true,false">
```

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```
    <!--ro, opt, bool, whether to enable green light running detection, attr:opt{req, string}-->true
  </runGreenLight>
  <runGreenLightParam>
    <!--ro, opt, object, parameters of green light running detection-->
    <detectEvent opt="runGreenLight,crossingStop">
      <!--ro, opt, enum, Detection Logic, subType:string, attr:opt{req, string}, desc:"runGreenLight" (running green light in congestion), "crossingStop"
(overstay at intersection)-->runGreenLight
    </detectEvent>
    <interval min="1" max="180">
      <!--ro, opt, int, capture interval, range:[1,180], unit:s, attr:min{req, int},max{req, int}, desc:capture interval-->15
    </interval>
    <secondLampState opt="0,1,2">
      <!--ro, opt, enum, traffic light status of the second capture times, subType:int, attr:opt{req, string}, desc:0 (red light), 1 (green light), 2
(unlimited)-->2
    </secondLampState>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </runGreenLightParam>
  <helmet>
    <!--ro, opt, bool, without helmet-->true
  </helmet>
  <helmetCapNo opt="1,2,3">
    <!--ro, opt, int, number of captured pictures in which the driver does not wear the helmet, range:[1,3], attr:opt{req, string}-->1
  </helmetCapNo>
  <nonMotorExist>
    <!--ro, opt, bool, non-motor vehicle on motor vehicle lane-->true
  </nonMotorExist>
  <nonMotorExistCapNo opt="2,3">
    <!--ro, opt, int, number of captured pictures in which the motor lane is occupied by non-motor vehicles, range:[2,3], attr:opt{req, string}-->2
  </nonMotorExistCapNo>
  <capType opt="vehicle,nonMotor,human,face,mixObject,motor,motorWithHum,nonMotorWithHum,motorNonmotor">
    <!--ro, opt, enum, capture types, subType:string, attr:opt{req, string}, desc:"vehWithHum" (motor vehicle, non-motor vehicle, and pedestrian), "vehicle"
(motor vehicle), "nonmotor" (non-motor vehicle), "human" (pedestrian), "motorWithHum" (motor vehicle and pedestrian), "nonmotorWithHum" (non-motor vehicle
and pedestrian), "motorNonmotor" (motor vehicle and non-motor vehicle)-->mixObject
  </capType>
  <deferredFrmNum min="0" max="25">
    <!--ro, opt, int, delay a few seconds to trigger the mode, range:[0,25], attr:min{req, int},max{req, int}-->0
  </deferredFrmNum>
  <redLightCapNo opt="3,4">
    <!--ro, opt, int, captured picture amount in red light, range:[3,4], attr:opt{req, string}-->3
  </redLightCapNo>
  <directionCapNo opt="3,4">
    <!--ro, opt, int, number of captured pictures of driving against direction guidance, range:[3,4], attr:opt{req, string}-->3
  </directionCapNo>
  <firstShowTailVep opt="0,1,2,3">
    <!--ro, opt, enum, place of the first capture time, subType:int, attr:opt{req, string}, desc:0 (none), 1 (expose vehicle tail), 2 (expose vehicle tail
and LPR success), 3(parking over stop line)-->0
  </firstShowTailVep>
  <banSignType opt="truck,yellowPlate,standard,bus">
    <!--ro, opt, enum, capture type list, subType:string, attr:opt{req, string}, desc:"truck", "yellowPlate" (yellow license plate vehicle), "standard",
"bus"-->truck
  </banSignType>
  <greenLightStopSensitivity min="1" max="100">
    <!--ro, opt, int, stopping sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->0
  </greenLightStopSensitivity>
  <nonMotorPassenger>
    <!--ro, opt, object, Manned Non-Motor Vehicle-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="1,2">
      <!--ro, req, int, the number of captured pictures, range:[1,2], attr:opt{req, string}-->0
    </capNo>
    <refineMannedIllegalCode opt="true,false">
      <!--ro, opt, bool, whether to enable detailed manned violation code, attr:opt{req, string}-->true
    </refineMannedIllegalCode>
  </nonMotorPassenger>
  <speedDetectMode opt="none,radar">
    <!--ro, opt, enum, speed detection mode, subType:string, attr:opt{req, string}, desc:"none", "radar"-->none
  </speedDetectMode>
  <BanMotorDet>
    <!--ro, opt, object, motorcycle driving in prohibited areas-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
    </enabled>
  </BanMotorDet>
  <Radar>
    <!--ro, opt, object, radar parameters-->
    <radarType opt="none,MultiLane,csr,IOBox,SSTK,custom" def="MultiLane">
      <!--ro, opt, string, radar type, attr:opt{req, string},def{req, string}, desc:"none", "MultiLane" (multi-lane radar), "csr" (TransMicrowave),"IOBox"
(radar connecting I/O expansion box), "SSTK" (Sensortech),"custom" (custom type), the default value is MultiLane-->MultiLane
    </radarType>
    <validRadarSpeedTime min="1" max="9000" def="2000">
      <!--ro, opt, int, radar speed valid time, range:[0,9000], unit:ms, attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist,
it indicates the function is supported. The default value is 2,000-->0
    </validRadarSpeedTime>
    <radarLinearCorrection min="0.0" max="2.0" def="1.0">
      <!--ro, opt, float, linear correction parameters, range:[0.0,2.0], attr:min{req, float},max{req, float},def{req, float}, desc:if this node does not
exist, it indicates the function is supported. The default value is 1.0, and the value should be accurate to 3 decimal places-->1.0
    </radarLinearCorrection>
    <radarConstantCorrection min="-100" max="100" def="0">
      <!--ro, opt, int, constant correction parameter,, range:[-100,100], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist,
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it indicates the function is supported. The default value is 0-->0
    </radarConstantCorrection>
    <radarRS485 min="1" max="5" def="2">
      <!--ro, opt, int, 485 radar No., range:[1,5], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist, it indicates the
function is supported. The default value is 2-->1
    </radarRS485>
  </Radar>
  <NotYieldToRightRoad>
    <!--ro, opt, object, not yield to vehicles from right roads-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <firstCapturePosition min="0" max="100">
      <!--ro, req, int, the capture position of the first picture, range:[0,100], step:1, attr:min{req, int},max{req, int}-->20
    </firstCapturePosition>
    <lateralDistance min="20" max="350">
      <!--ro, req, int, lateral distance between two vehicles, range:[20,350], step:1, attr:min{req, int},max{req, int}-->150
    </lateralDistance>
    <longitudinalDistance min="10" max="150">
      <!--ro, req, int, longitudinal distance between two vehicles, range:[10,150], step:1, attr:min{req, int},max{req, int}-->50
    </longitudinalDistance>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </NotYieldToRightRoad>
  <EnterNotYieldToInRoundabout>
    <!--ro, opt, object, not yield at roundabout intersection-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <firstCapturePosition min="0" max="100">
      <!--ro, req, int, the capture position of the first picture, range:[0,100], step:1, attr:min{req, int},max{req, int}-->20
    </firstCapturePosition>
    <lateralDistance min="20" max="350">
      <!--ro, req, int, lateral distance between two vehicles, range:[20,350], step:1, attr:min{req, int},max{req, int}-->150
    </lateralDistance>
    <longitudinalDistance min="10" max="150">
      <!--ro, req, int, longitudinal distance between two vehicles, range:[10,150], step:1, attr:min{req, int},max{req, int}-->50
    </longitudinalDistance>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </EnterNotYieldToInRoundabout>
  <BanSignTypeList size="10">
    <!--ro, opt, array, capture type list, subType:object, attr:size{req, int}-->
    <banSignType opt="standard,truck,yellowPlate,bus,yellowBus,dangerCar,instructionCar,pickupTruck,concreteMixer,slagTruck">
      <!--ro, opt, enum, prohibition violation type, subType:string, attr:opt{req, string}, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus), "dangercar" (dangerous goods transport vehicle), "instructioncar" (driver-training vehicle)--
>standard
    </banSignType>
  </BanSignTypeList>
  <lineType opt="unknown,white,singleYellow,doubleYellow,guardRail,noCross,fullDottedLine,dottedFullLine">
    <!--ro, opt, enum, line type, subType:string, attr:opt{req, string}, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle), "fullDottedLine" (double
yellow line (left: dotted, right: solid)), "dottedFullLine" (double yellow line (left: solid, right: dotted))-->unknown
  </lineType>
  <bigCarSignSpeed min="0" max="255">
    <!--ro, opt, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSignSpeed>
  <bigCarSpeedLimit min="0" max="60">
    <!--ro, opt, int, maximum speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSpeedLimit>
  <TruckTurnRightStop>
    <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->false
    </enabled>
    <capNo opt="3,5">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, stopping sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->50
    </sensitivity>
    <stopTime min="1" max="180">
      <!--ro, opt, int, Parking Duration, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->10
    </stopTime>
    <capInterval min="1" max="10000">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1000
    </capInterval>
  </TruckTurnRightStop>
  <carHighSpeed min="0" max="255">
    <!--ro, opt, int, abnormal speeding of small-sized vehicle, attr:min{req, int},max{req, int}-->0
  </carHighSpeed>
  <bigCarHighSpeed min="0" max="255">
    <!--ro, opt, int, abnormal speeding of large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarHighSpeed>
  <greenLightStopCapNo opt="3">
    <!--ro, opt, int, captured picture amount in green light, attr:opt{req, string}-->3
  </greenLightStopCapNo>
  <illegalTurnCapNo opt="3">
    <!--ro, req, int, number of captured violation turing picture(s), attr:opt{req, string}-->3
  </illegalTurnCapNo>
  <noBanLeftTypeList size="5">
    <!--ro, opt, array, allowlist of left-turn vehicle types, subType:object, attr:size{req, int}-->
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    <noBanLeftType opt="standard,truck,yellowPlate,bus,yellowBus">
      <!--ro, opt, enum, vehicle types which are allowed to turn left, subType:string, attr:opt{req, string}, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus", "yellowbus" (yellow license plate bus)-->standard
    </noBanLeftType>
  </noBanLeftTypeList>
  <noBanRightTypeList size="5">
    <!--ro, opt, array, allowlist of right-turn vehicle types, subType:object, attr:size{req, int}-->
    <noBanRightType opt="standard,truck,yellowPlate,bus,yellowBus">
      <!--ro, opt, enum, vehicle types which are allowed to left-turn, subType:string, attr:opt{req, string}, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus", "yellowbus" (yellow license plate bus)-->standard
    </noBanRightType>
  </noBanRightTypeList>
  <NoAlterTrafficDet>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="3">
      <!--ro, req, int, attr:opt{req, string}-->3
    </capNo>
    <ConvergenceLineList size="1">
      <!--ro, opt, array, subType:object, range:[1,1], attr:size{req, int}-->
      <ConvergenceLine>
        <!--ro, opt, object-->
        <lineId opt="1">
          <!--ro, req, int, attr:opt{req, string}-->1
        </lineId>
        <lineName>
          <!--ro, req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList size="2">
          <!--ro, opt, array, subType:object, range:[0,2], attr:size{req, int}-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ConvergenceLine>
    </ConvergenceLineList>
  </NoAlterTrafficDet>
  <nonMotorSignSpeed min="0" max="255">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </nonMotorSignSpeed>
  <nonMotorSpeedLimit min="0" max="255">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </nonMotorSpeedLimit>
  <nonMotorHighSpeed min="0" max="255">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </nonMotorHighSpeed>
  <freightManned>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="2,3">
      <!--ro, req, int, attr:opt{req, string}-->2
    </capNo>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->50
    </sensitivity>
  </freightManned>
  <busCapDirectionType opt="left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->leftRightStraight
  </busCapDirectionType>
  <reverseSensitivity min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </reverseSensitivity>
  <VirtualLaneList size="6">
    <!--ro, opt, object, attr:size{req, int}-->
    <VirtualLane>
      <!--ro, req, object-->
      <lineName opt="laneRightBoundaryLine,turnLeftLine,turnRightLine,topZebraLine,bottomZebraLine,triggerLine">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->laneRightBoundaryLine
      </lineName>
    </VirtualLane>
  </VirtualLaneList>
</VideoEpolice>
```

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### 14.5.1.29 Get the trigger mode parameters of video intersection violation system

**Request URL**

GET /ISAPI/ITC/TriggerMode/videoEpolice?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelVideoEpolice is "true" returned from /ISAPI/ITC/TriggerMode/capabilities). |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoEpolice xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the trigger mode parameters of video intersection violation system, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <relatedLaneCount>
    <!--ro, req, int, number of linked lanes, range:[1,4]-->0
  </relatedLaneCount>
  <TrafficLight>
    <!--ro, req, object, traffic lights-->
    <source>
      <!--ro, req, enum, signal source, subType:string, desc:"IO, RS485, videoDetect, network"-->IO
    </source>
    <IOLightList>
      <!--ro, opt, array, list, subType:object-->
      <IOLight>
        <!--ro, opt, object, I/O access-->
        <lightId>
          <!--ro, req, int, traffic light No., range:[1,6]-->1
        </lightId>
        <lightType>
          <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"Left, straight, right"-->straight
        </lightType>
        <relatedIO>
          <!--ro, req, enum, linked I/O input, subType:string, desc:"NON_IO, IO1, IO2, IO3, IO4, IO5, IO6, IO7, IO8"-->IO7
        </relatedIO>
        <redLightStatus>
          <!--ro, req, enum, red light level status, subType:string, desc:"low" (low-level red light), "high" (high-level red light)-->low
        </redLightStatus>
      </IOLight>
    </IOLightList>
    <RS485LightList>
      <!--ro, opt, array, list, subType:object-->
      <RS485Light>
        <!--ro, opt, object, RS-485 access-->
        <lightId>
          <!--ro, req, int, traffic light No., range:[1,6]-->0
        </lightId>
        <lightType>
          <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"left, right, straight"-->straight
        </lightType>
        <relatedLightChan>
          <!--ro, req, int, detector channel No. linked to red light, range:[0,16]-->0
        </relatedLightChan>
        <yellowLightChan>
          <!--ro, req, int, detector channel No. linked to yellow light, range:[0,16]-->0
        </yellowLightChan>
      </RS485Light>
    </RS485LightList>
    <NetworkLightList>
      <!--ro, opt, array, list, subType:object-->
      <NetworkLight>
        <!--ro, opt, object, network indicator light-->
        <lightId>
          <!--ro, req, int, traffic light No., range:[1,6]-->0
        </lightId>
        <lightType>
          <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
        </lightType>
        <relatedLightChan>
          <!--ro, req, int, detector channel No. linked to red light, range:[0,16]-->0
        </relatedLightChan>
        <yellowLightChan>
          <!--ro, req, int, detector channel No. linked to yellow light, range:[0,16]-->0
        </yellowLightChan>
      </NetworkLight>
    </NetworkLightList>
    <videoDetectLightList>
      <!--ro, opt, array, list, subType:object-->
      <videoDetectLight>
        <!--ro, opt, object, video detection access-->
        <videoDetectLightId>
          <!--ro, req, int, video detection signal No., range:[1,12]-->0
        </videoDetectLightId>
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      </videoDetectLightId>
      <lightNum>
        <!--ro, req, int, number of traffic lights, range:[0,3]-->0
      </lightNum>
      <straightLight>
        <!--ro, req, bool, whether the straight driving light is on-->true
      </straightLight>
      <leftLight>
        <!--ro, req, bool, whether the left-turn light is on-->true
      </leftLight>
      <rightLight>
        <!--ro, req, bool, whether the right-turn light is on-->true
      </rightLight>
      <redLight>
        <!--ro, req, bool, whether the red light is on-->true
      </redLight>
      <greenLight>
        <!--ro, req, bool, whether the green light is on-->true
      </greenLight>
      <yellowLight>
        <!--ro, req, bool, whether the yellow light is on-->true
      </yellowLight>
      <yellowLightTime>
        <!--ro, req, int, yellow light duration, range:[0,10], unit:s-->0
      </yellowLightTime>
      <rectRegion>
        <!--ro, req, object, traffic light region, desc:the origin is the upper-left corner of the screen-->
        <positionLeft>
          <!--ro, req, int, left boundary, range:[0,65535]-->0
        </positionLeft>
        <positionTop>
          <!--ro, req, int, top boundary, range:[0,65535]-->0
        </positionTop>
        <positionRight>
          <!--ro, req, int, right boundary, range:[0,65535]-->0
        </positionRight>
        <positionBottom>
          <!--ro, req, int, bottom boundary, range:[0,65535]-->0
        </positionBottom>
      </rectRegion>
    </videoDetectLight>
  </videoDetectLightList>
  <autoDetectTrafficLight>
    <!--ro, opt, object, dep:and,{$.VideoEpolice.TrafficLight.source,eq,autoDetectTrafficLight}-->
    <detectRegionList>
      <!--ro, opt, array, subType:object, range:[0,8]-->
      <detectRegion>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, range:[0,1]-->0.001
        </height>
        <width>
          <!--ro, req, float, range:[0,1]-->0.001
        </width>
        <x>
          <!--ro, req, float, range:[0,1]-->0.001
        </x>
        <y>
          <!--ro, req, float, range:[0,1]-->0.001
        </y>
      </detectRegion>
    </detectRegionList>
    <trafficLightPanelList>
      <!--ro, opt, array, subType:object, range:[0,8]-->
      <trafficLightPanel>
        <!--ro, opt, object-->
        <ruleID>
          <!--ro, req, int, range:[1,8]-->1
        </ruleID>
        <lampPanelDirection>
          <!--ro, opt, enum, subType:string-->level
        </lampPanelDirection>
        <lampPanelType>
          <!--ro, opt, enum, subType:string-->combination
        </lampPanelType>
        <lampSequence>
          <!--ro, opt, enum, subType:string-->redYellowGreen
        </lampSequence>
        <lampPanelRegion>
          <!--ro, opt, object-->
          <height>
            <!--ro, req, float, range:[0,1]-->0.001
          </height>
          <width>
            <!--ro, req, float, range:[0,1]-->0.001
          </width>
          <x>
            <!--ro, req, float, range:[0,1]-->0.001
          </x>
          <y>
            <!--ro, req, float, range:[0,1]-->0.001
          </y>
        </lampPanelRegion>
        <trafficLightList>
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            <!--ro, opt, array, subType:object, range:[0,8]-->
            <trafficLight>
              <!--ro, opt, object-->
              <trafficLightType>
                <!--ro, opt, enum, subType:string-->roundLight
              </trafficLightType>
              <lampPanelRegion>
                <!--ro, opt, object-->
                <height>
                  <!--ro, req, float, range:[0,1]-->0.001
                </height>
                <width>
                  <!--ro, req, float, range:[0,1]-->0.001
                </width>
                <x>
                  <!--ro, req, float, range:[0,1]-->0.001
                </x>
                <y>
                  <!--ro, req, float, range:[0,1]-->0.001
                </y>
              </lampPanelRegion>
            </trafficLight>
          </trafficLightList>
        </trafficLightPanel>
      </trafficLightPanelList>
    </autoDetectTrafficLight>
  </TrafficLight>
  <LaneParamList>
    <!--ro, req, array, lane parameter list, subType:object-->
    <LaneParam>
      <!--ro, req, object, lane parameters-->
      <laneId>
        <!--ro, req, int, lane No., range:[1,6]-->0
      </laneId>
      <laneDirectionType>
        <!--ro, req, enum, lane direction No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,99]-->0
      </relatedDriveWay>
      <recordEnable>
        <!--ro, req, bool, whether to enable recording when red light running is detected-->true
      </recordEnable>
      <recordType>
        <!--ro, req, enum, recording type, subType:string, desc:"preRecord, delayRecord"-->preRecord
      </recordType>
      <beginTime>
        <!--ro, req, enum, recording start time, subType:int, desc:"0, 1, 2, 3"-->0
      </beginTime>
      <preRecordTime>
        <!--ro, req, int, pre-record time, range:[0,100], unit:s-->0
      </preRecordTime>
      <recordDelayTime>
        <!--ro, req, int, post-record time, range:[0,100], unit:s-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--ro, req, int, recording timeout, range:[0,100], unit:s-->0
      </recordTimeOut>
      <signSpeed>
        <!--ro, opt, int, marked speed limit, range:[0,255], unit:km/h-->0
      </signSpeed>
      <speedLimit>
        <!--ro, opt, int, speed limit, range:[0,255], unit:km/h-->0
      </speedLimit>
      <usageType>
        <!--ro, req, enum, lane usage,only one of the options can be supported at a time, subType:string, desc:"carriageWay" (normal lane),"bus" (fast
lane),"slow" (slow lane), "nonmotor" (non-motor lane), "reverse" (reverse lane), "emergency" (emergency lane)-->carriageWay
      </usageType>
      <LaneUsageList>
        <!--ro, opt, array, lane usage, one or multiple options can be supported at a time, subType:object, dep:and,
{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special}, desc:lane usage, one or multiple options can be supported at a time-->
        <laneUsage>
          <!--ro, req, enum, lane usage, subType:string, desc:"bus" (bus lane), "banTrucks" (trucks prohibited lane), "banLeft" (left-turn prohibited lane),
"banRight" (right-turn prohibited lane), "motor" (motorcycle lane)-->bus
        </laneUsage>
      </LaneUsageList>
      <banTrucksTimeSwitchList>
        <!--ro, req, array, list of large vehicle prohibited time periods, subType:object-->
        <banTrucksTimeSwitch>
          <!--ro, req, object, large vehicle prohibited time periods-->
          <timeId>
            <!--ro, req, int, time period No., range:[1,4]-->0
          </timeId>
          <startHour>
            <!--ro, opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--ro, opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--ro, opt, int, end time (hour), range:[0,23]-->0
          </endHour>
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      <endMinute>
        <!--ro, opt, int, end time (minute), range:[0,59]-->0
      </endMinute>
    </banTrucksTimeSwitch>
  </banTrucksTimeSwitchList>
  <busWayTimeSwitchList>
    <!--ro, req, array, list, subType:object-->
    <busWayTimeSwitch>
      <!--ro, req, object, bus allowed passing time-->
      <timeId>
        <!--ro, req, int, time period No., range:[1,4]-->0
      </timeId>
      <startHour>
        <!--ro, opt, int, start time (hour), range:[0,23]-->0
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute), range:[0,59]-->0
      </startMinute>
      <endHour>
        <!--ro, opt, int, end time (hour), range:[0,23]-->0
      </endHour>
      <endMinute>
        <!--ro, opt, int, end time (minute), range:[0,59]-->0
      </endMinute>
    </busWayTimeSwitch>
  </busWayTimeSwitchList>
  <busCapDirectionType>
    <!--ro, opt, enum, subType:string-->leftRightStraight
  </busCapDirectionType>
  <emergencyTimeSwitchList>
    <!--ro, opt, array, time period for emergency lanes, subType:object-->
    <emergencyTimeSwitch>
      <!--ro, req, object, Emergency Lane Time-->
      <timeId>
        <!--ro, req, int, time period No., range:[1,4]-->0
      </timeId>
      <startHour>
        <!--ro, opt, int, start time (hour), range:[0,23]-->0
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute), range:[0,59]-->0
      </startMinute>
      <endHour>
        <!--ro, opt, int, end time (hour), range:[0,23]-->0
      </endHour>
      <endMinute>
        <!--ro, opt, int, end time (minute), range:[0,59]-->0
      </endMinute>
    </emergencyTimeSwitch>
  </emergencyTimeSwitchList>
  <banLeftTimeSwitchList>
    <!--ro, req, array, left-turn prohibited time period, subType:object-->
    <banLeftTimeSwitch>
      <!--ro, req, object, left-turn prohibited time-->
      <timeId>
        <!--ro, req, int, time period No., range:[1,4]-->0
      </timeId>
      <startHour>
        <!--ro, opt, int, start time (hour), range:[0,23]-->0
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute), range:[0,59]-->0
      </startMinute>
      <endHour>
        <!--ro, opt, int, end time (hour), range:[0,23]-->0
      </endHour>
      <endMinute>
        <!--ro, opt, int, end time (minute), range:[0,59]-->0
      </endMinute>
    </banLeftTimeSwitch>
  </banLeftTimeSwitchList>
  <banRightTimeSwitchList>
    <!--ro, req, array, right-turn prohibited time period, subType:object-->
    <banRightTimeSwitch>
      <!--ro, req, object, right-turn prohibited time-->
      <timeId>
        <!--ro, req, int, time period No., range:[1,4]-->0
      </timeId>
      <startHour>
        <!--ro, opt, int, start time (hour), range:[0,23]-->0
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute), range:[0,59]-->0
      </startMinute>
      <endHour>
        <!--ro, opt, int, end time (hour), range:[0,23]-->0
      </endHour>
      <endMinute>
        <!--ro, opt, int, end time (minute), range:[0,59]-->0
      </endMinute>
    </banRightTimeSwitch>
  </banRightTimeSwitchList>
  <banStraightTimeSwitchList>
    <!--ro, req, array, straight prohibited time period, subType:object-->
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```
        <banStraightTimeSwitch>
          <!--ro, req, object, straight prohibited time-->
          <timeId>
            <!--ro, req, int, index, range:[1,4]-->0
          </timeId>
          <startHour>
            <!--ro, opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--ro, opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--ro, opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
            <!--ro, opt, int, end time (minute), range:[0,59]-->0
          </endMinute>
        </banStraightTimeSwitch>
      </banStraightTimeSwitchList>
      <variableLaneTimeSwitchList>
        <!--ro, opt, array, time period for changeable lanes, subType:object, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special},
desc:time period for changeable lanes-->
        <variableLaneTimeSwitch>
          <!--ro, req, object, time for changeable lanes-->
          <timeId>
            <!--ro, req, int, index, range:[1,4]-->0
          </timeId>
          <startHour>
            <!--ro, opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--ro, opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--ro, opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
            <!--ro, opt, int, end time (minute), range:[0,59]-->0
          </endMinute>
          <laneDirection>
            <!--ro, req, enum, Driving Direction, subType:string, desc:"unknown", "left", "straight", "leftStraight", "right", "leftRight", "rightStraight",
"leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->unknown
          </laneDirection>
        </variableLaneTimeSwitch>
      </variableLaneTimeSwitchList>
      <variableLaneTimeSwitchType>
        <!--ro, opt, enum, subType:string, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special}-->week
      </variableLaneTimeSwitchType>
      <variableLaneWeekSwitchList>
        <!--ro, opt, array, subType:object, range:[0,7], dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special}-->
        <variableLaneWeekSwitch>
          <!--ro, req, object-->
          <dayOfWeek>
            <!--ro, req, enum, subType:int-->1
          </dayOfWeek>
          <timeRangeList>
            <!--ro, req, array, subType:object, range:[0,4]-->
            <timeRange>
              <!--ro, req, object-->
              <ID>
                <!--ro, req, int, range:[1,4]-->1
              </ID>
              <startHour>
                <!--ro, opt, int, range:[0,23]-->0
              </startHour>
              <startMinute>
                <!--ro, opt, int, range:[0,59]-->0
              </startMinute>
              <endHour>
                <!--ro, opt, int, range:[0,23]-->0
              </endHour>
              <endMinute>
                <!--ro, opt, int, range:[0,59]-->0
              </endMinute>
              <laneDirection>
                <!--ro, opt, enum, subType:string-->unknown
              </laneDirection>
            </timeRange>
          </timeRangeList>
        </variableLaneWeekSwitch>
      </variableLaneWeekSwitchList>
      <radarEnabled>
        <!--ro, opt, bool, whether to enable radar speed detection-->true
      </radarEnabled>
      <Radar>
        <!--ro, opt, object, radar parameters-->
        <radarType>
          <!--ro, req, enum, radar type, subType:string, desc:"none", "adr" (Andoray), "adr4Byte" (Andoray) (no radar controller), "olvia" (Olvia), "csr"
(TransMicrowave), "IOBox" (radar connecting I/O expansion box), "SSTK" (WHST), "custom" (custom type)-->none
        </radarType>
        <radarSensitivity>
          <!--ro, opt, int, radar sensitivity, range:[0,65535]-->0
        </radarSensitivity>
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          <radarAngle>
            <!--ro, opt, int, angle between the radar and the horizontal direction, range:[0,90]-->0
          </radarAngle>
          <validRadarSpeedTime>
            <!--ro, req, int, effective duration of radar speed, range:[0,2000]-->0
          </validRadarSpeedTime>
          <radarLinearCorrection>
            <!--ro, req, float, linear correction parameter, range:[0.0,2.0]-->0.000
          </radarLinearCorrection>
          <radarConstantCorrection>
            <!--ro, req, int, constant correction parameter, range:[-100,100]-->0
          </radarConstantCorrection>
          <radarRS485>
            <!--ro, req, int, radar linked RS-485 port No., range:[1,5]-->0
          </radarRS485>
        </Radar>
        <directionType>
          <!--ro, req, enum, lane direction, subType:string, desc:"unknown, left, straight, leftStraight, right, leftRight, rightStraight, leftRightStraight,
leftWait, straightWait, strightWaitRight"-->unknown
        </directionType>
        <stopLineDistance>
          <!--ro, req, int, distance between the vehicle and the stop line in the second captured picture of red light running violation, range:[0,300]-->0
        </stopLineDistance>
        <loopSensitivity>
          <!--ro, req, int, loop sensitivity, range:[1,100]-->1
        </loopSensitivity>
        <capPositionOffset>
          <!--ro, req, int, minimum offset between the capture positions of the second and the third picture, range:[0,1500]-->0
        </capPositionOffset>
        <detectSnapType>
          <!--ro, req, object, type of capture triggered by video detection-->
          <post>
            <!--ro, req, bool, checkpoint-->true
          </post>
          <postCapNo>
            <!--ro, req, int, checkpoint capture times, range:[1,3]-->0
          </postCapNo>
          <checkpointQuickTriggerEnabled>
            <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.post,eq,true}-->true
          </checkpointQuickTriggerEnabled>
          <overSpeed>
            <!--ro, req, bool, whether to enable speeding detection-->true
          </overSpeed>
          <overSpeedCapNo>
            <!--ro, req, int, number of captured pictures in which the speeding violation is detected, range:[2,3]-->0
          </overSpeedCapNo>
          <driveLine>
            <!--ro, req, bool, whether to enable driving on lane line detection-->true
          </driveLine>
          <driveLineCapNo>
            <!--ro, req, int, number of continuously captured pictures in which the driving on lane line happened, range:[2,3]-->0
          </driveLineCapNo>
          <driveLineSensitivity>
            <!--ro, req, int, sensitivity of driving on lane line detection, range:[0,100]-->0
          </driveLineSensitivity>
          <reverse>
            <!--ro, req, bool, whether to enable wrong-way driving detection-->true
          </reverse>
          <reverseCapNo>
            <!--ro, req, int, number of captured pictures in which the wrong-way driving is detected, range:[1,3]-->1
          </reverseCapNo>
          <redLight>
            <!--ro, req, bool, whether the red light is on-->true
          </redLight>
          <redLightMode>
            <!--ro, req, enum, red light running mode, subType:int, desc:0-by direction,1-by lane-->0
          </redLightMode>
          <nonMotorRedLightEnabled>
            <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.redLight,eq,true}-->true
          </nonMotorRedLightEnabled>
          <direction>
            <!--ro, req, bool, whether to enable detection of driving in wrong lane at intersection-->true
          </direction>
          <illegalPriority>
            <!--ro, req, enum, traffic violation priority, subType:int, desc:0-none, 1-red light running first-->0
          </illegalPriority>
          <intersectionCongest>
            <!--ro, req, bool, whether to enable intersection congestion detection-->true
          </intersectionCongest>
          <nonDriveWay>
            <!--ro, req, bool, whether to enable detection of motor vehicle on non-motor vehicle lane-->true
          </nonDriveWay>
          <nonDriveWayCapNo>
            <!--ro, req, int, number of captured pictures in which motor vehicle on non-motor vehicle lane happened, range:[2,3]-->0
          </nonDriveWayCapNo>
          <nonDriveWayTime>
            <!--ro, req, int, capture duration of motor vehicle on non-motor vehicle lane, range:[10,240], unit:s-->10
          </nonDriveWayTime>
          <changeLane>
            <!--ro, req, bool, whether to enable illegal lane change detection-->true
          </changeLane>
          <changeLaneCapNo>
            <!--ro, req, int, number of captured pictures in which illegal lane change is detected, range:[2,3]-->0
          </changeLaneCapNo>
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<changeLaneFirstImageReserveEnabled>
  <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.changeLane,eq,true}-->true
</changeLaneFirstImageReserveEnabled>
<banSign>
  <!--ro, req, bool, whether to enable ban breaking detection-->true
</banSign>
<banSignCapNo>
  <!--ro, req, int, number of continuously captured pictures in which the ban breaking happened, range:[2,3]-->0
</banSignCapNo>
<notCatchingReverseVehiclesEnabled>
  <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.banSign,eq,true}-->true
</notCatchingReverseVehiclesEnabled>
<intersectionStop>
  <!--ro, opt, object, whether to enable intersection parking detection-->
  <enable>
    <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
  <lastTime>
    <!--ro, req, int, congestion duration, range:[10,180], unit:s-->0
  </lastTime>
  <sensitivity>
    <!--ro, opt, int, detection sensitivity, range:[1,100]-->0
  </sensitivity>
  <bigCarRatio>
    <!--ro, opt, int, range:[1,100]-->0
  </bigCarRatio>
  <smallCarRatio>
    <!--ro, opt, int, range:[1,100]-->0
  </smallCarRatio>
  <capNo>
    <!--ro, opt, int, range:[3,3]-->3
  </capNo>
</intersectionStop>
<greenLightStop>
  <!--ro, req, bool, whether to enable detection of parking at green light-->true
</greenLightStop>
<illegalTurn>
  <!--ro, req, bool, illegal turning-->true
</illegalTurn>
<reverseLaneCheckpointFilteringEnabled>
  <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.illegalTurn,eq,true}-->true
</reverseLaneCheckpointFilteringEnabled>
<gasser>
  <!--ro, opt, object, queue jumping-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <sensitivity>
    <!--ro, req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <congestionThreshold>
    <!--ro, req, int, congestion threshold, range:[1,100]-->0
  </congestionThreshold>
  <capInterval>
    <!--ro, req, int, time interval of continuous capture (2 to 3 pictures), range:[0,6000], unit:ms-->0
  </capInterval>
  <capNo>
    <!--ro, opt, int, range:[3,3]-->3
  </capNo>
</gasser>
<congestion>
  <!--ro, opt, object, congestion-->
  <enabled>
    <!--ro, opt, bool, whether to enable intersection violation system-->true
  </enabled>
  <interval>
    <!--ro, req, int, alarm time interval, range:[1,10], unit:min-->0
  </interval>
  <sensitivity>
    <!--ro, req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <lastTime>
    <!--ro, req, int, congestion duration, range:[1,60], unit:s-->0
  </lastTime>
  <capNo>
    <!--ro, opt, int, range:[1,1]-->1
  </capNo>
</congestion>
<accident>
  <!--ro, opt, object, accident detection-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <lastTime>
    <!--ro, req, int, congestion duration, range:[1,180], unit:s-->0
  </lastTime>
  <interval>
    <!--ro, req, int, alarm time interval, range:[1,60], unit:min-->0
  </interval>
  <capNo>
    <!--ro, opt, int, range:[1,1]-->1
  </capNo>
</accident>
```

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```
        <nonMotorEnabled>
          <!--ro, opt, bool, whether to enable non-motor vehicle detection-->true
        </nonMotorEnabled>
        <runGreenLight>
          <!--ro, opt, bool, whether to enable green light running detection-->true
        </runGreenLight>
        <runGreenLightParam>
          <!--ro, opt, object, parameters of green light running detection-->
          <detectEvent>
            <!--ro, req, enum, Detection Logic, subType:string, desc:"runGreenLight" (running green light in congestion), "crossingStop" (overstay at
intersection)-->runGreenLight
          </detectEvent>
          <interval>
            <!--ro, req, int, capture interval, range:[1,180], unit:s, desc:capture interval-->15
          </interval>
          <secondLampState>
            <!--ro, opt, enum, traffic light status of the second capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
          </secondLampState>
          <threeLampState>
            <!--ro, opt, enum, traffic light status of the third capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
          </threeLampState>
          <capNo>
            <!--ro, opt, int, range:[3,3]-->3
          </capNo>
        </runGreenLightParam>
        <intervalType>
          <!--ro, req, enum, Burst Interval Type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
        </intervalType>
        <IntervalList>
          <!--ro, opt, array, list, subType:object-->
          <Interval>
            <!--ro, req, object, interval information-->
            <intervalValue>
              <!--ro, req, int, burst interval, desc:unit: ms/dm; capture interval by time is between 70 and 2000; capture interval by distance is between 1
and 20-->0
            </intervalValue>
          </Interval>
        </IntervalList>
        <redLightCapNo>
          <!--ro, opt, int, number of captured violation picture(s), range:[3,4]-->0
        </redLightCapNo>
        <directionCapNo>
          <!--ro, opt, int, number of captured pictures of driving against direction guidance, range:[3,4]-->0
        </directionCapNo>
        <banSignType>
          <!--ro, opt, enum, capture type list, subType:string, desc:"truck", "yellowPlate" (yellow license plate vehicle), "standard", "bus"-->truck
        </banSignType>
        <greenLightStopSensitivity>
          <!--ro, opt, int, stopping sensitivity, range:[1,100]-->0
        </greenLightStopSensitivity>
        <banNonMotorSign>
          <!--ro, opt, object, non-motor vehicle prohibition violation-->
          <enabled>
            <!--ro, req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--ro, req, int, the number of captured pictures, range:[2,3]-->0
          </capNo>
        </banNonMotorSign>
        <nonMotorPassenger>
          <!--ro, opt, object, Manned Non-Motor Vehicle-->
          <enabled>
            <!--ro, req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--ro, req, int, the number of captured pictures, range:[1,2]-->0
          </capNo>
          <refineMannedIllegalCode>
            <!--ro, opt, bool, whether to enable detailed manned violation code-->true
          </refineMannedIllegalCode>
        </nonMotorPassenger>
        <SuonaDet>
          <!--ro, opt, object, whistle detection-->
          <enabled>
            <!--ro, opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--ro, opt, int, the number of captured pictures, range:[2,3]-->0
          </capNo>
          <sensitivity>
            <!--ro, opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
        </SuonaDet>
        <BanMotorDet>
          <!--ro, opt, object, motorcycle driving in prohibited areas-->
          <enabled>
            <!--ro, opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <capNo>
            <!--ro, opt, int, the number of captured pictures, range:[1,3]-->2
          </capNo>
          <sensitivity>
            <!--ro, opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
```

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```
            </sensitivity>
          </BanMotorDet>
          <BanSignTypeList>
            <!--ro, opt, array, capture type list, subType:object, desc:capture type list-->
            <banSignType>
              <!--ro, opt, enum, prohibition violation type, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus",
"yellowbus" (yellow license plate bus), "dangercar" (dangerous goods transport vehicle), "instructioncar" (driver-training vehicle)-->standard
            </banSignType>
          </BanSignTypeList>
          <TurnSignalDet>
            <!--ro, opt, object, turning signal detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
            <enabled>
              <!--ro, opt, bool, whether to enable intersection violation system-->false
            </enabled>
            <turnFiliter>
              <!--ro, opt, bool, whether to filter left turn and right turn-->true
            </turnFiliter>
            <detectEvent>
              <!--ro, opt, enum, capture type, subType:string, desc:"turnLeft", "turnRight", "changeLane"-->changeLane
            </detectEvent>
            <detectEventList>
              <!--ro, opt, array, subType:object, range:[1,3]-->
              <detectEvent>
                <!--ro, opt, enum, subType:string-->changeLane
              </detectEvent>
            </detectEventList>
            <capNo>
              <!--ro, opt, int, range:[3,3]-->3
            </capNo>
          </TurnSignalDet>
          <ReflectiveTapeDet>
            <!--ro, opt, object, reflective stripe detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
            <enabled>
              <!--ro, opt, bool, whether to enable intersection violation system-->false
            </enabled>
            <capNo>
              <!--ro, opt, int, the number of captured pictures, range:[1,3]-->2
            </capNo>
          </ReflectiveTapeDet>
          <ZipperDriveDet>
            <!--ro, opt, object, pass alternatively detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
            <enabled>
              <!--ro, opt, bool, whether to enable intersection violation system-->true
            </enabled>
            <carNum>
              <!--ro, opt, int, capture triggered number of vehicles, range:[1,3]-->2
            </carNum>
            <parallDistance>
              <!--ro, opt, int, parallel distance, range:[0,200]-->0
            </parallDistance>
          </ZipperDriveDet>
          <reversing>
            <!--ro, opt, object-->
            <enabled>
              <!--ro, req, bool-->true
            </enabled>
            <snapTime>
              <!--ro, opt, int, range:[2,3], unit:s-->2
            </snapTime>
          </reversing>
          <NotYieldToRightRoad>
            <!--ro, opt, object-->
            <enabled>
              <!--ro, req, bool-->false
            </enabled>
            <firstCapturePosition>
              <!--ro, req, int, range:[0,100], step:1-->20
            </firstCapturePosition>
            <lateralDistance>
              <!--ro, req, int, range:[20,350], step:1-->150
            </lateralDistance>
            <longitudinalDistance>
              <!--ro, req, int, range:[10,150], step:1-->50
            </longitudinalDistance>
            <capNo>
              <!--ro, opt, int, range:[3,3]-->3
            </capNo>
          </NotYieldToRightRoad>
          <EnterNotYieldToInRoundabout>
            <!--ro, opt, object-->
            <enabled>
              <!--ro, req, bool-->false
            </enabled>
            <firstCapturePosition>
              <!--ro, req, int, range:[0,100], step:1-->20
            </firstCapturePosition>
            <lateralDistance>
              <!--ro, req, int, range:[20,350], step:1-->150
            </lateralDistance>
            <longitudinalDistance>
              <!--ro, req, int, range:[10,150], step:1-->50
            </longitudinalDistance>
            <capNo>
              <!--ro, opt, int, range:[3,3]-->3
```

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```
          </capNo>
        </EnterNotYieldToInRoundabout>
        <RampNotYieldToMainRoad>
          <!--ro, opt, object-->
          <enabled>
            <!--ro, req, bool-->false
          </enabled>
          <lateralDistance>
            <!--ro, req, int, range:[10,90], step:1-->60
          </lateralDistance>
          <capNo>
            <!--ro, opt, int, range:[3,3]-->3
          </capNo>
        </RampNotYieldToMainRoad>
        <reverseSensitivity>
          <!--ro, opt, int, range:[0,100]-->0
        </reverseSensitivity>
      </detectSnapType>
      <ViolationDetectLineList>
        <!--ro, req, array, list, subType:object-->
        <ViolationDetectLine>
          <!--ro, req, object, draw violation detection line for a single lane-->
          <lineName>
            <!--ro, req, enum, detection line name, subType:string, desc:"laneLine, stopLine, redlightLine, cancelLine, waitLine"-->laneLine
          </lineName>
          <lineType>
            <!--ro, opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
          </lineType>
          <customName>
            <!--ro, opt, string, custom triggering line name-->test
          </customName>
          <RegionCoordinatesList>
            <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </ViolationDetectLine>
      </ViolationDetectLineList>
      <firstShowTailVep>
        <!--ro, opt, enum, place of the first capture time, subType:int, desc:0 (none), 1 (expose vehicle tail), 2 (expose vehicle tail and LPR success),
3(parking over stop line)-->0
      </firstShowTailVep>
      <lowSpeedLimit>
        <!--ro, opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
      </lowSpeedLimit>
      <bigCarSignSpeed>
        <!--ro, opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSignSpeed>
      <bigCarSpeedLimit>
        <!--ro, opt, int, maximum speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSpeedLimit>
      <bigCarLowSpeedLimit>
        <!--ro, opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarLowSpeedLimit>
      <noBanLeftTypeList>
        <!--ro, opt, array, allowlist of left-turn vehicle types, subType:object-->
        <noBanLeftType>
          <!--ro, opt, enum, vehicle types which are allowed to turn left, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
        </noBanLeftType>
      </noBanLeftTypeList>
      <noBanRightTypeList>
        <!--ro, opt, array, allowlist of right-turn vehicle types, subType:object-->
        <noBanRightType>
          <!--ro, opt, enum, vehicle types which are allowed to left-turn, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
        </noBanRightType>
      </noBanRightTypeList>
      <dangerCarSignSpeed>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </dangerCarSignSpeed>
      <dangerCarSpeedLimit>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </dangerCarSpeedLimit>
      <dangerCarLowSpeedLimit>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </dangerCarLowSpeedLimit>
      <dangerCarHighSpeed>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </dangerCarHighSpeed>
      <carHighSpeed>
        <!--ro, opt, int, range:[0,255]-->0
      </carHighSpeed>
      <bigCarHighSpeed>
        <!--ro, opt, int, range:[0,255]-->0
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```
      </bigCarHighSpeed>
      <nonMotorSignSpeed>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </nonMotorSignSpeed>
      <nonMotorSpeedLimit>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </nonMotorSpeedLimit>
      <nonMotorLowSpeedLimit>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </nonMotorLowSpeedLimit>
      <nonMotorHighSpeed>
        <!--ro, opt, int, range:[0,255], unit:km/h-->0
      </nonMotorHighSpeed>
    </LaneParam>
  </LaneParamList>
  <VirtualLaneList>
    <!--ro, req, array, list, subType:object-->
    <VirtualLane>
      <!--ro, req, object, virtual lane line-->
      <lineName>
        <!--ro, req, enum, detection line name, subType:string, desc:"LaneLine, stopLine, redlightLine, cancelLine, waitLine"-->laneRightBoundaryLine
      </lineName>
      <lineType>
        <!--ro, opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (Lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
      </lineType>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </VirtualLane>
  </VirtualLaneList>
  <Posting>
    <!--ro, opt, object, positioning frame-->
    <postingEnable>
      <!--ro, req, bool, whether to enable positioning frame-->true
    </postingEnable>
    <postingSize>
      <!--ro, req, enum, positioning frame size, subType:string, desc:"small" (300 pixels), "medium" (400 pixels), "large" (500 pixels)-->small
    </postingSize>
  </Posting>
  <leftNotYieldStraight>
    <!--ro, opt, object, left turn not yield to straight-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </leftNotYieldStraight>
  <rightNotYieldLeft>
    <!--ro, opt, object, right turn not yield to left turn-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </rightNotYieldLeft>
  <uTurnNotYieldGoingStaight>
    <!--ro, opt, object, U-turn not Yield to Straight-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </uTurnNotYieldGoingStaight>
  <leftSideDrivingAfterLeftTurn>
    <!--ro, opt, object, turn left in short turn radius-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </leftSideDrivingAfterLeftTurn>
  <failYieldPed>
    <!--ro, opt, object, failing to yield for pedestrians-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <vehicleFromDirection>
      <!--ro, req, enum, driving direction of the vehicle which does not yield to pedestrian, subType:string, desc:"Front, Back"-->Front
    </vehicleFromDirection>
    <cameraPosition>
```

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```
      <!--ro, req, enum, camera position, subType:string, desc:"crossing, roadSection"-->crossing
    </cameraPosition>
    <noDisplacementPedEnabled>
      <!--ro, req, bool, whether to enable no displacement capture of pedestrian-->true
    </noDisplacementPedEnabled>
    <failYieldPedLineList>
      <!--ro, opt, array, list, subType:object-->
      <failYieldPedLine>
        <!--ro, opt, object, failing to yield for pedestrians line-->
        <lineId>
          <!--ro, req, int, line ID, range:[1,2]-->0
        </lineId>
        <lineName>
          <!--ro, req, string, detection line name, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </failYieldPedLine>
    </failYieldPedLineList>
    <notification>
      <!--ro, opt, object, capture linkage, desc:pedestrian linkage-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <ip>
        <!--ro, opt, string, IP address of traffic camera, range:[0,32]-->test
      </ip>
      <port>
        <!--ro, opt, int, port No. of traffic camera, range:[0,65535]-->1
      </port>
      <username>
        <!--ro, opt, string, user name of traffic camera, range:[0,32]-->test
      </username>
      <password>
        <!--ro, opt, string, password of traffic camera, range:[0,16]-->test
      </password>
      <sensitivity>
        <!--ro, req, int, detection sensitivity, range:[0,100]-->0
      </sensitivity>
      <RS485Port>
        <!--ro, req, int, RS-485 port No. for the traffic light to access the camera, range:[1,10]-->1
      </RS485Port>
    </notification>
    <failYieldPedSensitivity>
      <!--ro, req, int, Not Yielding to Pedestrian Sensitivity, range:[0,100]-->0
    </failYieldPedSensitivity>
    <failYieldnonMotor>
      <!--ro, opt, bool, whether to enable capturing not yielding to pedestrian-->true
    </failYieldnonMotor>
    <failYieldNum>
      <!--ro, opt, int, Number of Pedestrian Not Yielded to, range:[1,7]-->1
    </failYieldNum>
    <failYieldAreaType>
      <!--ro, opt, enum, detection area of failing to yield for pedestrians, subType:string, desc:"rectangle", "polygon"-->rectangle
    </failYieldAreaType>
    <failYieldDistanceMode>
      <!--ro, opt, enum, distance mode of failing to yield for pedestrians, subType:string, desc:"car" (distance from pedestrian to vehicle center), "lane"
(distance from pedestrian to lane center)-->car
    </failYieldDistanceMode>
    <failYieldPedSnapPosition>
      <!--ro, opt, int, distance range for capturing failing to yield for pedestrians, range:[0,100]-->20
    </failYieldPedSnapPosition>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </failYieldPed>
  <illegalParking>
    <!--ro, opt, object, illegal parking-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <sensitivity>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->0
    </sensitivity>
    <lastTime>
      <!--ro, req, int, congestion duration, range:[1,180], unit:s-->0
    </lastTime>
    <illegalParkingLineList>
      <!--ro, opt, array, list of illegal parking lines, subType:object, range:[0,3]-->
      <illegalParkingLine>
        <!--ro, opt, object, illegal parking lines-->
        <lineId>
          <!--ro, req, int, line ID, range:[1,3]-->0
```

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```
        </lineId>
        <lineName>
          <!--ro, req, string, detection line name, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </illegalParkingLine>
    </illegalParkingLineList>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </illegalParking>
  <detectMode>
    <!--ro, opt, enum, detection mode, subType:string, desc:"video", "coilLoop" (video and coil)-->video
  </detectMode>
  <helmet>
    <!--ro, opt, bool, Without Helmet-->true
  </helmet>
  <helmetCapNo>
    <!--ro, opt, int, number of captured pictures in which the driver does not wear the helmet, range:[1,3]-->1
  </helmetCapNo>
  <nonMotorExist>
    <!--ro, opt, bool, Non-Motor Vehicle on Motor Vehicle Lane-->true
  </nonMotorExist>
  <nonMotorExistCapNo>
    <!--ro, opt, int, number of captured pictures in which the motor lane is occupied by non-motor vehicle, range:[2,3]-->2
  </nonMotorExistCapNo>
  <motorIllegalCarrying>
    <!--ro, opt, bool, Illegal Manned Motor Vehicle-->true
  </motorIllegalCarrying>
  <nomotorIllegalCarrying>
    <!--ro, opt, bool, Illegal Manned Non-Motor Vehicle-->true
  </nomotorIllegalCarrying>
  <canopy>
    <!--ro, opt, bool, Umbrella Tent Installation Violation on Non-Motor Vehicle-->true
  </canopy>
  <pedRedLight>
    <!--ro, opt, bool, Pedestrian Red Light Running-->true
  </pedRedLight>
  <pedOnNomotorLane>
    <!--ro, opt, bool, whether to detect pedestrian walking on non-motor vehicle lane-->true
  </pedOnNomotorLane>
  <pedOnMotorLane>
    <!--ro, opt, bool, whether to detect pedestrian walking on motor vehicle lane-->true
  </pedOnMotorLane>
  <capType>
    <!--ro, opt, enum, "vehicle,nonMotor,human,face", subType:string, desc:"vehicle, nonMotor, human, face"-->mixObject
  </capType>
  <deferredFrmNum>
    <!--ro, req, int, delay a few seconds to trigger the mode, range:[0,25]-->0
  </deferredFrmNum>
  <parkingOverStopLine>
    <!--ro, opt, object, Over Stop Line-->
    <enabled>
      <!--ro, req, bool, whether to enable intersection violation system-->true
    </enabled>
  </parkingOverStopLine>
  <speedDetectMode>
    <!--ro, opt, enum, speed detection mode, subType:string, desc:"none", "radar"-->none
  </speedDetectMode>
  <NotSlowZebraCrossingEvent>
    <!--ro, opt, object, not slowing down at zebra crossing-->
    <enabled>
      <!--ro, req, bool, whether to enable intersection violation system-->true
    </enabled>
    <capNo>
      <!--ro, req, int, the number of captured pictures, range:[1,3], step:1-->1
    </capNo>
    <lowSpeed>
      <!--ro, req, int, lowest speed limit (km/h), range:[0,100], step:1-->0
    </lowSpeed>
    <highSpeed>
      <!--ro, req, int, highest speed limit (km/h), range:[0,100], step:1-->0
    </highSpeed>
    <LineInfoList>
      <!--ro, opt, object, list of lane line information-->
      <LineInfo>
        <!--ro, opt, object, single lane line information-->
        <lineId>
          <!--ro, req, int, lane index, range:[1,3]-->0
        </lineId>
        <RegionCoordinatesList>
          <!--ro, opt, array, list, subType:object, range:[0,2]-->
          <RegionCoordinates>
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```
            <!--ro, opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LineInfo>
    </LineInfoList>
  </NotSlowZebraCrossingEvent>
  <TurnRightStop>
    <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
    <enabled>
      <!--ro, req, bool, whether to enable intersection violation system-->true
    </enabled>
    <capNo>
      <!--ro, req, int, the number of captured pictures, range:[1,3], step:1-->3
    </capNo>
    <durationTime>
      <!--ro, req, int, duration, range:[1,600], unit:s-->5
    </durationTime>
    <slowUp>
      <!--ro, req, int, upper limit of deceleration, range:[0,100], step:1, unit:km/h-->40
    </slowUp>
    <slowStep>
      <!--ro, req, int, deceleration range, range:[0,50], step:1, unit:km/h-->5
    </slowStep>
    <slowDown>
      <!--ro, req, int, lower limit of deceleration, range:[0,100], step:1, unit:km/h-->10
    </slowDown>
    <detectMode>
      <!--ro, req, enum, deceleration mode, subType:string, desc:"stop", "slow"-->stop
    </detectMode>
    <SlowLane>
      <!--ro, opt, object, Deceleration Line-->
      <RegionCoordinatesList>
        <!--ro, opt, array, list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SlowLane>
  </TurnRightStop>
  <TruckRunARedLight>
    <!--ro, opt, object, red light running of large truck detection, desc:this node is used for linking evidence intersection violation cameras can link the checkpoints to capture the front license plate of the large trucks-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->false
    </enabled>
    <secondCam>
      <!--ro, req, int, the unique identifier for the linked cameras, range:[0,255], desc:all linked cameras share the same configuration, which is related to the uploaded value of secondCam in ANPR-->1
    </secondCam>
    <sceneMode>
      <!--ro, opt, enum, detection scene type, subType:string, desc:"singleLane", "multiLane"-->singleLane
    </sceneMode>
    <laneID>
      <!--ro, opt, int, Lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,singleLane}, desc:this node is valid when the camera is applied in single lane scene-->1
    </laneID>
    <bluePlateCarFilterEnabled>
      <!--ro, opt, bool, whether to enable filtering blue plate vehicles-->true
    </bluePlateCarFilterEnabled>
    <NotificationList>
      <!--ro, opt, array, list of linked camera, up to 3 linked cameras are supported, subType:object, range:[1,3]-->
      <notification>
        <!--ro, opt, object, picture capture-->
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <ipv4>
          <!--ro, req, string, IPv4 address, range:[0,32]-->test
        </ipv4>
        <port>
          <!--ro, req, int, port No., range:[0,66535]-->0
        </port>
        <username>
          <!--ro, opt, string, user name, range:[0,32], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
        </username>
        <password>
          <!--ro, req, string, password, range:[0,16], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
        </password>
        <laneID>
          <!--ro, opt, int, Lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,multiLane}, desc:this node is valid when the camera is applied in multiple lanes scene-->1
```

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```
        </laneID>
      </notification>
    </NotificationList>
    <linkSceneMode>
      <!--ro, opt, enum, detection scene type, subType:string, desc:"oneLinkMultiple"(one camera links with multiple cameras), "oneLinkOne" (one camera links with one camera), "multipleLinkOne" (multiple cameras link with multiple cameras)-->oneLinkOne
    </linkSceneMode>
  </TruckRunARedLight>
  <TruckTurnRightStop>
    <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <capNo>
      <!--ro, req, enum, number of captures, subType:int, desc:number of captures-->3
    </capNo>
    <sensitivity>
      <!--ro, req, int, stopping sensitivity, range:[1,100]-->50
    </sensitivity>
    <stopTime>
      <!--ro, opt, int, Parking Duration, range:[1,180], unit:s-->10
    </stopTime>
    <capInterval>
      <!--ro, opt, int, range:[1,10000], unit:ms-->1000
    </capInterval>
  </TruckTurnRightStop>
  <uphone>
    <!--ro, opt, object, making a phone call-->
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <capNo>
      <!--ro, req, int, number of captured pictures, range:[1,3]-->2
    </capNo>
  </uphone>
  <NoAlterTrafficDet>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <capNo>
      <!--ro, req, int, range:[3,3], step:1-->3
    </capNo>
    <ConvergenceLineList>
      <!--ro, opt, array, subType:object, range:[1,1]-->
      <ConvergenceLine>
        <!--ro, opt, object-->
        <lineId>
          <!--ro, req, int, range:[1,1]-->1
        </lineId>
        <lineName>
          <!--ro, req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ConvergenceLine>
    </ConvergenceLineList>
  </NoAlterTrafficDet>
  <greenLightStopCapNo>
    <!--ro, opt, int, range:[3,3]-->3
  </greenLightStopCapNo>
  <illegalTurnCapNo>
    <!--ro, opt, int, range:[3,3]-->3
  </illegalTurnCapNo>
  <nonMotorShed>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <capNo>
      <!--ro, req, int, range:[1,2]-->1
    </capNo>
  </nonMotorShed>
  <NonMotorCrossing>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <lastTime>
      <!--ro, req, int, range:[10,180], unit:s-->0
    </lastTime>
    <sensitivity>
      <!--ro, opt, int, range:[1,100]-->0
    </sensitivity>
```

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```
    </sensitivity>
    <nonMotorStopLine>
      <!--ro, opt, object-->
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </nonMotorStopLine>
  </NonMotorCrossing>
  <freightManned>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <capNo>
      <!--ro, req, int, range:[2,3], step:1-->2
    </capNo>
    <sensitivity>
      <!--ro, opt, int, range:[1,100]-->50
    </sensitivity>
  </freightManned>
  <vehicleRestrictions>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <capNo>
      <!--ro, opt, int, range:[2,3]-->2
    </capNo>
    <restrictionType>
      <!--ro, req, enum, subType:string-->region
    </restrictionType>
    <nonRestrictedRegionList>
      <!--ro, opt, array, subType:object, range:[0,64], dep:and,{$.VideoEpolice.vehicleRestrictions.restrictionType,eq,region}-->
      <nonRestrictedRegion>
        <!--ro, opt, object-->
        <nonRestrictedProvinces>
          <!--ro, req, enum, subType:string-->京
        </nonRestrictedProvinces>
        <nonRestrictedCityList>
          <!--ro, opt, array, subType:object, range:[0,64]-->
          <nonRestrictedCity>
            <!--ro, opt, object-->
            <nonRestrictedCityCode>
              <!--ro, req, enum, subType:string-->A
            </nonRestrictedCityCode>
          </nonRestrictedCity>
        </nonRestrictedCityList>
      </nonRestrictedRegion>
    </nonRestrictedRegionList>
    <timeSwitchList>
      <!--ro, opt, array, subType:object, range:[0,7]-->
      <timeSwitch>
        <!--ro, req, object-->
        <dayOfWeek>
          <!--ro, req, enum, subType:int-->1
        </dayOfWeek>
        <timeRangeList>
          <!--ro, req, array, subType:object, range:[0,4]-->
          <timeRange>
            <!--ro, req, object-->
            <ID>
              <!--ro, req, int, range:[1,4]-->1
            </ID>
            <startHour>
              <!--ro, opt, int, range:[0,23]-->0
            </startHour>
            <startMinute>
              <!--ro, opt, int, range:[0,59]-->0
            </startMinute>
            <endHour>
              <!--ro, opt, int, range:[0,23]-->0
            </endHour>
            <endMinute>
              <!--ro, opt, int, range:[0,59]-->0
            </endMinute>
          </timeRange>
        </timeRangeList>
      </timeSwitch>
    </timeSwitchList>
  </vehicleRestrictions>
  <nonMotorOccupyZebraCrossing>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
```

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```
    <capNo>
      <!--ro, opt, int, range:[1,3]-->1
    </capNo>
  </nonMotorOccupyZebraCrossing>
  <yellowGridParking>
    <!--ro, opt, object-->
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <lastTime>
      <!--ro, req, int, range:[1,180], unit:s-->0
    </lastTime>
    <yellowGridParkingLineList>
      <!--ro, opt, array, subType:object, range:[0,2]-->
      <yellowGridParkingLine>
        <!--ro, opt, object-->
        <lineId>
          <!--ro, req, int, range:[1,2]-->0
        </lineId>
        <lineName>
          <!--ro, req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </yellowGridParkingLine>
    </yellowGridParkingLineList>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </yellowGridParking>
  <cameraPosition>
    <!--ro, opt, enum, subType:string-->crossing
  </cameraPosition>
</VideoEpolice>
```

#### 14.5.1.30 Set the trigger mode parameters of video intersection violation system

#### Request URL

PUT /ISAPI/ITC/TriggerMode/videoEpolice?channelID=<channelID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoEpolice xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, trigger mode parameters of video intersection violation system, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable intersection violation system-->true
  </enabled>
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <TrafficLight>
    <!--req, object, traffic light-->
    <source>
      <!--req, enum, signal source, subType:string, desc:"IO", "RS485", "videoDetect" (video detection), "network"-->IO
    </source>
    <IOLightList>
      <!--opt, array, list, subType:object-->
      <IOLight>
        <!--opt, object, I/O access-->
        <lightId>
          <!--req, int, traffic light No., range:[1,3]-->1
        </lightId>
        <lightType>
          <!--req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
        </lightType>
        <relatedIO>
          <!--req, enum, linked I/O input, subType:string, desc:"IO1", "IO2", "IO3", "IO4", "IO5", "IO6", "IO7", "IO8"-->IO7
        </relatedIO>
```

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```
        <redLightStatus>
          <!--req, enum, red light level status, subType:string, desc:"low" (low-level red light), "high" (high-level red light)-->low
        </redLightStatus>
      </IOLight>
    </IOLightList>
    <RS485LightList>
      <!--opt, array, list, subType:object-->
      <RS485Light>
        <!--opt, object, RS-485 access-->
        <lightId>
          <!--req, int, traffic light No., range:[1,3]-->0
        </lightId>
        <lightType>
          <!--req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
        </lightType>
        <relatedLightChan>
          <!--req, int, detector channel No. linked to red light, range:[0,16]-->0
        </relatedLightChan>
        <yellowLightChan>
          <!--req, int, detector channel No. linked to yellow light, range:[0,16]-->0
        </yellowLightChan>
      </RS485Light>
    </RS485LightList>
    <NetworkLightList>
      <!--opt, array, list of network indicator light, subType:object-->
      <NetworkLight>
        <!--opt, object, network indicator light-->
        <lightId>
          <!--req, int, traffic light No., range:[1,3]-->0
        </lightId>
        <lightType>
          <!--req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
        </lightType>
        <relatedLightChan>
          <!--req, int, detector channel No. linked to red light, range:[0,16]-->0
        </relatedLightChan>
        <yellowLightChan>
          <!--req, int, detector channel No. linked to yellow light, range:[0,16]-->0
        </yellowLightChan>
      </NetworkLight>
    </NetworkLightList>
    <videoDetectLightList>
      <!--opt, array, list of video detection traffic lights, subType:object-->
      <videoDetectLight>
        <!--opt, object, video detection traffic light-->
        <videoDetectLightId>
          <!--req, int, video detection signal No., range:[1,2]-->0
        </videoDetectLightId>
        <lightNum>
          <!--req, int, number of traffic lights, range:[1,3]-->0
        </lightNum>
        <straightLight>
          <!--req, bool, whether the straight driving light is on-->true
        </straightLight>
        <leftLight>
          <!--req, bool, whether the left-turn light is on-->true
        </leftLight>
        <rightLight>
          <!--req, bool, whether the right-turn light is on-->true
        </rightLight>
        <redLight>
          <!--req, bool, whether the red light is on-->true
        </redLight>
        <greenLight>
          <!--req, bool, whether the green light is on-->true
        </greenLight>
        <yellowLight>
          <!--req, bool, whether the yellow light is on-->true
        </yellowLight>
        <yellowLightTime>
          <!--req, int, yellow light duration, range:[0,10], unit:s-->0
        </yellowLightTime>
        <rectRegion>
          <!--req, object, traffic light region, desc:the origin is the upper-left corner of the screen-->
          <positionLeft>
            <!--req, int, x-coordinate of upper left corner, range:[0,65535]-->0
          </positionLeft>
          <positionTop>
            <!--req, int, y-coordinate of upper left corner, range:[0,65535]-->0
          </positionTop>
          <positionRight>
            <!--req, int, x-coordinate of lower right corner, range:[0,65535]-->0
          </positionRight>
          <positionBottom>
            <!--req, int, y-coordinate of lower right corner, range:[0,65535]-->0
          </positionBottom>
        </rectRegion>
      </videoDetectLight>
    </videoDetectLightList>
  </TrafficLight>
  <carHighSpeed>
    <!--req, int, abnormal speeding of small-sized vehicle, range:[0,255]-->0
  </carHighSpeed>
```

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```
    <bigCarHighSpeed>
      <!--req, int, abnormal speeding of large-sized vehicle, range:[0,255]-->0
    </bigCarHighSpeed>
    <LaneParamList>
      <!--req, array, lane parameter list, subType:object-->
      <LaneParam>
        <!--req, object, lane parameters-->
        <laneId>
          <!--req, int, lane No., range:[1,6]-->0
        </laneId>
        <laneDirectionType>
          <!--req, enum, lane direction No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
        </laneDirectionType>
        <relatedDriveWay>
          <!--req, int, linked lane No., range:[1,99]-->0
        </relatedDriveWay>
        <recordEnable>
          <!--req, bool, whether to enable recording when red light running is detected-->true
        </recordEnable>
        <recordType>
          <!--req, enum, recording type, subType:string, desc:"preRecord", "delayRecord"-->preRecord
        </recordType>
        <beginTime>
          <!--req, enum, recording start time, subType:int, desc:recording start time,"1,2,3"-->0
        </beginTime>
        <preRecordTime>
          <!--req, int, pre-record time, range:[0,100], unit:s-->0
        </preRecordTime>
        <recordDelayTime>
          <!--req, int, post-record time, range:[0,100], unit:s-->0
        </recordDelayTime>
        <recordTimeOut>
          <!--req, int, recording timeout, range:[0,100], unit:s-->0
        </recordTimeOut>
        <signSpeed>
          <!--req, int, marked speed limit for small-sized vehicle, range:[0,255], unit:km/h-->0
        </signSpeed>
        <speedLimit>
          <!--req, int, speed limit for small-sized vehicle driving in high way, range:[0,255], unit:km/h-->0
        </speedLimit>
        <usageType>
          <!--req, enum, lane purpose, subType:string, desc:"carriageWay" (roadway), "bus" (bus lane), "slow" (slow lane), "nonmotor" (non-motor lane),
"reverse" (contraflow lane), "emergency" (emergency lane), "banNonMotor" (non-motor vehicle vehicles prohibited lane), "ramp", "fast" (overtaking lane)--
>carriageWay
        </usageType>
        <LaneUsageList>
          <!--opt, array, lane purpose list, subType:object-->
          <laneUsage>
            <!--req, enum, lane purpose, subType:string, desc:"bus" (bus lane), "banTrucks" (trucks prohibited lane), "banLeft" (left-turn prohibited lane),
"banRight" (right-turn prohibited lane), "motor" (motorcycle lane), "banStraight" (straight prohibited lane), "variableLane" (changeable lane)-->bus
          </laneUsage>
        </LaneUsageList>
        <banTrucksTimeSwitchList>
          <!--req, array, large-sized vehicle prohibited time period, subType:object-->
          <banTrucksTimeSwitch>
            <!--req, object, large-sized vehicle prohibited time-->
            <timeId>
              <!--req, int, time period No., range:[1,4]-->0
            </timeId>
            <startHour>
              <!--opt, int, start time (hour), range:[0,23]-->0
            </startHour>
            <startMinute>
              <!--opt, int, start time (minute), range:[0,59]-->0
            </startMinute>
            <endHour>
              <!--opt, int, end time (hour), range:[0,23]-->0
            </endHour>
            <endMinute>
              <!--opt, int, end time (minute), range:[0,59]-->0
            </endMinute>
          </banTrucksTimeSwitch>
        </banTrucksTimeSwitchList>
        <busWayTimeSwitchList>
          <!--req, array, bus allowed passing time period, subType:object-->
          <busWayTimeSwitch>
            <!--req, object, bus allowed passing time-->
            <timeId>
              <!--req, int, time period No., range:[1,4]-->0
            </timeId>
            <startHour>
              <!--opt, int, start time (hour), range:[0,23]-->0
            </startHour>
            <startMinute>
              <!--opt, int, start time (minute), range:[0,59]-->0
            </startMinute>
            <endHour>
              <!--opt, int, end time (hour), range:[0,23]-->0
            </endHour>
            <endMinute>
              <!--opt, int, end time (minute), range:[0,59]-->0
            </endMinute>
          </busWayTimeSwitch>
```

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```
    </busWayTimeSwitchList>
    <emergencyTimeSwitchList>
      <!--opt, array, time period for emergency lane, subType:object-->
      <emergencyTimeSwitch>
        <!--req, object, emergency lane time-->
        <timeId>
          <!--req, int, index, range:[1,4]-->0
        </timeId>
        <startHour>
          <!--opt, int, start time (hour), range:[0,23]-->0
        </startHour>
        <startMinute>
          <!--opt, int, start time (minute), range:[0,59]-->0
        </startMinute>
        <endHour>
          <!--opt, int, end time (hour), range:[0,23]-->0
        </endHour>
        <endMinute>
          <!--opt, int, end time (minute), range:[0,59]-->0
        </endMinute>
      </emergencyTimeSwitch>
    </emergencyTimeSwitchList>
    <banLeftTimeSwitchList>
      <!--req, array, left-turn prohibited time period, subType:object-->
      <banLeftTimeSwitch>
        <!--req, object, left-turn prohibited time-->
        <timeId>
          <!--req, int, time period No., range:[1,4]-->0
        </timeId>
        <startHour>
          <!--opt, int, start time (hour), range:[0,23]-->0
        </startHour>
        <startMinute>
          <!--opt, int, start time (minute), range:[0,59]-->0
        </startMinute>
        <endHour>
          <!--opt, int, end time (hour), range:[0,23]-->0
        </endHour>
        <endMinute>
          <!--opt, int, end time (minute), range:[0,59]-->0
        </endMinute>
      </banLeftTimeSwitch>
    </banLeftTimeSwitchList>
    <banRightTimeSwitchList>
      <!--req, array, right-turn prohibited time period, subType:object-->
      <banRightTimeSwitch>
        <!--req, object, right-turn prohibited time-->
        <timeId>
          <!--req, int, time period No., range:[1,4]-->0
        </timeId>
        <startHour>
          <!--opt, int, start time (hour), range:[0,23]-->0
        </startHour>
        <startMinute>
          <!--opt, int, start time (minute), range:[0,59]-->0
        </startMinute>
        <endHour>
          <!--opt, int, end time (hour), range:[0,23]-->0
        </endHour>
        <endMinute>
          <!--opt, int, end time (minute), range:[0,59]-->0
        </endMinute>
      </banRightTimeSwitch>
    </banRightTimeSwitchList>
    <banStraightTimeSwitchList>
      <!--req, array, straight prohibited time period, subType:object-->
      <banStraightTimeSwitch>
        <!--req, object, straight prohibited time-->
        <timeId>
          <!--req, int, index, range:[1,4]-->0
        </timeId>
        <startHour>
          <!--opt, int, start time (hour), range:[0,23]-->0
        </startHour>
        <startMinute>
          <!--opt, int, start time (minute), range:[0,59]-->0
        </startMinute>
        <endHour>
          <!--opt, int, end time (hour), range:[0,23]-->0
        </endHour>
        <endMinute>
          <!--opt, int, end time (minute), range:[0,59]-->0
        </endMinute>
      </banStraightTimeSwitch>
    </banStraightTimeSwitchList>
    <variableLaneTimeSwitchList>
      <!--req, array, time period for changeable lanes, subType:object-->
      <variableLaneTimeSwitch>
        <!--req, object, time for changeable lanes-->
        <timeId>
          <!--req, int, index, range:[1,4]-->0
        </timeId>
        <startHour>
```

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```
            <!--opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
            <!--opt, int, end time (minute), range:[0,59]-->0
          </endMinute>
          <laneDirection>
            <!--req, enum, driving direction, subType:string, desc:"unknown", "left", "straight", "LeftStraight", "right", "leftRight", "rightStraight",
"leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->unknown
          </laneDirection>
        </variableLaneTimeSwitch>
      </variableLaneTimeSwitchList>
      <radarEnabled>
        <!--opt, bool, whether to enable radar speed detection-->true
      </radarEnabled>
      <Radar>
        <!--opt, object, radar parameters-->
        <radarType>
          <!--req, enum, radar type, subType:string, desc:"none", "MultiLane" (multi-lane radar), "csr" (TransMicrowave),"IOBox" (radar connecting I/O
expansion box), "SSTK" (Sensortech),"custom" (custom type), "adr" (Andoray), "adr4Byte" (Andoray (no radar controller)),"olvia" (Olvia)-->none
        </radarType>
        <radarSensitivity>
          <!--req, int, sensitivity, range:[0,65535]-->0
        </radarSensitivity>
        <radarAngle>
          <!--req, int, angle between the radar and the horizontal direction, range:[0,90]-->0
        </radarAngle>
        <validRadarSpeedTime>
          <!--req, int, effective duration of radar speed, which is between 0 and 2000,unit: millisecond, range:[0,2000]-->0
        </validRadarSpeedTime>
        <radarLinearCorrection>
          <!--req, float, linear correction parameter, range:[0.0,2.0]-->0.000
        </radarLinearCorrection>
        <radarConstantCorrection>
          <!--req, int, constant correction parameter,, range:[-100,100]-->0
        </radarConstantCorrection>
        <radarRS485>
          <!--req, int, radar linked RS-485 port No., range:[1,5]-->0
        </radarRS485>
      </Radar>
      <directionType>
        <!--req, enum, lane direction, subType:string, desc:"unknown", "left", "straight", "leftStraight", "right", "leftRight", "rightStraight",
"leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->unknown
      </directionType>
      <stopLineDistance>
        <!--req, int, distance between the vehicle and the stop line in the second captured picture of red light running violation, range:[0,300]-->0
      </stopLineDistance>
      <loopSensitivity>
        <!--req, int, coil sensitivity, range:[1,100]-->1
      </loopSensitivity>
      <capPositionOffset>
        <!--req, int, minimum offset between the capture positions of the second and the third picture, range:[0,1500]-->0
      </capPositionOffset>
      <detectSnapType>
        <!--req, object, type of capture triggered by video detection-->
        <post>
          <!--req, bool, whether to enable checkpoint detection-->true
        </post>
        <postCapNo>
          <!--req, int, captured pictures of checkpoint, range:[2,3]-->0
        </postCapNo>
        <overSpeed>
          <!--req, bool, whether to enable speeding detection-->true
        </overSpeed>
        <overSpeedCapNo>
          <!--req, int, number of captured pictures in which the speeding violation is detected, range:[2,3]-->0
        </overSpeedCapNo>
        <driveLine>
          <!--req, bool, whether to enable driving on lane line detection-->true
        </driveLine>
        <driveLineCapNo>
          <!--req, int, number of continuously captured pictures in which the driving on lane line happened, range:[2,3]-->0
        </driveLineCapNo>
        <driveLineSensitivity>
          <!--req, int, sensitivity of driving on lane line detection, range:[0,100]-->0
        </driveLineSensitivity>
        <reverse>
          <!--req, bool, whether to enable wrong-way driving detection-->true
        </reverse>
        <reverseCapNo>
          <!--req, int, number of captured pictures in which the wrong-way driving is detected,which is between 2 and 3 and the default value is 2, range:
[0,100]-->0
        </reverseCapNo>
        <redLight>
          <!--req, bool, whether the red light is on-->true
        </redLight>
        <redLightMode>
          <!--req, enum, red light running mode,0-by direction,1-by lane, subType:int, desc:red light running mode,0-by direction,1-by lane-->0
        </redLightMode>
```

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```
<direction>
  <!--req, bool, whether to enable detection of driving in wrong lane at intersection-->true
</direction>
<illegalPriority>
  <!--req, enum, traffic violation priority, subType:int, desc:0 (by direction), 1 (by lane)-->0
</illegalPriority>
<intersectionCongest>
  <!--req, bool, whether to enable intersection congestion detection-->true
</intersectionCongest>
<nonDriveWay>
  <!--req, bool, whether to enable detection of motor vehicle on non-motor vehicle lane-->true
</nonDriveWay>
<nonDriveWayCapNo>
  <!--req, int, number of captured pictures in which motor vehicle on non-motor vehicle lane happened, range:[2,3]-->0
</nonDriveWayCapNo>
<nonDriveWayTime>
  <!--req, int, capture duration of motor vehicle on non-motor vehicle lane, range:[10,240], unit:s-->10
</nonDriveWayTime>
<changeLane>
  <!--req, bool, whether to enable illegal lane change detection-->true
</changeLane>
<changeLaneCapNo>
  <!--req, int, number of captured pictures in which illegal lane change is detected, range:[2,3]-->0
</changeLaneCapNo>
<banSign>
  <!--req, bool, whether to enable ban breaking detection-->true
</banSign>
<banSignCapNo>
  <!--req, int, number of continuously captured pictures in which the ban is broken, range:[2,3]-->0
</banSignCapNo>
<intersectionStop>
  <!--opt, object, whether to enable intersection parking detection-->
  <enable>
    <!--req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
  <lastTime>
    <!--req, int, congestion duration, range:[10,180], unit:s-->0
  </lastTime>
  <sensitivity>
    <!--opt, int, detection sensitivity, range:[1,100]-->0
  </sensitivity>
</intersectionStop>
<greenLightStop>
  <!--req, bool, whether to enable detection of parking at green light-->true
</greenLightStop>
<illegalTurn>
  <!--req, bool, illegal turning-->true
</illegalTurn>
<gasser>
  <!--opt, object, queue jumping-->
  <enabled>
    <!--req, bool, whether to enable intersection violation system-->true
  </enabled>
  <sensitivity>
    <!--req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <congestionThreshold>
    <!--req, int, congestion threshold, range:[1,100]-->0
  </congestionThreshold>
  <capInterval>
    <!--req, int, time interval of continuous capture (2 to 3 pictures), range:[0,6000], unit:ms-->0
  </capInterval>
</gasser>
<congestion>
  <!--opt, object, congestion-->
  <enabled>
    <!--opt, bool, whether to enable intersection violation system-->true
  </enabled>
  <interval>
    <!--req, int, alarm time interval, range:[1,10], unit:min-->0
  </interval>
  <sensitivity>
    <!--req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <lastTime>
    <!--req, int, congestion duration, range:[1,60], unit:s-->0
  </lastTime>
</congestion>
<accident>
  <!--opt, object, accident detection-->
  <enabled>
    <!--req, bool, whether to enable intersection violation system-->true
  </enabled>
  <lastTime>
    <!--req, int, congestion duration, unit: second, range:[1,180], unit:s-->0
  </lastTime>
  <interval>
    <!--req, int, alarm time interval, unit: minute, range:[1,60], unit:min-->0
  </interval>
</accident>
<nonMotorEnabled>
  <!--opt, bool, whether to enable non-motor vehicle detection-->true
</nonMotorEnabled>
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```
        <runGreenLight>
          <!--opt, bool, whether to enable green light running detection-->true
        </runGreenLight>
        <runGreenLightParam>
          <!--ro, opt, object, parameters of green light running detection-->
          <detectEvent>
            <!--req, enum, detection logic, subType:string, desc:"runGreenLight" (running green light in congestion), "crossingStop" (overstay at
intersection)-->runGreenLight
          </detectEvent>
          <interval>
            <!--req, int, capture interval, range:[1,180], unit:s, desc:capture interval-->15
          </interval>
          <secondLampState>
            <!--opt, enum, traffic light status of the second capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
          </secondLampState>
          <threeLampState>
            <!--opt, enum, traffic light status of the third capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
          </threeLampState>
        </runGreenLightParam>
        <intervalType>
          <!--req, enum, burst interval type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
        </intervalType>
        <IntervalList>
          <!--opt, array, list of time intervals, subType:object-->
          <Interval>
            <!--req, object, interval information-->
            <intervalValue>
              <!--req, int, burst interval, desc:unit: ms/dm; capture interval by time is between 70 and 2000; capture interval by distance is between 1 and
20-->0
            </intervalValue>
          </Interval>
        </IntervalList>
        <redLightCapNo>
          <!--opt, int, number of captured violation picture(s), range:[3,4]-->0
        </redLightCapNo>
        <directionCapNo>
          <!--opt, int, number of captured pictures of driving against direction guidance, range:[3,4]-->0
        </directionCapNo>
        <banSignType>
          <!--opt, enum, capture type list, subType:string, desc:"truck", "yellowPlate" (yellow license plate vehicle), "standard", "bus"-->truck
        </banSignType>
        <greenLightStopSensitivity>
          <!--opt, int, stopping sensitivity, range:[1,100]-->0
        </greenLightStopSensitivity>
        <nonMotorShed>
          <!--opt, object, non-motor vehicle umbrella tent-->
          <enabled>
            <!--req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--req, int, number of captures, range:[1,2]-->0
          </capNo>
        </nonMotorShed>
        <banNonMotorSign>
          <!--opt, object, non-motor vehicle prohibition violation-->
          <enabled>
            <!--req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--req, int, number of captures, range:[2,3]-->0
          </capNo>
        </banNonMotorSign>
        <nonMotorPassenger>
          <!--opt, object, manned non-motor vehicle-->
          <enabled>
            <!--req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--req, int, number of captures, range:[1,2]-->0
          </capNo>
          <refineMannedIllegalCode>
            <!--opt, bool, whether to enable detailed manned violation code-->true
          </refineMannedIllegalCode>
        </nonMotorPassenger>
        <SuonaDet>
          <!--opt, object, whistle detection-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--opt, int, number of captures, range:[2,3]-->0
          </capNo>
          <sensitivity>
            <!--opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
        </SuonaDet>
        <BanMotorDet>
          <!--opt, object, motorcycle driving in prohibited areas-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <capNo>
            <!--opt, int, number of captures, range:[1,3]-->2
          </capNo>
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          </capNo>
          <sensitivity>
            <!--opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
        </BanMotorDet>
        <BanSignTypeList>
          <!--opt, array, capture type list, subType:object-->
          <banSignType>
            <!--opt, enum, prohibition violation type, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus",
"yellowbus" (yellow license plate bus), "dangercar" (dangerous goods transport vehicle), "instructioncar" (driver-training vehicle)-->standard
          </banSignType>
        </BanSignTypeList>
        <TurnSignalDet>
          <!--opt, object, turning without signal detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <turnFiliter>
            <!--opt, bool, whether to filter left turn and right turn-->true
          </turnFiliter>
          <detectEvent>
            <!--opt, enum, capture type, subType:string, desc:"turnLeft", "turnRight", "changeLane"-->changeLane
          </detectEvent>
        </TurnSignalDet>
        <ReflectiveTapeDet>
          <!--opt, object, reflective stripe detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <capNo>
            <!--opt, int, number of captures, range:[1,3]-->2
          </capNo>
        </ReflectiveTapeDet>
        <ZipperDriveDet>
          <!--opt, object, pass alternatively detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <carNum>
            <!--opt, int, capture triggered number of vehicles, range:[1,3]-->2
          </carNum>
          <parallDistance>
            <!--opt, int, parallel distance, range:[0,200]-->0
          </parallDistance>
        </ZipperDriveDet>
        <NonMotorCrossing>
          <!--opt, object, non-motor vehicle parking over stop line, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <lastTime>
            <!--req, int, congestion duration, range:[10,180], unit:s-->0
          </lastTime>
          <sensitivity>
            <!--opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
          <nonMotorStopLine>
            <!--opt, object, non-motor vehicle parking over stop line-->
            <RegionCoordinatesList size="2">
              <!--opt, object, list, attr:size{req, int}-->
              <RegionCoordinates>
                <!--opt, object, point coordinates-->
                <positionX>
                  <!--req, int, x-coordinate, range:[0,1]-->0.1
                </positionX>
                <positionY>
                  <!--req, int, y-coordinate, range:[0,1]-->0.1
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </nonMotorStopLine>
        </NonMotorCrossing>
      </detectSnapType>
      <ViolationDetectLineList>
        <!--req, array, list, subType:object-->
        <ViolationDetectLine>
          <!--req, object, draw violation detection line for a single lane-->
          <lineName>
            <!--req, enum, line name, subType:string, desc:detection line name: "laneLine,stopLine,redlightLine,cancelLine,waitLine"-->laneLine
          </lineName>
          <lineType>
            <!--opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
          </lineType>
          <customName>
            <!--opt, string, custom triggering line name-->test
          </customName>
          <RegionCoordinatesList>
            <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
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            <positionY>
              <!--req, int, y-coordinate-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ViolationDetectLine>
    </ViolationDetectLineList>
    <firstShowTailVep>
      <!--opt, enum, place of the first capture time, subType:int, desc:0 (none), 1 (expose vehicle tail), 2 (expose vehicle tail and LPR success),
3(parking over stop line)-->0
    </firstShowTailVep>
    <lowSpeedLimit>
      <!--req, int, low speed limit for small-sized vehicle, range:[0,255]-->0
    </lowSpeedLimit>
    <bigCarSignSpeed>
      <!--req, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
    </bigCarSignSpeed>
    <bigCarSpeedLimit>
      <!--req, int, maximum speed limit for large-sized vehicle, range:[0,255]-->0
    </bigCarSpeedLimit>
    <bigCarLowSpeedLimit>
      <!--req, int, low speed limit for large-sized vehicle, range:[0,255]-->0
    </bigCarLowSpeedLimit>
    <noBanLeftTypeList>
      <!--opt, array, allowlist of left-turn vehicle types, subType:object-->
      <noBanLeftType>
        <!--opt, enum, vehicle types which are allowed to turn left, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
      </noBanLeftType>
    </noBanLeftTypeList>
    <noBanRightTypeList>
      <!--opt, array, allowlist of right-turn vehicle types, subType:object-->
      <noBanRightType>
        <!--opt, enum, vehicle types which are allowed to left-turn, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
      </noBanRightType>
    </noBanRightTypeList>
    <dangerCarSignSpeed>
      <!--opt, int, range:[0,255], unit:km/h-->0
    </dangerCarSignSpeed>
    <dangerCarSpeedLimit>
      <!--opt, int, range:[0,255], unit:km/h-->0
    </dangerCarSpeedLimit>
    <dangerCarLowSpeedLimit>
      <!--opt, int, range:[0,255], unit:km/h-->0
    </dangerCarLowSpeedLimit>
  </LaneParam>
 </LaneParamList>
 <VirtualLaneList>
  <!--req, array, list of virtual line, subType:object-->
  <VirtualLane>
   <!--req, object, virtual lane line-->
   <lineName>
    <!--req, enum, line name, subType:string, desc:detection line name: "laneLine,stopLine,redlightLine,cancelLine,waitLine"-->laneRightBoundaryLine
   </lineName>
   <lineType>
    <!--opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line), "doubleYellow"
(double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
   </lineType>
   <RegionCoordinatesList>
    <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
    <RegionCoordinates>
     <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
     <positionX>
      <!--req, int, x-coordinate, range:[0,1000]-->0
     </positionX>
     <positionY>
      <!--req, int, y-coordinate-->0
     </positionY>
    </RegionCoordinates>
   </RegionCoordinatesList>
  </VirtualLane>
 </VirtualLaneList>
 <Posting>
  <!--opt, object, positioning frame-->
  <postingEnable>
   <!--req, bool, whether to enable positioning frame-->true
  </postingEnable>
  <postingSize>
   <!--req, enum, positioning frame size, subType:string, desc:“small,middle,large”-->small
  </postingSize>
 </Posting>
 <leftNotYieldStraight>
  <!--opt, object, left turn not yield to straight-->
  <enable>
   <!--req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
 </leftNotYieldStraight>
 <rightNotYieldLeft>
  <!--opt, object, right turn not yield to left turn-->
  <enable>
   <!--req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
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    </rightNotYieldLeft>
    <uTurnNotYieldGoingStaight>
      <!--opt, object, U-turn not yield to straight-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
    </uTurnNotYieldGoingStaight>
    <leftSideDrivingAfterLeftTurn>
      <!--opt, object, turn left in short turn radius-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
    </leftSideDrivingAfterLeftTurn>
    <failYieldPed>
      <!--opt, object, failing to yield for pedestrians-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <vehicleFromDirection>
        <!--req, enum, driving direction of the vehicle which does not yield to pedestrian, subType:string, desc:"Front", "Back"-->Front
      </vehicleFromDirection>
      <cameraPosition>
        <!--req, enum, camera position, subType:string, desc:"crossing" (crossroad intersection), "roadsection"-->crossing
      </cameraPosition>
      <noDisplacementPedEnabled>
        <!--req, bool, whether to enable no displacement capture of pedestrian-->true
      </noDisplacementPedEnabled>
      <failYieldPedLineList>
        <!--opt, array, list, subType:object-->
        <failYieldPedLine>
          <!--opt, object, failing to yield for pedestrians line-->
          <lineId>
            <!--req, int, line ID, range:[1,2]-->0
          </lineId>
          <lineName>
            <!--req, string, detection line name, range:[0,64]-->test
          </lineName>
          <RegionCoordinatesList>
            <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </failYieldPedLine>
      </failYieldPedLineList>
      <notification>
        <!--req, object, capture linkage, desc:capture linkage via HCNetSDK-->
        <enable>
          <!--req, bool, whether to enable detection of left-turn without straight driving-->true
        </enable>
        <ip>
          <!--opt, string, IPv4 address, range:[0,32]-->test
        </ip>
        <port>
          <!--opt, int, port No. of traffic camera, range:[0,65535]-->1
        </port>
        <username>
          <!--opt, string, user name of traffic camera, range:[0,32]-->test
        </username>
        <password>
          <!--opt, string, password of traffic camera, range:[0,16]-->test
        </password>
        <sensitivity>
          <!--req, int, detection sensitivity, range:[0,100]-->0
        </sensitivity>
        <RS485Port>
          <!--req, int, RS-485 port No. for the traffic light to access the camera, range:[1,10]-->1
        </RS485Port>
      </notification>
      <failYieldPedSensitivity>
        <!--req, int, not yielding to pedestrian sensitivity, range:[0,100]-->0
      </failYieldPedSensitivity>
      <failYieldnonMotor>
        <!--req, bool, whether to enable capturing non-motor vehicle-->true
      </failYieldnonMotor>
      <failYieldNum>
        <!--opt, int, number of pedestrian not yielded to, range:[1,7]-->1
      </failYieldNum>
      <failYieldAreaType>
        <!--opt, enum, detection area of failing to yield for pedestrians, subType:string, desc:"rectangle", "polygon"-->rectangle
      </failYieldAreaType>
      <failYieldDistanceMode>
        <!--opt, enum, distance mode of failing to yield for pedestrians, subType:string, desc:"car" (distance from pedestrian to vehicle center), "lane"
(distance from pedestrian to lane center)-->car
      </failYieldDistanceMode>
      <failYieldPedSnapPosition>
        <!--opt, int, distance range for capturing failing to yield for pedestrians, range:[0,100]-->20
      </failYieldPedSnapPosition>
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</failYieldPed>
<illegalParking>
  <!--opt, object, illegal parking-->
  <enable>
    <!--req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
  <sensitivity>
    <!--req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <lastTime>
    <!--req, int, parking time, range:[1,180], unit:s-->0
  </lastTime>
  <illegalParkingLineList>
    <!--opt, array, list of illegal parking lines, subType:object-->
    <illegalParkingLine>
      <!--opt, object, illegal parking lines-->
      <lineId>
        <!--req, int, line ID, range:[1,3]-->0
      </lineId>
      <lineName>
        <!--req, string, line name, range:[0,64]-->test
      </lineName>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </illegalParkingLine>
  </illegalParkingLineList>
</illegalParking>
<detectMode>
  <!--req, enum, detection mode, subType:string, desc:"video", "coilLoop" (video and coil)-->video
</detectMode>
<helmet>
  <!--opt, bool, whether the driver is not wearing helmet-->true
</helmet>
<helmetCapNo>
  <!--opt, int, number of captured pictures in which the driver does not wear the helmet, range:[1,3]-->1
</helmetCapNo>
<nonMotorExist>
  <!--opt, bool, whether the non-motor vehicle is occupying non-motor vehicle lane-->true
</nonMotorExist>
<nonMotorExistCapNo>
  <!--opt, int, number of captured pictures in which the motor lane is occupied by non-motor vehicle, range:[2,3]-->2
</nonMotorExistCapNo>
<motorIllegalCarrying>
  <!--opt, bool, illegal manned motor vehicle-->true
</motorIllegalCarrying>
<nomotorIllegalCarrying>
  <!--opt, bool, illegal manned non-motor vehicle-->true
</nomotorIllegalCarrying>
<canopy>
  <!--opt, bool, umbrella tent installation violation on non-motor vehicle-->true
</canopy>
<pedRedLight>
  <!--opt, bool, pedestrian red light running-->true
</pedRedLight>
<pedOnNomotorLane>
  <!--opt, bool, whether to detect pedestrian walking on non-motor vehicle lane-->true
</pedOnNomotorLane>
<pedOnMotorLane>
  <!--opt, bool, whether to detect pedestrian walking on motor vehicle lane-->true
</pedOnMotorLane>
<capType>
  <!--opt, enum, Capture Type, subType:string, desc:"vehicle" (motor vehicle), "nonMotor" (non-motor vehicle), "human" (pedestrian), "face",-->mixObject
</capType>
<deferredFrmNum>
  <!--req, int, delay a few seconds to trigger the mode, range:[0,25]-->0
</deferredFrmNum>
<parkingOverStopLine>
  <!--opt, object, over stop line-->
  <enabled>
    <!--req, bool, whether to enable driving on the lane line trigger mode-->true
  </enabled>
</parkingOverStopLine>
<speedDetectMode>
  <!--opt, enum, speed detection mode, subType:string, desc:"none", "radar"-->none
</speedDetectMode>
<NotSlowZebraCrossingEvent>
  <!--opt, object, not slowing down at zebra crossing-->
  <enabled>
    <!--req, bool, whether to enable not slowing down at zebra crossing triggering mode-->true
  </enabled>
  <capNo>
    <!--req, int, number of captures, range:[1,3], step:1-->1
  </capNo>
```

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```
      <lowSpeed>
        <!--req, int, lowest speed limit (km/h), range:[0,100], step:1-->0
      </lowSpeed>
      <highSpeed>
        <!--req, int, highest speed limit (km/h), range:[0,100], step:1-->0
      </highSpeed>
      <LineInfoList>
        <!--opt, object, list of lane line information-->
        <LineInfo>
          <!--opt, object, single lane line information-->
          <lineId>
            <!--req, int, index, range:[1,3]-->0
          </lineId>
          <RegionCoordinatesList>
            <!--opt, array, list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, y-coordinate-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </LineInfo>
      </LineInfoList>
    </NotSlowZebraCrossingEvent>
    <TurnRightStop>
      <!--opt, object, large-sized vehicle not stopping before right turn-->
      <enabled>
        <!--req, bool, whether to enable intersection violation system-->true
      </enabled>
      <capNo>
        <!--req, int, number of captures, range:[1,3], step:1-->3
      </capNo>
      <durationTime>
        <!--req, int, duration, range:[1,600], unit:s-->5
      </durationTime>
      <slowUp>
        <!--req, int, upper limit of deceleration, range:[0,100], step:1, unit:km/h-->40
      </slowUp>
      <slowStep>
        <!--req, int, deceleration range, range:[0,50], step:1, unit:km/h-->5
      </slowStep>
      <slowDown>
        <!--req, int, lower limit of deceleration, range:[0,100], step:1, unit:km/h-->10
      </slowDown>
      <detectMode>
        <!--req, enum, deceleration mode, subType:string, desc:"stop", "slow"-->stop
      </detectMode>
      <SlowLane>
        <!--opt, object, deceleration line-->
        <RegionCoordinatesList>
          <!--opt, array, list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </SlowLane>
    </TurnRightStop>
    <NotYieldToRightRoad>
      <!--opt, object, not yield to vehicles from right roads-->
      <enabled>
        <!--req, bool, whether to enable not yield to vehicles from right roads triggering mode-->false
      </enabled>
      <firstCapturePosition>
        <!--req, int, capture position of first picture, range:[0,100], step:1-->20
      </firstCapturePosition>
      <lateralDistance>
        <!--req, int, lateral distance between two vehicles, range:[20,350], step:1-->150
      </lateralDistance>
      <longitudinalDistance>
        <!--req, int, longitudinal distance between two vehicles, range:[10,150], step:1-->50
      </longitudinalDistance>
    </NotYieldToRightRoad>
    <EnterNotYieldToInRoundabout>
      <!--opt, object, not yield at roundabout intersection-->
      <enabled>
        <!--req, bool, whether to enable not yield at roundabout intersection triggering mode-->false
      </enabled>
      <firstCapturePosition>
        <!--req, int, capture position of first picture, range:[0,100], step:1-->20
      </firstCapturePosition>
      <lateralDistance>
        <!--req, int, lateral distance between two vehicles, range:[20,350], step:1-->150
      </lateralDistance>
      <longitudinalDistance>
```

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```
      <!--req, int, longitudinal distance between two vehicles, range:[10,150], step:1-->50
    </longitudinalDistance>
  </EnterNotYieldToInRoundabout>
  <RampNotYieldToMainRoad>
    <!--opt, object, vehicle from ramp not yielding to vehicle on main road-->
    <enabled>
      <!--req, bool, whether to enable vehicle from ramp not yielding to vehicle on main road triggering mode-->false
    </enabled>
    <lateralDistance>
      <!--req, int, lateral distance of a vehicle when it drives from a ramp to a main road., range:[10,90], step:1-->60
    </lateralDistance>
  </RampNotYieldToMainRoad>
  <TruckRunARedLight>
    <!--opt, object, red light running of large truck detection, desc:this node is used for linking evidence intersection violation cameras can link the
checkpoints to capture the front license plate of the large trucks-->
    <enabled>
      <!--req, bool, whether to enable-->false
    </enabled>
    <secondCam>
      <!--req, int, the unique identifier for the linked cameras, range:[0,255], desc:all linked cameras share the same configuration, which is related to
the uploaded value of secondCam in ANPR-->1
    </secondCam>
    <sceneMode>
      <!--opt, enum, detection scene type, subType:string, desc:"singleLane", "multiLane"-->singleLane
    </sceneMode>
    <laneID>
      <!--opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,singleLane}, desc:this node is valid
when the camera is applied in single lane scene-->1
    </laneID>
    <bluePlateCarFilterEnabled>
      <!--opt, bool, whether to enable filtering blue plate vehicles-->true
    </bluePlateCarFilterEnabled>
    <NotificationList>
      <!--opt, array, list of linked camera, max. 3 linked cameras, subType:object, range:[1,3]-->
      <notification>
        <!--opt, object, picture capture-->
        <enabled>
          <!--req, bool, whether to enable-->true
        </enabled>
        <ipv4>
          <!--req, string, IPv4 address, range:[0,32]-->test
        </ipv4>
        <port>
          <!--req, int, port No., range:[0,66535]-->0
        </port>
        <username>
          <!--opt, string, user name, range:[0,32], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
        </username>
        <password>
          <!--req, string, password, range:[0,16], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
        </password>
        <laneID>
          <!--opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,multiLane}, desc:this node is
valid when the camera is applied in multiple lanes scene-->1
        </laneID>
      </notification>
    </NotificationList>
    <linkSceneMode>
      <!--ro, opt, enum, detection scene type, subType:string, desc:"oneLinkMultiple"(one camera links with multiple cameras), "oneLinkOne" (one camera
links with one camera), "multipleLinkOne" (multiple cameras link with multiple cameras)-->oneLinkOne
    </linkSceneMode>
  </TruckRunARedLight>
  <TruckTurnRightStop>
    <!--opt, object, large-sized vehicle not stopping before right turn-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <capNo>
      <!--req, int, number of captures, range:[2,3], step:1-->3
    </capNo>
    <sensitivity>
      <!--req, int, stopping sensitivity, range:[1,100]-->50
    </sensitivity>
    <stopTime>
      <!--opt, int, parking duration, range:[1,180], unit:s-->10
    </stopTime>
  </TruckTurnRightStop>
  <uphone>
    <!--ro, opt, object, making a phone call-->
    <enabled>
      <!--ro, req, bool, whether to enable phone call detection-->true
    </enabled>
    <capNo>
      <!--ro, req, int, number of captures, range:[1,3]-->2
    </capNo>
  </uphone>
  <NoAlterTrafficDet>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
    <capNo>
      <!--req, int, range:[3,3], step:1-->3
```

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```
    </capNo>
    <ConvergenceLineList>
      <!--opt, array, subType:object, range:[1,1]-->
      <ConvergenceLine>
        <!--opt, object-->
        <lineId>
          <!--req, int, range:[1,1]-->1
        </lineId>
        <lineName>
          <!--req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--opt, array, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object-->
            <positionX>
              <!--req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ConvergenceLine>
    </ConvergenceLineList>
  </NoAlterTrafficDet>
  <dangerCarHighSpeed>
    <!--opt, int, range:[0,255], unit:km/h-->0
  </dangerCarHighSpeed>
</VideoEpolice>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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## 14.5.1.31 Get the multiple-frame recognition mode parameters of smart video security

### Request URL

GET /ISAPI/ITC/TriggerMode/videoMonitor

### Query Parameter

None

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, trigger mode parameters of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, total number of linked lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--ro, req, enum, captured target type, subType:string, desc:"vehicle" (motor vehicle), "vehWithHum" (non-motor vehicle)-->vehicle
  </capType>
  <ViolationDetectLineList>
    <!--ro, req, array, list of violation detection line on the single lane, subType:object-->
    <ViolationDetectLine>
      <!--ro, opt, object, violation detection line on the single lane-->
      <lineName>
        <!--ro, req, enum, line name, subType:string, desc:"laneLine" (lane line)-->laneLine
      </lineName>
```

```
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,99]-->1
      </relatedDriveWay>
      <laneDirectionType>
        <!--ro, req, enum, lane direction type No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4
(from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)--
>1
      </laneDirectionType>
      <carDriveDirect>
        <!--ro, opt, enum, subType:string-->up2down
      </carDriveDirect>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <signSpeed>
        <!--ro, opt, int, marked speed limit, range:[0,255]-->0
      </signSpeed>
      <speedLimit>
        <!--ro, opt, int, speed limit, range:[0,255]-->0
      </speedLimit>
      <lowSpeedLimit>
        <!--ro, opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
      </lowSpeedLimit>
      <bigCarSignSpeed>
        <!--ro, opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSignSpeed>
      <bigCarSpeedLimit>
        <!--ro, opt, int, speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSpeedLimit>
      <bigCarLowSpeedLimit>
        <!--ro, opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarLowSpeedLimit>
    </ViolationDetectLine>
  </ViolationDetectLineList>
  <VirtualLane>
    <!--ro, req, object, draw violation detection line on all lanes-->
    <lineName>
      <!--ro, req, enum, line name, subType:string, desc:line name-->laneRightBoundaryLine
    </lineName>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </VirtualLane>
  <enabled>
    <!--ro, opt, bool, whether to enable smart video security, desc:if this node is not returned, smart video security of the device is enabled by default--
>true
  </enabled>
  <TriggerLine>
    <!--ro, opt, object, triggering line of smart video security capture-->
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TriggerLine>
  <capNo>
    <!--ro, opt, int, captured pictures of smart video security linked cameras, range:[1,2]-->1
  </capNo>
  <interval>
    <!--ro, opt, int, capture interval of smart video security, range:[1,3000], unit:ms-->1
  </interval>
  <carDriveDirect>
    <!--ro, opt, enum, vehicle detection type, subType:string, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
  </carDriveDirect>
  <sceneMode>
    <!--ro, opt, enum, scene mode, subType:string, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance" (entrance and exit
scene), "other" (other scenes)-->standardScene
  </sceneMode>
```

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```
  <overSpeed>
    <!--ro, opt, bool, whether to enable overspeed trigger mode, desc:whether to enable overspeed trigger mode-->true
  </overSpeed>
  <overSpeedCapNo>
    <!--ro, opt, int, capture times of overspeed, range:[1,3]-->1
  </overSpeedCapNo>
  <lowSpeed>
    <!--ro, opt, bool, whether to enable low speed trigger mode-->true
  </lowSpeed>
  <lowSpeedCapNo>
    <!--ro, opt, int, capture times of low speed, range:[1,3]-->1
  </lowSpeedCapNo>
  <carHighSpeed>
    <!--ro, opt, int, abnormal overspeed, range:[0,255]-->0
  </carHighSpeed>
  <carLowSpeed>
    <!--ro, opt, int, abnormal low speed, range:[0,255]-->0
  </carLowSpeed>
  <bigCarHighSpeed>
    <!--ro, opt, int, abnormal overspeed of large-sized vehicle, range:[0,255]-->0
  </bigCarHighSpeed>
  <bigCarLowSpeed>
    <!--ro, opt, int, abnormal low speed of large-sized vehicle, range:[0,255]-->0
  </bigCarLowSpeed>
  <nonmotorOverSpeed>
    <!--ro, opt, bool, whether to enable capturing non-motor vehicle-->true
  </nonmotorOverSpeed>
  <nonmotorHighSpeed>
    <!--ro, opt, int, speed limit for non-motor vehicle (unit: km/h), range:[0,255], unit:km/h-->0
  </nonmotorHighSpeed>
</VideoMonitor>
```

### 14.5.1.32 Set the multiple-frame recognition mode parameters of smart video security

#### Request URL

PUT /ISAPI/ITC/TriggerMode/videoMonitor

#### Query Parameter

None

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, trigger mode parameters of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--req, enum, captured target type, subType:string, desc:"vehicle" (motor vehicle), "vehWithHum" (non-motor vehicle)-->vehicle
  </capType>
  <ViolationDetectLineList>
    <!--req, array, list of violation detection line on the single lane, subType:object-->
    <ViolationDetectLine>
      <!--opt, object, violation detection line on the single lane-->
      <lineName>
        <!--req, enum, detection line name, subType:string, desc:"laneLine" (for a single lane only)-->laneLine
      </lineName>
      <relatedDriveWay>
        <!--req, int, linked lane No., range:[1,99]-->1
      </relatedDriveWay>
      <laneDirectionType>
        <!--req, enum, lane direction type No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <carDriveDirect>
        <!--opt, enum, subType:string-->up2down
      </carDriveDirect>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, only two coordinates are required, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <signSpeed>
        <!--opt, int, marked speed limit, range:[0,255]-->0
      </signSpeed>
      <speedLimit>
        <!--opt, int, speed limit, range:[0,255]-->0
      </speedLimit>
```

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```
        <lowSpeedLimit>
          <!--opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
        </lowSpeedLimit>
        <bigCarSignSpeed>
          <!--opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
        </bigCarSignSpeed>
        <bigCarSpeedLimit>
          <!--opt, int, speed limit for large-sized vehicle, range:[0,255]-->0
        </bigCarSpeedLimit>
        <bigCarLowSpeedLimit>
          <!--opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
        </bigCarLowSpeedLimit>
      </ViolationDetectLine>
    </ViolationDetectLineList>
    <VirtualLane>
      <!--req, object, draw violation detection line on all lanes-->
      <lineName>
        <!--req, enum, detection line name, subType:string, desc:"laneRightBoundaryLine" (parking space boundary line)-->laneRightBoundaryLine
      </lineName>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </VirtualLane>
    <enabled>
      <!--opt, bool, whether to enable smart video security, desc:if this node is not returned, smart video security of the device is enabled by default-->true
    </enabled>
    <TriggerLine>
      <!--opt, object, triggering line of smart video security capture-->
      <RegionCoordinatesList>
        <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TriggerLine>
    <capNo>
      <!--opt, int, captured pictures of smart video security linked cameras, range:[1,2]-->1
    </capNo>
    <interval>
      <!--opt, int, capture interval of smart video security, range:[1,3000], unit:ms-->1
    </interval>
    <carDriveDirect>
      <!--opt, enum, vehicle detection type, subType:string, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
    </carDriveDirect>
    <sceneMode>
      <!--opt, enum, scene mode, subType:string, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance" (entrance and exit scene), "other" (other scenes)-->standardScene
    </sceneMode>
    <overSpeed>
      <!--opt, bool, whether to enable overspeed trigger mode, desc:whether to enable overspeed trigger mode-->true
    </overSpeed>
    <overSpeedCapNo>
      <!--opt, int, capture times of overspeed, range:[1,3]-->1
    </overSpeedCapNo>
    <lowSpeed>
      <!--opt, bool, whether to enable low speed trigger mode-->true
    </lowSpeed>
    <lowSpeedCapNo>
      <!--opt, int, capture times of low speed, range:[1,3]-->1
    </lowSpeedCapNo>
    <carHighSpeed>
      <!--opt, int, abnormal overspeed, range:[0,255]-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--opt, int, abnormal low speed, range:[0,255]-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--opt, int, abnormal overspeed of large-sized vehicle, range:[0,255]-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--opt, int, abnormal low speed of large-sized vehicle, range:[0,255]-->0
    </bigCarLowSpeed>
    <nonmotorOverSpeed>
      <!--ro, opt, bool, whether to enable capturing non-motor vehicle-->true
    </nonmotorOverSpeed>
    <nonmotorHighSpeed>
      <!--ro, opt, int, speed limit for non-motor vehicle (unit: km/h), range:[0,255], unit:km/h-->0
    </nonmotorHighSpeed>
```

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```
</VideoMonitor>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

## 14.5.1.33 Get the multiple-frame recognition mode capability of smart video security

### Request URL

GET /ISAPI/ITC/TriggerMode/videoMonitor/capabilities

### Query Parameter

None

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, multiple-frame recognition mode capability of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="16">
    <!--ro, req, int, total number of linked lanes, attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <relatedDriveWay min="1" max="99">
    <!--ro, req, int, linked lane No., attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, lane direction, subType:int, attr:opt{req, string}, desc:0-unknown, 1-from east to west, 2-from west to east, 3-from south to north,
4-from north to south, 5-from southeast to northwest, 6-from northwest to southeast, 7-from northeast to southwest, 8-from southwest to northeast-->0
  </laneDirectionType>
  <capType opt="vehWithHum,vehicle,nonmotor,human,motorWithHum,nonmotorWithHum,motorWithNonmotor">
    <!--ro, req, enum, captured target type, subType:string, attr:opt{req, string}, desc:"vehWithHum" (motor vehicle, non-motor vehicle, and pedestrian),
"vehicle" (motor vehicle), "nonmotor" (non-motor vehicle), "human" (pedestrian), "motorWithHum" (motor vehicle and pedestrian), "nonmotorWithHum" (non-motor
vehicle and pedestrian), "motorNonmotor" (motor vehicle and non-motor vehicle)-->vehWithHum
  </capType>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable smart video security, attr:opt{req, string}-->true
  </enabled>
  <TriggerLine>
    <!--ro, opt, object, triggering line of smart video security capture-->
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TriggerLine>
  <capNo opt="1,2">
    <!--ro, opt, int, number of captured pictures, attr:opt{req, string}-->0
  </capNo>
  <interval min="1" max="3000">
    <!--ro, opt, int, capture time, unit:ms, attr:min{req, int},max{req, int}-->0
```

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```
    </interval>
    <carDriveDirect opt="up2down,down2up">
      <!--ro, req, enum, vehicle detection type, subType:string, attr:opt{req, string}, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
    </carDriveDirect>
    <sceneMode opt="standardScene ,lowPoleScene">
      <!--ro, opt, enum, scene mode, subType:string, attr:opt{req, string}, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance"
(entrance and exit scene), "other" (other scenes)-->standardScene
    </sceneMode>
    <overSpeed>
      <!--ro, opt, bool, whether to enable overspeed trigger mode-->true
    </overSpeed>
    <overSpeedCapNo opt="1,2,3">
      <!--ro, opt, int, capture times of overspeed, attr:opt{req, string}-->0
    </overSpeedCapNo>
    <lowSpeed>
      <!--ro, opt, bool, whether to enable low speed trigger mode-->true
    </lowSpeed>
    <lowSpeedCapNo opt="1,2,3">
      <!--ro, opt, int, capture times of low speed, attr:opt{req, string}-->0
    </lowSpeedCapNo>
    <signSpeed min="0" max="255">
      <!--ro, opt, int, marked speed limit, attr:min{req, int},max{req, int}-->0
    </signSpeed>
    <speedLimit min="0" max="255">
      <!--ro, opt, int, speed limit, attr:min{req, int},max{req, int}-->0
    </speedLimit>
    <bigCarSignSpeed min="0" max="255">
      <!--ro, opt, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarSignSpeed>
    <bigCarHighSpeedLimit min="0" max="255">
      <!--ro, opt, int, maximum speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarHighSpeedLimit>
    <bigCarLowSpeedLimit min="0" max="255">
      <!--ro, opt, int, low speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarLowSpeedLimit>
    <lowSpeedLimit min="0" max="60">
      <!--ro, opt, int, low speed limit for small-sized vehicle, attr:min{req, int},max{req, int}-->0
    </lowSpeedLimit>
    <bigCarSpeedLimit min="0" max="60">
      <!--ro, opt, int, speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarSpeedLimit>
    <carHighSpeed min="0" max="255">
      <!--ro, opt, int, abnormal overspeed, attr:min{req, int},max{req, int}-->0
    </carHighSpeed>
    <carLowSpeed min="0" max="255">
      <!--ro, opt, int, abnormal low speed, attr:min{req, int},max{req, int}-->0
    </carLowSpeed>
    <bigCarHighSpeed min="0" max="255">
      <!--ro, opt, int, abnormal overspeed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed min="0" max="255">
      <!--ro, opt, int, abnormal low speed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarLowSpeed>
    <nonmotorHighSpeedLimit min="0" max="255">
      <!--ro, opt, int, speed limit for non-motor vehicle, attr:min{req, int},max{req, int}-->0
    </nonmotorHighSpeedLimit>
  </VideoMonitor>
```

#### 14.5.1.34 Get the recommended value of smart video security multi-frame recognition parameter

**Request URL**

GET /ISAPI/ITC/TriggerMode/videoMonitor/recommendation?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, smart video security multi-frame recognition parameter, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, total number of linked lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--ro, req, enum, detection type, subType:string, desc:detection type-->vehicle
  </capType>
  <ViolationDetectLineList>
```

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```
<ViolationDetectLineList>
  <!--ro, req, array, subType:object-->
  <ViolationDetectLine>
    <!--ro, opt, object-->
    <lineName>
      <!--ro, req, enum, line name, subType:string, desc:line name-->laneLine
    </lineName>
    <relatedDriveWay>
      <!--ro, req, int, lane No., range:[1,99]-->1
    </relatedDriveWay>
    <laneDirectionType>
      <!--ro, req, enum, lane direction, subType:int, desc:0-unknown, 1-from east to west, 2-from west to east, 3-from south to north, 4-from north to
south, 5-from southeast to northwest, 6-from northwest to southeast, 7-from northeast to southwest, 8-from southwest to northeast-->1
    </laneDirectionType>
    <carDriveDirect>
      <!--ro, opt, enum, subType:string-->up2down
    </carDriveDirect>
    <RegionCoordinatesList>
      <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <signSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </signSpeed>
    <speedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </speedLimit>
    <lowSpeedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </lowSpeedLimit>
    <bigCarSignSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarSignSpeed>
    <bigCarSpeedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarSpeedLimit>
    <bigCarLowSpeedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarLowSpeedLimit>
  </ViolationDetectLine>
</ViolationDetectLineList>
<VirtualLane>
  <!--ro, req, object-->
  <lineName>
    <!--ro, req, enum, line name, subType:string, desc:line name-->laneRightBoundaryLine
  </lineName>
  <RegionCoordinatesList>
    <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
    <RegionCoordinates>
      <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</VirtualLane>
<enabled>
  <!--ro, opt, bool-->true
</enabled>
<TriggerLine>
  <!--ro, opt, object-->
  <RegionCoordinatesList>
    <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
    <RegionCoordinates>
      <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</TriggerLine>
<capNo>
  <!--ro, opt, int, range:[1,2]-->1
</capNo>
<interval>
  <!--ro, opt, int, range:[1,3000], unit:ms-->1
</interval>
<sceneMode>
  <!--ro, opt, enum, subType:string-->standardScene
</sceneMode>
```

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```
    <overSpeed>
      <!--ro, opt, bool-->true
    </overSpeed>
    <overSpeedCapNo>
      <!--ro, opt, int, range:[1,3]-->1
    </overSpeedCapNo>
    <lowSpeed>
      <!--ro, opt, bool-->true
    </lowSpeed>
    <lowSpeedCapNo>
      <!--ro, opt, int, range:[1,3]-->1
    </lowSpeedCapNo>
    <carHighSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarLowSpeed>
    <nonmotorOverSpeed>
      <!--ro, opt, bool-->true
    </nonmotorOverSpeed>
    <nonmotorHighSpeed>
      <!--ro, opt, int, range:[0,255], unit:km/h-->0
    </nonmotorHighSpeed>
    <channelId>
      <!--ro, opt, int, range:[1,5]-->1
    </channelId>
    <radarTrack>
      <!--ro, opt, bool-->true
    </radarTrack>
  </VideoMonitor>
```

### 14.5.1.35 Set the video mark parameters

### Request URL

PUT /ISAPI/ITC/videoMark

### Query Parameter

None

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMarkInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video mark information, attr:version{req, string, protocolVersion}-->
  <horizontalField>
    <!--req, int, horizontal FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </horizontalField>
  <verticalField>
    <!--req, int, vertical FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </verticalField>
  <autoGetGps>
    <!--req, bool, whether to get GPS information automatically, desc:the value of this node is false by default-->true
  </autoGetGps>
  <longitude>
    <!--req, string, longitude, range:[1,11], desc:longitude, which contains 11 digits-->W040D30M28S
  </longitude>
  <latitude>
    <!--req, string, latitude, range:[1,11], desc:latitude, which contains 11 digits-->N040D30M28S
  </latitude>
</VideoMarkInfo>
```

### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.5.1.36 Get the video mark parameters

**Request URL**

GET /ISAPI/ITC/videoMark

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMarkInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video mark information, attr:version{req, string, protocolVersion}-->
  <horizontalField>
    <!--ro, req, int, horizontal FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </horizontalField>
  <verticalField>
    <!--ro, req, int, vertical FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </verticalField>
  <autoGetGps>
    <!--ro, req, bool, whether to get GPS information automatically: "true"-yes, "false"-no (default), desc:the value of this node is false by default--
>true
  </autoGetGps>
  <longitude>
    <!--ro, req, string, longitude, range:[1,11], desc:longitude, which contains 11 digits-->W040D30M28S
  </longitude>
  <latitude>
    <!--ro, req, string, latitude, range:[1,11], desc:latitude, which contains 11 digits-->N040D30M28S
  </latitude>
</VideoMarkInfo>
```

#### 14.5.1.37 Set ANR parameters

**Request URL**

PUT /ISAPI/Traffic/ANR

**Query Parameter**

None

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ANRControl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ANR parameters, attr:version{req, string, protocolVersion}-->
  <uploadInterval>
    <!--req, int, upload interval, unit:ms-->0
  </uploadInterval>
  <uploadTimeout>
    <!--opt, int, uploading timed out, unit:ms-->0
  </uploadTimeout>
  <DeviceIpAddress>
    <!--opt, object, IP address of ANR arming server-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[0,64], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,32], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
  </DeviceIpAddress>
</ANRControl>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

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## 14.5.1.38 Set parameters of traffic picture composition

### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/imageMerge

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageMerge xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <isMerge>
    <!--req, int, whether to enable composition, range:[0,1]-->0
  </isMerge>
  <oneMergeType>
    <!--req, enum, subType:int-->101
  </oneMergeType>
  <twoMergeType>
    <!--req, enum, subType:int-->201
  </twoMergeType>
  <threeMergeType>
    <!--req, enum, subType:int-->301
  </threeMergeType>
  <fourMergeType>
    <!--req, enum, subType:int-->401
  </fourMergeType>
  <closeupIndex>
    <!--opt, int, range:[0,5]-->0
  </closeupIndex>
  <closeupScale>
    <!--opt, int, range:[0,100]-->0
  </closeupScale>
  <positionOffset>
    <!--opt, int, range:[50,2047]-->50
  </positionOffset>
  <closeUpCaptureEnabled>
    <!--opt, bool-->true
  </closeUpCaptureEnabled>
</ImageMerge>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.5.1.39 Get the picture composition parameters

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/imageMerge

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageMerge xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isMerge>
    <!--ro, req, int, range:[0,1]-->0
  </isMerge>
  <oneMergeType>
    <!--ro, req, enum, subType:string-->101
  </oneMergeType>
  <twoMergeType>
    <!--ro, req, enum, subType:string-->201
```

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```
    <!--ro, req, enum, subType:string-->201
  </twoMergeType>
  <threeMergeType>
    <!--ro, req, enum, subType:string-->301
  </threeMergeType>
  <fourMergeType>
    <!--ro, req, enum, subType:string-->401
  </fourMergeType>
  <closeupIndex>
    <!--ro, opt, int, range:[0,5]-->0
  </closeupIndex>
  <closeupScale>
    <!--ro, opt, int, range:[0,100]-->0
  </closeupScale>
  <positionOffset>
    <!--ro, opt, int, range:[0,2159]-->50
  </positionOffset>
  <featureEnabled>
    <!--ro, opt, bool-->true
  </featureEnabled>
  <closeUpCaptureEnabled>
    <!--ro, opt, bool-->true
  </closeUpCaptureEnabled>
  <featureOrder>
    <!--ro, opt, enum, subType:string-->after
  </featureOrder>
  <closeUpSelect>
    <!--ro, opt, enum, subType:string-->first
  </closeUpSelect>
  <IllegalImageMerge>
    <!--ro, opt, object-->
    <isIllegalMerge>
      <!--ro, opt, bool-->true
    </isIllegalMerge>
    <oneMergeType>
      <!--ro, opt, enum, subType:string-->101
    </oneMergeType>
    <twoMergeType>
      <!--ro, opt, enum, subType:string-->201
    </twoMergeType>
    <threeMergeType>
      <!--ro, opt, enum, subType:string-->301
    </threeMergeType>
    <fourMergeType>
      <!--ro, opt, enum, subType:string-->401
    </fourMergeType>
    <closeupIndex>
      <!--ro, opt, int, range:[0,5]-->0
    </closeupIndex>
    <closeupScale>
      <!--ro, opt, int, range:[0,100]-->0
    </closeupScale>
    <positionOffset>
      <!--ro, opt, int, range:[0,2159]-->50
    </positionOffset>
    <featureEnabled>
      <!--ro, opt, bool-->false
    </featureEnabled>
    <featureOrder>
      <!--ro, opt, enum, subType:string-->after
    </featureOrder>
    <closeUpCaptureEnabled>
      <!--ro, opt, bool-->true
    </closeUpCaptureEnabled>
    <staticPicFourMergeOrder>
      <!--ro, opt, int, range:[0,4]-->0
    </staticPicFourMergeOrder>
    <nonMotorCloseupScale>
      <!--ro, opt, int, range:[0,100]-->0
    </nonMotorCloseupScale>
    <closeUpSelect>
      <!--ro, opt, enum, subType:string-->first
    </closeUpSelect>
  </IllegalImageMerge>
  <staticPicFourMergeOrder>
    <!--ro, opt, int, range:[0,4]-->0
  </staticPicFourMergeOrder>
  <nonMotorCloseupScale>
    <!--ro, opt, int, range:[0,100]-->0
  </nonMotorCloseupScale>
</ImageMerge>
```

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#### 14.5.1.40 Get the capability of picture composition parameters

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/imageMerge/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<ImageMerge xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isMerge opt="0,1">
    <!--ro, req, int, range:[0,1], attr:opt{req, string}-->0
  </isMerge>
  <oneMergeType opt="101,102,103,104,0xff">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->101
  </oneMergeType>
  <twoMergeType opt="201,202,203,204,205,206,207,208,209,210,211,212,0xff">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->201
  </twoMergeType>
  <threeMergeType opt="301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,0xff">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->301
  </threeMergeType>
  <fourMergeType opt="401,402,403,0xff">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->401
  </fourMergeType>
  <closeupIndex min="0" max="5">
    <!--ro, opt, int, range:[0,5], attr:min{req, int},max{req, int}-->0
  </closeupIndex>
  <closeupScale min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </closeupScale>
  <positionOffset min="50" max="2047">
    <!--ro, opt, int, range:[50,2047], attr:min{req, int},max{req, int}-->50
  </positionOffset>
  <featureEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->test
  </featureEnabled>
  <closeUpCaptureEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </closeUpCaptureEnabled>
  <featureOrder opt="before,after">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->after
  </featureOrder>
  <IllegalImageMerge>
    <!--ro, opt, object-->
    <isIllegalMerge opt="true, false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </isIllegalMerge>
    <oneMergeType opt="101,102,103,104,0xff">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->101
    </oneMergeType>
    <twoMergeType opt="201,202,203,204,205,206,207,208,209,210,211,212,0xff">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->201
    </twoMergeType>
    <threeMergeType opt="301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,0xff">
      <!--ro, opt, enum, subType:int, attr:opt{req, string}-->301
    </threeMergeType>
    <fourMergeType opt="401,402,403,0xff">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->401
    </fourMergeType>
    <closeupIndex min="0" max="5">
      <!--ro, opt, int, range:[0,5], attr:min{req, int},max{req, int}-->0
    </closeupIndex>
    <closeupScale min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </closeupScale>
    <positionOffset min="50" max="2047">
      <!--ro, opt, int, range:[50,2047], attr:min{req, int},max{req, int}-->50
    </positionOffset>
    <featureEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->test
    </featureEnabled>
    <closeUpCaptureEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </closeUpCaptureEnabled>
    <featureOrder opt="before,after">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->after
    </featureOrder>
    <staticPicFourMergeOrder min="0" max="4">
      <!--ro, opt, int, range:[0,4], attr:min{req, int},max{req, int}-->0
    </staticPicFourMergeOrder>
    <nonMotorCloseupScale min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </nonMotorCloseupScale>
  </IllegalImageMerge>
  <nonMotorCloseupScale min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </nonMotorCloseupScale>
</ImageMerge>
```

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### 14.5.1.41 Set the parameters of vehicle detection captured pictures

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/picParam

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | channel No. |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of vehicle detection captured pictures, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--opt, object, picture parameters-->
    <mode>
      <!--req, enum, capture mode, subType:string, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality>
      <!--opt, int, picture quality, range:[1,100]-->50
    </pictureQuality>
    <pictureSize>
      <!--opt, int, picture size, unit:KB-->100
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--opt, object, picture overlay-->
    <enabled>
      <!--req, bool, whether to enable text overlay-->true
    </enabled>
    <OverlapItemList>
      <!--opt, array, overlay item list, subType:object-->
      <OverlapItem>
        <!--opt, object, overlay items-->
        <id>
          <!--req, int, index-->1
        </id>
        <item opt="positionNo">
          <!--req, string, overlay content, attr:opt{opt, enum, subType:string}-->test
        </item>
      </OverlapItem>
    </OverlapItemList>
    <fontColor>
      <!--opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--opt, object, Capture Overlay, desc:professional OSD configuration for picture capture, which is only supported by capture cameras now-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--opt, object, Composite Picture Overlay, desc:professional OSD configuration for picture composite, which is only supported by capture cameras now-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--opt, object, License Plate Enhancement-->
    <enabled>
      <!--opt, bool, whether to enable license plate enhancement on captured picture-->true
    </enabled>
    <level>
      <!--opt, int, set the level of license plate enhancement from 0 to 100, range:[0,100]-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList>
    <!--opt, array, the list of the types of captured pictures, subType:object, range:[1,3], desc:all types of picture shall be captured if this node is not
returned-->
    <capturePicType>
      <!--opt, enum, types of captured pictures, subType:string, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle picture),
"detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution>
    <!--opt, string, picture resolution, desc:subject to the returned range-->1920*1080
  </resolution>
  <capturePicIntervalEnabled>
    <!--ro, opt, bool, whether to enable picture capture interval-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime>
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled>
    <!--opt, bool, whether to enable background picture of panoramic channel, desc:a scene can be monitored by both channels of the front-end dual-channel
device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the panoramic-image channel-->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```

### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.5.1.42 Get parameters of vehicle detection captured pictures

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/picParam

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | channel No. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of vehicle detection captured pictures, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--ro, opt, object, picture parameters-->
    <mode>
      <!--ro, req, enum, capture mode, subType:string, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality>
      <!--ro, opt, int, picture quality, range:[1,100]-->50
    </pictureQuality>
    <pictureSize>
      <!--ro, opt, int, picture size, unit:KB-->100
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--ro, opt, object, picture overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable text overlay-->true
    </enabled>
    <OverlapItemList>
      <!--ro, opt, array, overlay item list, subType:object-->
      <OverlapItem>
        <!--ro, opt, object, overlay items-->
        <id>
          <!--ro, req, int, ID-->1
        </id>
        <item opt="positionNo">
          <!--ro, req, string, overlay content, attr:opt{opt, enum, subType:string}-->test
        </item>
      </OverlapItem>
    </OverlapItemList>
    <fontColor>
      <!--ro, opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--ro, opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--ro, opt, object, capture overlay, desc:professional OSD configuration for picture capture, which is only supported by capture cameras now-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--ro, opt, object, composite picture overlay, desc:professional OSD configuration for picture composite, which is only supported by capture cameras
now-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--ro, opt, object, license plate enhancement-->
    <enabled>
      <!--ro, opt, bool, whether to enable license plate enhancement on captured picture-->true
    </enabled>
    <level>
      <!--ro, opt, int, set the level of license plate enhancement from 0 to 100, range:[0,100]-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList>
    <!--ro, opt, array, the list of the types of captured pictures, subType:object, range:[1,3], desc:all types of picture shall be captured if this node is
not returned-->
    <capturePicType>
      <!--ro, opt, enum, types of captured pictures, subType:string, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle picture),
"detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution>
    <!--ro, opt, string, picture resolution, desc:subject to the returned range-->1920*1080
  </resolution>
  <capturePicIntervalEnabled>
    <!--ro, opt, bool, whether to enable picture capture interval-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime>
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled>
    <!--ro, opt, bool, whether to enable background picture of panoramic channel, desc:a scene can be monitored by both channels of the front-end dual-
channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the panoramic-image channel-
->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```

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### 14.5.1.43 Get captured picture configuration capability

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capabilities

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | channel No. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture capture configuration capability, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--ro, req, object, picture parameters-->
    <mode opt="quality,size">
      <!--ro, req, enum, capture mode, subType:string, attr:opt{opt, enum, subType:string}, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality min="1" max="100">
      <!--ro, opt, int, picture quality, range:[1,100], attr:min{opt, string},max{opt, string}-->1
    </pictureQuality>
    <pictureSize min="1" max="6553500000">
      <!--ro, opt, int, picture size, unit:KB, attr:min{req, int},max{req, int}-->1
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--ro, opt, object, picture overlay-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable text overlay, attr:opt{req, string}-->true
    </enabled>
    <OverlapItem opt="positionNo,positionInfo,cameraNo,captureTime,plateNo,vehicleColor,sceneName,carType,vehicleLogo,sceneNo,GPS,vehicleSpeed">
      <!--ro, req, enum, overlay items, subType:string, attr:opt{opt, enum}, desc:"positionNo" (detection point No.), "positionInfo" (detection point),
"cameraNo" (camera No.), "captureTime" (capture time), "plateNo" (license plate No.), "vehicleColor" (vehicle color), "sceneName" (scene name), "carType"
(vehicle type), "vehicleLogo" (main vehicle brand) , "sceneNo" (scene No.), "GPS" (GPS device)-->positionNo
    </OverlapItem>
    <fontColor>
      <!--ro, opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--ro, opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--ro, opt, object, professional OSD configuration for captured pictures,currently this configuration is only supported by capture cameras,
desc:professional OSD configuration for captured pictures,currently this configuration is only supported by capture cameras-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--ro, opt, object, professional OSD configuration for combined pictures,currently this configuration is only supported by capture cameras,
desc:professional OSD configuration for combined pictures,currently this configuration is only supported by capture cameras-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--ro, opt, object, license plate enhancement on captured picture-->
    <enabled opt="true,false" def="false">
      <!--ro, opt, bool, whether to enable license plate enhancement on captured picture, attr:opt{opt, string},def{opt, string}-->true
    </enabled>
    <level min="0" max="100" def="50">
      <!--ro, opt, int, license plate enhancement level range, range:[0,100], attr:min{opt, string},max{opt, string},def{opt, string}-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList size="3">
    <!--ro, opt, array, the list of the types of captured pictures, subType:object, attr:size{req, int}, desc:all types of picture shall be captured if this
node is not returned-->
    <capturePicType opt="licensePlatePicture,vehiclePicture,detectionPicture">
      <!--ro, opt, string, types of captured pictures, attr:opt{req, string}, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle
picture), "detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution opt="1920*1080,2688*1520,1280*720,704*576,352*288">
    <!--ro, opt, string, picture resolution, attr:opt{req, string}-->1920*1080
  </resolution>
  <capturePicIntervalEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable picture capture interval, attr:opt{req, string}-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime min="1" max="30" def="1">
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}, attr:min{req, int},max{req,
int},def{req, int}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable background picture of panoramic channel, attr:opt{req, string}, desc:a scene can be monitored by both channels of
the front-end dual-channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the
panoramic-image channel-->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```

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### 14.5.1.44 Get the capability of OSD overlay on a captured picture

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/capabilities

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled opt="true,false">
    <!--ro, req, bool, whether to enable text overlay, attr:opt{req, string}, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList size="50">
    <!--ro, req, array, subType:object, attr:size{opt, string}-->
    <OverlayInfo>
      <!--ro, req, object-->
      <itemType
opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53
,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,1
04,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119" min="1" max="88">
        <!--ro, req, enum, subType:int, attr:opt{opt, string},min{req, int},max{req, int}-->70
      </itemType>
      <editAble opt="true,false">
        <!--ro, req, bool, whether the overlaid information is editable, attr:opt{opt, string}-->true
      </editAble>
      <itemOverlayEnabled opt="true,false">
        <!--ro, req, bool, attr:opt{opt, string}-->true
      </itemOverlayEnabled>
      <customName min="0" max="32">
        <!--ro, req, string, self-defined name of the overlay information, attr:min{opt, string},max{opt, string}-->test
      </customName>
      <changeLineNum min="0" max="100">
        <!--ro, req, int, range:[0,100], attr:min{opt, string},max{opt, string}-->50
      </changeLineNum>
      <spaceNum min="0" max="255">
        <!--ro, req, int, range:[0,255], attr:min{opt, string},max{opt, string}-->1
      </spaceNum>
      <startPosEnable opt="true,false">
        <!--ro, opt, bool, whether to enable coordinate configuration, attr:opt{opt, string}-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point-->0
      </startPosLeft>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent min="0" max="100">
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], attr:min{opt, string},max{opt, string}-->100
  </linePercent>
  <itemsStlye opt=" horizontal,vertical">
    <!--ro, req, enum, overlay mode, subType:string, attr:opt{opt, string}, desc:"horizontal" (default),"vertical"-->vertical
  </itemsStlye>
  <charStyle opt="song_type,wei_type">
    <!--ro, req, enum, font type, subType:string, attr:opt{opt, string}, desc:font type-->song_type
  </charStyle>
  <charSize min="0" max="5">
    <!--ro, req, enum, character size, subType:int, attr:min{opt, string},max{opt, string}, desc:0-16*16(Chinese)/8*16(English),1-
32*32(Chinese)/16*32(English),2-48*48,3-64*64(Chinese)/32*64(English)-->1
  </charSize>
  <charPosition min="0" max="2">
    <!--ro, req, enum, text position overlaid on the picture, subType:int, attr:min{opt, string},max{opt, string}, desc:0 (in the picture),1 (outside the
top edge of the picture), 2 (outside the bottom edge of the picture)-->1
  </charPosition>
  <charInterval min="0" max="16">
    <!--ro, req, int, character separation distance, range:[0,16], unit:px, attr:min{opt, string},max{opt, string}-->1
  </charInterval>
  <foreColor min="0" max=" 0xffffff">
    <!--ro, req, int, foreground color, which is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value
is 0xffffff (white), attr:min{opt, string},max{opt, string}-->1
  </foreColor>
  <backColor min="0" max=" 0xffffff">
    <!--ro, req, int, background color, which is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value
is 0x0 (black), attr:min{opt, string},max{opt, string}-->1
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption: 0-no,1-yes-->true
  </colorAdapt>
```

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```
<zeroizeEnable opt="true,false">
<!--ro, opt, bool, whether to enable zero filling for OSD overlay, which is used to enable zero filling for vehicle speed, speed limit, overspeed ratio and lane No. Zero filling is enabled by default, attr:opt{opt, string}-->true
</zeroizeEnable>
<snapTimeType opt="uploadTime, generationTime, exposureTime">
<!--ro, opt, enum, subType:string, attr:opt{req, string}-->generationTime
</snapTimeType>
<charColorReverseEnabled opt="true, false">
<!--ro, opt, bool, attr:opt{req, string}-->true
</charColorReverseEnabled>
<platePicOverlay>
<!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
</platePicOverlay>
<platePicOverlay2>
<!--ro, opt, bool-->true
</platePicOverlay2>
<platePicOverlay3>
<!--ro, opt, bool-->true
</platePicOverlay3>
<platePicOverlayFeature>
<!--ro, opt, bool-->true
</platePicOverlayFeature>
<platePicOverlayMultiplier min="1" max="3">
<!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</platePicOverlayMultiplier>
<platePicPosTop>
<!--ro, req, int, start top coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
</platePicPosTop>
<platePicPosLeft>
<!--ro, req, int, start left coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
</platePicPosLeft>
<overlaidTextTransparency min="0" max="100">
<!--ro, opt, int, range:[0,100], unit:%, attr:min{req, int},max{req, int}-->1
</overlaidTextTransparency>
<featurePicOSDEnable opt="true,false">
<!--ro, opt, bool, attr:opt{req, string}-->true
</featurePicOSDEnable>
<dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
<!--ro, opt, string, attr:opt{req, string}-->YYYY-MM-DD
</dateStyle>
<timeStyle opt="12hour,24hour">
<!--ro, opt, string, attr:opt{req, string}-->12hour
</timeStyle>
<nonMotoPassengerNumEnabled opt="true,false">
<!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,60}, attr:opt{req, string}-->true
</nonMotoPassengerNumEnabled>
<prefixLanguage opt="English,Russian,Korean,Ukrainian,Croatian,TraditionalChinese,Spanish,Thai,Arabic">
<!--ro, opt, enum, subType:string, attr:opt{req, string}-->English
</prefixLanguage>
<PictureInfoList>
<!--ro, opt, array, subType:object-->
<PictureInfo>
<!--ro, opt, object, picture information-->
<pictureType opt="illegalPicture,timingPicture">
<!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:picture type-->illegalPicture
</pictureType>
<overlayInfoEnabled opt="true,false">
<!--ro, req, bool, attr:opt{req, string}-->true
</overlayInfoEnabled>
<OverlayInfoList size="50">
<!--ro, req, array, list of overlay information, subType:object, attr:size{opt, string}-->
<OverlayInfo>
<!--ro, req, object-->
<itemType opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109">
<!--ro, req, enum, subType:int, attr:opt{opt, string}-->70
</itemType>
<editAble opt="true,false">
<!--ro, req, bool, whether the overlaid information is editable, attr:opt{opt, string}-->true
</editAble>
<itemOverlayEnabled opt="true,false">
<!--ro, req, bool, attr:opt{opt, string}-->true
</itemOverlayEnabled>
<customName min="0" max="32">
<!--ro, req, string, self-defined name of the overlay information, attr:min{opt, string},max{opt, string}-->test
</customName>
<changeLineNum min="0" max="100">
<!--ro, req, int, range:[0,100], attr:min{opt, string},max{opt, string}-->50
</changeLineNum>
<spaceNum min="0" max="255">
<!--ro, req, int, range:[0,255], attr:min{opt, string},max{opt, string}-->1
</spaceNum>
<startPosEnable opt="true,false">
<!--ro, opt, bool, whether to enable coordinate configuration, attr:opt{opt, string}-->true
</startPosEnable>
<startPosTop>
<!--ro, req, int, top coordinate of the start point-->0
</startPosTop>
<startPosLeft>
<!--ro, req, int, left coordinate of the start point-->0
```

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```
            <!--ro, req, int, left coordinate of the start point-->0
          </startPosLeft>
        </OverlayInfo>
      </OverlayInfoList>
      <linePercent min="0" max="100">
        <!--ro, req, int, range:[0,100], attr:min{opt, string},max{opt, string}-->100
      </linePercent>
      <itemsStlye opt=" horizontal,vertical">
        <!--ro, req, enum, subType:string, attr:opt{opt, string}-->vertical
      </itemsStlye>
      <charStyle opt="song_type,wei_type">
        <!--ro, req, enum, font type, subType:string, attr:opt{opt, string}, desc:font type-->song_type
      </charStyle>
      <charSize min="0" max="3">
        <!--ro, req, enum, subType:int, attr:min{opt, string},max{opt, string}-->1
      </charSize>
      <charPosition min="0" max="2">
        <!--ro, req, enum, position of the text overlaid in the picture, subType:int, attr:min{opt, string},max{opt, string}, desc:0 (in the picture),1
(outside the top edge of the picture), 2 (outside the bottom edge of the picture)-->1
      </charPosition>
      <charInterval min="0" max="16">
        <!--ro, req, int, character pitch, range:[0,16], unit:px, attr:min{opt, string},max{opt, string}-->1
      </charInterval>
      <foreColor min="0" max=" 0xffffff">
        <!--ro, req, int, foreground color, attr:min{opt, string},max{opt, string}-->1
      </foreColor>
      <backColor min="0" max=" 0xffffff">
        <!--ro, req, int, background color, attr:min{opt, string},max{opt, string}-->1
      </backColor>
      <colorAdapt>
        <!--ro, opt, bool, whether to enable color self-adaption-->true
      </colorAdapt>
      <zeroizeEnable opt="true,false">
        <!--ro, opt, bool, whether to enable zero filling for OSD overlay, attr:opt{opt, string}-->true
      </zeroizeEnable>
      <snapTimeType opt="uploadTime, generationTime, exposureTime">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->generationTime
      </snapTimeType>
      <charColorReverseEnabled opt="true, false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </charColorReverseEnabled>
      <platePicOverlay>
        <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
      </platePicOverlay>
      <platePicOverlay2>
        <!--ro, opt, bool-->true
      </platePicOverlay2>
      <platePicOverlay3>
        <!--ro, opt, bool-->true
      </platePicOverlay3>
      <platePicOverlayFeature>
        <!--ro, opt, bool-->true
      </platePicOverlayFeature>
      <platePicOverlayMultiplier min="1" max="3">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </platePicOverlayMultiplier>
      <platePicPosTop>
        <!--ro, opt, int, top coordinate of the start point-->0
      </platePicPosTop>
      <platePicPosLeft>
        <!--ro, opt, int, left coordinate of the start point-->0
      </platePicPosLeft>
      <overlaidTextTransparency min="0" max="100">
        <!--ro, opt, int, range:[0,100], unit:%, attr:min{req, int},max{req, int}-->1
      </overlaidTextTransparency>
      <featurePicOSDEnable opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </featurePicOSDEnable>
      <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
        <!--ro, opt, string, attr:opt{req, string}-->YYYY-MM-DD
      </dateStyle>
      <timeStyle opt="12hour,24hour">
        <!--ro, opt, string, attr:opt{req, string}-->12hour
      </timeStyle>
      <nonMotoPassengerNumEnabled opt="true,false">
        <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,60}, attr:opt{req, string}-->true
      </nonMotoPassengerNumEnabled>
    </PictureInfo>
  </PictureInfoList>
  <multiCharPosition opt="0,1,2">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->0
  </multiCharPosition>
</CapturePicOverlays>
```

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### 14.5.1.45 Set the configuration parameters for overlaying all OSD text information on a captured picture

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_OverlayInfo, subType:object, attr:version{opt, string, protocolVersion}-->
  <OverlayInfo>
    <!--opt, object, overlay information-->
    <overlayInfoID>
      <!--req, int, No., range:[1,50]-->1
    </overlayInfoID>
    <itemType>
      <!--req, enum, the type of overlaid information on a picture, subType:int, desc:the type of overlaid information on a picture-->70
    </itemType>
    <itemOverlayEnabled>
      <!--req, bool-->true
    </itemOverlayEnabled>
    <customName>
      <!--req, string, self-defined name of the overlay information, range:[1,32]-->test
    </customName>
    <changeLineNum>
      <!--req, int, range:[0,100]-->2
    </changeLineNum>
    <spaceNum>
      <!--req, int, range:[0,255]-->1
    </spaceNum>
    <startPosEnable>
      <!--opt, bool, whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.5.1.46 Get the configuration parameters of overlaying all OSD text information on a captured picture

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, list of overlay information, subType:object, attr:version{opt, string, protocolVersion}-->
  <OverlayInfo>
    <!--ro, opt, object, overlay information-->
    <overlayInfoID>
      <!--ro, opt, int, range:[1,50]-->1
    </overlayInfoID>
    <itemType>
      <!--ro, req, enum, subType:int-->70
    </itemType>
    <itemOverlayEnabled>
      <!--ro, req, bool-->true
    </itemOverlayEnabled>
    <customName>
      <!--ro, req, string, self-defined name of the overlay information, range:[1,32]-->test
    </customName>
    <changeLineNum>
      <!--ro, req, int, range:[0,100]-->2
    </changeLineNum>
    <spaceNum>
      <!--ro, req, int, range:[0,255]-->1
    </spaceNum>
    <startPosEnable>
      <!--ro, opt, bool, whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--ro, opt, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--ro, opt, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--ro, opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--ro, opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

### 14.5.1.47 Get the configuration parameters of overlaying a piece of OSD text information on a captured picture

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info/<textID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| textID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, overlay information, attr:version{opt, string, protocolVersion}-->
  <overlayInfoID>
    <!--ro, opt, int, No., range:[1,50]-->1
  </overlayInfoID>
  <itemType>
    <!--ro, req, enum, overlayed information type, subType:int, desc:overlayed information type-->70
  </itemType>
  <itemOverlayEnabled>
    <!--ro, req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--ro, req, string, self-defined name of the overlay information, range:[1,32]-->test
  </customName>
  <changeLineNum>
    <!--ro, req, int, number of line feeds, range:[0,100]-->2
  </changeLineNum>
  <spaceNum>
    <!--ro, req, int, number of spaces, range:[0,255]-->1
  </spaceNum>
  <startPosEnable>
    <!--ro, opt, bool, whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--ro, opt, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--ro, opt, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--ro, opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--ro, opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
  </overlayInfoText2>
</OverlayInfo>
```

#### 14.5.1.48 Set the configuration parameters for overlaying a

piece of OSD text information on a captured picture **Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info/<textID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| textID | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, overlay information, attr:version{opt, string, protocolVersion}-->
  <overlayInfoID>
    <!--req, int, overlayed information ID, range:[1,50]-->1
  </overlayInfoID>
  <itemType>
    <!--req, enum, overlayed information type, subType:int, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line,etc.), 18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30/31/32-reserved, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-congestion-->70
  </itemType>
  <itemOverlayEnabled>
    <!--req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--req, string, custom overlaying name, range:[1,32]-->test
  </customName>
  <changeLineNum>
    <!--req, int, number of line feeds, range:[0,100]-->2
  </changeLineNum>
  <spaceNum>
    <!--req, int, number of spaces, range:[0,255]-->1
  </spaceNum>
  <startPosEnable>
    <!--opt, bool, whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--req, int, start top coordinate, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--req, int, start left coordinate, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--opt, string, overlay character string for camera 2, range:[1,32], desc:overlay character string for camera 2-->test
  </overlayInfoText2>
</OverlayInfo>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.5.1.49 Get overlay parameters of captured traffic pictures by channel

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays?multiCharPosition=<multiCharPosition>

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| multiCharPosition | enum | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_OverlayInfoList, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable text overlay, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--ro, opt, array, subType:object-->
    <OverlayInfo>
      <!--ro, opt, object-->
      <overlayInfoID>
        <!--ro, req, int-->1
      </overlayInfoID>
      <itemType>
        <!--ro, req, enum, overlay mode, subType:int, desc:overlay mode-->70
      </itemType>
      <itemOverlayEnabled>
        <!--ro, req, bool-->true
      </itemOverlayEnabled>
      <customName>
        <!--ro, req, string, self-defined name of the overlay information-->test
      </customName>
      <changeLineNum>
        <!--ro, req, int, range:[0,100]-->2
      </changeLineNum>
      <spaceNum>
        <!--ro, req, int, space number, range:[0,255]-->1
      </spaceNum>
      <startPosEnable>
        <!--ro, opt, bool, whether to enable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point-->0
      </startPosLeft>
      <overlayInfoText>
        <!--ro, opt, string, overlay character string, range:[0,128]-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--ro, opt, string, overlay character string, range:[0,32]-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--ro, req, int, percentage of overlaying lines, range:[0,100]-->1
  </linePercent>
  <itemsStlye>
    <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal", "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--ro, req, enum, font type, subType:string, desc:font type: 0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--ro, req, enum, character size, subType:int, desc:0 (8*16), 1 (16*32), 2 (48*48), 3 (32*64), 4 (64*128)-->0
  </charSize>
  <charPosition>
    <!--ro, req, enum, text position overlaid on the picture, subType:int, desc:0 (in the picture), 1 (outside the top edge of the picture), 2 (outside the
bottom edge of the picture)-->1
  </charPosition>
  <charInterval>
    <!--ro, req, int, character separation distance, range:[0,16]-->0
  </charInterval>
  <foreColor>
    <!--ro, req, int, foreground color, range:[0,16777215]-->1
  </foreColor>
  <backColorEnabled>
    <!--ro, opt, bool-->true
  </backColorEnabled>
  <backColor>
    <!--ro, opt, int, background color, range:[0,16777215], desc:background color-->1
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
```

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```
<zeroizeEnable>
  <!--ro, opt, bool, whether to enable zero filling for OSD overlay-->true
</zeroizeEnable>
<snapTimeType>
  <!--ro, opt, enum, subType:string-->generationTime
</snapTimeType>
<charColorReverseEnabled>
  <!--ro, opt, bool-->true
</charColorReverseEnabled>
<platePicOverlay>
  <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
</platePicOverlay>
<platePicOverlay2>
  <!--ro, opt, bool-->true
</platePicOverlay2>
<platePicOverlay3>
  <!--ro, opt, bool-->true
</platePicOverlay3>
<platePicOverlayFeature>
  <!--ro, opt, bool-->true
</platePicOverlayFeature>
<platePicOverlayMultiplier>
  <!--ro, opt, int-->1
</platePicOverlayMultiplier>
<platePicPosTop>
  <!--ro, req, int, start top coordinate-->0
</platePicPosTop>
<platePicPosLeft>
  <!--ro, req, int, start left coordinate-->0
</platePicPosLeft>
<posPercentMode>
  <!--ro, opt, bool, whether to enable percentage mode instead of pixel value mode-->true
</posPercentMode>
<startPosTopPercent>
  <!--ro, opt, int, this node is valid when <posPercentMode> is "true"-->0
</startPosTopPercent>
<startPosLeftPercent>
  <!--ro, opt, int, this node is valid when <posPercentMode> is "true"-->0
</startPosLeftPercent>
<itemsSpacesNumMode>
  <!--ro, opt, bool-->true
</itemsSpacesNumMode>
<itemsSpacesNum>
  <!--ro, opt, int, number of spaces-->1
</itemsSpacesNum>
<colOverDetModEnabled>
  <!--ro, opt, bool-->true
</colOverDetModEnabled>
<overlaidTextTransparency>
  <!--ro, opt, int, range:[0,100], unit:%-->1
</overlaidTextTransparency>
<featurePicOSDEnable>
  <!--ro, opt, bool-->true
</featurePicOSDEnable>
<dateStyle>
  <!--ro, opt, enum, subType:string-->YYYY-MM-DD
</dateStyle>
<timeStyle>
  <!--ro, opt, enum, subType:string-->12hour
</timeStyle>
<syncVideoGbOsdEnable>
  <!--ro, opt, bool-->true
</syncVideoGbOsdEnable>
<syncVideoGbOsd>
  <!--ro, opt, object-->
  <illegalType>
    <!--ro, opt, object-->
    <normalEnable>
      <!--ro, opt, bool-->true
    </normalEnable>
  </illegalType>
</syncVideoGbOsd>
<plateColorOverlayEnabled>
  <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,10}-->true
</plateColorOverlayEnabled>
<plateProvinceOverlayEnabled>
  <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,10}-->true
</plateProvinceOverlayEnabled>
<overSpeedPercentAccurateEnabled>
  <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,20}-->true
</overSpeedPercentAccurateEnabled>
<nonMotoPassengerNumEnabled>
  <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,60}-->true
</nonMotoPassengerNumEnabled>
<relateParkingNumEnabled>
  <!--ro, opt, bool-->true
</relateParkingNumEnabled>
<captureOverlayTargetFrame>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <overlayFrameColor>
```

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```
        <!--ro, opt, enum, subType:string-->red
      </overlayFrameColor>
      <overlayFrameLineWidth>
        <!--ro, opt, int, range:[1,10]-->1
      </overlayFrameLineWidth>
    </captureOverlayTargetFrame>
    <captureOverlayBerthFrame>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <overlayFrameColor>
        <!--ro, opt, enum, subType:string-->red
      </overlayFrameColor>
      <overlayFrameLineWidth>
        <!--ro, opt, int, range:[1,10]-->1
      </overlayFrameLineWidth>
    </captureOverlayBerthFrame>
    <prefixLanguage>
      <!--ro, opt, enum, subType:string-->English
    </prefixLanguage>
    <PictureInfoList>
      <!--ro, opt, array, subType:object-->
      <PictureInfo>
        <!--ro, opt, object-->
        <pictureType>
          <!--ro, req, enum, subType:string-->illegalPicture
        </pictureType>
        <overlayInfoEnabled>
          <!--ro, req, bool-->true
        </overlayInfoEnabled>
        <OverlayInfoList>
          <!--ro, opt, array, subType:object-->
          <OverlayInfo>
            <!--ro, opt, object-->
            <overlayInfoID>
              <!--ro, req, int-->1
            </overlayInfoID>
            <itemType>
              <!--ro, req, enum, overlay mode, subType:int, desc:overlay mode-->70
            </itemType>
            <itemOverlayEnabled>
              <!--ro, req, bool-->true
            </itemOverlayEnabled>
            <customName>
              <!--ro, req, string, self-defined name of the overlay information-->test
            </customName>
            <changeLineNum>
              <!--ro, req, int, range:[0,100]-->2
            </changeLineNum>
            <spaceNum>
              <!--ro, req, int, range:[0,255]-->1
            </spaceNum>
            <startPosEnable>
              <!--ro, opt, bool, whether to enable coordinate configuration-->true
            </startPosEnable>
            <startPosTop>
              <!--ro, req, int, top coordinate of the start point-->0
            </startPosTop>
            <startPosLeft>
              <!--ro, req, int, left coordinate of the start point-->0
            </startPosLeft>
            <overlayInfoText>
              <!--ro, opt, string, overlay character string, range:[0,128]-->test
            </overlayInfoText>
            <overlayInfoText2>
              <!--ro, opt, string, overlay character string, range:[0,32]-->test
            </overlayInfoText2>
          </OverlayInfo>
        </OverlayInfoList>
        <linePercent>
          <!--ro, req, int, range:[0,100]-->1
        </linePercent>
        <itemsStlye>
          <!--ro, req, enum, overlay mode, subType:string, desc:overlay mode-->horizontal
        </itemsStlye>
        <charStyle>
          <!--ro, req, enum, font type, subType:string, desc:font type-->song_type
        </charStyle>
        <charSize>
          <!--ro, req, enum, font size, subType:int, desc:font size-->0
        </charSize>
        <charPosition>
          <!--ro, req, enum, position of the text overlaid in the picture, subType:int, desc:0 (in the picture), 1 (outside the top edge of the picture), 2
(outside the bottom edge of the picture)-->1
        </charPosition>
        <charInterval>
          <!--ro, req, int, range:[0,16]-->0
        </charInterval>
        <foreColor>
          <!--ro, req, int, foreground color, range:[0,16777215]-->1
        </foreColor>
        <backColorEnabled>
          <!--ro, opt, bool-->true
```

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```
</backColorEnabled>
<backColor>
  <!--ro, opt, int, background color, range:[0,16777215], desc:background color-->1
</backColor>
<colorAdapt>
  <!--ro, opt, bool, whether to enable color self-adaption-->true
</colorAdapt>
<zeroizeEnable>
  <!--ro, opt, bool, whether to enable zero filling for OSD overlay-->true
</zeroizeEnable>
<snapTimeType>
  <!--ro, opt, enum, subType:string-->generationTime
</snapTimeType>
<charColorReverseEnabled>
  <!--ro, opt, bool-->true
</charColorReverseEnabled>
<platePicOverlay>
  <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
</platePicOverlay>
<platePicOverlay2>
  <!--ro, opt, bool-->true
</platePicOverlay2>
<platePicOverlay3>
  <!--ro, opt, bool-->true
</platePicOverlay3>
<platePicOverlayFeature>
  <!--ro, opt, bool-->true
</platePicOverlayFeature>
<platePicOverlayMultiplier>
  <!--ro, opt, int-->1
</platePicOverlayMultiplier>
<platePicPosTop>
  <!--ro, opt, int, top coordinate of the start point-->0
</platePicPosTop>
<platePicPosLeft>
  <!--ro, opt, int, left coordinate of the start point-->0
</platePicPosLeft>
<posPercentMode>
  <!--ro, opt, bool-->true
</posPercentMode>
<startPosTopPercent>
  <!--ro, opt, int-->0
</startPosTopPercent>
<startPosLeftPercent>
  <!--ro, opt, int-->0
</startPosLeftPercent>
<itemsSpacesNumMode>
  <!--ro, opt, bool-->true
</itemsSpacesNumMode>
<itemsSpacesNum>
  <!--ro, opt, int-->1
</itemsSpacesNum>
<colOverDetModEnabled>
  <!--ro, opt, bool-->true
</colOverDetModEnabled>
<overlaidTextTransparency>
  <!--ro, opt, int, range:[0,100], unit:%-->1
</overlaidTextTransparency>
<featurePicOSDEnable>
  <!--ro, opt, bool-->true
</featurePicOSDEnable>
<dateStyle>
  <!--ro, opt, enum, subType:string-->YYYY-MM-DD
</dateStyle>
<timeStyle>
  <!--ro, opt, enum, subType:string-->12hour
</timeStyle>
<syncVideoGbOsdEnable>
  <!--ro, opt, bool-->true
</syncVideoGbOsdEnable>
<syncVideoGbOsd>
  <!--ro, opt, object-->
  <illegalType>
    <!--ro, opt, object-->
    <normalEnable>
      <!--ro, opt, bool-->true
    </normalEnable>
    <illegalEnable>
      <!--ro, opt, bool-->true
    </illegalEnable>
  </illegalType>
  <targetType>
    <!--ro, opt, object-->
    <motorEnable>
      <!--ro, opt, bool-->true
    </motorEnable>
    <noMotorEnable>
      <!--ro, opt, bool-->true
    </noMotorEnable>
    <personEnable>
      <!--ro, opt, bool-->true
    </personEnable>
  </targetType>
```

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```
        </syncVideoGbOsd>
        <plateColorOverlayEnabled>
          <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,10}-->true
        </plateColorOverlayEnabled>
        <overSpeedPercentAccurateEnabled>
          <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,20}-->true
        </overSpeedPercentAccurateEnabled>
        <nonMotoPassengerNumEnabled>
          <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,60}-->true
        </nonMotoPassengerNumEnabled>
        <relateParkingNumEnabled>
          <!--ro, opt, bool-->true
        </relateParkingNumEnabled>
        <captureOverlayTargetFrame>
          <!--ro, opt, object-->
          <enabled>
            <!--ro, req, bool-->true
          </enabled>
          <overlayFrameColor>
            <!--ro, opt, enum, subType:string-->red
          </overlayFrameColor>
          <overlayFrameLineWidth>
            <!--ro, opt, int, range:[1,10]-->1
          </overlayFrameLineWidth>
        </captureOverlayTargetFrame>
        <captureOverlayBerthFrame>
          <!--ro, opt, object-->
          <enabled>
            <!--ro, req, bool-->true
          </enabled>
          <overlayFrameColor>
            <!--ro, opt, enum, subType:string-->red
          </overlayFrameColor>
          <overlayFrameLineWidth>
            <!--ro, opt, int, range:[1,10]-->1
          </overlayFrameLineWidth>
        </captureOverlayBerthFrame>
        <prefixLanguage>
          <!--ro, opt, enum, subType:string-->English
        </prefixLanguage>
      </PictureInfo>
    </PictureInfoList>
  </CapturePicOverlays>
```

#### 14.5.1.50 Set overlay parameters of captured traffic pictures by channel

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays?multiCharPosition=<multiCharPosition>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| multiCharPosition | enum | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, see details in the message of XML_OverlayInfoList, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--req, bool, whether to enable text overlay, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--opt, array, subType:object-->
    <OverlayInfo>
      <!--opt, object-->
      <overlayInfoID>
        <!--req, int-->1
      </overlayInfoID>
      <itemType>
        <!--req, enum, subType:int-->70
      </itemType>
      <itemOverlayEnabled>
        <!--req, bool-->true
      </itemOverlayEnabled>
      <customName>
        <!--req, string, self-defined name of the overlay information-->test
      </customName>
      <changeLineNum>
        <!--req, int, range:[0,100]-->2
      </changeLineNum>
      <spaceNum>
        <!--req, int, range:[0,255]-->1
      </spaceNum>
      <startPosEnable>
        <!--opt, bool, whether to enable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--req, int, left coordinate of the start point-->0
      </startPosLeft>
      <overlayInfoText>
        <!--opt, string, overlay character string-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--opt, string, overlay character string-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--req, int, percentage of overlaying lines, range:[0,100]-->1
  </linePercent>
  <itemsStlye>
    <!--req, enum, overlay mode, subType:string, desc:"horizontal" (default),"vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--req, enum, font type, subType:string, desc:0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--req, enum, character size, subType:int, desc:0-16*16(Chinese)/8*16(English),1-32*32(Chinese)/16*32(English),2-48*48,3-64*64(Chinese)/32*64(English)-
->0
  </charSize>
  <charPosition>
    <!--req, enum, text position overlayed on the picture, subType:int, desc:0 (in the picture),1 (outside the top edge of the picture), 2 (outside the
bottom edge of the picture)-->1
  </charPosition>
  <charInterval>
    <!--req, int, character separation distance, range:[0,16]-->0
  </charInterval>
  <colorAdapt>
    <!--opt, bool, whether to enable color self-adaption: 0-no,1-yes-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--opt, bool, whether to enable zero filling for OSD overlay, which is used to enable zero filling for vehicle speed, speed limit, overspeed ratio and
Lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
    <!--req, int, start top coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture height, and the
default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--req, int, start left coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture width, and the
default value is 0-->0
  </platePicPosLeft>
</CapturePicOverlays>
```

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| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| CapturePicOverlays | [Message content] | application/xml | -- | -- | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.5.1.51 Get the overlay capability in the composite captured pictures

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, composite picture overlay, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
  <OverlayInfoList size="50">
    <!--ro, req, array, list of overlay information, subType:object, attr:size{req, int}-->
    <OverlayInfo>
      <!--ro, req, object, overlay information-->
      <itemType opt="0,1,2,3,…,82,83,85">
        <!--ro, req, enum, overlay information type, subType:string, attr:opt{req, string}, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-
direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle
color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic
violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over
yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time,
24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow
label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making
a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box
detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed
```

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```
limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-
wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading
rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann
emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry &
exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license
plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-
reflective stripe-->0
      </itemType>
      <editAble opt="true,false">
        <!--ro, req, bool, whether the overlaid information is editable, attr:opt{req, string}-->true
      </editAble>
      <itemOverlayEnabled opt="true,false">
        <!--ro, req, bool, whether to overlay the item, attr:opt{req, string}-->true
      </itemOverlayEnabled>
      <customName min="0" max="32">
        <!--ro, req, string, custom overlaying name, range:[0,32], attr:min{req, int},max{req, int}, desc:the maximum length is 32 (including '\0'). If this
node is not configured, the default name will be overlaid-->test
      </customName>
      <changeLineNum min="0" max="100">
        <!--ro, req, int, number of line feeds, range:[0,100], attr:min{req, int},max{req, int}, desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum min="0" max="255">
        <!--ro, req, int, number of spaces, range:[0,255], attr:min{req, int},max{req, int}, desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable opt="true,false">
        <!--ro, opt, bool, whether to enable coordinate configuration, attr:opt{req, string}, desc:it is valid only for information overlay within the
picture. After enabled, the function of line feed and space will be invalid. You are recommended to disable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
      </startPosLeft>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent min="0" max="100">
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], attr:min{req, int},max{req, int}, desc:the default value is 100-->1
  </linePercent>
  <itemsStlye opt="horizontal,vertical">
    <!--ro, req, enum, overlay mode, subType:string, attr:opt{req, string}, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle opt="song_type,wei_type">
    <!--ro, req, enum, font type, subType:string, attr:opt{req, string}, desc:"song_type"-SimSun (default), "wei_type"-Wei-->song_type
  </charStyle>
  <charSize min="0" max="5">
    <!--ro, req, enum, character size, subType:string, attr:min{req, int},max{req, int}, desc:"0"-16*16(Chinese)/8*16(English), "1"-
32*32(Chinese)/16*32(English), "2"-48*48, "3"-64*64(Chinese)/32*64(English), "4"-128*128(Chinese)/64*128(English), "5"-96*96(Chinese)/48*96(English)-->0
  </charSize>
  <charPosition min="0" max="3">
    <!--ro, req, enum, position of the text overlaid on the picture, subType:int, attr:min{req, int},max{req, int}, desc:0 (in the picture),1 (outside the
top edge of the picture), 2 (outside the bottom edge of the picture)-->0
  </charPosition>
  <charInterval min="0" max="16">
    <!--ro, req, int, character pitch, range:[0,16], attr:min{req, int},max{req, int}, desc:character separation distance, which is between 0 and 16 and the
default value is 0, unit: pixel-->0
  </charInterval>
  <foreColor min="0" max="0xffffff">
    <!--ro, req, int, foreground color, attr:min{req, int},max{req, int}, desc:it is the RGB value directly obtained via the palette and its default value
is 0xffffff (white)-->0
  </foreColor>
  <backColor min="0" max="0xffffff">
    <!--ro, req, int, background color, attr:min{req, int},max{req, int}, desc:it is the RGB value directly obtained via the palette and its default value
is 0x0 (black)-->0
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <PictureInfoList>
    <!--ro, opt, array, overlay information list of other captured picture types, subType:object, desc:non-detection picture (background picture)-->
    <PictureInfo>
      <!--ro, opt, object, picture information-->
      <pictureType>
        <!--ro, req, enum, picture type, subType:string, desc:"illegalPicture" (violation capture, which corresponds to "compositePicture" returned by ANPR
for violation occurrence); whether violation occurred can be indicated by whether "illegalInfo" is returned by ANPR-->illegalPicture
      </pictureType>
      <overlayInfoEnabled>
        <!--ro, req, bool, whether to enable the function-->true
      </overlayInfoEnabled>
      <OverlayInfoList size="50">
        <!--ro, req, array, list of overlay information, subType:object, attr:size{req, int}-->
        <OverlayInfo>
          <!--ro, req, object, overlay information-->
          <itemType opt="0,1,2,3,…,82,83,85">
            <!--ro, req, enum, overlaid information type, subType:string, attr:opt{req, string}, desc:0-unknown, 1-place, 2-intersection No., 3-device No.,
4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-
vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code
(traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving
over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light
time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-
yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings,
```

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```
38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
          </itemType>
          <editAble opt="true,false">
            <!--ro, req, bool, whether the overlaid information is editable, attr:opt{req, string}-->true
          </editAble>
          <itemOverlayEnabled opt="true,false">
            <!--ro, req, bool, whether to overlay the information, attr:opt{req, string}-->true
          </itemOverlayEnabled>
          <customName min="0" max="32">
            <!--ro, req, string, custom overlaying name, range:[0,32], attr:min{req, int},max{req, int}, desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
          </customName>
          <changeLineNum min="0" max="100">
            <!--ro, req, int, number of line feeds, range:[0,100], attr:min{req, int},max{req, int}, desc:the default value is 0-->0
          </changeLineNum>
          <spaceNum min="0" max="255">
            <!--ro, req, int, number of spaces, range:[0,255], attr:min{req, int},max{req, int}, desc:the default value is 0-->0
          </spaceNum>
          <startPosEnable opt="true,false">
            <!--ro, opt, bool, whether to enable coordinate configuration, attr:opt{req, string}, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
          </startPosEnable>
          <startPosTop>
            <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
          </startPosTop>
          <startPosLeft>
            <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
          </startPosLeft>
        </OverlayInfo>
      </OverlayInfoList>
      <linePercent min="0" max="100">
        <!--ro, req, int, overlay line percentage, range:[0,100], attr:min{req, int},max{req, int}, desc:100 by default-->1
      </linePercent>
      <itemsStlye opt="horizontal,vertical">
        <!--ro, req, enum, overlay mode, subType:string, attr:opt{req, string}, desc:"horizontal" (default), "vertical"-->horizontal
      </itemsStlye>
      <charStyle opt="song_type,wei_type">
        <!--ro, req, enum, font type, subType:string, attr:opt{req, string}, desc:“song_type”-SimSun (default), “wei_type”-Wei-->song_type
      </charStyle>
      <charSize min="0" max="4">
        <!--ro, req, enum, character size, subType:string, attr:min{req, int},max{req, int}, desc:0 (8*16), 1 (16*32), 2 (48*48), 3 (32*64), 4 (64*128)-->0
      </charSize>
      <charPosition min="0" max="2">
        <!--ro, req, enum, position of the text overlaid in the picture, subType:int, attr:min{req, int},max{req, int}, desc:0 (in the picture),1 (outside the top edge of the picture), 2 (outside the bottom edge of the picture)-->0
      </charPosition>
      <charInterval min="0" max="16">
        <!--ro, req, int, character pitch, range:[0,16], attr:min{req, int},max{req, int}, desc:the default value is 0, unit: pixel-->0
      </charInterval>
      <foreColor min="0" max="0xffffff">
        <!--ro, req, int, foreground color, attr:min{req, int},max{req, int}, desc:it is the RGB value directly obtained via the palette and its default value is 0xffffff (white)-->0
      </foreColor>
      <backColor min="0" max="0xffffff">
        <!--ro, req, int, background color, attr:min{req, int},max{req, int}, desc:it is the RGB value directly obtained via the palette and its default value is 0x0 (black)-->0
      </backColor>
      <colorAdapt>
        <!--ro, opt, bool, whether to enable color self-adaption-->true
      </colorAdapt>
    </PictureInfo>
  </PictureInfoList>
</MergePicOverlays>
```

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### 14.5.1.52 Set the configuration parameters for overlaying all OSD text information on the composite captured picture

#### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

## Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, list of overlay information, subType:object, attr:version{req, string, protocolVersion}-->
  <OverlayInfo>
    <!--req, object, overlay information-->
    <overlayInfoID>
      <!--req, int, overlay information ID-->0
    </overlayInfoID>
    <itemType>
      <!--req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for parking space patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
    </itemType>
    <itemOverlayEnabled>
      <!--req, bool, whether to overlay the information-->true
    </itemOverlayEnabled>
    <customName>
      <!--req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
    </customName>
    <changeLineNum>
      <!--req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
    </changeLineNum>
    <spaceNum>
      <!--req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
    </spaceNum>
    <startPosEnable>
      <!--opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
    </startPosEnable>
    <startPosTop>
      <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

## Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

#### 14.5.1.53 Get parameters of overlaying characters on composite pictures

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, overlay information list, subType:object, attr:version{req, string, protocolVersion}-->
  <OverlayInfo>
    <!--ro, req, object, overlay information-->
    <overlayInfoID>
      <!--ro, req, int, overlay ID-->0
    </overlayInfoID>
    <itemType>
      <!--ro, req, enum, overlaid information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for parking space patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
    </itemType>
    <itemOverlayEnabled>
      <!--ro, req, bool, whether to overlay this item-->true
    </itemOverlayEnabled>
    <customName>
      <!--ro, req, string, custom overlaying name, range:[0,32], desc:customize name of the overlay information-->test
    </customName>
    <changeLineNum>
      <!--ro, req, int, the default value is 0, range:[0,100], desc:the default value is 0-->0
    </changeLineNum>
    <spaceNum>
      <!--ro, req, int, the number of spaces, range:[0,255], desc:the default value is 0-->0
    </spaceNum>
    <startPosEnable>
      <!--ro, opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--ro, opt, string, overlay characters, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

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### 14.5.1.54 Set the configuration parameters for overlaying a piece of OSD text information on the composite captured picture

#### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info/<textID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| textID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, overlay information, attr:version{req, string, protocolVersion}-->
  <overlayInfoID>
    <!--req, int, overlayed information ID-->0
  </overlayInfoID>
  <itemType>
    <!--req, enum, overlay information type, subType:string, desc:0-unknown,1-place,2-intersection No.,3-device No.,4-direction No.,5-direction,6-lane No.,7-lane,8-capture time (without millisecond),9-capture time (with millisecond),10-license plate number,11-vehicle color,12-vehicle type,13-vehicle brand,14-vehicle speed,15-speed limit sign,16-vehicle length (between 1 and 99 meters),17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information,19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30/31/32-reserved, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection,35-vehicle sub-brand detection, 36-vehicle direction,37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-congestion-->0
  </itemType>
  <itemOverlayEnabled>
    <!--req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--req, string, custom overlaying name, range:[0,32], desc:self-defined name of the overlay information-->test
  </customName>
  <changeLineNum>
    <!--req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
  </changeLineNum>
  <spaceNum>
    <!--req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
  </spaceNum>
  <startPosEnable>
    <!--opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
  </overlayInfoText2>
</OverlayInfo>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

## 14.5.1.55 Get the configuration parameters of overlaying a piece of OSD text information on the composite captured picture

### Request URL

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GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info/<textID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| textID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, overlay information, attr:version{req, string, protocolVersion}-->
  <overlayInfoID>
    <!--ro, req, int, overlayed information ID, which is between 1 and 50-->0
  </overlayInfoID>
  <itemType>
    <!--ro, req, enum, overlayed information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
  </itemType>
  <itemOverlayEnabled>
    <!--ro, req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--ro, req, string, custom overlaying name, range:[0,32], desc:self-defined name of the overlay information-->test
  </customName>
  <changeLineNum>
    <!--ro, req, int, number of line feeds, and the default value is 0, range:[0,100], desc:the default value is 0-->0
  </changeLineNum>
  <spaceNum>
    <!--ro, req, int, number of spaces, and the default value is 0, range:[0,255], desc:the default value is 0-->0
  </spaceNum>
  <startPosEnable>
    <!--ro, opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--ro, opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
  </overlayInfoText2>
</OverlayInfo>
```

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### 14.5.1.56 Set the overlay parameters in the composite captured pictures

### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays?multiCharPosition=<multiCharPosition>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| multiCharPosition | enum | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, see details in the message of XML_OverlayInfoList, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--req, array, list of overlay information, subType:object-->
    <OverlayInfo>
      <!--req, object, overlay information-->
      <overlayInfoID>
        <!--req, int, overlay ID-->0
      </overlayInfoID>
      <itemType>
        <!--req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
      </itemType>
      <itemOverlayEnabled>
        <!--req, bool, whether to overlay the information-->true
      </itemOverlayEnabled>
      <customName>
        <!--req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
      </customName>
      <changeLineNum>
        <!--req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum>
        <!--req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable>
        <!--opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
      </startPosEnable>
      <startPosTop>
        <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </startPosLeft>
      <overlayInfoText>
        <!--opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--req, int, percentage of overlaying lines, range:[0,100], desc:the default value is 100-->1
  </linePercent>
  <itemsStlye>
    <!--req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--req, enum, font type, subType:string, desc:font type: 0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--req, enum, character size, subType:string, desc:0 (8*16), 1 (16*32), 2 (48*48), 3 (32*64), 4 (64*128)-->0
  </charSize>
  <charPosition>
    <!--req, enum, position of the text overlaid in the picture, subType:int, desc:0 (in the picture),1 (outside the top edge of the picture), 2 (outside
```

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```
the bottom edge of the picture)-->0
  </charPosition>
  <charInterval>
    <!--req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
  </charInterval>
  <colorAdapt>
    <!--opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio,
and lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
    <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height,
and the default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width,
and the default value is 0-->0
  </platePicPosLeft>
</MergePicOverlays>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

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## 14.5.1.57 Get the parameters of OSD overlay on the composite captured picture

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays?multiCharPosition=<multiCharPosition>

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| multiCharPosition | enum | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, see details in the message of XML_OverlayInfoList, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--ro, req, array, list of overlay information, subType:object-->
    <OverlayInfo>
      <!--ro, req, object, overlay information-->
      <overlayInfoID>
```

```
        <!--ro, req, int, overlay ID-->0
      </overlayInfoID>
      <itemType>
        <!--ro, req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
      </itemType>
      <itemOverlayEnabled>
        <!--ro, req, bool, whether to overlay the information-->true
      </itemOverlayEnabled>
      <customName>
        <!--ro, req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
      </customName>
      <changeLineNum>
        <!--ro, req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum>
        <!--ro, req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable>
        <!--ro, opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </startPosLeft>
      <overlayInfoText>
        <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--ro, opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], desc:the default value is 100-->1
  </linePercent>
  <itemsStlye>
    <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--ro, req, enum, font type, subType:string, desc:“song_type”-SimSun (default), “wei_type”-Wei-->song_type
  </charStyle>
  <charSize>
    <!--ro, req, enum, font size, subType:string, desc:“0”-16*16(Chinese)/8*16(English), “1”-32*32(Chinese)/16*32(English), “2”-48*48, “3”-64*64(Chinese)/32*64(English)-->0
  </charSize>
  <charPosition>
    <!--ro, req, enum, position of the text overlaid in the picture, subType:int, desc:“0”-overlaid on the picture, "1”-overlaid outside the top edge of the picture, “2”-overlaid outside the bottom edge of the picture-->0
  </charPosition>
  <charInterval>
    <!--ro, req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
  </charInterval>
  <foreColor>
    <!--ro, req, int, foreground color, desc:it is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value is 0xffffff (white)-->0
  </foreColor>
  <backColor>
    <!--ro, req, int, background color, desc:it is the RGB value directly obtained via the palette and its default value is 0x0 (black)-->0
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--ro, opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio, and lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
```

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```
  <platePicPosTop>
    <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </platePicPosLeft>
  <PictureInfoList>
    <!--ro, opt, array, overlay information list of other captured picture types, subType:object, desc:non-detection picture (background picture)-->
    <PictureInfo>
      <!--ro, opt, object, picture information-->
      <pictureType>
        <!--ro, req, enum, picture type, subType:string, desc:"illegalPicture" (violation capture, which corresponds to "compositePicture" returned by ANPR for violation occurrence); whether violation occurred can be indicated by whether "illegalInfo" is returned by ANPR-->illegalPicture
      </pictureType>
      <overlayInfoEnabled>
        <!--ro, req, bool, whether to enable the function-->true
      </overlayInfoEnabled>
      <OverlayInfoList>
        <!--ro, req, array, list of overlay information, subType:object-->
        <OverlayInfo>
          <!--ro, req, object, overlay information-->
          <overlayInfoID>
            <!--ro, req, int, overlay ID-->0
          </overlayInfoID>
          <itemType>
            <!--ro, req, enum, overlaid information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-Lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying Lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
          </itemType>
          <itemOverlayEnabled>
            <!--ro, req, bool, whether to overlay the information-->true
          </itemOverlayEnabled>
          <customName>
            <!--ro, req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
          </customName>
          <changeLineNum>
            <!--ro, req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
          </changeLineNum>
          <spaceNum>
            <!--ro, req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
          </spaceNum>
          <startPosEnable>
            <!--ro, opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
          </startPosEnable>
          <startPosTop>
            <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
          </startPosTop>
          <startPosLeft>
            <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
          </startPosLeft>
          <overlayInfoText>
            <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, Lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
          </overlayInfoText>
          <overlayInfoText2>
            <!--ro, opt, string, overlay character string, range:[1,32], desc:for handling camera 2 when the overlay information type is camera information; the maximum length of camera 2 is 32 bytes-->test
          </overlayInfoText2>
        </OverlayInfo>
      </OverlayInfoList>
      <linePercent>
        <!--ro, req, int, overlay line percentage, range:[0,100], desc:the default value is 100-->1
      </linePercent>
      <itemsStlye>
        <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
      </itemsStlye>
      <charStyle>
        <!--ro, req, enum, font type, subType:string, desc:“song_type”-SimSun (default), “wei_type”-Wei-->song_type
      </charStyle>
      <charSize>
        <!--ro, req, enum, font size, subType:string, desc:“0”-16*16(Chinese)/8*16(English), “1”-32*32(Chinese)/16*32(English), “2”-48*48, “3”-64*64(Chinese)/32*64(English)-->0
```

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```
      </charSize>
      <charPosition>
        <!--ro, req, enum, position of the text overlaid in the picture, subType:int, desc:“0”-overlaid on the picture, "1”-overlaid outside the top edge of the picture, “2”-overlaid outside the bottom edge of the picture-->0
      </charPosition>
      <charInterval>
        <!--ro, req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
      </charInterval>
      <foreColor>
        <!--ro, req, int, foreground color, desc:it is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value is 0xffffff (white)-->0
      </foreColor>
      <backColor>
        <!--ro, req, int, background color, desc:it is the RGB value directly obtained via the palette and its default value is 0x0 (black)-->0
      </backColor>
      <colorAdapt>
        <!--ro, opt, bool, whether to enable color self-adaption-->true
      </colorAdapt>
      <zeroizeEnable>
        <!--ro, opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio, and lane No. Zero filling is enabled by default-->true
      </zeroizeEnable>
      <platePicOverlay>
        <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
      </platePicOverlay>
      <platePicPosTop>
        <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </platePicPosTop>
      <platePicPosLeft>
        <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </platePicPosLeft>
    </PictureInfo>
  </PictureInfoList>
</MergePicOverlays>
```

### 14.5.1.58 Set vehicle detection parameters

#### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | - |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
</VehicleDetectCfg>
```

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.5.1.59 Get vehicle detection parameters

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**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
</VehicleDetectCfg>
```

### 14.5.1.60 Get the capability of vehicle detection parameters

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</VehicleDetectCfg>
```

### 14.5.1.61 Get the recognition result of the license plate captured manually

**Request URL**

GET /ISAPI/Traffic/MNPR/channels/<channelID>?laneNo=<laneNo>&OSD=<OSDType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| laneNo | string | lane No. |
| OSDType | string | whether to enable OSD overlay for manual capture: enabled by default; not returned and not enabled when is 1 |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, req, enum, protocol type, subType:string, desc:transport protocol type: HTTP,HTTPS,EHome-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventType>
  <eventState>
    <!--ro, req, enum, continuous event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, enum, event description, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <ANPR>
    <!--ro, opt, object, information about license plate recognition alarm-->
    <region>
      <!--ro, req, enum, region, subType:string, desc:region-->ER
    </region>
    <country>
      <!--ro, req, enum, country or region, subType:int, desc:country or region-->253
    </country>
    <area>
      <!--ro, req, enum, province or state, subType:string, desc:province or state-->FJR
    </area>
    <licensePlate>
      <!--ro, req, enum, license plate number, subType:string, desc:license plate number,e.g.,"123456"-->noPlate
    </licensePlate>
    <line>
      <!--ro, req, int, recognized lane No.-->1
    </line>
    <direction>
      <!--ro, req, enum, target direction, subType:string, desc:"reverse", "forward", "unknown"-->reverse
    </direction>
    <confidenceLevel>
      <!--ro, req, int, confidence level, range:[0,100]-->50
    </confidenceLevel>
    <plateType>
      <!--ro, req, enum, license plate type, subType:string, desc:license plate type: "unknown", "arm"-police vehicle, "embassy"-embassy vehicle,
"motorola"-motorcycle, "coach"-driver-training vehicle, "tempTravl"-vehicle with temporary license plate, "trailer"-trailer, "consulate"-consular vehicle,
"tempEntry"-temporary vehicle, "newEnergy"-new energy vehicle-->unknown
    </plateType>
    <plateColor>
      <!--ro, req, enum, license plate color, subType:string, desc:license plate color: "white","yellow","blue","black","green","newEnergyGreen"-new energy
green,"newEnergyYellowGreen"-new energy flavogreen,"other"-other color-->black
    </plateColor>
    <licenseBright>
      <!--ro, opt, int, license plate brightness, which ranges from 0 to 255, range:[0,255]-->50
    </licenseBright>
    <Rect>
      <!--ro, opt, object, coordinates of the license plate thumbnail in the matched picture, desc:coordinates of the license plate thumbnail in the matched
picture-->
      <height>
        <!--ro, req, float, height, range:[0.000,1.000]-->1.000
      </height>
      <width>
        <!--ro, req, float, width, range:[0.000,1.000]-->1.000
      </width>
      <x>
        <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
      </x>
      <y>
        <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
      </y>
```

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```
      </y>
    </Rect>
    <pilotsafebelt>
      <!--ro, opt, enum, whether the driver is wearing a safety belt, subType:string, desc:whether the driver is wearing safety belt: "unknown,yes,no"--
>unknown
    </pilotsafebelt>
    <vicepilotsafebelt>
      <!--ro, opt, string, whether the co-driver is wearing a safety belt-->unknown
    </vicepilotsafebelt>
    <pilotsunvisor>
      <!--ro, opt, string, whether the driver room's sun visor is open-->unknown
    </pilotsunvisor>
    <vicepilotsunvisor>
      <!--ro, opt, string, whether the co-driver room's sun visor is open-->unknown
    </vicepilotsunvisor>
    <envprosign>
      <!--ro, req, enum, whether it is a yellow-label vehicle, subType:string, desc:"unknown" (unknown), "green" (green label), "yellow" (yellow label)--
>unknown
    </envprosign>
    <dangmark>
      <!--ro, req, enum, whether it is a dangerous goods vehicle, subType:string, desc:"unknown", "yes", "no"-->unknown
    </dangmark>
    <uphone>
      <!--ro, req, enum, whether the driver is making a phone call, subType:string, desc:"unknown", "yes", "no"-->unknown
    </uphone>
    <pendant>
      <!--ro, req, enum, whether there are window hangings detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pendant>
    <tissueBox>
      <!--ro, req, enum, whether there is a tissue box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </tissueBox>
    <frontChild>
      <!--ro, req, enum, whether the co-driver is with a baby in arm, subType:string, desc:"unknown", "yes", "no"-->unknown
    </frontChild>
    <label>
      <!--ro, req, enum, whether there are stickers detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </label>
    <decoration>
      <!--ro, req, enum, whether there are decorations detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </decoration>
    <smoking>
      <!--ro, req, enum, whether anyone is smoking, subType:string, desc:whether there is smoking detected: "unknown,yes,no"-->yes
    </smoking>
    <perfumeBox>
      <!--ro, opt, enum, whether there is perfume box detected, subType:string, desc:whether there is perfume box detected: "unknown,yes,no"-->unknown
    </perfumeBox>
    <pdvs>
      <!--ro, req, enum, whether there is a person sticking out of sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pdvs>
    <helmet>
      <!--ro, req, enum, whether the non-motor vehicle driver wears a helmet, subType:string, desc:"unknown", "yes", "no"-->no
    </helmet>
    <twoWheelVehicle>
      <!--ro, opt, enum, unknown, subType:string, desc:whether there is two-wheel detected: "unknown,yes,no"-->unknown
    </twoWheelVehicle>
    <threeWheelVehicle>
      <!--ro, opt, enum, whether there is three-wheel detected, subType:string, desc:whether there is three-wheel detected: "unknown,yes,no"-->unknown
    </threeWheelVehicle>
    <blackness>
      <!--ro, opt, int, Ringelmann emittance, desc:it is used for black smoke detection-->2
    </blackness>
    <plateCharBelieve>
      <!--ro, opt, string, confidence of each character in the recognized license plate, desc:confidence of each character in the recognized license plate--
>test
    </plateCharBelieve>
    <speedLimit>
      <!--ro, opt, int, upper speed limit, desc:it is valid only when overspeeding occurred-->50
    </speedLimit>
    <illegalInfo>
      <!--ro, opt, object, violation information-->
      <illegalCode>
        <!--ro, req, string, violation code, range:[0,64]-->1101
      </illegalCode>
      <illegalName>
        <!--ro, req, string, violation name, range:[0,128]-->test
      </illegalName>
      <illegalDescription>
        <!--ro, opt, string, violation description, range:[0,256]-->test
      </illegalDescription>
    </illegalInfo>
    <vehicleType>
      <!--ro, req, enum, vehicle type, subType:string, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->nonmotorVehicle
    </vehicleType>
    <postPicFileName>
      <!--ro, opt, string, name of the picture selected as the checkpoint picture when illegal action occurs, desc:name of the picture selected as the
checkpoint picture when illegal action occurs, "none" refers to not selecting any picture-->test
    </postPicFileName>
    <featurePicFileName>
      <!--ro, opt, string, red light running capture, desc:name of the picture selected as the close-up picture when running the red light in the
intersection violation system is detected, "none" refers to not selecting any picture-->test
    </featurePicFileName>
    <detectDir>
```

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```
      <!--ro, req, enum, detection direction, subType:int, desc:1 (upward), 2 (downward), 3 (bidirection), 4 (from east to west), 5 (from south to north), 6 (from west to east), 7 (from north to south), 8 (other)-->1
    </detectDir>
    <detectType>
      <!--ro, req, enum, detection type, subType:int, desc:1 (inductive loop trigger), 2 (video trigger), 3 (multiple-frame recognition), 4 (radar trigger)-->1
    </detectType>
    <barrierGateCtrlType>
      <!--ro, opt, enum, whether to enable elapsed time, subType:int, desc:0 (enabled), 1 (disabled)-->0
    </barrierGateCtrlType>
    <alarmDataType>
      <!--ro, opt, enum, alarm data type (real-time data or history data), subType:int, desc:1 (history data), 0 (real-time data)-->0
    </alarmDataType>
    <dwIllegalTime>
      <!--ro, opt, int, illegal action duration, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture-->100
    </dwIllegalTime>
    <vehicleInfo>
      <!--ro, opt, object, vehicle information-->
      <index>
        <!--ro, req, int, vehicle No.-->1
      </index>
      <vehicleType>
        <!--ro, req, enum, vehicle type, subType:int, desc:vehicle type: "unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickupTruck"-->0
      </vehicleType>
      <colorDepth>
        <!--ro, req, enum, shade of the vehicle color, subType:int, desc:0 (deep color), 1 (light color), 2 (unknown)-->0
      </colorDepth>
      <color>
        <!--ro, req, enum, vehicle color, subType:string, desc:"green", "brown", "pink", "purple", "deepGray" (dark gray), "cyan", "orange", "white", "silver" (silvery), "gray", "blacks", "red", "deepBlue" (dark blue), "blue", "yellow"-->green
      </color>
      <speed>
        <!--ro, req, int, target speed, unit:km/h-->1
      </speed>
      <length>
        <!--ro, req, int, length of the former vehicle, unit:dm-->10
      </length>
      <vehicleLogoRecog>
        <!--ro, req, int, vehicle main brand-->1
      </vehicleLogoRecog>
      <vehileSubLogoRecog>
        <!--ro, opt, int, vehicle sub-brand-->1
      </vehileSubLogoRecog>
      <vehileModel>
        <!--ro, opt, int, model year of vehicle sub-brand-->1
      </vehileModel>
      <vehicleTypeByWeight>
        <!--ro, opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1-class one vehicle (buses with seven or less seats,trucks with capacity of 2 tons or less), 2-class two vehicle (buses with 8 to 19 seats,trucks with capacity of 2 to 5 (included) tons), 3-class three vehicle (buses with 20 to 39 seats,trucks with capacity of 5 to 10 (included) tons), 4-class four vehicle (buses with 40 or more seats,trucks with capacity of 10 to 15 (included) tons), 5-class five vehicle (trucks with capacity of more than 15 tons)-->4
      </vehicleTypeByWeight>
    </vehicleInfo>
    <EntranceInfo>
      <!--ro, opt, object, entrance and exit information-->
      <parkingID>
        <!--ro, opt, string, parking space No.-->test
      </parkingID>
      <gateID>
        <!--ro, opt, string, entrance and exit No.-->test
      </gateID>
      <direction>
        <!--ro, req, enum, target direction, subType:string, desc:"enter", "leave"-->enter
      </direction>
      <cardNo>
        <!--ro, opt, string, card No.-->test
      </cardNo>
      <parkType>
        <!--ro, req, enum, parking type, subType:string, desc:"temporary", "permanent"-->temporary
      </parkType>
    </EntranceInfo>
    <pictureInfoList>
      <!--ro, req, array, picture list, subType:object-->
      <pictureInfo>
        <!--ro, req, object, picture information-->
        <fileName>
          <!--ro, req, enum, picture name, subType:string, desc:it must correspond to the picture name transmitted with the alarm message. "detectionPicture.jpg" (background picture), "licensePlatePicture.jpg" (license plate picture), "pilotPicture.jpg" (driver's picture matting), "copilotPicture.jpg" (co-driver's picture matting), "compositePicture.jpg" (composite picture), "plateBinaryPicture.jpg" (license plate binary picture), "nonMotorPicture.jpg" (non-motor vehicle picture matting), "pedestrianDetectionPicture.jpg" (pedestrian picture), "pedestrianPicture.jpg" (pedestrian's picture matting), "vehiclePicture.jpg" (vehicle picture).-->detectionPicture.jpg
        </fileName>
        <type>
          <!--ro, req, enum, parking type, subType:string, desc:"detectionPicture" (background picture), "licensePlatePicture" (license plate picture), "pilotPicture" (driver's picture matting), "copilotPicture" (co-driver's picture matting), "compositePicture" (composite picture), "plateBinaryPicture" (license plate binary picture), "nonMotorPicture" (non-motor vehicle picture matting), "pedestrianDetectionPicture" (pedestrian picture), "pedestrianPicture" (pedestrian's picture matting), "vehiclePicture" (vehicle picture)-->vehiclePicture
        </type>
        <dataType>
          <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
```

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```
        </dataType>
        <picRecogMode>
          <!--ro, opt, enum, license plate recognition mode, subType:int, desc:0 (front license plate recognition), 1 (rear license plate recognition)-->0
        </picRecogMode>
        <redLightTime>
          <!--ro, opt, int, red light time elapsed, unit:s-->60
        </redLightTime>
        <vehicleHead>
          <!--ro, req, enum, license plate recognition direction, subType:string, desc:"unknown", "forward" (front license plate recognition), "back" (rear
license plate recognition)-->unknown
        </vehicleHead>
        <absTime>
          <!--ro, opt, string, absolute time, desc:format: yyyyMMddHHmmssxxx, e.g.: 20090810235959999. the last three number is time in millisecond--
>20090810235959999
        </absTime>
        <plateRect>
          <!--ro, opt, object, license plate area coordinates, desc:this node is valid only when type is "detectionPicture"-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </plateRect>
        <vehicelRect>
          <!--ro, opt, object, vehicle area coordinates, desc:the normalized value is the current image size in percentage multiplying 1000. This node is
valid only when type is "detectionPicture"-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </vehicelRect>
        <pictureURL>
          <!--ro, opt, string, picture URL, desc:it is valid only when dataType is "URL"-->test
        </pictureURL>
        <pId>
          <!--ro, opt, string, strlen.max=32, range:[1,32], desc:the recommended generation rule is the device's serial number + the time elapsed since the
device was started + a random number-->test
        </pId>
      </pictureInfo>
    </pictureInfoList>
    <hasMoreData>
      <!--ro, opt, bool, whether there is more data, desc:this node is used to report the license plate information first, and then report XML message with
picture data; the XM message with picture data and license plate information are linked by UUID-->true
    </hasMoreData>
    <listType>
      <!--ro, opt, enum, name list attribute, subType:string, desc:"white" (allowlist), "black" (blocklist), "temporary" (temporary list)-->white
    </listType>
    <originalLicensePlate>
      <!--ro, opt, string, original license plate number, desc:when the license plate number is a minor language, return the original license plate number--
>test
    </originalLicensePlate>
    <CRIndex>
      <!--ro, req, enum, country/region, subType:int, desc:country or region index-->258
    </CRIndex>
    <VehicelGPSInfo>
      <!--ro, opt, object, GPS information of the vehicle-->
      <longitudeType>
        <!--ro, req, enum, W, subType:string, desc:longitude,"E,W"-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, N, subType:string, desc:latitude,"S,N"-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999]-->59.000000
        </sec>
      </Longitude>
      <Latitude>
        <!--ro, req, object, latitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
```

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```
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
        </sec>
      </Latitude>
    </VehicelGPSInfo>
    <vehiclePositionControl>
      <!--ro, opt, enum, vehicle arming, subType:string, desc:arming type: "vehicleMonitor"-intelligent arming of vehicle (PUT
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/<taskID>/startTask),"manualVehicleMonitor"-manual arming of vehicle (PUT
/ISAPI/Traffic/channels/<ID>/manualVehicleMonitor),"dailyVehicleMonitor"-daily arming of vehicle (you can check whether this arming type is supported via
the node isSupportDailyVehicleMonitor in the capability message returned by /ISAPI/Traffic/channels/<ID>/vehicleDetect/capabilities; when daily arming of
vehicle is enabled,both alarm of ANPR and intelligent arming of vehicle will be uploaded; if this node is not returned,it is normal vehicle detection--
>dailyVehicleMonitor
    </vehiclePositionControl>
    <vehicleMonitorTaskID>
      <!--ro, opt, string, task ID of intelligent arming of vehicle, range:[1,64], desc:this node is returned when the value of vehiclePositionControl is
"vehicleMonitor"-->test
    </vehicleMonitorTaskID>
    <vehicleListName>
      <!--ro, opt, string, vehicle list name, range:[0,128], desc:name of the list that the vehicle belongs to,the maximum size is 128 bytes-->test
    </vehicleListName>
    <vehicleThermometryEnabled>
      <!--ro, opt, bool, whether to enable vehicle temperature measurement-->true
    </vehicleThermometryEnabled>
    <currTemperature>
      <!--ro, opt, float, temperature-->36.5
    </currTemperature>
    <thermometryUnit>
      <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
    </thermometryUnit>
  </ANPR>
  <UUID>
    <!--ro, opt, string, UUID, desc:if the vehicle is captured many times, the UUID is the same-->test
  </UUID>
  <picNum>
    <!--ro, opt, int, number of pictures-->2
  </picNum>
  <monitoringSiteID>
    <!--ro, opt, string, camera number-->test
  </monitoringSiteID>
  <ePlateUUID>
    <!--ro, opt, string, electronic license plate ID, desc:If this node is configured with a value, it indicates that an electronic license plate is linked-
->test
  </ePlateUUID>
  <isDataRetransmission>
    <!--ro, opt, bool, data retransmission mark-->true
  </isDataRetransmission>
  <SceneInfo>
    <!--ro, opt, object, scene information-->
    <scenesID>
      <!--ro, opt, string, scene ID which is between 1 and 16-->test
    </scenesID>
    <sceneName>
      <!--ro, opt, string, scene name, range:[0,32]-->test
    </sceneName>
    <PTZPos>
      <!--ro, opt, object, PTZ position-->
      <elevation>
        <!--ro, opt, int, tilting angle, range:[-900,2700]-->0
      </elevation>
      <azimuth>
        <!--ro, opt, int, panning angle, range:[0,3600]-->0
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, int, zooming ratio, range:[0,1000]-->1
      </absoluteZoom>
    </PTZPos>
  </SceneInfo>
  <monitorDescription>
    <!--ro, opt, string, camera information, range:[0,256]-->test
  </monitorDescription>
  <DeviceGPSInfo>
    <!--ro, opt, object, device GPS information-->
    <longitudeType>
      <!--ro, req, enum, W, subType:string, desc:longitude,"E,W"-->E
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, N, subType:string, desc:latitude,"S,N"-->S
    </latitudeType>
    <Longitude>
      <!--ro, req, object, longitude-->
      <degree>
        <!--ro, req, int, degree-->60
      </degree>
      <minute>
        <!--ro, req, int, minute, range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second, range:[0,59.999999]-->59.000000
      </sec>
    </Longitude>
```

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```
      <Latitude>
        <!--ro, req, object, latitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
        </sec>
      </Latitude>
    </DeviceGPSInfo>
  </EventNotificationAlert>
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| EventNotificationAlert | [Message content] | application/xml | -- | -- | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

### 14.5.1.62 License plate recognition

**EventType:ANPR**

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate recognition result, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, req, enum, protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP"
when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
```

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```
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventType>
  <eventState>
    <!--ro, req, enum, continuous event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, enum, event description, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,128]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <ANPR>
    <!--ro, opt, object, information about license plate recognition alarm-->
    <region>
      <!--ro, opt, enum, region, subType:string, desc:"ER" (Russian-speaking region), "EU" (Europe), "EUandCIS" (Europe and Russia), "ME" (Middle East),
"other" (other regions), "APAC" (Asia-Pacific region), "AFandAM" (Africa and America), "THAandLA" (Thailand and Laos), "HKandMO" (Hong Kong China and Macao
China), "All" (all regions),-->ER
    </region>
    <country>
      <!--ro, opt, enum, country or region, subType:int, desc:253-invalid, 70-Philippines, 17-United Kingdom (previously Great Britain), 23-Macedonia (North
Macedonia since 2018), 188-Dominican, 227-Australia, 60-Bahrain, 95-Burma/Myanmar, 228-New Zealand, 28-Azerbaijan, 131-Gambia, 132-Mali, 159-Saint Helena,
133-Burkina Faso, 134-Guinea, 135-Guinea-Bissau, 136-Cape Verde, 137-Sierra Leone, 138-Liberia, 139-Ivory Coast, 140-Ghana, 141-Togo, 142-Benin, 143-Niger,
144-Zambia, 145-Angola, 146-Zimbabwe, 147-Malawi, 148-Mozambique, 149-Botswana, 150-Namibia, 151-South Africa, 152-Swaziland, 153-Lesotho, 154-Madagasca,
155-Comoros, 156-Mauritius, 157-Nigeria, 158-South Sudan, 160-Mayotte, 161-Reunion, 162-Canary Islands, 163-AZORES, 164-Madeira, 165-reserved, 166-reserved,
167-reserved, 168-reserved, 169-Canada, 170-Greenland Nuuk, 171-Pierre and Miquelon, 172-United States, 173-Bermuda, 174-Mexico, 175-Guatemala, 176-Belize,
177-El Salvador, 178-Honduras, 179-Nicaragua, 180-Costa Rica, 181-Panama, 182-Bahamas, 183-Turks and  Caicos Islands, 184-Cuba, 185-Jamaica, 186-Cayman
Islands, 187-Haiti, 189-Puerto Rico, 190-United States Virgin Islands, 191-British Virgin Islands, 192-Anguilla, 193-Antigua and Barbuda, 194-Collectivit de
Saint-Martin, 195-Autonomous country, 196-Saint-Barthlemy, 197-Saint Kitts and Nevis, 198-Montserrat, 199-Guadeloupe, 200-Dominica, 201-Martinique, 202-St.
Lucia, 203-Saint Vincent and the Grenadines, 204-Grenada, 205-Barbados, 206-Trinidad and Tobago, 207-Curaao, 0-Unsupported, 1-Czech Republic, 2-France, 3-
Germany, 4-Spain, 5-Italy, 6-Netherlands, 7-Poland, 8-Slovakia, 9-Belarus, 10-Moldova, 11-Russia, 12-Ukraine, 13-Belgium, 14-Bulgaria, 15-Denmark, 16-
Finland, 18-Greece, 19-Croatia, 20-Hungary, 21-Israel(), 22-Luxembourg, 24-Norway, 25-Portuga, 26-Romania, 27-Serbia, 29-Georgia(), 30-Kazakhstan(), 31-
Lithuania, 32-Turkmenistan(), 33-Uzbekistan(), 34-Latvia, 35-Estonia, 36-Albania, 37-Austria, 38-Bosnia and Herzegovina, 39-Ireland, 40-Iceland, 41-Vatican,
42-Malta, 43-Sweden, 44-Switzerland, 45-Cyprus, 46-Turkey, 47-Slovenia, 48-Montenegro, 49-Kosovo, 50-Andorra, 51-Armenia(), 52-Monaco, 53-Liechtenstein, 54-
San Marino, 55-reserved, 56-reserved, 90-Iraq, 57-reserved, 58-reserved, 59-China, 91-Vietnam, 61-South Korea, 62-Lebanon, 63-Nepal, 64-Thailand, 65-
Pakistan, 66-United Arab Emirates, 67-Bhutan, 68-Oman, 69-North Korea, 71-Cambodia, 72-Qatar, 73-Kyrgyzstan, 74-Maldives, 75-Malaysia, 76-Mongolia, 77-Saudi
Arabia, 78-Brunei, 79-Laos, 80-Japan, 81-Turkey, 82-Palestinian, 83-Tajikistan, 84-Kuwait, 85-Syria, 86-India, 87-Indonesia, 88-Afghanistan, 89-Sri Lanka,
92-Iran, 93-Yemen, 94-Jordan, 96-Sikkim, 97-Bangladesh, 98-Singapore, 99-Democratic Republic of Timor-Leste, 100-reserved, 101-reserved, 102-reserved, 103-
reserved, 104-Egypt, 105-Libya, 106-Sudan, 107-Tunisia, 108-Algeria, 109-Morocco, 110-Ethiopia, 111-Eritrea, 112-Somalia Democratic, 113-Djibouti, 114-
Kenya, 115-Tanzania, 116-Uganda, 117-Rwanda, 118-Burundi, 119-Seychelles, 120-Chad, 121-Central African, 122-Cameroon, 123-Equatorial Guinea, 124-Gabon,
125-Congo, 126-Democratic Republic of the Congo, 127-Sao Tome and Principe, 128-Mauritania, 129-Western Sahara, 130-Senegal, 208-Aruba, 209-Netherlands
Antilles, 210-Colombia, 211-Venezuela, 212-Guyana, 213-Suriname, 214-Guyane Francaise, 215-Ecuador, 216-Peru, 217-Bolivia, 218-Paraguay, 219-Chile, 220-
Brazil, 221-Uruguay, 222-Argentina, 223-reserved, 224-reserved, 225-reserved, 226-reserved, 229-Papua New Guinea, 230-Salomonen, 231-Vanuatu, 232-New
Caledonia, 233-Palau, 234-Federated States of Micronesia, 235-Marshall Island, 236-Northern Mariana Islands, 237-Guam, 238-Nauru, 239-Kiribati, 240-Fidschi,
241-Tonga, 242-Tuvalu, 243-Wallis et Futuna, 244-Samoa, 245-Eastern Samoa, 246-Tokelau, 247-Niue, 248-Cook Islands, 249-French Polynesia, 250-Pitcairn
Islands, 251-Hawaii State, 252-reserved, 254-unrecognized, 255-all-->253
    </country>
    <area>
      <!--ro, opt, enum, United Arab Emirates, subType:string, desc:"FJR" (Fujairah), "AD" (Abu Dhabi), "unknown", "UMW" (Umm al-Qaiwain), "other", "AM"
(Ajman), "RAK" (Ras al-Khaimah), "DB" (Dubai), "SJ" (Sharjah)-->FJR
    </area>
    <licensePlate>
      <!--ro, req, string, license plate number, range:[1,32], desc:noPlate (vehicle without license plate), unknown (no license plate recognized), XXXXXXXX
(specific license plate number; color information required by Chinese license plate; for motor vehicles, it is a 16-byte string; for non-motor vehicles, it
is a 48-byte string)-->A283KY77
    </licensePlate>
    <cameraNo>
      <!--ro, opt, string, device No., range:[1,48], desc:corresponds to <cameraNum> from /ISAPI/System/Video/inputs/channels/<ID>/cameraInfo (recommended
for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
    </cameraNo>
    <line>
      <!--ro, req, int, recognized lane No.-->1
    </line>
    <direction>
      <!--ro, opt, enum, license plate recognition direction, subType:string, desc:"reverse", "forward", "unknown"-->reverse
    </direction>
    <confidenceLevel>
      <!--ro, req, int, confidence level, range:[0,100]-->50
    </confidenceLevel>
    <plateType>
      <!--ro, opt, enum, license plate type, subType:string, desc:"unknown", "arm" (police vehicle), "92TypeArm" (type 92 armed police vehicle),
"02TypePersonalized" (type 02 custom vehicle),  "yellowTwoLine" (yellow two-line rear license plate), "embassy" (embassy vehicle), "oneLineArm"(new armed
police vehicle, one-line), "twoLineArm" (new armed police vehicle, two-line), "yellow1225FarmVehicle" (agricultural vehicle, yellow 1225),
"green1325FarmVehicle" (agricultural vehicle, green 1325), "yellow1325FarmVehicle" (agricultural vehicle, yellow 1325), "motorola" (motorcycle), "coach"
(driver-training vehicle), "tempTravl" (vehicle with temporary license plate), "trailer" (trailer), "consulate" (consular vehicle), "hongKongMacao" (Hong
Kong China/Macao China entrance vehicle), "tempEntry" (temporary vehicle), "civilAviation" (civil aviation license plate), "newEnergy" (new energy vehicle),
"92FarmVehicle" (two-line license plate civil vehicle), "emergency" (emergency license plate), "oneLineArmHeadquarters" (armed police headquarter license
plate, one-line), "twoLineArmHeadquarters" (armed police headquarter license plate, two-line), "twoWheelVehicle" (two wheeler)-->unknown
    </plateType>
    <plateColor>
      <!--ro, opt, enum, license plate color, subType:string, desc:"black", "blue","golden", "orange", "red", "yellow", "white", "unknown", "other",
"newEnergyYellowGreen"-new energy yellow-green,"civilAviationBlack", "civilAviationGreen" (civil aviation green), "green", "mixedColor", "newEnergyGreen"
(new energy green), "brown"-->black
    </plateColor>
    <licenseBright>
      <!--ro, opt, int, license plate brightness, which ranges from 0 to 255, range:[0,255]-->50
    </licenseBright>
    <Rect>
      <!--ro, opt, object, coordinates of the license plate thumbnail in the matched picture, desc:used by back-end DeepinMind; the origin is the upper-left
corner of the screen-->
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        <height>
          <!--ro, req, float, height, range:[0.000,1.000]-->1.000
        </height>
        <width>
          <!--ro, req, float, width, range:[0.000,1.000]-->1.000
        </width>
        <x>
          <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
        </x>
        <y>
          <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
        </y>
      </Rect>
      <pilotsafebelt>
        <!--ro, opt, enum, whether the driver is wearing a safety belt, subType:string, desc:"unknown", "yes", "no"-->unknown
      </pilotsafebelt>
      <vicepilotsafebelt>
        <!--ro, opt, enum, whether the co-driver is wearing a safety belt, subType:string, desc:"unknown", "yes", "no"-->unknown
      </vicepilotsafebelt>
      <pilotsunvisor>
        <!--ro, opt, enum, whether the driver room's sun visor is open, subType:string, desc:"unknown", "yes", "no"-->unknown
      </pilotsunvisor>
      <vicepilotsunvisor>
        <!--ro, opt, enum, whether the co-driver room's sun visor is open, subType:string, desc:"unknown", "yes", "no"-->unknown
      </vicepilotsunvisor>
      <envprosign>
        <!--ro, opt, enum, whether it is a yellow-label vehicle, subType:string, desc:"unknown", "green" (green-label), "yellow" (yellow-lable), "yes", "no"--
>unknown
      </envprosign>
      <dangmark>
        <!--ro, opt, enum, whether it is a dangerous goods vehicle, subType:string, desc:"unknown", "yes", "no"-->unknown
      </dangmark>
      <uphone>
        <!--ro, opt, enum, whether the driver is making a phone call, subType:string, desc:"unknown", "yes", "no"-->unknown
      </uphone>
      <pendant>
        <!--ro, opt, enum, whether there are window hangings detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </pendant>
      <tissueBox>
        <!--ro, opt, enum, whether there is a tissue box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </tissueBox>
      <frontChild>
        <!--ro, opt, enum, whether the co-driver is with a baby in arm, subType:string, desc:"unknown", "yes", "no"-->unknown
      </frontChild>
      <label>
        <!--ro, opt, enum, whether there are stickers detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </label>
      <decoration>
        <!--ro, opt, enum, whether there are decorations detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </decoration>
      <smoking>
        <!--ro, opt, enum, whether anyone is smoking, subType:string, desc:"unknown", "yes", "no"-->yes
      </smoking>
      <perfumeBox>
        <!--ro, opt, enum, whether there is perfume box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </perfumeBox>
      <pdvs>
        <!--ro, opt, enum, whether there is a person sticking out of sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
      </pdvs>
      <helmet>
        <!--ro, opt, enum, whether there is helmet detected, subType:string, desc:"unknown", "yes", "no"-->no
      </helmet>
      <twoWheelVehicle>
        <!--ro, opt, enum, whether there is a two-wheel vehicle detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </twoWheelVehicle>
      <threeWheelVehicle>
        <!--ro, opt, enum, whether there is a three-wheel vehicle detected, subType:string, desc:"unknown", "yes", "no"-->unknown
      </threeWheelVehicle>
      <blackness>
        <!--ro, opt, int, Ringelmann emittance, desc:it is used for black smoke detection-->2
      </blackness>
      <plateCharBelieve>
        <!--ro, opt, string, confidence of each character in the recognized license plate, desc:if "A12345" is detected as a license plate number and the
confidence is 10, 20, 30, 40, 50, 60, it indicates that the accuracy of "A" is 20%-->test
      </plateCharBelieve>
      <speedLimit>
        <!--ro, opt, int, maximum speed limit, desc:it is valid only when overspeeding occurred-->50
      </speedLimit>
      <illegalInfo>
        <!--ro, opt, object, vehicle traffic violation information, desc:illegal action information-->
        <illegalCode>
          <!--ro, req, string, illegal action code, range:[0,64]-->1301
        </illegalCode>
        <illegalName>
          <!--ro, req, string, illegal action name, range:[0,128]-->逆行
        </illegalName>
        <illegalDescription>
          <!--ro, opt, string, illegal action description, range:[0,256]-->车辆逆行
        </illegalDescription>
      </illegalInfo>
      <vehicleType>
        <!--ro, req, enum, vehicle type, subType:string, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->nonmotorVehicle
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    </vehicleType>
    <postPicFileName>
      <!--ro, opt, string, name of the picture selected as the checkpoint picture when illegal action occurs, desc:"none" (no picture selected)-->test
    </postPicFileName>
    <featurePicFileName>
      <!--ro, opt, string, name of the picture selected as the close-up picture when running the red light in the intersection violation system is detected,
desc:"none" (no picture selected)-->test
    </featurePicFileName>
    <detectDir>
      <!--ro, opt, enum, detection direction, subType:int, desc:1 (upward), 2 (downward), 3 (bidirectional), 4 (westward), 5 (northward), 6 (eastward), 7
(southward), 8 (other)-->1
    </detectDir>
    <detectType>
      <!--ro, opt, enum, detection type, subType:int, desc:0 (vehicle detection), 1 (inductive loop trigger), 2 (video trigger), 3 (multiple-frame
recognition), 4 (radar trigger), 5 (mixed-traffic detection)-->1
    </detectType>
    <barrierGateCtrlType>
      <!--ro, opt, enum, whether to enable elapsed time, subType:int, desc:0 (enabled), 1 (disabled)-->0
    </barrierGateCtrlType>
    <alarmDataType>
      <!--ro, opt, enum, alarm data type (real-time data or history data), subType:int, desc:0 (real-time data), 1 (history data)-->0
    </alarmDataType>
    <dwIllegalTime>
      <!--ro, opt, int, illegal action duration, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the
first picture-->100
    </dwIllegalTime>
    <vehicleInfo>
      <!--ro, opt, object, vehicle information-->
      <index>
        <!--ro, req, int, vehicle No.-->1
      </index>
      <vehicleType>
        <!--ro, opt, enum, vehicle type, subType:int, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->0
      </vehicleType>
      <colorDepth>
        <!--ro, req, enum, shade of the vehicle color, subType:int, desc:0 (deep color), 1 (light color), 2 (unknown)-->0
      </colorDepth>
      <color>
        <!--ro, req, enum, vehicle color, subType:string, desc:"unknown", "green", "brown", "pink", "purple", "deepGray" (dark gray), "cyan", "orange",
"white", "silver" (silvery), "gray", "blacks", "red", "deepBlue" (dark blue), "blue", "yellow"-->green
      </color>
      <speed>
        <!--ro, opt, int, vehicle speed, unit:km/h-->1
      </speed>
      <length>
        <!--ro, req, int, length of the former vehicle, unit:dm-->10
      </length>
      <vehicleLogoRecog>
        <!--ro, req, int, vehicle main brand-->1
      </vehicleLogoRecog>
      <vehileSubLogoRecog>
        <!--ro, opt, int, vehicle sub-brand-->1
      </vehileSubLogoRecog>
      <vehileModel>
        <!--ro, opt, int, model year of vehicle sub-brand-->1
      </vehileModel>
      <vehicleTypeByWeight>
        <!--ro, opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1 (class 1: buses with seven or less seats, trucks with capacity
of 2 tons or less), 2 (class 2: buses with 8 to 19 seats, trucks with capacity of 2 to 5 (included) tons), 3 (class 3: buses with 20 to 39 seats, trucks
with capacity of 5 to 10 (included) tons),4 (class 4: buses with 40 or more seats, trucks with capacity of 10 to 15 (included) tons), 5 (class 5: trucks
with capacity of more than 15 tons), 6 (class 6: special vehicle such as ambulance, fire truck, and garbage truck)-->4
      </vehicleTypeByWeight>
      <tollRoadVehicleSeries>
        <!--ro, opt, enum, vehicles on tollway, subType:int, desc:N/A-->2
      </tollRoadVehicleSeries>
      <tollRoadVehicleType>
        <!--ro, opt, enum, vehicle types on tollway, subType:int, desc:N/A-->4
      </tollRoadVehicleType>
      <CarWindowFeature>
        <!--ro, opt, object, vehicle window feature, desc:configured via <CarWindowFeature> from /ISAPI/ITC/carFeatureParam-->
        <tempPlate>
          <!--ro, opt, enum, whether there is temporary license plate, subType:string, desc:detect whether there is temporary license plate on the window:
"unknown", "yes", "no"-->unknown
        </tempPlate>
        <passCard>
          <!--ro, opt, enum, whether there is entry & exit pass, subType:string, desc:detect whether there is entry & exit pass on the window: "unknown",
"yes", "no"-->unknown
        </passCard>
        <carCard>
          <!--ro, opt, enum, whether there are cards detected, subType:string, desc:whether there is any card (business cards, flyers, etc.) attached to the
window: "unknown", "yes", "no"-->unknown
        </carCard>
      </CarWindowFeature>
      <CarBodyFeature>
        <!--ro, opt, object, bodywork feature, desc:configured via <CarBodyFeature> from /ISAPI/ITC/carFeatureParam-->
        <sparetire>
          <!--ro, opt, enum, whether it is with a spare tire, subType:string, desc:"unknown", "yes", "no"-->unknown
        </sparetire>
        <rack>
          <!--ro, opt, enum, whether it is with a luggage rack, subType:string, desc:"unknown", "yes", "no"-->unknown
        </rack>
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        <sunRoof>
          <!--ro, opt, enum, whether there is a sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
        </sunRoof>
        <words>
          <!--ro, opt, enum, whether the vehicle is painted, subType:string, desc:"unknown", "yes", "no"-->unknown
        </words>
        <slagTruckCoverPlate>
          <!--ro, opt, enum, whether the back cover of the dump truck is covered, subType:string, desc:"unknown", "yes", "no"-->unknown
        </slagTruckCoverPlate>
        <reflectiveTape>
          <!--ro, opt, enum, subType:string-->unknown
        </reflectiveTape>
        <closureWindow>
          <!--ro, opt, enum, subType:string-->unknown
        </closureWindow>
      </CarBodyFeature>
      <vehicleUseType>
        <!--ro, opt, enum, vehicle purpose type, subType:string, desc:configured via vehicleUseEnable from /ISAPI/ITC/carFeatureParam: "taxi", "ambulance",
"bus", "schoolbus", "coach", "unknown"-->taxi
      </vehicleUseType>
      <carWindowInfo>
        <!--ro, opt, object, vehicle window information, desc:configured via /ISAPI/ITC/TriggerMode/HOVLaneDetection?format=json-->
        <carWindowNum>
          <!--ro, opt, int, number of vehicle windows-->4
        </carWindowNum>
        <carWindowStatusList>
          <!--ro, opt, object, vehicle window status list-->
          <carWindowStatus>
            <!--ro, opt, object, vehicle window status-->
            <passengers>
              <!--ro, opt, int, number of persons behind the window, range:[0,10]-->1
            </passengers>
            <windowStatus>
              <!--ro, opt, enum, whether the vehicle windows are transparent, subType:string, desc:"unknown", "yes", "no"-->yes
            </windowStatus>
          </carWindowStatus>
        </carWindowStatusList>
      </carWindowInfo>
      <actualPassengers>
        <!--ro, opt, int, number of passengers in the vehicle, range:[0,100]-->1
      </actualPassengers>
      <vehiclePosition>
        <!--ro, opt, enum, detected vehicle body part, subType:string, desc:"leftSide", "rightSide", "headStock", "tailStock", "roof"-->leftSide
      </vehiclePosition>
      <containerInfo>
        <!--ro, opt, object, container information-->
        <containerNum>
          <!--ro, opt, int, number of transferred containers, range:[1,10]-->2
        </containerNum>
        <containerList>
          <!--ro, opt, array, list of container information, subType:object, range:[1,10]-->
          <container>
            <!--ro, opt, object, container information-->
            <containerMainNum>
              <!--ro, opt, string, major No. of captured container, range:[1,32]-->abcd1234
            </containerMainNum>
            <containerSubNum>
              <!--ro, opt, string, minor No. of captured container, range:[1,32]-->abcd1234
            </containerSubNum>
            <containerISONum>
              <!--ro, opt, string, ISO No. of captured container, range:[1,32]-->abcd1234
            </containerISONum>
            <containerNumConfidence>
              <!--ro, opt, int, confidence of captured container No., range:[0,100]-->0
            </containerNumConfidence>
          </container>
        </containerList>
      </containerInfo>
      <RFID>
        <!--ro, opt, string, link RFID, range:[1,64], dep:and,{$.EventNotificationAlert.ANPR.vehicleType,eq,nonmotorVehicle}-->12345
      </RFID>
      <vehicleCarryFeature>
        <!--ro, opt, object-->
        <isCarry>
          <!--ro, opt, enum, subType:string-->unknown
        </isCarry>
        <isTarpaulin>
          <!--ro, opt, enum, subType:string-->unknown
        </isTarpaulin>
        <isCoverplate>
          <!--ro, opt, enum, subType:string-->unknown
        </isCoverplate>
      </vehicleCarryFeature>
    </vehicleInfo>
    <EntranceInfo>
      <!--ro, opt, object, entrance and exit information-->
      <parkingID>
        <!--ro, opt, string, parking space No.-->test
      </parkingID>
      <gateID>
        <!--ro, opt, string, entrance and exit No.-->test
      </gateID>
      <direction>
        <!--ro, opt, enum, target direction, subType:string, desc:"enter", "leave"-->enter
```

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        <!--ro, opt, enum, target direction, subType:string, desc: enter ", "leave -->enter
      </direction>
      <cardNo>
        <!--ro, opt, string, card No.-->test
      </cardNo>
      <parkType>
        <!--ro, opt, enum, parking type, subType:string, desc:"temporary", "permanent"-->temporary
      </parkType>
    </EntranceInfo>
    <pictureInfoList>
      <!--ro, req, array, picture list, subType:object, range:[0,8]-->
      <pictureInfo>
        <!--ro, req, object, picture information-->
        <fileName>
          <!--ro, req, enum, picture name, subType:string, desc:"detectionPicture.jpg" (background picture), "licensePlatePicture.jpg" (license plate
picture), "pilotPicture.jpg" (driver's picture matting), "copilotPicture.jpg" (co-driver's picture matting), "compositePicture.jpg" (composite picture),
"plateBinaryPicture.jpg" (license plate binary picture), "nonMotorPicture.jpg" (non-motor vehicle picture matting), "pedestrianDetectionPicture.jpg"
(pedestrian picture), "pedestrianPicture.jpg" (pedestrian's picture matting), "vehiclePicture.jpg" (vehicle picture). Picture name, which must correspond to
the picture name transmitted with the alarm message-->detectionPicture.jpg
        </fileName>
        <type>
          <!--ro, req, enum, picture type, subType:string, desc:"detectionPicture" (background picture), "licensePlatePicture" (license plate picture),
"pilotPicture" (driver's picture matting), "copilotPicture" (co-driver's picture matting), "compositePicture" (composite picture), "plateBinaryPicture"
(license plate binary picture), "nonMotorPicture" (non-motor vehicle picture matting), "pedestrianDetectionPicture" (pedestrian picture),
"pedestrianPicture" (pedestrian's picture matting), "vehiclePicture" (vehicle picture)-->vehiclePicture
        </type>
        <dataType>
          <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
        </dataType>
        <picRecogMode>
          <!--ro, opt, enum, license plate recognition mode, subType:int, desc:0 (front license plate recognition), 1 (rear license plate recognition)-->0
        </picRecogMode>
        <redLightTime>
          <!--ro, opt, int, red light time elapsed, unit:s-->60
        </redLightTime>
        <vehicleHead>
          <!--ro, opt, enum, license plate recognition direction, subType:string, desc:"unknown", "forward" (front license plate recognition), "back" (rear
license plate recognition)-->unknown
        </vehicleHead>
        <absTime>
          <!--ro, opt, string, absolute time, desc:format: yyyyMMddHHmmssxxx, e.g., 20090810235959999, the last three digits are time in millisecond--
>20090810235959999
        </absTime>
        <plateRect>
          <!--ro, opt, object, license plate area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value
is the current image size in percentage multiplying 1000 and it is accurate to three decimal places. The origin is the upper-left corner of screen-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </plateRect>
        <vehicelRect>
          <!--ro, opt, object, the vehicle area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value is
the current image size in percentage multiplying 1000. The origin is the upper-left corner of screen-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </vehicelRect>
        <pictureURL>
          <!--ro, opt, string, picture URL, desc:it is valid when the value of dataType is "URL"-->test
        </pictureURL>
        <pId>
          <!--ro, opt, string, strlen.max=32, range:[1,32], desc:strlen.max=32-->test
        </pId>
        <province>
          <!--ro, opt, enum, province index, subType:int, desc:N/A-->0
        </province>
        <PilotRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
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          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </PilotRect>
        <VicepilotRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </VicepilotRect>
        <VehicelWindowRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </VehicelWindowRect>
        <extraPlateRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </extraPlateRect>
      </pictureInfo>
    </pictureInfoList>
    <hasMoreData>
      <!--ro, opt, bool, whether there is more data, desc:this field is used to report the license plate information first, and then report XML message with
picture data; the XM message with picture data and license plate information are linked by UUID-->true
    </hasMoreData>
    <listType>
      <!--ro, opt, enum, List Property, subType:string, desc:"white" (allowlist), "black" (blocklist), "temporary" (temporary list)-->white
    </listType>
    <originalLicensePlate>
      <!--ro, opt, string, original license plate number, desc:when the original license plate number contains non-English characters, the original license
plate number can be returned-->test
    </originalLicensePlate>
    <CRIndex>
      <!--ro, opt, enum, country or region index, subType:int, desc:country or region index-->258
    </CRIndex>
    <VehicelGPSInfo>
      <!--ro, opt, object, GPS information of the vehicle-->
      <longitudeType>
        <!--ro, req, enum, longitude, subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, latitude, subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude information-->
        <degree>
          <!--ro, req, int, degree(s)-->60
        </degree>
        <minute>
          <!--ro, req, int, minute(s), range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second(s), range:[0,59.999999]-->59.000000
        </sec>
      </Longitude>
      <Latitude>
        <!--ro, req, object, latitude information-->
        <degree>
          <!--ro, req, int, degree(s)-->60
        </degree>
        <minute>
          <!--ro, req, int, minute(s), range:[0,59]-->59
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        </minute>
        <sec>
          <!--ro, req, float, second(s), range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
        </sec>
      </Latitude>
    </VehicleGPSInfo>
    <vehiclePositionControl>
      <!--ro, opt, enum, vehicle arming type, subType:string, desc:"vehicleMonitor" (intelligent arming of vehicle, related URI: PUT
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/<taskID>/startTask), "manualVehicleMonitor" (manual arming of vehicle, related URI: PUT
/ISAPI/Traffic/channels/<ID>/manualVehicleMonitor), "dailyVehicleMonitor" (daily arming of vehicle)-->dailyVehicleMonitor
    </vehiclePositionControl>
    <vehicleMonitorTaskID>
      <!--ro, opt, string, task ID of intelligent arming of vehicle, range:[1,64], desc:the task ID is sent to the device by the upper layer, which should
guarantee the uniqueness of ID. It's valid when the value of vehiclePositionControl is "vehicleMonitor"-->test
    </vehicleMonitorTaskID>
    <vehicleListName>
      <!--ro, opt, string, name of the list that the vehicle belongs to, range:[0,128], desc:name of the list that the vehicle belongs to,the maximum size
is 128 bytes-->test
    </vehicleListName>
    <vehicleThermometryEnabled>
      <!--ro, opt, bool, whether to enable vehicle temperature measurement-->true
    </vehicleThermometryEnabled>
    <currTemperature>
      <!--ro, opt, float, temperature-->36.5
    </currTemperature>
    <thermometryUnit>
      <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
    </thermometryUnit>
    <plateCategory>
      <!--ro, opt, string, license plate additional information, range:[0,8], desc:license plate additional information-->test
    </plateCategory>
    <plateSize>
      <!--ro, opt, enum, license plate size, subType:int, desc:license plate size-->1
    </plateSize>
    <isNeedVerification>
      <!--ro, opt, bool, whether it requires verification, desc:false-do not verify-->true
    </isNeedVerification>
    <ISO-CR>
      <!--ro, opt, string, ISO 3166-1 standard country/region code, desc:refer to protocol dictionary-->CN
    </ISO-CR>
    <vehicelRectName>
      <!--ro, opt, string, country/region where the vehicle triggering events belongs, range:[0,64]-->11
    </vehicelRectName>
    <rodeType>
      <!--ro, opt, enum, road type, subType:string, desc:"entrance" (entrance and exit), "city", "custom", "alarmInput" (alarm input triggered by coil),
"post" (checkpoint), "publicSecurity" (general video security mode, in which the scene with mixed vehicles and without lane lines can be detected)--
>entrance
    </rodeType>
    <Custom>
      <!--ro, opt, object, custom parameters, dep:and,{$.EventNotificationAlert.ANPR.rodeType,eq,custom}-->
      <direction>
        <!--ro, opt, enum, target direction, subType:string, desc:"enter", "parkind", "leave"-->enter
      </direction>
    </Custom>
    <accelerationNoise>
      <!--ro, opt, object, vehicle revving behavior that causes huge noise-->
      <startTime>
        <!--ro, req, datetime, start time of the behavior-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time of the behavior-->1970-01-01T00:00:00+08:00
      </endTime>
      <noiseDecibel>
        <!--ro, req, float, noise decibel-->36.5
      </noiseDecibel>
    </accelerationNoise>
    <ADRNo>
      <!--ro, opt, string, hazardous material ID, range:[1,8]-->231965
    </ADRNo>
    <pilotmask>
      <!--ro, opt, enum, subType:string-->unknown
    </pilotmask>
    <vicepilotMask>
      <!--ro, opt, enum, subType:string-->unknown
    </vicepilotMask>
    <nonMotorMask>
      <!--ro, opt, enum, subType:string-->unknown
    </nonMotorMask>
    <pedestrianMask>
      <!--ro, opt, enum, subType:string-->unknown
    </pedestrianMask>
    <vehicleTemperature>
      <!--ro, opt, object-->
      <vehicleHeadTemperature>
        <!--ro, opt, float, range:[-20.0,550.0]-->50.2
      </vehicleHeadTemperature>
      <vehicleBodyTemperature>
        <!--ro, opt, float, range:[-20.0,550.0]-->35.2
      </vehicleBodyTemperature>
      <leftWheelHubList>
        <!--ro, opt, array, subType:object, range:[0,6]-->
        <leftWheelHub>
          <!--ro, opt, object-->
          <ID>
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            <!--ro, opt, int, range:[1,6]-->1
          </ID>
          <wheelHubTemperature>
            <!--ro, opt, float, range:[-20.0,550.0]-->38.2
          </wheelHubTemperature>
        </leftWheelHub>
      </leftWheelHubList>
      <rightWheelHubList>
        <!--ro, opt, array, subType:object, range:[0,6]-->
        <rightWheelHub>
          <!--ro, opt, object-->
          <ID>
            <!--ro, opt, int, range:[1,6]-->1
          </ID>
          <wheelHubTemperature>
            <!--ro, opt, float, range:[-20.0,550.0]-->38.2
          </wheelHubTemperature>
        </rightWheelHub>
      </rightWheelHubList>
    </vehicleTemperature>
    <deliveryVan>
      <!--ro, opt, enum, subType:string-->unknown
    </deliveryVan>
    <nonMotorShedUmbrella>
      <!--ro, opt, enum, subType:string-->unknown
    </nonMotorShedUmbrella>
    <nonMotorManned>
      <!--ro, opt, enum, subType:string-->unknown
    </nonMotorManned>
  </ANPR>
  <UUID>
    <!--ro, opt, string, unique common ID, which is used to link the same capture across multiple servers, desc:if the vehicle is captured many times, the
UUID is the same-->test
  </UUID>
  <picNum>
    <!--ro, opt, int, number of pictures-->2
  </picNum>
  <monitoringSiteID>
    <!--ro, opt, string, camera No., desc:corresponds to <positionNum> from /ISAPI/System/Video/inputs/channels/<channelID>/cameraInfo (recommended for new
devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
  </monitoringSiteID>
  <ePlateUUID>
    <!--ro, opt, string, electronic license plate ID, desc:if this field has a value, it indicates that an electronic license plate is linked-->test
  </ePlateUUID>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished-->true
  </isDataRetransmission>
  <SceneInfo>
    <!--ro, opt, object, scene information-->
    <scenesID>
      <!--ro, opt, string, scene ID, which is between 1 and 16-->test
    </scenesID>
    <sceneName>
      <!--ro, opt, string, scene name, range:[0,128]-->test
    </sceneName>
    <PTZPos>
      <!--ro, opt, object, PTZ information-->
      <elevation>
        <!--ro, opt, int, tilting angle, range:[-900,2700]-->0
      </elevation>
      <azimuth>
        <!--ro, opt, int, panning angle, range:[0,3600]-->0
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, int, zooming ratio, range:[0,1000]-->1
      </absoluteZoom>
    </PTZPos>
  </SceneInfo>
  <monitorDescription>
    <!--ro, opt, string, camera information, range:[0,256], desc:corresponds to <positionInfo> from /ISAPI/System/Video/inputs/channels/<ID>/cameraInfo
(recommended for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
  </monitorDescription>
  <DeviceGPSInfo>
    <!--ro, opt, object, device GPS information-->
    <longitudeType>
      <!--ro, req, enum, longitude, subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)-->E
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, latitude, subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)-->S
    </latitudeType>
    <Longitude>
      <!--ro, req, object, longitude information-->
      <degree>
        <!--ro, req, int, degree(s)-->60
      </degree>
      <minute>
        <!--ro, req, int, minute(s), range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second(s), range:[0,59.999999]-->59.000000
      </sec>
    </Longitude>
    <Latitude>
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      <!--ro, req, object, latitude information-->
      <degree>
        <!--ro, req, int, degree(s)-->60
      </degree>
      <minute>
        <!--ro, req, int, minute(s), range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second(s), range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
      </sec>
    </Latitude>
  </DeviceGPSInfo>
  <pilotStandardSafebelt>
    <!--ro, opt, enum, whether the driver wears the seat belt properly, subType:string, desc:"unknown", "yes", "no"-->yes
  </pilotStandardSafebelt>
  <vicepilotStandardSafebelt>
    <!--ro, opt, enum, whether the front passenger wears the seat belt properly, subType:string, desc:"unknown", "yes", "no"-->yes
  </vicepilotStandardSafebelt>
  <trafficLightSnap>
    <!--ro, opt, enum, whether it is captured at the traffic lights, subType:string, desc:"yes", "no"-->yes
  </trafficLightSnap>
  <sequence>
    <!--ro, opt, int, No. of vehicle capture from network-triggered burst, range:[1,4294967295], desc:command applied by network-triggered burst control;
related API: /ISAPI/Traffic/startRecognition; SDK interface: NET_DVR_ContinuousShoot-->0
  </sequence>
  <relaLaneDirectionType>
    <!--ro, opt, enum, linked lane direction, subType:int, desc:a camera can be linked with multiple lane directions, which is independent of the camera
setup angle <detectDir>; 0 (others),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to
northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
  </relaLaneDirectionType>
  <carDirectionType>
    <!--ro, opt, enum, vehicle running direction on the lane, subType:int, desc:independent of the camera setup angle <detectDir>; 0-from up to down, 1-from
down to up-->1
  </carDirectionType>
  <targetID>
    <!--ro, opt, string, target ID, range:[1,64], desc:UUID, of which the uniqueness is guaranteed by device, and corresponds to <dwMatchNo> in the private
protocol-->test
  </targetID>
  <isSecondCamera>
    <!--ro, opt, bool, whether to enable capture by the second camera, desc:the second camera: e.g., the perspective camera from distant/close-view capture
cameras, the back camera from front/back capture camera, etc.-->false
  </isSecondCamera>
  <dataAnalysisType>
    <!--ro, opt, enum, data analysis type, subType:int, desc:0-data not analyzed, 1-data analyzed-->1
  </dataAnalysisType>
  <RecordInfo>
    <!--ro, opt, object, vehicle video information-->
    <fileName>
      <!--ro, req, enum, video name, subType:string, desc:the video name should match <Content-ID> for binary transmission: "record.mp4" (vehicle video)--
>record.mp4
    </fileName>
    <dataType>
      <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
    </dataType>
    <URL>
      <!--ro, opt, string, URL, desc:it is valid when the value of dataType is "URL"-->test
    </URL>
  </RecordInfo>
  <VehicleWeightInfo>
    <!--ro, opt, object, vehicle weighing information, desc:this node is currently only supported by weighing data management server-->
    <isOverWeight>
      <!--ro, opt, bool, whether it is overweight, desc:set overweight reporting or overrun reporting as priority via <illegalPriorityType> from
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->false
    </isOverWeight>
    <axleNum>
      <!--ro, opt, int, number of axles, range:[1,10]-->4
    </axleNum>
    <axleModel>
      <!--ro, opt, enum, vehicle type by axle, subType:int, desc:0 (unknown), 512 (2-axle truck), 756 (3-axle centre-axle trailer combination), 766 (3-axle
articulated vehicle), 767 (3-axle truck), 768 (3-axle truck), 1024 (type-125 4-axle centre-axle trailer combination), 1025 (type-152 4-axle centre-axle
trailer combination), 1026 (4-axle articulated vehicle), 1027 (4-axle full trailer combination), 1028 (4-axle truck), 1280 (type-155 5-axle centre-axle
trailer combination), 1281 (type-1125 5-axle centre-axle trailer combination), 1282 (type-155 5-axle articulated vehicle), 1283 (type-1125 5-axle
articulated vehicle), 1284 (type-129 5-axle articulated vehicle), 1285 (type-1522 5-axle full trailer combination), 1286 (type-11222 5-axle full trailer
combination), 1536 (type-159 6-axle centre-axle trailer combination), 1537 (type-159-2 6-axle centre-axle trailer combination), 1538 (type-1155-1 6-axle
centre-axle trailer combination), 1539 (type-1155-2 6-axle centre-axle trailer combination), 1540 (type-159-3 6-axle articulated vehicle), 1541 (type-159-4
6-axle articulated vehicle), 1542 (type-1129 6-axle articulated vehicle), 1543 (type-11522-1 6-axle full trailer combination), 1544 (type-11522-2 6-axle
full trailer combination)-->0
    </axleModel>
    <overWeight>
      <!--ro, opt, float, over-limit weight (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </overWeight>
    <weight>
      <!--ro, opt, float, vehicle weight (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </weight>
    <limitWeight>
      <!--ro, opt, float, weight limit (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </limitWeight>
    <axleLen>
      <!--ro, opt, float, distance between tyres, range:[0.00,100.00], unit:吨, desc:accurate to two decimal places-->4.502
    </axleLen>
    <devDescInfo>
      <!--ro, opt, string, device description, range:[0,64]-->test
    </devDescInfo>
    <AxleInfoList>
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      <!--ro, opt, object, axle information list, desc:<axleNum>, number of axles-->
      <AxleInfo>
        <!--ro, opt, object, information of a single axle-->
        <axleWeight>
          <!--ro, opt, float, axle weight, range:[0.00,10000.00], unit:kg-->100.00
        </axleWeight>
        <axleDistance>
          <!--ro, opt, int, distance between axles, range:[0,100000], unit:mm, desc:the distance between the current axle and the next axle-->30000
        </axleDistance>
        <wheelNum>
          <!--ro, opt, int, range:[0,8]-->2
        </wheelNum>
      </AxleInfo>
    </AxleInfoList>
    <length>
      <!--ro, opt, int, vehicle length, range:[1,1000000], unit:cm-->4000
    </length>
    <width>
      <!--ro, opt, int, vehicle width, range:[1,1000000], unit:cm-->4000
    </width>
    <height>
      <!--ro, opt, int, vehicle height, range:[1,1000000], unit:cm-->4000
    </height>
    <tollwayVehicleType>
      <!--ro, opt, enum, vehicle types on tollway, subType:int, desc:0 (unknown), 1 (type-1 minibus), 2 (type-1 small-sized bus), 3 (type-2 medium-sized
bus), 4 (type-2 bus and passenger trailer combination), 5 (type-3 large-sized bus), 6 (type-4 large-sized bus), 7 (type-1 2-axle truck), 8 (type-2 2-axle
truck), 9 (type-3 3-axle truck), 10 (type-4 4-axle truck), 11 (type-5 5-axle truck), 12 (type-6 6-axle truck), 13 (type-1 2-axle operation vehicle), 14
(type-2 2-axle operation vehicle), 15 (type-3 3-axle operation vehicle), 16 (type-4 4-axle operation vehicle), 17 (type-5 5-axle operation vehicle), 18
(type-6 operation vehicle with 6 or more axles)-->1
    </tollwayVehicleType>
    <tiresNum>
      <!--ro, opt, int, number of tyres, range:[1,20]-->4
    </tiresNum>
    <approvedPassengers>
      <!--ro, opt, int, authorized passenger capacity, range:[1,100]-->7
    </approvedPassengers>
    <limitLength>
      <!--ro, opt, int, vehicle length limit, range:[1,1000000], unit:cm, desc:configured via
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitLength>
    <limitWidth>
      <!--ro, opt, int, vehicle width limit, range:[1,1000000], unit:cm, desc:configured via /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?
format=json-->4000
    </limitWidth>
    <limitHeight>
      <!--ro, opt, int, vehicle height limit, range:[1,1000000], unit:cm, desc:configured via
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitHeight>
    <overLength>
      <!--ro, opt, int, over-limit length, range:[1,1000000], unit:cm, desc:the over-limit part equals <length> minus <limitLength>-->4000
    </overLength>
    <overWidth>
      <!--ro, opt, int, over-limit width, range:[1,1000000], unit:cm, desc:the over-limit part equals <width> minus <limitWidth>-->4000
    </overWidth>
    <overHeight>
      <!--ro, opt, int, over-limit height, range:[1,1000000], unit:cm, desc:the over-limit part equals <height> minus <limitHeight>-->4000
    </overHeight>
    <isOverLimit>
      <!--ro, opt, bool, whether it is over-limit, desc:set overweight reporting or overrun reporting as priority via <illegalPriorityType> from
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->false
    </isOverLimit>
    <axleNumConfidence>
      <!--ro, opt, int, confidence of captured number of axles, range:[0,100]-->0
    </axleNumConfidence>
    <standardAxleTypeInfo>
      <!--ro, opt, object-->
      <axleType>
        <!--ro, req, enum, subType:int-->12
      </axleType>
      <axleDescribeInfo>
        <!--ro, req, string, range:[0,64]-->2轴载货汽车
      </axleDescribeInfo>
    </standardAxleTypeInfo>
  </VehicleWeightInfo>
  <isNotSlowZebraCrossing>
    <!--ro, opt, bool, whether the vehicle did not slow down at zebra crossing-->false
  </isNotSlowZebraCrossing>
  <isTurnRightStop>
    <!--ro, opt, bool, whether the vehicle did not stop at a right turn-->false
  </isTurnRightStop>
  <PlateInfoList>
    <!--ro, opt, object, this node is returned when license plates from Mainland, Hongkong, and Maokao are all supported-->
    <PlateInfo>
      <!--ro, opt, object, information of a single license plate-->
      <plateRect>
        <!--ro, opt, object, license plate area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value is
the current image size in percentage multiplying 1000 and it is accurate to three decimal places. The origin is the upper-left corner of screen-->
        <X>
          <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
        </X>
        <Y>
          <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
        </Y>
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        <width>
          <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
        </width>
        <height>
          <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
        </height>
      </plateRect>
      <plateColor>
        <!--ro, req, enum, license plate color, subType:string, desc:license plate color: "white","yellow","blue","black","green","newEnergyGreen"-new
energy green,"newEnergyYellowGreen"-new energy flavogreen,"other"-other color-->black
      </plateColor>
      <licensePlate>
        <!--ro, req, string, license plate number, range:[1,32], desc:license plate number,e.g.,"123456"-->A283KY77
      </licensePlate>
      <confidenceLevel>
        <!--ro, req, int, confidence level, range:[0,100]-->50
      </confidenceLevel>
      <province>
        <!--ro, opt, enum, province index, subType:int, desc:N/A-->0
      </province>
      <isExtraPlate>
        <!--ro, opt, bool-->true
      </isExtraPlate>
    </PlateInfo>
  </PlateInfoList>
  <deviceUUID>
    <!--ro, opt, string, device No., range:[1,32], desc:the device number is the device UUID which is the device's serial number and can be edited by the
node <deviceID> in the message of /ISAPI/System/deviceInfo-->12345
  </deviceUUID>
  <VehicleGATInfo>
    <!--ro, opt, object, N/A, desc:N/A-->
    <palteTypeByGAT>
      <!--ro, opt, enum, N/A, subType:int, desc:N/A-->1
    </palteTypeByGAT>
    <plateColorByGAT>
      <!--ro, opt, enum, N/A, subType:int, desc:0-white, 1-yellow, 2-blue, 3-black, 4-green, 5-not recognized, 9, other colors-->0
    </plateColorByGAT>
    <vehicleTypeByGAT>
      <!--ro, opt, enum, N/A, subType:string, desc:GA/T 16.4 complied: "K10" (large-sized bus), "K20" (medium-sized bus), "K30" (small-sized bus), "K33"
(small-sized car), "H10" (heavy truck), "H20" (medium-sized truck), "H30" (light truck), "M10" (three-wheel motorcycle), "M20" (two-wheel motorcycle), "X99"
(others)-->K10
    </vehicleTypeByGAT>
    <colorByGAT>
      <!--ro, opt, enum, N/A, subType:string, desc:GA/T 16.8 complied: A-white, B-gray, C-yellow, D-pink, E-red, F-purple, G-green, H-blue, I-brown, J-
black, K-not recognized, Z-other colors-->A
    </colorByGAT>
  </VehicleGATInfo>
  <EPlateInfo>
    <!--ro, opt, object, RFID detection result-->
    <EPlateChipNo>
      <!--ro, opt, string, chip No. of electronic license plate (detected by RFID), range:[0,16]-->E855000011F9AAA5
    </EPlateChipNo>
    <EPlateCardNo>
      <!--ro, opt, string, electronic license plate No. (detected by RFID), range:[0,12], desc:electronic license plate No. (12 digits; the first two
represent the province code-->E855000011F9AAA5
    </EPlateCardNo>
    <EPlateNo>
      <!--ro, opt, string, license plate No. (detected by RFID), range:[0,32], desc:"noPlate" (vehicle without license plate detected), "unknown" (license
plate not detected; temporary license plate detected)-->浙A12345
    </EPlateNo>
    <EPlateManufacturingDate>
      <!--ro, opt, string, production date (detected by RFID), range:[0,7], desc:format: yyyy-MM-->2014-01
    </EPlateManufacturingDate>
    <EPlateInspectionValidity>
      <!--ro, opt, string, inspection validity (detected by RFID), range:[0,7], desc:accurate to month; format: yyyy-MM-->2014-01
    </EPlateInspectionValidity>
    <EPlateMandaScrapPeriod>
      <!--ro, opt, int, compulsory scrapping period (detected by RFID), range:[0,100], unit:年, desc:from the start of inspection validity to the compulsory
scrapping time (unit: year)-->10
    </EPlateMandaScrapPeriod>
    <EPlatePassengersCapacity>
      <!--ro, opt, int, authorized passenger capacity (detected by RFID), range:[0,100], desc:e.g. 1-->1
    </EPlatePassengersCapacity>
    <EPlateCarryingCapacity>
      <!--ro, opt, float, tractive tonnage, range:[0,100], unit:吨, desc:tractive tonnage (unit: ton)-->2.0
    </EPlateCarryingCapacity>
    <EPlateVehicleColor>
      <!--ro, opt, enum, vehicle color (detected by RFID), subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue",
"yellow", "green", "brown", "pink", "purple", "deepGray", "cyan", "orange"-->black
    </EPlateVehicleColor>
    <EPlateNonMotorType>
      <!--ro, opt, enum, non-motor vehicle type (detected by RFID), subType:string, desc:"unknown", "threeWheelVehicle" (tricycle), "twoWheelVehicle" (two
wheeler)-->threeWheelVehicle
    </EPlateNonMotorType>
    <EPlateNonMotorPlateType>
      <!--ro, opt, enum, non-motor vehicle license plate type (detected by RFID), subType:string, desc:"unknown", "temporary", "normal"-->unknown
    </EPlateNonMotorPlateType>
    <EPlateColor>
      <!--ro, opt, enum, vehicle color (detected by RFID), subType:string, desc:"black", "blue","golden", "orange", "red", "yellow", "white", "unknown",
"other", "newEnergyYellowGreen"-new energy yellow-green,"civilAviationBlack", "civilAviationGreen" (civil aviation green), "green", "mixedColor",
"newEnergyGreen" (new energy green), "brown"-->black
    </EPlateColor>
    <EPlateUseType>
      <!--ro, opt, enum, non-motor vehicle purpose (detected by RFID), subType:string, desc:"unknown", "nonOperating" (non-operating), "takeOut" (take-out
```

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```
    <!--ro, opt, enum, non-motor vehicle purpose (detected by RFID), subType:string, desc: "unknown", "nonOperating" (non-operating), "takeOut" (take-out delivery), "express" (express delivery)-->nonOperating
    </EPlateUseType>
    <EPlateTriggerTime>
      <!--ro, req, datetime, device detection trigger time of electronic license plate (detected by RFID)-->1970-01-01T00:00:00+08:00
    </EPlateTriggerTime>
    <EPlateListenAuthCode>
      <!--ro, opt, string, licensing authority code of electronic license plate (detected by RFID)-->粤B
    </EPlateListenAuthCode>
  </EPlateInfo>
  <accelerationNoise>
    <!--ro, opt, object, vehicle revving behavior that causes huge noise-->
    <isExistsAccelerationNoise>
      <!--ro, req, bool, whether there is vehicle revving behavior that causes huge noise-->true
    </isExistsAccelerationNoise>
    <startTime>
      <!--ro, req, datetime, start time of the behavior, desc:accurate to millisecond-->1970-01-01T00:00:00.000+08:00
    </startTime>
    <endTime>
      <!--ro, req, datetime, end time of the behavior, desc:accurate to millisecond-->1970-01-01T00:00:00.000+08:00
    </endTime>
  </accelerationNoise>
  <noiseDecibel>
    <!--ro, opt, float, noise decibel of vehicle running, range:[0,300], unit:分贝, desc:accurate to one decimal place-->65.5
  </noiseDecibel>
  <exhaustPipeNum>
    <!--ro, opt, int, number of exhaust pipes, range:[0,4]-->2
  </exhaustPipeNum>
  <exhaustPipePositionType>
    <!--ro, opt, enum, exhaust pipe installation location, subType:string, desc:"unilateral", "bilateral", "middle"-->unilateral
  </exhaustPipePositionType>
  <openGateType>
    <!--ro, opt, enum, reasons for opening the barrier gate, subType:string, dep:and,{$.EventNotificationAlert.ANPR.barrierGateCtrlType,eq,0}, desc:"white" (auto-open barrier gate via device according to the allowlist), "manual" (manually open barrier gate via platform), "abnormal" (auto-open barrier gate via device for exceptional circumstances)-->white
  </openGateType>
  <plateStandardStatus>
    <!--ro, opt, enum, whether the license plate is valid, subType:string, desc:"yes", "no"-->yes
  </plateStandardStatus>
  <limitSurfaceSpeed>
    <!--ro, opt, int, cross-sectional device-detected speed limit (required for overspeed and violation), range:[0,255], unit:km/h-->1
  </limitSurfaceSpeed>
  <startCaptureSpeed>
    <!--ro, opt, int, device-detected capture triggered speed (required for overspeed and violation), range:[0,255], unit:km/h-->1
  </startCaptureSpeed>
  <redlightIllegalCode>
    <!--ro, opt, string, traffic lights and lane matching code for red light running, range:[0,4], desc:0-all lanes, 1-turn left, 2-go straight, 3-turn right, 4-go straight & turn left, 5-go straight & turn right-->1
  </redlightIllegalCode>
  <redlightSignalStartTime>
    <!--ro, opt, string, start time of the red light (required for red light running), desc:ISO 8601 format yyyy-MM-ddTHH:mm:ss+current zone, e.g., 2017-08-01T17:30:08+08:00-->"2004-05-03T17:30:08+08:00"
  </redlightSignalStartTime>
  <redlightSignalEndTime>
    <!--ro, opt, string, end time of the red light (required for red light running), desc:ISO 8601 format yyyy-MM-ddTHH:mm:ss+current zone, e.g., 2017-08-01T17:30:08+08:00-->"2004-05-03T17:30:08+08:00"
  </redlightSignalEndTime>
  <trafficControlTimeSpanList>
    <!--ro, opt, array, time period for traffic restriction, subType:object-->
    <TimeRange>
      <!--ro, req, object, time period-->
      <beginTime>
        <!--ro, req, time, start time-->10:00:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->10:00:00
      </endTime>
    </TimeRange>
  </trafficControlTimeSpanList>
  <detectionBackgroundImageResolution>
    <!--ro, opt, object, -ro, desc:corresponds to the picture represented by <pictureURL> and <pId> when <fileName> is "detectionPicture.jpg" and <type> is "detectionPicture" from <pictureInfoList> and <pictureInfo> in ANPR when the normalized coordinates of <Rect>, <pictureInfoList>, <pictureInfo>, <plateRect>, <vehicelRect>, <PlateInfoList>, <PlateInfo>, and <plateRect> are converted to the resolution coordinates, this node is required-->
    <height>
      <!--ro, req, int, resolution (height), range:[1,65535], unit:px-->1
    </height>
    <width>
      <!--ro, req, int, resolution (width), range:[1,65535], unit:px-->1
    </width>
  </detectionBackgroundImageResolution>
  <trafficSurveyVehicleType>
    <!--ro, opt, enum, subType:int-->1
  </trafficSurveyVehicleType>
  <doorTextInfoList>
    <!--ro, opt, array, subType:object, range:[0,2]-->
    <doorTextInfo>
      <!--ro, opt, object-->
      <doorID>
        <!--ro, opt, int, range:[0,2]-->1
      </doorID>
      <doorTextList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <doorText>
          <!--ro, opt, string, range:[0,128]-->载荷50人
```


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```
        </doorText>
      </doorTextList>
      <approvedPassengers>
        <!--ro, opt, int, range:[1,100]-->7
      </approvedPassengers>
      <doorTextConfidence>
        <!--ro, opt, int, range:[0,100]-->0
      </doorTextConfidence>
    </doorTextInfo>
  </doorTextInfoList>
  <splicingDetectInfo>
    <!--ro, opt, object-->
    <startSplicingTime>
      <!--ro, req, datetime-->1970-01-01T00:00:00 000+08:00
    </startSplicingTime>
    <endSplicingTime>
      <!--ro, req, datetime-->1970-01-01T00:00:00 000+08:00
    </endSplicingTime>
  </splicingDetectInfo>
</EventNotificationAlert>
```

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| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| anpr.xml | [Message content] | application/xml | -- | -- | -- |
| .value | [Binary picture data] | image/jpeg | detectionPicture | detectionPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | licensePlatePicture | licensePlatePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | pilotPicture | pilotPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | copilotPicture | copilotPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | compositePicture | compositePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | plateBinaryPicture | plateBinaryPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | nonMotorPicture | nonMotorPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | pedestrianDetectionPicture | pedestrianDetectionPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | vehiclePicture | vehiclePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | featurePicture | featurePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | panoramaPicture | panoramaPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | extraLicensePlatePicture | extraLicensePlatePicture.jpg | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

# 14.6 Traffic Control

## 14.6.1 Traffic Enforcement of Vehicles

### 14.6.1.1 Get the column speaker parameters

**Request URL**

GET /ISAPI/System/columnSpeakers?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
{
    "ColumnSpeakers": [
    /*ro, req, array, column speaker parameters, subType:object*/
        {
            "enabled":  true,
            /*ro, req, bool, whether to enable*/
            "connectionMode":  "network",
            /*ro, opt, enum, subType:string*/
            "485Port":  1,
            /*ro, opt, int, RS-485 port No., range:[1,4], desc:RS-485 port No.*/
            "ipv4":  "10.0.0.1",
            /*ro, opt, string, range:[1,17]*/
            "port":  34508,
            /*ro, opt, int, range:[1,65535]*/
            "authenticationEnabled":  false,
            /*ro, opt, bool*/
            "userName":  "test",
            /*ro, opt, string, range:[0,32], dep:and,{$.ColumnSpeakers[*].authenticationEnabled,eq,true}*/
            "485VoiceSpeed":  1,
            /*ro, opt, int, range:[1,9]*/
            "playFileList": ["1.mp3", "2.mp3", "3.wav"],
            /*ro, opt, array, subType:string, range:[0,16]*/
            "MessageList": [
            /*ro, opt, array, information list, subType:object, desc:information list*/
                {
                    "id":  1,
                    /*ro, req, int, ID, desc:it starts from 1*/
                    "message":  "test",
                    /*ro, req, string, text information, range:[0,128], desc:text information*/
                    "volume":  0,
                    /*ro, req, int, volume, range:[0,9]*/
                    "speaker":  1,
                    /*ro, req, enum, voice type, subType:int, desc:1 (male voice 1), 2 (male voice 2), 3 (female voice 1), 4 (female voice 2), 5 (child
voice)*/
                    "voiceSpeed":  1,
                    /*ro, opt, int, range:[0,9]*/
                    "repeatNum":  1,
                    /*ro, opt, int, range:[1,100]*/
                    "playType":  0,
                    /*ro, opt, enum, subType:int*/
                    "codes": [1, 2],
                    /*ro, req, array, violation code, subType:int, range:[0,2]*/
                    "channelID": [1, 2, 3]
                    /*ro, opt, array, subType:int, range:[1,64]*/
                }
            ],
            "speakerID":  1,
            /*ro, opt, int, column speaker ID, range:[1,6], desc:column speaker ID*/
            "schedule": {
            /*ro, opt, object*/
                "enabled":  true,
                /*ro, opt, bool*/
                "timeRange": [
                /*ro, opt, array, subType:object, range:[0,2]*/
                    {
                        "startTime":  "00:00",
                        /*ro, opt, string, range:[0,64]*/
                        "endTime":  "24:00"
                        /*ro, opt, string, range:[0,64]*/
                    }
                ]
            },
            "broadcastTimes":  5
            /*ro, opt, int, range:[0,15]*/
        }
    ]
}
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| | [Message content] | application/json | -- | -- | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit

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name.

- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

#### 14.6.1.2 Get the traffic service capability

**Request URL**

GET /ISAPI/Traffic/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<TrafficCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{opt, string, protocolVersion}-->
  <plateCap>
    <!--ro, opt, object, capability of license plate recognition-->
    <isSupportPlateList>
      <!--ro, opt, bool, whether it supports importing and exporting the license plate blocklist and allowlist, desc:whether it supports importing and exporting the license plate blocklist and allowlist-->true
    </isSupportPlateList>
    <isSupportPlateListAsync>
      <!--ro, opt, bool-->true
    </isSupportPlateListAsync>
  </plateCap>
  <isNotSupportImageMerge>
    <!--ro, opt, bool, whether it supports picture overlay, desc:whether it does not support picture overlay-->true
  </isNotSupportImageMerge>
</TrafficCap>
```

#### 14.6.1.3 Get the parameters of queue jumping detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/jamCrossLine

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<JamCrossLineEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the parameters of queue jumping detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--ro, opt, int, duration of fallen object detection, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--ro, opt, int, alarm interval, range:[1,120], unit:s-->30
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, range:[0,1]-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, req, int, capture interval, range:[1,3600]-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times, range:[0,1]-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, req, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, related lane No., range:[1,6]-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number, range:[1,8]-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <congestionThreshold>
    <!--ro, opt, int, range:[1,100]-->50
  </congestionThreshold>
  <sensitivity>
    <!--ro, opt, int, range:[1,100]-->50
  </sensitivity>
</JamCrossLineEvent>
```

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#### 14.6.1.4 Set the parameters of queue jumping detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/jamCrossLine

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

## Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<JamCrossLineEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, the parameters of queue jumping detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--req, int, duration of fallen object detection, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--opt, int, alarm interval, range:[1,120], unit:s-->30
  </filteringTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters, range:[0,1]-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval, range:[1,3600]-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:"far" (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times, range:[0,1]-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--req, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, related lane No., range:[1,6]-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space No., subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space No., range:[1,8]-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
</JamCrossLineEvent>
```

## Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.1.5 Get the configuration capability of queue jumping detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/jamCrossLine/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<JamCrossLineEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the parameters of queue jumping detection, attr:version{req, int, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false, true">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="60" def="10">
    <!--ro, opt, int, duration, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <filteringTime min="1" max="120" def="30">
    <!--ro, opt, int, alarm interval, unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="1">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="1" def="1">
          <!--ro, req, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:far (distant view)-->far
        </imgType>
        <times min="3" max="3" def="3">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->3
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, req, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, related lane No., subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, req, int, related lane No., attr:min{opt, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{opt, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <congestionThreshold min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->50
  </congestionThreshold>
  <sensitivity min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->50
  </sensitivity>
</JamCrossLineEvent>
```

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#### 14.6.1.6 Get the configuration capability of continuous lane change detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/changeLaneContinuously/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "ChangeLaneContinuouslyEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, range of scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "enabled": {
        /*ro, req, object, whether to enable continuous lane change detection*/
            "@opt": ["true", "false"]
            /*ro, req, array, options, subType:string*/
        },
        "vehicleTravelTime": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, unit:s*/
            "@max":  180,
            /*ro, req, int, unit:s*/
            "@def":  50
            /*ro, opt, int, unit:s*/
        },
        "SnapImageParam": {
        /*ro, opt, object, parameters for capturing pictures*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, req, object, capture times*/
                "@min":  3,
                /*ro, req, int, the minimum value*/
                "@max":  3,
                /*ro, req, int, the maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        },
        "vehicleDrivingDistance": {
        /*ro, opt, object, vehicle traveling distance*/
            "@min":  1,
            /*ro, req, int, the minimum value, range:[1,200], unit:m*/
            "@max":  200,
            /*ro, req, int, the maximum value, range:[1,200], unit:m*/
            "@def":  50
            /*ro, req, int, the default value, range:[1,200], unit:m*/
        },
        "triggerTimes": {
        /*ro, opt, object*/
            "@min":  2,
            /*ro, req, int, range:[2,4]*/
            "@max":  4,
            /*ro, req, int, range:[2,4]*/
            "@def":  2
            /*ro, opt, int, range:[2,4]*/
        }
    }
}
```

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### 14.6.1.7 Get parameters of continuous lane change detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/changeLaneContinuously?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "ChangeLaneContinuouslyEvent": {
    /*ro, req, object*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable continuous lane change detection*/
        "vehicleTravelTime":  0,
        /*ro, opt, int*/
        "SnapImageParam": {
        /*ro, opt, object, parameters for capturing pictures*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "vehicleDrivingDistance":  0,
        /*ro, opt, int, vehicle traveling distance, range:[1,200], unit:m*/
        "triggerTimes":  2
        /*ro, opt, int, range:[2,4]*/
    }
}
```

### 14.6.1.8 Set parameters of continuous lane change detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/changeLaneContinuously?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

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```
{
    "ChangeLaneContinuouslyEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable continuous lane change detection*/
        "vehicleTravelTime":  0,
        /*req, int*/
        "SnapImageParam": {
        /*req, object, parameters for capturing pictures*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        },
        "vehicleDrivingDistance":  0
        /*opt, int, vehicle traveling distance, unit:m*/
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

### 14.6.1.9 Get the configuration parameters of driving on the lane line detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/crossLane

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<CrossLaneEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration parameters of driving on the lane line detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--ro, opt, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, opt, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, opt, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:"far" (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <leanRoadSnapSentivity>
      <!--ro, opt, int, range:[1,100]-->1
    </leanRoadSnapSentivity>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, related lane No.-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space No., subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space No.-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <alertEnabled>
    <!--ro, opt, bool-->true
  </alertEnabled>
</CrossLaneEvent>
```

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### 14.6.1.10 Set the parameters for driving on the lane line detection

#### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/crossLane

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CrossLaneEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, coordinate ratio between the upper level and the device, the default value is 1000, see details in the message of
XML_NormalizedScreenSize, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:"far" (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <leanRoadSnapSentivity>
      <!--opt, int, range:[1,100]-->1
    </leanRoadSnapSentivity>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, related lane No.-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space No., subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space No.-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <alertEnabled>
    <!--opt, bool-->true
  </alertEnabled>
</CrossLaneEvent>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.1.11 Get the configuration capability of driving on the lane line detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/crossLane/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<CrossLaneEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of driving on the lane line detection, attr:version{req, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="60" def="10">
    <!--ro, opt, int, detection duration, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="1">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, opt, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="1" def="1">
          <!--ro, opt, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:"far" (distant view)-->far
        </imgType>
        <times min="3" max="3" def="3">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->3
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <leanRoadSnapSentivity min="1" max="100" def="50">
      <!--ro, opt, int, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
    </leanRoadSnapSentivity>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, req, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, related lane No. list, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, req, int, related lane No., attr:min{opt, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space No., subType:object, attr:size{opt, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space No., attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <alertEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
  </alertEnabled>
</CrossLaneEvent>
```

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### 14.6.1.12 Get the configuration capability of street racing detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/dragRacing/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "DragRacingEventCap": {
    /*ro, req, object, undefined*/
        "scenesID": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, minimum string length*/
            "@max":  16
            /*ro, req, int, maximum string length*/
        },
        "enabled": {
        /*ro, req, object, whether to enable the function*/
            "@opt": [true, false]
            /*ro, req, array, value range, subType:bool*/
        },
        "SnapImageParam": {
        /*ro, req, object, picture capturing parameters*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, minimum value*/
                "@max":  100,
                /*ro, req, int, maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, req, object, capture times*/
                "@min":  3,
                /*ro, req, int, minimum value*/
                "@max":  3,
                /*ro, req, int, maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        },
        "vehicleSpeed": {
        /*ro, opt, object, vehicle speed threshold*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  300,
            /*ro, req, int, maximum value*/
            "@def":  100
            /*ro, req, int, the default value*/
        },
        "overspeedVehicleNumber": {
        /*ro, opt, object, number of vehicles overspeeding simultaneously*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  10
            /*ro, req, int, maximum value*/
        }
    }
}
```

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#### 14.6.1.13 Get the parameters of street racing detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/dragRacing?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "DragRacingEvent": {
    /*ro, req, object*/
        "scenesID":  1,
        /*ro, req, int, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "SnapImageParam": {
        /*ro, req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "vehicleSpeed":  0,
        /*ro, opt, int, vehicle speed threshold, unit:km/h*/
        "overspeedVehicleNumber":  10
        /*ro, opt, int, number of vehicles overspeeding simultaneously*/
    }
}
```

#### 14.6.1.14 Set the parameters of street racing detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/dragRacing?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "DragRacingEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable street racing detection*/
        "SnapImageParam": {
        /*req, object, parameters of capturing pictures*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        },
        "vehicleSpeed":  0,
        /*opt, int, vehicle speed threshold, unit:km/h*/
        "overspeedVehicleNumber":  10
        /*opt, int, number of vehicles overspeeding simultaneously*/
    }
}
```

**Response Message**

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

#### 14.6.1.15 Set the parameters for illegal parking detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>
<IllegalParkingEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, the parameters of illegal parking detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <durationTime>
    <!--opt, int, parking duration, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--opt, int, filtering time, unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--opt, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string-->far
        </imgType>
        <times>
          <!--opt, int-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--opt, object, parameters of the linked target areas-->
  </RelatedAreaParam>
</IllegalParkingEvent>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.6.1.16 Get the parameters of illegal parking detection

#### Request URL

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IllegalParkingEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the parameters of illegal parking detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable the detection-->true
  </enabled>
  <amplifyDetectionEnabled>
    <!--ro, opt, bool-->true
  </amplifyDetectionEnabled>
  <durationTime>
    <!--ro, opt, int, duration of fallen object detection, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--ro, opt, int, filtering time, unit:min-->1
  </filteringTime>
  <noPlateEnable>
    <!--ro, opt, bool, whether to capture vehicles without license plate-->true
  </noPlateEnable>
  <plateMatchRatio>
    <!--ro, opt, int, license plate match ratio-->1
  </plateMatchRatio>
  <plateRecogEnabled>
    <!--ro, opt, bool, whether to enable license plate recognition for illegal parking events-->true
  </plateRecogEnabled>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, opt, array, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object-->
        <imgInterval>
          <!--ro, opt, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:picture type-->far
```

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```
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <congestionThreshold>
      <!--ro, opt, int, range:[0,100]-->70
    </congestionThreshold>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <DetectionRegionList>
    <!--ro, opt, array, area list for illegal parking detection, subType:object, range:[1,16]-->
    <DetectionRegion>
      <!--ro, opt, object, a single area for illegal parking detection-->
      <regionID>
        <!--ro, opt, int, range:[1,16]-->1
      </regionID>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegion>
  </DetectionRegionList>
  <RelatedAreaParam>
    <!--ro, opt, object, parameters of the linked target areas-->
    <relatedAreaNo>
      <!--ro, opt, string, linked area numbers, desc:multiple numbers should be separated by commas, e.g., 1,2,3,4.-->test
    </relatedAreaNo>
  </RelatedAreaParam>
  <illegalCodeIndex>
    <!--ro, opt, string-->test
  </illegalCodeIndex>
  <LinkPTZChannel>
    <!--ro, opt, object, whether to enable evidence capture of PTZ camera when illegal parking is detected by box camera (the PTZ camera will immediately be
switched to the configured scene)-->
    <enabled>
      <!--ro, req, bool, whether to enable the detection-->true
    </enabled>
    <scenesID>
      <!--ro, req, string, scene ID, range:[1,16]-->test
    </scenesID>
  </LinkPTZChannel>
  <congestionNoAlarmEnabled>
    <!--ro, opt, bool, whether to disable alarms during congestion-->true
  </congestionNoAlarmEnabled>
  <relatedShieldRegionNo>
    <!--ro, opt, string, numbers of linked shielded areas, desc:numbers of linked shielded areas-->1
  </relatedShieldRegionNo>
  <longDistanceTargetRuleRectEnabled>
    <!--ro, opt, bool-->false
  </longDistanceTargetRuleRectEnabled>
  <alertEnabled>
    <!--ro, opt, bool-->true
  </alertEnabled>
  <relatedPlateColor>
    <!--ro, opt, string-->other
  </relatedPlateColor>
  <relatedVehicleType>
    <!--ro, opt, string-->vehicle
  </relatedVehicleType>
  <alertUploadMoment>
```

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```
    <!--ro, opt, enum, subType:string, dep:and,{$.IllegalParkingEvent.alertEnabled,eq,true}-->accident
  </alertUploadMoment>
  <relatedType>
    <!--ro, opt, enum, subType:string-->area
  </relatedType>
  <quickCollectionEnabled>
    <!--ro, opt, bool-->false
  </quickCollectionEnabled>
  <quickCollectionImageList>
    <!--ro, opt, array, subType:object, dep:and,{$.IllegalParkingEvent.quickCollectionEnabled,eq,true}-->
    <quickCollectionImage>
      <!--ro, opt, object-->
      <imageID>
        <!--ro, req, int, range:[1,4]-->1
      </imageID>
      <imgInterval>
        <!--ro, req, int, range:[0,180]-->1
      </imgInterval>
      <imgType>
        <!--ro, req, enum, subType:string-->far
      </imgType>
    </quickCollectionImage>
  </quickCollectionImageList>
</IllegalParkingEvent>
```

### 14.6.1.17 Get the configuration capability of illegal parking detection

#### Request URL

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking/capabilities

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

#### Request Message

None

#### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IllegalParkingEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the parameters of illegal parking detection, attr:version{req, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <amplifyDetectionEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </amplifyDetectionEnabled>
  <durationTime min="1" max="60" def="10">
    <!--ro, opt, int, parking duration, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <filteringTime min="1" max="180" def="30">
    <!--ro, opt, int, filtering time, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <plateMatchRatio min="0" max="100">
    <!--ro, opt, int, plate matching ratio, attr:min{req, int},max{req, int}-->1
  </plateMatchRatio>
  <plateRecogEnabled opt="false,true">
    <!--ro, opt, bool, whether to enable license plate recognition for illegal parking event, attr:opt{req, string}-->true
  </plateRecogEnabled>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="4">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="4">
      <!--ro, opt, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="1" def="1">
          <!--ro, opt, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="near,far,feature,custom,middle">
```

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```
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:"near" (close view), "far" (distant view), "feature", "custom"-->far
        </imgType>
        <times min="1" max="3" def="2">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->3
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <congestionThreshold min="0" max="100" def="70">
      <!--ro, opt, int, range:[0,100], unit:s, attr:min{req, int},max{req, int},def{req, int}-->70
    </congestionThreshold>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, linked lane number, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, req, int, linked lane number, attr:min{opt, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{opt, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <DetectionRegionList size="16">
    <!--ro, opt, array, area list for illegal parking detection, subType:object, attr:size{req, int}-->
    <DetectionRegion>
      <!--ro, opt, object, a single area for illegal parking detection-->
      <regionID min="1" max="16">
        <!--ro, opt, int, range:[1,16], attr:min{req, int},max{req, int}-->1
      </regionID>
      <RegionCoordinatesList min="3" max="10">
        <!--ro, opt, array, area coordinates, subType:object, range:[3,10], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of
the screen-->
        <RegionCoordinates>
          <!--ro, opt, object, coordinates-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegion>
  </DetectionRegionList>
  <RelatedAreaParam>
    <!--ro, opt, object-->
    <relatedAreaNo opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16">
      <!--ro, opt, string, linked area No., attr:opt{req, string}-->test
    </relatedAreaNo>
  </RelatedAreaParam>
  <illegalCodeIndex opt="1,8,9,10">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </illegalCodeIndex>
  <LinkPTZChannel>
    <!--ro, opt, object, whether to enable evidence capture of PTZ camera when illegal parking is detected by box camera (the PTZ camera will immediately be
switched to the configured scene), desc:evidence capture of PTZ camera,when illegal parking is detected by box camera,the PTZ camera will immediately switch
to the configured scene; make sure the patrol schedule of PTZ camera has been configured and enabled,and make sure the scene has been added to the schedule
and illegal parking detection has been configured in this scene-->
    <enabled opt="false,true" def="false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string},def{req, string}-->true
    </enabled>
    <scenesID min="1" max="16">
      <!--ro, req, string, scene ID, range:[1,16], attr:min{req, string},max{req, string}-->test
    </scenesID>
  </LinkPTZChannel>
  <congestionNoAlarmEnabled opt="true,false" def="false">
    <!--ro, opt, bool, whether to disable alarms during congestion, attr:opt{req, string},def{req, string}, desc:whether to disable alarm during congestion:
true-disable alarm during congestion,false-enable alarm during congestion (default)-->true
  </congestionNoAlarmEnabled>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, linked shielded area No., attr:min{req, int},max{req, int}, desc:linked shielded area No.-->1
  </relatedShieldRegionNo>
  <longDistanceTargetRuleRectEnabled opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->false
  </longDistanceTargetRuleRectEnabled>
  <alertEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
  </alertEnabled>
  <relatedPlateColor opt="yellow,blue,green,other">
    <!--ro, opt, string, attr:opt{req, string}-->other
  </relatedPlateColor>
```

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```
    <relatedVehicleType opt="vehicle,van,truck,bus,unknown">
      <!--ro, opt, string, attr:opt{req, string}-->bus
    </relatedVehicleType>
    <alertUploadMoment opt="accident,imageOne,imageTwo,imageThree,imageFour">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->accident
    </alertUploadMoment>
    <relatedType opt="area,lane">
      <!--ro, opt, string, attr:opt{opt, string}-->area
    </relatedType>
    <quickCollectionEnabled opt="true,false" def="false">
      <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->false
    </quickCollectionEnabled>
    <quickCollectionImageList size="4">
      <!--ro, opt, object, dep:and,{$.IllegalParkingEvent.quickCollectionEnabled,eq,true}, attr:size{req, int}-->
      <quickCollectionImag>
        <!--ro, opt, object-->
        <imageID min="1" max="4">
          <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->1
        </imageID>
        <imgInterval min="0" max="180" def="0">
          <!--ro, req, int, range:[1,180], attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far,near,feature,middle">
          <!--ro, req, enum, subType:string, attr:opt{req, string}-->far
        </imgType>
      </quickCollectionImag>
    </quickCollectionImageList>
  </IllegalParkingEvent>
```

#### 14.6.1.18 Get the configuration parameters of illegal lane change detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/laneChange

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<LaneChangeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, ro, coordinate ratio between the upper level and the device, the default value is 1000, see details in the message of
XML_NormalizedScreenSize, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <durationTime>
    <!--ro, opt, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <congestionLevel>
    <!--ro, opt, int, degree of congestion, range:[0,5]-->1
  </congestionLevel>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, opt, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters, desc:linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <alertEnabled>
    <!--ro, opt, bool-->true
  </alertEnabled>
  <RelatedAreaParam>
    <!--ro, opt, object-->
    <relatedAreaNo>
      <!--ro, opt, string-->2,3
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedType>
    <!--ro, opt, enum, subType:string-->area
  </relatedType>
</LaneChangeEvent>
```

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### 14.6.1.19 Set the parameters for illegal lane change detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/laneChange

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LaneChangeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, illegal lane change detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID (1 to 16)-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--opt, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <congestionLevel>
    <!--opt, int, degree of congestion, range:[0,5]-->1
  </congestionLevel>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--opt, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:"far" (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, linked lane No.-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, subType:object-->
        <relatedParkingNo>
          <!--opt, int-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <alertEnabled>
    <!--opt, bool-->true
  </alertEnabled>
  <RelatedAreaParam>
    <!--opt, object-->
    <relatedAreaNo>
      <!--opt, string-->2,3
    </relatedAreaNo>
  </RelatedAreaParam>
</LaneChangeEvent>
```

### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.1.20 Get the configuration capability of illegal lane change detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/laneChange/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<LaneChangeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="60" def="10">
    <!--ro, opt, int, duration, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="4">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="4">
      <!--ro, opt, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="1" def="1">
          <!--ro, req, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:far (distant view)-->far
        </imgType>
        <times min="1" max="3" def="2">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->3
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, req, object, linked lane parameters, desc:linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, related lane No., subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, req, int, related lane No., attr:min{opt, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{opt, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <alertEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
  </alertEnabled>
  <RelatedAreaParam>
    <!--ro, opt, object-->
    <relatedAreaNo opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16">
      <!--ro, opt, string, attr:opt{req, string}-->2,3
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedType opt="area,lane">
    <!--ro, opt, string, attr:opt{opt, string}-->area
  </relatedType>
</LaneChangeEvent>
```

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### 14.6.1.21 Get the configuration capability of lane occupation by large-sized vehicle detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/largeVehicleOccupyLine/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "LargeVehicleOccupyLineEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "enabled": {
        /*ro, req, object, whether to enable lane occupation by large-sized vehicle detection*/
            "@opt": ["true", "false"]
            /*ro, req, array, subType:string*/
        },
        "SnapImageParam": {
        /*ro, req, object, picture capturing parameters*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, req, object, capture times*/
                "@min":  3,
                /*ro, req, int, the minimum value*/
                "@max":  3,
                /*ro, req, int, the maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        },
        "RelatedLaneParam": {
        /*ro, req, object, linked lane parameters*/
            "RelatedLaneNoList": {
            /*ro, opt, object*/
                "@size":  6,
                /*ro, req, int, number of linked lanes*/
                "relatedLaneNo": {
                /*ro, req, object, range of linked lane No.*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  6
                    /*ro, req, int, the maximum value*/
                },
                "RelatedParkingNoList": {
                /*ro, opt, object, list of linked parking space numbers*/
                    "@size":  8,
                    /*ro, req, int, number of linked lanes*/
                    "relatedParkingNo": {
                    /*ro, req, object, linked parking space number*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  8
                        /*ro, req, int, the maximum value*/
                    }
                }
            }
        }
    }
}
```

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### 14.6.1.22 Get the parameters of lane occupation by large-sized vehicle detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/largeVehicleOccupyLine?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "LargeVehicleOccupyLineEvent": {
    /*ro, req, object, undefined*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "SnapImageParam": {
        /*ro, req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "durationTime":  0,
        /*ro, opt, int, step:1, unit:s*/
        "RelatedLaneParam": {
        /*ro, req, object, linked lane parameters*/
            "RelatedLaneNoList": [
            /*ro, opt, array, subType:object*/
                {
                    "relatedLaneNo":  1,
                    /*ro, req, int, linked lane number*/
                    "RelatedParkingNoList": [
                    /*ro, opt, array, list of linked parking space numbers, subType:object*/
                        {
                            "relatedParkingNo":  1
                            /*ro, req, int, linked parking space number*/
                        }
                    ]
                }
            ]
        }
    }
}
```

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### 14.6.1.23 Set the parameters of the detection of lane occupation by large vehicles

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/largeVehicleOccupyLine?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "LargeVehicleOccupyLineEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable the detection of lane occupation by large vehicles*/
        "SnapImageParam": {
        /*req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        },
        "durationTime":  0,
        /*opt, int, step:1, unit:s*/
        "RelatedLaneParam": {
        /*req, object, linked lane parameters*/
            "RelatedLaneNoList": [
            /*opt, array, list of linked lane, subType:object*/
                {
                    "relatedLaneNo":  1,
                    /*req, int, linked lane No.*/
                    "RelatedParkingNoList": [
                    /*opt, array, list of linked parking space No., subType:object*/
                        {
                            "relatedParkingNo":  1
                            /*req, int, linked parking space No.*/
                        }
                    ]
                }
            ]
        }
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

### 14.6.1.24 Get the configuration capability of driving in low speed detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/lowSpeed/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "LowSpeedEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, scene ID*/
            "@min":  1,
```

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```
        /*ro, req, int, the minimum value*/
        "@max":  16
        /*ro, req, int, the maximum value*/
    },
    "enabled": {
    /*ro, req, object, whether to enable driving in low speed detection*/
        "@opt": ["true", "false"]
        /*ro, req, array, subType:string*/
    },
    "SnapImageParam": {
    /*ro, opt, object, parameters for capturing pictures*/
        "snapSentivity": {
        /*ro, req, object, capture sensitivity*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100,
            /*ro, req, int, the maximum value*/
            "@def":  50
            /*ro, req, int, the default value*/
        },
        "times": {
        /*ro, req, object, capture times*/
            "@min":  3,
            /*ro, req, int, the minimum value*/
            "@max":  3,
            /*ro, req, int, the maximum value*/
            "@def":  3
            /*ro, req, int, the default value*/
        }
    },
    "lowSpeed": {
    /*ro, opt, object, minimum vehicle speed threshold*/
        "@min":  0,
        /*ro, req, int, the minimum value*/
        "@max":  300,
        /*ro, req, int, the maximum value*/
        "@def":  100
        /*ro, req, int, the default value*/
    },
    "vehicleLaneParam": {
    /*ro, opt, object*/
        "laneList": {
        /*ro, opt, object*/
            "@size":  2,
            /*ro, opt, int*/
            "@minSize":  2,
            /*ro, opt, int*/
            "laneID": {
            /*ro, req, object*/
                "@def":  1,
                /*ro, opt, int*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  2
                /*ro, opt, int*/
            },
            "line": {
            /*ro, opt, object*/
                "@size":  2,
                /*ro, opt, int*/
                "@minSize":  1,
                /*ro, opt, int*/
                "positionX": {
                /*ro, req, object*/
                    "@def":  0.001,
                    /*ro, opt, float*/
                    "@min":  0.0,
                    /*ro, opt, float*/
                    "@max":  1.0
                    /*ro, opt, float*/
                },
                "positionY": {
                /*ro, req, object*/
                    "@def":  1.0,
                    /*ro, opt, float*/
                    "@min":  0.0,
                    /*ro, opt, float*/
                    "@max":  1.0
                    /*ro, opt, float*/
                }
            },
            "enabled": {
            /*ro, opt, object*/
                "@opt": ["true", "false"],
                /*ro, req, array, subType:string*/
                "@def":  true
                /*ro, opt, bool*/
            },
            "speedLimit": {
            /*ro, opt, object*/
                "@min":  20,
                /*ro, opt, int, range:[20,200], unit:km/h*/
                "@max":  200,
```

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```
                    /*ro, opt, int, range:[20,200], unit:km/h*/
                    "@def":  60
                    /*ro, opt, int, range:[20,200], unit:km/h*/
                }
            },
            "adjacentLaneDistance": {
            /*ro, opt, object*/
                "@def":  20,
                /*ro, opt, int*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  300
                /*ro, opt, int*/
            }
        }
    }
}
```

#### 14.6.1.25 Set parameters of driving in low speed detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/lowSpeed?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "LowSpeedEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable driving in low speed detection*/
        "SnapImageParam": {
        /*req, object, parameters for capturing pictures*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        },
        "lowSpeed":  0
        /*opt, int, minimum vehicle speed threshold, unit:km/h*/
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

#### 14.6.1.26 Get the parameters of driving in low speed detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/lowSpeed?format=json

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "LowSpeedEvent": {
    /*ro, req, object*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable driving in low speed detection*/
        "frontVehicleDistance":  0,
        /*ro, opt, int, step:1, unit:m*/
        "SnapImageParam": {
        /*ro, opt, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "lowSpeed":  0,
        /*ro, opt, int, minimum vehicle speed, unit:km/h*/
        "vehicleLaneParam": {
        /*ro, opt, object*/
            "laneList": [
            /*ro, opt, array, subType:object*/
                {
                    "laneID":  1,
                    /*ro, req, int, range:[1,2]*/
                    "line": [
                    /*ro, opt, array, subType:object*/
                        {
                            "positionX":  0.001,
                            /*ro, req, float, range:[0,1]*/
                            "positionY":  1.0
                            /*ro, req, float, range:[0,1]*/
                        }
                    ],
                    "enabled":  true,
                    /*ro, opt, bool*/
                    "speedLimit":  20
                    /*ro, opt, int, range:[20,200], unit:km/h*/
                }
            ],
            "adjacentLaneDistance": 20
            /*ro, opt, int, range:[1,300], unit:m*/
        }
    }
}
```

### 14.6.1.27 Get the configuration capability of not keeping a safe vehicle distance detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/notKeepDistance/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

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### Request Message

None

### Response Message

```
{
    "NotKeepDistanceEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "enabled": {
        /*ro, req, object, whether to enable not keeping a safe vehicle distance detection*/
            "@opt": ["true", "false"]
            /*ro, req, array, options, subType:string*/
        },
        "SnapImageParam": {
        /*ro, req, object, parameters for capturing pictures*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, req, object, capture times*/
                "@min":  3,
                /*ro, req, int, the minimum value*/
                "@max":  3,
                /*ro, req, int, the maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        },
        "vehicleSpeed": {
        /*ro, opt, object, vehicle speed threshold*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  300,
            /*ro, req, int, the maximum value*/
            "@def":  100
            /*ro, req, int, the default value*/
        },
        "lowSpeedDistance": {
        /*ro, opt, object, vehicle distance at a low speed*/
            "@min":  10,
            /*ro, req, int, the minimum value*/
            "@max":  200,
            /*ro, req, int, the maximum value*/
            "@def":  50
            /*ro, req, int, the default value*/
        },
        "highSpeedDistance": {
        /*ro, opt, object, vehicle distance at a high speed*/
            "@min":  10,
            /*ro, req, int, the minimum value*/
            "@max":  200,
            /*ro, req, int, the maximum value*/
            "@def":  100
            /*ro, req, int, the default value*/
        },
        "displayVehicleSpeed": {
        /*ro, opt, object, whether to display the vehicle distance*/
            "@opt": ["true", "false"]
            /*ro, req, array, options, subType:string*/
        }
    }
}
```

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#### 14.6.1.28 Set the parameters of not keeping vehicle distance detection

#### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/notKeepDistance?format=json

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "NotKeepDistanceEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, range of scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable not keeping vehicle distance detection*/
        "SnapImageParam": {
        /*req, object, parameters for capturing pictures*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        },
        "vehicleSpeed":  0,
        /*opt, int, vehicle speed threshold, unit:km/h*/
        "lowSpeedDistance":  10,
        /*opt, int, vehicle distance in low speed, unit:m*/
        "highSpeedDistance":  10,
        /*opt, int, vehicle distance in high speed, unit:m*/
        "displayVehicleSpeed":  true
        /*opt, bool, whether to display the vehicle distance*/
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
}
```

### 14.6.1.29 Get the parameters of not keeping a safe distance detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/notKeepDistance?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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dojo.jin@gmail.com

```
{
    "NotKeepDistanceEvent": {
    /*ro, req, object, not keeping a safe distance detection*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "SnapImageParam": {
        /*ro, req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "vehicleSpeed":  0,
        /*ro, opt, int, vehicle speed threshold, unit:km/h*/
        "lowSpeedDistance":  10,
        /*ro, opt, int, vehicle distance at low speed, unit:m*/
        "highSpeedDistance":  10,
        /*ro, opt, int, vehicle distance at high speed, unit:m*/
        "displayVehicleSpeed":  true
        /*ro, opt, bool, whether to display the vehicle distance*/
    }
}
```

#### 14.6.1.30 Get the configuration capability of not slowing down at zebra crossing detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/notSlowZebraCrossing/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

do jojo@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

```
{
    "NotSlowZebraCrossingEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "enabled": {
        /*ro, req, object, whether to enable the function*/
            "@opt": ["true", "false"]
            /*ro, req, array, range, subType:string*/
        },
        "SnapImageParam": {
        /*ro, req, object, picture capturing parameters*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, opt, object, capture times*/
                "@min":  3,
                /*ro, req, int, the minimum value*/
                "@max":  3,
                /*ro, req, int, the maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        },
        "VehicleLaneParam": {
        /*ro, req, object, lane line parameters*/
            "LaneNoList": {
            /*ro, opt, object*/
                "@size":  3,
                /*ro, req, int, number of lane lines*/
                "id": {
                /*ro, req, object, lane line No.*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  3
                    /*ro, req, int, the maximum value*/
                }
            }
        },
        "lowSpeed": {
        /*ro, opt, object, the minimum speed limit*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  200,
            /*ro, req, int, the maximum value*/
            "@def":  10
            /*ro, req, int, the default value*/
        },
        "highSpeed": {
        /*ro, opt, object, Maximum Speed*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  200,
            /*ro, req, int, the maximum value*/
            "@def":  40
            /*ro, req, int, the default value*/
        }
    }
}
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

#### 14.6.1.31 Get the parameters of not slowing down at zebra crossing detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/notSlowZebraCrossing?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "NotSlowZebraCrossingEvent": {
    /*ro, req, object*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable*/
        "SnapImageParam": {
        /*ro, req, object, parameters for capturing pictures*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "VehicleLaneParam": {
        /*ro, req, object, lane line parameters*/
            "LaneList": [
            /*ro, opt, array, lane list, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, lane line No.*/
                    "Line": [
                    /*ro, opt, array, coordinate information of lane lines, subType:object, range:[0,2], desc:the origin is the upper-left corner of the
screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            ]
        },
        "lowSpeed":  10,
        /*ro, opt, int, minimum speed, unit:km/h*/
        "highSpeed":  10
        /*ro, opt, int, maximum speed, unit:km/h*/
    }
}
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

#### 14.6.1.32 Set the parameters of not slowing down at zebra crossing detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/notSlowZebraCrossing?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "NotSlowZebraCrossingEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable the function*/
        "SnapImageParam": {
        /*req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        },
        "VehicleLaneParam": {
        /*req, object, lane line parameters*/
            "LaneList": [
            /*opt, array, subType:object*/
                {
                    "id":  1,
                    /*req, int, lane line No.*/
                    "Line": [
                    /*opt, array, coordinate information about lane lines, subType:object, range:[0,2], desc:the origin is the upper-left corner of the
screen*/
                        {
                            "x":  0.120,
                            /*req, float, x-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*req, float, y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            ]
        },
        "lowSpeed":  10,
        /*opt, int, minimum speed limit, unit:km/h*/
        "highSpeed":  10
        /*opt, int, maximum speed limit, unit:km/h*/
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

### 14.6.1.33 Get configuration capability of overtaking on the right detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/overtakeRightSide/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., obtained via GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. values of scenesID under SceneInfo in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

VISION AND SECURITY SYSTEM CO.,LTD. dejojin@gmail.com

```
{
    "OvertakeRightSideEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "enabled": {
        /*ro, req, object, whether to enable overtaking on the right detection*/
            "@opt": ["true", "false"]
            /*ro, req, array, subType:string*/
        },
        "rearVehicleDistance": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int, step:1*/
            "@max":  100,
            /*ro, opt, int, step:1*/
            "@def":  0
            /*ro, opt, int, step:1*/
        },
        "SnapImageParam": {
        /*ro, req, object, parameters of capturing pictures*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, req, object, capture times*/
                "@min":  3,
                /*ro, req, int, the minimum value*/
                "@max":  3,
                /*ro, req, int, the maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        }
    }
}
```

#### 14.6.1.34 Get the parameters of overtaking on the right detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/overtakeRightSide?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., obtained via GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. values of scenesID under SceneInfo in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "OvertakeRightSideEvent": {
    /*ro, req, object*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable overtaking on the right detection*/
        "SnapImageParam": {
        /*ro, req, object, parameters for capturing pictures*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        }
    }
}
```

#### 14.6.1.35 Set the parameters of overtaking on the right detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/overtakeRightSide?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "OvertakeRightSideEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable the function*/
        "SnapImageParam": {
        /*req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        }
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

#### 14.6.1.36 Get the configuration capability of slalom driving detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/SSharpDriving/capabilities?format=json

**Query Parameter**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "SSharpDrivingEventCap": {
    /*ro, req, object*/
        "scenesID": {
        /*ro, req, object, range of scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum length*/
            "@max":  16
            /*ro, req, int, the maximum length*/
        },
        "enabled": {
        /*ro, req, object, whether to enable slalom driving detection*/
            "@opt": ["true", "false"]
            /*ro, req, array, value range, subType:string*/
        },
        "SnapImageParam": {
        /*ro, opt, object, picture capturing parameters*/
            "snapSentivity": {
            /*ro, req, object, capture sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  50
                /*ro, req, int, the default value*/
            },
            "times": {
            /*ro, req, object, capture times*/
                "@min":  3,
                /*ro, req, int, the minimum value*/
                "@max":  3,
                /*ro, req, int, the maximum value*/
                "@def":  3
                /*ro, req, int, the default value*/
            }
        },
        "vehicleTravelTime": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,180], unit:s*/
            "@max":  180,
            /*ro, req, int, range:[1,180], unit:s*/
            "@def":  50
            /*ro, opt, int, range:[1,180], unit:s*/
        }
    }
}
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

### 14.6.1.37 Set the parameters of slalom driving detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/SSharpDriving?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

```
{
    "SSharpDrivingEvent": {
    /*req, object*/
        "scenesID":  "1",
        /*req, string, scene ID*/
        "enabled":  true,
        /*req, bool, whether to enable the function*/
        "SnapImageParam": {
        /*req, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*req, int, capture sensitivity*/
            "times":  3
            /*req, int, capture times*/
        }
    }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

### 14.6.1.38 Get the parameters of slalom driving detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/SSharpDriving?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

```
{
    "SSharpDrivingEvent": {
    /*ro, req, object*/
        "scenesID":  "1",
        /*ro, req, string, scene ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable slalom driving detection*/
        "SnapImageParam": {
        /*ro, opt, object, picture capturing parameters*/
            "snapSentivity":  1,
            /*ro, req, int, capture sensitivity*/
            "times":  3
            /*ro, req, int, capture times*/
        },
        "vehicleTravelTime":  1
        /*ro, opt, int, range:[1,180], unit:s*/
    }
}
```

### 14.6.1.39 Set parameters for wrong-way driving detection

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

### Request URL

PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/wrongDirection

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WrongDirectionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, wrong-way driving detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:far (distant view);-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <congestionThreshold>
      <!--opt, int, range:[0,100]-->60
    </congestionThreshold>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane No.s, subType:object-->
      <relatedLaneNo>
        <!--opt, int-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space No.s, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space No.-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <relatedShieldRegionNo>
    <!--opt, string, numbers of linked shielded areas, desc:numbers of linked shielded areas-->1
  </relatedShieldRegionNo>
</WrongDirectionEvent>
```

### Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.1.40 Get the configuration parameters of wrong-way driving detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/wrongDirection

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<WrongDirectionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, wrong-way driving detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--ro, req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view);-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <congestionThreshold>
      <!--ro, opt, int, range:[0,100]-->60
    </congestionThreshold>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, linked lane No.-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space No., subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space No.-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <relatedShieldRegionNo>
    <!--ro, opt, string, numbers of linked shielded areas, desc:numbers of linked shielded areas-->1
  </relatedShieldRegionNo>
</WrongDirectionEvent>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

### 14.6.1.41 Get the configuration capability of wrong-way driving detection

#### Request URL

GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/wrongDirection/capabilities

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

#### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WrongDirectionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, int, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="60" def="10">
    <!--ro, req, int, duration of wrong-way driving detection, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, parameters for capturing pictures-->
    <imgNum min="1" max="4">
      <!--ro, req, int, each rule corresponds to a group of parameters for capturing pictures, attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="4">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="1" def="1">
          <!--ro, req, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="near">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:"far" (distant view)-->far
        </imgType>
        <times min="1" max="3" def="2">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->3
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
    <congestionThreshold min="0" max="100" def="70">
      <!--ro, opt, int, range:[0,100], step:1, attr:min{req, int},max{req, int},def{req, int}-->70
    </congestionThreshold>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, list of linked lane numbers, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, req, int, related lane No., attr:min{opt, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space No., subType:object, attr:size{opt, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space No., attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, coordinate ratio between the upper level and the device-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, numbers of linked shielded areas, attr:min{req, int},max{req, int},
desc:1. this node is used to specify the area where the wrong-way driving is invalid;
2. multiple numbers of shielded areas are separated by commas, e.g., 1, 2, 3, 4-->1
  </relatedShieldRegionNo>
</WrongDirectionEvent>
```

## 14.6.2 Motor Vehicle Traffic Flow Statistics

### 14.6.2.1 Get the traffic flow statistics capability

### Request URL

GET /ISAPI/ITC/trafficParam/capabilities?channelID=<channelID>

### Query Parameter

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable opt="true,false">
    <!--ro, opt, bool, whether to enable traffic flow statistics and collection, attr:opt{req, string}, desc:whether to enable traffic flow statistics and
collection-->true
  </enable>
  <statisticsMinute min="0" max="65535">
    <!--ro, opt, int, range:[0,65535], unit:min, attr:min{req, int},max{req, int}-->0
  </statisticsMinute>
  <tpsStatistic>
    <!--ro, opt, object, traffic flow statistics-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}, desc:whether to enable the function-->true
    </enabled>
    <statisticsMinute min="0" max="65535">
      <!--ro, req, int, range:[0,65535], unit:min, attr:min{req, int},max{req, int}-->0
    </statisticsMinute>
  </tpsStatistic>
  <isSupportDetectionAreaParam>
    <!--ro, opt, bool-->true
  </isSupportDetectionAreaParam>
</TrafficParam>
```

### 14.6.2.2 Get the traffic flow statistics parameters

**Request URL**

GET /ISAPI/ITC/trafficParam?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the traffic flow statistics parameters, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, opt, bool, whether to enable traffic flow statistics and collection, desc:whether to enable traffic flow statistics and collection-->true
  </enable>
  <statisticsMinute>
    <!--ro, req, int, statistics interval, range:[0,65535], unit:min-->1
  </statisticsMinute>
  <tpsStatistic>
    <!--ro, opt, object, traffic flow statistics-->
    <enabled>
      <!--ro, req, bool, whether to enable the function, desc:whether to enable the function-->true
    </enabled>
    <statisticsMinute>
      <!--ro, req, int, statistics interval, range:[0,65535], unit:min-->0
    </statisticsMinute>
  </tpsStatistic>
</TrafficParam>
```

### 14.6.2.3 Set the traffic parameters of cameras

**Request URL**

PUT /ISAPI/ITC/trafficParam?channelID=<channelID>

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### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, the traffic flow statistics parameters, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--opt, bool, whether to enable traffic flow statistics and collection, desc:whether to enable traffic flow statistics and collection-->true
  </enable>
  <statisticsMinute>
    <!--req, int, statistics interval, range:[0,65535], unit:min-->1
  </statisticsMinute>
  <tpsStatistic>
    <!--opt, object, traffic flow statistics-->
    <enabled>
      <!--req, bool, whether to enable traffic flow statistics, desc:whether to enable traffic flow statistics-->true
    </enabled>
    <statisticsMinute>
      <!--req, int, statistics interval, range:[0,65535], unit:min-->0
    </statisticsMinute>
  </tpsStatistic>
</TrafficParam>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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## 14.6.2.4 Get the configuration capability of trigger mode for traffic data statistics

### Request URL

GET /ISAPI/ITC/TriggerMode/TPS/capabilities?channelID=<channelID>

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TPS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, traffic statistics parameters, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="10">
    <!--ro, req, int, number of supported lanes, range:[1,10], attr:min{req, int},max{req, int}-->1
  </relatedLaneCount>
  <relatedLaneEnabled opt="false,true">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </relatedLaneEnabled>
  <relatedDriveWay min="1" max="99">
    <!--ro, req, int, linked lane No., range:[1,99], attr:min{req, int},max{req, int}-->0
```

```
  </relatedDriveWay>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, types of lane directions, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from
south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from
southwest to northeast)-->0
  </laneDirectionType>
  <enRealtimeDataUpload>
    <!--ro, req, bool, whether it supports uploading real-time data-->true
  </enRealtimeDataUpload>
  <enStatisticalDataUpload>
    <!--ro, req, bool, whether it supports uploading statistics-->true
  </enStatisticalDataUpload>
  <statisticsTime min="0" max="999">
    <!--ro, req, int, statistics interval, range:[0,999], unit:min, attr:min{req, int},max{req, int}-->0
  </statisticsTime>
  <posEnable opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </posEnable>
  <CalibrationEnable opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </CalibrationEnable>
  <laneVolumeEnable>
    <!--ro, req, bool-->true
  </laneVolumeEnable>
  <laneAverageSpeedEnable>
    <!--ro, req, bool-->true
  </laneAverageSpeedEnable>
  <timeHeadwayEnable>
    <!--ro, req, bool-->true
  </timeHeadwayEnable>
  <spaceHeadwayEnable>
    <!--ro, req, bool-->true
  </spaceHeadwayEnable>
  <timeOccupyRationEnable>
    <!--ro, req, bool-->true
  </timeOccupyRationEnable>
  <spaceOccupyRationEnable>
    <!--ro, req, bool-->true
  </spaceOccupyRationEnable>
  <queueLengthEnable>
    <!--ro, req, bool-->true
  </queueLengthEnable>
  <vehicleTypeEnable>
    <!--ro, req, bool-->true
  </vehicleTypeEnable>
  <trafficStatusEnable>
    <!--ro, req, bool-->true
  </trafficStatusEnable>
  <leftTurnVolumeEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </leftTurnVolumeEnable>
  <rightTurnVolumeEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </rightTurnVolumeEnable>
  <straightForwardVolumeEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </straightForwardVolumeEnable>
  <bIsOverFlowEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </bIsOverFlowEnable>
  <laneVolumeCoilBEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </laneVolumeCoilBEnable>
  <laneVolumeCoilCEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </laneVolumeCoilCEnable>
  <DirectionLineEnable>
    <!--ro, opt, object-->
    <lineName opt="DirectionLine">
      <!--ro, req, enum, line name, subType:string, attr:opt{req, string}, desc:"DirectionLine"-->DirectionLine
    </lineName>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <laneDirection opt="upToDown,downToUp">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->upToDown
    </laneDirection>
  </DirectionLineEnable>
  <virtualRegionMode min="1" max="3">
    <!--ro, req, enum, protocol type, subType:int, attr:min{req, int},max{req, int}, desc:protocol type-->1
  </virtualRegionMode>
  <VirtualRegionList min="0" max="3">
    <!--ro, req, array, virtual coil list, subType:object, attr:min{req, int},max{req, int}-->
    <VirtualRegion>
      <!--ro, opt, object, virtual coil-->
```

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```
      <!--ro, opt, object, virtual coil-->
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </VirtualRegion>
  </VirtualRegionList>
  <LaneLine>
    <!--ro, opt, object, lane line-->
    <lineName min="0" max="64">
      <!--ro, req, string, line name, range:[1,64], attr:min{req, int},max{req, int}-->test
    </lineName>
    <RegionCoordinatesList num="2">
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,2], attr:num{opt, object}-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LaneLine>
  <CheckoutLineEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </CheckoutLineEnable>
  <LaneRightBoundaryLine>
    <!--ro, opt, object-->
    <lineName min="0" max="64">
      <!--ro, req, string, line name, range:[0,64], attr:min{req, int},max{req, int}-->test
    </lineName>
    <RegionCoordinatesList num="2">
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,2], attr:num{opt, object}-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LaneRightBoundaryLine>
  <isSupportFlowReset>
    <!--ro, opt, bool, whether it supports clearing traffic flow by one touch-->true
  </isSupportFlowReset>
  <statisticsTime2>
    <!--ro, opt, int, statistics time interval T2, range:[0,999], unit:min-->0
  </statisticsTime2>
  <enStatistical2DataUpload>
    <!--ro, opt, bool, whether it supports uploading traffic statistics every T2 time interval, desc:whether it supports uploading traffic statistics every
T2 time interval-->true
  </enStatistical2DataUpload>
  <enTTDUpload>
    <!--ro, opt, bool, whether it supports uploading instant traffic data, desc:whether it supports uploading instant traffic data-->true
  </enTTDUpload>
  <IntersectionParam>
    <!--ro, opt, object-->
    <enable opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enable>
  </IntersectionParam>
  <VehicleStatistics>
    <!--ro, opt, object, vehicle statistics-->
    <RegionList size="4">
      <!--ro, opt, object, attr:size{req, int}-->
      <Region>
        <!--ro, opt, object, region-->
        <id min="1" max="4">
          <!--ro, req, int, No., range:[1,4], attr:min{req, int},max{req, int}-->1
        </id>
        <enable opt="true,false">
          <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
        </enable>
        <RegionCoordinatesList size="4">
          <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
          <RegionCoordinates>
            <!--ro, req, object-->
            <positionX>
              <!--ro, req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
```

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```
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </VehicleStatistics>
  <IntersectionJam>
    <!--ro, opt, object, traffic jam over stop line in intersection, desc:traffic jam over stop line in intersection-->
    <enable opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enable>
    <threshold min="0" max="10">
      <!--ro, req, int, threshold, unit:m, attr:min{req, int},max{req, int}, desc:threshold-->0
    </threshold>
  </IntersectionJam>
  <virtualRegionWidth min="1" max="300">
    <!--ro, opt, int, range:[1,300], unit:min, attr:min{req, int},max{req, int}-->0
  </virtualRegionWidth>
  <isSupportGraphicSymbol>
    <!--ro, opt, bool-->true
  </isSupportGraphicSymbol>
  <polygonLane>
    <!--ro, opt, object-->
    <enable opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enable>
    <polygonRegionList size="16">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <polygonRegion>
        <!--ro, opt, object-->
        <id min="1" max="16">
          <!--ro, req, int, range:[1,16], attr:min{req, int},max{req, int}-->1
        </id>
        <RegionCoordinatesList min="4" max="10">
          <!--ro, opt, array, subType:object, range:[4,10], attr:min{req, int},max{req, int}-->
          <RegionCoordinates>
            <!--ro, req, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </polygonRegion>
    </polygonRegionList>
  </polygonLane>
</TPS>
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| TPS | [Message content] | application/xml | -- | -- | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

### 14.6.2.5 Traffic statistics (passing vehicle statistics)

**EventType:TPS**

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
```

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```
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo": 80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol": "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress": "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID": 1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime": "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount": 1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
    "eventType": "TPS",
    /*ro, req, string, event type, desc:"TPS" (passing vehicle statistics uploading)*/
    "eventState": "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the time headway,unit: secondheartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription": "traffic pass statistics",
    /*ro, req, string, event description*/
    "deviceID": "test0123",
    /*ro, opt, string, device No., desc:device No.*/
    "Target": [
    /*ro, opt, array, subType:object*/
        {
            "recognitionType": "vehicle",
            /*ro, req, enum, recognition type in traffic statistics, subType:string, desc:"vehicle"*/
            "TargetInfo": {
            /*ro, req, object, target information*/
                "recognition": "TPS",
                /*ro, req, string, specific recognition target*/
                "startTime": "2004-05-03T17:30:08.000+080:00",
                /*ro, req, date, statistics start time*/
                "stopTime": "2004-05-03T17:30:08.000+080:00",
                /*ro, req, date, statistics end time*/
                "samplePeriod": 3600,
                /*ro, req, int, statistics period, unit:s*/
                "totalLaneNum": 8,
                /*ro, req, int, number of valid lanes, range:[1,8]*/
                "LaneInfo": [
                /*ro, req, array, lane information, subType:object*/
                    {
                        "laneNo": 8,
                        /*ro, req, int, lane No., range:[1,8]*/
                        "aversgeSpeed": 60,
                        /*ro, req, int, average speed of vehicles traveling on the lane, unit:km/h*/
                        "smallCarNum": 20,
                        /*ro, req, int, number of small-sized vehicles*/
                        "midsizeCarNum": 20,
                        /*ro, req, int, number of medium-sized vehicles*/
                        "heavyVehicleNum": 20,
                        /*ro, req, int, number of heavy vehicles*/
                        "headTimeInterval": 20,
                        /*ro, req, int, time headway, unit:s*/
                        "headInterval": 20,
                        /*ro, req, int, space headway, unit:m*/
                        "spaceOccupyRation": 20.000,
                        /*ro, req, float, lane space occupancy, range:[0.000,100.000]*/
                        "timeOccupyRation": 20.000,
                        /*ro, req, float, lane time occupancy, range:[0.000,100.000]*/
                        "channelizationLaneNo": 2,
                        /*ro, opt, int, channelized lane No.*/
                        "averageParkingTime": 2,
                        /*ro, opt, int, average parking times*/
                        "averageDelay": 2,
                        /*ro, opt, int, average delay time, unit:s*/
                        "averageQueueLen": 2,
                        /*ro, opt, int, average queuing length, unit:m*/
                        "arrivalFlow": 20,
                        /*ro, opt, int*/
                        "vehicelNum": 20,
                        /*ro, opt, int, number of vehicles in area*/
                        "nonmotorVehicleNum": 20,
                        /*ro, opt, int*/
                        "oversizeVehicleNum": 20,
                        /*ro, opt, int, large-sized vehicle quantity, range:[1,10000]*/
                        "laneName": "车道1",
                        /*ro, opt, string, lane name, range:[1,32]*/
                        "laneType": 1,
                        /*ro, opt, enum, lane type, subType:int, desc:lane type*/
                        "laneState": 1,
                        /*ro, opt, enum, lane status, subType:int, desc:lane status*/
                        "varyType": 0,
                        /*ro, opt, enum, subType:int*/
                        "tpsType": 0,
                        /*ro, opt, enum, subType:int*/
                        "vehicleNum": 20,
                        /*ro, opt, int, range:[0,10000]*/
                        "QueueInfo": {
                        /*ro, opt, object*/
                            "HeadPosition": {
```

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```
                        "HeadPosition": {
                        /*ro, opt, object*/
                            "x": 0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y": 0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        },
                        "TailPosition": {
                        /*ro, opt, object*/
                            "x": 0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y": 0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        },
                        "queueLen": 2
                        /*ro, opt, int, range:[0,500], unit:m*/
                    },
                    "CoilPosition": {
                    /*ro, opt, object*/
                        "x": 0.120,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y": 0.120
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                }
            ],
            "totalCoilNum": 40,
            /*ro, opt, int, number of valid coils, range:[1,40]*/
            "CoilInfo": [
            /*ro, opt, array, coil information, subType:object*/
                {
                    "coilNo": 18,
                    /*ro, req, int, coil No.*/
                    "laneNo": 18,
                    /*ro, req, int, lane No., range:[1,18]*/
                    "aversgeSpeed": 60,
                    /*ro, req, int, average speed of vehicles traveling on the lane, unit:km/h*/
                    "smallCarNum": 20,
                    /*ro, req, int, number of small-sized vehicles*/
                    "midsizeCarNum": 20,
                    /*ro, req, int, number of medium-sized vehicles*/
                    "heavyVehicleNum": 20,
                    /*ro, req, int, number of heavy vehicles*/
                    "headTimeInterval": 20,
                    /*ro, req, int, time headway, unit:s*/
                    "headInterval": 20,
                    /*ro, req, int, space headway, unit:m*/
                    "channelizationLaneNo": 2,
                    /*ro, opt, int, channelized lane No.*/
                    "nonmotorVehicleNum": 20
                    /*ro, opt, int, number of non-motor vehicles*/
                }
            ]
        }
    }
],
"Location": {
/*ro, opt, object, device location*/
    "lng": 116.46,
    /*ro, req, float, longitude*/
    "lat": 39.92
    /*ro, opt, float, latitude*/
},
"algorithmInfo": {
/*ro, opt, object, algorithm version information, this field is required for secondary analysis*/
    "algorithmId": "test",
    /*ro, opt, string, algorithm package ID*/
    "algoVendorId": "test",
    /*ro, opt, string, manufacturer ID*/
    "algoVersionId": "test",
    /*ro, opt, string, algorithm version No.*/
    "algoChipType": "test",
    /*ro, opt, string, chip type*/
    "algoModelVersion": "test"
    /*ro, opt, string, model version*/
},
"detectDir": 1,
/*ro, opt, enum, subType:int*/
"detectType": 1,
/*ro, opt, enum, detection mode, subType:int, desc:detection mode*/
"monitoringSiteID": "test",
/*ro, opt, string, camera No., range:[0,48]*/
"startCode": "0xfe",
/*ro, opt, string, range:[1,8]*/
"CMD": "01",
/*ro, opt, enum, Command, subType:string*/
"deviceUUID": "12345"
/*ro, opt, string, device No., range:[1,32], desc:the device number is the device UUID which is the device's serial number and can be edited by the node <deviceID> in the message of /ISAPI/System/deviceInfo*/
}
```


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## 14.6.3 Traffic Incidents

### 14.6.3.1 Get the parameters of fallen object detection.

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/abandonedObject

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<AbandonedObjectEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, fallen object detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <amplifyDetectionEnabled>
    <!--ro, opt, bool-->true
  </amplifyDetectionEnabled>
  <durationTime>
    <!--ro, req, int, duration of fallen object detection, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, opt, int, capture interval, unit:s-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters, desc:Only one between RelatedLaneParam and RelatedAreaParam can be configured at the same time. That is,
you cannot configure the linked area if the linked lane has been configured, and vice versa.-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object, parameters of the linked target areas, desc:Only one between RelatedLaneParam and RelatedAreaParam can be configured at the same
time. That is, you cannot configure the linked area if the linked lane has been configured, and vice versa.-->
    <relatedAreaNo>
      <!--ro, opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--ro, opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
  <targetSizeRect>
    <!--ro, opt, object-->
    <displayEnabled>
      <!--ro, req, bool-->true
    </displayEnabled>
  </targetSizeRect>
  <relatedType>
    <!--ro, opt, enum, subType:string-->area
  </relatedType>
</AbandonedObjectEvent>
```

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### 14.6.3.2 Configure the parameters of fallen object detection.

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/abandonedObject

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AbandonedObjectEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, fallen object detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--req, int, duration of fallen object detection, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters, desc:Only one between RelatedLaneParam and RelatedAreaParam can be configured at the same time. That is, you
cannot configure the linked area if the linked lane has been configured, and vice versa.-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--opt, object, parameters of the linked target areas, desc:Only one between RelatedLaneParam and RelatedAreaParam can be configured at the same time.
That is, you cannot configure the linked area if the linked lane has been configured, and vice versa.-->
    <relatedAreaNo>
      <!--opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
</AbandonedObjectEvent>
```

### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0:OK;1:OK;2:Device Busy;3:Device Error;4:Invalid Operation;5:Invalid XML Format;6:Invalid XML
Content;7:Reboot Required;-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, detailed description of the error code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.3.3 Get the capability of configuring parameters of fallen object detection.

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/abandonedObject/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |
| SID | string | Scene No., by calling GET /ISAPI/Traffic/channels//sceneinfo/capabilities protocol, the min. and max. attributes of SceneInfo->scenesID in the response message indicate the value range of scene No. If the device does not support this protocol, the default value is 1. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<AbandonedObjectEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of fallen object detection, attr:version{req, int, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, range:[1,16], attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false, true">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <amplifyDetectionEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </amplifyDetectionEnabled>
  <durationTime min="1" max="60" def="10">
    <!--ro, req, int, duration of fallen object detection, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="1">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="1" def="1">
          <!--ro, opt, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:far (distant view)-->far
        </imgType>
        <times min="3" max="3" def="3">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->3
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, req, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, linked lane number, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, req, int, linked lane number, attr:min{req, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{req, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object, linked areas-->
    <relatedAreaNo opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16">
      <!--ro, opt, string, linked area numbers, attr:opt{req, string}, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, numbers of linked shielded areas, attr:min{req, int},max{req, int}, desc:Multiple numbers should be separated by commas, e.g.,
1,2,3,4.-->1
  </relatedShieldRegionNo>
  <targetSizeRect>
    <!--ro, opt, object-->
    <displayEnabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </displayEnabled>
    <high>
      <!--ro, req, int, range:[0,1000]-->1
    </high>
    <width>
      <!--ro, req, int, range:[0,1000]-->1
    </width>
  </targetSizeRect>
  <relatedType opt="area,lane">
    <!--ro, opt, string, attr:opt{opt, string}-->area
  </relatedType>
</AbandonedObjectEvent>
```

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#### 14.6.3.4 Configure the event rule parameters of congestion detection.

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/congestion

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<CongestionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--opt, object, congestion detection, attr:version{opt, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,600], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--req, int, filtering time, range:[1,120], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval, range:[1,3600], unit:s-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--req, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <roadSmoothWaterLine>
    <!--opt, int, road smoothness threshold, range:[1,80], desc:This node specifies the road smoothness threshold. For example, if the value of this node is
30, the road will be considered smooth if the degree of road congestion is lower than 30.-->30
  </roadSmoothWaterLine>
  <roadCongestionWaterLine>
    <!--opt, int, road congestion threshold, range:[50,100],
desc:1. This node specifies the road congestion threshold. For example, if the value of this node is 80, the road will be considered congested if the degree
of road congestion is higher than 80.
2. If the degree of road congestion is between the threshold of road smoothness and that of road congestion, vehicles can move slowly on the road.-->70
  </roadCongestionWaterLine>
  <lowSpeedVehicleCongestionDistance>
    <!--opt, int, traffic congestion length of low speed vehicles, range:[50,100], unit:m, desc:This node specifies the traffic congestion length of low
speed vehicles. For example, if the value of this node is 20, when the traffic congestion length of low speed vehicles is shorter than 20 meters, the road
will not be considered congested and the congestion alarm will not be uploaded.-->70
  </lowSpeedVehicleCongestionDistance>
  <repeatAlarmEnable>
    <!--req, bool, whether to enable uploading alarms many times before the road is not congested, desc:This node specifies whether to upload the congestion
alarms many times before the road is not congested. For example, if the congestion event occurred and it lasts for 10 minutes, the value of this node will
determine whether to upload the congestion alarms many times during congestion.-->false
  </repeatAlarmEnable>
</CongestionEvent>
```

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## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0:OK;1:OK;2:Device Busy;3:Device Error;4:Invalid Operation;5:Invalid XML Format;6:Invalid XML
Content;7:Reboot Required;-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, detailed description of the error code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.3.5 Get the event rule parameters of congestion detection.

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/congestion

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<CongestionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, opt, object, congestion detection, attr:version{opt, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--ro, req, int, detection duration, range:[1,600], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--ro, req, int, filtering time, range:[1,120], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, req, int, capture interval, range:[1,3600], unit:s-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <roadSmoothWaterLine>
    <!--ro, opt, int, road smoothness threshold, range:[1,80], desc:This node specifies the road smoothness threshold. For example, if the value of this
node is 30, the road will be considered smooth if the degree of road congestion is lower than 30.-->30
  </roadSmoothWaterLine>
  <roadCongestionWaterLine>
    <!--ro, opt, int, road congestion threshold, range:[50,100],
desc:1. This node specifies the road congestion threshold. For example, if the value of this node is 80, the road will be considered congested if the degree
of road congestion is higher than 80.
2. If the degree of road congestion is between the threshold of road smoothness and that of road congestion, vehicles can move slowly on the road.-->70
  </roadCongestionWaterLine>
  <lowSpeedVehicleCongestionDistance>
    <!--ro, opt, int, traffic congestion length of low speed vehicles, range:[50,100], unit:m, desc:This node specifies the traffic congestion length of low
speed vehicles. For example, if the value of this node is 20, when the traffic congestion length of low speed vehicles is shorter than 20 meters, the road
will not be considered congested and the congestion alarm will not be uploaded.-->70
  </lowSpeedVehicleCongestionDistance>
  <repeatAlarmEnable>
    <!--ro, opt, bool, whether to enable uploading alarms many times before the road is not congested, desc:This node specifies whether to upload the
congestion alarms many times before the road is not congested. For example, if the congestion event occurred and it lasts for 10 minutes, the value of this
node will determine whether to upload the congestion alarms many times during congestion.-->false
  </repeatAlarmEnable>
</CongestionEvent>
```

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### 14.6.3.6 Get the capability of configuring the event rule parameters of congestion detection.

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/congestion/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<CongestionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, opt, object, congestion detection event, attr:version{opt, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="600" def="30">
    <!--ro, req, int, detection duration, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <filteringTime min="1" max="10" def="2">
    <!--ro, req, int, Filtering time, unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="1">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{opt, int},max{opt, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="1">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="3600" def="600">
          <!--ro, req, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:far (distant view)-->far
        </imgType>
        <times min="1" max="1" def="1">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, list of linked lane numbers, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, opt, int, linked lane number, attr:min{req, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{opt, string}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <roadSmoothWaterLine min="1" max="60" def="30">
    <!--ro, opt, int, road smoothness threshold, unit:s, attr:min{req, int},max{req, int},def{req, int}, desc:This node specifies the road smoothness
threshold. For example, if the value of this node is 30, the road will be considered smooth if the degree of road congestion is lower than 30.-->1
  </roadSmoothWaterLine>
  <roadCongestionWaterLine min="30" max="100" def="70">
    <!--ro, opt, int, road congestion threshold, unit:s, attr:min{req, int},max{req, int},def{req, int},
desc:1. This node specifies the road congestion threshold. For example, if the value of this node is 80, the road will be considered congested if the degree
of road congestion is higher than 80.
2. If the degree of road congestion is between the threshold of road smoothness and that of road congestion, vehicles can move slowly on the road.-->70
  </roadCongestionWaterLine>
  <lowSpeedVehicleCongestionDistance min="1" max="200" def="50">
    <!--ro, opt, int, traffic congestion length of low speed vehicles, unit:m, attr:min{req, int},max{req, int},def{req, int}, desc:This node specifies the
traffic congestion length of low speed vehicles. For example, if the value of this node is 20, when the traffic congestion length of low speed vehicles is
shorter than 20 meters, the road will not be considered congested and the congestion alarm will not be uploaded.-->50
  </lowSpeedVehicleCongestionDistance>
  <repeatAlarmEnable opt="false,true" def="false">
    <!--ro, opt, bool, whether to enable uploading alarms many times before the road is not congested, attr:opt{req, string},def{req, string}, desc:This
node specifies whether to upload the congestion alarms many times before the road is not congested. For example, if the congestion event occurred and it
lasts for 10 minutes, the value of this node will determine whether to upload the congestion alarms many times during congestion.-->false
  </repeatAlarmEnable>
</CongestionEvent>
```

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### 14.6.3.7 Get the parameters of construction detection.

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/construction

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ConstructionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, construction detection, attr:version{opt, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--ro, req, int, detection duration, range:[1,30], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--ro, req, int, filtering time, range:[1,120], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, opt, int, capture interval, range:[1,3600], unit:s-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object, parameters of the linked target areas-->
    <relatedAreaNo>
      <!--ro, opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--ro, opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
  <onlyReportOnceBeforeRevocationEnable>
    <!--ro, opt, bool-->true
  </onlyReportOnceBeforeRevocationEnable>
  <filterOtherEventsEnable>
    <!--ro, opt, bool-->true
  </filterOtherEventsEnable>
  <relatedType>
    <!--ro, opt, enum, subType:string-->area
  </relatedType>
</ConstructionEvent>
```

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### 14.6.3.8 Configure parameters of construction detection.

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/construction

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ConstructionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, construction detection, attr:version{opt, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,30], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--req, int, filtering time, range:[1,120], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval, range:[1,3600], unit:s-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--opt, object, parameters of the linked target areas-->
    <relatedAreaNo>
      <!--opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
</ConstructionEvent>
```

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### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0:OK;1:OK;2:Device Busy;3:Device Error;4:Invalid Operation;5:Invalid XML Format;6:Invalid XML
Content;7:Reboot Required;-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, detailed description of the error code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.6.3.9 Get the capability of configuring parameters of construction detection.

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/construction/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ConstructionEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, Get the capability of configuring parameters of construction detection., attr:version{req, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="3600" def="600">
    <!--ro, req, int, detection duration, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <filteringTime min="1" max="120" def="30">
    <!--ro, req, int, filtering time, unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="1">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, attr:min{opt, int},max{opt, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="1">
      <!--ro, req, int, capture sensitivity, attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="3600" def="600">
          <!--ro, opt, int, capture interval, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:far (distant view)-->far
        </imgType>
        <times min="1" max="1" def="1">
          <!--ro, opt, int, capture times, attr:min{req, int},max{req, int},def{req, int}-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, list of linked lane numbers, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, opt, int, linked lane number, attr:min{req, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{opt, string}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object, linked areas-->
    <relatedAreaNo opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16">
      <!--ro, opt, string, linked area numbers, attr:opt{req, string}, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, numbers of linked shielded areas, attr:min{req, int},max{req, int}, desc:Multiple numbers should be separated by commas, e.g.,
1,2,3,4.-->1
  </relatedShieldRegionNo>
  <onlyReportOnceBeforeRevocationEnable opt="false,true" def="true">
    <!--ro, opt, bool, attr:opt{req, string},def{req, int}-->true
  </onlyReportOnceBeforeRevocationEnable>
  <filterOtherEventsEnable opt="false,true" def="true">
    <!--ro, opt, bool, attr:opt{req, string},def{req, int}-->true
  </filterOtherEventsEnable>
  <relatedType opt="area,lane">
    <!--ro, opt, string, attr:opt{opt, string}-->area
  </relatedType>
</ConstructionEvent>
```

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### 14.6.3.10 Get the rule parameters of pedestrian detection.

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/pedestrian

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PedestrianEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, rule configuration of pedestrian detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <amplifyDetectionEnabled>
    <!--opt, bool-->true
  </amplifyDetectionEnabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--opt, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--opt, object, parameters of the linked target areas-->
    <relatedAreaNo>
      <!--opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <targetType>
    <!--opt, enum, detection target, subType:string, desc:pedestrian, TwoWheeler (two wheeler), Tricycle (tricycle), police (police officer),
constructionStaff (construction staff)-->pedestrian
```

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```
constructionStaff (construction staff)-->pedestrian
  </targetType>
  <TargetTypeList>
    <!--opt, array, list of target types, subType:string, desc:Multiple target types are supported.-->
    <targetType>
      <!--opt, enum, detection target, subType:string, desc:The default value is pedestrian. If the device does not support this node, the default value is
also pedestrian. pedestrian, TwoWheeler (two wheeler), Tricycle (tricycle), police (police officer), constructionStaff (construction staff)-->pedestrian
    </targetType>
  </TargetTypeList>
  <filteringTime>
    <!--opt, int, filtering time, range:[0,120], unit:min-->1
  </filteringTime>
  <relatedShieldRegionNo>
    <!--opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
  <relatedType>
    <!--opt, enum, subType:string-->area
  </relatedType>
</PedestrianEvent>
```

### 14.6.3.11 Configure the rule parameters of pedestrian detection.

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/pedestrian

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<PedestrianEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, rule configuration of pedestrian detection, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, linked lane number-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--opt, object, parameters of the linked target areas-->
    <relatedAreaNo>
      <!--opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <targetType>
    <!--opt, enum, detection target, subType:string, desc:pedestrian, TwoWheeler (two wheeler), Tricycle (tricycle), police (police officer),
constructionStaff (construction staff)-->pedestrian
  </targetType>
  <TargetTypeList>
    <!--opt, array, list of target types, subType:object, desc:Multiple target types are supported.-->
    <targetType>
      <!--opt, enum, detection target, subType:string, desc:The default value is pedestrian. If the device does not support this node, the default value is
also pedestrian. pedestrian, TwoWheeler (two wheeler), Tricycle (tricycle), police (police officer), constructionStaff (construction staff)-->pedestrian
    </targetType>
  </TargetTypeList>
  <filteringTime>
    <!--opt, int, filtering time, range:[0,120], unit:min-->1
  </filteringTime>
  <relatedShieldRegionNo>
    <!--opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
</PedestrianEvent>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0:OK;1:OK;2:Device Busy;3:Device Error;4:Invalid Operation;5:Invalid XML Format;6:Invalid XML
Content;7:Reboot Required;-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, detailed description of the error code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.6.3.12 Get the capability of configuring the parameters of the pedestrian detection event.

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/pedestrian/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<PedestrianEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of the pedestrian detection event, attr:version{req, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, range:[1,16], attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <amplifyDetectionEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </amplifyDetectionEnabled>
  <durationTime min="0" max="60" def="1">
    <!--ro, req, int, duration of pedestrian detection, range:[0,60], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, opt, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times, range:[0,1]-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters, desc:Only one between RelatedLaneParam and RelatedAreaParam can be configured at the same time. That is,
you cannot configure the linked area if the linked lane has been configured, and vice versa.-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, linked lane number, range:[1,8]-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number, range:[1,255]-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[0,1000]-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[0,1000]-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object, parameters of the linked target areas, desc:Only one between RelatedLaneParam and RelatedAreaParam can be configured at the same
time. That is, you cannot configure the linked area if the linked lane has been configured, and vice versa.-->
    <relatedAreaNo>
      <!--ro, opt, string, linked area numbers, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <targetType opt="pedestrian,TwoWheeler,Tricycle,police,constructionStaff">
    <!--ro, opt, enum, target type, subType:string, attr:opt{req, string}, desc:pedestrian, TwoWheeler (two wheeler), Tricycle (tricycle), police (police
officer), constructionStaff (construction staff)-->pedestrian
  </targetType>
  <filteringTime min="0" max="180" def="30">
    <!--ro, opt, int, filtering time, step:1, unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, numbers of linked shielded areas, attr:min{req, int},max{req, int},
desc:1. This node is used to specify the area where the pedestrian detection is invalid.
2. Multiple numbers of shielded areas are separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
  <relatedType opt="area,lane">
    <!--ro, opt, string, attr:opt{opt, string}-->area
  </relatedType>
</PedestrianEvent>
```

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dojojing@gmail.com

#### 14.6.3.13 Get the configuration parameters of roadblock detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/roadBlock

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<RoadBlockEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, ro, coordinate ratio between the upper level and the device, the default value is 1000, see details in the message of
XML_NormalizedScreenSize, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID,ranges from 1 to 16, range:[1,16]-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <amplifyDetectionEnabled>
    <!--ro, opt, bool-->true
  </amplifyDetectionEnabled>
  <durationTime>
    <!--ro, req, int, duration of roadblock detection,ranges from 1 to 60,the default value is 10,unit: second, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--ro, req, int, filtering time,ranges from 1 to 60,the default value is 30,unit: minute, range:[0,120], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object-->
    <imgNum>
      <!--ro, req, int-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object-->
        <imgInterval>
          <!--ro, opt, int-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, subType:string-->far
        </imgType>
        <times>
          <!--ro, opt, int-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object-->
    <RelatedLaneNoList>
      <!--ro, req, array, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object-->
    <relatedAreaNo>
      <!--ro, opt, string-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--ro, opt, string-->1
  </relatedShieldRegionNo>
  <onlyReportOnceBeforeRevocationEnable>
    <!--ro, opt, bool-->true
  </onlyReportOnceBeforeRevocationEnable>
  <relatedType>
    <!--ro, opt, enum, subType:string-->area
  </relatedType>
</RoadBlockEvent>
```

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### 14.6.3.14 Get the configuration capability of roadblock detection

**Request URL**

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/roadBlock/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<RoadBlockEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{opt, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, which is between 1 and 16, attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="false,true">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <amplifyDetectionEnabled opt="false,true" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </amplifyDetectionEnabled>
  <durationTime min="1" max="60" def="30">
    <!--ro, req, int, duration of roadblock detection, range:[0,60], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <filteringTime min="0" max="120" def="2">
    <!--ro, req, int, filtering time, range:[0,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, parameters for capturing pictures-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of parameters for capturing pictures-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, opt, int, capture interval-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:"far" (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="4">
      <!--ro, req, array, list of linked lane numbers, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="16">
        <!--ro, opt, int, related lane No., attr:min{req, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space No., subType:object, attr:size{req, int}-->
        <relatedParkingNo min="1" max="16">
          <!--ro, opt, int, linked parking space No., attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, coordinate ratio between the upper level and the device-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <RelatedAreaParam>
    <!--ro, opt, object, linked areas-->
    <relatedAreaNo opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16">
      <!--ro, opt, string, linked area No., attr:opt{req, string}, desc:multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, numbers of linked shielded areas, attr:min{req, int},max{req, int}, desc:multiple numbers should be separated by commas, e.g.,
1,2,3,4.-->1
  </relatedShieldRegionNo>
  <onlyReportOnceBeforeRevocationEnable opt="false,true" def="true">
    <!--ro, opt, bool, attr:opt{req, string},def{req, int}-->true
  </onlyReportOnceBeforeRevocationEnable>
  <relatedType opt="area,lane">
    <!--ro, opt, string, attr:opt{opt, string}-->area
  </relatedType>
</RoadBlockEvent>
```

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### 14.6.3.15 Get the configuration parameters of traffic accident detection

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/trafficAccident

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

None

**Response Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficAccidentEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--ro, req, string, scene ID, which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <durationTime>
    <!--ro, req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--ro, req, int, filtering time, range:[1,60], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum>
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--ro, req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--ro, req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval>
          <!--ro, opt, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--ro, req, enum, picture type, subType:string, desc:far (distant view)-->far
        </imgType>
        <times>
          <!--ro, opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--ro, req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--ro, opt, int, related lane No.-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--ro, opt, array, list of linked parking space numbers, subType:object-->
        <relatedParkingNo>
          <!--ro, opt, int, linked parking space number-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <mode>
    <!--ro, opt, enum, mode, subType:string, desc:mode-->single
  </mode>
  <RelatedAreaParam>
    <!--ro, opt, object-->
    <relatedAreaNo>
      <!--ro, opt, string, linked area No., desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->test
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--ro, opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
  <roadConstructionAlarmEnabled>
    <!--ro, opt, bool-->false
  </roadConstructionAlarmEnabled>
  <relatedType>
    <!--ro, opt, enum, subType:string-->area
  </relatedType>
</TrafficAccidentEvent>
```

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### 14.6.3.16 Set the parameters for traffic accident detection

**Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/AID/<SID>/trafficAccident

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| SID | string | -- |

**Request Message**

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficAccidentEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, ro, coordinate ratio between the upper level and the device, the default value is 1000, see details in the message of
XML_NormalizedScreenSize, attr:version{req, string, protocolVersion}-->
  <scenesID>
    <!--req, string, scene ID, which is between 1 and 16-->1
  </scenesID>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <durationTime>
    <!--req, int, detection duration, range:[1,60], unit:s-->1
  </durationTime>
  <filteringTime>
    <!--req, int, filtering time, range:[1,60], unit:min-->1
  </filteringTime>
  <SnapImageParam>
    <!--opt, object, picture capturing parameters-->
    <imgNum>
      <!--req, int, each rule corresponds to a group of picture capturing parameters-->1
    </imgNum>
    <snapSentivity>
      <!--req, int, capture sensitivity, range:[1,100]-->1
    </snapSentivity>
    <ImgGroupInfoList>
      <!--req, array, list of the picture group information, subType:object-->
      <ImgGroupInfo>
        <!--opt, object, picture group information-->
        <imgInterval>
          <!--req, int, capture interval-->1
        </imgInterval>
        <imgType>
          <!--req, enum, picture type, subType:string, desc:"far" (distant view)-->far
        </imgType>
        <times>
          <!--opt, int, capture times-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, list of linked lane numbers, subType:object-->
      <relatedLaneNo>
        <!--opt, int, related lane No.-->1
      </relatedLaneNo>
      <RelatedParkingNoList>
        <!--opt, array, list of linked parking space No., subType:object-->
        <relatedParkingNo>
          <!--opt, int, linked parking space No.-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <mode>
    <!--opt, enum, mode, subType:string, desc:mode-->single
  </mode>
  <RelatedAreaParam>
    <!--opt, object-->
    <relatedAreaNo>
      <!--opt, string, linked area No., desc:Multiple numbers should be separated by commas, e.g., 1, 2, 3, 4.-->2,3
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo>
    <!--opt, string, numbers of linked shielded areas, desc:Multiple numbers should be separated by commas, e.g., 1, 2, 3, 4.-->1,4
  </relatedShieldRegionNo>
  <roadConstructionAlarmEnabled>
    <!--opt, bool-->false
  </roadConstructionAlarmEnabled>
</TrafficAccidentEvent>
```

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dojojin@gmail.com

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

### 14.6.3.17 Get the configuration capability of traffic accident detection

#### Request URL

GET /ISAPI/Traffic/channels/<channelID>/AID/<SID>/trafficAccident/capabilities

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

#### Request Message

None

#### Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficAccidentEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <scenesID min="1" max="16">
    <!--ro, req, string, scene ID, range:[1,16], attr:min{req, int},max{req, int}-->1
  </scenesID>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <durationTime min="1" max="180" def="10">
    <!--ro, req, int, detection duration, range:[1,180], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
  </durationTime>
  <filteringTime min="1" max="60" def="10">
    <!--ro, req, int, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </filteringTime>
  <SnapImageParam>
    <!--ro, opt, object, picture capturing parameters-->
    <imgNum min="1" max="1">
      <!--ro, req, int, each rule corresponds to a group of picture capturing parameters, range:[0,1], attr:min{req, int},max{req, int}-->1
    </imgNum>
    <snapSentivity min="1" max="100" def="50">
      <!--ro, req, int, capture sensitivity, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->1
    </snapSentivity>
    <ImgGroupInfoList size="1">
      <!--ro, req, array, list of the picture group information, subType:object, attr:size{req, int}-->
      <ImgGroupInfo>
        <!--ro, opt, object, picture group information-->
        <imgInterval min="1" max="3600" def="600">
          <!--ro, opt, int, capture interval, attr:min{req, int},max{req, int},def{req, int}-->1
        </imgInterval>
        <imgType opt="far">
          <!--ro, req, enum, picture type, subType:string, attr:opt{req, string}, desc:far (distant view)-->far
        </imgType>
        <times min="1" max="1" def="1">
          <!--ro, opt, int, capture times, range:[0,1], attr:min{req, int},max{req, int},def{req, int}-->1
        </times>
      </ImgGroupInfo>
    </ImgGroupInfoList>
  </SnapImageParam>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="4">
      <!--ro, req, array, list of linked lane numbers, subType:object, attr:size{req, int}-->
      <relatedLaneNo min="1" max="6">
        <!--ro, opt, int, related lane No., range:[1,6], attr:min{req, int},max{req, int}-->1
      </relatedLaneNo>
      <RelatedParkingNoList size="8">
        <!--ro, opt, array, list of linked parking space numbers, subType:object, attr:size{req, int}-->
        <relatedParkingNo min="1" max="8">
          <!--ro, opt, int, linked parking space number, range:[1,8], attr:min{req, int},max{req, int}-->1
        </relatedParkingNo>
      </RelatedParkingNoList>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <NormalizedScreenSize>
    <!--ro, opt, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[0,1000]-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[0,1000]-->1000
    </normalizedScreenHeight>
  </NormalizedScreenSize>
  <mode opt="single,multiple,singleAndMultiple">
    <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:mode-->single
  </mode>
  <RelatedAreaParam>
    <!--ro, opt, object-->
    <relatedAreaNo opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16">
      <!--ro, opt, string, linked area No., attr:opt{req, string}, desc:Multiple numbers should be separated by commas, e.g., 1,2,3,4.-->2,3
    </relatedAreaNo>
  </RelatedAreaParam>
  <relatedShieldRegionNo min="1" max="6">
    <!--ro, opt, string, numbers of linked shielded areas, attr:min{req, int},max{req, int},
desc:1. This node is used to specify the area where the pedestrian detection is invalid.
2. Multiple numbers of shielded areas are separated by commas, e.g., 1,2,3,4.-->1
  </relatedShieldRegionNo>
  <roadConstructionAlarmEnabled opt="true,false" def="false">
    <!--ro, req, bool, attr:opt{req, string},def{req, string}-->false
  </roadConstructionAlarmEnabled>
  <relatedType opt="area,lane">
    <!--ro, opt, string, attr:opt{opt, string}-->area
  </relatedType>
</TrafficAccidentEvent>
```

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### 14.6.3.18 Get the traffic channel capabilities

### Request URL

GET /ISAPI/Traffic/channels/<channelID>/capabilities

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TrafficChannelCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <deviceTypeSupport opt="0,1,2,3,4,5">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->0
  </deviceTypeSupport>
  <isEvidenceGettingSupport>
    <!--ro, opt, bool-->true
  </isEvidenceGettingSupport>
  <isBasicSupport>
    <!--ro, opt, bool-->true
  </isBasicSupport>
  <isImageMergeSupport>
    <!--ro, opt, bool-->true
  </isImageMergeSupport>
  <isOverlapSupport>
    <!--ro, opt, bool-->true
  </isOverlapSupport>
  <isEdfAlgSupport>
    <!--ro, opt, bool-->true
  </isEdfAlgSupport>
  <isAutoTraceSupport>
    <!--ro, req, bool-->true
  </isAutoTraceSupport>
  <isEdfManualItsCapSupport>
    <!--ro, opt, bool-->true
  </isEdfManualItsCapSupport>
  <isViolationTypeStdSupport>
    <!--ro, opt, bool-->true
  </isViolationTypeStdSupport>
  <isEDFRemoteHostSupport>
    <!--ro, opt, bool-->true
  </isEDFRemoteHostSupport>
  <isANRSupport>
    <!--ro, opt, bool-->true
  </isANRSupport>
  <isvoiceTriggerSupport>
    <!--ro, opt, bool-->true
  </isvoiceTriggerSupport>
  <uploadDataTypesSupport>
    <!--ro, opt, enum, subType:string-->parking
  </uploadDataTypesSupport>
  <aidTypeSupport>
    <!--ro, opt, enum, subType:string-->parking
  </aidTypeSupport>
  <isSupportVehicleRecognitionPosParams>
    <!--ro, opt, bool-->true
  </isSupportVehicleRecognitionPosParams>
  <isSupportVehicleLaneGenerateStatusOverlay>
    <!--ro, opt, bool-->true
  </isSupportVehicleLaneGenerateStatusOverlay>
  <isSupportStopCurrentIllegalEventNotification>
    <!--ro, opt, bool-->true
  </isSupportStopCurrentIllegalEventNotification>
</TrafficChannelCap>
```

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### 14.6.3.19 Traffic Incident Detection

### EventType:AID

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, result, attr:version{req, string}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
```

```
  <portNo>
    <!--ro, opt, int, communication port number of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP.-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel number of the device that triggers the alarm, desc:video channel number that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:AID (traffic incident detection)-->AID
  </eventType>
  <eventState>
    <!--ro, req, enum, continuous event status, subType:string, desc:active (valid event); inactive (invalid event)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->automatic incident detection
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:It should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0).-->12345
  </deviceID>
  <AID>
    <!--ro, opt, object, information about traffic incident detection-->
    <illegalTrafficEvent>
      <!--ro, req, enum, traffic violation events, subType:string, desc:abandonedObject (fallen object), checkPoint (checkpoint detection), conflagration (fire), congestion, construction, crossLane (weaving out of the lane), edfManual (manual enforcement), fogDetection (fog detection), gasser (queue jumping), group (people gathering), illegalParking (illegal parking), intersectionStranded (vehicle overstay), laneChange (illegal lane change), objectDroppedDown (object falling down), occupyOvertakingLane (occupy overtaking lane), parallelParking (parallel parking), pedestrian (pedestrian detection), polyJam (polygonal congestion area), prohibitionMarkViolation (prohibition violation), roadBlock (roadblock), slowMoving (slow moving), smoke, speed (speeding), suddenSpeedDrop (sudden speed drop), trafficAccident (traffic incident detection), trafficConflict (traffic flow conflict), turnRound (illegal U-turn), unknown, vehicleexist (vehicle on non-motor vehicle lane), wrongDirection (wrong-way driving), singleVehicleBreakdown (single vehicle breakdown, the event that one vehicle stops due to breakdown and a person gets off the vehicle to check the vehicle), vehNoYieldPedest (motor vehicles failing to yield for pedestrians), illegalMannedVeh (illegal manned motor vehicle), illegalMannedNonMotorVeh (illegal manned non-motor vehicle), umbrellaTentInstall (umbrella tent installation violation on non-motor vehicle), nonMotorVehOnVehLane (non-motor vehicle on motor vehicle lane), wearingNoHelmet (wearing no helmet), pedestRedLightRunning (pedestrian red light running), pedestOnNonMotorVehLane (pedestrian walking on non-motor vehicle lane), pedestOnVehLane (pedestrian walking on motor vehicle lane)-->abandonedObject
    </illegalTrafficEvent>
    <VehicleInfo>
      <!--ro, opt, object, vehicle information-->
      <vehicleSpeed>
        <!--ro, opt, int, vehicle speed, range:[0,255], unit:km/h-->1
      </vehicleSpeed>
      <vehicleEnterState>
        <!--ro, opt, enum, vehicle entering status, subType:string, desc:vehicleEnter (vehicle entered), vehicleExit (vehicle exited)-->vehicleEnter
      </vehicleEnterState>
      <vehicleConfidence>
        <!--ro, opt, float, vehicle confidence, range:[0.0,100.0]-->0.0
      </vehicleConfidence>
      <vehicleType>
        <!--ro, req, enum, vehicle type, subType:string, desc:SUVMPV (SUV/MPV), buggy (small-sized truck), bus, concreteMixer (concrete mixer), containerTruck (container truck), coupe, crane, hatchback, largeBus (large-sized bus), lightTruck (light truck), mediumBus (middle-sized bus), mediumHeavyTruck (medium and heavy truck), miniCar (mini sedan (it will be converted to vehicle)), minibus, minitruck, motorVehicle (motor vehicle (it will be converted to vehicle)), nonmotorVehicle (non-motor vehicle (it will be converted to threeWheelVehicle)), oilTankTruck (oil tank truck), pedestrian, pickupTruck (pickup truck (it will be converted to buggy)), platformTrailer (platform trailer), saloon, slagTruck (dump truck), smallCar (small sedan (it will be converted to vehicle)), threeWheelVehicle (tricycle), truck, twoWheelVehicle (two wheeler), unknown, van, vehicle (sedan)-->SUVMPV
      </vehicleType>
      <vehicleColor>
        <!--ro, req, enum, vehicle color, subType:string, desc:black, blue, brown, cyan, deepBlue (dark blue), deepGray (dark gray), gray, green, orange, pink, purple, red, silver, unknown, white, yellow-->black
      </vehicleColor>
    </VehicleInfo>
    <PlateInfo>
      <!--ro, opt, object, license plate information-->
      <plate>
        <!--ro, req, string, license plate number, desc:noPlate (vehicle without license plate), unknown (no license plate recognized), XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string)-->蓝浙A12345
      </plate>
      <plateType>
        <!--ro, req, enum, license plate type, subType:string, desc:02TypePersonalized (type 02 custom vehicle), 92FarmVehicle (two-line license plate civil vehicle), 92TypeArm (type 92 armed police vehicle), 92TypeCivil (type 92 civil vehicle), arm (police vehicle), civilAviation (civil aviation license plate), coach (driver-training vehicle), consulate (consular vehicle), embassy (embassy vehicle), emergency (emergency license plate), green1325FarmVehicle (agricultural vehicle (green 1325)), hongKongMacao (Hong Kong/Macao entrance vehicle), motorola (motorcycle), newEnergy (new energy vehicle license plate), oneLineArm (new armed police vehicle (one-line)), oneLineArmHeadquarters (armed police headquarter license plate (one-line)), tempEntry (temporary entry car), tempTravl (temporary license plate car), trailer, twoLineArm (new armed police vehicle (two-line)), twoLineArmHeadquarters (armed police headquarter license plate (two-line)), unknown, yellow1225FarmVehicle (agricultural vehicle (yellow 1225)), yellow1325FarmVehicle (agricultural vehicle (yellow 1325)), yellowTwoLine (yellow two-line rear license plate)-->02TypePersonalized
      </plateType>
```

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```
      <plateColor>
        <!--ro, req, enum, license plate color, subType:string, desc:black, blue, civilAviationBlack (civil aviation black), civilAviationGreen (civil aviation green), golden, green, mixedColor (mixed color), newEnergyGreen (new energy green), newEnergyYellowGreen (new energy green and yellow), orange, other (other color), red, unknown, white, yellow-->black
      </plateColor>
      <licenseBright>
        <!--ro, opt, int, license plate brightness, range:[0,255]-->1
      </licenseBright>
      <confidence>
        <!--ro, opt, float, license plate confidence, range:[0.0,100.0]-->0.0
      </confidence>
      <plateCharacterConfidenceList>
        <!--ro, opt, array, subType:object-->
        <plateCharacterConfidence>
          <!--ro, opt, float, range:[0.00,100.00], step:0.01-->0.01
        </plateCharacterConfidence>
      </plateCharacterConfidenceList>
      <comprehensiveEvaluation>
        <!--ro, opt, int, range:[0,1000]-->1
      </comprehensiveEvaluation>
    </PlateInfo>
    <region>
      <!--ro, opt, enum, region, subType:string, desc:AFandAM (Africa and America), APAC (Asia-Pacific Region), ALl (all regions), ER (Russian Region), EU (Europe Region), EUandCIS (Europe and Russia), HKandMO (Hong Kong and Macao China), ME (Middle East), THAandLA (Thailand and Laos), other (other)-->AFandAM
    </region>
    <country>
      <!--ro, opt, enum, country or region, subType:string, desc:0 (Unsupported. It indicates that the algorithm library does not support recognizing license plates of this country), 1 (Czech Republic), 10 (Moldova), 100 (Reserved), 101 (Reserved), 102 (Reserved), 103 (Reserved), 104 (Egypt), 105 (Libya), 106 (Sudan), 107 (Tunisia), 108 (Algeria), 109 (Morocco), 11 (Russia), 110 (Ethiopia), 111 (Eritrea), 112 (Somalia Democratic), 113 (Djibouti), 114 (Kenya), 115 (Tanzania), 116 (Uganda), 117 (Rwanda), 118 (Burundi), 119 (Seychelles), 12 (Ukraine), 120 (Chad), 121 (Central African), 122 (Cameroon), 123 (Equatorial Guinea), 124 (Gabon), 125 (Republic of the Congo (Congo-Brazzaville)), 126 (Democratic Republic of the Congo (Congo-Kinshasa)), 127 (São Tomé and Príncipe), 128 (Mauritania), 129 (Western Sahara (Saharaui)), 13 (Belgium), 130 (Senegal), 131 (Gambia), 132 (Mali), 133 (Burkina Faso), 134 (Guinea), 135 (Guinea-Bissau), 136 (Cape Verde), 137 (Sierra Leone), 138 (Liberia), 139 (Ivory Coast), 14 (Bulgaria), 140 (Ghana), 141 (Togo), 142 (Benin), 143 (Niger), 144 (Zambia), 145 (Angola), 146 (Zimbabwe), 147 (Malawi), 148 (Mozambique), 149 (Botswana), 15 (Denmark), 150 (Namibia), 151 (South Africa), 152 (Swaziland), 153 (Lesotho), 154 (Madagascar), 155 (Comoros), 156 (Mauritius), 157 (Nigeria), 158 (South Sudan), 159 (Saint Helena), 16 (Finland), 160 (Mayotte), 161 (Reunion), 162 (Canary Islands), 163 (AZORES), 164 (Madeira), 165 (Reserved), 166 (Reserved), 167 (Reserved), 168 (Reserved), 169 (Canada), 17 (United Kingdom Great Britain), 170 (Greenland), 171 (St. Pierre and Miquelon), 172 (United State), 173 (Bermuda), 174 (Mexico), 175 (Guatemala), 176 (Belize), 177 (El Salvador), 178 (Honduras), 179 (Nicaragua), 18 (Greece), 180 (Costa Rica), 181 (Panama), 182 (Bahamas), 183 (Turks and  Caicos Islands), 184 (Cuba), 185 (Jamaica), 186 (Cayman Islands), 187 (Haiti), 188 (Dominican), 189 (Puerto Rico), 19 (Croatia), 190 (United States Virgin Islands), 191 (British Virgin Islands), 192 (Anguilla), 193 (Antigua and Barbuda), 194 (Collectivité de Saint-Martin), 195 (Autonomous country), 196 (Saint-Barthélemy), 197 (Saint Kitts and Nevis), 198 (Montserrat), 199 (Guadeloupe), 2 (France), 20 (Hungary), 200 (Dominica), 201 (Martinique), 202 (St. Lucia), 203 (Saint Vincent and the Grenadines), 204 (Grenada), 205 (Barbados), 206 (Trinidad and Tobago), 207 (Curaçao), 208 (Aruba), 209 (Netherlands Antilles), 21 (Israel), 210 (Colombia), 211 (Venezuela), 212 (Guyana), 213 (Suriname), 214 (French Guiana), 215 (Ecuador), 216 (Peru), 217 (Bolivia), 218 (Paraguay), 219 (Chile), 22 (Luxembourg), 220 (Brazil), 221 (Uruguay), 222 (Argentina), 223 (Reserved), 224 (Reserved), 225 (Reserved), 226 (Reserved), 227 (Australia), 228 (New Zealand ), 229 (Papua New Guinea), 23 (Macedonia, North Macedonia after 2018), 230 (Solomon Islands), 231 (Vanuatu), 232 (New Caledonia), 233 (Palau), 234 (Federated States of Micronesia), 235 (Marshall Island), 236 (Northern Mariana Islands), 237 (Guam), 238 (Nauru), 239 (Kiribati), 24 (Norway), 240 (Fiji Islands), 241 (Tonga), 242 (Tuvalu), 243 (Wallis et Futuna), 244 (Samoa), 245 (Eastern Samoa), 246 (Tokelau), 247 (Niue), 248 (Cook Islands), 249 (French Polynesia), 25 (Portugal), 250 (Pitcairn Islands), 251 (Hawaii State), 252 (Reserved), 253 (invalid), 254 (Unrecognized), 255 (ALl), 26 (Romania), 27 (Serbia), 28 (Azerbaijan), 29 (Georgia), 3 (Germany), 30 (Kazakhstan), 31 (Lithuania), 32 (Turkmenistan), 33 (Uzbekistan), 34 (Latvia), 35 (Estonia), 36 (Albania), 37 (Austria), 38 (Bosnia and Herzegovina), 39 (Ireland), 4 (Spain), 40 (Iceland), 41 (Vatican), 42 (Malta), 43 (Sweden), 44 (Switzerland), 45 (Cyprus), 46 (Turkey), 47 (SLovenia), 48 (Montenegro), 49 (Kosovo), 5 (Italy), 50 (Andorra), 51 (Armenia), 52 (Monaco), 53 (Liechtenstein), 54 (San Marino), 55 (Reserved), 56 (Reserved), 57 (Reserved), 58 (Reserved), 59 (China), 6 (Netherlands), 60 (Bahrain), 61 (South Korea), 62 (Lebanon), 63 (Nepal), 64 (Thailand), 65 (Pakistan), 66 (United Arab Emirates), 67 (Bhutan), 68 (Oman), 69 (North Korea), 7 (Poland), 70 (Philippines), 71 (Cambodia), 72 (Qatar), 73 (Kyrgyzstan), 74 (Maldives), 75 (Malaysia), 76 (Mongolia), 77 (Saudi Arabia), 78 (Brunei), 79 (Laos), 8 (Slovakia), 80 (Japan), 81 (Turkey), 82 (Palestinian), 83 (Tajikistan), 84 (Kuwait), 85 (Syria), 86 (India), 87 (Indonesia), 88 (Afghanistan), 89 (Sri Lanka), 9 (Belarus), 90 (Iraq), 91 (Vietnam), 92 (Iran), 93 (Yemen), 94 (Jordan), 95 (Burma/Myanmar), 96 (Sikkim), 97 (Bangladesh), 98 (Singapore), 99 (Democratic Republic of Timor-Leste)-->0
    </country>
    <parkingSerialNo>
      <!--ro, opt, string, parking serial number, range:[1,16], desc:This node is valid when the value of <illegalTrafficEvent> is parallelParking, and the related URI is /ISAPI/Traffic/channels/<ID>/AID/<SID>/parallelParking.-->1
    </parkingSerialNo>
    <relatedLaneNo>
      <!--ro, opt, string, linked lane number-->test
    </relatedLaneNo>
    <targetType>
      <!--ro, opt, enum, target type, subType:string, desc:unknown, invalidTarget (invalid target), largeSizedVehicle (large-sized vehicle), smallSizedVehicle (small-sized vehicle), vehicle, nonMotor (non-motor vehicle), motorcycle, pedestrian, nonQuadricycle (non-quadricycle), twoWheelVehicle (two wheeler), threeWheelVehicle (tricycle), face, triangleRack (triangle rack), bus, truck, smoke, fire, banner, stall, headandShoulder (head and shoulders), head, trafficSign (traffic sign (speed limit sign, weight limit sign, etc.)), trafficLight (traffic light), laneLine (lane line), curb, roadSurfaceMarking (road marking), wheel, vehicleHead (vehicle head), vehicleTail (vehicle tail), vehicleFront (vehicle front), vehicleRear (vehicle rear), licensePlate (license plate), vehicleWindow (vehicle window), roadSurface (road surface), zebraCrossing (zebra crossing), generalObstacle (general obstacle), airplane, UAV, ship, police, policeVehicle (police vehicle), constructionStaff (construction worker), constructionVehicle (construction vehicle)-->unknown
    </targetType>
    <CRIndex>
      <!--ro, opt, enum, country/region, subType:int, desc:0 (Unsupported. It indicates that the algorithm library does not support recognizing license plates of this country), 1 (Czech Republic), 10 (Moldova), 100 (Reserved), 101 (Reserved), 102 (Reserved), 103 (Reserved), 104 (Egypt), 105 (Libya), 106 (Sudan), 107 (Tunisia), 108 (Algeria), 109 (Morocco), 11 (Russia), 110 (Ethiopia), 111 (Eritrea), 112 (Somalia Democratic), 113 (Djibouti), 114 (Kenya), 115 (Tanzania), 116 (Uganda), 117 (Rwanda), 118 (Burundi), 119 (Seychelles), 12 (Ukraine), 120 (Chad), 121 (Central African), 122 (Cameroon), 123 (Equatorial Guinea), 124 (Gabon), 125 (Republic of the Congo (Congo-Brazzaville)), 126 (Democratic Republic of the Congo (Congo-Kinshasa)), 127 (São Tomé and Príncipe), 128 (Mauritania), 129 (Western Sahara (Saharaui)), 13 (Belgium), 130 (Senegal), 131 (Gambia), 132 (Mali), 133 (Burkina Faso), 134 (Guinea), 135 (Guinea-Bissau), 136 (Cape Verde), 137 (Sierra Leone), 138 (Liberia), 139 (Ivory Coast), 14 (Bulgaria), 140 (Ghana), 141 (Togo), 142 (Benin), 143 (Niger), 144 (Zambia), 145 (Angola), 146 (Zimbabwe), 147 (Malawi), 148 (Mozambique), 149 (Botswana), 15 (Denmark), 150 (Namibia), 151 (South Africa), 152 (Swaziland), 153 (Lesotho), 154 (Madagascar), 155 (Comoros), 156 (Mauritius), 157 (Nigeria), 158 (South Sudan), 159 (Saint Helena), 16 (Finland), 160 (Mayotte), 161 (Reunion), 162 (Canary Islands), 163 (AZORES), 164 (Madeira), 165 (Reserved), 166 (Reserved), 167 (Reserved), 168 (Reserved), 169 (Canada), 17 (United Kingdom Great Britain), 170 (Greenland), 171 (St. Pierre and Miquelon), 172 (United State), 173 (Bermuda), 174 (Mexico), 175 (Guatemala), 176 (Belize), 177 (El Salvador), 178 (Honduras), 179 (Nicaragua), 18 (Greece), 180 (Costa Rica), 181 (Panama), 182 (Bahamas), 183 (Turks and  Caicos Islands), 184 (Cuba), 185 (Jamaica), 186 (Cayman Islands), 187 (Haiti), 188 (Dominican), 189 (Puerto Rico), 19 (Croatia), 190 (United States Virgin Islands), 191 (British Virgin Islands), 192 (Anguilla), 193 (Antigua and Barbuda), 194 (Collectivité de Saint-Martin), 195 (Autonomous country), 196 (Saint-Barthélemy), 197 (Saint Kitts and Nevis), 198 (Montserrat), 199 (Guadeloupe), 2 (France), 20 (Hungary), 200 (Dominica), 201 (Martinique), 202 (St. Lucia), 203 (Saint Vincent and the Grenadines), 204 (Grenada), 205 (Barbados), 206 (Trinidad and Tobago), 207 (Curaçao), 208 (Aruba), 209 (Netherlands Antilles), 21 (Israel), 210 (Colombia), 211 (Venezuela), 212 (Guyana), 213 (Suriname), 214 (French Guiana), 215 (Ecuador), 216 (Peru), 217 (Bolivia), 218 (Paraguay), 219 (Chile), 22 (Luxembourg), 220 (Brazil), 221 (Uruguay), 222 (Argentina), 223 (Reserved), 224 (Reserved), 225 (Reserved), 226 (Reserved), 227 (Australia), 228 (New Zealand ), 229
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```
(Papua New Guinea), 23 (Macedonia, North Macedonia after 2018), 230 (Solomon Islands), 231 (Vanuatu), 232 (New Caledonia), 233 (Palau), 234 (Federated
States of Micronesia), 235 (Marshall Island), 236 (Northern Mariana Islands), 237 (Guam), 238 (Nauru), 239 (Kiribati), 24 (Norway), 240 (Fiji Islands), 241
(Tonga), 242 (Tuvalu), 243 (Wallis et Futuna), 244 (Samoa), 245 (Eastern Samoa), 246 (Tokelau), 247 (Niue), 248 (Cook Islands), 249 (French Polynesia), 25
(Portugal), 250 (Pitcairn Islands), 251 (Hawaii State), 252 (Reserved), 253 (invalid), 254 (Unrecognized), 255 (All), 26 (Romania), 27 (Serbia), 28
(Azerbaijan), 29 (Georgia), 3 (Germany), 30 (Kazakhstan), 31 (Lithuania), 32 (Turkmenistan), 33 (Uzbekistan), 34 (Latvia), 35 (Estonia), 36 (Albania), 37
(Austria), 38 (Bosnia and Herzegovina), 39 (Ireland), 4 (Spain), 40 (Iceland), 41 (Vatican), 42 (Malta), 43 (Sweden), 44 (Switzerland), 45 (Cyprus), 46
(Turkey), 47 (Slovenia), 48 (Montenegro), 49 (Kosovo), 5 (Italy), 50 (Andorra), 51 (Armenia), 52 (Monaco), 53 (Liechtenstein), 54 (San Marino), 55
(Reserved), 56 (Reserved), 57 (Reserved), 58 (Reserved), 59 (China), 6 (Netherlands), 60 (Bahrain), 61 (South Korea), 62 (Lebanon), 63 (Nepal), 64
(Thailand), 65 (Pakistan), 66 (United Arab Emirates), 67 (Bhutan), 68 (Oman), 69 (North Korea), 7 (Poland), 70 (Philippines), 71 (Cambodia), 72 (Qatar), 73
(Kyrgyzstan), 74 (Maldives), 75 (Malaysia), 76 (Mongolia), 77 (Saudi Arabia), 78 (Brunei), 79 (Laos), 8 (Slovakia), 80 (Japan), 81 (Turkey), 82
(Palestinian), 83 (Tajikistan), 84 (Kuwait), 85 (Syria), 86 (India), 87 (Indonesia), 88 (Afghanistan), 89 (Sri Lanka), 9 (Belarus), 90 (Iraq), 91 (Vietnam),
92 (Iran), 93 (Yemen), 94 (Jordan), 95 (Burma/Myanmar), 96 (Sikkim), 97 (Bangladesh), 98 (Singapore), 99 (Democratic Republic of Timor-Leste)-->0
    </CRIndex>
    <alarmLevel>
      <!--ro, opt, enum, alarm priority, subType:string, desc:alert (pre-alarm)-->alert
    </alarmLevel>
    <alertStartTime>
      <!--ro, opt, datetime, pre-alarm start time-->1970-01-01T00:00:00+08:00
    </alertStartTime>
    <alertEndTime>
      <!--ro, opt, datetime, pre-alarm end time-->1970-01-01T00:00:00+08:00
    </alertEndTime>
    <uid>
      <!--ro, opt, string, 64-byte unique identifier of VCA event uploading linkage, which can be generated by the time (it is accurate to millisecond) and
a random number, range:[1,64], desc:The unique identifier indicates the uniqueness of the recognized target in a capture event.-->test
    </uid>
    <targetID>
      <!--ro, opt, int, ID of the captured target, desc:The ID indicates the uniqueness of the recognized target in a capture event.-->0
    </targetID>
    <laneNo>
      <!--ro, opt, int, number of the lane where the target is located-->1
    </laneNo>
    <illegalType>
      <!--ro, opt, string, violation type, desc:N/A-->1001A
    </illegalType>
    <isOverSpeed>
      <!--ro, opt, bool, whether the vehicle is speeding-->true
    </isOverSpeed>
    <isRapidDeceleration>
      <!--ro, opt, bool, whether the vehicle is decelerating suddenly-->true
    </isRapidDeceleration>
    <illegalParkingEvent>
      <!--ro, opt, enum, illegal parking event of the vehicle, subType:string, desc:crossLane (parking over the lane line), crossBerth (occupying two
parking spaces), crossLaneBerth (parking over the lane line and occupying two parking spaces), illegalParking (illegal parking)-->crossLane
    </illegalParkingEvent>
    <illegalTrafficEventSubType>
      <!--ro, opt, enum, event sub-type, subType:string, desc:passengers, abandonedObject (fallen object), construction, trafficAccident (traffic accident),
roadBlock (roadblock), unknown-->passengers
    </illegalTrafficEventSubType>
    <Position>
      <!--ro, opt, object, offset of the target center relative to the camera (unit: meter), desc:An X-coordinate is positive if it is to the right of the
Y-axis and negative if it is to the left of the Y-axis. A Y-coordinate is positive if it is above of the X-axis and negative values are not supported. The
origin is the center point (the center point of all lane lines) of the horizontal line where the camera is located and which is perpendicular to the lane
line.-->
      <x>
        <!--ro, req, float, X-coordinate-->0.000
      </x>
      <y>
        <!--ro, req, float, Y-coordinate-->0.000
      </y>
    </Position>
    <AIDGPSInfo>
      <!--ro, opt, object, GPS information of the location where the AID event (except the congestion event) occurred-->
      <longitudeType>
        <!--ro, req, enum, longitude, subType:string, desc:E (eastern longitude), W (western longitude)-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, latitude, subType:string, desc:S (southern latitude), N (northern latitude)-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude-->
        <degree>
          <!--ro, req, int, degree-->1
        </degree>
        <minute>
          <!--ro, req, int, minute-->1
        </minute>
        <sec>
          <!--ro, req, float, description, desc:The value is accurate to six decimal places.-->0.000000
        </sec>
      </Longitude>
      <Latitude>
        <!--ro, req, object, latitude-->
        <degree>
          <!--ro, req, int, degree-->1
        </degree>
        <minute>
          <!--ro, req, int, minute-->1
        </minute>
        <sec>
          <!--ro, req, float, description, desc:The value is accurate to six decimal places.-->0.000000
        </sec>
      </Latitude>
    </AIDGPSInfo>
```

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```
<CongestionHeadGPSInfo>
  <!--ro, opt, object-->
  <longitudeType>
    <!--ro, req, enum, subType:string-->E
  </longitudeType>
  <latitudeType>
    <!--ro, req, enum, subType:string-->S
  </latitudeType>
  <Longitude>
    <!--ro, req, object-->
    <degree>
      <!--ro, req, int-->1
    </degree>
    <minute>
      <!--ro, req, int-->1
    </minute>
    <sec>
      <!--ro, req, float-->0.000000
    </sec>
  </Longitude>
  <Latitude>
    <!--ro, req, object-->
    <degree>
      <!--ro, req, int-->1
    </degree>
    <minute>
      <!--ro, req, int-->1
    </minute>
    <sec>
      <!--ro, req, float-->0.000000
    </sec>
  </Latitude>
</CongestionHeadGPSInfo>
<CongestionTailGPSInfo>
  <!--ro, opt, object-->
  <longitudeType>
    <!--ro, req, enum, subType:string-->E
  </longitudeType>
  <latitudeType>
    <!--ro, req, enum, subType:string-->S
  </latitudeType>
  <Longitude>
    <!--ro, req, object-->
    <degree>
      <!--ro, req, int-->1
    </degree>
    <minute>
      <!--ro, req, int-->1
    </minute>
    <sec>
      <!--ro, req, float-->0.000000
    </sec>
  </Longitude>
  <Latitude>
    <!--ro, req, object-->
    <degree>
      <!--ro, req, int-->1
    </degree>
    <minute>
      <!--ro, req, int-->1
    </minute>
    <sec>
      <!--ro, req, float-->0.000000
    </sec>
  </Latitude>
</CongestionTailGPSInfo>
<CongestionLaneGPSInfo>
  <!--ro, opt, object, GPS location information of the center point of the lane in which the congestion event occurred-->
  <longitudeType>
    <!--ro, req, enum, longitude, subType:string, desc:E (eastern longitude), W (western longitude)-->E
  </longitudeType>
  <latitudeType>
    <!--ro, req, enum, latitude, subType:string, desc:S (southern latitude), N (northern latitude)-->S
  </latitudeType>
  <Longitude>
    <!--ro, req, object, longitude-->
    <degree>
      <!--ro, req, int, degree-->1
    </degree>
    <minute>
      <!--ro, req, int, minute-->1
    </minute>
    <sec>
      <!--ro, req, float, description, desc:The value is accurate to six decimal places.-->0.000000
    </sec>
  </Longitude>
  <Latitude>
    <!--ro, req, object, latitude-->
    <degree>
      <!--ro, req, int, degree-->1
    </degree>
    <minute>
      <!--ro, req, int, minute-->1
```

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        </minute>
        <sec>
          <!--ro, req, float, description, desc:The value is accurate to six decimal places.-->0.000000
        </sec>
      </Latitude>
    </CongestionLaneGPSInfo>
    <visibility>
      <!--ro, opt, int, visibility in thick fog, range:[0,30000], unit:m, dep:and,{$.EventNotificationAlert.AID.illegalTrafficEvent,eq,fogDetection}-->30
    </visibility>
    <RuleInfo>
      <!--ro, opt, object, AID rule information-->
      <ruleID>
        <!--ro, opt, int, ID of the rule according to which the event is detected, range:[1,16]-->1
      </ruleID>
      <ruleName>
        <!--ro, opt, string, name of the rule according to which the event is detected, range:[1,32]-->name
      </ruleName>
    </RuleInfo>
    <DirectPosition>
      <!--ro, opt, object, direction line of alarm detection, desc:The origin is the upper-left corner of the screen.-->
      <StartPoint>
        <!--ro, opt, object, coordinates of the start point-->
        <x>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </x>
        <y>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </y>
      </StartPoint>
      <EndPoint>
        <!--ro, opt, object, coordinates of the end point-->
        <x>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </x>
        <y>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </y>
      </EndPoint>
    </DirectPosition>
    <speedLimit>
      <!--ro, opt, int, current speed limit, range:[0,255], unit:km/h-->100
    </speedLimit>
    <state>
      <!--ro, opt, enum, event uploading status, subType:int, desc:0 (upload changes), 1 (upload auto-switch), 2 (upload the data captured by the current device by schedule). The value 2 is valid when the value of dwAIDType (alarm/event type) is the parking event.-->0
    </state>
    <fogUploadMode>
      <!--ro, opt, enum, method of uploading the thick fog event, subType:string, dep:and,
{$.EventNotificationAlert.AID.illegalTrafficEvent,eq,fogDetection}, desc:trigger (upload the alarm when it is triggered), timing (upload the alarm by schedule). If this node is not returned, the default value is trigger.-->trigger
    </fogUploadMode>
    <trafficAccidentInfoList>
      <!--ro, opt, array, subType:object, dep:or,{$.EventNotificationAlert.TFS.illegalTrafficEvent,eq,trafficAccident}-->
      <trafficAccidentInfo>
        <!--ro, opt, object-->
        <PlateInfo>
          <!--ro, opt, object-->
          <plate>
            <!--ro, req, string-->蓝浙A12345
          </plate>
          <plateType>
            <!--ro, req, enum, subType:string-->02TypePersonalized
          </plateType>
          <plateColor>
            <!--ro, req, enum, subType:string-->black
          </plateColor>
        </PlateInfo>
        <trafficAccidentStartDateTime>
          <!--ro, opt, datetime-->2004-05-03T17:30:08+08:00
        </trafficAccidentStartDateTime>
        <trafficAccidentGPSInfo>
          <!--ro, opt, object-->
          <longitudeType>
            <!--ro, req, enum, subType:string-->E
          </longitudeType>
          <latitudeType>
            <!--ro, req, enum, subType:string-->S
          </latitudeType>
          <Longitude>
            <!--ro, req, object-->
            <degree>
              <!--ro, req, int-->1
            </degree>
            <minute>
              <!--ro, req, int-->1
            </minute>
            <sec>
              <!--ro, req, float-->0.000000
            </sec>
          </Longitude>
          <Latitude>
            <!--ro, req, object-->
            <degree>
              <!--ro, req, int-->1
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            <!--ro, req, int-->1
          </degree>
          <minute>
            <!--ro, req, int-->1
          </minute>
          <sec>
            <!--ro, req, float-->0.000000
          </sec>
        </Latitude>
      </trafficAccidentGPSInfo>
      <targetMovementInfoList>
        <!--ro, opt, array, subType:object-->
        <targetMovementInfo>
          <!--ro, opt, object-->
          <longitudeType>
            <!--ro, req, enum, subType:string-->E
          </longitudeType>
          <latitudeType>
            <!--ro, req, enum, subType:string-->S
          </latitudeType>
          <Longitude>
            <!--ro, req, object-->
            <degree>
              <!--ro, req, int-->1
            </degree>
            <minute>
              <!--ro, req, int-->1
            </minute>
            <sec>
              <!--ro, req, float-->0.000000
            </sec>
          </Longitude>
          <Latitude>
            <!--ro, req, object-->
            <degree>
              <!--ro, req, int-->1
            </degree>
            <minute>
              <!--ro, req, int-->1
            </minute>
            <sec>
              <!--ro, req, float-->0.000000
            </sec>
          </Latitude>
          <targetMovementDateTime>
            <!--ro, opt, datetime-->2004-05-03T17:30:08+08:00
          </targetMovementDateTime>
        </targetMovementInfo>
      </targetMovementInfoList>
    </trafficAccidentInfo>
  </trafficAccidentInfoList>
  <alarmDistance>
    <!--ro, opt, int, range:[20,220], unit:cm-->20
  </alarmDistance>
  <lowSpeedLimit>
    <!--ro, opt, int, range:[20,200], unit:km/h-->20
  </lowSpeedLimit>
  <averageSpeed>
    <!--ro, opt, float, range:[0,200], unit:km/h-->60.0
  </averageSpeed>
  <congestionLevel>
    <!--ro, opt, int, range:[1,3]-->3
  </congestionLevel>
  <congestionQueueLen>
    <!--ro, opt, int, range:[0,10000], unit:m-->10
  </congestionQueueLen>
  <visibilityLevel>
    <!--ro, opt, int, range:[0,4]-->3
  </visibilityLevel>
  <abandonedObjectVehicleProportion>
    <!--ro, opt, float, range:[0.00,10.00]-->0.500
  </abandonedObjectVehicleProportion>
  <relateObjectInfoList>
    <!--ro, opt, array, subType:object, range:[0,32]-->
    <relateObjectInfo>
      <!--ro, opt, object-->
      <targetType>
        <!--ro, req, enum, subType:string-->triangleRack
      </targetType>
      <objectRect>
        <!--ro, opt, object-->
        <X>
          <!--ro, req, int, range:[0,1000]-->1
        </X>
        <Y>
          <!--ro, req, int, range:[0,1000]-->1
        </Y>
        <width>
          <!--ro, req, int, range:[0,1000]-->1
        </width>
        <height>
          <!--ro, req, int, range:[0,1000]-->1
        </height>
      </objectRect>
```

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```
        <targetID>
          <!--ro, opt, int, range:[1,1000000]-->1
        </targetID>
      </relateObjectInfo>
    </relateObjectInfoList>
    <multiTargetList>
      <!--ro, opt, array, subType:object, range:[0,8]-->
      <multiTarget>
        <!--ro, opt, object-->
        <targetType>
          <!--ro, opt, enum, subType:string-->unknown
        </targetType>
      </multiTarget>
    </multiTargetList>
  </AID>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission method, subType:string, desc:url (picture URL), binary (binary picture data)-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode-->1
  </detectionPicturesNumber>
  <contentIDSence1>
    <!--ro, opt, string, scene picture ID of the vehicle (it indicates the uniqueness of the picture data), range:[1,32], desc:The generation rule is the
device's serial number + the time elapsed since the device was started + a random number-->D6A18E41308CE9934DCDDD1111
  </contentIDSence1>
  <contentIDSence2>
    <!--ro, opt, string, scene picture ID of the vehicle (it indicates the uniqueness of the picture data), range:[1,32], desc:The generation rule is the
device's serial number + the time elapsed since the device was started + a random number-->D6A18E41308CE9934DCDDD1111
  </contentIDSence2>
  <contentIDSence3>
    <!--ro, opt, string, scene picture ID of the vehicle (it indicates the uniqueness of the picture data), range:[1,32], desc:The generation rule is the
device's serial number + the time elapsed since the device was started + a random number-->D6A18E41308CE9934DCDDD1111
  </contentIDSence3>
  <contentIDVehiclePlate>
    <!--ro, opt, string, vehicle picture ID (it indicates the uniqueness of the picture data), range:[1,32], desc:The generation rule is the device's serial
number + the time elapsed since the device was started + a random number-->D6A18E41308CE9934DCDDD2222
  </contentIDVehiclePlate>
  <contentIDComposite>
    <!--ro, opt, string, composite vehicle picture (it indicates the uniqueness of the picture data), range:[1,32], desc:The generation rule is the device's
serial number + the time elapsed since the device was started + a random number-->D6A18E41308CE9934DCDDD2222
  </contentIDComposite>
  <PictureURLInfoList>
    <!--ro, opt, array, list of picture URL information, subType:object, desc:This node is returned when the value of <detectionPictureTransType> is url.-->
    <PictureURLInfo>
      <!--ro, opt, object, picture URL information-->
      <pictureName>
        <!--ro, req, enum, picture name, subType:string, desc:detectionPicture (captured background picture), licensePlatePicture (license plate close-up)--
>detectionPicture
      </pictureName>
      <pictureURL>
        <!--ro, req, string, picture URL-->http://127.0.0.1
      </pictureURL>
      <plateRect>
        <!--ro, opt, object, coordinates of the license plate frame, desc:This node is valid when the value of <type> is detectionPicture. The origin is the
upper-left corner of the screen.-->
        <X>
          <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1
        </X>
        <Y>
          <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1
        </Y>
        <width>
          <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1
        </width>
        <height>
          <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1
        </height>
      </plateRect>
      <objectInfo>
        <!--ro, opt, object, target information-->
        <targetType>
          <!--ro, req, enum, target type, subType:string, desc:unknown, invalidTarget (invalid target), largeSizedVehicle (large-sized vehicle),
smallSizedVehicle (small-sized vehicle), vehicle, nonMotor (non-motor vehicle), motorcycle, pedestrian, nonQuadricycle (non-quadricycle), twoWheelVehicle
(two wheeler), threeWheelVehicle (tricycle), face, triangleRack (triangle rack), bus, truck, smoke, fire, banner, stall, headandShoulder (head and
shoulders), head, trafficSign (traffic sign (speed limit sign, weight limit sign, etc.)), trafficLight (traffic light), laneLine (lane line), curb,
roadSurfaceMarking (road marking), wheel, vehicleHead (vehicle head), vehicleTail (vehicle tail), vehicleFront (vehicle front), vehicleRear (vehicle rear),
licensePlate (license plate), vehicleWindow (vehicle window), roadSurface (road surface), zebraCrossing (zebra crossing), generalObstacle (general
obstacle), airplane, UAV, ship-->unknown
        </targetType>
        <confidence>
          <!--ro, opt, float, confidence level, range:[0.00,100.00]-->0.00
        </confidence>
        <objectRect>
          <!--ro, opt, object, coordinates of the target, desc:The origin is the upper-left corner of the screen.-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1
          </width>
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            <height>
              <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1
            </height>
          </objectRect>
        </objectInfo>
      </PictureURLInfo>
    </PictureURLInfoList>
    <UUID>
      <!--ro, opt, string, UUID, desc:If the vehicle is captured many times, the UUID is the same.-->test
    </UUID>
    <targetAttrs>
      <!--ro, opt, string, target attributes. This node is only supported by data center products and will be transparently transmitted., desc:target attributes. This node is only supported by data center products and will be transparently transmitted.-->{"deviceId": "1","deviceChannel": 1,"deviceName": "shebei"}
    </targetAttrs>
    <monitoringSiteID>
      <!--ro, opt, string, camera number-->test
    </monitoringSiteID>
    <monitorDescription>
      <!--ro, opt, string, camera information-->test
    </monitorDescription>
    <SceneInfo>
      <!--ro, opt, object, scene information-->
      <scenesID>
        <!--ro, opt, string, scene ID, range:[1,16]-->test
      </scenesID>
      <sceneName>
        <!--ro, opt, string, scene name, range:[0,128]-->test
      </sceneName>
      <PTZPos>
        <!--ro, opt, object, PTZ position-->
        <elevation>
          <!--ro, opt, int, elevation, range:[-900,2700]-->1
        </elevation>
        <azimuth>
          <!--ro, opt, int, azimuth, range:[0,3600]-->1
        </azimuth>
        <absoluteZoom>
          <!--ro, opt, int, zooming ratio, range:[1,1000]-->1
        </absoluteZoom>
      </PTZPos>
      <relatedAreaNo>
        <!--ro, opt, string, linked area number, range:[1,8], desc:This number is generated by calling /ISAPI/Traffic/channels/<channelID>/eventRule/<SID> and corresponds to <ruleId>. The URI for configuring congestion event parameters is /ISAPI/Traffic/channels/<channelID>/AID/<SID>/ployCongestion in which the linked area number (<relatedAreaNo>) needs to be configured.-->1
      </relatedAreaNo>
      <RelatedRegionCoordinatesList>
        <!--ro, opt, array, coordinate list of linked areas, subType:object, range:[3,10], desc:This list contains coordinates of areas in <relatedAreaNo>.-->
        <RegionCoordinates>
          <!--ro, opt, object, coordinates of the area, desc:The origin is the upper-left corner of the screen.-->
          <positionX>
            <!--ro, req, float, X-coordinate, range:[0.000,1.000]-->0.000
          </positionX>
          <positionY>
            <!--ro, req, float, Y-coordinate, range:[0.000,1.000]-->0.000
          </positionY>
        </RegionCoordinates>
      </RelatedRegionCoordinatesList>
      <regionType>
        <!--ro, opt, enum, subType:string-->normal
      </regionType>
      <regionName>
        <!--ro, opt, string, range:[0,128]-->test
      </regionName>
    </SceneInfo>
    <detectDir>
      <!--ro, opt, enum, detection direction, subType:int, desc:1 (upward), 2 (downward), 3 (bidirection), 4 (from east to west), 5 (from south to north), 6 (from west to east), 7 (from north to south), 8 (other)-->1
    </detectDir>
    <dateMilliSecondTime>
      <!--ro, opt, datetime, time when the alarm is triggered. The time can be accurate to millisecond.-->2004-05-03T17:30:08.000+08:00
    </dateMilliSecondTime>
    <deviceUUID>
      <!--ro, opt, string, device number, range:[1,32], desc:The device number is the device UUID which is the device's serial number and can be edited by the node <deviceID> in the message of /ISAPI/System/deviceInfo.-->12345
    </deviceUUID>
    <vehiclePicture>
      <!--ro, opt, string, The generation rule is the device's serial number + the time elapsed since the device was started + a random number, range:[1,32], desc:The generation rule is the device's serial number + the time elapsed since the device was started + a random number-->D6A18E41308CE9934DCDDD1111
    </vehiclePicture>
    <batteryInfo>
      <!--ro, opt, object-->
      <batterStatus>
        <!--ro, opt, enum, subType:string-->normal
      </batterStatus>
      <batteryPower>
        <!--ro, opt, int, range:[0,100], step:1, unit:%, dep:and,{$.EventNotificationAlert.batteryInfo.batterStatus,eq,normal}-->1
      </batteryPower>
    </batteryInfo>
    <isAlert>
      <!--ro, opt, bool-->true
    </isAlert>
    <binaryPictureObjectInfo>
      <!--ro, opt, object, dep:and,{$.EventNotificationAlert.detectionPictureTransType,eq,binary}-->
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      <targetType>
        <!--ro, req, enum, subType:string-->unknown
      </targetType>
      <confidence>
        <!--ro, opt, float, range:[0.00,100.00]-->0.00
      </confidence>
      <objectRect>
        <!--ro, opt, object-->
        <X>
          <!--ro, req, int, range:[0,1000]-->1
        </X>
        <Y>
          <!--ro, req, int, range:[0,1000]-->1
        </Y>
        <width>
          <!--ro, req, int, range:[0,1000]-->1
        </width>
        <height>
          <!--ro, req, int, range:[0,1000]-->1
        </height>
      </objectRect>
    </binaryPictureObjectInfo>
    <AIDBackgroundImageResolution>
      <!--ro, opt, object-->
      <height>
        <!--ro, req, int, unit:px-->1
      </height>
      <width>
        <!--ro, req, int, unit:px-->1
      </width>
    </AIDBackgroundImageResolution>
    <plateInfoList>
      <!--ro, opt, object-->
      <plateInfo>
        <!--ro, opt, object-->
        <plateRect>
          <!--ro, opt, object-->
          <X>
            <!--ro, req, int, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </plateRect>
        <plateColor>
          <!--ro, opt, enum, subType:string-->black
        </plateColor>
        <plateNo>
          <!--ro, req, string, range:[1,32]-->蓝浙A12345
        </plateNo>
        <confidence>
          <!--ro, opt, float, range:[0.0,100.0]-->0.0
        </confidence>
        <province>
          <!--ro, opt, enum, subType:int-->0
        </province>
      </plateInfo>
    </plateInfoList>
    <isDataRetransmission>
      <!--ro, opt, bool-->true
    </isDataRetransmission>
    <ultrasonicRawData>
      <!--ro, opt, string-->test
    </ultrasonicRawData>
    <relaLaneDirectionType>
      <!--ro, opt, enum, subType:int-->1
    </relaLaneDirectionType>
    <regionTrackDir>
      <!--ro, opt, enum, subType:string-->leftToRight
    </regionTrackDir>
    <congestionVehicleNum>
      <!--ro, opt, int, range:[0,10000]-->100
    </congestionVehicleNum>
    <congestionOverTimeVehicleNum>
      <!--ro, opt, int, range:[0,10000]-->80
    </congestionOverTimeVehicleNum>
  </EventNotificationAlert>
```

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| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| AID | [Message content] | application/xml | -- | -- | -- |
| .value | [Binary picture data] | image/jpeg | detectionPicture | detectionPicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | licensePlatePicture | licensePlatePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | compositePicture | compositePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | vehiclePicture | vehiclePicture.jpg | -- |
| .value | [Binary picture data] | image/jpeg | contentIDSence1 | contentIDSence1.jpg | -- |
| .value | [Binary picture data] | image/jpeg | contentIDSence2 | contentIDSence2.jpg | -- |
| .value | [Binary picture data] | image/jpeg | contentIDSence3 | contentIDSence3.jpg | -- |

**Note:** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

## 14.6.4 Radar-Assisted Traffic Target Detection

### 14.6.4.1 Get the capability of lane parameters

**Request URL**

GET /ISAPI/Radar/laneParam/capabilities?format=json&radarChannelID=<radarChannelID>&childDevID=<devIndex>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| radarChannelID | string | Radar channel No., which can be appended to URL (e.g., /ISAPI/Radar/radarInfo?format=json&radarChannelID=1). If the parameter is absent, it defaults to radar channel 1. |
| devIndex | string | Sub device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json. |

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### Request Message

None

### Response Message

```
{
    "LanParam": {
    /*ro, req, object, lane parameters*/
        "direction": {
        /*ro, opt, object, lane direction of intersection upper border*/
            "@opt": ["forward", "backward", "all"]
            /*ro, req, array, the value range, subType:string, desc:"forward", "backward", "all"*/
        },
        "horzDistCorrect": {
        /*ro, opt, object, horizontal distance correction, desc:unit: meter*/
            "@min":  -25,
            /*ro, req, int, the minimum value*/
            "@max":  25
            /*ro, req, int, the maximum value*/
        },
        "vertDistCorrect": {
        /*ro, opt, object, vertical distance correction, desc:unit: meter*/
            "@min":  -25,
            /*ro, req, int, the minimum value*/
            "@max":  25
            /*ro, req, int, the maximum value*/
        },
        "angleCorrect": {
        /*ro, opt, object, angle correction, desc:unit: degree*/
            "@min":  -12,
            /*ro, req, int, the minimum value*/
            "@max":  12
            /*ro, req, int, the maximum value*/
        },
        "laneWidth": {
        /*ro, opt, object, lane width, desc:unit: meter*/
            "@min":  2.00,
            /*ro, req, float, the minimum value*/
            "@max":  6.00,
            /*ro, req, float, the maximum value*/
        },
        "setupHeight": {
        /*ro, opt, object, assumed height, desc:unit: meter*/
            "@min":  0.00,
            /*ro, req, float, the minimum value*/
            "@max":  10.00
            /*ro, req, float, the maximum value*/
        },
        "stopLineDistance": {
        /*ro, opt, object, distance to stop line, desc:unit: meter*/
            "@min":  0.0,
            /*ro, req, float, the minimum value*/
            "@max":  350.0
            /*ro, req, float, the maximum value*/
        },
        "sensitivity": {
        /*ro, opt, object, sensitivity*/
            "sensitivity": {
            /*ro, opt, object, sensitivity level*/
                "@min":  0.0,
                /*ro, req, float, the minimum value*/
                "@max":  12.0,
                /*ro, req, float, the maximum value*/
            },
            "advancedSensitivity": {
            /*ro, opt, object, valid distance range of advanced sensitivity, desc:it defaults to 0 (meter). And it requires only one distance parameter,
according to which the radar automatically choose the closest distance*/
                "@size":  4,
                /*ro, req, int, maximum number of groups of advanced sensitivity settings*/
                "effectDistance": {
                /*ro, opt, object, valid distance*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  250
                    /*ro, req, int, the maximum value*/
                },
                "sensitivity": {
                /*ro, opt, object, sensitivity*/
                    "@min":  0.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  12.0,
                    /*ro, req, float, the maximum value*/
                }
            }
        },
        "debugMode": {
        /*ro, opt, object, whether to enable entry/exit debugging mode*/
            "@opt": [true, false]
            /*ro, req, array, the value range, subType:bool*/
        },
        "angleCorrectEnabled": {
```

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```
        "angleCorrectEnabled" : {
        /*ro, opt, object, auto angle correction, desc:this is valid only under the debug mode*/
            "@opt": [true, false]
            /*ro, req, array, the value range, subType:bool*/
        },
        "sceneMode": {
        /*ro, opt, object, scene mode*/
        "radarPredictionEnabled": {
        /*ro, opt, object, radar prediction, desc:known as radar extrapolation. It is the switch of radar target prediction, which is used for debugging
radar parameters during installation. Radar prediction: when radar detects no target, it predicts the target's possible position based on the previous
movement direction and speed, and reports it to the camera*/
            "@opt": [true, false]
            /*ro, opt, array, options, subType:bool*/
        },
        "radarDetectDistance": {
        /*ro, opt, object, radar detection distance*/
            "@opt": [100, 150, 200, 250, 300, 350],
            /*ro, opt, array, options, subType:int*/
            "#text":  100
            /*ro, opt, int, current value*/
        },
    }
}
```


## 14.6.5 Traffic Signal Control Station

### 14.6.5.1 Get the capability of traffic light correction

**Request URL**

GET /ISAPI/ITC/lightCorrect/capabilities?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelLightCorrect is "true" returned from GET /ISAPI/ITC/capabilities). |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<LightCorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, correction parameters of traffic lights, attr:version{req, string, protocolVersion}-->
  <isSupportBCorrect>
    <!--ro, req, bool, whether to enable traffic light correction-->true
  </isSupportBCorrect>
  <mode opt="normal,intelligently">
    <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:"normal" (normal mode), "intelligently" (intelligent mode)-->normal
  </mode>
  <lightNum min="1" max="6">
    <!--ro, req, string, number of areas of traffic light correction, range:[1,6], attr:min{req, int},max{req, int}-->test
  </lightNum>
  <lightCorrcntLevel min="0" max="100">
    <!--ro, req, int, traffic light correction level, range:[0,100], attr:min{req, int},max{req, int}-->0
  </lightCorrcntLevel>
  <CorrcntLevel min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
  </CorrcntLevel>
  <lightCorrectVersion min="1" max="2">
    <!--ro, req, int, XML version of traffic light correction, range:[1,2], attr:min{req, int},max{req, int}, desc:if this node does not exist or this not
is 1,it indicates that the value is normalized to 1000x1000; if this node is 2,it indicates that the value is normalized to 10000x10000-->0
  </lightCorrectVersion>
  <LightAdvanParam>
    <!--ro, req, object, advanced red parameters of the red light-->
    <halLevel min="0" max="100" def="50">
      <!--ro, req, int, halo suppression grade, range:[0,100], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the value is,the better the
effect will be-->0
    </halLevel>
    <redLevel min="0" max="100" def="50">
      <!--ro, req, int, red intensity, range:[0,100], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the value is,the greater the intensity
will be-->0
    </redLevel>
    <whiteClip min="0" max="100" def="50">
      <!--ro, req, int, red adjustment level of too bright red light, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->0
    </whiteClip>
    <sensitivity min="0" max="100">
      <!--ro, req, int, detection sensitivity, range:[0,100], attr:min{req, int},max{req, int}, desc:detection sensitivity-->0
    </sensitivity>
    <drawRect opt="smallGreenFrame,medWhiteFrame,bigRedFrame,trackBlueFrame,notDisplay">
      <!--ro, opt, enum, display frame, subType:string, attr:opt{req, string}, desc:":smallGreenFrame" (small green frame), "medWhiteFrame" (middle white
frame), "bigRedFrame" (big red frame), "trackBlueFrame" (blue tracking frame), "notDisplay" (not display the frame)-->smallGreenFrame
    </drawRect>
    <extenPixel min="0" max="20" def="50">
      <!--ro, opt, int, number of extended pixels, range:[0,20], attr:min{req, int},max{req, int},def{req, int}-->0
    </extenPixel>
    <halExtenPixel min="0" max="40">
      <!--ro, opt, int, range:[0,40], attr:min{req, int},max{req, int}-->1
    </halExtenPixel>
    <maskExtenPixel min="0" max="10">
      <!--ro, opt, int, range:[0,40], attr:min{req, int},max{req, int}-->1
    </maskExtenPixel>
    <blendEn>
      <!--ro, opt, bool, whether to enable smoothing the red effect, desc:whether to enable smoothing the red effect-->true
    </blendEn>
    <SignalLightParamList>
      <!--ro, opt, array, list of traffic light parameters, subType:object, range:[1,6], desc:list of traffic light parameters-->
      <signalLightParam>
        <!--ro, opt, object, traffic light parameters-->
        <yellowLightDelay min="0" max="20">
          <!--ro, req, int, delay time of the yellow light, range:[0,30], unit:s, attr:min{req, int},max{req, int}-->0
        </yellowLightDelay>
        <digitalLight>
          <!--ro, opt, bool, whether it is a digital light-->true
        </digitalLight>
        <signallighttype opt="roundLight,digitalLight,arrowLight,leftTurnLight,longStraightBarLight,other">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->roundLight
        </signallighttype>
      </signalLightParam>
    </SignalLightParamList>
    <templateAdjRation min="64" max="128">
      <!--ro, opt, int, shape change extension, range:[64,128], attr:min{req, int},max{req, int}, desc:for the name of this node,refer to the device
variable name. Default: 83-->64
    </templateAdjRation>
    <maskExtenPixelDay min="0" max="10">
      <!--ro, opt, int, range:[0,40], attr:min{req, int},max{req, int}-->1
    </maskExtenPixelDay>
    <lightShapeRefine min="0" max="100">
      <!--ro, opt, int, range:[0,40], attr:min{req, int},max{req, int}-->1
    </lightShapeRefine>
    <shortFrameBrightTh min="0" max="100">
      <!--ro, opt, int, range:[0,40], attr:min{req, int},max{req, int}-->1
    </shortFrameBrightTh>
    <realTimeViewEnabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </realTimeViewEnabled>
  </LightAdvanParam>
</LightCorParam>
```

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#### 14.6.5.2 Get the correction parameters of traffic lights

**Request URL**

GET /ISAPI/ITC/lightCorrect?channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Camera ID (this optional parameter is supported when isSupportChannelLightCorrect is "true" returned from GET /ISAPI/ITC/capabilities). |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<LightCorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, correction parameters of traffic lights, attr:version{req, string, protocolVersion}-->
  <bCorrect>
    <!--ro, opt, bool, whether to enable traffic light correction-->true
  </bCorrect>
  <mode>
    <!--ro, opt, enum, mode, subType:string, desc:"normal" (normal mode), "intelligently" (intelligent mode)-->normal
  </mode>
  <lightCorrectNum>
    <!--ro, req, string, number of areas of traffic light correction, range:[1,6]-->1
  </lightCorrectNum>
  <LightCorrectAreaList>
    <!--ro, req, array, area list of traffic light correction, subType:object-->
    <lightCorrectArea>
      <!--ro, opt, object, area of traffic light correction-->
      <lightCorrectAreaXStart>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </lightCorrectAreaXStart>
      <lightCorrectAreaYStart>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </lightCorrectAreaYStart>
      <lightCorrectAreaWidth>
        <!--ro, req, int, width, range:[1,512]-->1
      </lightCorrectAreaWidth>
      <lightCorrectAreaHeight>
        <!--ro, req, int, height, range:[1,512]-->1
      </lightCorrectAreaHeight>
    </lightCorrectArea>
  </LightCorrectAreaList>
  <lightCorrectLevel>
    <!--ro, req, int, traffic light correction level, range:[0,100], desc:the default value is 50-->0
  </lightCorrectLevel>
  <CorrectLevelSwitchList>
    <!--ro, opt, array, subType:object-->
    <correctLevelTimeSwitch>
      <!--ro, opt, object-->
      <timeId>
        <!--ro, opt, int-->1
      </timeId>
      <startHour>
        <!--ro, opt, int, start hour, range:[0,12]-->0
      </startHour>
      <startMinute>
        <!--ro, opt, int, start minute, range:[0,59]-->0
      </startMinute>
      <endHour>
        <!--ro, opt, int, end hour, range:[0,12]-->23
      </endHour>
      <endMinute>
        <!--ro, opt, int, end minute, range:[0,59]-->59
      </endMinute>
      <correctLevel>
        <!--ro, opt, int, range:[0,100]-->50
      </correctLevel>
    </correctLevelTimeSwitch>
  </CorrectLevelSwitchList>
  <lightCorrectVersion>
    <!--ro, req, enum, XML version of traffic light correction, subType:string, desc:if this node does not exist or if the value is 1, it indicates that the
value is normalized to 1000x1000; if this node is 2, it indicates that the value is normalized to 10000x10000-->1
  </lightCorrectVersion>
  <LightAdvanParam>
    <!--ro, req, object, advanced red parameters of the red light-->
    <halLevel>
      <!--ro, req, int, halo suppression grade, range:[0,100], desc:the default value is 50. The larger the value is, the better the effect will be-->0
    </halLevel>
```

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```
    <redLevel>
      <!--ro, req, int, red intensity, range:[0,100], desc:the default value is 50. The larger the value is, the greater the intensity will be-->0
    </redLevel>
    <whiteClip>
      <!--ro, req, int, red adjustment level of too bright red light, range:[0,100], desc:the default value is 50-->0
    </whiteClip>
    <sensitivity>
      <!--ro, req, int, detection sensitivity, range:[1,100], desc:the larger the value is, the more easily that the red light and yellow light will be
detected-->1
    </sensitivity>
    <drawRect>
      <!--ro, opt, enum, display frame, subType:string, desc:"smallGreenFrame" (small green frame), "medWhiteFrame" (middle white frame), "bigRedFrame" (big
red frame), "trackBlueFrame" (blue tracking frame), "notDisplay" (not display the frame)-->smallGreenFrame
    </drawRect>
    <extenPixel>
      <!--ro, opt, int, number of extended pixels, range:[0,20], desc:the default value is 3-->0
    </extenPixel>
    <halExtenPixel>
      <!--ro, opt, int, range:[0,40]-->1
    </halExtenPixel>
    <maskExtenPixel>
      <!--ro, opt, int, range:[0,40]-->1
    </maskExtenPixel>
    <blendEn>
      <!--ro, opt, bool, whether to enable smoothing the red effect, desc:the default value is true-->true
    </blendEn>
    <SignalLightParamList>
      <!--ro, req, array, list of traffic light parameters, subType:object, desc:the number of traffic lights depends on the node lightCorrectNum and is
between 1 and 6-->
      <signalLightParam>
        <!--ro, opt, object, traffic light parameters-->
        <yellowLightDelay>
          <!--ro, req, int, delay time of the yellow light, range:[0,30], unit:s-->0
        </yellowLightDelay>
        <digitalLight>
          <!--ro, opt, bool, whether it is a digital light-->true
        </digitalLight>
      </signalLightParam>
    </SignalLightParamList>
    <templateAdjRation>
      <!--ro, opt, int, shape change extension, range:[64,128], desc:for the name of this node, refer to the device variable name. The default value is 83--
>64
    </templateAdjRation>
    <maskExtenPixelDay>
      <!--ro, opt, int, range:[0,40]-->1
    </maskExtenPixelDay>
    <lightShapeRefine>
      <!--ro, opt, int, range:[0,40]-->1
    </lightShapeRefine>
    <shortFrameBrightTh>
      <!--ro, opt, int, range:[0,40]-->1
    </shortFrameBrightTh>
    <realTimeViewEnabled>
      <!--ro, opt, bool-->true
    </realTimeViewEnabled>
  </LightAdvanParam>
</LightCorParam>
```

### 14.6.5.3 Set the parameters of traffic light correction

#### Request URL

PUT /ISAPI/ITC/lightCorrect?channelID=<channelID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LightCorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, correction parameters of traffic lights, attr:version{req, string, protocolVersion}-->
  <bCorrect>
    <!--opt, bool, whether to enable traffic light correction-->true
  </bCorrect>
  <mode>
    <!--opt, enum, mode, subType:string, desc:"normal" (normal mode), "intelligently" (intelligent mode)-->normal
  </mode>
  <lightCorrectNum>
    <!--req, string, number of areas of traffic light correction, range:[1,6]-->1
  </lightCorrectNum>
  <LightCorrectAreaList>
    <!--req, array, area list of traffic light correction, subType:object-->
```

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```
    <lightCorrectArea>
      <!--opt, object, area of traffic light correction-->
      <lightCorrectAreaXStart>
        <!--req, int, x-coordinate, range:[0,1000]-->0
      </lightCorrectAreaXStart>
      <lightCorrectAreaYStart>
        <!--req, int, y-coordinate, range:[0,1000]-->0
      </lightCorrectAreaYStart>
      <lightCorrectAreaWidth>
        <!--req, int, width, range:[1,512]-->1
      </lightCorrectAreaWidth>
      <lightCorrectAreaHeight>
        <!--req, int, height, range:[1,512]-->1
      </lightCorrectAreaHeight>
    </lightCorrectArea>
  </LightCorrectAreaList>
  <lightCorrectLevel>
    <!--req, int, traffic light correction level, range:[0,100], desc:traffic light correction level,value range:[0,100],default: 50-->0
  </lightCorrectLevel>
  <CorrectLevelSwitchList>
    <!--opt, array, subType:object-->
    <correctLevelTimeSwitch>
      <!--opt, object-->
      <timeId>
        <!--opt, int-->1
      </timeId>
      <startHour>
        <!--opt, int, range:[0,12]-->0
      </startHour>
      <startMinute>
        <!--opt, int, range:[0,59]-->0
      </startMinute>
      <endHour>
        <!--opt, int, range:[0,12]-->23
      </endHour>
      <endMinute>
        <!--opt, int, range:[0,59]-->59
      </endMinute>
      <correctLevel>
        <!--opt, int, range:[0,100]-->50
      </correctLevel>
    </correctLevelTimeSwitch>
  </CorrectLevelSwitchList>
  <lightCorrectVersion>
    <!--req, enum, XML version of traffic light correction, subType:string, desc:if this node does not exist or this it is 1, it indicates that the value is
normalized to 1000x1000; if this node is 2, it indicates that the value is normalized to 10000x10000-->1
  </lightCorrectVersion>
  <LightAdvanParam>
    <!--req, object, advanced red parameters of the red light-->
    <halLevel>
      <!--req, int, halo suppression grade, range:[0,100], desc:the default value is 50. The larger the value is, the better the effect will be-->0
    </halLevel>
    <redLevel>
      <!--req, int, red intensity, range:[0,100], desc:the default value is 50. The larger the value is, the greater the intensity will be-->0
    </redLevel>
    <whiteClip>
      <!--req, int, red adjustment level of too bright red light, range:[0,100], desc:default: 50-->0
    </whiteClip>
    <sensitivity>
      <!--req, int, detection sensitivity, range:[1,100], desc:the larger the value is, the more easily that the red light and yellow light will be
detected-->1
    </sensitivity>
    <drawRect>
      <!--opt, enum, display frame, subType:string, desc:“smallGreenFrame” (small green frame), “medWhiteFrame” (middle white frame), “bigRedFrame” (big red
frame), “trackBlueFrame” (blue tracking frame), “notDisplay” (not display the frame)-->smallGreenFrame
    </drawRect>
    <extenPixel>
      <!--opt, int, number of extended pixels, range:[0,20], desc:the default value is 3-->0
    </extenPixel>
    <halExtenPixel>
      <!--opt, int, range:[0,40]-->1
    </halExtenPixel>
    <maskExtenPixel>
      <!--opt, int, range:[0,40]-->1
    </maskExtenPixel>
    <blendEn>
      <!--opt, bool, whether to enable smoothing the red effect, desc:default value: true-->true
    </blendEn>
    <SignalLightParamList>
      <!--req, array, list of traffic light parameters, subType:object, desc:the number of traffic lights depends on the node lihgtCorrectNum and is between
1 and 6-->
      <signalLightParam>
        <!--opt, object, traffic light parameters-->
        <yellowLightDelay>
          <!--req, int, delay time of the yellow light, range:[0,30], unit:s-->0
        </yellowLightDelay>
        <digitalLight>
          <!--opt, bool, whether it is a digital light-->true
        </digitalLight>
      </signalLightParam>
    </SignalLightParamList>
    <templateAdjRation>
      <!--opt, int, shape change extension, range:[64,128], desc:for the name of this node, refer to the device variable name. Default: 83-->64
    </templateAdjRation>
```

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```
  </LightAdvanParam>
</LightCorParam>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

# 14.7 Multi-Target-Type Detection

## 14.7.1 Multi-Target-Type Detection

### 14.7.1.1 Multi-target-type detection

### EventType:mixedTargetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The
value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video
channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "dateTimeMillisecond":  999,
    /*ro, opt, int, the milliseconds of the alarm trigger time, range:[0,999], desc:it should be used together with field dateTime*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is the event which is triggered
continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object
recognition occurs, and the recognition of each object will trigger the event once, e.g., the facial recognition. When the pulse events are triggered, this
node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously
according to the device detection frequency; if it changes to another source, then the counting will start again*/
    "eventType":  "mixedTargetDetection",
    /*ro, req, enum, event type, subType:string, desc:"mixedTargetDetection" (multi-target-type detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event
ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Mixed target detection",
    /*ro, req, enum, event description, subType:string, desc:"Mixed target detection" (multi-target-type detection event)*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it can be configured via URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:it can be configured via URI (/ISAPI/System/Network/Ehome)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:whether it is a resending data (if the real-time detection data upload fails due to the network
condition, the data will be uploaded again after the device recovers.)*/
    "customChanID":  "test",
    /*ro, opt, string, custom camera channel No., desc:the maximum string size is 64. The value of this node is that of the node <customChanID> configured
by /ISAPI/ContentMgmt/InputProxy/channels/<ID>*/
    "faceBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"faceBackgroundImage" (face background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
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        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanBackgroundImage" (human body background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanImage" (human body thumbnail)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "nonMotorBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"nonMotorBackgroundImage" (background picture of the non-motor vehicle)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleBackgroundImage" (vehicle background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleImage" (vehicle captured picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "CaptureResult": [
    /*ro, opt, array, captured results of detected target, subType:object*/
        {
            "targetID":  1,
            /*ro, req, int, captured target ID*/
            "Face": {
            /*ro, opt, object, face attribute*/
                "Rect": {
                /*ro, opt, object, face thumbnail coordinates, desc:rectangular frame coordinates of the face thumbnail, which is based on the background
picture (faceBackgroundImage). The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, face attribute details, subType:object*/
                    {
                        "description":  "age",
                        /*ro, opt, enum, attributes description, subType:string, desc:"age" (age), "ageGroup" (age group), "gender" (gender), "glass"
(whether the person wears glasses), "smile" (whether the person wears a smile), "mask" (whether the person wears a mask), "faceExpression" (face
expression), "hat" (whether the person wears a hat), "score" (face score)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share
one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "faceBackgroundImage",
                /*ro, opt, enum, "faceBackgroundImage" (background picture of the face), subType:string, desc:"faceBackgroundImage" (face background
picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle
license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and
non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "FacePictureRect": {
                /*ro, opt, object, coordinates of face picture containing head and shoulder, desc:it is based on the face background picture matting and
equals to the face thumbnail uploaded by the device. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, face thumbnail (contains head and shoulder) ID, which indicates the uniqueness of the picture data, range:[0,32],
desc:both "contentID" and "pId" are available for the data transmission format Content-Type: multipart/form-data, and implement the relation between picture
data and structured data of corresponding contents. "contentID1" and "contentID2" correspond to "contentID", "pId1" and "pId2" correspond to "name" and
"filename". The advantage of "pId" is that it has better compatibility, it can match data via "name" and indicate the file name and format via "filename".
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Both "contentID" and "pId" need to be returned, while "faceImageURL" and "faceBackgroundImageURL" are not required*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, face background picture ID*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "faceBackgroundImageURL":  "test",
                /*ro, opt, string, face background picture URL, desc:face background picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, face capture time*/
                "snapTimeMillisecond":  999
                /*ro, opt, int, the milliseconds of face capture time, range:[0,999], desc:it should be used together with the field snapTime*/
            },
            "Human": {
            /*ro, opt, object, human body attributes*/
                "Rect": {
                /*ro, req, object, target frame (based on the background picture), desc:coordinates of the human body target frame, which is based on the
background picture matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, attributes description, subType:string, desc:"jacketColor" (top color), "trousersColor" (bottoms color), "ageGroup"
(age group), "gender", "glass" (whether the person wears glasses), "ride" (whether the person is riding a bike), "bag" (whether the person is carrying a
bag), "things"(whether the person is carrying things), "hat" (whether the person wears a hat), "trousersType" (bottoms type), "mask" (whether the person
wears a mask), "hairStyle" (hairstyle), "cyclingPersonNumber" (number of riders), "direction" (target direction), "score" (human body grading), "hatStyle"
(hat style), "cyclingType" (riding type), "jacketType" (top type)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "modeldata":  "test",
                /*ro, opt, string, human body model data (encoded by base64)*/
                "contentID1":  "humanImage",
                /*ro, opt, enum, human body thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail),
"nonMotorImage" (non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple
properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be
returned*/
                "contentID2":  "humanBackgroundImage",
                /*ro, opt, enum, human body background picture, subType:string, desc:"faceBackgroundImage" (face background picture), "humanBackgroundImage"
(human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle license plate thumbnail). When
there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected,
these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, human body thumbnail ID, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are
detected at the same, these properties can share one thumbnail. The value of pId1 is same as that of pId1 in nonMotor, i.e., human body and non-motor
vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a
target are detected at the same, these properties can share one background picture. The value of pId2 is same as that of pId2 in nonMotor, i.e., human body
and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanImageURL":  "test",
                /*ro, opt, string, human body thumbnail URL, desc:when properties of both human body and non-motor vehicle of a target are detected at the
same, these properties can share one picture. The value of humanImageURL is same as that of nonMotorImageURL, i.e., human body and non-motor vehicle share
the binary picture data of one non-motor vehicle thumbnail*/
                "humanBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the human body, desc:when properties of both human body and non-motor vehicle of a target are
detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, human body capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, milliseconds of human body capture time, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "humanMergeImage",
                /*ro, opt, string, "humanMergeImage" (picture of human body merging details)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of the picture of human body merging details, range:[0,32]*/
                "contentID4":  "humanBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of human body, subType:string, desc:"humanBackgroundImage2" (2nd background picture of human body)*/
                "contentID5":  "humanBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of human body, subType:string, desc:"humanBackgroundImage3" (3rd background picture of human body)*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 2nd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle
of a target are detected at the same, these properties can share one background picture. The value of pId4 is same as that of pId2 in nonMotor, i.e., human
body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 3rd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle
of a target are detected at the same, these properties can share one background picture. The value of pId5 is same as that of pId2 in nonMotor, i.e., human
body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of 2nd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are
detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "humanBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of 3rd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are
detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "HumanImageRect": {
                /*ro, opt, object, target rectangle frame based on the human body thumbnail, desc:the coordinates of the human body target frame, which is
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based on the human body thumbnail matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, rectangle frame of human body thumbnail (based on the background picture), desc:the rectangle frame is the area of the
small human body picture based on humanBackgroundImage. The matted picture is the same as humanImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of
mixedTargetDetection (multi-target-type detection event). when a device uploads human body properties, other related targets attributes will be linked, and
the client software can get the information about targets around the captured human body*/
                "contentID6":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test"
                /*ro, opt, string, close-up picture URL*/
            },
            "NonMotor": {
            /*ro, opt, object, non-motor vehicle attributes*/
                "Rect": {
                /*ro, req, object, frame information of non-motor vehicle thumbnail, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "nonMotorType",
                        /*ro, opt, enum, attributes description, subType:string, desc:"detectType" (device type), "nonMotorType" (non-motor vehicle type),
"plateColor" (license plate color of non-motor vehicle), "plateNo" (license plate number of non-motor vehicle), "directionIndex" (vehicle driving
direction), "laneNo" (lane number), "RFID" (linked RFID), "EPlateChipNo" (chip number of electronic license plate), "EPlateCardNo" (card number of
electronic license plate), "EPlateListenAuthCode" (licensing authority code of electronic license plate), "EPlateNo" (electronic license plate number),
"EPlateManufacturingDate" (data of production of electronic license plate), "EPlateInspectionValidity" (inspection validity of electronic license plate),
"EPlateMandaScrapPeriod" (compulsory scrapping period of electronic license plate), "EPlatePassengersCapacity" (authorized passenger capacity),
"EPlateCarryingCapacity" (tractive tonnage), "EPlateVehicleColor" (non-motor vehicle color), "EPlateNonMotorType" (non-motor vehicle type),
"EPlateNonMotorPlateType" (license plate type of non-motor vehicle), "EPlateColor" (license plate color of non-motor vehicle), "EPlateUseType" (function
properties of non-motor vehicle), "EPlateTriggerTime" (detection trigger time), "retrograde" (wrong-way driving of non-motor vehicle), "manned" (carrying
passengers on non-motor vehicle), "hatStyle" (hat style)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "nonMotorImage",
                /*ro, opt, enum, face picture ID, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body
and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "nonMotorBackgroundImage",
                /*ro, opt, enum, background picture of the non-motor vehicle, subType:string, desc:"faceBackgroundImage" (face background picture),
"humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture). When there are multiple properties
for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one
picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, ID of the non-motor vehicle thumbnail, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the non-motor vehicle, range:[0,32]*/
                "nonMotorImageURL":  "test",
                /*ro, opt, string, URL of the non-motor vehicle thumbnail*/
                "nonMotorBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the non-motor vehicle*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, non-motor vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "nonMotorBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage2" (2nd background picture of
the non-motor vehicle)*/
                "contentID4":  "nonMotorBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage3" (3rd background picture of
the non-motor vehicle)*/
                "contentID5":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail)*/
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                "contentID6":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2223",
                /*ro, opt, string, 2nd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2224",
                /*ro, opt, string, 3rd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2225",
                /*ro, opt, string, face thumbnail ID, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2226",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "nonMotorBackgroundImage2URL":  "test",
                /*ro, opt, string, 2nd background picture URL of the non-motor vehicle*/
                "nonMotorBackgroundImage3URL":  "test",
                /*ro, opt, string, 3rd background picture URL of the non-motor vehicle*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture*/
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of mixedTargetDetection (multi-target-type detection event). when a device uploads non-motor vehicle properties, other related targets attributes will be linked, and the client software can get the information about targets around the captured non-motor vehicle*/
                "contentID7":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId7":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "nonMotorSpeed":  23.4
                /*ro, opt, float, driving speed of non-motor vehicle, range:[0,1024], unit:km/h*/
            },
            "Vehicle": {
            /*ro, opt, object, vehicle attributes*/
                "VehicleRect": {
                /*ro, req, object, target frame information of vehicle picture (based on background picture), desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "PlateRect": {
                /*ro, req, object, frame information of license plate picture, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "plateNo",
                        /*ro, opt, enum, attributes description, subType:string, desc:"plateNo" (license plate No.), "plateType" (license plate type), "plateColor" (license plate color), "vehicleType"(vehicle type), "vehicleColor" (vehicle color), "vehicleLogo" (vehicle brand), "vehicleSublogo" (vehicle sub-brand), "vehicleSize" (vehicle size)*/
                        "value":  "23",
                        /*ro, opt, string, it depends on the value of description, desc:it depends on the value of description*/
                        "confidence":  99.00
                        /*ro, opt, float, confidence, range:[0.00,100.00]*/
                    }
                ],
                "contentID1":  "vehicleImage",
                /*ro, opt, enum, vehicle thumbnail (on the basis of the target frame, a certain proportion is expanded), subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "contentID3":  "vehicleBackgroundImage",
                /*ro, opt, enum, vehicleBackgroundImage, subType:string, desc:"vehicleBackgroundImage" (vehicle background picture 1)*/
                "contentID4":  "vehicleBackgroundImage2",
                /*ro, opt, enum, vehicleBackgroundImage2, subType:string, desc:"vehicleBackgroundImage2" (vehicle background picture 2)*/
                "contentID5":  "vehicleBackgroundImage3",
                /*ro, opt, enum, vehicleBackgroundImage3, subType:string, desc:"vehicleBackgroundImage3" (vehicle background picture 3)*/
                "contentID6":  "vehicleBackgroundImage4",
                /*ro, opt, enum, vehicleBackgroundImage4, subType:string, desc:"vehicleBackgroundImage4" (vehicle background picture 4)*/
                "contentID8":  "vehicleBackgroundImage5",
```

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```
                /*ro, opt, enum, vehicleBackgroundImage5, subType:string, desc:"vehicleBackgroundImage5" (vehicle background picture 5)*/
                "contentID9":  "pilotPicture",
                /*ro, opt, enum, pilotPicture, subType:string, desc:"pilotPicture" (driver's picture)*/
                "contentID10":  "copilotPicture",
                /*ro, opt, enum, copilotPicture, subType:string, desc:"copilotPicture" (co-driver's picture)*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, vehicle thumbnail ID, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, vehicle background picture ID, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 2nd vehicle background picture, range:[0,32]*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 3rd vehicle background picture, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 4th vehicle background picture, range:[0,32]*/
                "pId8":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 5th vehicle background picture, range:[0,32]*/
                "pId9":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, driver's picture, range:[0,32]*/
                "pId10":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, co-driver's picture, range:[0,32]*/
                "vehicleImageURL":  "test",
                /*ro, opt, string, vehicle thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "vehicleBackgroundImageURL":  "test",
                /*ro, opt, string, vehicle background picture URL*/
                "vehicleBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of the 2nd vehicle background picture*/
                "vehicleBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of the 3rd vehicle background picture*/
                "vehicleBackgroundImage4URL":  "test",
                /*ro, opt, string, URL of the 4th vehicle background picture*/
                "vehicleBackgroundImage5URL":  "test",
                /*ro, opt, string, URL of the 5th vehicle background picture*/
                "pilotPictureURL":  "test",
                /*ro, opt, string, driver's picture URL*/
                "copilotPictureURL":  "test",
                /*ro, opt, string, co-driver's picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "humanMergeImageURL":  "test",
                /*ro, opt, string, URL of the picture of human body merging details*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code, range:[0,64]*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture
time of the first picture*/
                "VehicleImageRect": {
                /*ro, opt, object, rectangle target frame in the vehicle thumbnail, desc:the frame is the target area based on vehicleImage. The origin is
the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "SmallImageRectInBackground": {
                /*ro, opt, object, vehicle thumbnail frame (based on the background picture), desc:the rectangle frame of vehicle thumbnail is based on
vehicleBackgroundImage. The matted picture equals to vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MultiPlateInfo": [
                /*ro, opt, array, information about multiple license plates, subType:object*/
                    {
                        "MultiPlateRect": {
                        /*ro, opt, object, license plate thumbnail frame, desc:based on the vehicle background picture*/
                            "height":  1.000,
                            /*ro, req, float, height, range:[0.000,1.000]*/
                            "width":  1.000,
                            /*ro, req, float, width, range:[0.000,1.000]*/
                            "x":  0.000,
                            /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                            "y":  0.000
                            /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
```


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```
                    },
                    "plateNo":  "noPlate",
                    /*ro, req, enum, license plate number, subType:string, desc:"noPlate" (vehicle without license plate), "unknown" (no license plate recognized), XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string*/
                    "plateColor":  "black",
                    /*ro, req, enum, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow), "orange", "other" (other color), "red", "unknown", "white", "yellow"*/
                    "confidence":  100.0,
                    /*ro, opt, float, confidence, range:[0,100.0]*/
                    "province":  0,
                    /*ro, req, enum, province index, subType:int*/
                    "contentID7":  "HongkongPlateImage",
                    /*ro, opt, enum, license plate thumbnail, subType:string, desc:license plate thumbnail*/
                    "pId7":  "F4F665D6A18E41308CE9934DCDDD3333",
                    /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                    "multiplateImageURL":  "test"
                    /*ro, opt, string, thumbnail URL of multiple license plates*/
                }
            ],
            "carDriveDirect":  "forward",
            /*ro, opt, enum, vehicle driving direction, subType:string, desc:"back" (backward), "eastWest" (from east to west), "forward" (forward), "northSouth" (from north to south), "northeastToSouthwest" (from northeast to southwest), "northwestToSoutheast" (from northwest to southeast), "other" (other), "southNorth" (from south to north), "southeastToNorthwest" (from southeast to northwest), "southwestToNortheast" (from southwest to northeas), "unknown" (unknown), "westEast" (from west to east)*/
            "contentID11":  "featurePicture",
            /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
            "pId11":  "F4F665D6A18E41308CE9934DCDDD2222",
            /*ro, opt, string, close-up picture ID, range:[0,32]*/
            "featurePictureURL":  "test",
            /*ro, opt, string, close-up picture URL*/
            "vehicleSpeed":  23.4
            /*ro, opt, float, vehicle driving speed, range:[0,1024], unit:km/h*/
        },
        "AbsoluteHigh": {
        /*ro, opt, object, PTZ position information*/
            "elevation":  11,
            /*ro, opt, int, tilting angle, desc:tilting angle*/
            "azimuth":  22,
            /*ro, opt, int, azimuth, desc:panning angle*/
            "absoluteZoom":  33
            /*ro, opt, int, zooming ratio, desc:zooming rate,value range: 1 to 1000*/
        },
        "GPS": {
        /*ro, opt, object, device GPS information*/
            "longitude":  -180.000000,
            /*ro, req, float, longitude*/
            "latitude":  -90.000000
            /*ro, req, float, latitude*/
        },
        "FaceContrastResult": [
        /*ro, opt, array, face picture comparison result, one result can contains comparison results of multiple persons, the captured picture with highest comparison similarity is adopted, subType:object*/
            {
                "errorCode":  1,
                /*ro, opt, int, error code, desc:error information returned when analyzing face picture failed*/
                "errorMsg":  "ok",
                /*ro, opt, string, the modeling data of base picture in the face picture library, desc:when analyzing face picture failed, the corresponding error information will be returned*/
                "modelData":  "test",
                /*ro, opt, string, the modeling data of base picture in the face picture library, range:[0,2048], desc:base64 encoded*/
                "faces": [
                /*ro, opt, array, matched face information, subType:object, desc:the information includes detection results, properties, modeling data, and matched list. One picture can contains multiple faces*/
                    {
                        "faceId":  1,
                        /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                        "identify": [
                        /*ro, opt, array, features, subType:object*/
                            {
                                "relationId":  "9887",
                                /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                                "maxsimilarity":  0.204000,
                                /*ro, opt, float, the maximum similarity, range:[0.000000,1.000000], desc:the maximum similarity in the matched results the similarity is between 0 and 1, and it corrects to two decimals*/
                                "candidate": [
                                /*ro, opt, array, person information list, subType:object*/
                                    {
                                        "blacklist_id":  "2323",
                                        /*ro, opt, string, alarm list ID, range:[0,64], desc:it corresponds to the FDID of face picture library. For cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                        "human_data": [
                                        /*ro, opt, array, face picture information list, subType:object*/
                                            {
                                                "face_id":  "123445",
                                                /*ro, opt, string, face ID*/
                                                "contentID":  "faceLibImage",
                                                /*ro, opt, string, picture in the face picture library*/
                                                "similarity":  0.204000,
                                                /*ro, opt, float, face picture similarity, range:[0.000000,1.000000]*/
                                                "isNoSaveFDPicture":  true,
                                                /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (not save). If the function of not saving original pictures during picture importing or modeling is enabled, true should be returned, otherwise, this field will not be returned*/
```


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```
returned*/
                                                "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                                /*ro, opt, string, picture ID in face picture library, range:[0,32], desc:picture ID in face picture
library*/
                                                "face_picurl":  "http://127.0.0.1:6120/face_picurl"
                                                /*ro, opt, string, face thumbnail URL*/
                                            }
                                        ],
                                        "human_id":  "123",
                                        /*ro, opt, string, corresponds to the FPID in the face picture library, range:[0,64], desc:person ID in the
list, which corresponds to FPID in the face picture library*/
                                        "reserve_field": {
                                        /*ro, req, object, transmitted data*/
                                            "name":  "张三",
                                            /*ro, opt, string, name, range:[0,96]*/
                                            "gender":  "any",
                                            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition), "unknown", "male", "female"*/
                                            "bornTime":  "1970-01-01+08:00",
                                            /*ro, opt, date, date of birth*/
                                            "city":  "130100",
                                            /*ro, opt, string, city code of person birth, desc:see the data dictionary for details*/
                                            "certificateType":  "officerID",
                                            /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card),
"passport" (passport), "other" (other certificate)*/
                                            "certificateNumber":  "test",
                                            /*ro, opt, string, certificate No., range:[0,32]*/
                                            "ageGroup":  "unknown",
                                            /*ro, opt, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged”
(the middle age), “old”, “prime”, “teenager”, “unknown”, “young”*/
                                            "phoneNumber":  "test",
                                            /*ro, opt, string, phone number, range:[0,64]*/
                                            "cityName":  "石家庄市",
                                            /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the
field "city)*/
                                            "provinceName":  "河北省"
                                            /*ro, opt, string, province, range:[0,32], desc:province*/
                                        },
                                        "similarity":  0.780,
                                        /*ro, opt, float, the maximum similarity, range:[0.000,1.000], desc:the maximum similarity of all matched
pictures for the person*/
                                        "listType":  "blackList",
                                        /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                        "extendData": [
                                        /*ro, opt, array, person extended information, subType:object*/
                                            {
                                                "extendID":  1,
                                                /*ro, opt, int, ID of the extended person information*/
                                                "enable":  true,
                                                /*ro, opt, bool, whether to enable the person extended information*/
                                                "name":  "test",
                                                /*ro, opt, string, name of person tag extended information, range:[0,96]*/
                                                "value":  "test"
                                                /*ro, opt, string, person tag extended information, range:[0,64]*/
                                            }
                                        ],
                                        "FDLibName":  "facelib1",
                                        /*ro, opt, string, detection threshold of face picture library*/
                                        "FDLibThreshold":  50,
                                        /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                        "customFaceLibID":  "test",
                                        /*ro, opt, string, custom face picture library ID*/
                                        "customHumanID":  "test"
                                        /*ro, opt, string, custom person ID*/
                                    }
                                ]
                            }
                        ],
                        "stayDuration":  20
                        /*ro, opt, int, dwell time duration in the image, unit:ms*/
                    }
                ]
            }
        ],
        "ModelingResult": {
        /*ro, opt, object, modeling result of captured face picture*/
            "modelingStatus":  "succeeded",
            /*ro, req, enum, modeling status, subType:string, desc:"succeeded" (modeling succeeded), "resourceInsuf" (insufficient engine capacity),
"faceDetectingFail" (face detection failed), "faceModelingFail" (face modeling failed), "faceGradingFail" (face quality grading), "featureExtractFail"
(extracting features failed), "featureAnalyzeFail" (features analysis failed), "otherErrors" (other error of the engine), "DSPInternalError" (DSP internal
error)*/
            "facePicQuality":  "lower",
            /*ro, opt, enum, quality level of face thumbnail, subType:string, desc:"lower", "medium", "high"*/
            "stayDuration":  20,
            /*ro, opt, int, dwell time duration in the image, unit:ms*/
            "modeldata":  "test"
            /*ro, opt, string, model data (encoded by base64)*/
        },
        "sid":  "123",
        /*ro, opt, string, scene ID*/
        "sceneName":  "场景1",
        /*ro, opt, string, scene name*/
        "uid":  "test",
        /*ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random
digits*/
```


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```
            "behaviorLinkageUid":  "test",
            /*ro, opt, string, which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes, range:[0,64],
desc:which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes*/
            "position": [500, 500],
            /*ro, opt, array, offset position coordinates of target center point, subType:int, desc:an X-coordinate is positive if it is to the right of the
Y-axis and negative if it is to the left of the Y-axis. A Y-coordinate is positive if it is above of the X-axis and negative values are not supported. The
origin is the center point (the center point of all lane lines) of the horizontal line where the camera is located and which is perpendicular to the lane
line*/
            "laneNo":  1,
            /*ro, opt, int, number of the lane where the target is located*/
            "speed":  23.4,
            /*ro, opt, float, vehicle speed*/
            "targetType":  "human",
            /*ro, opt, enum, vehicle type, subType:string, desc:"vehicle", "human", "nonMotor" (non-motor vehicles)*/
            "horizonSpeed":  33.4,
            /*ro, opt, float, horizontal speed of target*/
            "NormalizationPosition": [500, 500],
            /*ro, opt, array, normalized coordinates, subType:int, desc:normalized position coordinates,the value of x-coordinate is between 0 and 1000,and
the value of y-coordinate is between -500 and 1500,e.g.,[500,500]*/
            "targetMaxY":  340,
            /*ro, opt, int, the maximum Y-coordinate of target, desc:the value should be smaller than 1500*/
            "TargetGPSInfo": [
            /*ro, opt, array, GPS information of the target, subType:object*/
                {
                    "longitude":  -180.000000,
                    /*ro, req, float, longitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                    "latitude":  -90.000000
                    /*ro, req, float, latitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                }
            ],
            "perimeterLinkageUid":  "test",
            /*ro, opt, string, UID of linked perimeter events uploading, desc:64-byte unique identifier, which can be composed of time (accurate to the
millisecond) and random digits*/
            "MEFData": [
            /*ro, opt, array, mobile electronic fence data corresponding to the target, subType:object, desc:mobile electronic fence data of multiple users
is supported (the data needs to be encrypted)*/
                {
                    "time":  "2004-05-03T17:30:08+08:00",
                    /*ro, req, datetime, electronic fence data acquired time*/
                    "IMSI":  "460001357924680",
                    /*ro, req, string, SIM card number (international mobile subscriber identity), range:[1,15]*/
                    "IMEI":  "359355041886388",
                    /*ro, opt, string, international mobile equipment identity, range:[0,17]*/
                    "bandIdx":  11
                    /*ro, opt, enum, frequency band type used by the current user, subType:int, desc:0 (China Unicom B3), 1 (China Telecom B3), 2 (China
Mobile B39), 3 (China Mobile B38), 4 (China Mobile B40), 5 (China Mobile B41), 6 (China Unicom B1), 7 (China Telecom B1), 8 (China Mobile 2G), 9 (China
Unicom 2G), 10 (China Unicom WCDMA), 11 (China Telecom CDMA), 12 (China Telecom CDMA2000), 13 (China Mobile TD), 19 (China Mobile B1), 20 (China Mobile B3),
21 (China Mobile B8)*/
                }
            ]
        }
    ],
    "captureResultTargetNum":  10,
    /*ro, opt, int, number of uploaded alarm targets*/
    "URLCertificationType":  "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "triggerSourceType":  "linkedTrackingCapture"
    /*ro, opt, enum, alarm source type, subType:string, desc:"ruleTrigger" (rule parameters triggered alarm), "linkedTrackingCapture" (alarm triggered by
linked tracking capture). If this field is not returned or the returned value is "ruleTrigger", it indicates that the alarm is triggered by multi-target-
type detection rules (corresponding URL for rule configurations: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection?format=json). If the returned value
is "linkedTrackingCapture", it indicates that the alarm is triggered by linked tracking capture (corresponding URL for configurations:
/ISAPI/MasterSlaveTracking/linkedTracking?format=json)*/
}
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name |
|---|---|---|---|---|
| mixedTargetDetection | [Message content] | application/json | -- | -- |
| ..value | [Binary picture data] | image/jpeg | faceImage | faceImage.jpg |
| ..value | [Binary picture data] | image/jpeg | faceBackgroundImage | faceBackgroundImage.jpg |
| ..value | [Binary picture | image/jpeg | humanImage | humanImage.jpg |

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| | | | | |
|---|---|---|---|---|
| | data] | | | |
| ..value | [Binary picture data] | image/jpeg | humanBackgroundImage | humanBackgroundImage.jpg |
| ..value | [Binary picture data] | image/jpeg | humanMergeImage | humanMergeImage.jpg |
| ..value | [Binary picture data] | image/jpeg | humanBackgroundImage2 | humanBackgroundImage2.jpg |
| ..value | [Binary picture data] | image/jpeg | humanBackgroundImage3 | humanBackgroundImage3.jpg |
| ..value | [Binary picture data] | image/jpeg | nonMotorImage | nonMotorImage.jpg |
| ..value | [Binary picture data] | image/jpeg | nonMotorBackgroundImage | nonMotorBackgroundImage.jpg |
| ..value | [Binary picture data] | image/jpeg | nonMotorBackgroundImage2 | nonMotorBackgroundImage2.jpg |
| ..value | [Binary picture data] | image/jpeg | nonMotorBackgroundImage3 | nonMotorBackgroundImage3.jpg |
| ..value | [Binary picture data] | image/jpeg | faceImage | faceImage.jpg |
| ..value | [Binary picture data] | image/jpeg | plateImage | plateImage.jpg |
| ..value | [Binary picture data] | image/jpeg | vehicleImage | vehicleImage.jpg |
| ..value | [Binary picture data] | image/jpeg | plateImage | plateImage.jpg |
| ..value | [Binary picture data] | image/jpeg | vehicleBackgroundImage | vehicleBackgroundImage.jpg |
| ..value | [Binary picture data] | image/jpeg | vehicleBackgroundImage2 | vehicleBackgroundImage2.jpg |
| ..value | [Binary picture data] | image/jpeg | vehicleBackgroundImage3 | vehicleBackgroundImage3.jpg |

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| | | | | |
|---|---|---|---|---|
| ..value | [Binary picture data] | image/jpeg | vehicleBackgroundImage4 | vehicleBackgroundImage4.jpg |
| ..value | [Binary picture data] | image/jpeg | HongkongPlateImage | HongkongPlateImage.jpg |
| ..value | [Binary picture data] | image/jpeg | MacaoPlateImage | MacaoPlateImage.jpg |
| ……value | [Binary picture data] | image/jpeg | faceLibImage | faceLibImage.jpg |
| ..value | [Binary picture data] | image/jpeg | vehicleBackgroundImage5 | vehicleBackgroundImage.jpg5 |
| ..value | [Binary picture data] | image/jpeg | pilotPicture | pilotPicture.jpg |
| ..value | [Binary picture data] | image/jpeg | copilotPicture | copilotPicture.jpg |
| ..value | [Binary picture data] | image/jpeg | featurePicture | featurePicture |
| ..value | [Binary picture data] | image/jpeg | featurePicture | featurePicture |
| ..value | [Binary picture data] | image/jpeg | featurePicture | featurePicture |
| ..value | [Binary picture data] | image/jpeg | compositePicture | compositePicture.jpg |
| ..value | [Binary picture data] | image/jpeg | compositePicture | compositePicture.jpg |
| ..value | [Binary picture data] | image/jpeg | compositePicture | compositePicture.jpg |
| ..value | [Binary picture data] | image/jpeg | copilotPicture | copilotPicture.jpg |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

# 14.8 Video Recognition

## 14.8.1 (Professional) Smart Abnormal Event Detection

### 14.8.1.1 Get the intelligent capability of device

**Request URL**

GET /ISAPI/Intelligent/channels/<channelID>/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isFaceSupport>
    <!--ro, req, bool, whether it supports face detection, desc:whether it supports face detection-->true
  </isFaceSupport>
  <isFieldDetectionSupport>
    <!--ro, req, bool, whether it supports intrusion detection, desc:whether it supports intrusion detection-->true
  </isFieldDetectionSupport>
  <isSupportAlgParam>
    <!--ro, opt, bool-->true
  </isSupportAlgParam>
  <isSupportFaceRule>
    <!--ro, opt, bool-->true
  </isSupportFaceRule>
  <isSupportIntelliResource>
    <!--ro, opt, bool-->true
  </isSupportIntelliResource>
  <isSupportShield>
    <!--ro, opt, bool-->true
  </isSupportShield>
  <isLeavePositionSupport>
    <!--ro, opt, bool-->true
  </isLeavePositionSupport>
  <isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports people density detection, desc:whether the device supports people density detection-->true
  </isSupportPersonDensityDetection>
  <isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events, desc:whether the device supports multi-target-type detection events--
>true
  </isSupportMixedTargetDetection>
  <isSupportRadarVideoTargetUpload>
    <!--ro, opt, bool-->true
  </isSupportRadarVideoTargetUpload>
</IntelliCap>
```

### 14.8.1.2 Get the capability of picture overlay parameters of smart event

**Request URL**

dojojin@gmail.com
HUBEI JOJIN AND SECURITY SYSTEM CO.,LTD.

GET /ISAPI/Intelligent/channels/<channelID>/picInfoOverlap/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |

**Request Message**

None

**Response Message**

```
{
    "PicInfoOverlapCap": {
    /*ro, req, object*/
        "enabled": [true, false],
        /*ro, req, array, subType:bool*/
        "overlapItemNum": {
        /*ro, req, object*/
            "@max":  5
            /*ro, opt, int*/
        },
        "overlapItemId": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  3
            /*ro, req, int*/
        },
        "itemType": ["positionInfo", "cameraNo", "captureTime", "sex", "age", "GPS", "ruleName", "count"],
        /*ro, req, array, subType:string*/
        "eventType": ["faceSnap", "behaviorAnalysis", "judiciary"]
        /*ro, req, array, subType:string*/
    }
}
```

### 14.8.1.3 Get the shielded area of a specified channel

**Request URL**

GET /ISAPI/Intelligent/channels/<channelID>/Shield?SID=<SID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |
| SID | string | -- |

**Request Message**

None

**Response Message**

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dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, opt, object, shielded area configuration, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, enum, whether to enable the function, subType:string, desc:true (yes), false (no)-->true
  </enabled>
  <RegionList>
    <!--ro, opt, array, list of shielded areas, subType:object-->
    <Region>
      <!--ro, opt, object, shielded area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <enabled>
        <!--ro, req, enum, whether to enable the function, subType:string, desc:true (yes), false (no)-->true
      </enabled>
      <name>
        <!--ro, opt, string, range:[0,32]-->test
      </name>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <regionCoordinatesHighPrecisionList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,10000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,10000]-->1
          </positionY>
        </RegionCoordinates>
      </regionCoordinatesHighPrecisionList>
    </Region>
  </RegionList>
</Shield>
```

#### 14.8.1.4 Set the shielded area of all events

**Request URL**

PUT /ISAPI/Intelligent/channels/<channelID>/Shield?SID=<SID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| SID | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, enum, whether enables the shield function, subType:string, desc:whether enables the shield function-->true
  </enabled>
  <RegionList>
    <!--opt, array, subType:object-->
    <Region>
      <!--opt, object-->
      <id>
        <!--req, string, area ID-->1
      </id>
      <enabled>
        <!--req, enum, whether enables the shield function, subType:string, desc:whether enables the shield function-->true
      </enabled>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

# 14.9 Two-Way Audio

## 14.9.1 Two-Way Audio

### 14.9.1.1 Get the parameters of all two-way audio channels

**Request URL**

GET /ISAPI/System/TwoWayAudio/channels

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, configuration of the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType>
      <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--ro, opt, int, input volume, range:[1,100]-->100
    </speakerVolume>
    <microphoneVolume>
      <!--ro, opt, int, output volume, range:[1,100]-->100
    </microphoneVolume>
    <noisereduce>
      <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--ro, opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-
adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled>
        <!--ro, req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--ro, opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--ro, opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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#### 14.9.1.2 Set audio parameters for all two-way audio channels

**Request URL**

PUT /ISAPI/System/TwoWayAudio/channels

**Query Parameter**

None

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--opt, object, configuration of the two-way audio channel-->
    <id>
      <!--req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function or not-->true
    </enabled>
    <audioCompressionType>
      <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--req, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--opt, int, input volume-->100
    </speakerVolume>
    <microphoneVolume>
      <!--opt, int, output volume-->100
    </microphoneVolume>
    <noisereduce>
      <!--opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--req, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--opt, object, linked video channel-->
      <enabled>
        <!--req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

## Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.9.1.3 Receive two-way audio data

**Request URL**

GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| audioID | string | -- |
| audioSessionID | string | -- |

**Request Message**

Binary Data

**Response Message**

Binary Data

#### 14.9.1.4 Send two-way audio data

**Request URL**

PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>&type=<type>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| audioID | string | -- |
| audioSessionID | string | Two-way audio session ID |
| type | enum | If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub-device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub-devices, with multiple childDevID values configured in /ISAPI/System/TwoWayAudio/channels//open. |

**Request Message**

Binary Data

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

#### 14.9.1.5 Set audio parameters of a specified two-way audio channel

**Request URL**

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| twoWayAudioChannelID | string | -- |

**Request Message**

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dojo jin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--opt, object, linked video channel-->
    <enabled>
      <!--req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the value
is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the value
is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
</TwoWayAudioChannel>
```

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dojojin@gmail.com

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.9.1.6 Get the capability of configuring audio parameters for a specified two-way audio channel

**Request URL**

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/capabilities?audioInputType=<audioInputType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| twoWayAudioChannelID | string | Two-way audio channel No. |
| audioInputType | string | Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type. |

**Request Message**

None

**Response Message**

dojo jin@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of configuring audios for the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus,G.722.1">
    <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,G.722.1">
    <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume min="0" max="100">
    <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
  </speakerVolume>
  <microphoneVolume min="0" max="100">
    <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
  </noisereduce>
  <audioBitRate min="1" max="100">
    <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, int},max{opt, int}-->100
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp,doubleMic,networkPickUp">
    <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID min="1" max="2048">
        <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate min="1.00" max="48.00">
    <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
  </audioSamplingRate>
  <lineOutForbidden opt="true,false">
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
  </lineOutForbidden>
  <muteDuringPanTilt opt="true,false">
    <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
  </muteDuringPanTilt>
  <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker,HDMI" def="LineOut">
    <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <isSupportEchoCancellation>
    <!--ro, opt, bool, whether the device supports echo cancellation-->true
  </isSupportEchoCancellation>
  <timeOut min="10" max="600">
    <!--ro, opt, int, timeout during two-way audio, attr:min{req, int},max{req, int}, desc:when the Web Client initiates the two-way audio actively towards
the radar device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some
noise. When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client
stops speaking or not.-->300
  </timeOut>
  <matrixPickUp opt="true,false">
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it is
valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
  </matrixPickUp>
  <isSupportRestore>
    <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
  </isSupportRestore>
  <networkPickUpStatus opt="online,offline">
    <!--ro, opt, enum, network pick-up status, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}, attr:opt{opt, string},
desc:network status of the network pickup linked with the audio channel: "online", "offline"-->online
  </networkPickUpStatus>
  <noisereduceType opt="normal,smart">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

### 14.9.1.7 Stop two-way audio

### Request URL

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PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/close?sessionId=<audioSessionID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| twoWayAudioChannelID | string | Two-Way Audio Channel ID |
| audioSessionID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
</ResponseStatus>
```

### 14.9.1.8 Start two-way audio

**Request URL**

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/open?type=<type>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| twoWayAudioChannelID | string | Two-way audio channel No. |
| type | enum | If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub devices, with multiple childDevID values specified in the message, and the audioID in the URL is invalid. |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudio xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters for this two-way audio, attr:version{req, string, protocolVersion}-->
  <audioLevel>
    <!--opt, int, audio level, range:[0,15], desc:range: [0,15]-->0
  </audioLevel>
  <microphoneVolume>
    <!--opt, int, output volume-->100
  </microphoneVolume>
  <audioCompressionType>
    <!--opt, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <isBroadcast>
    <!--opt, bool-->true
  </isBroadcast>
  <broadcastType>
    <!--opt, enum, subType:string-->alarm
  </broadcastType>
  <childList>
    <!--opt, array, subType:object, range:[0,8], dep:and,{$.TwoWayAudio.isBroadCast,eq,true}-->
    <childDevID>
      <!--opt, string, range:[9,32]-->test
    </childDevID>
    <voiceTalkIndex>
      <!--opt, int, range:[1,256]-->test
    </voiceTalkIndex>
  </childList>
</TwoWayAudio>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioSession xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, result returned when starting two-way audio succeeds, attr:version{opt, string, protocolVersion}-->
  <sessionId>
    <!--ro, req, string, two-way audio session ID-->1
  </sessionId>
</TwoWayAudioSession>
```

### 14.9.1.9 Get the audio parameters of a specified two-way audio channel

**Request URL**

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>?audioInputType=<audioInputType>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| twoWayAudioChannelID | string | Two-way audio channel No. |
| audioInputType | string | Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--ro, opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--ro, opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--ro, opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled>
      <!--ro, req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--ro, opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--ro, opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the
value is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--ro, opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--ro, opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
  <networkPickUpStatus>
    <!--ro, opt, enum, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}-->online
  </networkPickUpStatus>
  <noisereduceType>
    <!--ro, opt, enum, subType:string-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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# 14.10 Entrance & Exit Management and Control

## 14.10.1 Access and Exit Cabinet

### 14.10.1.1 Get the cabinet parameters

**Request URL**

GET /ISAPI/ITC/cabinetParam

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<CabinetParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <cabinetValidEnable>
    <!--ro, req, bool-->true
  </cabinetValidEnable>
  <CabinetParamList>
    <!--ro, opt, array, subType:object-->
    <singleCabinetParam>
      <!--ro, opt, object-->
      <cabinetId>
        <!--ro, req, int, range:[1,4]-->0
      </cabinetId>
      <associateIO>
        <!--ro, req, int, range:[1,255]-->0
      </associateIO>
      <cabinetName>
        <!--ro, opt, string, range:[0,32]-->test
      </cabinetName>
      <cabinetStatus>
        <!--ro, req, enum, subType:int-->0
      </cabinetStatus>
      <alarmIntervalTime>
        <!--ro, req, int, range:[1,300], step:30, unit:s-->1
      </alarmIntervalTime>
    </singleCabinetParam>
  </CabinetParamList>
</CabinetParam>
```

### 14.10.1.2 Set the cabinet parameters

**Request URL**

PUT /ISAPI/ITC/cabinetParam

**Query Parameter**

None

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<CabinetParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <cabinetValidEnable>
    <!--req, bool-->true
  </cabinetValidEnable>
  <CabinetParamList>
    <!--opt, array, subType:object-->
    <singleCabinetParam>
      <!--opt, object-->
      <cabinetId>
        <!--req, int, range:[1,4]-->0
      </cabinetId>
      <associateIO>
        <!--req, int, range:[1,255]-->0
      </associateIO>
      <cabinetName>
        <!--opt, string, range:[0,32]-->test
      </cabinetName>
      <cabinetStatus>
        <!--req, enum, subType:int-->0
      </cabinetStatus>
      <alarmIntervalTime>
        <!--req, int, range:[1,300], step:30, unit:s-->1
      </alarmIntervalTime>
    </singleCabinetParam>
  </CabinetParamList>
</CabinetParam>
```

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**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.10.1.3 Get the capability of cabinet parameters

**Request URL**

GET /ISAPI/ITC/cabinetParam/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<CabinetParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isSupportCabinetEnable>
    <!--ro, req, bool-->true
  </isSupportCabinetEnable>
  <cabinetId min="1" max="4">
    <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->0
  </cabinetId>
  <associateIO opt="1,2,3,4,5,6,7,8,0xff">
    <!--ro, req, int, range:[1,255], attr:opt{req, string}-->0
  </associateIO>
  <cabinetName min="0" max="32">
    <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
  </cabinetName>
  <cabinetStatus min="0" max="1">
    <!--ro, req, int, range:[0,1], attr:min{req, int},max{req, int}-->0
  </cabinetStatus>
  <alarmIntervalTime min="1" max="300">
    <!--ro, req, int, range:[1,300], step:30, attr:min{req, int},max{req, int}-->1
  </alarmIntervalTime>
</CabinetParam>
```

### 14.10.2 Parameters Management of Entrance and Exit Display Screen

#### 14.10.2.1 Import the LCD (Liquid Crystal Display) media data

**Request URL**

POST /ISAPI/Parking/channels/<channelID>/LCD/mediaData/import?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

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```
{
    "MediaDataInfo": {
    /*req, object*/
        "dataType":  "picture",
        /*req, enum, type of the imported data: "picture", "video", subType:string, desc:"picture", "video"*/
        "pictureType":  "backgroundPicture"
        /*opt, enum, picture type, subType:string, desc:"backgroundPicture" (background picture), "contentPicture" (content picture), "logo" (logo
picture)*/
    }
}
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| MediaDataInfo | [Message content] | application/json | -- | -- | -- |
| picture | [Binary picture data] | image/jpeg | picture | 图片 | -- |
| video | [Stream data] | application/octet-stream | | 视频 | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

**Response Message**

```
{
    "id":  1
    /*ro, req, int, media data ID*/
}
```

#### 14.10.2.2 Get the configuration capability of LED screen display

**Request URL**

GET /ISAPI/Parking/channels/<channelID>/LEDConfiguration/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<LEDConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the configuration capability of LED screen display, attr:version{req, string, protocolVersion}-->
  <information min="0" max="512">
    <!--ro, opt, string, LED screen display content, attr:min{req, int},max{req, int}-->test
  </information>
  <displayMode opt="left,right,immediate">
    <!--ro, opt, enum, display mode, subType:string, attr:opt{req, string}, desc:display mode: "left"-move left,"right"-move right,"immediate"-display
immediately-->left
  </displayMode>
  <speedType opt="fast,medium,slow">
    <!--ro, opt, enum, speed type, subType:string, attr:opt{req, string}, desc:"fast", "medium", "slow"-->fast
  </speedType>
  <showTime min="1" max="300">
    <!--ro, req, int, display duration, range:[1,300], unit:s, attr:min{req, int},max{req, int}-->1
  </showTime>
  <showPlate>
    <!--ro, opt, bool, whether to display temporary license plate-->true
  </showPlate>
  <ledCount min="1" max="10">
    <!--ro, opt, int, number of LED screens. An LED screen is a logic unit which can be controlled independently, attr:min{req, int},max{req, int}, desc:the
default value is 1-->1
  </ledCount>
  <fontSize min="1" max="10">
    <!--ro, opt, int, font size, attr:min{req, int},max{req, int}-->1
  </fontSize>
  <fontColor min="0" max="10">
    <!--ro, opt, int, RGB value of the font color, attr:min{req, int},max{req, int}-->1
  </fontColor>
  <laneID min="1" max="32">
    <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}-->1
  </laneID>
  <passingIconType opt="passing,noPassing">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->passing
  </passingIconType>
  <QRCodeInfo>
    <!--ro, opt, object-->
    <QRCode min="1" max="128">
      <!--ro, req, string, range:[1,128], attr:min{req, int},max{req, int}-->test
    </QRCode>
    <QRSize min="12" max="256">
      <!--ro, opt, int, range:[12,256], attr:min{req, int},max{req, int}-->16
    </QRSize>
    <QRColor opt="red,green,yellow,blue,cyan,purple,white">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->green
    </QRColor>
  </QRCodeInfo>
</LEDConfiguration>
```

### 14.10.2.3 Set the display parameters of the LED screen

#### Request URL

PUT /ISAPI/Parking/channels/<channelID>/LEDConfiguration?id=<netScreenID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| netScreenID | string | -- |

#### Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LEDConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int-->1
  </id>
  <information>
    <!--req, string-->test
  </information>
  <displayMode>
    <!--req, enum, subType:string-->left
  </displayMode>
  <speedType>
    <!--req, enum, subType:string-->fast
  </speedType>
  <showTime>
    <!--req, int, range:[1,60], unit:s-->1
  </showTime>
  <showPlate>
    <!--opt, bool-->true
  </showPlate>
  <fontSize>
    <!--opt, int-->1
  </fontSize>
  <fontColor>
    <!--opt, int-->1
  </fontColor>
</LEDConfiguration>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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## 14.10.2.4 Get the parameters of displaying multiple scenes on the LED screen

### Request URL

GET /ISAPI/Parking/channels/<channelID>/LEDConfigurations/multiScene?format=json

### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

### Request Message

None

### Response Message

```
{
    "MultiSceneLEDConfigurations": [
    /*ro, opt, array, parameters of displaying multiple scenes on the LED screen, subType:object*/
        {
            "SingleSceneLEDConfigurations": {
            /*ro, opt, object, parameters of displaying a single scene on the LED screen*/
                "sid":  1,
                /*ro, req, int, scene ID*/
                "mode":  "passingVehicle",
                /*ro, req, enum, scene modes, subType:string, desc:"passingVehicle" (passing-vehicle scene), "noVehicle" (no-passing-vehicle scene)*/
                "showFreeEnabled":  true,
                /*ro, opt, bool, whether to enable vacancy display mode, dep:and,
{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
                "displayTime":  1,
                /*ro, opt, int, display duration, range:[1,300], unit:s, dep:and
```

```
                /*ro, opt, int, display duration, range:[1,300], unit:s, dep:and,
{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
                "ctrlMode":  "cameraAndplatform",
                /*ro, opt, enum, subType:string*/
                "vehicleDisplayEnabled":  true,
                /*ro, opt, bool*/
                "allowListDisplayEnabled":  false,
                /*ro, opt, bool*/
                "blockListDisplayEnabled":  false,
                /*ro, opt, bool*/
                "temporaryListDisplayEnabled":  false,
                /*ro, opt, bool*/
                "LEDConfigurationList": [
                /*ro, opt, array, the list of LED configuration parameters, subType:object*/
                    {
                        "LEDConfiguration": {
                        /*ro, opt, object, LED configuration parameters*/
                            "id":  1,
                            /*ro, req, int, display screen ID*/
                            "enabled":  true,
                            /*ro, req, bool, enable the LED screen to display multiple lines*/
                            "ShowInfoList": [
                            /*ro, opt, array, list of display information, subType:object*/
                                {
                                    "ShowInfo": {
                                    /*ro, opt, object, display information*/
                                        "id":  1,
                                        /*ro, req, int, display screen ID, desc:read-only,display screen ID*/
                                        "fontSize":  16,
                                        /*ro, opt, enum, the types of font sizes, subType:int, desc:"16" (16 × 16 dot matrix), "32" (32 × 32 dot matrix)*/
                                        "fontColor":  "red",
                                        /*ro, opt, enum, font color, subType:string, desc:read-only,the types of font colors: "red","green","yellow"*/
                                        "speedType":  "fast",
                                        /*ro, req, enum, speed type, subType:string, desc:"fast", "medium", "slow"*/
                                        "displayMode":  "left",
                                        /*ro, req, enum, display modes, subType:string, desc:"left", "right", "instant"*/
                                        "LineInfoList": [
                                        /*ro, opt, array, the list of line information, subType:object*/
                                            {
                                                "LineInfo": {
                                                /*ro, opt, object, line information, desc:value and customValue cannot be configured simultaneously,one ID
can only be configured with one of these two parameters*/
                                                    "id":  1,
                                                    /*ro, req, int, display screen ID, desc:read-only,display screen ID*/
                                                    "value":  "plate",
                                                    /*ro, opt, enum, display information, subType:string, desc:read-only,display information: "plate"
(license plate number),"YYYY-MM-DD" (date displayed in this format),"hh-mm-ss" (time displayed in this format)*/
                                                    "customValue":  "欢迎光临",
                                                    /*ro, opt, string, custom display information*/
                                                }
                                            }
                                        ]
                                    }
                                }
                            ]
                        }
                    }
                ],
                "LEDGeneralConfiguration": {
                /*ro, opt, object*/
                    "FontSize":  16,
                    /*ro, opt, enum, font size, subType:int*/
                    "FontColor":  "red",
                    /*ro, opt, enum, font color, subType:string*/
                    "Frequency":  5,
                    /*ro, req, int*/
                    "FreeTimeShowInfo": {
                    /*ro, opt, object*/
                        "CustomContentList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "CustomContent": {
                                /*ro, opt, object*/
                                    "id":  1,
                                    /*ro, req, int, display screen ID, range:[1,5], desc:read-only,display screen ID*/
                                    "content":  "test"
                                    /*ro, opt, string*/
                                }
                            }
                        ],
                        "TimeBlockList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "TimeBlock": {
                                /*ro, opt, object, time period*/
                                    "id":  1,
                                    /*ro, opt, int, read-only,display screen ID*/
                                    "beginTime":  "00:00",
                                    /*ro, opt, string, start time*/
                                    "endTime":  "24:00"
                                    /*ro, opt, string*/
                                }
                            }
                        ],
```

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```
                    "showTime":  15
                    /*ro, req, int*/
                },
                "IllegalShowInfo": {
                /*ro, opt, object*/
                    "IllegalList": [
                    /*ro, opt, array, subType:object*/
                        {
                            "IllegalContent": {
                            /*ro, opt, object*/
                                "id":  1,
                                /*ro, req, int, read-only,display screen ID, range:[1,5], desc:read-only,display screen ID*/
                                "content":  "test",
                                /*ro, opt, string*/
                                "contentType":  "nonHelmet"
                                /*ro, opt, enum, subType:string*/
                            }
                        }
                    ],
                    "TimeBlockList": [
                    /*ro, opt, array, subType:object*/
                        {
                            "TimeBlock": {
                            /*ro, opt, object, time period*/
                                "id":  1,
                                /*ro, opt, int, display screen ID*/
                                "beginTime":  "00:00",
                                /*ro, opt, string, start time*/
                                "endTime":  "24:00"
                                /*ro, opt, string*/
                            }
                        }
                    ],
                    "showTime":  15
                    /*ro, req, int*/
                }
            }
        }
    }
]
}
```

#### 14.10.2.5 Get the capability set of displaying information on the LCD screen

**Request URL**

GET /ISAPI/System/LCDScreen/capabilities?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
{
    "LCDScreenCap": {
    /*ro, opt, object, capability set of displaying information on the LCD screen*/
        "DisplayInfoCap": {
        /*ro, opt, object, capability of the LCD screen*/
            "QRCodeImg": {
            /*ro, opt, object, QR code information*/
                "QRCodeNum":  "1",
                /*ro, opt, string, the maximum number of QR code pictures that can be displayed*/
                "QRCodeImgFormat": {
                /*ro, opt, object, payment method and picture type of the QR code defined by the file name and the extension name*/
                    "@opt":  "weixin.jpg,zhifubao.bmp"
                    /*ro, req, string, optional values*/
                },
                "paymentType": {
                /*ro, opt, object, payment method (multiple methods can be separated by commas)*/
                    "@opt":  "WeChatPay,Alipay"
                    /*ro, req, string, optional values*/
                },
                "preDisPlayEnabled": {
                /*ro, opt, object, whether to enable pre-display, desc:true (device stores the QR code and displays it in advance when a corresponding scene
detected, allowing users to scan for registration), false or field not exist (device displays the QR code instantly and does not store the QR code)*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool, default value*/
                }
            },
            "contentUrl": {
            /*ro, opt, object, picture URL, desc:the device downloads the picture via URL to display it*/
                "@min":  0,
```

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```
                /*ro, req, int, minimum string length*/
                "@max":  256
                /*ro, req, int, maximum string length*/
            },
            "backgroundPicture": {
            /*ro, opt, object, background picture content, desc:a binary picture string encoded by Base64*/
                "@min":  0,
                /*ro, req, int, minimum string length*/
                "@max":  40960
                /*ro, req, int, maximum string length*/
            },
            "sence": {
            /*ro, opt, object, scene*/
                "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15"
                /*ro, req, string, optional values, desc:"1" (entered vehicles with license plate), "2" (entered vehicles without license plate), "3"
(exited vehicles with license plate and without unpaid fee), "4" (exited vehicles with license plate and unpaid fee), "5" (exited vehicles without license
plate)*/
            },
            "amounts":  true,
            /*ro, opt, bool, whether it supports displaying the account*/
            "licenseLen":  64,
            /*ro, opt, int, the maximum string length of the license plate characters, desc:If this node exists, it indicates that displaying the license
plate information is supported*/
            "phoneNumberLen":  20,
            /*ro, opt, int, the maximum string length of the phone No., desc:If this node exists, it indicates that displaying the contact information is
supported*/
            "noticeLen":  256,
            /*ro, opt, int, the maximum string length of the notice information*/
            "enterTime":  true,
            /*ro, opt, bool, whether it supports displaying the entering time*/
            "leaveTime":  true,
            /*ro, opt, bool, whether it supports displaying the exiting time*/
            "customInfoLen":  128,
            /*ro, opt, int, the maximum string length of the displayed custom information, desc:If this node exists, it indicates that displaying the custom
information is supported*/
            "customInfoList": {
            /*ro, opt, object*/
                "@size":  4,
                /*ro, req, int*/
                "customInfo": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int*/
                    "@max":  64
                    /*ro, req, int*/
                },
                "fontSize": {
                /*ro, opt, object*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  100
                    /*ro, opt, int*/
                },
                "fontColor": {
                /*ro, opt, object*/
                    "R": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  255
                        /*ro, opt, int*/
                    },
                    "G": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  255
                        /*ro, opt, int*/
                    },
                    "B": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  255
                        /*ro, opt, int*/
                    }
                }
            },
            "plateType": {
            /*ro, opt, object, license plate type*/
                "@opt":  "fixedPlate,temporaryPlate,blackPlate,payByPeriodPlate,noPlate"
                /*ro, req, string, optional values, desc:"fixedPlate" (registered vehicle), "temporaryPlate" (temperatory vehicle). "blackPlate" (vehicles
in the blocklist)*/
            },
            "validityPeriod": {
            /*ro, opt, object, validity period*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  29
                /*ro, req, int, the maximum value*/
            },
            "remainingCharge":  "30",
            /*ro, opt, string, account balance*/
```

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```
            "parkingLot":  2,
            /*ro, opt, int, number of vacant parking spaces*/
            "titleLen":  128,
            /*ro, opt, int, the maximum string length of the title, desc:if this node exists, it indicates that setting the title is supported*/
            "laneID": {
            /*ro, opt, object, lane No., desc:the traffic server connects cameras and link cameras to lane*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  32
                /*ro, req, int, the maximum value*/
            },
            "senceDisplayTime": {
            /*ro, opt, object, display duration of a scene, desc:the display duration of a scene before refresh. Unit: second*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  600
                /*ro, req, int, the maximum value*/
            }
        }
    }
}
```

| Parameter Name | Parameter Value | Parameter Type(Content-Type) | Content-ID | File Name | Description |
|---|---|---|---|---|---|
| | [Message content] | application/json | -- | -- | -- |

**Note：** The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

#### 14.10.2.6 Configure the information to be displayed on the screen.

**Request URL**

POST /ISAPI/System/LCDScreen/displayInfo?format=json

**Query Parameter**

None

**Request Message**

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```
{
    "DisplayInfo": {
    /*opt, object, displayed information*/
        "QRCodeImg": [
        /*opt, array, array of object, list of QR codes, subType:object*/
            {
                "QRCodeBase64":  "abcdeadfadsf==",
                /*req, string, QR code encoded via Base64*/
                "QRCodeFileName":  "weixin.jpg",
                /*req, string, picture format of the QR code, desc:“weixin.jpg” (WeChat), “zhifubao.bmp” (Alipay)*/
                "paymentType":  "WeChatPay,Alipay",
                /*req, string, payment method, desc:"WeChatPay", "Alipay". Multiple methods can be separated by commas*/
                "QRContent":  "http://",
                /*opt, string, QR code content, desc:When the node QRCodeBase64 is not configured, the device will generate a QR code using this node*/
                "preDisPlayEnabled":  false
                /*opt, bool, whether to enable, desc:true (device stores the QR code and displays it in advance when a corresponding scene detected,
allowing users to scan for registration), false or field not exist (device displays the QR code instantly and does not store the QR code)*/
            }
        ],
        "contentUrl":  "http://xxx.jpg",
        /*opt, string, picture URL, range:[0,256], desc:the device downloads the picture via URL to display it*/
        "backgroundPicture":  "abcdeadfadsf=="
        /*opt, string, background picture content, range:[0,40960], desc:a binary picture string encoded by Base64*/
    },
    "sence":  1,
    /*req, enum, scene, subType:int, desc:1 (entered vehicles with license plate),2 (entered vehicles without license plate),3 (exited vehicles without
unpaid fee),4 (exited vehicles with unpaid fee),5 (exited vehicles without license plate),6 (the fee is paid),7 (exited vehicles without license plate and
unpaid fee),8 (exited vehicles without license plate and with unpaid fee),9 (paying failed),10 (free). This node is used to control the content and layout
displayed on the screen*/
    "amounts":  30.0,
    /*opt, float, float,amount, desc:float,amount*/
    "license":  "ABC12345",
    /*opt, string, license plate number*/
    "phoneNumber":  "13012345678",
    /*opt, string, phone number*/
    "notice":  "abcdefg",
    /*opt, string, notice information*/
    "enterTime":  "1970-01-01T00:00:00+08:00",
    /*opt, datetime, entering time*/
    "leaveTime":  "1970-01-01T00:00:00+08:00",
    /*opt, datetime, exiting time*/
    "customInfo":  "abcdefg",
    /*opt, string, displayed custom information*/
    "customInfoList": [
    /*opt, array, subType:object, range:[0,4]*/
        {
            "customInfo":  "abcdefg",
            /*req, string, range:[0,64]*/
            "fontSize":  1,
            /*opt, int, range:[1,100]*/
            "fontColor": {
            /*opt, object*/
                "R":  1,
                /*opt, int, range:[0,255]*/
                "G":  1,
                /*opt, int, range:[0,255]*/
                "B":  1
                /*opt, int, range:[0,255]*/
            }
        }
    ],
    "plateType":  "fixedPlate",
    /*opt, enum, license plate type, subType:string, desc:"fixedPlate" (registered vehicle), "temporaryPlate" (temporary vehicle), "blackPlate" (vehicles in
the blocklist)*/
    "validityPeriod":  "2020年11月12日-2020年12月11日",
    /*opt, string, validity period*/
    "remainingCharge":  "30",
    /*opt, string, the account balance, desc:the account balance*/
    "parkingLot":  2,
    /*opt, int, number of vacant parking spaces*/
    "tideLaneStatus":  "opening",
    /*opt, enum, reversible lane status, subType:string, desc:"opening", "closed"*/
    "title":  "abc",
    /*opt, string, displayed title*/
    "laneID":  "1",
    /*opt, string, lane No., range:[1,32], desc:the traffic server connects cameras and link cameras to lane*/
    "senceDisplayTime":  60
    /*opt, int, range:[1,600], unit:s*/
}
```

## Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

## 14.10.3 Parking Lot

### 14.10.3.1 Get the parking lot service capability

**Request URL**

GET /ISAPI/Parking/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

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```
<?xml version="1.0" encoding="UTF-8"?>
<ParkingCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parking lot service capability, attr:version{req, string, protocolVersion}-->
  <isSupportTME>
    <!--ro, opt, bool, whether it supports TME nodes-->true
  </isSupportTME>
  <isSupportLampCtrl>
    <!--ro, opt, bool, whether it supports configuring and controlling parking space indicators-->true
  </isSupportLampCtrl>
  <isSupportparkingStatus>
    <!--ro, opt, bool, whether it supports getting parking space status-->true
  </isSupportparkingStatus>
  <isSupportPicDevision>
    <!--ro, opt, bool, whether it supports parking space picture division-->true
  </isSupportPicDevision>
  <isSupportAlterCtrlLampCap>
    <!--ro, opt, bool, whether it supports alternate indicator control-->true
  </isSupportAlterCtrlLampCap>
  <isSupportParkingSpace>
    <!--ro, opt, bool, whether it supports parking area configuration-->true
  </isSupportParkingSpace>
  <isSupportVideoParkingPole>
    <!--ro, opt, bool, whether it supports parking video pole-->true
  </isSupportVideoParkingPole>
  <isSupportBollard>
    <!--ro, opt, bool-->true
  </isSupportBollard>
  <isSupportLEDConfigurationsMultiScene>
    <!--ro, opt, bool-->true
  </isSupportLEDConfigurationsMultiScene>
  <isSupportVoiceCombinateBroadcast>
    <!--ro, opt, bool-->true
  </isSupportVoiceCombinateBroadcast>
  <isSupportLCD>
    <!--ro, opt, bool-->true
  </isSupportLCD>
  <isSupportPackingSpaceRecognition>
    <!--ro, opt, bool, whether it supports parking space detection, desc:corresponding URL: /ISAPI/Parking/channels/<channelID>/packingSpaceRecognition?
format=json-->true
  </isSupportPackingSpaceRecognition>
  <isSupportPackingSpaceIndicatorLamp>
    <!--ro, opt, bool-->true
  </isSupportPackingSpaceIndicatorLamp>
  <isSupportManualPackingLock>
    <!--ro, opt, bool-->true
  </isSupportManualPackingLock>
  <isSupportAutoPackingLock>
    <!--ro, opt, bool-->true
  </isSupportAutoPackingLock>
</ParkingCap>
```

#### 14.10.3.2 Get the capability of configuring the parameters of combined voice broadcasts

**Request URL**

GET /ISAPI/Parking/channels/<channelID>/voiceBroadcastInfo/combinateBroadcast/capabilities?format=json

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

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```
{
    "CombinateBroadcastCap": {
    /*ro, opt, object, configuration capability of combined broadcast*/
        "enabled": {
        /*ro, req, object, whether to enable combined broadcasts*/
            "@opt": [true, false]
            /*ro, req, array, range, subType:bool*/
        },
        "volume": {
        /*ro, opt, object, voice volume*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  10
            /*ro, req, int, the maximum value*/
        },
        "volumeTimeEnabled": {
        /*ro, opt, object, whether to enable volume time control*/
            "@opt": [true, false]
            /*ro, req, array, range, subType:bool*/
        },
        "VolumeTimeBlockList": {
        /*ro, opt, object, the list of volume time control configuration information*/
            "TimeBlock": {
            /*ro, opt, object, time period*/
                "id": {
                /*ro, req, object, No., desc:object,ID*/
                    "@maxSize":  5
                    /*ro, req, int, maximum number of elements in the array*/
                },
                "volume": {
                /*ro, opt, object, voice volume*/
                    "@min":  0,
                    /*ro, req, int, the minimum value*/
                    "@max":  10
                    /*ro, req, int, the maximum value*/
                },
                "beginTime": {
                /*ro, opt, object, start time, desc:different time periods for control during 24 hours,it should be accurate to minute and in hh:MM format*/
                    "@min":  "00:00",
                    /*ro, req, string, the minimum value*/
                    "@max":  "24:00"
                    /*ro, req, string, maximum value*/
                },
                "endTime": {
                /*ro, opt, object, end time, desc:different time periods for control during 24 hours,it should be accurate to minute and in hh:MM format*/
                    "@min":  "00:00",
                    /*ro, req, string, the minimum value*/
                    "@max":  "24:00"
                    /*ro, req, string, maximum value*/
                }
            }
        },
    }
}
```

### 14.10.3.3 Set the parameters of combined voice broadcast

#### Request URL

PUT /ISAPI/Parking/channels/<channelID>/voiceBroadcastInfo/combinateBroadcast?format=json&powerOffSaveEnabled=<powerOffSaveEnabled>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |
| powerOffSaveEnabled | string | -- |

#### Request Message

```
{
    "CombinateBroadcast": {
    /*req, object, combined broadcasts*/
        "enabled":  true,
        /*req, bool, whether to enable combined broadcasts*/
        "volume":  1,
        /*opt, int, voice volume, range:[0,10]*/
        "ctrlMode":  "cameraAndplatform",
        /*ro, opt, enum, subType:string*/
        "vehicleBroadcastEnabled":  true,
        /*opt, bool*/
        "allowListBroadcastEnabled":  false,
```

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```
        /*opt, bool*/
        "blockListBroadcastEnabled":  false,
        /*opt, bool*/
        "temporaryListBroadcastEnabled":  false,
        /*opt, bool*/
        "BroadcastInfoList": [
        /*opt, array, list of combined broadcasts information, subType:object, range:[1,8]*/
            {
                "BroadcastInfo": {
                /*opt, object, combined broadcasts information*/
                    "id":  1,
                    /*req, int, ID, range:[1,8], desc:it increases from 1*/
                    "value":  "plate",
                    /*opt, enum, options, subType:string, desc:"plate" (license plate number prompt)*/
                    "customValue":  "欢迎光临"
                    /*opt, string, custom broadcast information, range:[1,100], desc:value and customValue cannot be configured simultaneously, and one ID
can only be configured with one of these two parameters*/
                }
            }
        ],
        "volumeTimeEnabled":  true,
        /*opt, bool, whether to enable volume time control*/
        "VolumeTimeBlockList": [
        /*opt, array, the list of volume time control configuration information, subType:object, range:[1,5]*/
            {
                "TimeBlock": {
                /*opt, object, time period*/
                    "id":  1,
                    /*req, int, ID, range:[1,5], desc:it increases from 1, and supports up to 5 time periods*/
                    "volume":  1,
                    /*opt, int, volume, range:[0,10]*/
                    "beginTime":  "00:00",
                    /*opt, time, start time, desc:different time periods for control during 24 hours, it should be accurate to minute and in hh:MM format*/
                    "endTime":  "00:00"
                    /*opt, time, end time, desc:different time periods for control during 24 hours, it should be accurate to minute and in hh:MM format*/
                }
            }
        ],
        "soundMode":  "child",
        /*opt, enum, types of voice mode, subType:string, desc:"child", "cartoon", "alto", "bass", "soprano", "tenor"*/
        "broadcastPort":  "audioPort",
        /*ro, opt, enum, broadcast port, subType:string, desc:"rs485", "audioPort"*/
        "Illegal": {
        /*opt, object, illegal*/
            "broadcastInfo":  "test"
            /*opt, string, broadcast information, range:[0,64]*/
        },
        "PedestrianAlert": {
        /*opt, object, pedestrian pre-alarm*/
            "enabled":  true,
            /*req, bool, whether to enable combined broadcasts*/
            "broadcastInfo":  "test"
            /*opt, string, broadcast information, range:[0,64]*/
        },
        "TrafficLight": {
        /*opt, object, traffic light*/
            "enabled":  true,
            /*req, bool, whether to enable combined broadcasts*/
            "broadcastMode":  "single"
            /*opt, enum, broadcast mode, subType:string, desc:"single", "cycle"*/
        },
        "FreeTimeBroadcast": {
        /*opt, object*/
            "CustomContentList": [
            /*ro, opt, array, subType:object*/
                {
                    "CustomContent": {
                    /*ro, opt, object, broadcast information*/
                        "id":  1,
                        /*ro, req, int, ID, range:[1,5], desc:it increases from 1*/
                        "content":  "test"
                        /*ro, opt, string, broadcast information*/
                    }
                }
            ],
            "TimeBlockList": [
            /*ro, opt, array, subType:object*/
                {
                    "TimeBlock": {
                    /*ro, opt, object, time period*/
                        "id":  1,
                        /*ro, opt, int, ID*/
                        "beginTime":  "00:00",
                        /*ro, opt, string, start time, desc:different time periods for control during 24 hours, it should be accurate to minute and in hh:MM
format*/
                        "endTime":  "24:00"
                        /*ro, opt, string, end time, desc:different time periods for control during 24 hours, it should be accurate to minute and in hh:MM
format*/
                    }
                }
            ],
            "times":  1,
            /*ro, req, int*/
            "cycleIntervalTimes":  15
```

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```
            /*ro, req, int, loop interval*/
        },
        "IllegalBroadcast": {
        /*opt, object*/
            "IllegalList": [
            /*ro, opt, array, subType:object*/
                {
                    "IllegalContent": {
                    /*ro, opt, object, broadcast information*/
                        "id": 1,
                        /*ro, req, int, ID, range:[1,5], desc:it increases from 1*/
                        "content": "test",
                        /*ro, opt, string, broadcast information*/
                        "contentType": "nonHelmet"
                        /*opt, enum, content type, subType:string*/
                    }
                }
            ],
            "TimeBlockList": [
            /*ro, opt, array, subType:object*/
                {
                    "TimeBlock": {
                    /*ro, opt, object, time period*/
                        "id": 1,
                        /*ro, opt, int, ID*/
                        "beginTime": "00:00",
                        /*ro, opt, string, start time, desc:different time periods for control during 24 hours, it should be accurate to minute and in hh:MM
format*/
                        "endTime": "24:00"
                        /*ro, opt, string, end time, desc:different time periods for control during 24 hours, it should be accurate to minute and in hh:MM
format*/
                    }
                }
            ],
            "times": 1,
            /*ro, req, int*/
            "cycleIntervalTimes": 15
            /*ro, req, int, loop interval*/
        }
    }
}
```

**Response Message**

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

### 14.10.3.4 Get the parameters of combined voice broadcast

**Request URL**

GET /ISAPI/Parking/channels/<channelID>/voiceBroadcastInfo/combinateBroadcast?format=json&powerOffSaveEnabled=<powerOffSaveEnabled>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |
| powerOffSaveEnabled | string | Whether to save parameter settings in flash after power off; if this field is not applied, the parameter setting will not be saved in flash. If it is false, the device will restore it to the default value after rebooting. For example, /ISAPI/Parking/channels//voiceBroadcastInfo/combinateBroadcast?format=json&powerOffSaveEnabled=true. |

**Request Message**

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None

### Response Message

```
{
    "CombinateBroadcast": {
    /*ro, req, object, combined broadcasts*/
        "enabled":  true,
        /*ro, req, bool, whether to enable combined broadcasts*/
        "volume":  1,
        /*ro, opt, int, voice volume, range:[0,10]*/
        "volumeTimeEnabled":  true,
        /*ro, opt, bool, whether to enable volume time control*/
        "VolumeTimeBlockList": [
        /*ro, opt, array, the list of volume time control configuration information, subType:object, range:[1,5]*/
            {
                "TimeBlock": {
                /*ro, opt, object, time period*/
                    "id":  1,
                    /*ro, req, int, No., range:[1,5], desc:read-only,ID,value range: [1,8],starts from one*/
                    "volume":  1,
                    /*ro, opt, int, voice volume, range:[0,10]*/
                    "beginTime":  "00:00",
                    /*ro, opt, time, Start time, desc:different time periods for control during 24 hours,it should be accurate to minute and in hh:MM
format*/
                    "endTime":  "00:00"
                    /*ro, opt, time, end time, desc:different time periods for control during 24 hours,it should be accurate to minute and in hh:MM format*/
                }
            }
        ],
    }
}
```

## 14.10.4 Entrance/Exit Ticket Station Management

### 14.10.4.1 Get entrance & exit station parameters

### Request URL

GET /ISAPI/Parking/channels/<channelID>/TMEConfiguration

### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TMEConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configurations of entrance & exit station, attr:version{req, string, protocolVersion}-->
  <illegalCardFilteringEnabled>
    <!--ro, opt, bool, whether to enable invalid card filtering-->true
  </illegalCardFilteringEnabled>
  <LEDDefaultInfo min="0" max="512">
    <!--ro, opt, string, default display content on LED, attr:min{req, int},max{req, int}-->test
  </LEDDefaultInfo>
  <sendCardSensingCoilEnabled>
    <!--ro, opt, bool, whether to link the card dispenser with inductive loop detector-->true
  </sendCardSensingCoilEnabled>
  <wiegendSensingCoilEnabled>
    <!--ro, opt, bool, whether to link long-range card with inductive loop detector-->true
  </wiegendSensingCoilEnabled>
  <gateSwitchSignalEnabled>
    <!--ro, opt, bool-->true
  </gateSwitchSignalEnabled>
  <verifyKeyWriteCardEnabled>
    <!--ro, opt, bool, whether to enable key verification and writing data to cards-->true
  </verifyKeyWriteCardEnabled>
  <noplateTakePermissionEnabled>
    <!--ro, opt, bool, whether to enable non-license plate entry with voucher (ticket or temporary card)-->true
  </noplateTakePermissionEnabled>
  <scancodeCoilEnabled>
    <!--ro, opt, bool, whether to link code scanner with inductive loop detector-->true
  </scancodeCoilEnabled>
  <noPaperDisplayEnabled>
    <!--ro, opt, bool, whether to enable out of paper prompt-->true
  </noPaperDisplayEnabled>
```

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```
<fullParkingNoTakeCard>
<!--ro, opt, bool, whether to stop issuing card when no vacant parking space-->true
</fullParkingNoTakeCard>
<ctrlMode>
<!--ro, opt, enum, control mode of card No., subType:int, desc:0 (controlled by device), 1 (controlled by platform). Controlled by Device: The device reads the card number, processes and prompts according to the local event rule. If the platform is connected, the device uploads the card number and result to the platform. Controlled by Platform: The device reads the card number and uploads to the platform, which determines whether the vehicle is allowed to enter/exit-->1
</ctrlMode>
<wiegendFilterMode>
<!--ro, opt, enum, card No. filtering mode, subType:int, desc:0 (card reader mode), 1 (control box mode). Card reader mode: each card reader filters its own duplicate card numbers within a time period. Control box mode: control box filters duplicate card numbers from all card readers within a time period-->1
</wiegendFilterMode>
<wiegendFilterTime>
<!--ro, opt, int, Wiegand filtering time, range:[1,600], step:1, unit:s, desc:within the time period (defaults to 3 seconds), duplicate card-swiping events only trigger one event upload-->3
</wiegendFilterTime>
<wiegendMode>
<!--ro, opt, enum, Wiegand format, subType:int, desc:0 (26-bit), 1 (34-bit)-->1
</wiegendMode>
<cardNoMode>
<!--ro, opt, enum, card No. format, subType:int, desc:0 (decimal), 1 (hexadecimal). Changing the card number format will affect adding/deleting/editing/searching of vehicle card information. For example, if the card number format is edited to hexadecimal, you need to enter a hexadecimal format card number when adding a vehicle card-->0
</cardNoMode>
<openCarHit>
<!--ro, opt, bool, keep boom pole rising for vehicle queue-->true
</openCarHit>
<codeBarDisplaySeconds>
<!--ro, opt, int, the maximum lighting duration for QR code scanning, range:[1,600], step:1, unit:s, desc:it defaults to 300 seconds. The supplement light will be turned on when QR code scanning starts-->3
</codeBarDisplaySeconds>
<LCDDisplayTime>
<!--ro, opt, int, display duration of vehicle passing scene information on LCD, range:[1,600], step:1, unit:s, desc:it defaults to 300 seconds. When the senceDisplayTime of URL (POST /ISAPI/System/LCDScreen/displayInfo?format=json) is absent, this field takes effect-->3
</LCDDisplayTime>
<paperTicketPrintCtrlMode>
<!--ro, opt, enum, ticket printing control mode, subType:string, desc:"platform" (the platform applies the printing command, then the device prints the ticket automatically), "button" (press the ticket-taking button to print the ticket)-->button
</paperTicketPrintCtrlMode>
<singleTicketSensingCoilEnabled>
<!--ro, opt, bool, whether to enable single ticket-taking triggered by radar coil-->true
</singleTicketSensingCoilEnabled>
<paperTicketPrintInterval>
<!--ro, opt, int, ticket printing interval, range:[1,600], step:1, unit:s, desc:you can only get the ticket once in this time period (defaults to 3 seconds)-->3
</paperTicketPrintInterval>
<scancodeCtrlMode>
<!--ro, opt, enum, scanning control mode, subType:string, desc:"sensing" (sensing mode, the scanner automatically detects and scans the ticket), "singleTrigger" (single trigger mode, the coil triggers the scanning or the platform controls the device to scan)-->singleTrigger
</scancodeCtrlMode>
<scancodeFilterTime>
<!--ro, opt, int, scanning filter time, range:[1,600], step:1, unit:s, desc:within the time period (defaults to 3 seconds), duplicate scans will only trigger one scan event upload-->3
</scancodeFilterTime>
<laneInfo>
<!--ro, opt, object, linked lane information-->
<relateLaneType>
<!--ro, opt, enum, type of lane where the device is installed, subType:string, desc:"motorVehicle" (motor vehicle lane), "twoWheelVehicle" (non-motor vehicle lane), "HVT" (mixed lane)-->motorVehicle
</relateLaneType>
<HVTLaneMode>
<!--ro, opt, enum, mixed lane mode, subType:string, dep:and,{$.TMEConfiguration.laneInfo.laneType,eq,HVT}, desc:for the mixed lane, the following vehicle detection modes are supported: "camera" (recognized by camera, and valid only when the lane capture camera is added), "coil" (determine the vehicle type via single/double coil. Non-motor vehicles trigger unicoil detection, and motor vehicles trigger double-coil detection), "notDistinguish" (no distinction. The vehicle type is unknown)-->camera
</HVTLaneMode>
</laneInfo>
<releaseMode>
<!--ro, opt, enum, entry/exit control mode, subType:string, desc:"platform" (controlled by the platform), "allowList" (devices allow the entry/exit according to the vehicle list), "ticket" (devices allow the entry according to the ticket taking action)-->platform
</releaseMode>
<LCDAutoRefreshEnabled>
<!--ro, opt, bool, whether to enable automatically refreshing the LCD display content to idle scene information, desc:when enabled, if the latest content applied by platform has displayed for the duration of LCDDisplayTime, the device will automatically refresh to the idle scene. When disabled, LCD will not refresh the interface-->false
</LCDAutoRefreshEnabled>
<LCDCharMoveMode>
<!--ro, opt, enum, subType:string-->scrolling
</LCDCharMoveMode>
</TMEConfiguration>
```

### 14.10.4.2 Get capabilities of entrance & exit station parameters

**Request URL**

GET /ISAPI/Parking/channels/<channelID>/TMEConfiguration/capabilities

**Query Parameter**

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| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | Channel No. |

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TMEConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configurations of entrance & exit station, attr:version{req, string, protocolVersion}-->
  <illegalCardFilteringEnabled>
    <!--ro, opt, bool, whether to enable invalid card filtering-->true
  </illegalCardFilteringEnabled>
  <LEDDefaultInfo min="0" max="512">
    <!--ro, opt, string, default display content on LED, attr:min{req, int},max{req, int}-->test
  </LEDDefaultInfo>
  <sendCardSensingCoilEnabled>
    <!--ro, opt, bool, whether to link the card dispenser with inductive loop detector-->true
  </sendCardSensingCoilEnabled>
  <wiegendSensingCoilEnabled>
    <!--ro, opt, bool, whether to link long-range card with inductive loop detector-->true
  </wiegendSensingCoilEnabled>
  <gateSwitchSignalEnabled>
    <!--ro, opt, bool-->true
  </gateSwitchSignalEnabled>
  <verifyKeyWriteCardEnabled>
    <!--ro, opt, bool, whether to enable key verification and writing data to cards-->true
  </verifyKeyWriteCardEnabled>
  <noplateTakePermissionEnabled>
    <!--ro, opt, bool, whether to enable non-license plate entry with voucher (ticket or temporary card)-->true
  </noplateTakePermissionEnabled>
  <scancodeCoilEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to link code scanner with inductive loop detector, attr:opt{req, string},def{req, bool}-->true
  </scancodeCoilEnabled>
  <noPaperDisplayEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable out of paper prompt, attr:opt{req, string},def{req, bool}-->true
  </noPaperDisplayEnabled>
  <fullParkingNoTakeCard opt="true,false" def="true">
    <!--ro, opt, bool, whether to stop issuing card when no vacant parking space, attr:opt{req, string},def{req, bool}-->true
  </fullParkingNoTakeCard>
  <ctrlMode opt="0,1">
    <!--ro, opt, enum, control mode of card No., subType:int, attr:opt{req, string}, desc:0 (controlled by device), 1 (controlled by platform)-->0
  </ctrlMode>
  <wiegendFilterMode opt="0,1">
    <!--ro, opt, enum, card No. filtering mode, subType:int, attr:opt{req, string}, desc:0 (card reader mode), 1 (control box mode)-->0
  </wiegendFilterMode>
  <wiegendFilterTime min="1" max="600">
    <!--ro, opt, int, Wiegand filtering time, range:[1,600], step:1, unit:s, attr:min{req, int},max{req, int}, desc:within the time period (defaults to 3
seconds), duplicate card-swiping events only trigger one event upload-->3
  </wiegendFilterTime>
  <wiegendMode opt="0,1">
    <!--ro, opt, enum, Wiegand format, subType:int, attr:opt{req, string}, desc:0 (26-bit), 1 (34-bit)-->1
  </wiegendMode>
  <cardNoMode opt="0,1">
    <!--ro, opt, enum, card No. format, subType:int, attr:opt{req, string}, desc:0 (decimal), 1 (hexadecimal). Changing the card number format will affect
adding/deleting/editing/searching of vehicle card information. For example, if the card number format is edited to hexadecimal, you need to enter a
hexadecimal format card number when adding a vehicle card-->0
  </cardNoMode>
  <openCarHit opt="true,false" def="true">
    <!--ro, opt, bool, keep boom pole rising for vehicle queue, attr:opt{req, string},def{req, bool}-->true
  </openCarHit>
  <codeBarDisplaySeconds min="1" max="600">
    <!--ro, opt, int, the maximum lighting duration for QR code scanning, range:[1,600], step:1, unit:s, attr:min{req, int},max{req, int}, desc:it defaults
to 300 seconds. The supplement light will be turned on when QR code scanning starts-->3
  </codeBarDisplaySeconds>
  <LCDDisplayTime min="1" max="600">
    <!--ro, opt, int, display duration of vehicle passing scene information on LCD, range:[1,600], step:1, unit:s, attr:min{req, int},max{req, int}, desc:it
defaults to 300 seconds. When the senceDisplayTime of URL (POST /ISAPI/System/LCDScreen/displayInfo?format=json) is absent, this field takes effect-->3
  </LCDDisplayTime>
  <paperTicketPrintCtrlMode opt="platform,button">
    <!--ro, opt, enum, ticket printing control mode, subType:string, attr:opt{req, string}, desc:"platform" (the platform applies the printing command, then
the device prints the ticket automatically), "button" (press the ticket-taking button to print the ticket)-->button
  </paperTicketPrintCtrlMode>
  <singleTicketSensingCoilEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable single ticket-taking triggered by radar coil, attr:opt{req, string},def{req, bool}-->true
  </singleTicketSensingCoilEnabled>
  <paperTicketPrintInterval min="1" max="600">
    <!--ro, opt, int, ticket printing interval, range:[1,600], step:1, unit:s, attr:min{req, int},max{req, int}, desc:you can only get the ticket once in
this time period (defaults to 3 seconds)-->3
  </paperTicketPrintInterval>
  <scancodeCtrlMode opt="sensing,singleTrigger">
    <!--ro, opt, enum, scanning control mode, subType:string, attr:opt{req, string}, desc:"sensing" (sensing mode, the scanner automatically detects and
scans the ticket), "singleTrigger" (single trigger mode, the coil triggers the scanning or the platform controls the device to scan)-->singleTrigger
  </scancodeCtrlMode>
  <scancodeFilterTime min="1" max="600">
    <!--ro, opt, int, scanning filter time, range:[1,600], step:1, unit:s, attr:min{req, int},max{req, int}, desc:within the time period (defaults to 3
```

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```
    <!--ro, opt, int, scanning filter time, range:[1,600], step:1, unit:s, attr:min{req, int},max{req, int}, desc:within the time period (defaults to 3
seconds), duplicate scans will only trigger one scan event upload-->3
  </scancodeFilterTime>
  <laneInfo>
    <!--ro, opt, object, linked lane information-->
    <relateLaneType opt="motorVehicle,twoWheelVehicle,HVT">
      <!--ro, opt, enum, type of lane where the device is installed, subType:string, attr:opt{req, string}, desc:"motorVehicle" (motor vehicle lane),
"twoWheelVehicle" (non-motor vehicle lane), "HVT" (mixed lane)-->motorVehicle
    </relateLaneType>
    <HVTLaneMode opt="camera,coil,notDistinguish">
      <!--ro, opt, enum, mixed lane mode, subType:string, dep:and,{$.TMEConfiguration.laneInfo.laneType,eq,HVT}, attr:opt{req, string}, desc:for the mixed
lane, the following vehicle detection modes are supported: "camera" (recognized by camera, and valid only when the lane capture camera is added), "coil"
(determine the vehicle type via single/double coil. Non-motor vehicles trigger unicoil detection, and motor vehicles trigger double-coil detection),
"notDistinguish" (no distinction. The vehicle type is unknown)-->camera
    </HVTLaneMode>
  </laneInfo>
  <releaseMode opt="platform,allowList,ticket">
    <!--ro, opt, enum, entry/exit control mode, subType:string, attr:opt{req, string}, desc:"platform" (controlled by the platform), "allowList" (devices
allow the entry/exit according to the vehicle list), "ticket" (devices allow the entry according to the ticket taking action)-->platform
  </releaseMode>
  <LCDAutoRefreshEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable automatically refreshing the LCD display content to idle scene information, attr:opt{req, string}, desc:when
enabled, if the latest content applied by platform has displayed for the duration of LCDDisplayTime, the device will automatically refresh to the idle
scene. When disabled, LCD will not refresh the interface-->false
  </LCDAutoRefreshEnabled>
</TMEConfiguration>
```

### 14.10.4.3 Set the control parameters of audio announcement

#### Request URL

PUT /ISAPI/Parking/channels/<channelID>/TMEVoiceConfiguration

#### Query Parameter

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| channelID | string | -- |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TMEVoiceConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, audio prompt control parameters, attr:version{req, string, protocolVersion}-->
  <voiceSpeed>
    <!--req, int, speed, range:[0,100]-->0
  </voiceSpeed>
  <voicePitch>
    <!--req, int, which is between 0 and 100, range:[0,100]-->0
  </voicePitch>
  <voiceVolum>
    <!--req, int, volume, range:[0,100]-->0
  </voiceVolum>
  <voiceRole>
    <!--req, int, speaker, desc:options: "3", "51", "52", "53", "54", "55"-->0
  </voiceRole>
  <voiceContent>
    <!--req, string, audio announcement content-->test
  </voiceContent>
  <voiceFileName>
    <!--req, string, audio file name-->test
  </voiceFileName>
  <voicePlate>
    <!--opt, bool, whether to enable audio prompt for license plates-->true
  </voicePlate>
</TMEVoiceConfiguration>
```

#### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

#### 14.10.4.4 Get total capability of entrance & exit station

**Request URL**

GET /ISAPI/Parking/TME/capabilities

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<TMECap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the total capability of entrance & exit station, attr:version{req, string, protocolVersion}-->
  <isSupportTMEChargeRule>
    <!--ro, opt, bool, related URL: /ISAPI/Parking/channels/<ID>/TMEChargeRule/capabilities, desc:related URL:
/ISAPI/Parking/channels/<ID>/TMEChargeRule/capabilities-->true
  </isSupportTMEChargeRule>
  <isSupportPaperCharge>
    <!--ro, opt, bool, whether the device supports configuring charging payment information, desc:related URL:
/ISAPI/Parking/channels/<ID>/paperChargeInfo/capabilities-->true
  </isSupportPaperCharge>
  <isSupportParkingSpace>
    <!--ro, opt, bool, whether the device supports configuring the parking space information, desc:related URL:
/ISAPI/Parking/channels/<ID>/parkingSpaceInfo/capabilities-->true
  </isSupportParkingSpace>
  <isSupportTMEOffline>
    <!--ro, opt, bool, whether the device supports configuring offline mode parameters, desc:related URL:
/ISAPI/Parking/channels/<ID>/TMEOfflineConfiguration/capabilities-->true
  </isSupportTMEOffline>
  <isSupportTMEMultiCtrl>
    <!--ro, opt, bool, related URL: /ISAPI/Parking/channels/<ID>/TMEMultiCtrlConfiguration/capabilities, desc:related URL:
/ISAPI/Parking/channels/<ID>/TMEMultiCtrlConfiguration/capabilities-->true
  </isSupportTMEMultiCtrl>
  <isSupportTMECFG>
    <!--ro, opt, bool, whether the device supports configuring entrance & exit station parameters, desc:related URL:
/ISAPI/Parking/channels/<ID>/TMEConfiguration/capabilities-->true
  </isSupportTMECFG>
  <isSupportChargeAmount>
    <!--ro, opt, bool, whether the device supports charging account configurations, desc:related URL:
/ISAPI/Parking/channels/<ID>/chargeAmount/capabilities-->true
  </isSupportChargeAmount>
  <isSupportVehicleCtrl>
    <!--ro, opt, bool, whether the device supports vehicle arming configurations, desc:related URL:
/ISAPI/Parking/channels/<ID>/vehicleControl/capabilities-->true
  </isSupportVehicleCtrl>
  <isSupportBarrierGate>
    <!--ro, opt, bool, whether the device supports barrier gate control, desc:related URL: /ISAPI/Parking/channels/<ID>/barrierGate/capabilities-->true
  </isSupportBarrierGate>
  <isSupportTemporaryParkingCard>
    <!--ro, opt, bool, whether the device supports temporary cards-->true
  </isSupportTemporaryParkingCard>
  <isSupportLEDConfiguration>
    <!--ro, opt, bool, whether the device supports LED display parameter configuration, desc:related URL:
/ISAPI/Parking/channels/<ID>/LEDConfiguration/capabilities-->true
  </isSupportLEDConfiguration>
  <isSupportVoiceBroadcastInfo>
    <!--ro, opt, bool, whether the device supports audio prompt control, desc:related URL: /ISAPI/Parking/channels/<ID>/voiceBroadcastInfo/capabilities--
>true
  </isSupportVoiceBroadcastInfo>
  <isSupportPaperPrintFormatInfo>
```

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```
    <!--ro, opt, bool, whether the device supports configuring the ticket printing format, desc:related URL:
/ISAPI/Parking/channels/<ID>/paperPrintFormatInfo/capabilities-->true
  </isSupportPaperPrintFormatInfo>
  <isSupportLockGateConfig>
    <!--ro, opt, bool, whether the device supports intelligent barrier gate configurations, desc:related URL:
/ISAPI/Parking/channels/<ID>/lockGateConfiguration/capabilities-->true
  </isSupportLockGateConfig>
  <isSupportTMEVoiceConfiguration>
    <!--ro, opt, bool, whether the device supports configuring audio parameters, desc:related URL: /ISAPI/Parking/channels/<ID>/TMEVoiceConfiguration-->true
  </isSupportTMEVoiceConfiguration>
  <isSupportVoiceFileUpload>
    <!--ro, opt, bool, whether the device supports importing audio files, desc:related URL: /ISAPI/Parking/channels/<ID>/VoiceFileUpload/<NAME>-->true
  </isSupportVoiceFileUpload>
  <isSupportVoiceFileList>
    <!--ro, opt, bool, whether the device supports getting audio file list, desc:related URL: /ISAPI/Parking/channels/<ID>/VoiceFileList-->true
  </isSupportVoiceFileList>
  <isSupportRequestCardOpenBarrierOperationResult>
    <!--ro, opt, bool, whether the device supports searching for results of pass by swiping card, desc:POST
/ISAPI/Parking/TME/RequestCardOpenBarrierOperationResult?format=json-->true
  </isSupportRequestCardOpenBarrierOperationResult>
  <isSupportCardReaderTriggerParams>
    <!--ro, opt, bool, whether the device supports configuring card reader linkage parameters, desc:/ISAPI/Parking/TME/CardReaderTriggerParams?format=json--
>true
  </isSupportCardReaderTriggerParams>
  <isSupportSearchSwipeCardRecord>
    <!--ro, opt, bool, whether the device supports searching for card swiping records, desc:POST
/ISAPI/Parking/TME/SwipeCardRecordMgr/SearchSwipeCardRecord?format=json-->true
  </isSupportSearchSwipeCardRecord>
  <isSupportTMECardVehicleDelete>
    <!--ro, opt, bool-->true
  </isSupportTMECardVehicleDelete>
  <isSupportAddVehicleCardInfo>
    <!--ro, opt, bool, whether the device supports adding vehicle card information of entrance/exit, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/AddVehicleCardInfo?format=json-->true
  </isSupportAddVehicleCardInfo>
  <isSupportModifyVehicleCardInfo>
    <!--ro, opt, bool, whether the device supports editing vehicle card information of entrance/exit, desc:PUT
/ISAPI/Parking/TME/VehicleCardInfoMgr/ModifyVehicleCardInfo?format=json-->true
  </isSupportModifyVehicleCardInfo>
  <isSupportSearchCardInfoByVehicle>
    <!--ro, opt, bool, whether the device supports searching for vehicle card information by vehicle, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/SearchCardInfoByVehicle?format=json-->true
  </isSupportSearchCardInfoByVehicle>
  <isSupportSearchVehicleCardInfo>
    <!--ro, opt, bool, whether the device supports searching for vehicle card information, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/SearchVehicleCardInfo?format=json-->true
  </isSupportSearchVehicleCardInfo>
  <isSupportDeleteCardVehicleInfo>
    <!--ro, opt, bool, whether the device supports deleting vehicle card information of entrance/exit, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/DeleteCardVehicleInfo?format=json-->true
  </isSupportDeleteCardVehicleInfo>
  <isSupportCreateNewVehicleCard>
    <!--ro, opt, bool, whether the device supports adding vehicle card of entrance/exit, desc:GET
/ISAPI/Parking/TME/VehicleCardInfoMgr/CreateNewVehicleCard?format=json-->true
  </isSupportCreateNewVehicleCard>
  <isSupportWriteEncryptCardCfg>
    <!--ro, opt, bool, whether the device supports encryption configuration for writing data to cards,
desc:/ISAPI/Parking/TME/VehicleCardInfoMgr/WriteEncryptCardCfg?format=json-->true
  </isSupportWriteEncryptCardCfg>
  <isSupportCustomPaperTicketPrintInfo>
    <!--ro, opt, bool, whether the device supports customizing ticket printing content, desc:/ISAPI/Parking/TME/CustomPaperTicketPrintInfo?format=json--
>true
  </isSupportCustomPaperTicketPrintInfo>
  <isSupportPaperPrintTest>
    <!--ro, opt, bool, whether the device supports ticket printing test, desc:POST /ISAPI/Parking/TME/PaperPrintTest?format=json-->true
  </isSupportPaperPrintTest>
</TMECap>
```

## 14.10.5 Vehicle Access Control

### 14.10.5.1 Set parameters to open/close and lock/unlock barrier gate

**Request URL**

PUT /ISAPI/ITC/Entrance/barrierGateCtrl

**Query Parameter**

None

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<BarrierGateCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <barrietGateNum>
    <!--opt, int, barrier gate No.-->1
  </barrietGateNum>
  <BarrierGateCtrlList>
    <!--opt, array, subType:object-->
    <barrietGateOper>
      <!--opt, enum, barrier gate control type, subType:string, desc:"off"-fall, "on"-rise, "stop"-stop at a certain position, "locked"-Lock;-->on
    </barrietGateOper>
  </BarrierGateCtrlList>
</BarrierGateCtrl>
```

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.10.5.2 Get the entrance and exit capability

### Request URL

GET /ISAPI/ITC/Entrance/capabilities

### Query Parameter

None

### Request Message

None

### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EntranceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <supportEntrance>
    <!--ro, opt, bool, whether it supports entrance and exit control-->true
  </supportEntrance>
  <supportBarrierGateNum>
    <!--ro, opt, int-->1
  </supportBarrierGateNum>
  <supportAlarmINNum>
    <!--ro, opt, int-->1
  </supportAlarmINNum>
  <supportRelayNum>
    <!--ro, opt, int-->1
  </supportRelayNum>
  <isSupportWizardDeviceInfo>
    <!--ro, req, bool, whether to support configuring device information in wizard-->true
  </isSupportWizardDeviceInfo>
  <isSupportWizardFirstLogin>
    <!--ro, req, bool, whether to support judging if it is login for the first time in wizard-->true
  </isSupportWizardFirstLogin>
  <supportEntranceParam>
    <!--ro, opt, int-->1
  </supportEntranceParam>
</EntranceCap>
```

## 14.10.5.3 Set the supplement light control parameters of the entrance and exit device

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VISION AND SECURITY SYSTEM CO.,LTD.

**Request URL**

PUT /ISAPI/ITC/Entrance/lampCtrl

**Query Parameter**

None

**Request Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<LampCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <barrietGateNum>
    <!--opt, int, barrier gate No.-->0
  </barrietGateNum>
  <LampCtrlList>
    <!--opt, array, subType:object-->
    <lampOper>
      <!--opt, enum, solid light operation, subType:string, desc:"off" (turn off solid light), "on" (turn on solid light)-->off
    </lampOper>
  </LampCtrlList>
</LampCtrl>
```

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

## 14.10.6 Vehicle List Comparison

### 14.10.6.1 Search the vehicle blocklist and allowlist

**Request URL**

POST /ISAPI/Traffic/channels/<channelID>/searchLPListAudit

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

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```
<?xml version="1.0" encoding="UTF-8"?>

<LPListAuditSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, whether the device supports searching the license plate blocklist and allowlist, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the number of search results this time, desc:no more than 2000 result items for one time-->1
  </maxResults>
  <type>
    <!--opt, enum, list type, subType:string, desc:"blackList" (license plates in blocklist), "whitelist" (license plates in allowlist), "allVehicleList" (license plates in all lists), "otherVehicleList" (license plates in other lists)-->blackList
  </type>
  <LicensePlate>
    <!--opt, string, license plate number, fuzzy search is supported, desc:license plate number,fuzzy search is supported-->test
  </LicensePlate>
  <cardNo>
    <!--opt, string, card No.-->test
  </cardNo>
  <cardID>
    <!--opt, string, card ID (Wiegand protocol), range:[1,9]-->test
  </cardID>
  <plateColor>
    <!--opt, enum, license plate color, subType:string, desc:black, blue, civilAviationBlack (civil aviation black), civilAviationGreen (civil aviation green), golden, green, mixedColor (mixed color), newEnergyGreen (new energy green), newEnergyYellowGreen (new energy green and yellow), orange, other (other color), red, unknown, white, yellow-->black
  </plateColor>
  <plateType>
    <!--opt, enum, license plate type, subType:string, desc:02TypePersonalized (type 02 custom vehicle), 92FarmVehicle (two-line license plate civil vehicle), 92TypeArm (type 92 armed police vehicle), 92TypeCivil (type 92 civil vehicle), arm (police vehicle), civilAviation (civil aviation license plate), coach (driver-training vehicle), consulate (consular vehicle), embassy (embassy vehicle), emergency (emergency license plate), green1325FarmVehicle (agricultural vehicle (green 1325)), hongKongMacao (Hong Kong China / Macao China entrance vehicle), motorola (motorcycle), newEnergy (new energy vehicle license plate), oneLineArm (new armed police vehicle (one-line)), oneLineArmHeadquarters (armed police headquarter license plate (one-line)), tempEntry (temporary entry car), tempTravl (temporary license plate car), trailer, twoLineArm (new armed police vehicle (two-line)), twoLineArmHeadquarters (armed police headquarter license plate (two-line)), unknown, yellow1225FarmVehicle (agricultural vehicle (yellow 1225)), yellow1325FarmVehicle (agricultural vehicle (yellow 1325)), yellowTwoLine (yellow two-line rear license plate)-->02TypePersonalized
  </plateType>
  <name>
    <!--opt, string, range:[1,32]-->张三
  </name>
  <certificateType>
    <!--opt, enum, subType:string-->officerID
  </certificateType>
  <certificateNumber>
    <!--opt, string, range:[1,32]-->123456
  </certificateNumber>
  <virtualParkingNum>
    <!--opt, string, range:[1,16]-->1
  </virtualParkingNum>
  <groupName>
    <!--opt, string, range:[1,64]-->test
  </groupName>
</LPListAuditSearchDescription>
```

## Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LPListAuditSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search result of vehicle blocklist and allowlist, attr:version{req, string}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool, status search-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO_MATCHES" (no matched data)-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of returned records-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <LicensePlateInfoList>
    <!--ro, opt, object, license plate information list-->
    <LicensePlateInfo>
      <!--ro, opt, object, license plate information-->
      <id>
        <!--ro, req, string, ID-->test
      </id>
```

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```
        <LicensePlate>
          <!--ro, opt, string, license plate number-->test
        </LicensePlate>
        <type>
          <!--ro, opt, enum, list type, subType:string, desc:"blackList" (license plates in blocklist), "whitelist" (license plates in allowlist),
"allVehicleList" (license plates in all lists), "otherVehicleList" (license plates in other lists)-->blackList
        </type>
        <createTime>
          <!--ro, opt, datetime, creation time-->1970-01-01T00:00:00+08:00
        </createTime>
        <direction>
          <!--ro, opt, enum, direction, subType:string, desc:"forward,reverse,unknown"-->forward
        </direction>
        <laneNo>
          <!--ro, opt, int, lane No.-->1
        </laneNo>
        <plateCategory>
          <!--ro, opt, string, license plate type-->test
        </plateCategory>
        <country>
          <!--ro, opt, string, country or region-->BLR
        </country>
        <area>
          <!--ro, opt, int, region-->1
        </area>
        <effectiveTime>
          <!--ro, opt, date, validity period (date) of the blocklist and allowlist-->1970-01-01+08:00
        </effectiveTime>
        <countryIndex>
          <!--ro, opt, int, country/region index-->1
        </countryIndex>
        <cardID>
          <!--ro, opt, string, card ID (Wiegand protocol), range:[1,9]-->test
        </cardID>
        <effectiveDateTime>
          <!--ro, opt, datetime, validity period of the blocklist and allowlist-->1970-01-01T00:00:00+08:00
        </effectiveDateTime>
        <effectiveStartDate>
          <!--ro, opt, string, start date of the validity period-->1970-01-01
        </effectiveStartDate>
        <cardNo>
          <!--ro, opt, string, card No.-->test
        </cardNo>
        <plateColor>
          <!--ro, opt, string, license plate color-->test
        </plateColor>
        <name>
          <!--ro, opt, string, range:[1,32]-->张三
        </name>
        <certificateType>
          <!--ro, opt, enum-->officerID
        </certificateType>
        <certificateNumber>
          <!--ro, opt, string, range:[1,32]-->123456
        </certificateNumber>
        <virtualParkingNum>
          <!--ro, opt, string, range:[1,16]-->1
        </virtualParkingNum>
        <plateDescription>
          <!--ro, opt, string, range:[0,128]-->abc
        </plateDescription>
        <groupName>
          <!--ro, opt, string, range:[1,64]-->test
        </groupName>
      </LicensePlateInfo>
    </LicensePlateInfoList>
  </LPListAuditSearchResult>
```

## 14.10.7 Settings Wizard

### 14.10.7.1 Get wizard parameters

**Request URL**

GET /ISAPI/ITC/Entrance/WizardDeviceInfo/<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

**Request Message**

None

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### Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WizardDeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, channel No.-->1
  </id>
  <defaultCHN>
    <!--ro, req, string, province name-->test
  </defaultCHN>
  <ipVersion opt="IPv4,IPv6,dual">
    <!--ro, req, enum, IP address version, subType:string, attr:opt{req, string}, desc:IP version,only IPv4 is supported-->IPv4
  </ipVersion>
  <ipAddress min="0" max="40">
    <!--ro, opt, string, IPv4 address, attr:min{req, int},max{req, int}-->test
  </ipAddress>
  <subnetMask min="0" max="15">
    <!--ro, opt, string, IPv4 subnet mask, attr:min{req, int},max{req, int}-->test
  </subnetMask>
  <DefaultGateway>
    <!--ro, opt, object, default gateway-->
    <ipAddress>
      <!--ro, opt, string, IPv4 address-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address-->test
    </ipv6Address>
  </DefaultGateway>
  <capMode>
    <!--ro, req, enum, capture mode, subType:string, desc:capture mode: "strobe"-strobe light mode,"flashing"-flashing light mode-->strobe
  </capMode>
  <sceneMode>
    <!--ro, req, enum, scene mode, subType:string, desc:"commonEntrance" (common entrance and exit), "tollGate" (toll station (vehicles will stay for longer
time)), "ParkingEntrance"-entrance and exit of underground parking lot (there will be dark day and night)-->commonEntrance
  </sceneMode>
</WizardDeviceInfo>
```

### 14.10.7.2 Get the configuration capability of device information in wizard for entrance and exit capture unit

### Request URL

GET /ISAPI/ITC/Entrance/WizardDeviceInfo/capabilities

### Query Parameter

None

### Request Message

None

### Response Message

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dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<WizardDeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id min="0" max="10000">
    <!--ro, req, int, channel No., attr:min{req, int},max{req, int}-->1
  </id>
  <defaultCHN>
    <!--ro, req, enum, province abbreviation, subType:string, desc:province abbreviation-->云
  </defaultCHN>
  <ipVersion opt="IPv4">
    <!--ro, req, enum, IP version, subType:string, attr:opt{req, string}, desc:only IPv4 is supported-->IPv4
  </ipVersion>
  <ipAddress min="0" max="40">
    <!--ro, opt, string, IP address, attr:min{req, int},max{req, int}-->test
  </ipAddress>
  <subnetMask min="0" max="15">
    <!--ro, opt, string, subnet mask for IPv4 address, attr:min{req, int},max{req, int}-->test
  </subnetMask>
  <DefaultGateway>
    <!--ro, opt, object, default gateway, currently only IPv4 is supported-->
    <ipAddress min="0" max="40">
      <!--ro, opt, string, IP address, attr:min{req, int},max{req, int}-->test
    </ipAddress>
    <ipv6Address min="0" max="40">
      <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->test
    </ipv6Address>
  </DefaultGateway>
  <capMode opt="strobe,flashing">
    <!--ro, req, enum, capture mode, subType:string, attr:opt{req, string}, desc:"strobe" (strobe light mode), "flashing" (flashing light mode)-->strobe
  </capMode>
  <sceneMode opt="commonEntrance,tollGate,ParkingEntrance">
    <!--ro, req, enum, scene mode, subType:string, attr:opt{req, string}, desc:"commonEntrance" (common entrance and exit), "tollGate" (toll station
(vehicles will stay for longer time)), "ParkingEntrance" (entrance and exit of underground parking lot (there will be dark day and night))-->commonEntrance
  </sceneMode>
</WizardDeviceInfo>
```

### 14.10.7.3 Set the wizard parameters of login for the first time

#### Request URL

PUT /ISAPI/ITC/Entrance/WizardFirstLogin/<channelID>

#### Query Parameter

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | Channel No. |

#### Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WizardFirstLogin xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, channel No.-->1
  </id>
  <isFirstLogin>
    <!--opt, bool, the default value is "true", desc:it is true by default-->true
  </isFirstLogin>
  <isReboot>
    <!--opt, bool, whether the device supports rebooting, desc:it is true by default-->true
  </isReboot>
</WizardFirstLogin>
```

#### Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

#### 14.10.7.4 Get the wizard capability of logging in for the first time

**Request URL**

GET /ISAPI/ITC/Entrance/WizardFirstLogin/<channelID>/capabilities

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| channelID | string | -- |

**Request Message**

None

**Response Message**

```
<?xml version="1.0" encoding="UTF-8"?>

<WizardFirstLogin xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id min="0" max="100000">
    <!--ro, req, int, channel No., attr:min{req, int},max{req, int}-->1
  </id>
  <isFirstLogin>
    <!--ro, opt, bool, the default value is "true"-->true
  </isFirstLogin>
  <isReboot>
    <!--ro, opt, bool, whether the device supports rebooting-->true
  </isReboot>
</WizardFirstLogin>
```

# 14.11 Parking Space Detection

## 14.11.1 Parking Space Status Detection

#### 14.11.1.1 Set parking space detection parameters for all channels

**Request URL**

PUT /ISAPI/Parking/channels/packingSpaceRecognition?format=json

**Query Parameter**

None

**Request Message**

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "PackingSpaceRecognitionList": [
    /*req, array, parameters of parking space detection, subType:object, range:[,]*/
        {
            "channelID":  1,
            /*req, int, device channel No., range:[,], step:, unit:, unitType:*/
            "channelAnalysisEnabled":  false,
            /*opt, bool, whether to enable channel analysis algorithom, desc:returned by fisheye cameras, for four-channel, enabling the analysis algorithm
of only two channels is required.*/
            "packingSpaceTotalNum":  1,
            /*opt, int, total number of parking spaces, range:[0,4], step:, unit:, unitType:*/
            "PackingSpaceInfoList": [
            /*opt, array, parking space information list, subType:object, range:[0,4]*/
                {
                    "PackingSpaceInfo": {
                    /*opt, object, parking space information*/
                        "absoulteParkingNum":  "123",
                        /*req, string, parking space No., range:[1,16], desc:the parking space No. is the actual parking space No. drawn on the group, which
can be obtained by cameras*/
                        "virtualParkingNum":  "123",
                        /*opt, string, virtual parking space No., range:[1,16], desc:the virtual parking space refers to the user-defined parking space No.
when configuring parking space list for the device; this field will not be uploaded if it is not configured*/
                        "parkingType":  1,
                        /*opt, enum, parking space type, subType:int, [], desc:the special parking space will be configured with indicators which are
different with that configured for the normal parking spaces, e.g., VIP parking space;  0: normal parking space; 1: special parking space*/
                        "Region": [
                        /*opt, array, area coordinates of a specified parking space, subType:object, range:[1,4]*/
                            {
                                "Point": {
                                /*ro, opt, object, point coordinates*/
                                    "positionX":  0.123,
                                    /*ro, req, float, x-coordinate, range:[0,1], step:, unit:, unitType:, desc:the accuracy: 0.001, the origin of coordinate
is in the left corner of image*/
                                    "positionY":  0.123
                                    /*ro, req, float, y-coordinate, range:[0,1], step:, unit:, unitType:, desc:the accuracy: 0.001, the origin of coordinate
is in the left corner of image*/
                                }
                            }
                        ]
                    }
                }
            ]
        }
    ]
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, range:[,], step:, unit:, unitType:, desc:it must be returned if it cannot be represented by 1 (succeeded)*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:it must be returned if it cannot be represented by ok (succeeded)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:it must be returned if it cannot be represented by ok (succeeded)*/
    "errorCode":  1,
    /*ro, req, int, error code, range:[,], step:, unit:, unitType:, desc:it corresponds to subStatusCode when the value of statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, range:[,], desc:it can be extended based on the protocol constraints when the statusCode is not 1*/
}
```

# 14.12 Radar (General)

## 14.12.1 Radar Parameters Management

### 14.12.1.1 Get security radar capability set

**Request URL**

GET /ISAPI/Radar/capabilities?format=json

**Query Parameter**

None

**Request Message**

None

**Response Message**

```
{
```

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```
    "RadarCap": {
    /*ro, opt, object, security radar capability set*/
        "isSupportSensitivity": true,
        /*ro, opt, bool, whether it supports setting sensitivity*/
        "isSupportSensitivityList": true,
        /*ro, opt, bool, whether it supports getting all microwave radar sensitivity parameters, desc:/ISAPI/Radar/Configuration/sensitivityList?
format=json*/
        "isSupportDetectInfo": true,
        /*ro, opt, bool, whether it supports setting detection angle and range*/
        "isSupportEnvironmentManualLearning": true,
        /*ro, opt, bool, whether it supports auto environment learning*/
        "isSupportZoneDrawingMode": true,
        /*ro, opt, bool, whether it supports setting zone drawing mode*/
        "isSupportZonesCoordinateInfo": true,
        /*ro, opt, bool, whether it supports setting zone coordinates*/
        "isSupportCalibratingMode": true,
        /*ro, opt, bool, whether it supports setting calibration mode*/
        "isSupportManualCalibrating": true,
        /*ro, opt, bool, whether it supports manual calibration*/
        "isSupportMasterSlaveTracking": true,
        /*ro, opt, bool, whether it supports smart linkage*/
        "isSupportTrailInfoSearch": true,
        /*ro, opt, bool, whether it supports radar pattern search*/
        "isSupportRadarInfo": true,
        /*ro, opt, bool, whether it supports getting the radar information*/
        "isSupportUpgrade": true,
        /*ro, opt, bool, whether it supports upgrading the radar's firmware*/
        "isSupportLaneParam": true,
        /*ro, opt, bool, whether it supports configuring lane parameters*/
        "isSupportCalibration": true,
        /*ro, opt, bool, whether it supports radar-camera calibration*/
        "isSupportChannelCalibration": true,
        /*ro, opt, bool, whether it supports radar-camera calibration (video channel), desc:related URI (/ISAPI/Radar/calibration?format=json&radarChannel=
<radarChannelID>&channelID=<channelID>), channelID is supported*/
        "isSupportRadarChannelCalibration": true,
        /*ro, opt, bool, whether it supports radar-camera calibration (radar channel), desc:related URI (/ISAPI/Radar/calibration?format=json&radarChannel=
<radarChannelID>&channelID=<channelID>), radarChannel is supported*/
        "isSupportFrequencyRange": true,
        /*ro, opt, bool, whether it supports setting frequency range*/
        "isSupportFrequencyRangeList": true,
        /*ro, opt, bool, whether it supports getting configuration of all radar channel frequency bands, desc:/ISAPI/Radar/Configuration/requencyRangeList?
format=json*/
        "isSupportAlarmLine": true,
        /*ro, opt, bool, whether it supports setting detection line*/
        "isSupportAlarmLineRule": true,
        /*ro, opt, bool, whether it supports setting detection rule*/
        "isSupportMasterSlaveCfg": true,
        /*ro, opt, bool, whether it supports setting smart linkage parameters*/
        "isSupportSlaveRadar": true,
        /*ro, opt, bool, whether it supports adding sub radar*/
        "isSupportRadarList": true,
        /*ro, opt, bool, whether it supports getting radar list*/
        "isSupportRadarTrailInfoBriefSearch": true,
        /*ro, opt, bool, whether it supports searching for brief information of radar patterns*/
        "isSupportTrailDetailInfo": true,
        /*ro, opt, bool, whether it supports searching radar pattern details by condition*/
        "isSupportForceTrackCfg": true,
        /*ro, opt, bool, whether it supports configuring mandatory tracking*/
        "isSupportForceTrackCtrl": true,
        /*ro, opt, bool, whether it supports controlling mandatory tracking*/
        "isSupportSpeedThresholdCfg": true,
        /*ro, opt, bool, whether it supports configuring speed threshold*/
        "isSupportRelatedIPCInfo": true,
        /*ro, opt, bool, whether it supports getting relative position of the speed dome linked to the radar*/
        "isSupportRadarLED": true,
        /*ro, opt, bool, whether it supports configuring radar LED indicator*/
        "isSupportBackgroundPic": true,
        /*ro, opt, bool, whether it supports managing the background picture of live view of the radar, desc:whether it supports managing the background
picture of live view of the radar*/
        "isSupportBackgroundParam": true,
        /*ro, opt, bool, whether it supports configuring the background area parameters of the radar, desc:whether it supports configuring the background
area parameters of the radar*/
        "isSupportScaleCfg": true,
        /*ro, opt, bool, whether it supports configuring radar scale parameters, desc:whether it supports configuring radar scale parameters*/
        "isSupportAlarmLineCfg": true,
        /*ro, opt, bool, whether it supports configuring trigger line, desc:/ISAPI/Radar/Configuration/alarmLineCfg?format=json*/
        "isSupportAlarmLineRuleCfg": true,
        /*ro, opt, bool, whether it supports configuring alarm rule, desc:/ISAPI/Radar/Configuration/alarmLineRuleCfg/<ID>?format=json*/
        "isSupportRadarTrail": true,
        /*ro, opt, bool, whether it supports configuring radar pattern, desc:/ISAPI/Radar/Configuration/radarTrail/capabilities?format=json*/
        "isSupportRadarFieldPreview": true,
        /*ro, opt, bool, whether it supports live view in the radar FOV, desc:/ISAPI/Radar/Configuration/radarFieldPreview/capabilities?format=json*/
        "isSupportPreManualCalibrating": true,
        /*ro, opt, bool, whether it supports configuring radar pre-calibration, desc:/ISAPI/Radar/Configuration/preManualCalibrating/capabilities*/
        "isSupportRadarSurface": true,
        /*ro, opt, bool, whether it supports configuring radar arrays, desc:/ISAPI/Radar/Configuration/radarSurface/capabilities?format=json*/
        "isSupportLevelCalibration": true,
        /*ro, opt, bool, whether it supports level calibration, desc:/ISAPI/Radar/Configuration/levelCalibrating/radarChannels/<ID>/goto?format=json*/
        "isSupportManualTrace": true,
        /*ro, opt, bool, whether it supports tracking manually, desc:/ISAPI/Radar/Configuration/radarChannels/<ID>/manualTrace?format=json*/
        "isSupportLevelPosition": true,
        /*ro, opt, bool, whether it supports speed dome turning with radar's level calibration,
desc:/ISAPI/Radar/Configuration/levelCalibrating/channels/<ID>/position/<radarChannelID>?format=json*/
```

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```
        "isSupportPosition": true,
        /*ro, opt, bool, whether it supports controlling the speed dome to turn to the current radar's direction after switching radar,
desc:/ISAPI/Radar/Configuration/channels/<ID>/position/<radarChannelID>?format=json*/
        "isSupportShield": true,
        /*ro, opt, bool, whether it supports configuring shielded areas for the radar,
desc:/ISAPI/Radar/Configuration/radarChannels/<ID>/shield/capabilities?format=json*/
        "isSupportvirtualAlarmMarkCfg": true,
        /*ro, opt, bool, whether it supports configuring false track mark, desc:/ISAPI/Radar/Configuration/virtualAlarmMarkCfg?format=json*/
        "isSupportLandformLearning": true,
        /*ro, opt, bool, whether it supports terrain learning, desc:/ISAPI/Radar/Configuration/landformsLearningCtrl/capabilities?format=json*/
        "isSupportIntelligentAnalysis": true,
        /*ro, opt, bool, whether it supports configuring radar intelligent analysis, desc:/ISAPI/Radar/intelligentAnalysis/channels/<ID>?format=json*/
        "isSupportIntelligentAnalysisLinkage": true,
        /*ro, opt, bool, whether it supports trigger configuration of radar intelligent analysis, desc:/ISAPI/Radar/intelligentAnalysisLinkage?format=json*/
        "isSupportIntelligentAnalysisInfo": true,
        /*ro, opt, bool, whether it supports getting basic information of radar intelligent analysis, desc:/ISAPI/Radar/intelligentAnalysisInfo?
format=json*/
        "isSupportIntelligentAnalysisWeekPlan": true,
        /*ro, opt, bool, whether it supports weekly schedule of intelligent analysis of radar, desc:desc: /ISAPI/Radar/intelligentAnalysisWeekPlan?
format=json*/
        "isSptOverStayTimeCfg": true,
        /*ro, opt, bool, whether it supports the overstay duration configuration, desc:/ISAPI/Radar/Configuration/zones/overStayTimeCfg/capabilities?
format=json*/
        "isSptBaseStationCfg": true,
        /*ro, opt, bool, whether it supports ranging base station configuration, desc:/ISAPI/Radar/Configuration/baseStationCfg/capabilities?format=json*/
        "isSupportDisplayMode": true,
        /*ro, opt, bool, whether it supports configuring the display mode of radar detection area, desc:/ISAPI/Radar/Configuration/displayMode?format=json*/
        "isSupportLidarSensitivity": true,
        /*ro, opt, bool, whether it supports configuring laser radar sensitivity, desc:/ISAPI/Radar/Configuration/lidarSensitivity?format=json*/
        "isSupportSizeThresholdCfg": true,
        /*ro, opt, bool, whether it supports size threshold configuration, desc:/ISAPI/Radar/Configuration/sizeThresholdCfg?format=json*/
        "isSupportRadarRelay": true,
        /*ro, opt, bool, whether it supports radar relay, desc:/ISAPI/Radar/relay*/
        "isSupportDetectSwitch": true,
        /*ro, opt, bool, whether it supports radar detection switch, desc:/ISAPI/Radar/Configuration/detectSwitch/capabilities?format=json*/
        "RadarMisalignmentCap": {
        /*ro, opt, object, configuration capability of radar direction deviation, desc:/ISAPI/Radar/Configuration/misalignment*/
            "misalignmentAngle": {
            /*ro, opt, object, radar direction angle deviation, desc:radar's level calibration inaccuracy due to the horizontal direction deviation between
the external component and the camera caused during device delivery or by external damage*/
                "@min": -5.0,
                /*ro, opt, float, the minimum value*/
                "@max": 5.0,
                /*ro, opt, float, the maximum value*/
                "@def": 0.0
                /*ro, opt, float, the default value*/
            }
        },
        "isSupportPointCloudFilterAreaCfg": true,
        /*ro, opt, bool, whether it supports configuring point cloud filtering area, desc:/ISAPI/Radar/Configuration/pointCloudFilterAreaCfg/capabilities?
format=json*/
        "isSupportGetEnvironmentInfo": true,
        /*ro, opt, bool, whether it supports getting manual learning results of radar environment, desc:/ISAPI/Radar/Control/environmentInfo?format=json*/
        "isSupportAddCustomAudio": true,
        /*ro, opt, bool, whether it supports importing custom audio files, desc:/ISAPI/Radar/customAudio/addCustomAudio?format=json*/
        "isSupportDeleteCustomAudio": true,
        /*ro, opt, bool, whether it supports deleting custom audio files, desc:/ISAPI/Radar/customAudio/deleteCustomAudio?format=json*/
        "isSupportPlayCustomAudio": true,
        /*ro, opt, bool, whether it supports playing custom audio files, desc:/ISAPI/Radar/customAudio?format=json*/
        "isSupportGPSInfoCfg": true,
        /*ro, opt, bool, whether it supports configuring radar GPS information, desc:/ISAPI/Radar/Configuration/coordinateCfg/capabilities?format=json*/
        "isSupportOneWaySpeaking": true,
        /*ro, opt, bool, whether it supports one-way speaking, desc:the web client speaks to the radar and the device will play the received audio data. The
web client only supports one-way speaking and cannot receive anything*/
        "isSupportUpgradeProgress": true,
        /*ro, opt, bool, whether it supports getting the radar firmware upgrade progress, desc:/ISAPI/Radar/upgrade/progress?format=json*/
        "isSupportIntegratedCalibratingCfg": true,
        /*ro, opt, bool, whether it supports configuring integrated radar calibration, desc:true (support), this node is not returned (not support); related
URI: /ISAPI/Radar/integratedCalibrating/paramCfg/capabilities?format=json*/
        "isSupportImagingRadarPanelROIParamCfg": true,
        /*ro, opt, bool, whether it supports configuring ROI of radar imaging host, desc:true (support), this node is not returned (not support); related
URI: /ISAPI/Radar/ImagingRadarPanel/ROIParamCfg/capabilities?format=json*/
        "isSupportImagingRadarPanelShowParamCfg": true,
        /*ro, opt, bool, whether it supports configuring display parameters of radar imaging host, desc:true (support), this node is not returned (not
support); related URI: /ISAPI/Radar/ImagingRadarPanel/ShowParamCfg/capabilities?format=json&devIndex=<deviceID>*/
        "isSupportImagingRadarPanelPositionParamCfg": true,
        /*ro, opt, bool, whether it supports configuring radar imaging host location, desc:true (support), this node is not returned (not support); related
URI: /ISAPI/Radar/ImagingRadarPanel/PositionParamCfg/capabilities?format=json&devIndex=<deviceID>*/
        "isSupportImagingRadarPanelScanParamCfg": true,
        /*ro, opt, bool, whether it supports configuring material scanning of radar imaging host, desc:true (support), this node is not returned (not
support); related URI: /ISAPI/Radar/ImagingRadarPanel/Scan/ParamCfg/capabilities?format=json&devIndex=<deviceID>*/
        "isSupportImagingRadarPanelScanTask": true,
        /*ro, opt, bool, whether it supports material scanning tasks of radar imaging host, desc:true (support), this node is not returned (not support);
related URI: /ISAPI/Radar/ImagingRadarPanel/Scan/Task/capabilities?format=json*/
        "isSupportImagingRadarPanelSearchScanRecord": true,
        /*ro, opt, bool, whether it supports searching for material scanning records of radar imaging host, desc:true (support), this node is not returned
(not support); related URI: /ISAPI/Radar/ImagingRadarPanel/Scan/SearchRecord/capabilities?format=json*/
        "isSupportImagingRadarPanelBasicParamCfg": true,
        /*ro, opt, bool, whether it supports configuring basic parameters of radar imaging host, desc:true (support), this node is not returned (not
support); related URI: /ISAPI/Radar/ImagingRadarPanel/BasicParamCfg/capabilities?format=json&devIndex=<deviceID>*/
        "isSupportImagingRadarPanelCustomRegionParamCfg": true,
        /*ro, opt, bool, whether it supports configuring custom area of radar imaging host point cloud, desc:true (support), this node is not returned (not
support); related URI: /ISAPI/Radar/ImagingRadarPanel/CustomRegionParamCfg/capabilities?format=json*/
        "isSupportImagingRadarPanelVolumeCorrectionParamCfg": true,
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```
        "isSupportImagingRadarPanelVolumeCorrectionParamCfg": true,
        /*ro, opt, bool, whether it supports configuring material volume correction of radar imaging host, desc:true (support), this node is not returned (not support); related URI: /ISAPI/Radar/ImagingRadarPanel/Scan/VolumeCorrectionParamCfg/capabilities?format=json*/
        "isSupportImagingRadarPanelThresholdValue": true,
        /*ro, opt, bool, whether it supports configuring material threshold of radar imaging host, desc:true (support), this node is not returned (not support); related URI: /ISAPI/Radar/ImagingRadarPanel/Scan/ThresholdValue/capabilities?format=json*/
        "isSupportScreenPlanCfg": true,
        /*ro, opt, bool, whether it supports configuring radar screen display schedule, desc:true (support), this node is not returned (not support); related URI: /ISAPI/Radar/ScreenPlan/capabilities?format=json*/
        "isSupportBackgroundPoint": true,
        /*ro, opt, bool, whether it supports importing coordinate information of the background picture, desc:true (support), this node is not returned (not support); related URI: /ISAPI/Radar/BackgroundPoint/capabilities?format=json*/
        "isSupportRadarVideoDetectionCapturePicture": true,
        /*ro, opt, bool, whether it supports configuring radar-assisted capture parameters, desc:true (support), this node is not returned (not support); related URI: /ISAPI/Intelligent/channels/<channelID>/radarVideoDetection/capturePicture/capabilities?format=json*/
        "isSupportDeleteAutoCorrectionData": true,
        /*ro, opt, bool, whether it supports deleting radar auto-correction data, desc:true (support), this node is not returned (not support); related URI: Delete /ISAPI/Radar/AutoCorrectionData?format=json*/
        "isSupportLEDDisplayCfg": true,
        /*ro, opt, bool, whether it supports LED display configuration, desc:true (support), this node is not returned (not support); related URI: /ISAPI/Radar/Configuration/LEDDisplayCfg/capabilities?format=json*/
        "isSupportSetCameraControlStrategy": true,
        /*ro, opt, bool, whether it supports configuring control strategy parameters of radar linked with camera for the specified channel, desc:POST /ISAPI/Radar/RadarLinkageCameraMgr/channels/<channelID>/SetCameraControlStrategy?format=json*/
        "isSupportGetCameraControlStrategyList": true,
        /*ro, opt, bool, whether it supports getting control strategy parameters of radar linked with camera for all channels, desc:POST /ISAPI/Radar/RadarLinkageCameraMgr/GetCameraControlStrategyList?format=json*/
        "isSupportGetAutoCorrectionParamList": true,
        /*ro, opt, bool, whether it supports getting auto-correction parameters of all channels, desc:POST /ISAPI/Radar/TargetTracking/GetAutoCorrectionParamList?format=json*/
        "isSupportSetAutoCorrectionParam": true,
        /*ro, opt, bool, whether it supports auto-correction configuration of the specified channel, desc:POST /ISAPI/Radar/TargetTracking/channels/<channelID>/SetAutoCorrectionParam?format=json*/
        "isSupportRadarFrontInfo": true,
        /*ro, opt, bool, whether it supports radar array information, desc:GET /ISAPI/Radar/RadarParam/RadarFrontInfo?format=json*/
        "isSupportSetRadarModuleDebugCommand": true,
        /*ro, opt, bool, whether the device supports configuring debugging commands of radar module, desc:POST /ISAPI/Radar/RadarParam/SetRadarModuleDebugCommand?format=json*/
        "isSupportPersonLocationDetectionRule": true,
        /*ro, opt, bool, whether the device supports parameters of person location detection rules, desc:GET/PUT /ISAPI/Radar/PersonLocationDetectionRule?format=json*/
        "isSupportPeopleFallingDownDetectionRule": true,
        /*ro, opt, bool, whether the device supports parameters of person fall detection rules, desc:GET/PUT /ISAPI/Radar/PeopleFallingDownDetectionRule?format=json*/
        "isSupportPeopleVitalSignsDetectionRule": true,
        /*ro, opt, bool, whether the device supports parameters of vital sign detection rules, desc:GET/PUT /ISAPI/Radar/PeopleVitalSignsDetectionRule?format=json*/
        "isSupportPeopleGetUpRule": true,
        /*ro, opt, bool, whether the device supports parameters of person get-up detection rules, desc:GET/PUT /ISAPI/Radar/PeopleGetUpRule?format=json*/
        "isSupportPeopleBodyMovingRule": true,
        /*ro, opt, bool, whether the device supports parameters of body movement detection rules, desc:GET/PUT /ISAPI/Radar/PeopleBodyMovingRule?format=json*/
        "isSupportRadarModuleDebug": true,
        /*ro, opt, bool, whether the device supports radar module debugging, desc:POST /ISAPI/Radar/RadarParam/RadarModuleDebug?format=json*/
        "isSupportGetTargetLocationInfo": true,
        /*ro, opt, bool*/
        "isSupportGetFilterModelParamList": true,
        /*ro, opt, bool*/
        "isSupportSetFilterModelParam": true,
        /*ro, opt, bool*/
        "isSupportSearchLANRadarModuleList": true,
        /*ro, opt, bool*/
        "isSupportSetLANRadarModule": true,
        /*ro, opt, bool*/
        "isSupportSendCalibrationObjectList": true,
        /*ro, opt, bool*/
        "isSupportExportRadarDebugFile": true,
        /*ro, opt, bool*/
        "isSupportSearchVolumeCorrectionParam": true,
        /*ro, opt, bool*/
        "isSupportSearchCalibrationObject": true,
        /*ro, opt, bool*/
        "isSupportGetRadarDisplayRange": true,
        /*ro, opt, bool*/
        "isSupportSetScanDataSamplingParam": true,
        /*ro, opt, bool*/
        "isSupportGetScanDataSamplingParam": true,
        /*ro, opt, bool*/
        "isSupportDeleteCalibrationObject": true,
        /*ro, opt, bool*/
        "isSupportExecuteCalibrationObject": true,
        /*ro, opt, bool*/
        "isSupportSubCameraManualCalibrating": true,
        /*ro, opt, bool*/
        "isSupportPeopleFallingDownDetectionTriggerParams": true,
        /*ro, opt, bool*/
        "isSupportPeopleFallingDownDetectionScheduleParams": true,
        /*ro, opt, bool*/
        "isSupportModifyCalibrationObject": true,
        /*ro, opt, bool*/
        "isSupportExecuteRadarSetUpCalibration": true,
        /*ro, opt, bool*/
        "isSupportRadarShieldAreaCalibrationParams": true,
        /*ro, opt, bool*/
```


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        "isSupportRadarGoodsLocationCalibrationParams": true,
        /*ro, opt, bool*/
        "isSupportPeopleSedentaryScheduleParams": true,
        /*ro, opt, bool, whether the device supports arming schedule parameters of sitting detection, desc:GET or PUT
/ISAPI/Radar/PeopleSedentaryScheduleParams?format=json*/
        "isSupportPeopleSedentaryRuleParams": true,
        /*ro, opt, bool, whether the device supports arming rule parameters of sitting detection, desc:GET or PUT /ISAPI/Radar/PeopleSedentaryRuleParams?
format=json*/
        "isSupportPeopleSedentaryTriggerParam": true,
        /*ro, opt, bool, whether the device supports linkage method parameters of sitting detection, desc:GET or PUT
/ISAPI/Radar/PeopleSedentaryTriggerParam?format=json*/
        "isSupportGetRadarSetUpCalibrationStatus": true,
        /*ro, opt, bool, whether the device supports getting radar installation calibration status, desc:GET
/ISAPI/Radar/RadarCalibration/GetRadarSetUpCalibrationStatus?format=json*/
        "isSupportRadarHeatmap": true,
        /*ro, opt, bool*/
        "isSupportSendRadarModulePowerSupplyCmd": true,
        /*ro, opt, bool*/
        "isSupportBarrierDetectionOverlayParams": true,
        /*ro, opt, bool*/
        "isSupportRadarRelayMapTargetTrajectoryParams": true,
        /*ro, opt, bool*/
        "isSupportSearchTargetHistoryTrajectory": true,
        /*ro, opt, bool*/
        "isSupportRadarChannelVCAResource": true,
        /*ro, opt, bool*/
        "isSupportSearchTargetTrajectoryStatistics": true,
        /*ro, opt, bool*/
        "isSupportRadarRelayMapSectionParams": true,
        /*ro, opt, bool*/
        "isSupportRadarRelayMapCrossingParams": true,
        /*ro, opt, bool*/
        "isSupportRadarDetectFrameRate": true,
        /*ro, opt, bool*/
        "isSupportGetPTZInfoToRadarPosition": true,
        /*ro, opt, bool*/
        "isSupportGetRadarPositionToPTZInfo": true,
        /*ro, opt, bool*/
        "isSupportGotoLinkageCameraScene": true,
        /*ro, opt, bool*/
        "isSupportLinkageCameraSceneList": true,
        /*ro, opt, bool*/
        "isSupportLinkageCameraScene": true,
        /*ro, opt, bool*/
        "isSupportRadarFallenObjectsParams": true,
        /*ro, opt, bool*/
        "isSupportSearchTargetHistoryTrajectoryStatistics": true,
        /*ro, opt, bool*/
        "isSupportExportMaterialPointCloudFile": true,
        /*ro, opt, bool*/
        "isSupportSearchMaterialPointCloudFile": true,
        /*ro, opt, bool*/
        "isSupportImportMaterialPointCloudFile": true,
        /*ro, opt, bool*/
        "isSupportRecalculateTargetVolume": true,
        /*ro, opt, bool*/
        "isSupportRadarTrackRatio": true,
        /*ro, opt, bool*/
        "isSupportRadarFeaturePointCalibration": true,
        /*ro, opt, bool*/
        "isSupportImportRadarDebugFile": true,
        /*ro, opt, bool*/
        "isSupportClutterAutoLearningCtrl": true,
        /*ro, opt, bool*/
        "isSupportClutterPatternCalibrationParams": true,
        /*ro, opt, bool*/
        "isSupportGetGroundModelingParamsList": true,
        /*ro, opt, bool*/
        "isSupportSetGroundModelingParams": true,
        /*ro, opt, bool*/
        "isSupportModifyMaterialVolumeCalculateScene": true,
        /*ro, opt, bool*/
        "isSupportAddMaterialVolumeCalculateScene": true,
        /*ro, opt, bool*/
        "isSupportGetMaterialVolumeCalculateSceneList": true,
        /*ro, opt, bool*/
        "isSupportDeleteMaterialVolumeCalculateScene": true,
        /*ro, opt, bool*/
        "isSupportStartCollectCalibrationMaterial": true,
        /*ro, opt, bool*/
        "isSupportStopCollectCalibrationMaterial": true,
        /*ro, opt, bool*/
        "isSupportImportCalibrationFile": true,
        /*ro, opt, bool*/
        "isSupportExportRadarDebugFileByBinary": true,
        /*ro, opt, bool*/
        "isSupportGetRadarCalibrateParams": true,
        /*ro, opt, bool*/
        "isSupportEnvironmentMisParams": true,
        /*ro, opt, bool*/
        "isSupportEnvironmentManualLearningByRadarID": true,
        /*ro, opt, bool*/
        "isSupportTimingScanParam": true,
        /*ro, opt, bool*/
```

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```
        /*ro, opt, bool*/
        "isSupportRadarCalibrationAutoAmendStatus": true,
        /*ro, opt, bool*/
        "isSupportRadarCalibrationAngleParam": true,
        /*ro, opt, bool*/
        "isSupportRadarTargetDetectionParams": true,
        /*ro, opt, bool*/
        "isSupportRadarPerimeterOverlapParams": true,
        /*ro, opt, bool*/
        "isSupportPeopleVitalSignsAbnormalRuleParams": true,
        /*ro, opt, bool, whether the device supports vital signs exception detection parameters, desc:GET or PUT
/ISAPI/Radar/PeopleVitalSignsAbnormalRuleParams?format=json*/
        "isSupportPeopleLongTimeLeaveScheduleParams": true,
        /*ro, opt, bool, whether the device supports arming schedule parameters of person long-time off-bed detection, desc:GET or PUT
/ISAPI/Radar/PeopleLongTimeLeaveScheduleParams?format=json*/
        "isSupportPeopleLongTimeLeaveRuleParams": true,
        /*ro, opt, bool, whether the device supports arming rule parameters of person long-time off-bed detection, desc:GET or PUT
/ISAPI/Radar/PeopleLongTimeLeaveRuleParams?format=json*/
        "isSupportSearchPeopleVitalSignsData": true,
        /*ro, opt, bool, whether the device supports searching for vital sign data, desc:POST /ISAPI/Radar/SearchPeopleVitalSignsData?format=json*/
        "isSupportCreateExportPeopleVitalSignsTask": true,
        /*ro, opt, bool, whether the device supports creating vital sign data export tasks, desc:POST /ISAPI/Radar/CreateExportPeopleVitalSignsTask?
format=json*/
        "isSupportGetPeopleVitalSignsExportTask": true,
        /*ro, opt, bool, whether the device supports getting details of vital sign data export tasks, desc:POST /ISAPI/Radar/GetPeopleVitalSignsExportTask?
format=json*/
        "isSupportPersonRetentionRuleParams": true,
        /*ro, opt, bool, whether the device supports people overstay detection, desc:GET/PUT /ISAPI/Radar/PersonRetentionRuleParams?format=json*/
        "isSupportMaterialVolumeCalculateAlgParams": true,
        /*ro, opt, bool, whether the device supports object volume calculation method parameters, desc:GET or PUT
/ISAPI/Radar/MaterialVolumeCalculate/MaterialVolumeCalculateAlgParams?format=json*/
        "isSupportSearchPeopleBodyMovingData": true,
        /*ro, opt, bool, whether the device supports searching for body movement data, desc:POST /ISAPI/Radar/SearchPeopleBodyMovingData?format=json*/
        "isSupportSearchPeopleLongTimeLeaveData": true,
        /*ro, opt, bool, whether the device supports searching person long-time off-bed data, desc:POST /ISAPI/Radar/SearchPeopleLongTimeLeaveData?
format=json*/
        "isSupportCreateExportPeopleLongTimeLeaveTask": true,
        /*ro, opt, bool, whether the device supports creating tasks of exporting person long-time off-bed data, desc:POST
/ISAPI/Radar/CreateExportPeopleLongTimeLeaveTask?format=json*/
        "isSupportGetPeopleLongTimeLeaveExportTask": true,
        /*ro, opt, bool, whether the device supports getting export task details of long-time off-bed data, desc:POST
/ISAPI/Radar/GetPeopleLongTimeLeaveExportTask?format=json*/
        "isSupportGetPeopleBodyMovingExportTask": true,
        /*ro, opt, bool, whether the device supports getting details of body movement data export tasks, desc:POST
/ISAPI/Radar/GetPeopleBodyMovingExportTask?format=json*/
        "isSupportCreateExportPeopleBodyMovingTask": true,
        /*ro, opt, bool, whether the device supports creating tasks of exporting body movement data, desc:POST
/ISAPI/Radar/CreateExportPeopleBodyMovingTask?format=json*/
        "isSupportPeopleVitalSignsStatus": true,
        /*ro, opt, bool, whether the device supports vital signs status, desc:GET /ISAPI/Radar/PeopleVitalSignsStatus?format=json*/
        "isSupportSleepingDetectionParams": true,
        /*ro, opt, bool*/
        "isSupportPeopleSleepingDetectionScheduleParams": true,
        /*ro, opt, bool*/
        "isSupportSearchPeopleSleepingDetectionReport": true,
        /*ro, opt, bool*/
        "isSupportGetWeekTimingScanPlan": true,
        /*ro, opt, bool*/
        "isSupportModifyWeekTimingScanPlan": true,
        /*ro, opt, bool*/
        "isSupportImportMotionTransformerModel": true,
        /*ro, opt, bool*/
        "isSupportExportMotionTransformerModel": true,
        /*ro, opt, bool*/
        "isSupportImportRadarRelayChannelizedDataFile": true,
        /*ro, opt, bool*/
        "isSupportExportRadarRelayChannelizedDataFile": true,
        /*ro, opt, bool*/
        "isSupportClearPointCloudICPIRectification": true,
        /*ro, opt, bool*/
        "isSupportPointCloudICPRectification": true,
        /*ro, opt, bool*/
        "isSupportPointCloudCameraManualCalibrationStatus": true,
        /*ro, opt, bool*/
        "isSupportPointCloudCameraManualCalibration": true,
        /*ro, opt, bool*/
        "isSupportRadarFallenObjectsAdvancedParams": true,
        /*ro, opt, bool*/
        "isSupportGetCalibratingStatus": true,
        /*ro, opt, bool*/
        "isSupportModifySinglePeopleLongTimeLeaveRule": true
        /*ro, opt, bool*/
    }
}
```

#### 14.12.1.2 Get the capability of getting the radar information

**Request URL**

GET /ISAPI/Radar/radarInfo/capabilities?format=json&radarChannelID=<radarChannelID>

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**Query Parameter**

| Parameter Name | Parameter Type | Description |
| --- | --- | --- |
| radarChannelID | string | URL is applied with the radarChannelID parameter configured: /ISAPI/Radar/radarInfo/capabilities?format=json&radarChannelID=1. If the parameter is not configured, it is the radar channel 1 by default. |

**Request Message**

None

**Response Message**

```
{
    "RadarInfoCap": {
    /*ro, opt, object*/
        "firmwareVersion":  true,
        /*ro, opt, bool, whether it supports getting the software version information*/
        "hardwareVersion":  true,
        /*ro, opt, bool, whether it supports getting the hardware version information*/
        "status": {
        /*ro, opt, object, status*/
            "@opt": ["normal", "abnormal", "updating", "rebooting"]
            /*ro, req, array, options, subType:string, desc:"normal", "abnormal"*/
        },
        "serialNumber": {
        /*ro, opt, object, radar serial No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  128
            /*ro, req, int, the maximum value*/
        },
        "thermometryUnit": {
        /*ro, opt, object, temperature unit*/
            "@opt": ["celsius", "fahrenheit", "kelvin"]
            /*ro, opt, array, options, subType:string, desc:"celsius" (Celsius), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
        },
        "currTemperature": {
        /*ro, opt, object, it should be accurate to one decimal place, desc:it should be accurate to one decimal place*/
            "@min":  0.0,
            /*ro, opt, float, the minimum value*/
            "@max":  0.0
            /*ro, opt, float, the maximum value*/
        },
        "detailFirmwareVersion": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "temperatureStatus": {
        /*ro, opt, object*/
            "@opt": ["normal", "highOrLow", "extremelyHighOrLow"]
            /*ro, req, array, subType:string*/
        },
        "voltageStatus": {
        /*ro, opt, object*/
            "@opt": ["normal", "relativelyHigh", "extremelyHigh"]
            /*ro, req, array, subType:string*/
        },
        "engineStatus": {
        /*ro, opt, object*/
            "opt": ["normal", "warning", "error"]
            /*ro, req, array, subType:string*/
        },
        "dirtyObstruction": {
        /*ro, opt, object*/
            "@opt": ["yes", "no"]
            /*ro, req, array, subType:string*/
        },
        "life": {
        /*ro, opt, object*/
            "opt": ["normal", "low"]
            /*ro, req, array, subType:string*/
        }
    }
}
```

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### 14.12.1.3 Get the radar information

**Request URL**

GET /ISAPI/Radar/radarInfo?format=json&radarChannelID=<radarChannelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| radarChannelID | string | URL is applied with the radarChannelID parameter configured: /ISAPI/Radar/radarInfo?format=json&radarChannelID=1. If the parameter is not configured, it is the radar channel 1 by default. |

**Request Message**

None

**Response Message**

```
{
    "RadarInfo": {
    /*ro, opt, object*/
        "firmwareVersion":  "v1.1.0 221010",
        /*ro, opt, string, radar software version information, range:[1,64]*/
        "detailFirmwareVersion":  "XXXXXX",
        /*ro, opt, string, range:[1,64]*/
        "hardwareVersion":  "V1.0.0",
        /*ro, opt, string, radar hardware version information, range:[1,64]*/
        "algorithmVersionList": [
        /*ro, opt, array, subType:object, range:[0,1]*/
            {
                "algorithmName":  "traffic target detection",
                /*ro, opt, string, range:[0,64]*/
                "algorithmVersion":  "V2.0.1"
                /*ro, req, string, range:[0,64]*/
            }
        ],
        "status":  "normal",
        /*ro, opt, enum, radar status information, subType:string, desc:"normal", "abnormal"*/
        "serialNumber":  "iDS-9632NX-I8 / S",
        /*ro, opt, string, radar serial No., range:[1,128]*/
        "thermometryUnit":  "celsius",
        /*ro, opt, enum, temperature unit, subType:string, desc:temperature unit*/
        "currTemperature":  50.0,
        /*ro, opt, float, it should be accurate to one decimal place, desc:it should be accurate to one decimal place*/
        "temperatureStatus":  "normal",
        /*ro, opt, enum, subType:string*/
        "voltageStatus":  "normal",
        /*ro, opt, enum, subType:string*/
        "engineStatus":  "normal",
        /*ro, opt, enum, subType:string*/
        "dirtyObstruction":  "yes",
        /*ro, opt, enum, subType:string*/
        "life":  "normal"
        /*ro, opt, enum, subType:string*/
    }
}
```

## 14.12.2 Radar Correction

### 14.12.2.1 Get the capability of coordinate calibration parameters

**Request URL**

GET /ISAPI/Radar/calibration/capabilities?format=json&radarChannelID=<radarChannelID>&channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| radarChannelID | string | The radar channel No. If "isSupportRadarChannelCalibration" is returned with true value in the response of the API (/ISAPI/Radar/capabilities?format=json), the device supports this parameter. |
| channelID | string | The channel No. If "isSupportChannelCalibration" is returned with true value in the response of the API (/ISAPI/Radar/capabilities?format=json), the device supports this parameter. |

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### Request Message

None

### Response Message

```
{
    "RadarCalibration": {
    /*ro, req, object, radar coordinate calibration*/
        "mode": {
        /*ro, opt, object, calibration mode, desc:"mobileDev" (mobile device calibration mode, which requires to configure parameters: totalLaneCount, LaneParamList, and LaneRightBoundaryLine), "coordinates" (coordinates mode), "angel" (angel mode), "auto" (auto calibration mode), "polarCoordinates" (polar coordinate mode, the coordinates on radar are polar coordinates), "mandatoryAuto" (forced auto calibration mode)*/
            "@opt": ["coordinates", "angel", "auto", "polarCoordinates", "mobileDev", "mandatoryAuto"]
            /*ro, req, array, the value range, subType:string*/
        },
        "setupHeight": {
        /*ro, opt, object, construction height*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  15.0
            /*ro, req, float, the maximum value*/
        },
        "coordinates": {
        /*ro, opt, object, coordinate mode*/
            "@size":  4,
            /*ro, req, int, supported number of calibration points*/
            "coordinate": {
            /*ro, opt, object, coordinate mode*/
                "radarX": {
                /*ro, req, object, x-coordinate*/
                    "@min":  0.000,
                    /*ro, req, float, minimum value of radar x-coordinate*/
                    "@max":  350.000
                    /*ro, req, float, maximum value of radar x-coordinate*/
                },
                "radarY": {
                /*ro, req, object, y-coordinate*/
                    "@min":  0.000,
                    /*ro, req, float, minimum value of radar y-coordinate*/
                    "@max":  350.000
                    /*ro, req, float, maximum value of radar y-coordinate*/
                },
                "positionX": {
                /*ro, req, object, corresponding video x-coordinate, desc:the value = normalized coordinates x 1000*/
                    "@min":  0,
                    /*ro, req, int, minimum value of video x-coordinate*/
                    "@max":  1000
                    /*ro, req, int, maximum value of video x-coordinate*/
                },
                "positionY": {
                /*ro, req, object, corresponding video y-coordinate, desc:the value = normalized coordinates x 1000*/
                    "@min":  0,
                    /*ro, req, int, minimum value of video y-coordinate*/
                    "@max":  1000
                    /*ro, req, int, maximum value of video y-coordinate*/
                }
            }
        },
        "angel": {
        /*ro, opt, object, angle mode*/
            "horzDistCorrect": {
            /*ro, opt, object, horizontal distance correction*/
                "@min":  -25,
                /*ro, req, int, the minimum value*/
                "@max":  25
                /*ro, req, int, the maximum value*/
            },
            "vertDistCorrect": {
            /*ro, opt, object, vertical distance correction*/
                "@min":  -25,
                /*ro, req, int, the minimum value*/
                "@max":  25
                /*ro, req, int, the maximum value*/
            },
            "horizontalRotation": {
            /*ro, req, object, horizontal rotation angle*/
                "@min":  -15.0,
                /*ro, req, float, the minimum value*/
                "@max":  15.0
                /*ro, req, float, the maximum value*/
            },
            "pitchRotation": {
            /*ro, req, object, tilted rotation angle*/
                "@min":  -15.0,
                /*ro, req, float, the minimum value*/
                "@max":  15.0
                /*ro, req, float, the maximum value*/
            }
        },
        "auto": {
        /*ro, opt, object, auto calibration mode*/
```

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```
        /*ro, opt, object, auto calibration mode*/
            "calibrationOpt": {
            /*ro, opt, object, calibration operation*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  2
                /*ro, req, int, the maximum value*/
            },
            "calibrationState":  "unknown",
            /*ro, req, string, automatic calibration status*/
            "calibrationProgress": {
            /*ro, opt, object, automatic calibration progress*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  100
                /*ro, req, int, the maximum value*/
            },
        },
    }
}
```

#### 14.12.2.2 Set the coordinates calibration parameters

**Request URL**

PUT /ISAPI/Radar/calibration?format=json&radarChannelID=<radarChannelID>&channelID=<channelID>

**Query Parameter**

| Parameter Name | Parameter Type | Description |
|---|---|---|
| radarChannelID | string | The radar channel No. If "isSupportRadarChannelCalibration" is returned with true value in the response of the API (/ISAPI/Radar/capabilities?format=json), the device supports this parameter. |
| channelID | string | The channel No. If "isSupportChannelCalibration" is returned with true value in the response of the API (/ISAPI/Radar/capabilities?format=json), the device supports this parameter. |

**Request Message**

```
{
    "RadarCalibration": {
    /*req, object, radar coordinate calibration*/
        "mode":  "mobileDev",
        /*opt, enum, calibration mode, subType:string, desc:"mobileDev" (mobile device calibration mode, which requires to configure parameters:
totalLaneCount, LaneParamList, and LaneRightBoundaryLine), "coordinates" (coordinates mode), "angel" (angel mode), "auto" (auto calibration mode),
"polarCoordinates" (polar coordinate mode, the coordinates on radar are polar coordinates), "mandatoryAuto" (forced auto calibration mode)*/
        "setupHeight":  1.00,
        /*opt, float, construction height, unit:m*/
        "coordinates": [
        /*opt, array, coordinate mode, subType:object*/
            {
                "radarX":  100.001,
                /*req, float, x-coordinate, range:[0.000,350.000], unit:m*/
                "radarY":  100.001,
                /*req, float, y-coordinate, range:[0.000,350.000], unit:m*/
                "positionX":  100,
                /*ro, req, int, corresponding video x-coordinate, range:[0,1000], step:1, desc:the value = normalized coordinates x 1000*/
                "positionY":  100
                /*ro, opt, int, corresponding video y-coordinate, range:[0,1000], step:1, desc:the value = normalized coordinates x 1000*/
            }
        ],
        "angel": {
        /*opt, object, angle mode*/
            "horzDistCorrect":  0,
            /*opt, int, horizontal distance correction, unit:m*/
            "vertDistCorrect":  0,
            /*opt, int, vertical distance correction, unit:m*/
            "horizontalRotation":  15,
            /*req, int, horizontal rotation angle*/
            "pitchRotation":  15
            /*req, int, tilted rotation angle*/
        },
        "auto": {
        /*opt, object, auto calibration mode*/
            "calibrationOpt":  1,
            /*req, enum, calibration operation, subType:int, desc:1 (enable auto calibration), 2 (manually stop calibration). The calibration value applied
by API of setting coordinates calibration parameters can be 1 and 2, while returned by API of getting coordinates calibration parameters is 0*/
            "calibrationState":  "unknown",
            /*ro, opt, string, automatic calibration status*/
            "calibrationProgress":  50,
            /*ro, opt, int, automatic calibration progress, range:[0,100]*/
```

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```
            /*ro, opt, int, automatic calibration progress, range:[0,100]*/
            "calibrationLoss": [0, 0, 0, 0, 0, 0, 0, 0],
            /*ro, opt, array, calibration section error, subType:int, range:[0,8], desc:read-only. Error of the calibration result returned by the algorithm after automatic calibration. Unit: dm. One error value is outputted per 50 meters*/
            "calibrationFinishTime":  "2024-05-03T17:30:08+08:00"
            /*ro, opt, datetime, calibration latest completion time*/
        },
        "polarCoordinates": [
        /*opt, array, polar coordinates mode, subType:object*/
            {
                "angle":  120.00,
                /*req, float, radar's polar coordinates angle, range:[0,360]*/
                "distance":  20.00,
                /*req, float, radar's polar coordinates distance, range:[0,100], unit:m*/
                "positionX":  100,
                /*ro, req, int, corresponding video x-coordinate, range:[0,1000], step:1, desc:the value = normalized coordinates x 1000*/
                "positionY":  100
                /*ro, opt, int, corresponding video y-coordinate, range:[0,1000], step:1, desc:the value = normalized coordinates x 1000*/
            }
        ],
        "mandatoryAuto": {
        /*ro, opt, object, forced auto calibration mode*/
            "calibrationOpt":  0,
            /*ro, opt, int, calibration operation, desc:the calibration value applied by API of setting coordinates calibration parameters can be 1 and 2, while returned by API of getting coordinates calibration parameters is 0*/
            "calibrationState":  "unknown",
            /*ro, req, string, automatic calibration status*/
            "calibrationProgress":  50,
            /*ro, opt, int, automatic calibration progress, range:[0,100]*/
            "calibrationLoss": [0, 0, 0, 0, 0, 0, 0, 0],
            /*ro, opt, array, calibration section error, subType:int, range:[0,8], desc:read-only. Error of the calibration result returned by the algorithm after automatic calibration. Unit: dm. One error value is outputted per 50 meters*/
            "calibrationFinishTime":  "2024-05-03T17:30:08+08:00"
            /*ro, opt, datetime, calibration latest completion time*/
        },
        "laneConfigAloneEnabled":  false,
        /*req, bool, whether to enable the separate configuration of lane*/
        "totalLaneCount":  10,
        /*req, int, linked lane amount, range:[1,10]*/
        "LaneParamList": [
        /*opt, array, lane parameter list, subType:object*/
            {
                "LaneParam": {
                /*opt, object, lane parameters*/
                    "laneId":  1,
                    /*req, int, linked lane no., range:[1,99]*/
                    "LaneLine": {
                    /*opt, object, lane line*/
                        "lineName":  "laneLine",
                        /*req, string, lane line*/
                        "RegionCoordinatesList": [
                        /*opt, array, area coordinate list, subType:object*/
                            {
                                "RegionCoordinates": {
                                /*opt, object, coordinates*/
                                    "positionX":  50,
                                    /*req, int, x-coordinate, range:[1,1000]*/
                                    "positionY":  300
                                    /*req, int, y coordinate, range:[1,1000]*/
                                }
                            }
                        ]
                    }
                }
            }
        ],
        "LaneRightBoundaryLine": {
        /*opt, object, lane right boundary line*/
            "lineName":  "laneRightBoundaryLine",
            /*req, string, lane line name*/
            "RegionCoordinatesList": [
            /*opt, array, area coordinate list, subType:object*/
                {
                    "RegionCoordinates": {
                    /*opt, object, coordinates*/
                        "positionX":  50,
                        /*req, int, x-coordinate, range:[1,1000]*/
                        "positionY":  300
                        /*req, int, y coordinate, range:[1,1000]*/
                    }
                }
            ]
        },
        "laneConfigurationType":  "lineType",
        /*ro, opt, enum, lane configuration mode, subType:string, desc:"lineType" (lane line), "polygonalType" (polygon lane)*/
        "polygonLaneParam": [
        /*ro, opt, array, polygon lane, subType:object*/
            {
                "id":  1,
                /*ro, req, int, lane number, range:[1,16]*/
                "regionCoordinatesList": [
                /*ro, req, array, area position, subType:object*/
                    {
                        "regionCoordinates": {
```


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```
                    /*ro, opt, object, coordinates*/
                        "positionX":  0,
                        /*ro, req, int, x-coordinate, range:[0,1000]*/
                        "positionY":  0
                        /*ro, req, int, y coordinate, range:[0,1000]*/
                    }
                }
            ]
        }
    ]
  }
}
```

**Response Message**

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

# 15 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.

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